NARUTO IDLE RPG

Master Game Theory & Design Bible

An evolving, evidence-aware design specification with prior reports preserved in appendices

VERSION 0.10.32 | 02 NOVEMBER 2026 | PRIORITY 5C ACCESSIBLE RECOVERY AND TRANSACTION INTEGRITY

Design goal: build a game where a team wins because its members work together—not merely because the player owns the five highest-power shinobi.

WORKING DOCUMENT • RESEARCH/CONCEPT PHASE ONLY • LIVE TESTING NOT AUTHORIZED YET

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Single-source-of-truth rule

Future updates should edit this same master document and increment its version/change log. Do not create a separate competing design bible. Every new figure must be tagged as a verified game fact, a project decision, an illustrative proposal, or an open hypothesis.

- 16-19. Consolidation audit, integrated concept, resolved fairness questions and change log; Appendices A-D preserve four recovered reports, Appendix E tracks archive status, Appendix F preserves the user-supplied research text verbatim, and Appendix G contains maintenance rules.
- 29–57. Whole-game research through mobile UX information architecture: economy, recruitment, shared Hall growth, tactical encounter proof, the 15-node opening campaign, first-session narrative, and four-destination phone navigation.
- 58. Compact-phone content hierarchy, truthful microcopy and accessible recovery states
- 59. Accessible recovery, transaction uncertainty and first-loss integrity
- 20-28. Mobile-first combat engine, fairness model, research audits, combat rules/roster matrix, edge-case closure, and the research/concept-phase boundary with future validation gates.

1. How to use this master file

This is the continuous, editable design bible for the game project. It consolidates the design direction and requirements stated in our conversation, includes benchmark information that can currently be cited, and starts the first concrete fairness/composition specification. It is designed to evolve: later research should refine sections in place, not create a new master file.

- Decision status tags: [FACT] = supported by a cited source; [PROJECT DECISION] = direction explicitly selected by the project owner; [PROPOSAL] = recommended design setting not yet accepted; [HYPOTHESIS] = plausible but unvalidated; [OPEN] = still needs a decision or evidence; [FUTURE VALIDATION REQUIREMENT] = a future test gate not to be executed until the owner explicitly starts live testing.
- Numbers marked illustrative are not claimed to be industry standards. They are theoretical proposals or hypothetical starting points for a future prototype; do not present them as validated or run simulation/live-playtest tasks until the project owner explicitly starts that phase.
- When a future revision changes a settled proposal, record the previous value, new value, reason, and data supporting the change in the decision log.
- Sources are recorded at the end with access dates and a note on their reliability. Fan guides are not treated as authoritative proof of hidden or internal game systems.

2. Project brief and non-negotiable design principles

2.1 Product vision

Build a Naruto-inspired anime idle RPG with turn-based or auto-resolving squad combat, character collection, long-term progression, multiple modes, and a team-builder where composition, roles, synergy, weaknesses, and counters matter. The intended style is close enough to the genre to feel familiar, but the mechanics should form one internally consistent system rather than a pile of unrelated boosts.

- [PROJECT DECISION] The design must be friendlier to free-to-play players than systems that make spending feel required for ordinary progress.
- [PROJECT DECISION] Spending should be primarily pay-to-progress: save time, add convenience, buy cosmetics, or widen roster options—not purchase unavoidable competitive power.
- [PROJECT DECISION] Study many relevant games, not only AFK Arena and Mythic Heroes. Benchmark similar auto-battlers, turn-based hero collectors, and hybrid games separately.
- [PROJECT DECISION] Current research priority is team building, composition, synergy, roles, weaknesses, counters, gameplay, and game modes. Economy research is included when it affects access to viable team pieces.
- [LEGAL / PRODUCTION NOTE] A commercial game using Naruto characters, names, designs, story, symbols, or other protected elements requires appropriate rights/licensing. Until licensed, Naruto character examples in this document are hypothetical design illustrations, not a claim of authorization.

2.2 Fairness constitution (draft)

Principle	Draft rule
Viable roles	Every role required to answer common tactics is available through starters, standard acquisition, deterministic rewards, or an equivalent low-rarity unit.
Synergy at baseline	A character's defining tactical function works at base copy/rank. Duplicates may improve reliability or stats modestly but must not unlock the only way the kit functions.
No mandatory limited unit	No campaign or competitive archetype should require one limited hero, premium signature item, paid bond, or one specific duplicate threshold.
Counter access	Each strong strategy has at least two practical answers from different functional categories; one answer should be broadly obtainable.
Spend advantage	Purchases may accelerate development or widen roster choice. In normalized PvP, spending must not create a direct, unanswerable stat advantage.
Transparent rules	Summon odds, pity, pity carryover, duplicate conversion, event requirements, and PvP normalization are shown clearly in the game.
Variety over power score	The game must reward tactical substitutions and adaptation, not teach players to always field the highest displayed power score.
No assumed validation	All proposed formulas and fairness numbers remain provisional until instrumented simulations and representative player tests confirm them.

3. Research archive status and evidence rules

Important research-integrity note — corrected in Version 0.2. Four completed report outputs were recovered from the project Library (Executive Summary; Executive Summary (1); Executive Summary (2); Economy & Summoning Mechanics) and are reproduced in full in Appendices A–D. The first build omitted these reports; that was an error. The main body is the current synthesis, while appendices preserve the prior report text. Historical claims and citation markers are research leads, not automatic approval or proof of current game mechanics.

3.1 Evidence labels

Label	Meaning	How we use it
[FACT]	A specific mechanic is directly stated by an official source or supported by a current guide with clear scope.	Can inform benchmark comparisons; official sources receive higher confidence.
[COMMUNITY GUIDE]	A strategy or mechanic is documented by a third-party game guide or community resource.	Useful for practical team-building insights; verify if the mechanic is patch-sensitive.
[PROJECT DECISION]	The owner has stated the desired product direction.	Treat as a requirement unless later changed.
[PROPOSAL]	Our proposed rule or initial number.	Use in prototype and balance tests, not as a factual statement about other games.
[HYPOTHESIS]	A causal claim that needs observed data.	Design a test before relying on it.

3.2 What remains unverified

- We have not yet inspected every benchmark game hands-on in the same current patch, nor collected consistent win-rate or revenue data. Claims about long-term success or failure must therefore remain cautious.
- We have not validated the proposed currency income, pity budget, role power budget, PvP normalization, or exact damage formula in a playable prototype.
- Community tier lists and recommended teams are snapshots of a particular patch, mode, roster and investment level—not universal proof that a character or role is mandatory.

4. What benchmark games teach us so far

The goal of benchmarking is not to copy another game's roster or proprietary kit. It is to identify recurring design patterns, understand their trade-offs, and adapt only the parts that serve this game's fairness and team-identity goals.

Game	Subgenre / structure	Observed pattern	Transferable design lesson	Evidence status
AFK Arena	Idle auto-battler; five-hero lineup and faction/role framework.	Faction advantage and lineup bonuses create team-level choices. A current community guide describes the core faction wheel and bonuses that can reach +25% ATK/HP for five heroes of one core faction; patch notes also show faction-specific event variants.	Use factions as a mild build incentive, not as an overwhelming requirement. Build enough cross-faction utility that the best tactical answer can beat a slightly larger faction bonus. [S1, S2]	Community guide / patch notes; faction tuning is patch-sensitive.
AFK Journey	Idle squad RPG with map positioning, auto-combat, multiple modes.	Current guide material describes a common pattern of a three-unit faction core with two flexible synergy slots; mono-faction bonuses are not always worth losing utility. A current team builder supports up to five heroes and formation placement.	Keep a readable soft faction core, but avoid a mandatory five-stack. Make the flex slots meaningful. [S3, S4]	Third-party guides and team builder; verify mode-by-mode.
Mythic Heroes: Idle RPG	Idle RPG; five-hero team, front/back formation, faction aura/counter system.	Official beginner guides describe four factions, tactical advantages, a free daily summon, arena, and post-battle statistics. The official guide also directs players to adapt factions when stuck on a stage.	A team builder should show why the squad works, and post-battle stats should reveal who dealt damage and who protected allies. [S5, S6]	Official Mythic Heroes guides; exact economy/rank details need current in-game audit.
Goddess of Victory: NIKKE	Hybrid real-time squad shooter / hero collector, not a turn-based idle RPG.	A current community guide documents role-like Burst I → Burst II → Burst III sequencing and team construction around cooldowns, burst generation, boss needs, and role replacements.	Synergy can be a sequence or engine, not just a stat aura. Make the team's setup-and-payoff legible and provide budget substitutes. [S7]	Community guide updated Aug 2026; not directly comparable to turn-based combat.

Game	Subgenre / structure	Observed pattern	Transferable design lesson	Evidence status
Honkai: Star Rail	Turn-based hero collector, not an idle auto-battler.	Official HoYoWiki tutorial explains shared Skill Points, energy accumulation, ultimates as extra actions, and weakness/toughness breaks that delay enemies. Official support documents banner guarantee/carryover rules, illustrating explicit communication of pity. [S8, S9]	Shared team resources create real opportunity cost. Counterplay can be based on enemy traits/weakness instead of only raw stats. Explain pity and combat rules transparently.	Official game reference/help; exact gacha rules can change, re-check notices.
Cookie Run: Kingdom	Team-based action / strategy RPG with front-middle-rear placement.	Third-party guides emphasize that team function changes by mode, that position and cooldown timing matter, and that a role may cover several jobs.	Use role coverage as a flexible tool, not a rigid five-slot recipe. Different modes should ask different tactical questions. [S10]	Third-party guide; not official. Verify current patch details before using as data.
Idle Heroes / Epic Seven / Summoners War / RAID / Mobile Legends: Adventure / Reverse: 1999 / Brown Dust 2	Adjacent or partially overlapping hero-collector, tactical RPG, or idle RPG references.	Included in the research backlog for role design, counterplay, equipment investment, duplicate pressure, status engines, and mode-specific team requirements. This initial master build does not assert detailed mechanics where current source verification is incomplete.	Research in targeted passes; distinguish game subgenre and patch context instead of flattening all examples into “idle RPG”.	Pending deeper sourced audit.

Cross-game conclusion (early, not final)

The strongest transferable pattern is functional interdependence: heroes contribute different jobs, resources, timing, target access, protection, or counters. Faction bonuses can help players form identities, but they should not be so strong that players must sacrifice the unit that answers the encounter. Different modes should test different functions. These are design lessons inferred from the cited systems—not claims that any one benchmark has a perfect fairness model.

5. Team-composition theory: what makes a squad coherent

5.1 A useful definition

A composition is a group of units with a shared win condition, a believable path to execute it, enough role coverage to survive common interruptions, and identifiable weaknesses that opponents can exploit. “Five high-power heroes” is a roster. “A protected carry with a mark-setter, a control opener, and a cleanser against counter-control” is a composition.

5.2 Team identity has six parts

Part	Question it answers	Example
Win condition	How does this team usually win?	Build energy and let a single finisher remove the most dangerous enemy.
Engine / enablers	What makes the win condition reliable?	A mark, speed boost, energy support, or control that creates a safe damage window.
Role coverage	What essential problem can it solve?	Survive opening burst; reach a backline target; interrupt an ultimate.
Tempo	When does it become dangerous?	First-turn burst, mid-fight ramp, or attrition after the enemy exhausts cooldowns.
Counterplay	What can stop it?	Cleanse and focus-fire disrupt a setup-heavy control team.
Investment / access	Can ordinary players build it and does it work at baseline?	A free starter can set marks; a rare hero may improve the plan but does not own the only mark mechanic.

5.3 Team size decision: recommend five

Team size	Advantages	Costs / risks	Decision
4 units	Readable, fast battles; easier to balance; each slot matters a great deal.	Less room for role hybrids, tactical counters, or a distinct synergy enabler. May collapse toward one carry + three mandatory supports.	Viable alternative, but not first choice for this project.

Team size	Advantages	Costs / risks	Decision
5 units	Enough room for carry, protection, enabler, utility/counter, and a flex role; familiar from several idle team games.	More hero kits and balancing combinations; can create support tax if every carry needs the same pair.	Recommended MVP baseline. Test before final lock.
6 units	More room for combinations and faction/group identity.	Higher mental load and effect interactions; weakens individual slot trade-offs; broadens balance/testing matrix and may make battles visually noisy.	Defer unless testing shows five slots cannot express the desired depth.

Formation proposal

Use five deployment slots: two frontline positions and three rear positions. Some mobile or assassin-style units may begin in a flank slot or bypass to a rear target, but movement/bypass should be an explicit ability, not the default. Five slots and 2+3 positioning are a hypothesis to prototype, not a final balance fact.

5.4 Role coverage, not a rigid recipe

Do not force every team into “one tank + one healer + three DPS.” Instead, require a sensible portfolio of functions based on the strategy. A burst team may trade healing for disruption and win before sustain matters. A boss team may need single-target throughput and cleanse rather than wide control. A counterattack team may rely on shields and retaliation instead of a conventional healer.

Coverage axis	Minimum design question	Common ways to satisfy it
Damage path	What reliably removes enemy HP or defeats the encounter?	Single-target burst, sustained carry, DoT engine, AoE clear, execute, summon damage.
Survival path	How does this squad avoid being eliminated before its plan works?	Tank, shield, damage reduction, healing, control, evasion with counter-risk, revive (rare and capped).
Enabler / tempo	What makes the main plan happen on time?	Speed, energy/chakra, cooldown reduction, mark, grouping, debuff, turn advance.
Utility / answer	What response prevents a common failure?	Cleanse, control resistance, interrupt, dispel, anti-heal, anti-summon, reveal.
Reliability	Does the plan still work if one opener misses or one hero dies?	Redundant trigger source, backup damage, reasonable proc chance, fallback basic abilities.

6. Proposed roster roles and role coverage

Roles should be tags, not mutually exclusive classes. One shinobi may be a bruiser + counterattacker; another may be a healer + chakra battery. Role tags are useful to the team-builder, but a unit’s skill interactions—not its label—determine its actual performance.

Role tag	Primary job	Natural counter categories	Balance guardrail
Guardian / tank	Absorb, redirect, or reduce incoming damage; protect rear units.	Bypass/mark, anti-shield, sustained damage, control resistance.	Must not make rear-line access impossible. A tank should protect, not erase all tactical targeting.
Bruiser / off-tank	Deal damage while holding space or surviving focus.	Kiting, control, sustained range damage, anti-heal if sustain-based.	Avoid making bruisers strictly better than tanks and carries.
Single-target finisher	Convert a setup window into a kill or near-kill.	Shields, bodyguard/redirect, evade, interrupt, reserve revive if allowed.	Its burst should depend on target setup or cooldown, not always one-shot.
AoE / wave clearer	Remove several weak enemies or pressure the entire team.	Spread formation, shields, damage mitigation, fast removal of the caster.	Against bosses, trade some single-target efficiency for wave value.

Role tag	Primary job	Natural counter categories	Balance guardrail
Sustained / ramping DPS	Become more dangerous over several turns.	Burst before ramp, cleanse of stacks, focus-fire, anti-heal where thematic.	Must have a ramp limit and a way to catch up if one stack is removed.
Controller / disruptor	Delay, interrupt, silence, pull, root, or redirect actions.	Tenacity, cleanse, immunity windows, control diminishing returns.	No infinite lock: hard CC uptime and repeat control must be capped.
Buffer / enabler	Make a carry or engine more effective.	Dispel, target the buffer, duration expiry, buff-type caps.	Buffer should not amplify every damage category multiplicatively.
Debuffer / mark setter	Expose a target, reduce defense, apply DoT, or make focus-fire easier.	Cleanse, decoy/clone, resistance, target swap, shorter debuff duration on bosses.	Marks should create a window, not permanent unavoidable vulnerability.
Healer / protector	Restore HP, shield allies, or reduce damage.	Anti-heal, burst, interrupt, shield-break, repeated pressure.	Healing should be valuable but not turn the fight into indefinite stalling.
Cleanser / counter-support	Remove key negative effects or protect against control.	Bait cleanse, staggered debuffs, burst after cleanse cooldown.	Keep cleanse timing meaningful; do not grant blanket immunity at all times.
Chakra battery / resource engine	Help allies access skills/ultimates sooner.	Energy drain, pressure on battery, cap per turn, interrupt.	Energy generation must have a hard ceiling per round.
Summoner / clone specialist	Create extra bodies, decoys, attacks, or tactical targeting choices.	AoE, anti-summon, reveal, summon cap and duration.	Summons should not multiply action economy without cost.
Counterattacker / disruptor	Punish attacks or convert enemy actions into value.	DoT, indirect damage, debuffs, delayed attacks, baited reactions.	Cap reactions per action; avoid recursive counter loops.
Reviver	Recover a fallen ally or prevent a loss.	Wounds, anti-revive, interrupt, limit to once per unit/battle.	Revival must be a rare tempo swing, not an undo button.

7. Synergy system and stack rules

7.1 Synergy families

Synergy type	Example concept	Design note
Role synergy	A protector makes a carry safer; a control unit creates an opening for burst.	Core; works across many rosters.
Condition / status engine	One unit applies a mark; another consumes or detonates it.	Core; should have visible trigger text and a counter.
Action-chain / follow-up	A tagged target triggers one ally follow-up after an attack.	Use with explicit once-per-turn limits.
Turn order / speed	A fast controller opens; a slower finisher acts before the debuff expires.	Powerful, so speed thresholds and turn manipulation need caps.
Protection loop	A shield or taunt buys time for ramping damage; healing sustains the window.	Avoid endless damage prevention loops.
Faction / village / clan	A small team bonus or role-flavored passive for common identity.	Soft bonus only; no mandatory five-stack.
Relationship / bond	A pair has a small optional interaction or cosmetic narrative reward.	Never make essential counters dependent on a specific canon relationship.
Formation / positioning	Frontline guards intercept; flankers reach a vulnerable target with a cost.	Must be understandable before battle resolves.
Nature / element counter	A tactical bonus against a type or enemy trait.	Use narrow, previewable modifiers, not a broad hard counter matrix at MVP.
Counter-synergy	A unit breaks shields, reveals stealth, interrupts a channel, or weakens summons.	Counter tools should exist at accessible rarity.

7.2 Recommended rules for MVP stacking

- [PROPOSAL] Same named buff: do not stack its magnitude. Keep the strongest active instance and refresh duration, unless the tooltip explicitly says it stacks.
- [PROPOSAL] Different buffs in the same category add together before the category cap. Attack buffs from different sources therefore do not multiply each other.
- [PROPOSAL] Use separate categories only when they represent truly distinct mechanics: Attack, Skill Damage, Final Vulnerability, Crit, Defense Reduction, and Speed. Keep “Final Damage” modifiers rare and tightly capped.
- [PROPOSAL] Team-wide faction/clan/relationship bonuses should add modest stats or unlock tactical choices, not add multiple hidden final multipliers.
- [PROPOSAL] Each unit may trigger at most one automatic follow-up per action. Follow-up attacks cannot trigger another follow-up. Summons do not recursively trigger skill chains unless a skill explicitly pays a resource and has a once-per-round cap.
- [SUPERSEDED RULE — DO NOT IMPLEMENT] Fractional hard-control duration (50% or 25% of a turn) is not implementable in this round-based battle. Use Section 25.8 and Chapter 27: Stun denies at most one action and repeated applications use shared probability tiers.

7.3 Illustrative damage and buff model

[HISTORICAL DRAFT — NOT FOR IMPLEMENTATION] The formula below used an earlier stat scale. The authoritative prototype formula is in Section 24.6 and the reconciliation is in Section 25.9.

[HISTORICAL FORMULA — SUPERSEDED; DO NOT IMPLEMENT]
This active-body draft used Effective DEF = max(0, Target DEF × (1 − Armor Penetration)), a 100/(100 + Effective DEF) mitigation multiplier, and a +35% Skill Damage cap. Those values conflict with the current prototype. The sole authoritative formula is Section 24.6: DEF Multiplier = max(0.35, 1000/(1000 + Effective DEF)); Effective DEF reduction = min(40%, sum of DEF Down and DEF Penetration percentages); Skill/Type Bonus cap = +25%. See Section 25.9 for the reconciliation. The older original reports are retained in Appendices A-D and F.

[HISTORICAL EXAMPLE — SUPERSEDED] This earlier example used DEF=100 and a 100/(100+DEF) scale. Do not mix it with the current 1,000-reference DEF multiplier in Section 24.6. The principle that a buff is counted in one named bucket only remains valid.

Stacking example

If two different allies grant +20% Attack and +15% Attack, the combined Attack increase is +35% before the proposed +40% Attack-buff cap. It is not 1.20 × 1.15 = +38%. If they both grant the same named +20% Attack buff, only one instance applies (or the stronger one), preventing duplicate buffers from scaling without limit.

7.4 Suggested initial caps (to test)

System	Initial test value	Reason / measurement
Combined Attack buffs	Cap at +40% from temporary Attack Bonus effects; permanent progression stats are separate.	Prevents double-buffer dominance; tune damage time-to-kill in the simulator.
Skill-damage bonus	Cap at +25% in the combined Skill/Type Bonus category.	Supersedes the historical +35% value; keeps damage buckets bounded.
Final vulnerability	Cap at +15%; same-type sources add before the cap and the Mark effect in Section 20.7 uses this same bucket.	Makes focus-fire useful without an uncapped multiplier.
Crit chance	Player Crit Chance is capped at 75% in all MVP modes, not only normalized PvP.	Stops guaranteed-crit loops and leaves defensive counterplay.
Crit damage	Critical damage is capped at 2.0× final non-critical damage.	Makes burst legible and limits one-shot spikes.
Defense reduction	Combined effective-DEF reduction from DEF Down and penetration caps at 40%. No additional flat-PEN bucket in MVP.	Maintains armor value and prevents several systems bypassing the same layer.
Speed manipulation	Speed changes are capped at ±20% and affect next-round initiative only. No turn meter, action advance or extra normal action in MVP.	Prevents perpetual turn loops; supersedes the former turn-meter allowance.

System	Initial test value	Reason / measurement
Hard CC	Use shared Stun application-chance tiers in Sections 25.8 and 27; no fractional lost actions.	A discrete lost action cannot be halved into 50% or 25% of a turn.
Revive	Revive is excluded from MVP. If later introduced, default once per unit per battle at 20% Max HP, with no revive chain.	Prevents unproductive stalling and inconsistent defeat triggers.
Faction / bond combat bonus	Village/faction threshold bonus uses only the highest active tier and caps at +5% ATK/HP total. Relationship effects should average no more than a small optional utility swing and cannot unlock a required role.	Keeps tactical choice stronger than a mandatory roster-tag requirement.
Attack Down	Use the strongest active Attack Down reduction, capped at 40%; Attack Down effects do not stack magnitude. Apply before temporary Attack Bonus.	Keeps the offense-reduction rule simple and readable; temporary Attack buffs are still handled in their own +40% bucket.

8. F2P / pay-to-progress fairness decisions

The aim is not to make progression free or remove monetization. The aim is to avoid a design where ordinary progress or viable team building is deliberately obstructed until the player pays. Paid options should shorten a wait, expand choice, or provide expression; they should not supply a unique required answer to a combat problem.

8.1 Character availability and duplicate power

Decision area	Recommended rule	Why it protects team-building
Core roles	Every role and at least two practical answers to common control, burst, sustain, stealth/decoy, and summon strategies are accessible without payment.	Players can learn counters without owning premium units.
Base kit	The character's key synergy is operational at the first copy and baseline rank.	A new hero can join a working team immediately.
Duplicates	Duplicates add small stat growth, resource conversion, cosmetics, or build flexibility—not new indispensable mechanics.	Avoids forcing repeated pulls to complete a role.
Duplicate power budget	Target no more than about 5-8% combat effectiveness difference between base copy and max duplicate investment in normalized PvP; PvE may allow a somewhat larger but bounded speed-up.	A paid duplicate can feel useful without being a different tier of character.
Premium signature gear	Sidegrade or specialization option, not a required kit function. Equivalent function can be earned on a generic gear path with lower convenience or different trade-offs.	No "character is incomplete without the paid weapon" trap.
Limited units	May be time-limited in first release, but enter a deterministic selection path later; no limited unit should be the sole counter or only enabler for an archetype.	Preserves collection excitement without permanent competitive exclusion.
Accessibility	Starters and early quest rewards should collectively assemble at least: one protector, one damage carry, one controller, one support/energy source, and one cleanse/anti-control answer.	Players can test several team plans before relying on summon luck.

8.2 Illustrative summon and fragment model

These are proposed prototype values, not facts about the benchmark games. The numbers should be replaced after economy simulation, retention analysis, and content cost estimation.

Mechanic	Example starting value	Fairness purpose
Free pull income	Target average: 45-50 standard/featured pulls per 4-week period once daily, weekly, event, and mode rewards are combined.	A predictable income baseline; do not count one-time launch bonuses as recurring income.
Featured guarantee	Hard pity at 70 pulls for the featured hero; no 50/50 loss at hard pity. Counter carries across featured banners.	The player can calculate the cost of a guaranteed target.
Pity visibility	Show current pity, remaining pulls to guarantee, probability/rate table, and carryover rules on the summon screen.	Reduces uncertainty and perceived bait.
Universal fragments	Target 15-20 universal hero fragments per week from ordinary play; 100 fragments = selector for an eligible standard-pool hero.	A deterministic fallback that helps build missing roles.
New hero access	New releases get trial access immediately. After a limited first window, add a deterministic token/selector route; avoid permanent role-exclusive limited units.	Players can test the mechanic before investing and eventually complete teams.
Base functioning team	The starter set should already form one complete team. A player should not need a lucky pull to unlock frontline protection, damage, and at least one utility/counter tool.	Team building is playable from the start.
Full alternative archetype	Target 4-8 weeks of ordinary play to unlock the missing pieces for one additional archetype, assuming the player uses targeted selectors instead of spreading resources randomly.	Creates collection goals without prolonged dead time.

Pity math example

At 48 pulls per 28 days, a 70-pull guarantee takes about 41 days if the player spends all those pulls on one banner (70 ÷ 48 × 28 ≈ 40.8 days). A new player's one-time rewards may shorten the first guarantee; if free pulls are split across several banners, the target takes longer. This is a planning model, not a promise of the eventual live economy.

[ECONOMY CLARIFICATION ADDED IN v0.10.7] The 45-50 pull target in Section 8.2 is not yet defined as featured-banner-eligible income. Because the source map distinguishes standard recruitment resources, featured-banner resources and universal fragments, do not infer a 39-44 day featured guarantee cadence until the share of pulls actually usable on the featured banner is defined. See Section 36. Historical illustrative values are retained for traceability.

8.3 "Pay to progress" guardrails

- Monetization may sell cosmetics, skins, battle-pass cosmetics/materials, inventory convenience, extra saved team presets, optional auto-clear convenience for already-cleared stages, and capped progression acceleration.
- Avoid selling exclusive combat counters, direct PvP stat boosts, paid-only core roles, or a second duplicate gate that unlocks the hero's actual signature synergy.
- Optional acceleration should not widen the permanent power gap indefinitely. If a spender reaches a new level earlier, the level ceiling and content tuning should let an active F2P player catch up over a reasonable horizon.
- Do not put the best competitive rewards behind a threshold that assumes paid characters, high duplicate ranks, or daily attendance with no grace days.
- A paid character may be a new option, not a universal replacement for older characters. Require counter coverage and role diversity before release.

9. PvP fairness proposal

Ranked PvP is where optional spending most easily feels like power purchase. The recommended design separates team-building expression from raw account-spending advantage.

9.1 Two-lane PvP structure

Mode	Power rules	Primary rewards	Purpose
Normalized Arena	Normalize hero level, core skill rank, gear baseline, and duplicate stat gains; preserve kit choice, tactical builds, and modest investment expression.	Season currency, cosmetics, title, profile frame; rank rewards are capped and not exclusive sources of combat-critical power.	Competitive fairness and composition testing.
Progression Challenge / PvE ladder	Account investment can matter; stages have visible mechanics and power estimates.	Ordinary progression materials, limited weekly milestone rewards.	Let collection and long-term progression feel worthwhile without using it as the sole competitive measure.
Casual / practice battle	Optional rulesets, draft/sandbox, or friend battles; no loss of meaningful rank.	Small capped participation rewards, if any.	Let players try team identity and counterplay without punishing experimentation.

9.2 Illustrative weekly rewards

This table is an example reward structure to test. Quantities are intentionally abstracted as “Arena Tokens” until the broader currency economy is designed.

Activity	Weekly example reward	Guardrail
Complete 5 matches	80 Arena Tokens	Participation matters; losses still count.
Win 3 matches	+40 Arena Tokens	Skill is rewarded, but a streak is not required.
Reach a rating band	0-60 extra Arena Tokens by band	Keep the rank differential modest; avoid creating an endless power compounding loop.
Season participation milestones	Cosmetic token / frame / title and a modest one-time selector-fragment reward	A new player can earn some rewards without reaching the top ranks.
Top leaderboard	Prestige cosmetics and non-exclusive status rewards	No exclusive combat unit or irreplaceable equipment.

9.3 Matchmaking, reset, and spend gap

- [PROPOSAL] Match by rating first, then use normalized roster power only as a tie-break or to protect brand-new accounts from extreme investment gaps.
- [PROPOSAL] Soft season reset: compress ratings toward the middle band rather than erase all progress. Use placement matches to reduce mismatches after reset.
- [PROPOSAL] Do not require daily matches. Weekly cap-based milestones should be completable over several sessions with missed-day forgiveness.
- [PROPOSAL] In normalized PvP, target a 45-55% win-rate band for balanced matchups before mastery differences, with no purchased upgrade contributing a direct >5% normalized stat increase. Any roster-breadth advantage must still be answerable through accessible counters and matchmaking.
- [HYPOTHESIS] A separate unnormalized “account power” event could satisfy players who enjoy progression, but its rewards must not be required to keep up in normalized Arena. Test whether this adds fun or simply reproduces pay-to-win pressure.

10. Sample five-shinobi compositions

The following are hypothetical role designs to illustrate the framework. They are not assertions about any character's canon abilities or existing Naruto game kits. Names are used only as thematic placeholders pending licensing.

Archetype	Illustrative roster / roles	Win condition / loop	Main weakness	Counterplay	Fairness note
A. Starter balanced squad	Naruto (flex / clone decoy), Sakura (healer), Shikamaru (controller), Kakashi (single-target DPS), Rock Lee (bruiser).	Control slows the enemy opener; protector/sustain keeps the carry alive; decoy splits target priority; Kakashi converts an opening into a kill.	Too slow against high burst if control resists; limited AoE.	Accessible cleanser or control-resistant unit; early pressure on the controller.	Teaches role coverage and target priority.
B. Protect-the-carry	Sasuke (main carry), Sakura (healer), Shikamaru (control), Hinata (protector / mark), Naruto (decoy / flex).	Mark or control the highest threat while protecting Sasuke long enough to execute a damage cycle.	Anti-shield/anti-heal and backline bypass can break the protection chain.	Interrupt the carry window; use a flank finisher or anti-heal.	All roles should have common-pool substitutes.
C. Speed / tempo	Minato (tempo enabler), Kakashi (finisher), Shikamaru (control), Sai (mark setter), Sakura (emergency sustain).	Win turn order, apply a visible mark, burst before enemy support cycles.	Speed denial, durable frontline, missed setup, or control immunity.	Slow/energy disruption or frontline guard with a cleanse.	Speed boosts must be capped; no endless extra turns.
D. DoT / attrition	Shino (DoT engine), Sakura (sustain), Hinata (protector), Shikamaru (control), Sai (debuffer).	Stack limited damage-over-time effects while control reduces the enemy's ability to burst the team.	Cleanse, fast focus-fire, boss DoT resistance, or time limits.	Cleanse on key stacks; eliminate the engine before ramp.	Make DoT a distinct strategy, not universally optimal.
E. Summon / clone tactics	Naruto (clone generator), Sai (decoy / mark), Kakashi (finisher), Hinata (protector), Sakura (healer).	Use bounded decoys/summons to manipulate targeting and create attack windows.	Area damage, anti-summon effects, reveal, or summon cap.	AoE and reveal; bypass clones if a real target has a mark.	Summons must not multiply full actions recursively.
F. Boss slayer	Sasuke (single target), Rock Lee (sustained DPS), Shikamaru (defense debuff / interrupt), Sakura (sustain), Minato (resource engine).	Maintain stable damage uptime, coordinate resource generation, and interrupt predictable boss actions.	Boss immunity to control, high retaliation, or damage windows that punish ramp.	In PvE, boss mechanics can require timing rather than simple stat checks.	Do not make boss success depend on a limited hero.
G. Control and counter	Shikamaru (controller), Hinata (protection), Sakura (cleanse), Kakashi (pickoff), Naruto (disruption).	Deny one key enemy action, survive the response, then remove an exposed unit.	Tenacity, control diminishing returns, cleanse timing, broad AoE damage.	Bait the cleanse, then stagger a second debuff; use resilient damage sources.	CC must have clear diminishing returns and boss-specific limits.
H. Sustain / counterattack	Hinata (protector / counter), Sakura (healer), Rock Lee (bruiser), Shino (DoT), Shikamaru (disruptor).	Absorb predictable enemy actions, punish attacks, then win through repeated damage.	Burst before setup, anti-heal, indirect damage, or a hard time limit.	Avoid attacking into counter windows; focus the healer or use anti-heal.	Counterattack cannot recursively trigger counterattack.

10.1 One worked example: synergy versus raw power

Illustrative encounter: the opponent fields a high-stat carry, a healer, and three damage units with no cleanse or protection. Our lower-power team uses a mark setter, controller, protector, finisher, and healer. On turn 1 the controller slows the enemy carry; the mark setter marks the healer; the protector absorbs a flank attack; on turn 2 the finisher gains bonus damage against the marked healer and removes the sustain source. The opponent has higher raw stats, but its win engine is disrupted. The result is not guaranteed: if the mark is resisted, the controller is interrupted, or the opponent has an accessible cleanser, our setup may fail. That uncertainty is the point—synergy creates a plan, not an automatic victory.

11. Validation plan and success metrics

FUTURE PHASE ONLY — This section preserves the proposed validation method and success metrics for later use. It is not an instruction to build a simulator, run seeded outcomes, recruit testers, conduct live tests, or report results now. In the current phase, use hypothetical scenarios and written reasoning to check internal consistency. A composition score can help players understand a lineup, but it cannot certify balance.

11.1 Composition-review rubric

Metric	Score / measurement	Prototype target (provisional)

Metric	Score / measurement	Prototype target (provisional)
Win condition clarity	Can testers explain how the team wins in one sentence?	At least 80% of testers identify it without outside explanation.
Role coverage	Damage path, survival path, enabler/tempo, utility/counter, reliability.	No archetype should require all five roles to be filled by one specific rarity band.
Synergy reliability	How often does the chain activate under realistic opponent responses?	Core engine activates in 60-85% of intended lines, depending on counterplay; not 100% guaranteed.
Counter resilience	Number of practical accessible answers; win rate when countered.	Every strong strategy has >=2 practical answers; no single counter suppresses every version of the archetype.
Team diversity	Share of battles by archetype across rating bands.	No single archetype >35% sustained pick rate without a clear format-specific reason.
Matchup balance	Win rate for archetype A vs B across equalized power.	Most non-specialist pairings 45-55%; deliberate counters may reach 40-60, but no hard lockout.
Replacement flexibility	Result when one non-core unit is replaced by accessible alternative with same function.	Core archetype win rate should not collapse >8 percentage points from reasonable replacement.
Duplicate sensitivity	Compare same roster base-copy vs max duplicate under normalized PvP.	Target <=5-8% combat effectiveness difference.
First-death timing	Turn / second of first knockout by role and matchup.	Detect carries or support units being deleted before their first meaningful action too often.
CC uptime	Share of enemy action opportunities removed by hard CC.	Monitor high-percentile CC lock; no unanswerable repeated-lock pattern.
Spend-power curve	Normalized PvP outcome by spending band while controlling rating and skill.	No purchased stat path should create >5% direct normalized power; investigate win-rate gap >5 percentage points after controls.

11.2 Simulation and playtest process

1. Build a deterministic combat simulator with seed control, full action logs, turn order, target selection, damage, buffs/debuffs, CC resistance, and trigger counts.
2. Run automated mirror matches and all archetype-vs-archetype pairings at equal levels, then repeat with reasonable gear and investment variance.
3. Run “removal tests”: remove each unit from its team and measure the win-rate drop. If removing one hero destroys the archetype and there is no accessible substitute, the design may have created a mandatory unit.
4. Run “substitution tests”: replace a unit with an obtainable hero that covers the same role. Compare win rate, time-to-win, and synergy activation.
5. Run “counter tests”: intentionally pit each archetype against the two best practical counters, plus a random composition, across several seeds.
6. Review battle logs and ask players what they think happened. A synergy that works numerically but cannot be understood is not yet successful.
7. Change one balance variable at a time where feasible; record the patch, the predicted outcome, the observed outcome, and any unintended metagame shifts.

12. Versioned decision log and risk register

ID	Decision	Rationale	Status	Alternative rejected / trade-off	Confidence
D-001	Project should reward composition over raw power.	Owner explicitly requested synergy, roles, weaknesses and counterplay rather than random high-power teams.	Settled direction	Do not make higher rarity the only way to get a required tactical function.	High
D-002	F2P users must build viable archetypes without paid-only counters.	Core fairness requirement.	Settled direction	Paid-only exclusive answers would monetize frustration and destroy team-building fairness.	High
D-003	Five-unit team is the MVP hypothesis.	Four slots may feel tight; six increase balance and cognitive cost; common benchmark examples support five-unit teams.	Proposed / test	Could revise to four if battle readability and team depth test better.	Medium
D-004	Use 2 front + 3 rear formation.	Creates frontline protection and distinct backline access.	Proposed / test	Free-position grid offers more tactical depth but costs UI and AI complexity.	Medium
D-005	Use additive buff categories with caps; same named buffs do not stack magnitude.	Controls exponential synergy and reduces double-buffer dominance.	Proposed / test	Purely multiplicative stacking is simpler to explain in some contexts but risky for balance.	Medium
D-006	Synergy should be functional and bounded; faction/bond bonuses stay small.	Avoids mono-faction mandate or paywalled relationship teams.	Proposed / test	Large faction bonuses create easy identity but can invalidate stronger tactical choices.	High
D-007	Pity and character access should be predictable.	User prioritizes generosity and pay-to-progress over pay-to-win.	Proposed / test	A high-uncertainty 50/50 system may create excitement but makes target costs less predictable.	High
D-008	Normalized PvP should prioritize kit choice and tactical play over duplicate/stat spend.	Competitive fairness requirement.	Proposed / test	Fully unnormalized PvP can preserve progression fantasy but risks purchase-dominant outcomes.	High
D-009	This document is the project's single master file.	Owner explicitly asked for one continuously edited editable file.	Settled workflow	Separate reports can fragment decisions and introduce contradictory mechanics.	High

12.1 Risk register

Risk	Warning sign	Mitigation / test
One mandatory meta team	A single composition is picked or wins far above others across modes.	Use archetype matrix, role substitutes, removal tests, patch cadence, and non-identical encounter goals.
Power creep	New hero's kit does old jobs strictly better and adds another unique effect.	Define each new unit's trade-off and compare against accessible same-role alternatives.
Buff multiplication runaway	Double buffers or repeated triggers cause nonlinear kill-time collapse.	Separate additive categories, caps, once-per-action triggers, and combat-log trigger counts.
Paid counter lock	A strategy has only one answer and that answer is premium/limited.	Require at least two practical counters, one broadly accessible; audit every new release.
Over-control / stunlock	Units lose repeated turns and cannot respond.	CC family diminishing returns, immunity windows, control-resistance access, turn-lock metric.
Support tax	Every good carry requires the same rare buffer and battery.	Test diverse support shells; add low-rarity alternatives with changed trade-offs.
Long-term grind fatigue	Ordinary players need weeks to unlock basic team functionality.	Starter completeness, deterministic fragments, milestone selectors, catch-up tests.
Unclear auto-combat	Players cannot explain why they lost or why a combo triggered.	Readable targeting, concise log, after-battle breakdown, replay, team builder warnings.

13. Remaining research priorities

Priority	Work item	Output expected	Depends on
P0	Implement deterministic combat simulator / reference engine.	Executable reference model with seeded rolls, event log, action order, modifier trace, end-state check and regression tests for every rule in Sections 24-27.	Written spec in Section 27 is internally reconciled.
P0	Tune skill coefficient ranges against normalized role packet.	Use the new base stats/coefficient budgets; report time-to-KO, role contribution, p95 burst, fight duration and counter effects for each archetype.	Deterministic simulator.
P0	Run counter/edge-case matchup regression suite.	At least 1,000 seeded matches per directed archetype pairing with side-priority swaps, removal/substitution tests and status logs.	Reference engine + first-pass playable kits.
P1	Audit benchmark games mode by mode.	Current-source comparison for AFK Arena, AFK Journey, Mythic Heroes, Idle Heroes, Epic Seven, Summoners War, RAID, Nikke, HSR, Reverse: 1999, CRK, Brown Dust 2, MLA.	Web research and hands-on screenshots/gameplay if available.
P1	Simulate resource and duplicate model.	Expected pulls/month, featured acquisition, fragment income, base-to-max power, PvP spend gap.	Initial values in Section 8.
P1	Specify modes around team questions.	Campaign, wave clear, boss, PvP, tower/faction challenge, guild boss with unique build requirements.	Combat engine and archetypes.
P2	Design UI/UX and readable combat recap.	Team builder tooltips, predicted role coverage, synergy display, loss-reason analysis, replay.	Stable combat rules.
P2	Estimate sustainable monetization.	Cosmetics/convenience mix, live-ops cost model, payer value without direct competitive power sale.	Economy model and audience assumptions.

14. How the next revision should proceed

[SUPERSEDED HISTORICAL DIRECTION] Version 0.6 closed the proposed core combat rules at the written-specification level and earlier revisions recommended immediate implementation/validation before other feature research. That sequence remains relevant to later implementation and empirical balance decisions, but it is not the current work instruction.

[CURRENT DIRECTION — VERSION 0.9.1] Stay in the research and game-concept phase. Continue researching what defines an idle RPG, its core loop and idle/active balance, team-combat theory, collection, progression, currencies/economy, monetization/F2P fairness, conceptual game modes, onboarding/mobile UX, retention/live-ops patterns, and how these systems interact. Work through examples and hypothetical scenarios only. Do not implement a prototype, build/run a simulator, perform seeded-match testing, recruit live testers, or claim any criterion has passed until the project owner explicitly says live testing is starting. See Section 28 for the governing workflow.

Revision protocol

When adding new research, keep the current decisions unless new evidence or playtests justify changing them. Add the new content in the relevant section, update the version and change log, and preserve the old value and reason for any balance change. If a claim cannot be verified, write “unverified” rather than filling the gap with a plausible-sounding number.

15. Sources and evidence notes

Accessed 27 October 2026 unless a publication/update date is noted. Current live-game values can change; revisit patch-sensitive claims before implementation. In-text markers [S#] refer to this source list.

[S1] Factions, hero types and roles in AFK Arena. AFK Global (third-party guide); Published 21 Oct 2026. Used for a current description of faction advantage, formation bonuses, hero roles and faction identity. It is a community guide, not an official publisher source.
https://www.afk.global/afk-arena/guides/factions-and-hero-types

[S2] AFK Arena Version 1.202 Patch Notes. AFK Global (patch-note mirror / guide site); Published Sep 2026. Used only to note that faction-specific team events and formation systems continue to exist; check official in-game patch notes before copying any detail.
https://www.afk.global/afk-arena/patch-notes/1-202

[S3] AFK Progression Teams. Prydwen Institute; Crawled Oct 2026; guide date shown on site. Used for five-unit role coverage, three-unit faction core plus two flexible picks, and the caveat that faction bonuses may not outweigh utility.
https://www.prydwen.gg/afk-journey/guides/afk-progression-teams

[S4] AFK Journey Team Builder. AFK Global (third-party interactive tool); Crawled Oct 2026. Used for current team size, positional placement, faction bonus and mode-specific formation context. Values may change with seasons.
https://www.afk.global/afk-journey/team-builder

[S5] Day 3 Guide. Mythic Heroes official website / IGG; Official guide; current page. Used for four factions, tactical advantage, Arena access, Arena Badges, and battle statistics.
https://www.mythicheroes.com/day-3-guide/

[S6] Day 2 Guide. Mythic Heroes official website / IGG; Official guide; current page. Used for the official guide's daily free summon, role example of a healing/protective support, Pantheon, and guild modes.
https://www.mythicheroes.com/day2-guide/

[S7] Team Building Guide - Nikke: Goddess of Victory. Nikke.gg (community guide); Updated 26 Aug 2026. Used as an adjacent-game example of sequence-based synergy, burst categories, cooldowns, roles, boss targeting, and budget replacements. This is not a turn-based idle game.
https://nikke.gg/team-building-guide/

[S8] Base Mechanism - Honkai: Star Rail. HoYoWiki / HoYoverse; Current official game wiki page. Used for shared Skill Points, energy/ultimate rules and Weakness Break. The page is a game reference and may be updated.
https://wiki.hoyolab.com/pc/hsr/entry/1618?lang=en-us

[S9] How does the Wish guarantee system work?. HoYoverse Help Center; Updated 17 Sep 2025. Used as an example of explicit published hard pity and carryover information. Its numbers describe Honkai: Star Rail/Genshin Wish systems, not our proposed system.
https://support.hoyoverse.com/hc/en-us/articles/50333940684953-How-does-the-Wish-guarantee-system-work

[S10] Cookie Run: Kingdom Best Team: Top Comps for Every Mode. CookieRun Kingdom Wiki (third-party community guide); Published Oct 2026. Used only for the general observation that teams and role priorities can vary by mode. Not an official source; patch-specific details are not treated as facts about our game.
https://www.cookierunkingdom.wiki/teams/cookie-run-kingdom-best-team

[S11] Mythic Heroes: Idle RPG - Google Play listing. Google Play / IGG; Listing updated 14 May 2026. Used to confirm the game is positioned and listed as an Idle RPG with hero collection and competitive multiplayer features.
https://play.google.com/store/apps/details?hl=en-US&id=com.igg.android.mythicheroes

Source confidence notes

- Highest confidence in this starter compilation: the official Mythic Heroes guides, HoYoWiki/HoYoverse Help Center, and official store listing for the claims they directly state.
- Moderate confidence: current third-party benchmark guides and tools for visible game systems; these can be patch-sensitive and may reflect a specific competitive context.

- Low confidence for broad causal claims such as “this monetization model makes games die” unless supported by longitudinal player, revenue, or retention data. No causal market conclusion is asserted here.

End of the Version 0.2 consolidated main body. Appendices A–D preserve four recovered prior completed research reports. All proposed numerical systems remain subject to simulation and playtesting.

16. Consolidation audit: what was missing and what is now included

16.1 Correction to Version 0.1

The first master-file build was incomplete. It summarized the live design conversation and a selection of benchmark notes, but it did not merge several completed research reports already saved in the project Library. Its research-status note incorrectly implied that full prior reports were unavailable. That statement is corrected in this version.

Four prior completed report outputs have now been recovered from the Library and are reproduced in Appendices A–D, using their saved report text. Together they contain approximately 13,262 words before conversion to Word formatting. The existing 20-page master body remains the current synthesis; the recovered reports are preserved as a historical research archive so no earlier research text is discarded.

Important distinction: the prior reports contain a mix of useful design hypotheses, illustrative numbers, community claims, and benchmark-game statements. Keeping a statement in the archive does not mean it is verified or approved for implementation. Where a historical recommendation conflicts with a newer decision in the main body, use the main body’s current decision register and the evidence labels below. Re-verify patch-sensitive claims before implementing them.

16.2 Recovered report register

Archive: Appendix A - **Report title as stored:** Executive Summary - **Approx. words:** 5,462 - **Main coverage:** Core loop, auto-battle, progression, collection, currencies, monetization, retention, pacing, UX, analytics, anti-grind, IP, benchmark comparison - **Status in this master:** Preserved in full; broad systems summarized and cross-linked in Sections 17–18

Archive: Appendix B - **Report title as stored:** Executive Summary (1) - **Approx. words:** 3,184 - **Main coverage:** AFK Arena, Mythic Heroes and Idle Heroes comparison; gacha/pity; duplicates; economy; monetization; F2P issues; event/PvP concepts; decision notes - **Status in this master:** Preserved in full; historical game claims are not automatically treated as verified facts

Archive: Appendix C - **Report title as stored:** Executive Summary (2) - **Approx. words:** 3,485 - **Main coverage:** Roles, synergy, counterplay, duplicates/awakening, F2P balance, PvP, Naruto-style example teams, formulas, unresolved questions - **Status in this master:** Preserved in full; design decisions are reconciled against the current proposal

Archive: Appendix D - **Report title as stored:** Economy & Summoning Mechanics - **Approx. words:** 1,131 - **Main coverage:** Pity/spark, fragments, duplicates, currencies, progression, team synergy, event design, open questions - **Status in this master:** Preserved in full; current working values are in Sections 8–9 and 18

The word counts are approximate counts of the saved report message text, not claims about the original Deep Research application’s internal source database. The latest rate-limited attempts did not produce another completed report output that could be added here.

16.3 Coverage matrix: nothing should disappear between research rounds

Research area: Product vision and non-negotiable F2P direction - **Where it is now represented:** Sections 2 and 8–9; Appendix B–D - **Evidence / maturity status:** User direction is settled; exact implementation numbers are proposals

Research area: Core idle gameplay loop and offline rewards - **Where it is now represented:** Appendix A; summarized in Section 17.1 - **Evidence / maturity status:** Genre pattern, not yet tuned for this specific game’s economy

Research area: Auto-battle, speed, skills, formation and combat clarity - **Where it is now represented:** Sections 5–7 and 10–11; Appendix A and C - **Evidence / maturity status:** Proposed combat model; needs a playable simulator and UX tests

Research area: Team roles, synergy, counters, archetypes and sample teams - **Where it is now represented:** Sections 5–7 and 10–11; Appendix C - **Evidence / maturity status:** Central current design focus; most exact values remain unvalidated

Research area: Hero collection, gacha, duplicates, awakening and fragments - **Where it is now represented:** Section 8; Appendices A–D - **Evidence / maturity status:** Example economy exists; live benchmark details require current source audits

Research area: Currencies, progression, upgrade sinks and pacing - **Where it is now represented:** Section 8 and Appendix A–D; summarized in Section 17.2 - **Evidence / maturity status:** System concepts recorded; costs and income need economy simulation

Research area: Monetization and pay-to-progress model - **Where it is now represented:** Sections 2, 8–9 and Appendix A–D - **Evidence / maturity status:** Strategic constraint is settled; revenue sustainability is still a hypothesis

Research area: PvP fairness and seasonal rewards - **Where it is now represented:** Section 9; Appendices B–C - **Evidence / maturity status:** Proposed normalization and reward concepts; must be tested with a simulator and playtests

Research area: Modes, social features, retention, UX, analytics - **Where it is now represented:** Section 17; Appendix A - **Evidence / maturity status:** Concept inventory, not a validated release plan

Research area: Benchmark evidence - **Where it is now represented:** Section 4 and Section 15; historical reports in Appendices A–D - **Evidence / maturity status:** Main-body source list is the citation starting point; legacy citation IDs in appendices may not resolve outside the original tool view

Research area: Legal / IP - **Where it is now represented:** Section 2 and Appendix A - **Evidence / maturity status:** Naruto commercial use requires appropriate licensing; no license is assumed

16.4 Evidence hierarchy and conflict-resolution rule

When two claims conflict, apply this order:

1. **Current official source or current in-game inspection** for the exact benchmark mechanic.
2. **Current, well-supported third-party guide** for strategies or player-facing systems that the guide can actually observe.
3. **Archived report claim** as a research lead, not conclusive proof.
4. **Design proposal** as an intentional rule for our game, never evidence that another game uses it.
5. **Hypothesis** only after it has a test plan.

Examples of conflicts that must not be silently blended:

1. Historical reports suggest varied hard-pity ranges—from roughly 80–100 to 100–200 pulls—while the current project proposal uses 70 pulls for a featured guarantee. The 70-pull value is a **prototype choice**, not a statement about industry standards or a validated final rate.
2. Earlier reports explore upgrades requiring several duplicate copies, including examples such as five copies or 10–20 copies. The current fairness rule supersedes those examples: a character's defining kit and core team function must work at baseline copy/rank; duplicates may improve power only modestly and should not unlock mandatory team mechanics.
3. Historical notes describe specific benchmark systems (such as rates, pity, faction counts, VIP boosts or daily rewards) that may be patch-sensitive or sourced from community reports. Do not use those as exact live-game facts until re-checked against current official information.
4. "Generous events improve retention" and "pay-to-win causes a game to die" are causal claims that require longitudinal player, retention and revenue evidence. This project treats generosity and fairness as product requirements, but does not claim that generosity alone guarantees business sustainability.

17. Integrated game concept: the systems outside team building

This section brings together the broad game concepts from the earlier research so the project remains holistic while the next development phase focuses on combat composition.

17.1 Core loop and player cadence

Core short loop: claim idle rewards → inspect progress or an obstacle → upgrade a hero, skill or item → refine the five-unit squad → run campaign or a mode → earn resources, fragments and tactical information → repeat.

Core long loop: unlock roster options → specialize several heroes → form alternative compositions → tackle new encounter rules and harder modes → earn deterministic progress toward missing roles or characters → refine teams and return to content.

Idle progress should lower routine friction rather than replace all decisions. The player should be able to make visible progress in a short session, while team-building choices create longer-term goals. The offline reward cap, claim frequency and resource rates are not locked; earlier reports mentioned ranges such as 8–12 hours as an example, but that is only a tuning starting point.

17.2 Proposed game modes and the team-building question each asks

Mode: Campaign / story - **Main question it tests:** Can the player handle varied enemy patterns while progressing steadily? - **Composition implication:** Teach role coverage, target priority, formation, marks, control and sustain progressively - **Fairness guardrail:** Every chapter has more than one viable solution; do not require a banner-only unit

Mode: Idle reward stage - **Main question it tests:** Can a team clear reliably enough to advance its resource baseline? - **Composition implication:** Favors consistent clears and survivability more than peak one-battle burst - **Fairness guardrail:** A temporary progress wall cannot be bypassed only by buying power; offline rewards remain useful while stuck

Mode: Wave clear / trials - **Main question it tests:** Can AoE, tempo and efficient cooldown use remove many enemies? - **Composition implication:** Rewards target distribution, AoE, reset mechanics and protection from backline threats - **Fairness guardrail:** Avoid making one limited AoE unit the only practical answer

Mode: Boss / raid - **Main question it tests:** Can the squad maintain single-target output, survive telegraphed attacks and interrupt or cleanse important effects? - **Composition implication:** Rewards sustained damage, defense reduction, resource management and timed utility - **Fairness guardrail:** Boss immunity/weakness rules are previewed; no paid unit is the only boss solution

Mode: Tower / endurance - **Main question it tests:** Can the squad preserve health and adapt across consecutive fights? - **Composition implication:** Rewards sustain, flexible utility, resource efficiency and replacement depth - **Fairness guardrail:** Give retry or recovery systems; avoid an arbitrary gear wall disguised as strategy

Mode: Faction / village challenge - **Main question it tests:** Can players use a themed roster or work around a roster restriction? - **Composition implication:** Builds collection depth and encourages secondary squads - **Fairness guardrail:** Restricted rosters must have obtainable units and a parallel unrestricted progression path

Mode: Normalized Arena - **Main question it tests:** Can the player build and pilot a strong composition against comparable opponents? - **Composition implication:** Rewards role interaction, counters and matchup knowledge - **Fairness guardrail:** Normalize core stats and duplicate power; seasonal rewards mostly cosmetic/status with capped currency

Mode: Progression / account-power arena - **Main question it tests:** How much has the player grown their roster and account? - **Composition implication:** Lets gear, account development and breadth matter - **Fairness guardrail:** Optional lane only; rewards must not be required for competitive viability

Mode: Guild / cooperative boss - **Main question it tests:** Can players coordinate contribution, timing or role diversity over shared progress? - **Composition implication:** Encourages varied damage, support, debuff and protection squads - **Fairness guardrail:** Avoid mandatory purchases and top-rank rewards that compound exclusive combat power

Mode: Practice / friendly battle - **Main question it tests:** Can players test substitutions, synergies and counters with low risk? - **Composition implication:** Essential for learning and meta exploration - **Fairness guardrail:** No meaningful loss for experimenting; provide optional small capped participation rewards

Mode: Character trial / rotating scenario - **Main question it tests:** Can the player understand a new kit or archetype in a controlled environment? - **Composition implication:** Demonstrates a new tactical identity and counterplay - **Fairness guardrail:** Trial access is free; trials should not imply the hero is mandatory

These are a mode inventory, not an instruction to ship every mode in the MVP. Start with campaign, idle rewards, one boss, normalized Arena prototype, and a sandbox/replay path. Add modes only when they test a different team-building skill and can be maintained with available content resources.

17.3 Collection and growth systems: design principles inherited from previous research

The original reports identify familiar growth dimensions: hero level and experience, skill upgrades, equipment, promotion/awakening, faction or relationship identity, idle resources, event tokens and duplicate conversion. The current design should avoid allowing every dimension to multiply every other dimension without limit.

Use the following provisional split:

1. **Character level:** modest, predictable base-stat growth.
2. **Skill rank:** improves defined coefficients or cooldowns in bounded steps; do not unlock the only working form of a signature combo at a high paid rank.
3. **Equipment:** changes a few stats or tactical options; generic gear must cover essential defensive and offensive needs.
4. **Promotion/awakening:** mostly horizontal customization, a modest stat bump, or a small reliability improvement. Base kit identity stays intact.
5. **Account-wide progression:** unlocks modes, presets, training quality-of-life and catch-up benefits; avoid hidden bonuses that create a permanent PvP power gulf.
6. **Duplicates:** convert to a universal resource, modest stat growth or elective customization. A duplicate is never the sole way to make a core synergy function.
7. **Relationship/faction/village tags:** provide modest bonuses or alternate triggers. A team must not be punished so severely for mixing factions that the best counter or role coverage becomes irrational.

Every upgrade needs both a **source** (where the resource comes from) and a **sink** (what consumes it). If an upgrade requires a resource that comes mainly from the same high-level mode the upgrade is needed to beat, it creates a circular gate. The player should have at least one viable source that remains available while stuck.

17.4 Currency, events and retention: what the archived reports proposed

The recovered reports proposed a small set of familiar currency roles: premium summoning currency; common gold/coins for levels and gear; hero fragments; skill materials; guild/arena tokens; and event-specific exchange currency. This is a concept, not a finalized currency list. Use as few currencies as possible until a real economy model shows why a separate currency is needed. Every currency should have a readable purpose, a visible earn path, and a non-confusing sink.

Useful event shapes include cumulative milestone tracks, login calendars, weekly challenge towers, bosses, limited story chapters, guild events and free/premium passes. The fairness constraint supersedes any incentive to create punitive FOMO:

1. Most valuable progression rewards should be cumulative, not dependent on leaderboard placement.
2. Missed days should be recoverable within a weekly window.
3. Limited banners may create a first-release moment, but the unit must eventually be obtainable through a deterministic path.
4. Premium passes can add cosmetics, optional convenience and a modest amount of non-exclusive materials, but should not hold the only copy of a combat-critical character.

5. A returning player should receive catch-up help, not a guilt message or an impossible backlog.

17.5 Sustainable monetization hypothesis

Earlier reports explored cosmetics, skins, battle passes, optional subscriptions, capped acceleration, inventory/preset convenience and rewarded ads. The recommended direction is to prioritize cosmetic identity, optional convenience and bounded time-saving. These are hypotheses for a sustainable product, not proof that any particular product mix will fund development.

A payer should be able to say “I saved time” or “I have more ways to build teams,” rather than “I bought the only viable answer.” To test actual sustainability, model server/content/staff costs, payer conversion, repeat purchase, platform fees, price-region differences, and revenue concentration separately from combat balance. Monitor whether high spenders are funding cosmetics and convenience or acquiring an ever-growing share of ranked wins. Do not use a fairness charter as a substitute for a business model.

18. Resolved questions: fairness decisions for F2P and pay-to-progress

This chapter turns the earlier “Unresolved Questions and Considerations” into a consistent working answer. All quantities are **proposed prototype targets**, not benchmark-game facts.

18.1 Buff and debuff stacking

Decision: same-named buffs do not stack their magnitude; different sources of the same category add together up to a category cap; distinct categories may combine multiplicatively only when they represent materially different mechanics and each is capped. Current authoritative caps and interaction rules are in Sections 24.6 and 25.9; older figures in this section are historical where they conflict.

Example for Attack: a unit with 1,000 ATK receives +20% Attack and +15% Attack from different sources. Effective ATK = 1,000 × (1 + 0.20 + 0.15) = 1,350, before any other modifiers. If both effects are the same named +20% buff, only the stronger active instance applies; the result is 1,200, not 1,440. Temporary Attack buffs cap at +40% in the MVP.

Authoritative prototype caps are in Section 24.6: Attack Bonus +40%; Skill/Type Bonus +25%; Vulnerability +15%; combined effective-DEF reduction +40%; non-DEF damage reduction +30%; player Crit Chance +75%; Crit Damage up to 2.0×; total active Shield capacity +40% of max HP. The earlier +35% skill bonus and 20% turn-meter advance are superseded/deferred. MVP has no turn meter, action advance or extra-turn effects. These remain proposals until tested.

Crowd control rule: the first hard-CC effect in a family uses full duration; a repeat in the same family within two turns applies 50% duration; the next applies 25%; then the target gains one turn of immunity to that family. The meter is per target, not per caster. Bosses may use separate scripted resistance rules, but those rules must be visible.

18.2 Team size and role access

Decision: prototype five deployed shinobi with two frontline positions and three rear positions. Five slots give room for damage, survival, an enabler, a counter/utility tool, and one flex role without forcing a fixed formula. Four units remain a possible alternative if five slots produce slow, noisy battles. Six units are deferred because they multiply interactions and increase cognitive/balance load.

Do not hard-code “one tank, one healer, three DPS.” Evaluate a composition using five coverage questions: damage path, survival path, engine/tempo, utility/answer, and reliability. A fast burst squad may trade healing for control; a boss team may trade AoE for sustained single-target damage; a counterattack team may use protection instead of a conventional healer.

The starter roster must provide, without lucky pulls: one protector, one reliable damage source, one control or interruption tool, one support/energy option, and one cleanse or anti-control answer. Every potent strategy needs at least two practical counter categories, including one accessible answer. No role should exist only on a limited, paid or duplicate-heavy character.

18.3 Fragments and new-character access

Working weekly income target: 45–50 pulls per 28 days from recurring daily, weekly, event and mode sources, plus 15–20 universal fragments per week. At 100 universal fragments per eligible standard-pool selector, that gives approximately one selector every 5–7 weeks. The exact rates must be simulated against content release cadence, duplicate conversion and the number of new heroes per patch.

Featured guarantee proposal: guarantee the selected featured hero by 70 pulls with no hard-pity coin flip; pity progress carries across banners in the same banner family. Display odds, current pity, remaining pulls and carryover rules in the UI. At 48 recurring pulls per 28 days, a player who saves every pull for one banner reaches 70 pulls in about 40.8 days (70 ÷ 48 × 28), before one-time beginner rewards. Splitting spending across several banners delays a guarantee, so the UI should make this trade-off visible.

A base hero must perform their intended role at first acquisition. Duplicate upgrades may contribute only a small bounded PvP power advantage; the proposed target is no more than about 5–8% combat effectiveness between a base-copy hero and a max-duplicate equivalent under normalized PvP. Prefer universal fragments, selectors and duplicate conversion over multi-copy gates. A new archetype should be buildable in roughly 4–8 weeks of ordinary targeted play, not only through luck. This is a target to test, not a promise until the full income model is built.

18.4 Paywall placement and premium design

Never paywall the only cleanser, only anti-summon, only control answer, only backline-access tool, or only role that makes a mode functional. Premium/limited units may offer different execution, a new playstyle, or a broader roster; they must not be a mandatory key. Their base kit should work at base rank. Signature equipment should specialize a unit or create a sidegrade; an earnable generic option must remain viable.

Paid acceleration needs a ceiling. If it lets a spender reach a power milestone earlier, content caps and ongoing normal-play income should allow an active F2P player to close the relevant gap within a defined horizon. Paid stat paths should not compound forever. In normalized PvP, purchased upgrades should not provide more than a 5% direct stat/power advantage; any observed win-rate gap must be investigated while controlling for roster choice, experience and match volume.

18.5 PvP fairness

Use two lanes only if there is enough audience/content capacity: (1) **Normalized Arena**, where levels, gear baseline and duplicate stat effects are normalized, with team kit, composition, draft and player choices preserved; (2) optional **Progression Challenge**, where account investment matters but rewards are non-essential to competitive parity.

Illustrative weekly normalized-Arena rewards: complete 5 matches = 80 Arena Tokens; win 3 matches = +40; reach a rating band = up to +60 extra; a seasonal participation milestone grants a small one-time selector-fragment reward and a cosmetic token. Top leaderboard rewards should be prestige cosmetics, titles or frames, not exclusive counter units or irreplaceable combat gear. Losses should still count toward the participation milestone. The values are placeholders pending the wider economy.

Matchmaking: rating first; normalized power and account experience can protect new players from extreme mismatches but should not simply mirror spend. Use soft season compression instead of a hard reset, followed by placement matches. Weekly milestones should be completable across several sessions with missed-day forgiveness; daily attendance is not a competitive tax.

18.6 Fairness constitution and red lines

1. Core kit functions work at base copy/rank.
2. Core roles and at least two answers to major tactics are obtainable without payment.
3. No single limited hero or paid signature item is the sole answer to a combat problem.
4. No permanent duplicate gate unlocks essential synergy.
5. Same-category buffs are additive and capped; recursive triggers have explicit action/round limits.
6. Normalized PvP protects kit expression while bounding stat and duplicate gaps.

7. PvE progression rewards more than one archetype and offers a source for resources while stuck.
8. Ranked rewards do not make top placement necessary to obtain combat-critical resources.
9. New heroes add a trade-off or new tactical choice, not a strict upgrade to every old unit.
10. Every patch is reviewed for F2P-accessible counters, power creep, role coverage and dominant compositions.
11. Odds, pity, carryover, shard conversion and normalization are transparent in the game UI.
12. If monetization tests show that purchasers feel forced to buy power to clear ordinary content, the design is not meeting the project goal, even if short-term revenue rises.

18.7 Acceptance metrics

All thresholds below are test gates for a prototype, not universal industry benchmarks.

Measure: Win-condition comprehension - **Initial acceptance target:** At least 80% of test players can state a squad's intended win condition after one tutorial and one battle - **Action if failed:** Rewrite role explanations and battle recap before adding more systems

Measure: Archetype diversity - **Initial acceptance target:** At least 4–6 archetypes are viable across intended modes; no archetype over 35% sustained pick share without a mode-specific reason - **Action if failed:** Check unit power, mode design and support tax; rebalance or improve alternatives

Measure: Equal-power matchup - **Initial acceptance target:** Most non-specialist archetype pairings reach about 45–55%; deliberate counters may fall outside this but should not create hard locks - **Action if failed:** Inspect damage, target access, speed and counter availability

Measure: Strategy answerability - **Initial acceptance target:** At least two practical counter categories per strong strategy; one broadly accessible - **Action if failed:** Do not release the strategy or release it with accessible answers

Measure: Replacement flexibility - **Initial acceptance target:** Substituting a reasonable accessible unit of the same function should not collapse an archetype by more than 8 percentage points win rate - **Action if failed:** Adjust role kit, synergy dependence, or substitute roster

Measure: Synergy reliability - **Initial acceptance target:** Core intended combo activates in roughly 60–85% of intended lines depending on counterplay, not 100% regardless of response - **Action if failed:** Adjust trigger clarity, failure paths or counter windows

Measure: Duplicate sensitivity - **Initial acceptance target:** Base-copy versus max-duplicate equivalent differs by no more than about 5–8% combat effectiveness in normalized PvP - **Action if failed:** Reduce stat growth or move duplicate rewards toward customization/resources

Measure: Spend gap - **Initial acceptance target:** Investigate normalized-PvP win-rate gaps above 5 percentage points after controlling for rating, match count and roster choice - **Action if failed:** Separate skill/roster effects from spend; normalize further or change sales

Measure: Free archetype build time - **Initial acceptance target:** One alternative archetype becomes buildable in about 4–8 weeks through targeted play - **Action if failed:** Increase deterministic fragment/selector income or reduce the number of required pieces

Measure: Anti-stunlock - **Initial acceptance target:** No unit can be denied meaningful actions indefinitely by repeat hard-CC in standard PvP - **Action if failed:** Fix shared diminishing returns, immunity or action economy before release

Measure: Auto-battle readability - **Initial acceptance target:** At least 80% of test players can identify the main reason for a loss from recap/log - **Action if failed:** Improve effect icons, trigger history, target intention and recap language

18.8 Validation cadence

Before closed testing: deterministic simulator with seeded randomness, complete action logs, and automated mirror/matchup testing. During each balance pass: run all archetype-vs-archetype matchups at equal power; perform removal, substitution, counter and investment-sensitivity tests. In playtests: collect observations from beginners, F2P-focused players and paying players separately; ask players to explain why the team won or lost. On each content update:

audit the new unit against the starter roster, every core counter category, and at least two relevant archetypes. In live operations: track pick rate, win rate, first-death timing, crowd-control uptime, action share, healing efficiency, average battle duration, retry count, spend band, progression velocity and churn around progression gates.

Never treat a power score as a balance result. A score may summarize known stats, role coverage and synergy, but it cannot replace matchup simulation or playtesting.

19. Version 0.2 change log and remaining work

19.1 Changes made in this consolidation edition

Change: Corrected the old archive-status note - **Why it was made:** Four completed prior reports were recovered from the Library; the old note incorrectly implied full reports were absent - **Status:** Corrected

Change: Added complete historical report appendices - **Why it was made:** Preserve the actual prior research text, including details not present in the 20-page synthesis - **Status:** Completed in this file

Change: Added a coverage matrix - **Why it was made:** Makes it possible to audit whether a previous research area is missing from the living file - **Status:** Completed

Change: Reconciled contradictory historical values - **Why it was made:** Older examples such as 90–200 pulls and duplicate-heavy awakening are not silently mixed with the current 70-pull / low-duplicate-power proposal - **Status:** Working design decision; requires simulation

Change: Explicitly resolved the fairness section - **Why it was made:** Added concrete answer for stacking, fragments, team size, acquisition, paywalls, PvP, acceptance metrics and test cadence - **Status:** Proposed; not playtest-validated

Change: Added broad modes/progression/retention/monetization overview - **Why it was made:** Preserves the holistic RPG concept while keeping the next development phase focused on team combat - **Status:** Concept level; detailed design remains future work

19.2 Remaining high-priority work

P0 — Implement and validate the completed combat specification in Section 24. The rules are now specified; the open work is simulation/playtesting, not inventing more systems before the base engine works.

P0 — Build the starter roster matrix from Section 24.11: 12–15 original or licensed character concepts, primary/secondary roles, target access, combo trigger, counter category, accessibility tier and at least one substitute. Do not implement power-up add-ons yet.

P0 — Run the matchup and counter matrix in Section 24.12 across the ten archetypes; use results to tune the base combat kit before designing game modes or additive power systems.

P1 — Re-audit benchmark games mode by mode: verify current systems and date the observation, beginning with AFK Arena, AFK Journey, Mythic Heroes, Idle Heroes and then selected adjacent games. Separate game fact from community strategy and monetization opinion. Avoid comparing exact economy values across games with different currencies or modes.

P1 — Continue the paper acquisition economy by validating the source-level reliability of the recruitment ledger in Section 37: conservative recurring floor vs effort-dependent, seasonal and one-time rewards; test whether the proposed 4–8 week target-to-field-ready horizon is arithmetically possible. No prototype, simulator, seeded run or live/player test is authorized until the owner explicitly approves that phase.

P1 — Design first release modes: implement only modes that test distinct team questions; ensure each main role and counter appears in ordinary play before it becomes necessary in competitive content.

P2 — Plan sustainable monetization and live operations: estimate costs and test cosmetic/convenience demand; avoid assuming that generous pity alone creates revenue or retention.

19.3 Current open questions (not hidden)

1. Is five units and 2-front/3-rear the right battle size after a prototype? Keep four units as the primary alternative to test.
2. Does the proposed damage formula produce useful differences between tanks, bruisers and damage dealers without making armor/penetration mandatory?
3. Is a 70-pull no-coin-flip featured guarantee affordable at 45–50 recurring pulls per 28 days when the design team adds 1–2 heroes per release window? This depends on release cadence and must be modeled.
4. Are 15–20 universal fragments per week and a 100-fragment selector enough to keep team experimentation alive without making all collection goals trivial?
5. How much of a spender's value should come from wider roster choice versus convenience, cosmetics and bounded acceleration? Requires pricing research, not just combat theory.
6. Which benchmark mechanics remain current in October 2026? Re-check official patch notes and in-game screens before borrowing exact numerical values.
7. What does player testing show about battle readability, willingness to experiment, session length, retention and actual payer value? No prototype data exists yet.

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Status: implementable prototype proposal, not validated in a playable build.

20. Mobile-first combat engine: proposed implementation specification

Purpose. Earlier chapters establish the north star and role/synergy theory. This chapter turns the next layer into deterministic rules that a small mobile-game prototype could implement and log. Every number below is a starting hypothesis, not an industry standard. It must be checked by simulation and playtests. It supersedes an older example only for new prototype tests; historical formulas remain preserved in Appendices A-D and F.

Benchmark translation (freshly checked 28 October 2026)

These observations are used as design inputs, not instructions to copy another title. AFK Arena's current community guide describes role labels, faction advantages and lineup bonuses [M1]; we borrow the idea that team tags create a trade-off, but deliberately cap our proposed village bonus at +5% ATK/HP to avoid making a five-person faction stack mandatory. AFK Journey's current community team-builder shows up to five heroes, map-dependent placements and faction-bonus context [M2], supporting a five-unit prototype as a reasonable hypothesis—not proof that five is universally best.

Mythic Heroes' official beginner guide describes four factions, a +25% attack advantage against tactically vulnerable opponents and post-battle performance statistics [M3]. The transferable lessons are explicit counters and useful battle recaps. We should not copy its exact bonus: a +25% attack swing may be too large for our fairness goals and needs its own game-specific test. NIKKE's current community team guide describes a Burst I → II → III sequence that can make some team slots structurally necessary [M4]. That shows how an action sequence creates a real composition problem; our version should provide several obtainable units for each sequence function rather than one mandatory premium unit.

Honkai: Star Rail is not an idle RPG, but its official update notes show rotating endgame modes with different encounter modifiers [M8]. The transferable lesson is to let modes ask different tactical questions, not to demand the same five heroes everywhere. All community-guide observations can change with patches; these references are snapshots. Proposed Naruto-style rules remain original design proposals, not claims about the benchmark games.

20.1 Mobile design constraints

The game is a mobile Naruto idle RPG, not a PC auto-battler squeezed onto a phone. The battle must remain legible on a small display, continue to work when the player is not watching, and let a player understand the result without studying a

dense combat log. Default assumption for the prototype: portrait orientation, one-thumb navigation, concise skill descriptions, and a battle viewport that keeps all five allies and the current target readable. Treat orientation as an early usability test, not an irreversible commitment.

Constraint	Prototype rule	Why it supports fairness and mobile use
Short, low-friction sessions	A normal check-in should let the player claim idle rewards, make an upgrade, adjust a team and start a battle in roughly 2-5 minutes.	Progress does not require a long daily grind; short sessions remain meaningful.
Auto-battle is complete	Daily/campaign auto mode uses the full base kit. No paid item or premium-only unit is required to make automation work.	Convenience is not held hostage to spending.
Optional intervention	Use automatic skill priority by default. Manual targeting/ultimate timing may be available in practice, boss or story encounters, but must not be a paid-only advantage.	Depth can coexist with idle play and accessibility.
Readable effects	Display at most a small stack count plus one icon per effect family; expand details on tap. Clearly label target intent, active guard/mark, ultimate readiness and control resistance.	Players can understand cause and effect without visual overload.
Low-end-device support	Provide 30 FPS low-effects mode, animation speed/skip options and reduced particles; aim for 60 FPS on supported devices. Measure on representative low/mid/high devices.	The game must not reward players solely because they own faster hardware.
No mandatory attendance tax	Idle rewards accumulate continuously to a proposed 12-hour cap; missed daily tasks can contribute to a weekly catch-up track. The cap and rates remain tunable.	A player who misses a day loses less than a player who misses an exclusive power reward.
Server-authoritative paid items	Purchases, pity state, balances and ranked results are verified server-side. Offline rewards are calculated with sanity checks when reconnecting.	Protects purchases, fairness and player trust from basic client manipulation.

20.2 Team size and formation

[PROPOSAL] Launch the prototype with five deployed shinobi: two front positions and three rear positions. Five allows a recognizable team identity while leaving space for protection, damage conversion, enablers and counterplay. Six units are deferred because each added unit multiplies targeting, status, combo and interface complexity. Four units remain the main alternative if five produces slow fights or obscures decisions on phone screens.

Formation rules for MVP:

- Front positions are the default targets for melee single-target basics. A rear target can still be hit by a skill that explicitly reaches the rear, by a mark/focus effect, or when a guard/intercept ability permits it.
- Frontline status is not a permanent shield over the entire rear row. Avoid a system where rear assassins, marks or ranged attacks cannot matter until two front units die; that would make formation overly deterministic.
- Formation must affect decisions but not demand pixel-perfect placement. The two front and three rear slots are fixed and clearly labelled for MVP; free-grid tile placement is deferred.
- Team presets: at least 5 named saves available through normal play, so players can keep a campaign team, boss team, wave-clear team, anti-control team and test team without repeatedly rebuilding manually.
- Before battle, show the enemy threat preview (for example, “rear-row assassin”, “high DoT”, “boss charges ultimate”). Do not hide a required counter behind unknown or paywalled information.

20.3 Team composition is coverage, not a rigid recipe

Every team must answer five questions: (1) What is its win condition? (2) How does it stay alive long enough to execute it? (3) What converts its setup into meaningful progress? (4) How does it answer the opponent’s main threat? (5) How reliable is the plan when a key unit is controlled or defeated? These questions replace a compulsory “one tank, one healer, three DPS” template. A composition may trade a healer for disruption, a tank for tempo, or AoE for boss damage—but it must name the risk that trade creates.

Role / tag	Primary job	Typical enabler	Main weakness / answer	MVP budget guidance
Guardian / Protector	Redirects, intercepts or reduces incoming damage.	Carry, healer, counterattacker.	Defense bypass, marks that expose a different target, anti-shield tools.	Usually one protection specialist or two lighter protectors; full immunity is not the role.
Bruiser / Off-tank	Deals steady damage while surviving contact.	Healer, guard, counterattack.	Kiting, control, high single-target focus.	Can cover survival and damage without being best at both.
Burst finisher	Removes one priority enemy in a short window.	Mark setter, defense-down, speed/order support.	Guard, decoy, cleanse, defensive cooldown.	High peak, lower uptime or higher setup requirement.
AoE / Wave clearer	Removes or weakens groups.	Gather/mark/defense-down, shield support.	Bosses, split formations, AoE reduction, staggered waves.	Lower single-target efficiency than a dedicated boss finisher.
Ramping / DoT damage	Wins through damage over time or increasing output.	Protection, defense-down, anti-cleanse bait.	Cleanse, fast burst, short time limits.	DoT stacks have a per-target cap and clear readable timers.
Controller / Disruptor	Delays or restricts a high-value enemy action.	Speed opener, focus-fire, follow-up.	Tenacity, control DR, cleanse, alternate damage plan.	Hard control must never create a permanent turn lock.
Buffer / Tempo enabler	Improves attack, speed, energy or action quality.	A carry with compatible timing.	Dispel, interrupt, eliminating the support.	Must have a useful low-rarity alternative; same-category buffs cap.
Debuffer / Mark setter	Exposes a target or reduces defense/accuracy.	Single-target finisher or AoE follow-through.	Cleanse, decoy, target protection, resistance.	Mark cannot force every ally to retarget when doing so would be harmful.
Healer / Cleanser	Restores HP or removes harmful effects.	Tank, DoT-heavy team, endurance archetype.	Anti-heal, burst, cleanse bait, interrupt.	Avoid allowing healing to erase all damage without a time cost.
Chakra / energy support	Brings an ultimate or important technique online earlier.	High-value ultimate users.	Energy drain, slow, killing or controlling support.	Accelerates a plan; must not be mandatory for the plan to function.
Scout / Anti-stealth	Reveals, marks or improves target access.	Burst, rear access, anti-evasion.	Protection, decoys, alternate win condition.	Counter available in starter or standard pool.
Counterattacker / Reflector	Punishes enemy aggression or shields.	Guard, sustain, clear attack forecast.	Unanswerable burst, non-attack effects, dispel.	Reflect caps per action and cannot recursively trigger itself.
Summoner / Clone maker	Adds decoys, bodies or extra tactical triggers.	AoE, turn protection, summon interactions.	AoE, anti-summon, target selection.	MVP uses a simple capped clone object; advanced summon behavior is phase 2.
Flex / Hybrid	Covers two functions with a clear trade-off.	Compositions short on one type of coverage.	Lower peak than a specialist, kit overload.	A hybrid must not outperform specialists at every job.

20.4 Round order, actions, technique and chakra

[PROPOSAL] Use a round-based initiative queue for the first prototype rather than a free-form action bar. Each living unit receives one normal action per round. Units act in descending Speed order; ties are settled by a deterministic battle-seed rule and displayed in replay. Recompute order at the start of the next round. Speed buffs/debuffs change initiative but do not create extra actions on their own.

Rule	MVP value / behavior	Anti-exploit condition
Normal action	One action per living unit per round: Ultimate if ready and AI condition is met; otherwise Technique if off cooldown and its condition is met; otherwise Basic.	The player can inspect each unit's priority setting; AI never silently skips a ready action without a logged reason.

Rule	MVP value / behavior	Anti-exploit condition
Technique cooldown	Default cooldown is 2 full subsequent owner action slots after use; it is unavailable on each of those two slots and becomes eligible on the following owner slot. Cooldown decrements at end of subsequent owner slots, never on the cast slot.	Every kit declares its cooldown and AI eligibility condition. No hidden cooldown reduction or reset loop in MVP.
Personal Chakra	Each unit has 0–100 personal Chakra. Basic grants +25; Technique grants +15; Ultimate spends 100. A defender gains +5 once per distinct direct enemy attack instance only if that attack produces positive post-DEF/post-DR damage before Shield, up to +15 per defender per round. A root action that KOs at least one enemy grants its actor +10 once, even if the KO is made by that root action's bounded follow-up.	No Chakra from DoT/Wound, Shield generation, clone/object attacks, follow-up events, reaction hits or Wound KOs. Shield absorption does not erase the incoming-damage Chakra event if pre-Shield mitigated damage was positive. Chakra caps at 100; overflow is lost and logged.
Ultimate trigger	At 100 Chakra, an Ultimate becomes available. Auto mode uses it next action unless a visible saved AI rule says to hold it for a condition.	No premium-only "smarter auto". Manual hold may be available for practice/story, not a ranked spending advantage.
Follow-up event	A follow-up is a bounded damage/utility event, not a full normal action. Max one follow-up per acting unit per round and two team-wide follow-ups per round.	Follow-ups cannot crit, generate Chakra, create a takedown-trigger chain, lifesteal/reflect recursively or trigger another follow-up.
Speed / turn advance	Speed modifiers are capped at ±20% and reorder actions at the beginning of the next round only. Turn-meter advance/delay and extra normal actions are excluded from MVP.	Revisit dynamic turn meters only as a new combat redesign after the round-based model is tested; never layer both systems together.

Why personal Chakra for MVP: a single shared team gauge risks making "Chakra battery" mandatory. Personal meters let every character execute a baseline ultimate plan. Chakra support remains valuable because it changes timing, not because every team becomes nonfunctional without it. If testing later favors a shared gauge for the Naruto fantasy, it must be tested as an alternative, not quietly layered on top.

20.5 Targeting rules and battle readability

- [SUPERSEDED BY SECTION 24.5] Standard Basic target order: eligible Taunt target first; then the chosen legal squad-policy priority; then the frontline enemy with the lowest current HP percentage; if the frontline is empty, the lowest-HP-percentage legal rear target; ties resolve left-to-right. Skill-authored legal target rules take precedence.
- Taunt: overrides target selection only for attacks that can legally target the taunter. It does not redirect global AoE, unavoidable boss mechanics or explicitly marked attacks.
- Mark: a marked target gains the specified vulnerability or focus interaction. A mark does not automatically override all target rules unless the skill tooltip says "retarget to marked target".
- Rear access: assassination, ranged and dash skills explicitly declare whether they can hit the rear. Do not hide rear access in ambiguous animation cues.
- AoE: MVP shapes are "single target", "front row", "rear row", "all enemies" and "adjacent ally" only if adjacency is implemented. Avoid grid geometry until tested.

- Boss telegraphs: show the target or area and the round in which a major attack will resolve. A boss should test a team's response, not require a premium unit whose kit is unknown.
- Post-battle recap: show damage dealt, healing, damage prevented, control actions, Chakra contribution, first knockout, and the top 1-2 reasons the team lost. Display synergy triggers and the effect each trigger had.

20.6 Prototype damage and defensive formula

[HISTORICAL DRAFT — SUPERSEDED BY SECTION 24.6] This older v0.5 paragraph used a flat DEF−PEN expression. The current prototype uses a percentage-based shared DEF-reduction bucket with a 40% cap. Keep one pipeline for ordinary direct attacks and explicit exceptions for DoT, healing and rare true damage.

[HISTORICAL VARIABLE DEFINITIONS — SUPERSEDED] Current definitions: effective DEF is target DEF multiplied by (1 − capped effective DEF reduction). Do not introduce flat PEN as a second independent defense bypass in MVP.

[HISTORICAL FORMULA — SUPERSEDED] Use Section 24.6: DEF Multiplier = max(0.35, 1000/(1000 + Effective DEF)); Effective DEF = max(0, target DEF × (1 − min(40%, sum of DEF Down and DEF Penetration percentages))).

[CURRENT CONSTRAINTS] The explanatory principles remain; for authoritative calculations and caps, use Section 24.6. Flat defense penetration and percentage DEF reduction are not separate MVP buckets; they share one 40% effective-DEF-reduction cap.

Worked example: ATK 1,200; C = 1.6; attack bonus +20%; skill bonus +10%; DEF = 1,000; PEN = 0; vulnerability +10%; no critical hit. Damage = floor(1,200 × 1.6 × 1.20 × 1.10 × 0.50 × 1.10) = 1,393. The rounded value is illustrative; implementation precision must be consistent across platforms.

Buff stacking: identical named effects do not stack strength; keep the stronger active instance and refresh duration only if the refreshing effect is at least equal. Different sources in the same named category add before that category's cap. Multiplicative categories are allowed only when they represent distinct mechanics and are separately capped. Every passive that triggers from damage, healing, a kill, a summon or an extra action must declare its trigger cap.

20.7 Defensive rules and status effects for MVP

[PROPOSAL — LOCKED FOR PROTOTYPE] Use the following ten named conditions only in the first build. Every condition has a visible icon, a concise tooltip, a precise start/end timing rule, and at least one answer available without premium acquisition. There is no hidden Accuracy or Tenacity stat in MVP; each skill shows its chance, and any special boss resistance is explicit in the encounter preview.

Condition	Rule for MVP	Counter / stacking rule
Mark	After-damage by default: target takes +10% final damage for its next 2 action slots. If applied before its slot, it can affect that slot; if applied after it acted, its duration starts next slot. Does not itself alter target selection.	Cleanse, interception, decoy or change target can answer it. PRE-DAMAGE effects must be explicitly tagged; Mark's +10% shares the total +15% Vulnerability cap.
Wound (DoT)	At the start of each of the target's next 2 action slots, deal 18% of the applier's snapshot ATK (after Attack Down, before temporary Attack Bonus). Uses normal DEF and non-DEF damage reduction and is absorbed by Shield; max 20% target Max HP from all Wound sources per slot before mitigation.	Cannot crit, generate Chakra/takedown Chakra, or trigger on-hit, reflect, lifesteal, or follow-up effects. Same source refreshes; max 2 distinct sources. Cleanse one source, eliminate applier, or win before tick value compounds.
Attack Down	Reduce ATK by the strongest active Attack Down effect (default 15%) for 2 target action slots; combined magnitude does not stack and is capped at 40%. It applies before temporary Attack Bonus.	Cleanse, prevent the applier acting, or wait out duration. Log the effective ATK used for each damage event.
Defense Down	Reduce effective DEF by 15% for 2 target action slots. DEF Down and DEF Penetration share one additive effective-DEF-reduction bucket capped at 40%.	Cleanse or eliminate the debuffer; show the combined DEF-reduction contribution in the damage recap.

Condition	Rule for MVP	Counter / stacking rule
Slow	Reduce Speed by 15% for the next round's initiative calculation only; it expires at the end of that round. Speed changes affect ordering next round, never move a unit inside the current queue.	Cleanse, apply Speed-up or rely on higher base Speed. Same-named Slow does not stack.
Stun	On a successful application, the target loses its next normal action once. Repeated attempts use the shared target-specific 60%/30%/15% tiers, subject to displayed base chance and the status chance clamp. A Stun never extends another Stun.	Cleanse before the target's slot, prevent the controller acting, or use previewed Stun resistance. An attempt on an already-Stunned target cannot cause another lost action or increase the tier.
Silence	Blocks Technique and Ultimate for the target's next action slot; Basic remains available and earns its normal +25 Chakra. Applied after damage unless the skill is explicitly PRE-DAMAGE.	Cleanse or wait out one action; does not raise/reset Stun DR and cannot cancel a Charging skill unless it also has the Interrupt tag.
Healing Suppression	Reduce healing received by up to 40% for the target's next 2 action slots. Strongest active same-category suppression applies; it does not reduce Shield unless the skill explicitly says so.	Cleanse, use Shield/mitigation or defeat the applier. Healing formula and suppression amount are shown in recap.
Shield	Absorbs damage after DEF and non-DEF Damage Reduction. Each shield lasts through 2 recipient action slots; same-named shield magnitude does not stack, different shield sources share a 40% Max HP active-capacity ceiling.	Shield generation grants no Chakra. An incoming direct hit can grant defender Chakra if its pre-Shield mitigated damage was positive, even when Shield absorbs all remaining damage. Dispel/break must be named and visible.
Taunt	Redirects the next 2 eligible enemy single-target actions to the Taunter, or expires when the Taunter completes its second owner action after application, whichever occurs first. Ends immediately if the Taunter is defeated.	AoE, explicit Ignore Taunt and scripted unavoidable attacks do not redirect. Mark cannot override Taunt unless the skill explicitly says so.

Timing contract: Wound ticks before the afflicted unit chooses an action. If the tick defeats the unit, it takes no action. Stun is checked after start-of-turn damage and before AI action selection. Other negative durations tick down after the afflicted unit completes or loses its turn. A new copy of the same named effect never increases magnitude; it may refresh duration only when the new instance is at least as strong and does not shorten the existing duration.

[SUPERSEDED BY SECTION 25.8] MVP has Stun as the only action-denying hard control. Because a turn cannot be divided into 50% or 25%, diminishing returns reduce application chance by control tier instead of duration.

Boss exception: do not give bosses hidden full immunity to every useful tactic. A boss may reduce hard-control duration or reject action denial, but its resistance must be visible. A distinct Interrupt tag may cancel a clearly telegraphed charge without deleting a whole normal turn; every such charge must have at least one broadly obtainable counter and one alternative response (guard/mitigation, cleanse or target defeat).

Effect	MVP rule	Counterplay / clarity
Shield	Shield absorbs post-DEF and post-non-DEF-DR damage. Same-named magnitude does not stack; all active Shields share a 40% recipient Max HP capacity cap. Default duration is 2 recipient action slots.	Shield does not itself create Chakra. The target still gains incoming-damage Chakra only when the incoming direct hit produced positive damage before Shield.

Effect	MVP rule	Counterplay / clarity
Guard / Intercept	May redirect only the specific eligible hit/action named by its skill. At most one eligible hit per Intercept effect; no re-interception. AoE may hit its declared targets normally and cannot be globally redirected.	Row/AoE, rear-access and Ignore-Intercept tags provide counterplay; all target changes are logged.
Mark	Default Mark is +10% Vulnerability for the next 2 target action slots; shares the overall +15% vulnerability cap. It lands after the triggering hit by default.	A skill tagged PRE-DAMAGE may apply it before the hit; the tag must be visible. Otherwise it cannot amplify the attack that applied it.
Defense Down	Reduces DEF, but total DEF reduction caps at 40%. Same-named sources do not stack strength.	Cleanse/Dispel or killing the debuffer. Boss immunity, where needed, must be previewed.
Wound (MVP DoT)	Snapshot applier ATK after Attack Down, before temporary Attack Bonus. Tick at the beginning of each of the target's next 2 action slots; normal DEF/non-DEF reduction, then Shield. Total Wound ticks capped at 20% target Max HP per slot before mitigation.	No Crit, Chakra, takedown Chakra, lifesteal, reflect, on-hit or follow-up triggers; max 2 sources, same source refreshes.
Slow	Lowers initiative by a proposed 15% for 1 round; multiple slows refresh duration, not magnitude.	Cleanse, immunity window or a comp that wins by setup rather than speed.
Stun / Hard CC	Stun denies one normal action. Repeated Stun uses the shared target-specific application-chance tiers in Section 25.8; fractional duration is not allowed. Stun never extends an active Stun.	Cleanse before the target acts, prevent the controller acting, or use an explicitly previewed resistance. A target already Stunned cannot lose a second action from another Stun before acting.
Silence / Ability lock	Blocks Technique and Ultimate for the target's next action; Basic remains available and gains its normal Chakra. Silence is soft control and does not raise or reset the Stun diminishing-return tier.	Cleanse, wait out one action, or use a Basic-focused plan. It cannot cancel a charged move unless the effect is explicitly tagged Interrupt.
Cleanse / Dispel	Cleanse removes harmful effects from allies; Dispel removes positive effects from enemies. They are separate keyword actions. A single effect cannot be both immune to and removable by the same rule.	High-value control or buff/debuff strategies must have at least one broadly obtainable answer.
Revive	Deferred from MVP. If later introduced, the default is once per unit per battle at 20% Max HP, 0 Chakra, no self-revive, no replay of start-of-battle effects and no recursive revive chain.	A future revive requires its own balance pass and an accessible counter; it must not become a required role or unending reset loop.

Use 8-10 statuses in the first combat prototype; defer elemental reactions, persistent terrain, long summon skill trees, revive chains and compound debuff inheritance. More systems do not necessarily create more strategy if players cannot read them on a phone.

20.8 Synergy: make the combo legible, bounded and accessible

A strong synergy should have three visible stages: SETUP → CONVERSION → RESPONSE WINDOW. Example: a scout marks a rear enemy (setup), a finisher gains a small damage bonus against that marked target (conversion), and the opponent can cleanse the mark, guard the target or control the finisher (response window). That is stronger design than giving five related characters a huge invisible attack multiplier.

Synergy family	Recommended implementation	Fairness guardrail
Role / functional	Protection keeps a carry alive; mark amplifies a finisher; cleanse answers control; DoT gains value from a defense shred.	Any essential function has at least one accessible substitute.
Action-chain	One explicit follow-up when a condition is met, such as “after a marked target is hit, this ally performs a small strike once per round”.	Follow-ups do not recursively trigger follow-ups; one extra action per unit per round.
Village / faction	2 members: proposed +3% ATK and HP; 3+ members: +5% ATK and HP total; no higher stat tier for 4 or 5 members.	Faction is a mild bonus, not a 5-unit lock. Utility must often beat raw bonus.
Clan / relationship	A tag or pair may add a small tactical interaction, such as +5 Chakra once per round to one partner or a one-time assist.	Avoid large always-on stat premiums. The interaction still has a non-bond alternative.
Formation	Frontline defender creates a guard window; rear scout reaches hidden threats; adjacent protection is reserved for later if it increases interface burden.	Positional bonuses must be explainable and not require pixel accuracy.
Status engine	Poison setter plus an ally who gains a small effect against poisoned targets.	Hard cap stack count and frequency; cleanse must be a real answer.
Counter-synergy	An anti-shield or reveal tool gains value against one archetype, while remaining useful in at least one other matchup.	Avoid single-purpose units that are useless unless a specific opponent appears.

Pairing or bond bonuses should usually modify a trigger, timing or resource—not award a large permanent multiplier. A proposed maximum: faction/village stats contribute no more than +5% ATK/HP; relationship effects should be measured by expected win-rate impact rather than their text number. If a bond produces a larger win-rate swing than the rest of a hero’s kit, tone it down or make its counter more accessible. These are test gates, not truths derived from other games.

20.9 Ten archetypes and their answers

Archetypes are tactical plans, not fixed classes or mandatory five-hero recipes. The same hero may fit more than one archetype; what matters is whether the lineup can execute its win condition, survives the counter-window, and has at least one accessible substitute for each critical function.

Archetype	Win condition / minimum functions	Weakness and practical counters
Protect-the-carry	Keep one scaling damage dealer alive; use protector/guard plus one enabler and enough utility to stop a rush.	Weak to rear access, mark-and-focus and anti-heal. Answers: guard the carry or control the diver; accessible AoE can pressure the protection shell.
Speed / tempo	Act earlier, apply Slow/Mark/control, and win before enemy setup. Needs Speed/initiative function, conversion damage and a fallback if first turn fails.	Weak to resilient teams and Speed-down. Answers: slow/control the initiator; shield or punish the first burst and win the longer fight.
Burst pickoff	Mark or expose a high-value target, then focus a finisher into a knockout before enemy support responds.	Weak to shields, intercept, cleanse and decoys. Answers: protect/intercept the target; remove Mark or control the finisher.
Control tempo	Use Stun/Silence/Slow to interrupt one key enemy action, then convert the gained window into damage or sustain advantage.	Weak to cleanse, hard-control DR, immunity windows and long fights. Answers: cleanse/tenacity-like kit; burst the controller before repeated control is possible.
Wound / attrition	Apply bounded DoT, reduce enemy efficiency, and survive long enough for repeated ticks.	Weak to cleanse, burst and shielded opening turns. Answers: cleanse Wound; focus the applier or use tempo to end the fight before ticks compound.
Counterattack / trap	Invite eligible direct attacks into guard/counter triggers, then punish repeated enemy actions.	Weak to AoE, DoT and effects that do not trigger counters. Answers: use AoE/DoT; bypass the guarded front with declared rear access.

Archetype	Win condition / minimum functions	Weakness and practical counters
Clone / decoy	Use capped decoys to absorb single-target attacks or misdirect targeting while the real team sets up.	Weak to AoE and reveal/mark effects. Answers: use AoE; select a real target using the published target policy. Clones never create full extra turns.
Sustain / grind	Combine healer/cleanser, protector or bruiser and reliable damage; win by surviving opponent burst.	Weak to Healing Suppression, focused pickoff and time limit. Answers: accessible anti-heal; coordinate burst on healer or damage engine.
Mark and focus	A scout/debuffer marks a target; finisher, ranged attacker or follow-up converts the vulnerability.	Weak to cleanse, intercept and redundant enemy threats. Answers: cleanse/protect the target; disrupt the finisher or switch focus to a secondary target.
AoE / wave pressure	Spread efficient damage across enemy positions and remove fragile units or decoys; includes an answer to the enemy carry.	Weak to front-loaded protection and durable single-target fighters. Answers: guard a key unit and focus the AoE attacker; use resilient bruisers or quick pickoff.
Boss / single-target execution	Stack reliable single-target output, bounded DEF reduction, sustain and an interrupt/mitigation answer to a telegraphed action.	Weak to setup denial, damage caps and over-investment in debuffs. Answers: protect the debuffer or use raw sustained damage; interrupt/mitigate the charged attack using an accessible option.

Answerability rule: each strategy must have at least two counter categories from different functional sources. One counter must be available in the starter roster; the second must be in the ordinary non-paid pool or a deterministic earn path. A “counter” that is just a specific higher-rarity hero is not a counter category.

Archetype / win condition	Useful role coverage	Primary weaknesses	Two practical counter categories	F2P-access rule
Protect the Carry: keep one damage dealer alive through two ultimate windows.	Protector, carry, enabler, flexible control or cleanse.	Assassin access, control on carry, anti-shield.	Mark/rear access; dispel or anti-shield; burst before protection setup.	Starter protector and at least two obtainable carries.
Speed / Tempo: act first, disrupt setup and snowball initiative.	Speed support, controller, quick finisher, fallback sustain.	Slow/turn-order denial, durable team that survives opener.	Accessible slow/control resistance; defensive opening or cleanse.	Speed support not required for all other teams.
Burst Pickoff: mark and remove one high-value enemy.	Scout/mark, finisher, turn enabler, protection.	Guard, decoy, cleanse, low burst uptime.	Intercept/guard; mark cleanse/decoy; control finisher.	Basic mark and guard options in standard acquisition.
Control: buy turns to win through actions denied.	Controller, speed support, follow-up damage, cleanse or energy.	Tenacity, cleansing, control DR, alternate damage.	Control resistance/cleanse; AoE or DoT that contributes through control.	Never create paid-only control immunity.
DoT / Attrition: survive while damage accumulates.	DoT applier, protector, support, cleanse denial or focus.	Fast burst, cleanse, short boss timer.	Cleanse/dispel; burst/pickoff before stacks.	At least one standard-pool DoT and cleanse.
Counterattack: survive enemy actions and punish them.	Guard, counterattacker, sustain, debuffer.	Non-attack effects, anti-heal, burst outside the guarded target.	Target bypass/mark; silence/anti-heal or burst window.	Counterattack cannot require premium gear.
Clone / Summon: use decoys or extra bodies to confuse targeting.	Summoner, main carry, protect, AoE or disruption.	AoE, anti-summon, target selection, time limit.	Broadly obtainable AoE; anti-summon or mark/reveal.	Clone core mechanic exists on a basic-access unit.
Sustain / Grind: outlast burst and win on resource efficiency.	Healer, protection, steady damage, cleanse.	Anti-heal, target priority, timed objectives.	Anti-heal/interrupt; burst focusing healer.	No exclusive healer; support alternatives trade cleanse vs raw heal.
Mark and Focus Fire: concentrate actions on one target.	Mark setter, two damage converters, protector, utility.	Dispel, decoy, guard, target redirection.	Cleanse/decoy; guard/intercept or forced retarget.	Mark does not require a named canon bond.
Boss Specialist: maximize reliable single-target output.	Single-target DPS, defense-down, Chakra/tempo, cleanse/protect.	Multi-wave fights, boss phase interrupts, sustain pressure.	Boss mechanic tests; defensive swaps and alternate damage source.	Boss control immunity is visible; no one premium unit is the only solution.

20.10 Worked example: synergy beats a raw-power pile

Hypothetical test, not a Naruto canon skill claim: both teams have equal total normalized stats. Team A spends all five slots on high personal damage. Team B uses a guard, mark setter, finisher, controller and flex damage. Enemy preview

announces a dangerous rear-row ultimate on round 2. Team A focuses the front tank and fails to interrupt the threat; Team B marks the enemy ultimate user, uses control or guard to survive the first exchange, then spends the finisher's ultimate on the marked threat. Team B can win despite lower total damage dealt because it prevented one enemy ultimate and converted the target vulnerability into a knockout.

The point is not that Team B must win. It is that composition should create a plausible path to win. If Team A still wins nearly every equal-power simulation because of raw stat coefficients, role interactions are too weak. If Team B wins regardless of whether its mark lands or the enemy has a counter, the synergy is too strong. Test both failure paths.

21. Fairness-by-design for mobile F2P and pay-to-progress

The project requirement is not "no monetization"; it is a clear boundary on what monetization buys. Mobile players may spend to save time, obtain more cosmetic expression, or broaden their roster earlier. They must not be forced to purchase the only functional guard, cleanser, counter, energy enabler or damage pattern. Fairness needs measurable thresholds and progression math, not just a kind-sounding promise.

21.1 Access rules for team-critical functions

Function / system	Minimum access rule	Reject if...
Core roles and counters	Starter team or guaranteed early selectors cover protection, reliable damage, AoE or multi-target damage, mark/priority targeting, one disruption tool, sustain, and a cleanse/anti-control route. These need not all be seven separate heroes; starter kits may cover more than one role.	A normal campaign boss or common PvP archetype is practically unbeatable without a paid/limited unit.
Base hero identity	At base copy, a hero has its entire tactical identity: signature mechanic, core synergy trigger and counterplay. Rank-ups may improve reliability or options, not unlock the only version that functions.	The base copy is deliberately incomplete and needs many dupes to activate the advertised role.
Signature gear	Core kit works without signature equipment. A signature item can provide alternate tuning or convenience, not an exclusive hard counter or a huge multiplicative damage jump.	A carry needs a paid weapon to reach ordinary expected output, or only a paid weapon can answer a strategy.
Limited units	They may debut on a limited banner, but core roles and counters are not limited to that banner. Offer a predictable future standard/selector path or functional equivalent after a stated delay.	A limited character becomes mandatory in an endgame mode before F2P players can access an equivalent.
Duplicates	Duplicates can yield universal fragments, cosmetics, sidegrade options or bounded mastery. They do not unlock essential synergy. Normalized-PvP base-copy vs max-copy effectiveness target ≤5%; investigate and revise above 8%.	A team cannot function at one copy, or max duplicates add multiple extra actions/turn locks.
Automation and retries	Core auto battle, speed controls, saved teams and sufficient daily PvP attempts are baseline features. Convenience monetization cannot make the free client painful.	A paid pass is required to avoid repetitive 1x battles or to get a fair competitive attempt count.

21.2 Illustrative summon budget and deterministic paths

These are project targets for paper budgeting, not facts about benchmark games. A "pull-equivalent" is the free value needed for one ordinary summon, but each source must be tagged as standard-only, featured-eligible, fungible between banner families, variable event income, or one-time/progression milestone income. The table below is a gross 28-day planning window whose source reliability differs; Section 36 reconciles its categories. It is not a statement that all listed pulls can be spent on the featured banner.

Illustrative acquisition source	Pull-equivalents per 28-day window	Design note
Login/attendance rewards	8	Use flexible claim windows; do not erase most of the reward for missing one day.
Daily activity, accumulated through weekly completion	14	A weekly catch-up track can replace a perfect streak; no need to log in at exact reset times.
Weekly objectives	12	Four weeks of moderate tasks; avoid thresholds that require spending or hoarding.
Campaign/tower milestone income	10	Mostly one-time or progression-driven; model separately for new accounts and veterans.
Monthly event participation milestones	10	Core rewards should be reachable without a limited unit or paid stamina.
Guild/social/rotating activity	6	Use participation rewards instead of top-rank-only payouts.
TOTAL GROSS TARGET	60 per 28 days	Includes progression/milestone value; not a sustainable recurring floor.

Current accounting note (v0.10.11): Section 40 supplies a new, explicitly proposed non-overlapping core reward schedule. The older 14 daily / 12 weekly rows in this section remain historical assumptions; their independence was never established. The new schedule is a design alternative, not evidence that those original totals were correct.

Featured character access proposal: 70 pulls gives one guaranteed copy of the featured hero, with no 50/50 failure step. The guarantee counter carries to the next featured banner in the same banner family; spark tokens do not expire. The previous 32.7-day illustration at 60 pull-equivalents per 28 days applies only if all 60 are usable on that featured banner and sustainably earned. Section 36 now separates that optimistic all-featured case from the unresolved cross-banner source allocation.

Universal character fragment proposal: 100 universal fragments unlock one selectable standard-pool hero. Target income is 15-20 fragments per week across multiple modes, so a player can target an alternative archetype in roughly 5-7 weeks if all weekly sources are completed. Add a new-player selector that grants the missing basic function—not a randomly chosen DPS—when the account's starter roster is incomplete. Limited heroes should later enter a predictable selector/rotation or be replaceable by functionally equivalent units.

New-unit cadence is a major dependency [HYPOTHESIS, NOT VALIDATED]. At one hero every 4-6 weeks, a 70-pull guarantee and 45-50 recurring gross pull-equivalents could let a focused player target a featured hero roughly every 39-44 days only if the full recurring amount is featured-eligible. If income must also fund standard recruitment, or the monthly total includes variable sources, the guarantee cadence is slower. Therefore, do not claim that the current income target funds most new releases until the acquisition ledger and release cadence are reconciled. Content must remain answerable without collecting every new unit.

21.3 What paying users may buy

Purchase category	Recommended policy	Bound / excluded benefit
Cosmetics and expression	Primary monetization: costumes, alternate VFX, profile frames, emotes, village décor, victory poses. For licensed Naruto content, all assets require approval.	No hidden stats. Competitive VFX must remain readable and may be simplified in ranked mode.
Optional progress bundles	Resources that shorten leveling or gear crafting.	PvE speed is acceptable only if no ordinary story or core role is paywalled. Ranked PvP normalizes these stats.
Roster breadth	Summons, selectable heroes and extra build options can be sold, provided the same core role/counter functions have accessible earnable routes.	No character is the only answer to a common strategy. Banner pity and odds must be shown clearly.

Purchase category	Recommended policy	Bound / excluded benefit
Convenience / subscription	Cosmetic track, optional reward convenience or extra team preset slots after baseline presets are useful.	Do not sell basic auto, default sweep for already-cleared grind, or extra rated PvP attempts.
Battle pass	Free and paid tracks; paid track adds cosmetics and modest catch-up materials.	No mandatory character, unique tactical effect or elite PvP-only gear on the paid track.
Paid competitive power	Reject as a core product strategy.	No direct purchase of permanent ranked stats, exclusive counters, extra turns or immunity that cannot be earned/answered.

Sustainability caveat: “fair” does not automatically mean financially sustainable. Model team/staff/content costs, platform fees, payer conversion, repeat purchasing, regional pricing, and revenue concentration separately. Use cosmetics and optional acceleration as hypotheses to test with players, not as guaranteed revenue. If the model only works when a small number of users buy a permanent competitive advantage, the business model conflicts with the project’s design constitution.

21.4 PvP rules that prevent spending from buying the result

- Primary ranked Arena: normalize character level, skill tier and gear power to the current ranked template. Preserve unit kit choice, formation and player drafting. Allow only a small bounded investment difference; test the target of ≤5% combat-effectiveness difference between base-copy and max-duplicate versions.
- Optional unranked “Power Arena”: may let account progression show more strongly, but the reward ceiling must not include exclusive competitive power. Use cosmetics, titles or prestige points that do not create a compounding power gap.
- Matchmaking: rating first, with account age/power band only as a safeguard for new players. Do not match primarily by spend. Use soft seasonal compression and placement matches rather than a hard reset that forces a daily climb.
- Rated attempts: baseline attempts are free and sufficient to reach ordinary seasonal milestones. Purchases cannot buy more rating attempts or reset a loss. Extra practice is unranked.
- Rewards: most combat-critical resources come from participation and cumulative seasonal milestones. Rank rewards add cosmetics and a modest amount of general currency; top-rank placement is not required to build viable teams.
- Anti-FOMO: the weekly reward path should allow skipped days, and a seasonal character should not become permanently unavailable if a player misses one month. Do not place a key counter only inside a short high-rank or paid event.

21.5 Mobile store transparency and player trust

For a mobile launch, treat odds disclosure as a visible product requirement, not an obscure legal checkbox. Apple’s App Review Guidelines state that randomized virtual-item purchases (“loot boxes”) must disclose odds before purchase [M5]. Google Play’s developer policy similarly requires odds to be clearly disclosed in advance and in close proximity to the purchase [M6]. The UK ASA/CAP published a February 2026 enforcement notice calling for a prominent statement when listings contain paid loot boxes or randomized items bought with purchased-only currency [M7]. Requirements differ by jurisdiction and may change; legal review is required before launch in each target market.

Product requirement: every banner panel displays per-rarity and featured rates, guarantee counter, whether pity carries over, spark cost, fragment conversion, and what happens to duplicate copies before the player spends. Show purchase prices in local currency, never rely on a vague “best value” label, and provide a purchase history and clear refund/support process.

22. Prototype test plan, fairness gates and live balance governance

22.1 Minimum prototype before content expansion

- Five allies and five enemies; two front and three rear positions; one basic, one technique and one ultimate per hero; personal Chakra; round-based initiative.

- Roles represented by at least 12–15 original character concepts so every core role has a substitute. Use placeholder art and abstract ninja archetypes until the licensing path is established.
- Ten MVP conditions; shared hard-control diminishing returns; damage formula v0.6; no recursive triggers; readable auto-battle recap. Full rules are in Section 24.
- Ten archetypes and a matchup matrix; use only three small encounter scripts as combat test fixtures (durable target, multiple enemies, and a dangerous rear threat). Do not build the full campaign, wave-clear, boss, or ranked modes until the combat exit gate passes.
- Logging of every action, target choice, buff application, damage modifier, Chakra change, control resistance tier, shield absorption, and reason an AI rule selected a skill.

22.2 Required tests and acceptance gates

FUTURE IMPLEMENTATION / VALIDATION PHASE ONLY — These are proposed release gates for the baseline combat engine, not tasks for the current research phase and not promises about current balance. If a gate fails later, adjust the combat rule or kit coefficient before introducing extra progression power systems.

Gate	Proposed acceptance threshold	Required response if it fails
Archetype viability	At equal normalized power, 8–10 archetypes each have at least 2 plausible favorable matchups and no archetype has an unexplained overall win rate above 60% across the test matrix.	Audit role budgets, matchup assumptions and direct-damage coefficients; do not solve with rarity gating.
Matchup balance	For intended even matchups, target 45–55% win rate. Intended counter matchups may reach 55–65%; above 65% requires an explicit opposite counter and evidence it is still answerable.	Check seeded side advantage, targeting loops, control uptime and single-unit dominance; rebalance before adding modes.
Substitution robustness	Replacing one critical unit with an accessible unit covering the same function changes win rate by no more than 8 percentage points in its intended archetype role.	Create a second kit path or reduce the first unit's unique effect; do not make a character mandatory.
Counter availability	Every potent archetype has 2 distinct counter categories; at least one is in the starter roster and the second can be obtained through ordinary play without limited banner timing.	Add a low-rarity functionally equivalent counter; re-test all affected matchups.
Hard-control safety	No standard PvP target is denied more than 1 normal action in any 3 of its own turns by repeat control. Shared DR and immunity are logged correctly.	Fix the shared control meter or duration; do not rely on a rare cleanse to repair the design.
Chain safety	No event can create an infinite trigger loop. Follow-ups are capped at 1 per acting unit and 2 per team per round; follow-ups do not generate Chakra or further follow-ups.	Break the recursion path in engine rules, not just in a single character's kit.
Battle duration	Target 5–8 rounds for ordinary equal-power battles; no PvP battle exceeds 10 rounds; no PvE test exceeds 12 rounds without a defined encounter rule.	Audit shield/heal loops, low-damage supports, target access and action count before increasing damage globally.
Spend/duplicate gap	In normalized PvP, base-copy versus max-duplicate versions differ by no more than 5% effective combat output; investigate spend-segment win-rate gaps above 5 percentage points after MMR/roster controls.	Increase normalization or move value from combat stats to cosmetic/convenience/breadth.
Synergy clarity	At least 80% of test players can explain the key combo trigger and primary reason for a win/loss using the battle recap.	Rewrite telegraph, icon, tooltip and recap; remove hidden chains before adding new systems.
Mobile readability	At least 80% of testers can correctly identify frontline/backline, current target, active control, Chakra readiness and the next major enemy threat on a phone-sized screen.	Reduce visual effects/labels, strengthen icons and test portrait versus landscape; follow platform touch-target guidance for tappable UI.
F2P composition access	A fresh account can build a coherent 5-unit baseline team and answer the first control, sustain, protection and backline-pressure encounters without requiring a paid or limited hero.	Revise starter roster and deterministic unlocks; do not tune difficulty around paid access.

Test method: run at least 1,000 seeded outcomes per directed archetype matchup at equal normalized stats, swap starting-side priority, and report confidence intervals rather than only point estimates. For eight archetypes, the full 8×8 directed matrix requires 64,000 simulations at this sample size. Repeat finalists with the ten-archetype set and then with realistic, but separately logged, investment variance.

These gates are design targets, not industry standards or evidence that the current proposal is already balanced. Combat is not considered cleared until the simulator and phone playtests pass these gates or a documented exception is approved in the decision log.

Test / telemetry	Initial gate	Action if it fails

Test / telemetry	Initial gate	Action if it fails
Equal-power archetype matchups	For non-specialist matchups, most win rates should fall near 45-55%; deliberate counters may deviate but must not become hard locks. Run at least 200 seeded simulations per pairing first, then scale sample size for confidence.	Inspect damage coefficients, initiative, synergy reliability and counter access; do not solve by adding arbitrary power score.
Archetype diversity	At least 4-6 viable archetypes across intended modes; no single archetype exceeds 35% sustained pick share across relevant ranked bands without a mode-specific reason.	Check over-efficient generalists, support tax and encounter design.
Counter answerability	Every strong strategy has at least two functional counter categories; at least one is available through starter/standard/deterministic access.	Do not release the strategy as-is.
Starter account completeness	An account with no lucky pull can clear tutorial and early campaign with role coverage; can form at least two coherent variants by early game.	Change starter characters/selectors before making content easier by brute force.
F2P target build time	One alternate archetype can be built in 4-8 weeks from deterministic sources; first team is functional in the tutorial/first few days.	Increase universal fragments/selectors, simplify required components or improve starter coverage.
Duplicate sensitivity	Target ≤5% normalized PvP effectiveness gap between base copy and max duplicate; investigate above 8%.	Reduce duplicate stats or move reward to cosmetics/sidegrades/resources.
Spend gap	Investigate normalized-PvP win-rate gap above 5 percentage points after controlling for rating, match count and unit selection.	Separate roster-choice/skill effects; normalize more or remove paid power.
Control lock	No unit loses meaningful actions indefinitely. Simulate extreme speed and repeated-CC teams.	Fix control counter, immunity windows and extra-action caps before release.
Battle understanding	At least 80% of test players can explain the main reason for a battle result after one recap.	Improve target-intent labels, effect icons and recap; avoid adding more status types.
Low-end mobile performance	No crash or severe stutter under repeated 5v5 auto-battles; effects can be reduced with settings. Define frame-time budgets per target device.	Reduce effects, batch updates, simplify animation and profile on physical devices.
Paid value without paid necessity	Spenders report time saved or cosmetic/roster choice; F2P testers can clear all core modes and counters.	Redesign bundle/value proposition if spending is required for ordinary progress.

22.3 Balance procedure

8. Run deterministic simulations first, then randomized seeded matches for each archetype pairing at equal power. Record both win rates and how/why a match ended.
9. Run removal and substitution tests. Remove the synergy enabler or replace a unit with an accessible unit that performs the same role. If win rate collapses by more than 8 percentage points, the team may be too dependent on one character.
10. Run investment sensitivity: same team at base copy, moderate build and max duplicate. Separate “time saved” in PvE from normalized competitive power in PvP.
11. Run enemy counter tests: try two practical answer categories against each archetype. If both are premium or both depend on the same rare unit, the archetype has not met fairness rules.
12. After a patch, publish the intended adjustment and measure pick rate, win rate by rank band, average battle duration, first-death timing, control uptime, Chakra source contribution, healing efficiency, damage share, disconnect/retry rates and churn near progression walls.
13. Do not buff a rare hero solely because it is underused. It may have an unclear role, low accessibility, or a substitute that is stronger. Decide whether the cause is power, usability, acquisition or mode relevance before changing numbers.

23. Design continuity card: rules to carry into every future research round

Current research checkpoint (Version 0.10.32). Section 59 formalizes recovery and accessibility: pending vs authoritative posted/rejected states, account/source identity, guarded recruitment and Hall requests, separate Google Play verification, screen-reader reading order and live announcements, first-loss explanation honesty, 72-hour idle return and 320dp/200% text paper cases. No numeric rewards, combat stats, Hall costs or four-root routes changed. NEXT: Priority 5D localization, low-end device budget and mobile UX exit-gate research. Recommended research level HIGH unless system architecture needs reopening. Concept phase only; no implementation or empirical validation.

[PROJECT DECISION] We are developing the game concept on paper before production. When a design question is open, the research lead chooses a working proposal using the project's priorities. Each significant choice records the alternative, the reason for the choice, what it costs the player or the design, which connected systems depend on it, and what would justify changing it. Carry the proposal into the next paper case. Keep owner decisions, assumptions, calculated findings and questions for future playtesting distinguishable.

[PROJECT DECISION] Before each research pass, tell the owner which performance level to use. At every handoff, name the next research question, recommend one level from light, medium, high, extrahigh or max, and give one sentence explaining the choice. Record it here so the owner sees it before the work begins. Update the recommendation in chat and in this checkpoint if a change in scope calls for a different level.

This card is the stable project memory inside the master document. On every research integration, read this card first, update the relevant chapters in place, add sources, append a dated change record, and keep unresolved questions visible. Do not rely only on conversational recall.

Persistent requirement	Interpretation for future work
Platform	Mobile-first Naruto idle RPG for phones; prioritize readable small-screen UI, stable auto-combat, low-friction sessions and low/mid-range device performance.
Game identity	Turn-based / auto-resolving squad combat. A coherent team has a win condition, role coverage, synergies, weaknesses and answers; not merely five high-power units.
Fairness principle	Balance fairness for both F2P and pay-to-progress. Spending may save time, add convenience, cosmetics or widen the roster earlier, but must not buy unanswerable mandatory combat power.
F2P viability	All essential roles and counter categories have accessible paths; no base kit requires duplicates to function; free players can build several archetypes over time.
Research breadth	Benchmark popular idle games and adjacent hero collectors beyond AFK Arena and Mythic Heroes. Correctly distinguish idle auto-battlers, turn-based collectors, and hybrids.
Evidence integrity	Separate verified current mechanics, historical report claims, community strategy, project decisions, proposals and hypotheses. Never invent findings for sessions that did not produce a report.
Single source of truth	Update /Game Design/Naruto_Idle_RPG_Master_Design_Bible.docx. Do not create a competing master. Keep original supplied reports in appendices, and update the archive and version log.
Legal/IP	Naruto character, name, art, story or other protected use requires appropriate licensing. Until licensed, Naruto examples are hypothetical illustrations only.
Validation	No proposed number is "balanced" until tested by simulator and playtest. Keep acceptance metrics, confidence and unanswered questions next to decisions.

23.1 Version 0.5 decision register

ID	Decision / proposed value	Rationale	Confidence / status	Still to test

ID	Decision / proposed value	Rationale	Confidence / status	Still to test
D-010	Treat small-screen/mobile performance and clarity as combat requirements.	Auto-battle should be readable and useful in short check-in sessions.	High as project requirement; design rules are proposed.	Portrait vs landscape; device matrix; session observation.
D-011	Keep five deployed units, 2 front + 3 rear for MVP.	Allows enough role coverage without six-unit complexity.	Medium; proposed.	Compare four vs five units on readability, time-to-resolution and archetype variety.
D-012	Use one normal action per living unit per round; Speed determines order; extra action chains capped.	Prevent speed stacking and infinite recursion from dominating.	Medium; proposed.	Simulation under extreme speed, control and kill-reset cases.
D-013	Personal Chakra: Basic +25; Technique +15; incoming direct-damage event +5 once per enemy root action per defender (up to +15 per defender per round); allied battery effects +15 per recipient per round; takedown +10 once per root action that KOs at least one enemy; Ultimate costs 100.	Keeps Ultimate cadence understandable without making a shared-gauge battery mandatory; caps all non-action energy sources and prevents multi-hit/follow-up farming.	Written proposal; no simulation yet.	Measure Ultimate turns by role; incoming-hit frequency; ally-generated Chakra; single/multi-hit interaction; time-to-Ultimate.
D-014	One authoritative damage formula in Section 24.6; shared additive capped buckets; Stun chance-tier diminishing returns in Section 25.8; no action advance in MVP.	Predictability and bounded combinations are easier to audit and explain.	High confidence in consistency rules; numeric balance untested.	Damage distributions and action denial in a seeded simulator.
D-015	Village stat bonus capped at +5% ATK/HP at 3+; bonds focus on modest utility.	Avoids five-of-a-kind faction lock.	Medium; proposed.	Compare mixed faction and mono faction win rates.
D-016	60 free pull-equivalents per 28 days; guaranteed featured copy at 70 pulls; no 50/50; pity and spark carry over.	Gives an understandable upper bound on acquisition while retaining collection goals.	Low-medium; economy hypothesis.	Real completion-rate income; release cadence; cost/revenue model.
D-017	100 universal fragments per standard selector; 15-20 earned weekly.	Target one alternate archetype in roughly 5-7 weeks.	Low-medium; proposed.	Measure completion source distribution, selector demand and roster breadth.
D-018	Normalized ranked PvP; target ≤5% base-vs-max-duplicate effectiveness gap, investigate above 8%.	Let skill and roster choices matter without direct purchase dominance.	High as project principle; numerical threshold proposed.	Test normalization impact, match quality and perceived progression value.

ID	Decision / proposed value	Rationale	Confidence / status	Still to test
D-019	Treat this document's continuity card as project memory and update it every integration.	Preserves constraints across conversations and research iterations.	High; process requirement.	Audit every update against the card and section register.

23.2 Source register for the mobile/combat revision

[M1] AFK Arena Guide, "Factions, hero types and roles in AFK Arena" (third-party current guide, dated 21 October 2026). It reports faction advantage, formation bonuses and role labels; use as a snapshot, not an official source. https://www.afk.global/afk-arena/guides/factions-and-hero-types

[M2] AFK Global, "AFK Journey Team Builder" (third-party interactive tool, accessed 28 October 2026). It shows map-specific deployment, teams up to five heroes and faction bonus context; it is a dynamic guide/tool. https://www.afk.global/afk-journey/team-builder

[M3] Mythic Heroes official site, "Day 3 Guide" (official beginner guide, accessed 28 October 2026). It describes four major factions, a +25% attack advantage against tactically vulnerable opponents, Arena Badges and post-battle statistics. https://www.mythicheroes.com/day-3-guide/

[M4] Nikke.gg, "Team Building Guide" (community guide, updated 26 August 2026). NIKKE is an adjacent real-time shooter/hero collector, not a turn-based idle RPG; its burst phases show how a required action sequence can create team architecture, but exact implementation should not be copied. https://nikke.gg/team-building-guide/

[M5] Apple Developer, "App Review Guidelines," section 3.1.1 (official platform policy; accessed 28 October 2026). Randomized virtual-item purchases must disclose odds before purchase. https://developer.apple.com/app-store/review/guidelines/

[M6] Google Play Console Help, "Developer Program Policy" (official platform policy; accessed 28 October 2026). Loot-box odds must be clearly disclosed in advance and close to purchase. https://support.google.com/googleplay/android-developer/answer/18258653?hl=en

[M7] UK Advertising Standards Authority / CAP, "Enforcement Notice: Disclosure of loot boxes in app stores" (published 26 February 2026). A jurisdiction-specific note on prominence of loot-box disclosure in app listings. https://www.asa.org.uk/resource/enforcement-notice-disclosure-of-loot-boxes-in-app-stores.html

[M8] HoYoverse, "Version 4.6 'Dance With the Beast Before Moonrise' Update Details" (official Honkai: Star Rail news page, accessed 28 October 2026). Useful adjacent evidence for rotating endgame modes and explicit mode-specific battle modifiers; it is not an idle RPG and is not a direct model for our combat. https://hsr.hoyoverse.com/en-us/news/166468

Confidence notes: M1, M2 and M4 are community sources and can be patch-specific; M3 is publisher-authored but describes its own tutorial rather than exposing all live formulas. M5-M7 are platform/advertising rules with jurisdictional scope. The numeric rules in Sections 20-22 are original proposals and are not sourced from these games.

23.3 Design decisions now fixed; empirical validation still required

- Decision fixed for prototype: portrait-first and five-unit squads; compare portrait vs landscape and four vs five in playtests before production lock. This is validation, not a reason to expand to other systems now.
- Decision fixed for prototype: round-based Speed order, one normal action per unit per round, deterministic target logic and no extra turns. Test whether players can understand the order and fight outcome; do not layer an action-bar mechanic on top before evidence.
- Decision fixed for prototype: personal Chakra; Basic +25, Technique +15, taking direct damage +5 per hit up to +15 per round, one takedown bonus +10 per acting unit turn; Ultimate costs 100 and grants no Chakra. Simulate ultimate frequency and fight duration, then tune values.
- Deferred to the later economy research phase: banner income, 70-pull guarantee, featured acquisition cost, and live-operations funding are not combat-engine decisions. Do not let these assumptions alter baseline team power or counter availability.

- Validation required: phone aspect ratio, low-end device performance, text scale, color-independent status icons and tap-target layout. Use real Android/iOS devices; the mobile-first UX constraints and sources are recorded in Section 24.
- Separate production/legal track: Naruto names, characters, visual designs and story use require appropriate rights. Until licensed, combat tests should use original placeholder shinobi concepts; this does not block development of the mechanics.

24. Core team-combat closure specification (Version 0.6)

Purpose and status. This chapter closes the design choices required to make a coherent five-shinobi team battle work. It deliberately does not design the wider game modes or battle-power add-ons yet. Rules below are the authoritative prototype specification unless later simulations or playtests justify a recorded change. Benchmark observations are cited separately from original recommendations.

24.1 Scope boundary: what we are finishing now

- [IN SCOPE] Squad size and formation; unit roles; action/turn order; Chakra; target selection; damage/healing/shield calculations; buffs/debuffs; hard control and counterplay; combo/summon rules; auto-battle AI; team archetypes; role/counter accessibility; mobile readability; balance testing and the criteria that let this research phase end.
- [DEFERRED ON PURPOSE] Full campaign, tower, boss/raid, guild, event, ranked season and other mode designs; equipment rarity/set bonuses; pets/companions; artifacts/relics; signature weapons; additional awakening passives; power-score/rating ladders; idle-reward pacing; complete banner economics and live-operations pricing. These must not mask a broken base combat system.
- [NON-NEGOTIABLE] Free players must obtain viable baseline examples of all core tactical functions. Paid/limited units, duplicates and future power-up addons cannot be the only way to enable a role, execute a synergy or answer a common threat.
- [IP NOTE] Naruto character names in the examples below are mechanical placeholders only. Commercial use of protected characters, names, art, music, story and designs requires the appropriate license. Otherwise, use original ninja-world characters and names.

24.2 Battle structure and state machine

Battle format: five allies versus the encounter's enemy formation. For the core 1v1 test harness, use five enemy slots as well so composition matrices are comparable. Combat is fully automated in MVP; players make decisions in team building, formation, team tactic preset and investment, not by tapping tiny mid-battle buttons. Optional manual skill casting is not part of the baseline balance target.

1. Setup: validate five distinct deployed units, formation and the selected squad tactic. Show enemy threat tags and telegraphed charge effects before the first action.
2. Start-of-battle effects resolve in fixed slot order: front-left, front-right, rear-left, rear-middle, rear-right; side priority alternates by a recorded match seed if both teams have same-priority effects. Each effect is logged.
3. Start each round: list living units in descending Speed order. Speed is sampled once at round start; Speed changes take effect in the next round and never grant an extra action by themselves. Exact ties are resolved by the seeded side-priority token, alternating sides by round, then by slot order.
4. At each unit turn: check defeat; apply Wound tick; if defeated, stop this turn. Check Stun; if stunned, the unit loses its normal action. Otherwise select Ultimate, Technique or Basic according to the AI rules in Section 24.7.
5. Resolve the selected action in this order: target selector → Taunt/guard interception checks → hit and status chance → damage modifiers → shield absorption → HP change/defeat → non-recursive follow-up → trigger effects → end-of-turn duration updates. Each stage is written into the combat log.
6. At round end, resolve end-of-round effects and check victory. MVP PvE fights end at 12 rounds; PvP simulation fights end at 10 rounds. If a PvP fight reaches the cap, the winner is the team with the higher summed percentage of HP remaining; tie-break with total damage dealt; an exact tie is a draw and grants no rating change. Do not create an infinite sustain loop.

24.3 Unit stats and role budget

Only the following primary stats are exposed in MVP: HP, ATK, DEF, Speed, Crit Chance and Crit Damage. There is no hidden Accuracy, Evasion or Tenacity stat. Every status application has a visible base chance; bosses can have explicit, previewed resistance tags. Skill/role tags (control, mark, Wound, protection, healing, summon) describe interactions but do not create an invisible elemental multiplier.

Role profile	Expected baseline trade-off at equal level	Balance guardrail
Guardian / Protector	Highest durability; low-to-moderate personal damage; can shield, taunt or intercept.	Protection must have a window/cap; does not make the back row invulnerable.
Bruiser / Off-tank	Above-average durability and useful sustained damage.	Should not out-damage a pure Finisher and out-survive a Guardian simultaneously.
Finisher / Duelist	High single-target output and target access; fragile when isolated.	Needs setup or exposure window; burst has to be answerable by Shield/Guard/Control.
AoE / Wave attacker	Lower damage per target than a single-target specialist but efficient across multiple targets.	Five-target total cannot trivially erase a team; test per-target coefficient against number of targets.
Sustained / Wound DPS	Lower initial burst; accumulates damage over multiple target turns.	Wound cap and cleanse access prevent scaling without limit.
Controller / Tempo	Lower personal damage when supplying reliable control or initiative disruption.	One hard-control family meter; no permanent action denial.
Debuffer / Marker	Moderate output with Mark, DEF Down, ATK Down or Healing Suppression.	Debuffs share caps; should still deal some value when cleansed early.
Buffer / Chakra support	Lower direct damage; improves timing/reliability for one or more allies.	Must not be mandatory for a baseline Ultimate rotation.
Healer / Cleanser	Low direct damage; prevents a defeat or removes a meaningful condition.	Healing cannot outpace all damage indefinitely; Anti-heal cannot be paywalled.
Scout / Rear access	Enables focus on a chosen threat or bypasses frontline at a stated cost.	Every team must have at least one available way to answer rear pressure or protect against it.
Counterattacker / Trapper	Punishes eligible direct hits or predictable actions.	DoT, AoE and non-triggering actions must be valid counter categories.
Summoner / Clone specialist	Creates capped decoys or one-time follow-up pressure, not extra full turns.	Summons never enter faction counts or recursively trigger summons/follow-ups.
Hybrid / Flex	Covers two functions with weaker per-function output than a specialist.	A hybrid must pay an opportunity cost; it should not be best-in-slot for both roles.

Role is not rarity. A lower-rarity or starter unit may be easier to use and less specialized, but it must perform its advertised job at base copy. High rarity may express a distinctive tactical variant, animation, or broader flexibility; it cannot own the only cleanser, only interrupt, only rear-access attacker or only practical anti-sustain tool.

24.4 Turn order, action economy and personal Chakra

Rule	Prototype value	Reason / constraint
Normal actions	At most 1 normal action per living unit per round.	No extra turns, turn theft or repeated action loops in MVP.
Basic Attack	Always available; generates +25 personal Chakra after resolving.	Guarantees every unit can reach its Ultimate without a premium battery.
Technique	One per unit kit; no Chakra cost; cooldown = 2 full subsequent own turns after use; generates +15 Chakra after resolving.	Useful role identity without a shared energy tax. On the next two own turns it is unavailable; on the following own turn it may be used.
Ultimate	Requires 100 personal Chakra and consumes the normal action; Chakra resets to 0; the Ultimate itself generates no Chakra.	Strong payoff but no free extra turn.
Taking damage	Gain +5 Chakra once per enemy root action per defender if at least one direct hit in that action produces positive post-DEF/post-non-DEF-DR damage before Shield; maximum +15 per unit per round. Multi-hit hits and follow-up subevents in the same root action count once for that defender.	Zero pre-Shield damage grants none. Shield may absorb all remaining damage without canceling the Chakra event. Wound/DoT, object, follow-up and reaction hits do not create separate events.

Rule	Prototype value	Reason / constraint
Takedown	Gain +10 Chakra once per root action if its direct hit or bounded follow-up resolves a KO; multiple KOs count once. The reward is credited to the root action's actor, not separately to the follow-up owner.	Wound/DoT KOs do not award takedown Chakra. Follow-ups cannot independently trigger another reward.
Follow-up / clone hit	Does not grant independent Chakra, Crit, lifesteal or additional follow-ups. A follow-up KO may satisfy the root actor's single takedown award, but not create a second award.	Prevents recursive action economy and makes logs deterministic.
Speed modifier	Speed changes are capped at ±20%; applied when the next round order is calculated.	Tempo matters, but Speed buffs cannot silently multiply actions per battle.

AI priority: if alive, not Stunned and not Silenced, use Ultimate when Chakra is 100 and its target conditions are legal; otherwise use Technique if cooldown is zero and its stated target/condition is satisfied; otherwise Basic. Each unit has a documented fallback target and never wastes a Technique on an impossible target. Skill designers may define one visible condition (for example, "prefer a marked enemy" or "use when at least two enemies are alive") but cannot hide a multi-step decision tree in a short tooltip.

Squad tactic preset, selected before battle: (1) Frontline First — apply base targeting rules; (2) Focus Mark — prioritize legal marked targets; (3) Break Threat — prioritize an enemy visibly charging a high-impact action if the chosen ability can interrupt or reach it. These presets change targeting only, not stats, odds or action counts; all are free and available to all players. In PvP the chosen preset is shown on the pre-battle preview.

24.5 Target selection, formation and protection

1. Skill-authored target rules take precedence when valid. Every skill specifies Single, Front Row, Rear Row, All Enemies, Lowest HP%, Highest ATK, Marked, Ally Lowest HP%, or a visible role/threat tag. A skill cannot silently retarget outside its declared legal range.
2. For a normal single-target Basic: eligible Taunt target first; then the selected squad tactic's legal priority; then front-row enemy with lowest current HP percentage; ties use left-to-right slot order. If the front row is empty, choose the lowest-HP-percentage rear target. All tie rules are shown in the help panel and recorded in logs.
3. Formation is two front slots and three rear slots. Any hero may be placed in any slot; no role is hard-locked to a row. The front row has targeting priority but no invisible default stat bonus. This keeps formation meaningful without making the rear row immune until both front units die.
4. Taunt redirects only eligible single-target actions. Global/row AoE, explicitly marked unavoidable attacks and Ignore Taunt skills do not redirect. Mark never overrides Taunt unless the skill explicitly says otherwise.
5. Interception redirects at most one eligible hit to the protector. The redirected hit cannot be intercepted twice. Damage is then calculated against the final defender and follows the normal damage pipeline.
6. AoE does not disappear just because a Guardian is alive. Rear-access skills, mark/focus attacks and line/row attacks are explicit tools with visible opportunity costs; at least two accessible team functions must be able to pressure or counter a protected rear line.

24.6 Final damage, healing and mitigation calculations

Use one deterministic pipeline for all ordinary direct attacks. Damage calculations use full precision internally and round down once at the end. Every modifier identifies its named bucket; a new skill cannot secretly bypass DEF, Shield, vulnerability and damage-reduction caps simultaneously.

Effective DEF reduction = min(40%, sum of active DEF Down and DEF Penetration percentages). Effective DEF = max(0, target DEF × (1 − Effective DEF reduction)). DEF multiplier = max(0.35, 1000 / (1000 + Effective DEF)).

ATK after Attack Down = base in-battle ATK × (1 − the strongest active Attack Down reduction, capped at 40%; Attack Down effects do not stack magnitude). Direct Damage = floor(ATK after Attack Down × Skill Coefficient × (1 + min(40%, Attack

Bonus Sum)) × (1 + min(25%, Skill/Type Bonus Sum)) × DEF Multiplier × (1 + min(15%, Vulnerability Sum)) × Crit Factor). Crit Factor is 1 for a non-critical hit and the attacker's Crit Damage multiplier for a critical hit. Player Crit Chance is capped at 75% in all modes in the MVP; Crit Damage is capped at 2.0×. No ordinary attack misses in MVP; status application is a separate roll.

Example: ATK 1,200; coefficient 1.6; Attack Bonus +20%; Skill Bonus +10%; DEF 1,000; Effective DEF Reduction 0%; Vulnerability +10%; no critical. Damage = floor(1,200 × 1.6 × 1.20 × 1.10 × [1000/2000] × 1.10) = 1,393. With 20% DEF reduction, Effective DEF becomes 800 and DEF multiplier becomes 1000/1800 ≈ 0.5556; otherwise identical damage becomes approximately 1,549. Values demonstrate formula behavior only and are not tuned production numbers.

Mitigation after DEF: separate non-DEF Damage Reduction effects combine additively and cap at 30%; apply this layer after DEF and before Shield. Shield absorbs the remaining damage; HP receives any remainder. True Damage is excluded from the MVP. A future exception requires a named damage tag, explicit counterplay, and a separate balance review; it cannot silently bypass Shield or caps.

Healing = floor(Healing ATK × Heal Coefficient × (1 + min(25%, Healing Bonus Sum)) × (1 − Healing Suppression)). Healing cannot exceed missing HP. The Healing Suppression multiplier is at most 40% reduction. Healing cannot crit in MVP. Shields use their explicit coefficient or stated max-HP percentage; all active shields sum to no more than 40% of recipient maximum HP. Each shield retains its own expiry; the earliest-expiring shield is consumed first.

Wound is the MVP DoT and uses a snapshot of the applier's ATK after Attack Down but before temporary Attack Bonus effects at application. It cannot crit and does not trigger on-hit, Chakra, takedown Chakra, reflect, lifesteal or follow-up events. It uses the ordinary DEF and non-DEF Damage Reduction layers and is absorbed by Shield. Total Wound damage from all sources is capped at 20% of the target's Max HP per target action slot before mitigation; Wound kills do not award takedown Chakra.

24.7 Status timing, control and interactions

Use the ten conditions specified in Section 20.7. No second status system or independent control-resistance meter is allowed in MVP. Hard-control diminishing returns are global per target and shared by all future action-denial effects. Silence remains a soft control: it constrains skill choice but the target still takes a Basic action and gains Chakra.

Control/application chance: each skill displays an explicit base chance; default ordinary debuffs may begin at 70%, Stun at 60%, and a skill can state a different value. Clamp player-applied status chances between 10% and 90% after all declared modifiers. Stun then uses the shared tier multiplier in Section 25.8. There is no hidden Accuracy or Evasion stat; direct attacks land on a legal target, while the status roll may fail. Special resistance must be a named trait, previewed before battle, and logged when it blocks an effect.

Interrupt is separate from Stun. Only a skill with the Interrupt tag can cancel an enemy ability shown as Charging. Interrupt prevents that specific charged effect and applies the stated interrupt cooldown; it does not also remove the enemy's next ordinary turn unless the skill explicitly pays for that effect. Every important telegraphed attack must have multiple answers: interrupt, protection/mitigation, or removal of the caster before resolution.

24.8 Synergy taxonomy and strict stack limits

Synergy family	Permitted effect	Fairness / recursion rule
Role / functional	Protector buys a window; marker exposes a target; finisher converts the opening; cleanse answers debuffs.	Prefer a clear setup → conversion → response window; each role has accessible substitutes.
Skill-condition / status engine	One skill applies Mark/Wound/DEF Down; another gains a limited bonus against that condition.	Same effect does not multiply; use shared category caps and at most 2 distinct Wound sources.
Follow-up / action chain	A trigger produces a small secondary hit or one small utility effect.	Max 1 follow-up per acting unit per round and 2 follow-ups per team per round; no crit, Chakra, on-kill proc or recursive trigger from follow-up.
Speed / tempo	Speed orders actions earlier or applies Slow to move priority next round.	No extra actions, turn-stealing or cooldown reset in MVP; Speed change cap ±20%.
Protection / sustain	Shield/Heal/Interception lets the team survive to execute a plan.	Shield cap 40% max HP; non-DEF damage reduction cap 30%; healing cannot loop into action generation.
Mark / focus	Mark raises vulnerability and/or changes target priority if the policy/skill says so.	Vulnerability cap +15%; cleanse/intercept is an accessible answer.

Synergy family	Permitted effect	Fairness / recursion rule
Formation / position	Front targeting, rear access, guards and row-specific skills change risk.	No hidden front-row stat buffs; AoE rules are declared; every rear strategy has a counter.
Village / clan / faction	At 2 matching tags +2% ATK/HP; at 3 +4%; at 4 or 5 +5% total ATK/HP.	Use only the highest reached tier, no additive tier stacking. All tag types share a total teamwide set-bonus ceiling of +5% ATK/HP in MVP.
Relationship / signature duo	Optional bond triggers one small follow-up or resource nudge once per battle.	Expected total team output swing ≤3% in normalized testing; cannot unlock core role or be the only counter.
Nature / elemental tags	Tags may be referenced by a specific skill or encounter rule.	No universal rock-paper-scissors damage multiplier in MVP; revisit only after accessible counter coverage is proven.
Clone / decoy	A temporary decoy redirects one eligible single-target attack or supports one explicitly written effect.	Clones do not take full turns, cast Ultimate, count for faction/relationship bonuses, or recursively trigger more clones.

Stacking rule across all synergies: same-named positive or negative effects do not stack magnitude; the strongest instance remains and may refresh duration only if the incoming effect is at least as strong. Different sources in the same category add up to the bucket cap. Distinct damage buckets multiply only where listed in Section 24.6. Trigger caps are engine-level and cannot be bypassed by a new character, summon, artifact or passive.

Relationship bonuses may be flavorful, but every pair has to pass a substitution test. A player who lacks one half of a featured pair must still assemble the archetype with a functionally equivalent unit; losing the bond may change the best route, not make the team type impossible.

24.9 Clone, summon, revive and reaction rules

- Clone/decoy is an object or tag, not a sixth full hero. It cannot act, use a skill, be healed, gain Chakra, count toward formation/faction bonuses or start another trigger. A clone intercepts at most one eligible single-target attack, expires after that intercept or at the end of the following round, and does not protect against AoE or Ignore Decoy effects. A side may have at most 2 active clones.
- Summons that deal a single attack are treated as follow-ups for action economy: no crit, no Chakra, no follow-up recursion, no additional turn. Summon effects state their target, duration, targetability, maximum count and expiry. If a summon must take real turns to express its identity, it is outside MVP and must be separately prototyped after baseline combat passes.
- Revive is excluded from the first MVP roster because revive chains make defeat, damage and action economy hard to understand. If a later kit introduces revive, the default contract is: once per unit per battle; revive at 20% maximum HP; 0 Chakra; no self-revive trigger; revived units do not rerun start-of-battle effects; summons cannot revive; revival cannot recursively trigger further revives. Any exception must be declared and tested separately.
- Reactive counters can trigger only from a listed event (direct eligible attack, marked-target hit, ally defeat or enemy charge). DoT ticks, shield absorption, follow-up attacks and reaction attacks do not trigger a second reaction unless a skill says so; engine caps still apply.

24.10 Composition design: how to build a team that is not random

For every five-slot composition, document all five functions—not necessarily five separate roles: (1) Win condition: what ends the fight in your favor? (2) Setup: what enables it? (3) Conversion: which unit/action turns setup into HP or action advantage? (4) Survival: how does the team live through the opponent's response? (5) Answer/fallback: what can it do if its main combo fails or its enabler is defeated? A unit may cover two functions, but the design must name the trade-off.

A team does not need one Tank, one Healer and three DPS. A burst team may replace the healer with tempo; a boss-focused team may trade AoE for sustained damage and interrupt; a counterattack team may use mitigation instead of repeated healing. The mandatory requirement is not a rigid class recipe—it is a credible win condition, a survival path, a way to convert setup, and an accessible answer to at least one likely counter.

Review dimension	Score 0	Score 1	Score 2
Win condition	No stated way to win except higher stats.	Win path exists but needs ideal conditions.	Clear and repeatable plan with a defined window.
Setup → conversion	No useful interaction between kits.	One combo, but timing or target access is unreliable.	Readable chain with a response window and fallback.
Survival	Team dies before operating or heals without damage.	Can survive some matchups with a clear sacrifice.	Survives likely opening threat without endless sustain.

Review dimension	Score 0	Score 1	Score 2
Counterplay / coverage	No answer to common control/burst/rear threats.	One answer or only a rare unit.	Two answer categories; at least one accessible.
Reliability / redundancy	One unit's death removes all function.	One substitute but meaningful drop.	Role function survives one plausible disruption/substitution.
Target access	Can only hit an irrelevant frontline target.	Can access priority target under one condition.	Has legal plan against both frontline and rear threat.

Use the rubric to identify missing functions, not to produce a single all-purpose “best team” score. Any numerical composition score is advisory. The simulator and player tests decide whether a lineup is strong, understandable and answerable.

24.11 Sample cohesive five-shinobi squads

The following are illustrative Naruto-universe role assignments, not claims about actual canon abilities or existing game kits. Each can be remapped to original characters if licensing is unavailable. Every role can be replaced by another accessible unit with the same tactical function.

Squad concept	Five slots and tactical loop	Main weakness, answers and replacement principle
1. Balanced starter	Naruto-style striker (reliable damage); Sakura-style healer/cleanser; Shikamaru-style controller; Hinata-style protector; Rock Lee-style finisher. Protect the opening, control one threat, then convert a mark/control window into focused damage.	Weak if the controller is cleansed and the finisher is focused. Answer: accessible AoE or anti-control. Replacement principle: any starter striker + cleaner + control + protection + finisher; no Team 7 lock.
2. Mark-and-execute	Shino-style marker; Sasuke-style finisher; Kakashi-style rear-access debuffer; Gaara-style protector; Sakura-style cleanser. Mark the priority target, survive the return hit, and use rear access or finisher to remove it.	Weak to cleanse/intercept and backup carries. Answers: remove the marker or protect the marked target; control the finisher. Substitute any marker and any single-target conversion unit.
3. Wound attrition	Shino-style Wound applier; Kiba-style sustained chaser; Tsunade-style healer; Choji-style bruiser; Ino-style utility/cleanser. Apply at most two distinct Wound sources, remain alive, and force enemy turns to pay the tick cost.	Weak to early burst and cleansing. Answers: remove the Wound appliers; use a cleanser or faster pickoff. A second DoT source is optional, not mandatory.
4. Anti-control / response	Sakura-style cleaner; Kakashi-style interrupt/debuffer; Gaara-style protector; Rock Lee-style finisher; Naruto-style flexible damage. Mitigate the first control, interrupt a telegraphed action and counter-burst during the opening.	Weak to repeated pressure that forces cleanse cooldowns or AoE that ignores protection. Answers: pressure the cleaner; sustain through the first counter-burst. Cleanse/interrupt are starter-accessible functions.
5. Boss-style single target	Sasuke-style sustained finisher; Kakashi-style DEF reducer; Shikamaru-style Chakra/tempo support; Sakura-style healer/cleanser; Gaara-style protector. Maintain safe uptime and interrupt or mitigate the one visible high-threat attack.	Weak to DEF-reduction caps or fast target removal. Answers: use a raw sustained-DPS alternative; protect/interrupt the charge. Boss output cannot rely on a single limited debuffer.
6. Multi-target pressure	Naruto-style bounded clone/AoE; Temari-style AoE; Shikamaru-style controller; Gaara-style protector; Sakura-style sustain. Spread damage, slow a threat and remove weak units without needing a top-rarity AoE carry.	Weak to front-loaded protection and single-target punishes. Answers: protect a key unit, interrupt the AoE unit, focus-fire its low-defense row. Every account should have a basic multi-target tool.
7. Counterattack shell	Neji-style counterattacker; Hinata-style protector; Tsunade-style healer; Rock Lee-style bruiser; Shino-style debuffer. Invite eligible single-target attacks, punish them, and sustain through predictable bursts.	Weak to DoT, AoE and enemy attacks that do not trigger counters. Answers: use AoE/Wound; target the healer or break formation via declared rear access. Counterattack must not be the only tank answer.
8. Clone / misdirection	Naruto-style clone creator; Kabuto-style summon/utility; Shino-style scout; Hinata-style protector; Sasuke-style finisher. Use a capped decoy to absorb one single-target action while scout/finisher converts target priority.	Weak to AoE, reveal and target switching. Answers: AoE removes the value of decoy; mark/real-target policy bypasses confusion. Clone strength is capped by the global summon rule.
9. Tempo pickoff	Minato-style fast opener; Shisui-style controller; Sasuke-style finisher; Sakura-style cleaner; Hinata-style protector. Go early, control the key response and execute one target; if the first window fails, use the protector and fall back to normal actions.	Weak to Slow, shields and durable teams. Answers: speed reduction/guard; survive the first Ultimate window and win the long fight. Speed cannot grant extra actions.
10. No-healer disruption	Shikamaru-style controller; Kakashi-style interrupt; Gaara-style protector; Naruto-style AoE; Sasuke-style finisher. Trade repeated healing for denial and fast removal of the enemy engine.	Weak to cleanse-resistant sustained output and long battles. Answers: diversify threats; protect the enemy controller/carry from single-target pickoff. This is a tactical option, not a universal replacement for healing.

24.12 Counter matrix and minimum roster accessibility

Threat strategy	Counter category A	Counter category B	Access requirement
Burst pickoff	Shield/Interception	Control or force target swap	At least one starter protector; another low-rarity response.
Protect-the-carry	Rear access/Mark	Anti-heal or control the enabler	Rear access available by starter or deterministic unlock.
Control tempo	Cleanse	Hard-control DR/interrupt before control chain	Cleaner and interrupt/resilience are not premium-exclusive.
Wound attrition	Cleanse	Remove applier / win before ticks	No need to own an exact matching healer.
Sustain/grind	Healing Suppression	Burst focus on healer/engine	Anti-heal is available in ordinary pool; a clean burst path exists.
Mark-and-focus	Cleanse Mark	Guard/Interception	Mark can be answered by multiple kits, not one hero.
Speed/tempo	Slow/initiative response	Shield and survive opening	Slow/defense accessible; no Speed-only premium lock.
Counterattack	AoE/Wound	Rear access / non-triggering damage	Counters do not require repeated direct attacks.
Clone/decoy	AoE	Reveal/Mark/target policy	AoE and honest target rules available in starter/normal pool.
Heavy protection	DEF reduction up to shared cap	Bypass/rear access or attrition	Defense bypass is a tactical tool with caps, not a paid raw-stat check.

Starter roster floor: the tutorial account obtains five functioning units immediately, plus enough deterministic early recruitment to reach at least 8–10 choices. By early account progression, the free roster should supply one additional option for each critical function: protector, cleaner/anti-control, rear access, control/interrupt, anti-sustain and single-target finisher. The design target is 12–15 core character concepts in the initial roster, with at least one broadly obtainable substitute for each role. This is a composition-access requirement, not an economy-design decision.

24.13 Mobile readability and auto-combat UX

- Keep the active battle HUD to the essentials: team HP, current target, visible control/status icons, Chakra readiness, round/turn order and any telegraphed enemy action. Put secondary stats in a pause/recap panel rather than covering the battlefield.
- Use icon + text label + shape/animation; never communicate an essential status by color alone. Show the target arrow/outline and a concise tooltip for every special target rule. Statuses need consistent placement and time remaining in target turns.
- For tap targets such as speed, pause, recap and formation controls, follow platform guidance: Android recommends at least 48×48 dp touch targets; Apple recommends 44×44 pt for iOS game controls. These are interface guidelines, not combat rules [C6][C7].
- Provide 1× and 2× speed from the start; defer 4× and skip only after logs remain understandable. Battle can be watched or left to auto-resolve; the player should not have to tap every Ultimate to win.
- Post-battle recap must show: damage, healing, shield absorbed, damage prevented, Wound total, control actions and resisted/shortened control, Chakra created by source, first knockout, combo triggers, number of triggers capped, and the two most material reasons for the result.
- At least 80% of user testers should identify the main reason for a win/loss and the next tactical adjustment using the recap alone. If not, simplify the battle feedback before adding more statuses or special effects.

24.14 How we prevent one mandatory meta composition

1. Keep at least eight of the ten archetypes viable under the acceptance criteria in Section 22.2, across both beginner and experienced test cohorts.
2. Every new unit must state its role, win condition, opportunity cost, intended counter(s), two accessible substitutes and the synergy category it affects. “More stats plus unique utility” is not an acceptable design pitch by itself.
3. Apply removal and substitution tests. If removing a particular unit destroys the strategy by more than 8 percentage points and there is no accessible replacement, the strategy is too unit-dependent.

4. Apply a pairwise synergy budget. Estimate a combo's expected contribution across a battle at equal stats. Bonds/faction tags must not exceed the defined caps; no pair may create unlimited turns, unbounded Chakra, stacking vulnerability or repeated reset loops.
5. Use Pareto comparison instead of one linear power ranking: compare damage, survival, tempo, control, target access and counter-resilience. A unit may be best in one dimension, but no unit should strictly dominate another accessible unit in all dimensions.
6. Separate matchup-specific strength from general dominance. A protect-the-carry team may beat burst but lose to Wound/anti-heal; a speed team may win first initiative but lose to guards; a control team may win once but lose when cleansed. The counter must be practical and accessible, not merely theoretical.
7. No global elemental advantage triangle in MVP. Village/faction tags offer at most +5% ATK/HP, with one threshold tier active. This avoids asking players to pull an exact faction just to avoid an automatic disadvantage.
8. No unbounded power from duplicates, gear or future add-ons may be used to validate a broken composition. First compare base kits at normalized stats; future progression layers are added one at a time after base combat passes.

24.15 Future implementation and empirical validation gate (not a current research-phase blocker)

This is a future gate for moving from the written combat specification into implementation and empirical balance claims; it is not a gate that blocks desk research or conceptual design of other idle-RPG systems. The combat rules may be specified on paper without being proven balanced. Current research may continue across genre identity, idle rewards, core loop, modes at the concept level, progression, economy, collection, monetization and UX, while dependencies on untested combat rules remain labelled provisional.

- [SPECIFIED, NOT EMPIRICALLY VALIDATED] Five-unit formation, initiative, action economy, Chakra, action priority, targeting, Taunt/interception, damage/healing/shield, statuses, CC DR, Wound, clone/decoy and follow-up/revive rules are specified. Section 25 adds a research audit and resolves contradictions; no value is considered balanced until tested.
- Roster complete: at least 12–15 character concepts; every primary role and counter category has a base-copy accessible unit and at least one substitute; no essential synergy requires duplicates or a limited banner.
- Simulator complete: all actions and calculations are deterministic under a recorded seed; all modifiers and trigger caps appear in logs; same inputs produce same output; no infinite recursion or hidden tie-break exists.
- Matrix complete: run ≥1,000 seeded outcomes per directed archetype matchup at normalized power, with side-priority swaps; report confidence intervals, battle length, first death, action share, CC uptime, healing/shield efficiency and win rate.
- Fairness gate passed: each strategy has two different counter categories; one available from starter roster; normalized duplicate power gap ≤5% effective output; investigate spend-segment win gaps over 5 percentage points after controlling for rank and roster.
- Readability gate passed: ≥80% of phone testers explain target choice, key synergy trigger and primary loss reason; active status/Chakra/telegraph indicators remain legible under 2× battle speed.
- Diversity gate passed: at least eight archetypes satisfy viability targets; no unexplained archetype exceeds 60% equalized overall win rate, and no single unit substitution collapses multiple team types.
- Only after these future-phase gates pass should the project treat the combat engine as empirically validated or finalize additional power layers whose balance depends on measured combat outcomes. They do not prohibit conceptual research now. Equipment, awakenings, pets, artifacts and similar combat-power add-ons may be described at the concept/theory level, but should not be finalized as balance-safe systems or introduced into a tested build before the combat gates pass. Game modes may be researched conceptually now; their exact tuning and release readiness remain future validation questions.

24.16 Source findings vs project decisions

[SOURCE FINDING] AFK Arena's current community guide documents substantial faction-count bonuses, with the displayed maximum 5-of-a-kind formation bonus at +25% ATK and HP in its current guide snapshot [C1]. That makes

faction composition meaningful but risks turning faction count into a raw power tax. Our +5% cap is deliberately smaller and original.

[SOURCE FINDING] AFK Journey's current team-builder exposes placement on map-specific tiles, up to five deployed heroes, faction bonuses beginning at three heroes, and mode-specific map structures [C2]. This supports formation as a real team-building variable, while we defer a free-grid MVP because its UI and AI cost needs evidence.

[SOURCE FINDING] Mythic Heroes' official beginner guide says its faction tactical advantage adds +25% attack against vulnerable factions and provides post-battle performance statistics [C3]. The useful lesson is previewable counters and battle recaps; we do not adopt a 25% matchup swing because it is too large for this fairness target unless tests prove otherwise.

[SOURCE FINDING] NIKKE team-building guides describe a Burst I → II → III chain and cooldown interactions that can make certain team functions structurally required [C4]. This is a good illustration of intentional team architecture, but it also demonstrates why each required function in our game must have several accessible providers and should not be tied to one limited unit.

[SOURCE FINDING] HoYoWiki's Honkai: Star Rail combat explanations describe action-order visibility, follow-up attacks, weakness-break action delay, dispels and interrupting charged actions [C5]. HSR is not an idle RPG; the transferable pattern is to signal enemy intent and make actions/interruption legible, not to copy its resource system.

[SOURCE FINDING] Android recommends minimum 48×48 dp touch targets [C6], and Apple's game-control guidance recommends 44×44 pt for iOS controls [C7]. These support the mobile touch-target standard used in this chapter.

[PROJECT DECISION] Use these findings as benchmark observations, not templates. No third-party source establishes that our formula, role budgets or thresholds are correct. Our numerical recommendations are original design hypotheses and must pass our prototype gates.

24.17 Version 0.6 decision register

ID	Decision locked for prototype	Rationale / alternative not chosen	Status
C-001	Five allies; 2 front + 3 rear.	Room for protection, finisher, enabler, answer and flex; free grid/six-unit roster deferred due mobile readability and balance cost.	Chosen; compare 4 vs 5 in tests.
C-002	Round-based Speed order; one normal action per unit per round.	Easier to audit than a continuously moving action bar; Speed cannot manufacture extra actions.	Chosen; validate pacing.
C-003	Personal Chakra; Basic +25, Technique +15, being hit +5 capped +15/round, takedown +10 once/action, Ultimate 100.	Avoids shared-gauge mandatory-battery problem; values require simulation.	Chosen; tune from logs.
C-004	Automated action priority: Ultimate, eligible Technique, otherwise Basic.	Keeps idle promise and prevents manual micro or payer reflex advantage; manual casting deferred.	Chosen.
C-005	Use Section 24.6 as the sole damage pipeline; older Section 7 / 20 formula examples are historical. Capped additive buckets are defined there.	Prevents mixing stat scales and flat penetration with percentage DEF reduction.	Written rule selected; numerical balance pending.
C-006	Ten named conditions only; Stun application-chance DR in Section 25.8; Silence is soft control; no fractional action duration.	Avoids overlapping control systems and impossible fractional turns.	Written rule selected; probabilities pending simulator.
C-007	Follow-up/clone attacks cannot generate Chakra, crit, recurse or grant extra turns.	Breaks infinite triggers and limits action-economy advantage.	Chosen.
C-008	Village/faction bonus 2/3/4+ = +2/+4/+5% ATK/HP; only highest tier active.	Recognizes theme without forcing a mono-faction lineup or copy gate.	Chosen; playtest.
C-009	No global elemental multiplier in the first combat prototype.	Prevents blind hard-counter lock and roster/acquisition tax before base kits are balanced.	Chosen; revisit after MVP.
C-010	Core counters and role identities work at base copy; paid/limited units are variants, not keys.	Consistent with the project's fairness constitution.	Settled product constraint.

ID	Decision locked for prototype	Rationale / alternative not chosen	Status
C-011	No gear/awakening/signature/pet/artifact additions until the combat exit gate passes.	Prevents a progression layer from masking a weak base engine or paid power gap.	Settled sequencing constraint.

24.18 Research and implementation record

The next iteration must update this same live master, starting with the exit-gate results. Add the simulator version, test seeds, roster version, patch of every benchmark game inspected, results tables and any changed number. When changing a rule, preserve the previous rule in the decision log and state the evidence for the change. Do not silently rewrite old findings or claim that untested values are “balanced.”

New sources for this revision (accessed 28 October 2026):

- [C1] AFK Arena Guide, “Factions, hero types and roles in AFK Arena,” current community guide snapshot, updated October 2026. https://www.afk.global/afk-arena/guides/factions-and-hero-types
- [C2] AFK Global, “AFK Journey Team Builder,” dynamic community tool, accessed 28 October 2026. https://www.afk.global/afk-journey/team-builder
- [C3] Mythic Heroes / IGG, “Day 3 Guide,” official beginner guide, accessed 28 October 2026. https://www.mythicheroes.com/day-3-guide/
- [C4] Nikke.gg, “Team Building Guide,” community guide; use for the structural Burst-chain example only and verify current kit details before implementation. https://nikke.gg/team-building-guide/
- [C5] HoYoWiki, Honkai: Star Rail “Base Mechanism” and “Special Mechanism” entries; adjacent turn-based RPG source, accessed 28 October 2026. https://wiki.hoyolab.com/pc/hsr/entry/1618?lang=en-us and https://wiki.hoyolab.com/pc/hsr/entry/1620?lang=en-us
- [C6] Android Developers, “Make apps more accessible (Views),” last updated 21 April 2026. https://developer.android.com/guide/topics/ui/accessibility/views/apps-views
- [C7] Apple Developer, “Game controls” human-interface guidance, accessed 28 October 2026. https://developer.apple.com/design/human-interface-guidelines/game-controls

Evidence confidence: C3, C6 are official primary sources for the stated facts. C1, C2 and C4 are third-party/community references and should be rechecked for patch changes. C5 is an official game knowledge resource for a different genre. All numbers, formulas, team examples and fairness gates in Section 24 are our proposals, not source-derived game facts.

25. Deep research review of the core combat system (Version 0.7 foundation; expanded in v0.8-v0.9)

Status and purpose. This chapter deepens the team-combat research before any move to game modes, gear sets, artifacts, pets, awakening add-ons or power-rating systems. It combines verified or source-attributed benchmark observations with original design reasoning, resolves contradictions in the active draft, and identifies what remains research-open. It does not claim that untested numbers are balanced. The project remains mobile-first and F2P-fair: purchases must not be the only way to assemble a functional team or counter a dominant strategy.

Direct answer to the review question: the prior chapter was a much more complete specification, but it had not yet been deeply reconciled against enough comparative combat systems. This version adds that review. The theory can be specified more completely now; balance, comprehension and fairness still require future simulation and device playtests.

25.1 Evidence labels used in this review

- [OBSERVED FACT] A source explicitly documents a feature of a named game. This establishes what that source says, not why the game succeeded commercially.
- [TRANSFERABLE INFERENCE] A design pattern appears useful for our system, but the source does not prove it is universally best.
- [PROJECT DECISION] A constraint requested by the game owner, such as viable F2P teams and spending that does not buy an unanswerable ranked advantage.
- [PROPOSAL] A concrete starting rule for our prototype. It must be versioned if later simulation or research supports a change.
- [OPEN] Research or implementation is not yet complete. Do not represent it as solved.

25.2 Cross-game combat research: mechanisms worth learning from

These games are not all the same subgenre. AFK Arena, AFK Journey, Mythic Heroes and Idle Heroes are idle/auto-battle hero collectors; NIKKE is a hybrid real-time squad shooter; Honkai: Star Rail and Reverse: 1999 are turn-based hero collectors; Brown Dust 2 is more explicitly tactical and player-directed; RAID: Shadow Legends and Summoners War use speed/turn-meter systems. The comparison is about combat architecture and team construction, not a claim that every title is an idle RPG.

Game / source type	Documented combat pattern	Useful lesson for our game	Do not copy blindly
AFK Arena — community reference [R1]	Role labels coexist with faction matchups and team-wide formation bonuses; the current guide describes faction advantage and a 3/4/5-hero bonus ladder, with a five-of-a-kind bonus reaching +25% ATK and HP in that snapshot.	Role coverage and team tags are separate design dimensions. A team can be encouraged by tags without relying on tags alone.	Large flat faction boosts can create a collection tax: players lose substantial stats for using the tactically correct off-faction unit. Our proposed village bonus stays much smaller.
AFK Journey — dynamic team-builder and guide [R2, R3]	The team-builder supports up to five heroes and map-specific deployment tiles; its current page reports faction bonuses and different team contexts. The guide notes faction bonuses may not be worth sacrificing important utility.	Make deployment and faction effects visible before battle; treat placement as a choice with an opportunity cost. Evaluate compositions by context.	Do not add free-grid placement, special pets or mode-specific formation layers to the first combat model before basic target rules are clear.
Mythic Heroes — official beginner guide [R4]	Its official Day 3 guide documents four major factions, a +25% ATK tactical advantage against vulnerable factions, Arena, and post-battle statistics showing damage and protection.	Readable matchup hints and a post-battle recap help players learn why a team succeeds or fails.	A 25% matchup swing is too strong to adopt without testing in a fairness-first game. Use counters as meaningful edges, not automatic wins.
NIKKE — community strategy guide [R5]	Burst I → Burst II → Burst III is a required sequence for Full Burst; cooldown lengths and auto-selection order influence team composition. The guide explains that this leaves a flexible slot after the chain is completed.	A composition can be an engine or schedule, not merely five isolated stat blocks. Define enabler, conversion and flex roles clearly.	A hard sequence can make a type of unit mandatory. Every required function in our game must have multiple accessible providers and a fallback when one is disabled.

Game / source type	Documented combat pattern	Useful lesson for our game	Do not copy blindly
Honkai: Star Rail — official HoYoWiki [R6, R7]	The combat tutorial documents speed/action order, shared Skill Points, energy/Ultimate use, weakness break, action delay, follow-up attacks, buff/debuff dispels and interruption of charged attacks.	Telegraphed enemy actions and explicit interruption/cleanse rules create tactical counterplay. Action economy and resources must have visible rules.	It is not an idle auto-battler. Its shared resource/action-advance complexity should not be layered into our MVP while personal Chakra and deterministic auto-actions are still being defined.
Brown Dust 2 — game guideline [R8]	The guideline describes attack/defend phases and allows the active side to move companions, reserve skills and change attack order; positioning can be changed defensively.	Positioning and action order can create real choices if each has a visible effect.	The manual tactical layer does not translate directly to a fully automated idle battle. Borrow readable positioning and target rules, not repeated taps.
Summoners War / RAID — Speed/turn-meter community guides; Summoners War official Challenge Battle notice [R9, R10, R21]	Community references describe Speed/attack-meter systems in these games. Separately, the official Summoners War v8.8.5 notice describes Challenge Battle using preset Monsters, a pick/ban phase and normalized Summoner Skills.	Speed tuning creates meaningful tempo but can snowball; shared/preset-unit draft modes can isolate composition decisions from roster investment in a later fair PvP lane.	Do not infer that all Ranked modes are normalized. Community meter descriptions can be patch-sensitive; our MVP deliberately excludes dynamic meter/action advance.
Reverse: 1999 — adjacent turn-based collector [R11]	The guide describes incantation-card actions, action points tied to the active team, visible enemy action information, and management of skills/energy. It also notes that the game does not use a PvP competitive mode in the referenced guide.	Shared action resources create opportunity cost; previewing enemy intent makes tactical trade-offs legible.	A shared action-point or card-hand system is a different combat identity and would require a separate prototype. It is not needed to make our idle squad compositions meaningful.
Idle Heroes — official publisher site + community Tower wiki [R12, R18, R19]	The publisher page showcases hero-specific skills and a large roster. A community Tower of Oblivion page documents a six-hero enemy lineup, two-front/four-rear layout and faction aura in that mode.	Six units can widen role coverage and allow more tactical layers, but increase visible links and target candidates. Our five-unit choice is a mobile-readability hypothesis, not proof that five is universally optimal.	The six-unit/faction-aura details are mode/wiki observations, not an official universal rule. Do not infer that Idle Heroes is more or less F2P-friendly from team size.
Epic Seven — community combat/effect references [R20]	Community reference pages document Speed-based initial order, Combat Readiness increases/decreases, buff/debuff immunity and the fact that some turn manipulation can be resisted or blocked by encounter traits.	The value of a “turn controller” depends on how turn order, resistance and cleanse interact; if our Speed only orders next-round actions, that is a deliberate simplification and should not be described as Epic Seven-like action-economy depth.	Community pages are not authoritative publisher documentation and exact character/skill lists are patch-sensitive. Do not copy hidden resistance or broad immunity assumptions.

Game / source type	Documented combat pattern	Useful lesson for our game	Do not copy blindly
Cookie Run: Kingdom — official update + storefront role guide [R15, R16]	The 26 August 2026 update notice gives teams a preview of each wave boss's Weakness, revises Cookie Alliance from five teams to three, and describes composition-specific buffs, including a Fire-skill count threshold at 1/3/5 allies. Google Play's editorial guide separates frontline, damage and support/healing roles.	Encounter requirements can create useful draft/build choices when players can preview the condition and the effect is tied to an understandable role or tag. The number of required teams should be justified by mode purpose, not by roster inflation.	A team-specific weakness buff can quietly turn an element specialist into a mandatory roster tax. Provide at least two accessible answers and keep such mode rules out of the base PvP formula.
Mobile Legends: Adventure — official store listing [R17]	The publisher's Google Play listing describes an idle auto-battler with 100+ heroes, seven hero types, composition strategy for bosses/PvP, and level transfer/level sharing for building new lineups.	Separate role identity from investment friction. Level sharing/transfer is a useful design pattern to reduce the cost of trying replacement units and secondary archetypes.	The store listing does not prove all teams are equally competitive or that this model guarantees F2P fairness. Avoid copying exact hero count or progression model blindly.
Summoners War — official Challenge Battle update [R21]	The official v8.8.5 update notice (30 September 2025) describes a Challenge Battle using a provided preset monster pool, pick/ban, and max-level Summoner Skills regardless of account progression.	A future opt-in ranked/skill lane can normalize investment to foreground composition and decisions, while a separate progression lane can preserve collection growth.	This proves the described alternate mode exists, not its player outcomes or fairness over time. Do not copy its pool size or draft rules without a separate design review.

Research interpretation: the strongest transferable pattern across these examples is that team construction depends on at least three connected layers: (1) what a unit does, (2) when/where its action happens, and (3) what team-level condition enables or counters it. The weakness pattern is equally consistent: when one of those layers becomes an exact roster requirement, a rare unit or heavily invested stat can turn a strategy into a gate instead of a choice.

25.3 What makes a team coherent: an implementable composition model

Treat a team as a dependency graph. Units are nodes; links describe one of four things: enabling (A creates a condition B uses), conversion (B turns the condition into damage or action advantage), protection (C keeps A/B alive through the response), or counter/fallback (D prevents the plan from collapsing if A/B fails). A line-up that has five individually strong nodes but no useful links can underperform a lower-power team whose links form a reliable plan.

Required function	Question the team must answer	Examples of valid solutions	Failure to watch for
Win condition	What eventually wins this fight?	Pick off a carry, outlast with Wound, clear the back line with AoE, protect a ramping damage dealer.	"We have high power" is not a win condition.
Setup	What must happen before the payoff?	Mark a target, apply Defense Down, build Chakra, force a predictable attack, survive a charge.	If setup requires a specific limited unit, the archetype is access-gated.
Conversion	Which action turns setup into a meaningful lead?	Finisher, AoE follow-up, Wound tick, interrupt plus focus fire, counterattack.	A combo that creates effects but never changes damage, survival or tempo is cosmetic.
Survival	What lets the plan survive the opponent's response?	Protector, shield, healing, control, decoy, faster opener, interrupt or cleanse.	Five damage dealers may get one strong opening but lack a second plan.

Required function	Question the team must answer	Examples of valid solutions	Failure to watch for
Fallback / answer	What if the enabler dies, misses, is silenced or gets countered?	Second target path, backup control, basic attack value, a second source of damage, alternative protect/cleanse.	If one unit's removal collapses the entire composition, it is too mandatory or too fragile.

A role is a job description, not a mandatory slot. A bruiser can provide both damage and survival; a healer can also cleanse; a controller can be a secondary damage source. For the initial roster, give each hero one primary role and no more than two secondary role tags. The skill rules, not the role label, decide actual performance. If a unit is best-in-slot at damage, control and protection all at once, it is probably too broad unless its costs, cooldowns or conditions make those jobs compete with each other.

Composition contract for every hero kit: record (a) primary and secondary roles, (b) intended archetypes, (c) trigger and target rules, (d) expected payoff, (e) opportunity cost, (f) at least two counter categories, (g) baseline-copy operation, and (h) one accessible replacement. The contract is required before a new character enters the roster, not after players discover the meta.

25.4 Five units versus four or six: rationale and limits

Team size	Pairwise relationships	Triple relationships	Main design consequence
4 units	4×3/2 = 6	4	Cleaner and faster to read; fewer slots mean each unit must cover more functions. This can make roles and replacements feel mandatory.
5 units — selected MVP	5×4/2 = 10	10	Room for a win condition, enabler, converter, survival and fallback/answer without requiring five separate role classes. Manageable with strict trigger caps.
6 units — deferred	6×5/2 = 15	20	Adds 50% more pairwise links and doubles the number of possible triples compared with five. It can make combo stacking, target selection, visual clarity and balance harder.

The combinatorics do not prove five is objectively best; they explain why six costs more to balance. Five is a prototype choice because the game needs meaningful synergy and enough room for counterplay. Four must remain an explicit comparison in a future usability test: if mobile readability or battle duration is poor, test four without quietly expanding to six. In the meantime, enforce the fairness side of a five-slot design: the starter roster must be able to cover all critical functions, and there must be substitutes for every unit with an unusually high dependency score.

25.5 Synergy: how to make the whole stronger without creating a mandatory pair

Use a controlled mix of kit synergy and small identity bonuses. Most useful synergy should arise from a visible condition-and-payoff interaction; clan/village/relationship tags should add flavor or modest flexibility rather than serve as the whole power budget.

Synergy family	Recommended MVP treatment	Required counter / guardrail
Role interaction	Protector keeps a carry active; cleanser removes a key debuff; finisher converts a mark.	Do not let one unit perform every role at full strength.
Status engine	Mark, Defense Down or Wound sets a condition that a second unit benefits from.	Same effect does not stack magnitude; one cleanse or prevention route must be accessible.
Action sequence / follow-up	A listed event enables one limited follow-up or extra effect.	Follow-up cannot recursively trigger follow-up, gain Chakra or create a full extra turn in MVP.
Tempo / turn order	Speed changes initiative order; control or interrupt changes whether a key action resolves.	No action bar, meter advance or extra-turn mechanics in MVP.
Protection / sustain	Taunt, intercept, Shield, healing and counterattack support a trade-off between survival and offense.	Protection must not make the rear row unreachable. Anti-heal does not secretly cancel shields.
Formation / target access	Frontline priority is predictable; rear access, Mark, Taunt and row attacks have explicit rules.	Position must change a choice, not hide a target-selection rule in an animation.
Village / faction / clan	One highest active threshold, with modest bonus and no stacked family of bonuses.	Total ATK/HP contribution from the tag system is capped at +5%; no compulsory mono-faction five-stack.
Canon relationship / bond	Prefer changing a trigger or providing a utility variant over a large permanent multiplier.	Baseline hero kit works without the partner; the pair must pass removal and substitute tests.

Synergy family	Recommended MVP treatment	Required counter / guardrail
Counter-synergy	AoE pressures clones; cleanse answers debuff engines; direct pressure challenges unprotected buffers.	Counter categories must be functional and obtainable; no single paid character is the only answer.

Synergy measurement rule: compare a team with the intended combo against the same team with a functionally equivalent, equally invested replacement. Record win probability, first-knockout timing, fight duration, damage prevented, damage converted, control uptime, Chakra source and health remaining. A synergy is not proven by its tooltip or its animation; it is proven by a repeatable change in tactical outcomes that remains answerable.

Proposed synergy budget: a combo should usually improve a relevant outcome by about 3–7 percentage points of win rate versus the equivalent-function substitute, not dominate all matchups. If removing one hero reduces win rate by more than 8 percentage points and no broadly obtainable substitute restores the plan, the composition is too unit-dependent. These are internal guardrails, not observed industry averages. For faction/village/relationship stats, keep the total contribution at or below +5% ATK/HP and activate only the highest eligible tier.

25.6 Action economy and Speed: a deliberate departure from speed-tuning games

The source comparison makes one trade-off clear. Games with continuously filling action/turn meters can make Speed affect action frequency, turn order and extra-turn chains. That creates deep speed tuning, but it also creates a risk that a small stat lead multiplies into more turns, more energy and more control before the opponent can respond. Our mobile idle design should not inherit that complexity before its base roles work.

Authoritative MVP decision: each living unit gets exactly one normal action per round. Sort actions by Speed at the beginning of the round. Recompute next round, not during the current queue. Speed changes reorder actions but never grant extra normal actions. No action advance, turn-meter manipulation, cooldown reset into extra action, or skill that lets a unit take a second full turn exists in MVP. Follow-up attacks are bounded damage events, not full turns.

This simplifies fairness but makes Speed mostly an initiative-order stat rather than an action-frequency stat. That is an intentional trade-off. Therefore: do not give Speed enormous scaling; ensure it has understandable value when order matters (interrupt before charge, control before an enemy opener, shield before burst). Slow only reorders the next round; it never skips a turn. Any proposal to return to a dynamic action bar must be a separate, evidence-backed redesign after the round-based model is understood—not an added subsystem.

25.7 Chakra and technique timing: remove hidden edge cases

The current proposal uses personal Chakra rather than one shared squad gauge. This prevents the “Chakra battery or the team does not function” failure mode: every hero can build toward an Ultimate independently, while batteries remain accelerators rather than mandatory keys.

Event	MVP rule	Exploit guard
Basic action	+25 Chakra to the acting unit after the action resolves.	Chakra is capped at 100; overflow is discarded, not stored.
Technique action	+15 Chakra to the acting unit after the action resolves.	The Technique must offer a meaningful utility/damage advantage over Basic to justify its lower Chakra yield.
Direct damage taken	+5 once per enemy root action per defender when any direct hit in that root action produces positive post-DEF/post-non-DEF-DR damage before Shield; cap +15 per defender per round. Multi-hits/follow-ups count once per target/root action.	Zero pre-Shield damage grants none. Shield may absorb all post-DR damage and the event still qualifies. DoT/Wound, objects and reaction/follow-up hits are not separate Chakra sources.
Takedown	+10 Chakra to the root action’s actor once if the root action or its bounded follow-up KOs at least one enemy; multi-KOs count once.	Wound/DoT KOs do not count; follow-up owner receives no extra award from the root actor’s takedown trigger.
Ultimate	Available at 100 Chakra; activating it spends 100 and grants no immediate Chakra.	No overflow or recursive Ultimate loop.

Event	MVP rule	Exploit guard
Chakra battery effect	Ally-origin Chakra gifts share a cap of +15 Chakra per recipient per round across all abilities. The effect must state amount, target, trigger and once-per-round limit.	Does not trigger from Chakra created by itself; no target exceeds 100; no battery refresh/repeat loop; no paid-only start-Chakra advantage in normalized PvP.

Technique cooldown convention: each Technique must declare a cooldown in owner-turns. Prototype default is 2 full owner turns after casting: if a Technique is used on turn 1 and has cooldown 2, it is unavailable on the next two turns of that owner and becomes eligible on the following turn. Reduce cooldown only at the end of subsequent owner turns, not on the turn the Technique is cast. A kit can use cooldown 1 or 3 if needed, but the rule must be explicit. The auto-AI may use a Technique when ready and its one stated condition is met; otherwise it falls back to Basic. No hidden multi-step priority tree.

25.8 Crowd-control diminishing returns: correct the fractional-turn problem

The earlier proposal reduced a repeated Stun from one target turn to 50% or 25% of a turn. In a discrete round-based game, that is not implementable: the target either loses its normal action or does not. This ambiguity is resolved here. Stun is the only action-denying control in MVP; Silence blocks Techniques and Ultimate but still permits Basic and normal Chakra generation.

Shared hard-control tier on target	Effective Stun chance	After a successful Stun
Tier 0 — fresh	Use the skill's displayed base chance (default proposal: 60%).	Increase target tier to 1.
Tier 1 — recently controlled	Half the displayed base chance (e.g. 60% becomes 30%).	Increase tier to 2, capped at tier 2.
Tier 2 — repeatedly controlled	Quarter the displayed base chance (e.g. 60% becomes 15%).	On successful Stun, target gains hard-control immunity until it completes its next non-Stunned turn; after that, reset to tier 1.
Decay	A failed application does not increase the tier.	After a target completes a full turn without being Stunned, reduce tier by one. Silence does not increase or reset the hard-control tier.

Additional Stun rules: Stun cannot extend an active Stun; two Stuns resolving before the same target's next action cannot cause two lost actions. An attempt against an already-Stunned target does not extend Stun or increase the diminishing-return tier, though its ordinary damage still resolves. Only a successful transition from not-Stunned to Stunned raises the shared tier. On a successful Tier-2 Stun, hard-control immunity starts immediately after that Stun; it lasts until the target completes its next non-Stunned action, then immunity ends and the tier becomes Tier 1. A full non-Stunned action slot without a successful Stun reduces a nonzero tier by one, except that the immunity-release rule above sets the tier to Tier 1. Interrupt is separate: it cancels only a specifically telegraphed Charging effect and does not remove the target's next ordinary action.

This chance-tier model is a clearer prototype candidate than fractional duration, but its 60%/30%/15% values are still proposals. The simulator must measure lost actions per 100 target turns, win rate by control team, and whether accessible cleanse/protection keeps control meaningful without making it the only viable strategy.

25.9 Formula reconciliation and damage budget

There were multiple active draft versions of the formula. Section 7.3 used a 100/(100+DEF) example with DEF=100; Section 20.6 used a flat DEF−PEN expression; Section 24.6 uses a 1,000 reference scale and percentage DEF reduction. This version resolves the conflict: Section 24.6 is the only authoritative implementation formula for MVP. Sections 7.3 and 20.6 are now explicitly marked historical drafts; their numeric examples are not to be combined with current values.

Authoritative calculation order: (1) choose a legal target; (2) resolve Taunt and one eligible interception; (3) for ordinary attacks, damage always lands unless the target is no longer legal; roll status chance separately; (4) calculate ATK after Attack Down, Attack Bonus, Skill/Type Bonus, effective DEF, Vulnerability and Crit using Section 24.6; (5) apply non-DEF

Damage Reduction up to 30%; (6) absorb with Shield up to its capacity; (7) apply remaining damage to HP; (8) apply on-hit effects after damage unless the ability declares PRE-DAMAGE; (9) process defeat and only listed non-recursive reactions; (10) award Chakra/takedown resources under the caps; (11) update action-slot durations and write the log. All steps are observable in a combat replay.

Authoritative caps: Attack Down uses the strongest active reduction, capped at 40%; Attack Bonus +40%; Skill/Type Bonus +25%; Vulnerability +15%; effective-DEF reduction +40%; non-DEF damage reduction +30%; player Crit Chance +75% in all MVP modes; Crit Damage up to 2.0×; total active Shield capacity no more than 40% of Max HP; Healing Bonus +25%; Healing Suppression up to 40%. Flat DEF penetration is not a separate additional bucket in MVP; DEF Down and penetration contribute to one 40% percentage-reduction cap.

One-shot guardrail: do not hard-clamp every damage number in the first implementation, which can make combat feel arbitrary. Instead, tune coefficients and normalized stats so an ordinary non-execute action cannot one-shot a full-health non-tank of equivalent investment in ranked PvP, including a critical hit. Any specialist finisher that approaches this limit must require visible setup, have a cooldown/condition and be answered by at least two accessible categories (for example, Shield/intercept or cleanse/disable). Log p95 damage per action and test against low-HP, mid-HP and tank targets. If coefficient tuning cannot preserve that guardrail, consider a transparent PvP-only per-action cap as a fallback design proposal.

Wound budget: current rule is 18% of source ATK at the start of the target's turn for two target turns, cannot crit, and does not generate on-hit/Chakra/reflect/follow-up triggers. Same-source Wound refreshes rather than stacks; max two distinct sources. Add a test guardrail that total Wound damage from all sources does not exceed 20% of a normal target's max HP in any target turn under normalized PvP settings. If it does, reduce coefficient or stack count rather than making cleanse mandatory.

25.10 Target selection and formation: one source of truth

The earlier mobile chapter and the closure specification used two different normal-Basic targeting statements ("front-most legal enemy" versus "frontline enemy with lowest HP percentage"). Keep the latter as the authoritative rule because it creates understandable focus-fire while maintaining frontline priority. The final precedence is:

1. A valid skill-authored target selector takes precedence (Single, Front Row, Rear Row, All Enemies, Lowest HP%, Highest ATK, Marked, Ally Lowest HP%, or visible threat/role tag). The skill cannot select outside its legal range.
2. For Basic attacks: if an eligible Taunt exists, target it; otherwise apply the chosen squad policy if it produces a legal target; otherwise target the frontline enemy with the lowest current HP percentage; if no legal frontline target remains, target the lowest-HP-percentage legal rear target.
3. Tie-breaks are deterministic and visible: left-to-right within row after the side-priority seed has been applied to exact initiative ties. An animation never changes target legality.
4. Taunt redirects eligible single-target actions only. AoE, explicit Ignore-Taunt skills and visibly unavoidable scripted actions are not redirected. Mark creates its stated vulnerability but does not silently override target logic. Rear-access and interception are named skill rules.

The two-front/three-rear formation is a prototype default, not a statement that frontline units must be tanks. Any unit can occupy any slot. Frontline position creates predictable target priority but no hidden defensive stat bonus, and the rear is never absolutely untouchable. At least two broadly accessible functions must be able to pressure or counter protected rear units.

25.11 Role and archetype coverage: what must exist at baseline rarity

Core function	Example role providers	At least one accessible answer	Design risk
Damage conversion	Single-target finisher, sustained carry, AoE, Wound/attrition dealer.	At least one reliable damage unit plus a different damage route in the normal acquisition pool.	Premium carries become the only way to finish fights.
Protection / sustain	Guardian, bruiser, shield, healer, decoy/intercept.	A starter protector and a non-premium sustain or mitigation alternative.	Healer/tank tax makes every team follow the same recipe.
Tempo / control	Speed support, Slow, Stun, Silence, interrupt, cooldown enabler.	A starter/early earnable control or interrupt tool and a way to answer control.	A premium opener repeatedly denies the opponent's first action.

Core function	Example role providers	At least one accessible answer	Design risk
Cleanse / anti-control	Cleaner, protective buffer, control-resistant unit.	One broadly obtainable cleanse/prevention answer; a second answer through a different category.	The only cleanse is on a limited banner, forcing players to summon defensively.
Target access / anti-protection	Mark, rear access, AoE, Taunt bypass, anti-shield if needed.	At least two practical ways to challenge rear protection across accessible units.	Tanks turn the rear row into an unreachable paywall or assassins ignore formation at no cost.
Anti-sustain	Healing Suppression, focus-fire on healer, interrupt/kill the enabler.	At least two counter categories; one should not require a specific premium unit.	Sustain has no answer, or anti-heal cancels shields/revives without warning.
Anti-summon / anti-decoy	AoE, reveal/mark, published targeting or object-clearing effect.	AoE and deterministic target rules available without payment.	Summons add extra bodies and extra turns indefinitely.

For the starter roster, target 12–15 original placeholder shinobi concepts, not 12–15 premium units. Every core function has one baseline accessible provider and at least one alternative. The first copy must express the character's advertised tactical identity; duplicates may improve efficiency or customization, but must not unlock the only version of the essential combo. Canon names and skills in archived drafts remain illustrative pending appropriate rights/licensing.

25.12 Composition quality and anti-mandatory-meta tests

Test	Method	Proposed pass signal	If it fails
Synergy reliability	Count how often the intended setup → conversion sequence occurs under normal matchups and counterplay.	Usually 60–85% of intended opportunities depending on how interruptible the plan is.	Clarify trigger, improve fallback or add a meaningful response window.
Ablation / replacement	Remove one synergy unit; replace with an equally invested accessible unit of similar function.	No unexplained drop above 8 percentage points in win rate.	Reduce exact-pair dependency, improve substitutes or lower bond power.
Archetype viability	Run directed archetype-vs-archetype matches with equalized stats and side-priority swaps.	At least 8 of 10 planned archetypes viable overall; no unexplained overall win rate above 60% or below 40%.	Inspect universal stat edge, matchup coverage, target rules and counters.
Counter availability	For each archetype, verify two distinct functional counter categories, at least one in starter roster.	Every strong strategy is answerable without a limited/paid-only hero.	Add accessible provider or redesign the strategy.
Control health	Measure Stun attempts, successful Stuns, lost actions per 100 target turns and outcome swing.	No indefinite lock; control does not dominate wins across all matchups.	Adjust chance tiers, duration, availability or cleanse.
One-shot safety	Test standard full-health non-tanks under p50/p95 damage, crit, full buffs and setup combos in normalized PvP.	No ordinary single action kills a full-health equivalent non-tank.	Reduce coefficient or use a transparent PvP-only cap.
Readable causality	Phone testers review the recap and explain target choice, combo trigger, and why the team lost.	At least 80% can identify the main cause and one reasonable adjustment.	Reduce statuses, improve recaps, expose target rules.
Robust sample size	Run 1,000 outcomes per directed matchup, then increase finalists to 2,500 outcomes.	Initial 1,000-match 95% margin at p≈0.5 is about ±3.1 percentage points; finalists target about ±2 pp.	Do not overreact to noisy 1–2 pp differences; collect more runs.

The confidence-interval figures above are statistical approximations for a binomial win-rate estimate, not game industry benchmarks. At p=0.5, standard error is sqrt(0.25/n); for n=1,000 the approximate 95% margin is 1.96×sqrt(0.25/1000) ≈ 3.1 percentage points. At n=2,500 it is about 2.0 points. Use exact binomial confidence intervals in the simulator report.

Do not collapse all measures into one team-power score. Compare the Pareto profile: damage, survival, tempo/order, control, targeting access, reliability, and counter-resilience. A team can be strong at one dimension and weak at another; a single score can conceal that trade-off.

25.13 Mobile auto-battle readability: requirements derived from the combat model

- Battle-start screen: show the five deployed shinobi, formation, selected tactic preset, enemy threat tags, obvious counter opportunities, and whether a special charge is telegraphed.
- During battle: show the current action/turn queue, current target highlight, Chakra meter and Ultimate-ready state, status icon plus text label and duration, active Taunt/intercept, and the visible telegraph. Do not communicate essential effects by color alone.
- Skill details: every skill names legal target range, trigger condition, cooldown in owner-turns, effect chance, duration, stacking behavior, whether it bypasses Taunt/DEF/Shield, and whether the event generates Chakra or follow-ups.

- Auto-AI: deterministic rules first. Ultimate when Chakra=100 and target legality/skill condition is satisfied; otherwise Technique when its cooldown and one stated condition permit; otherwise Basic. A unit must never waste an attack on an impossible target. Skill AI may not hide a complex if/then tree in a short tooltip.
- Recap: show damage, healing, Shield absorbed, damage prevented, Wound total, control attempts/successes/resists, Chakra by source, first knockout, combo trigger/cap count and the two most material reasons for the result. Use this recap as a teaching system, not a decorative scoreboard.

25.14 Fairness constraints that belong inside combat—not only in the economy

- Base-copy viability: a character performs its advertised primary job at the first copy and baseline rank. Extra copies cannot be required to activate the core combo.
- No paywalled answer: every common strategy (burst, control, sustain, rear pressure, summons/decoys, Wound/DoT) has two practical counter categories; at least one provider of each category is available to all players through the starter roster or deterministic progression.
- No rare-only role: no premium/limited unit owns the only cleanse, only charge interrupt, only rear access, only anti-summon answer, or only anti-sustain path.
- Normalized ranked comparisons: when ranked PvP is designed later, normalize levels, duplicate stat gains and baseline equipment enough to prevent spending from deciding outcomes directly. Roster breadth can remain meaningful, but must not produce an unanswerable matchup.
- Future power-ups are quarantined: gear, artifacts, signature equipment, pets and awakening passives must not be used to “fix” a broken base kit. First demonstrate that the base-kit matchup matrix has several viable archetypes; then add each power layer one at a time with its own power budget and F2P source.

25.15 Research closure status (updated through Version 0.9)

[HISTORICAL SEQUENCING NOTE — SUPERSEDED BY SECTION 28] Version 0.7 made the combat specification more deeply researched and internally reconciled. It did not establish that every benchmark title had been exhaustively inspected or that proposed numbers were optimal. The combat specification remains an important current design pillar, but this older statement is not a reason to pause research on other game systems.

Area	Status after v0.9	What must happen before moving to other major systems
Combat rules	Sections 24-27 now specify the MVP action engine, damage/mitigation formula, status timing, target selection, Chakra events, follow-up limits, CC DR, clone/taunt/shield behavior and edge-case packet. All numbers remain untested proposals.	Implement a deterministic seeded simulator and pass the regression/edge-case pack in Section 27. No modes/add-ons until that gate is met.
Cross-game benchmark	Desk-based source comparison across AFK Arena/Journey, Mythic Heroes, Idle Heroes, Epic Seven, Summoners War, RAID, NIKKE, HSR, Reverse: 1999, Brown Dust 2, Cookie Run: Kingdom and Mobile Legends: Adventure; source types are explicitly labeled.	Not a full hands-on patch survey. Check exact current UI/rules in clients when feasible; do not infer hidden formulas, win rates, F2P fairness or business results from descriptions.
Roster-to-role map	Section 26 defines 19 placeholder kits, 10 archetype shells, accessible substitutes and role coverage. Section 27 adds normalized stat packet and ability coefficient budgets.	Select final/original character identities and implement kits. Verify that each base-copy kit does its advertised job and replacement teams remain viable.

Area	Status after v0.9	What must happen before moving to other major systems
Synergy/counter matrix	Section 26 defines the directed matchup/counter matrix; Section 27 adds edge-case stress tests and decision thresholds.	No matchups have been simulated; run 1,000+ seeded outcomes per directed pairing and check confidence intervals/side bias.
Numerical balance	A normalized baseline stat packet and first skill coefficient budget now give the simulator reproducible starting inputs.	The exact values remain hypotheses. Test for one-shots, role dominance, long fights, control denial and damage/effect mix before freezing values.
Other game systems	Intentionally deferred by project decision.	No detailed modes, gear/artifact/pet/awakening/signature power-up or Battle Power rating research until the core combat simulator, matrix and phone readability exit gates pass.

“Nothing open” remains the intended documentation standard for each system: every core rule should be defined or visibly assigned to an open-decision/evidence register. It is not a requirement to finish every combat detail before beginning other desk research. The integrated concept should advance steadily while dependencies and unresolved combat questions remain explicit.

25.16 Combat research decision register (updated through Version 0.9)

ID	Decision / correction	Rationale	Confidence / status
C-010	Section 24.6 is the one authoritative damage formula; older Section 7 and 20 formulas are marked historical.	Prevents mixing stat scales and flat PEN with percentage defense reduction.	High confidence as an editorial consistency fix; damage values untested.
C-011	Hard-CC DR reduces application chance by tier rather than creating fractional turns.	Discrete one-action Stun cannot be halved to half a turn.	High confidence that the old rule was ambiguous; new percentages are a proposal.
C-012	Standard Basic targeting follows the precedence in Section 24.5; older front-most wording is superseded.	One deterministic policy is necessary for explainable auto-combat.	High confidence as rule consolidation; target policy still needs player comprehension tests.
C-013	MVP contains no dynamic turn meter, action advance or full extra turns.	Avoids speed/action economy runaway and makes combat easier to debug on mobile.	Medium; purposeful simplification to compare against later if needed.
C-014	Personal Chakra values and cooldown convention are explicitly stated.	Prevents hidden triggers, energy loops and ambiguous Technique timing.	Medium; frequency requires simulation.
C-015	Synergy is evaluated against an equal-function replacement; pair/bond effects remain small.	Avoids one specific unit becoming a mandatory key to an archetype.	High as fairness direction; quantitative threshold unvalidated.
C-016	The combat research phase stays active; no transition to modes/add-ons yet.	The benchmark, roster mapping and empirical balance gates remain incomplete.	Settled project workflow.

25.17 Sources used for the Version 0.7-0.9 combat research refinement

Sources are classified by what they directly support. Community guides can accurately describe visible mechanics but may be patch-specific; they are not proof of hidden formulas or commercial outcomes. Accessed 28 October 2026 unless a publication date is stated.

- [R1] AFK Global, “Factions, hero types and roles in AFK Arena,” community guide snapshot dated 21 October 2026. Used for faction matchup, formation bonus and role taxonomy examples. https://www.afk.global/afk-arena/guides/factions-and-hero-types
- [R2] AFK Global, “AFK Journey Team Builder,” dynamic community tool. Used for five-hero deployments, map-specific placement, faction bonus and mode-specific formation examples. https://www.afk.global/afk-journey/team-builder

- [R3] Prydwen Institute, “AFK Progression Teams,” third-party strategy guide. Used for the warning that faction bonus is not always worth sacrificing utility. https://www.prydwen.gg/afk-journey/guides/afk-progression-teams
- [R4] Mythic Heroes / IGG, “Day 3 Guide,” official beginner guide. It states a +25% attack faction advantage against tactically vulnerable units and describes post-battle damage/protection statistics. https://www.mythicheroes.com/day-3-guide/
- [R5] Nikke.gg, “Team Building Guide,” community guide; the visible page lists an original publication date of 12 November 2023. Used only for the Burst I→II→III sequence and cooldown-dependent team structure; confirm live skill details in-game. https://nikke.gg/team-building-guide/
- [R6] HoYoWiki, Honkai: Star Rail, “Base Mechanism.” Official tutorial source for shared Skill Points, energy/Ultimate, weakness break and action delay. https://wiki.hoyolab.com/pc/hsr/entry/1618?lang=en-us
- [R7] HoYoWiki, Honkai: Star Rail, “Special Mechanism.” Official tutorial source for action order, follow-up attacks, action advance/delay, buff/debuff dispel and charged-attack interruption. https://wiki.hoyolab.com/pc/hsr/entry/1620?lang=en-us
- [R8] Brown Dust 2, “Game Guideline — Battle” and “Placement.” Game guideline describes turns, skill order, and positioning changes. https://browndust2.gitbook.io/guide_en/game-guideline/battle/battle and https://browndust2.gitbook.io/guide_en/game-guideline/battle/placement
- [R9] SWGT.io, “Speed Tick Bracket Calculator FAQ,” community reference updated 27 October 2026. Used for the described Summoners War attack-bar/Speed model; not an official publisher source. https://swgt.io/faq/39
- [R10] RAID: Shadow Legends, “Game Overview,” publisher game guide. Used for the Turn Meter and Speed interaction and the existence of turn-advance/extra-turn abilities. https://raidshadowlegends.com/game-guide/
- [R11] Prydwen Institute, “Introduction to the game — Reverse: 1999,” community guide. Used for incantation-card actions, team action points and the distinction from idle PvP combat. https://www.prydwen.gg/re1999/guides/introduction-to-the-game/
- [R12] DHGames, “Idle Heroes,” official App Store listing. Used only to confirm the publisher markets idle training and a large hero roster with factions and specific skills; it does not establish current detailed team formulas. https://apps.apple.com/ph/app/idle-heroes-idle-games/id1153461915
- [R13] Epic Helper, “Buffs & Debuffs Reference,” community database for Epic Seven effects and counters. Useful as a taxonomy research lead, not an official source or proof of present balance. https://www.epic7helper.com/effects
- [R14] Prydwen Institute, “Introduction to the game — Reverse: 1999” and beginner guide, used for the four-character roster / three-active plus substitute pattern and limited shared Action Points; adjacent turn-based collector, not a direct idle model. https://www.prydwen.gg/re1999/guides/introduction-to-the-game/ and https://www.prydwen.gg/re1999/guides/beginner-guide/

Confidence note: R4, R6, R7, R10 and R12 are publisher/official-guide sources for the limited facts they state. R1–R3, R5, R8–R9 and R11/R13–R14 are third-party or community references; use them to identify transferable visible mechanics, then re-check live game state if an exact value matters. No source above validates our custom damage formula, Chakra rates, five-unit target policy, CC tiers or F2P acceptance thresholds.

Version 0.7 work record. Updated the live master with a cross-game combat research comparison; marked older active-body mechanics as superseded where they conflicted; resolved fractional-turn crowd-control ambiguity; formalized team dependency/synergy evaluation; specified Chakra/Technique timing and cooldown conventions; expanded role/counter accessibility and mobile readability requirements; and added a research closure register. The historical appendices were not rewritten. Next work must continue this combat research and roster mapping only.

26. Starter roster-to-archetype matrix and combat closure gate (Version 0.8 baseline; clarified in 0.9)

Purpose. This chapter fills the major design gap that remained after the combat rules were written: we had defined roles and 10+ archetypes, but had not yet mapped them into a coherent, replaceable starter roster. The roster below is made of functional placeholders, not final licensed Naruto kits. Canon names, likenesses, story elements and trademark use require appropriate rights. The important deliverable is the role-and-counter architecture; characters and animations can be mapped later.

Fairness requirement. The first copy of every essential function must work. Duplicates, rare variants and premium skins may offer customization, alternate conditions or modest efficiency, but must not unlock a character's only viable skill rotation, required counter, or the only route to a tactical archetype. No single placeholder is mandatory: every roster function has a substitute pathway.

26.1 A character kit contract: fields every future shinobi must specify

Field	Required definition	Why it matters for team-building / F2P fairness
Role identity	One primary role and no more than two secondary roles; name the explicit opportunity cost of each secondary function.	Prevents a hybrid from being best-in-slot as damage dealer, protector and controller at once.
Basic / Technique / Ultimate	State target legality, exact effects, damage bucket, duration timing, Chakra gain, cooldown and whether it is a normal action or a bounded follow-up.	Every hero can function at baseline copy; effects cannot hide triggers or create recursive actions.
Auto decision	One visible condition for Technique and a deterministic Basic fallback; Ultimate priority is explicit.	The idle promise stays complete and players do not buy a better combat AI.
Setup and conversion	Write "This kit sets up X; it converts X into Y"; if it does not need setup, name its reliable fallback.	A skill's synergy is testable rather than just a vague description.
Weakness and counter	Name at least two distinct counter categories that can challenge its strongest line.	Stops one dominant kit becoming an unanswerable gacha gate.
Replacement	Name at least one accessible unit that can perform the same function with a different trade-off.	The team archetype remains buildable without one named hero or duplicate.
Investment scaling	Separate what level/gear change from what rank/duplicate changes; cap duplicate-only combat advantage in normalized PvP.	Avoids investing into a hero's basic identity being a paid requirement.
Logging / readability	Specify trigger log text, visible status icons, target rule, counter window and result recap.	The player can learn why an auto-battle succeeded or failed on a phone screen.

26.2 Nineteen placeholder kits for the first coherent roster

Numbers in this roster are either references to existing authoritative prototype rules (for example Mark +10%, Wound 18% source ATK, personal Chakra) or deliberately qualitative kit definitions. Do not treat unspecified skill coefficients as final values. The role shells are intended to prove roster coverage, not a final hero tier list.

ID / placeholder kit	Primary / secondary role	Functional kit contract	Natural weaknesses / counter categories	Accessible substitute / fairness rule
K-01 Guardian	Primary: protector; secondary: control	Taunt eligible single-target actions for the stated window; modest damage; protects an exposed ally through a visible redirect rule.	AoE, rear access, Ignore-Taunt skills, or focusing the guardian before the window.	K-02 Interceptor or a common Shield provider. Must be obtainable early; no stars needed to Taunt.
K-02 Interceptor Bruiser	Primary: off-tank; secondary: counterattack	Redirect one explicitly eligible hit to self or reduce it; retaliates only if the hit qualifies.	AoE, Wound/DoT, attacks tagged to bypass interception, control.	K-01 Guardian or K-10 Healer/Protector; the substitute trades retaliation for more reliable protection.
K-03 Single-Target Finisher	Primary: burst conversion; secondary: marked-target specialist	High single-target action converts Mark/DEF Down or a low-HP window into a knockout threat; never a free unconditional full-health one-shot.	Shield/intercept, cleanse Mark, disrupt finisher, force a bad target.	K-05 Sustained Carry or a second accessible burst unit; replacement is slower, not unusable.
K-04 AoE Formation Breaker	Primary: AoE; secondary: decoy pressure	Hits a row or all legal enemies at a lower per-target coefficient than a dedicated finisher; clears decoys and pressures guards.	Shields, spread/defensive formation, high single-target threat to the caster.	K-16 Rear-Access unit plus a normal damage dealer; single-target path covers some but not all wave-clearing value.
K-05 Sustained / Ramping Carry	Primary: sustained damage; secondary: boss focus	Gains bounded ramp value when it acts or survives; payoff improves with protection and time, with a visible stack cap.	Burst before ramp, remove/cleanse stacks if specified, control, focus-fire.	K-03 Finisher or K-06 Wound unit; substitute changes the win condition and timing.

ID / placeholder kit	Primary / secondary role	Functional kit contract	Natural weaknesses / counter categories	Accessible substitute / fairness rule
K-06 Wound / Attrition Specialist	Primary: DoT; secondary: debuff pressure	Applies the current Wound rule: 18% of source ATK at target turn start for two target turns; maximum two distinct sources; same-source refresh.	Cleanse Wound, defeat applier, burst before tick value compounds, mitigation using normal DEF formula.	K-05 Sustained Carry or K-17 Debuffer; no team must own a specific DoT source.
K-07 Stun Controller	Primary: single-target control; secondary: setup	Uses the shared Stun probability tiers in Section 25.8 to deny one normal action once; control creates a defined conversion window.	Cleanse before action, hard-control tier, interrupt/remove controller, visible resistance.	K-08 Disruptor or a low-rarity Silence/Slow control unit; no premium-only Stun counter.
K-08 Ability Disruptor	Primary: soft control / interrupt; secondary: tempo	Either applies Silence for one target action or interrupts a visible Charging move; each skill declares which effect it performs.	Cleanse, use Basic while Silenced, bait the interrupt, use non-Charging pressure.	A cleanser plus protector can replace its function through two units if a starter has no interrupt yet.
K-09 Cleanser	Primary: cleanse; secondary: anti-control	Removes one or more explicitly named harmful effects or grants one short protection window; has meaningful timing and cooldown.	Force the cleanse early, apply staggered different debuffs, then attack after it expires.	K-10 Protective Support or K-01 Guardian with control prevention; no single cleanse kit may be the only answer.
K-10 Healer / Protector	Primary: sustain; secondary: shield	Heals missing HP or supplies a bounded Shield; follows Section 24.6 healing and 40% active-shield cap.	Healing Suppression, burst, interrupt/defeat the enabler, shield break if explicitly available.	K-01 Guardian or K-09 Cleanser can partly replace survival by preventing damage / removing the disabling effect.
K-11 Buffer / Tactical Enabler	Primary: buffer; secondary: tempo	Raises one named modifier bucket or improves an ally's next action; contributes little direct damage and exposes a counter window.	Dispel/buff removal, target the enabler, wait out duration, prevent the follow-up.	K-12 Marker can enable a finisher through vulnerability instead of Attack buffs.
K-12 Scout / Mark Setter	Primary: mark/focus; secondary: reveal	Applies default Mark (+10% vulnerability for two target turns) subject to the shared +15% cap; may reveal a decoy only if tagged.	Cleanse, intercept, guard the target, switch to a second threat.	K-17 DEF/Healing debuffer or a tactic preset that targets low-HP/visible threat without Mark.
K-13 Chakra Support	Primary: resource utility; secondary: support	Grants a small explicit amount of personal Chakra to one ally; all ally-origin Chakra gifts share +15 per recipient per round. It cannot exceed 100 and never triggers from its own generated Chakra.	Pressure/silence the battery, deny its action, spend control on the carry, or use a non-Ultimate win condition.	Basic generation means no team needs a battery to reach Ultimate; another support can replace it but will change timing.
K-14 Clone / Decoy Specialist	Primary: decoy; secondary: misdirection	Creates a targetable object that intercepts at most one eligible single-target attack or expires at the end of the next round; no action, HP healing or faction count.	AoE, reveal/mark, wait out expiry, use an eligible Ignore-Decoy attack.	K-04 AoE Breaker or a published target policy; no paid anti-clone required.
K-15 Counterattacker / Trapper	Primary: counter; secondary: protection	Punishes one eligible direct attack or a visible enemy event; reaction is capped and does not trigger a second reaction.	AoE/DoT, use non-triggering actions, bait the counter, remove the trap setter.	K-06 Wound/Attrition or K-04 AoE; both can produce value without granting the enemy counter.
K-16 Rear-Access Infiltrator	Primary: target access; secondary: finisher	May choose a legal rear target with a stated coefficient/cooldown opportunity cost; does not silently ignore Taunt unless tagged Ignore-Taunt.	Guard/intercept, Mark a different target, control the infiltrator, protect the rear target.	K-12 Mark + K-03 Finisher can focus a frontline opening; no one unit should be the only rear-pressure tool.
K-17 Defense / Sustain Debuffer	Primary: debuff; secondary: anti-sustain	Applies DEF Down, Attack Down or Healing Suppression as separate named choices; DEF Down shares the 40% effective-DEF cap.	Cleanse, remove applier, protect the target, use raw damage that needs less setup.	K-12 Mark or K-03 Finisher provides an alternate conversion line. Anti-heal does not silently cancel Shields/revives.
K-18 Buff / Shield Breaker	Primary: utility counter; secondary: debuff	Removes one positive effect or reduces a Shield only when the skill explicitly says so; can also remain useful through a small base attack.	Protect the caster, use a team whose win condition is not the removed effect, interrupt before breaker acts.	AoE or direct focus can handle a decoy/protected target; anti-shield is not mandatory in every match.
K-19 Reliable Basic / Flexible Striker	Primary: reliable damage; secondary: flex	Has a dependable Basic and a Technique that remains useful when no synergy setup is present; provides a fallback when specialist combos are countered.	Outscaled in a narrow matchup by a specialist; can be controlled or protected against like any other damage source.	Any accessible damage role can provide the fallback; this role exists to make teams resilient rather than combo-dependent.

26.3 Ten coherent five-shinobi archetypes and their tactical loops

These are team shells, not mandatory five-character recipes. K-IDs refer to the placeholder kits in Section 26.2. A functioning roster should offer more than one viable unit for each essential function as it grows; the starter roster must include at least one base-copy provider for each required answer category.

Archetype	Five functional slots	Win condition / loop	Main weakness	Two counter routes / access rule

Archetype	Five functional slots	Win condition / loop	Main weakness	Two counter routes / access rule
Balanced starter	K-01, K-03, K-09, K-12, K-19	Protect a reliable striker, Mark a legal target, then convert into single-target damage; Cleanser is the fallback against control.	Weak to broad AoE or if the Mark is cleansed before conversion.	AoE plus focus the finisher; or use a cleanse-resistant backup damage path. Accessible counter: K-04 and any common damage role.
Protect-the-carry	K-01, K-05, K-09, K-10, K-11	Survive the opening while a ramping carry builds power; cleanser preserves the window and buffer accelerates the payoff.	Weak to Wound/attrition, Healing Suppression and pressure that forces protection cooldowns.	Use K-17 healing suppression or remove K-11/K-05 early; both are accessible tactics, not a single rare counter.
Burst pickoff	K-12, K-03, K-16, K-01, K-09	Mark or access a target, survive the response with one protector, and turn the opening into a knockout.	Weak to interception, decoy, cleanse, and a second enemy carry after the first attack is spent.	Intercept or cleanse Mark; keep a second threat so the team's turn is not wasted. Two answers are common-pool functions.
Tempo / control	K-07, K-08, K-03, K-12, K-01	Deny one key action or interrupt one charge, then use initiative order to convert before the enemy response.	Weak to Stun diminishing returns, cleanser, resistance and a long fight where low direct output matters.	Use K-09 cleanse and/or protect/defeat the controller; no chain can deny actions indefinitely.
Wound / attrition	K-06, K-10, K-01, K-09, K-19	Apply capped Wound pressure and survive long enough for enemy turns to pay the tick cost.	Weak to cleanse, early burst, time-limited fights, or defeating the applier.	Cleanse one Wound source or focus K-06; raw burst or an accessible cleanser can answer.
Clone / misdirection	K-14, K-04, K-16, K-01, K-09	Use a decoy to absorb one predictable single-target attack while AoE and rear access create a choice about real targets.	Weak to AoE, reveal, target rules that expose the real threat, or a short decoy duration.	K-04 AoE is the direct answer; deterministic targeting and waiting out the decoy are the second answer.
Counterattack / trap	K-15, K-02, K-10, K-06, K-19	Invite predictable eligible hits into limited counter value while sustain and Wound punish a long direct-attack fight.	Weak to AoE, DoT/indirect attacks, control, and attackers who can wait out the counter window.	AoE/Wound routes avoid direct trigger events; remove or control the trap setter.
Sustain / grind	K-01, K-10, K-09, K-05, K-19	Prevent a burst knockout, recover from a key status, and win through reliable sustained damage.	Weak to healing suppression, early focus on sustain source, or excessive duration limits.	K-17 suppression and K-03 burst are separate practical routes; anti-heal does not remove Shield automatically.
AoE / multi-target pressure	K-04, K-11, K-10, K-01, K-19	Apply broad damage, clear decoys and punish fragile groups while protection keeps the AoE unit alive.	Weak to durable single-target guards and a protected carry that survives the broad attack.	Guard the key target and focus K-04; single-target burst offers another approach.
Boss / single-target execution shell	K-05, K-17, K-10, K-01, K-08	Maintain reliable single-target output, bounded DEF Down, sustain, and a tool to interrupt or mitigate a previewed charge.	Weak to setup denial, debuff limits, fast defeat of the debuffer or a boss with explicit resistance.	Use a raw sustained carry instead of K-17; mitigate a charge or interrupt it using accessible tools.

26.4 Directed matchup and counter coverage matrix

Use this as a hypothesis sheet for simulation, not as a claim of proven win rates. "Counter A/B" must be two different functional categories. A new unit cannot be released as the only answer to an existing strategy; any replacement must be accessible through starter, standard acquisition or a deterministic earnable route.

Threat archetype	Counter category A	Counter category B	Fairness implementation rule
Burst pickoff	Shield / interception / decoy on the target	Cleanse Mark or control the finisher	Use two of these; do not require an exact defender.
Protect-the-carry	Reach the rear row with a declared opportunity cost	Apply Healing Suppression or defeat its protector/support	At least one rear-pressure route is normal-pool and one is obtainable early.
Tempo / control	Cleanse or brief control prevention	Shared Stun diminishing returns / interrupt before charge resolves	No hard-control unit or resistance counter is paid-only.
Wound / attrition	Cleanse Wound sources	Burst / focus the applier / win before the next tick	DoT and cleanse are both available outside limited banners.
Sustain / grind	Healing Suppression	Burst focus on the healer / interrupt sustain action	Anti-heal does not also erase shields or revive by default.
Clone / decoy	AoE that can hit the decoy and other targets	Reveal/Mark the true target or use deterministic priority after expiry	No unique premium "true sight" requirement.
Counterattack / trap	AoE, Wound or other non-triggering damage	Bait the counter window or remove/control the trap setter	Core damage route must not depend on triggering the opponent.
AoE pressure	Protect key targets with guard/shield and avoid clumping if placement supports it	Focus the AoE caster before the next cycle	Protectors and focus-damage units are both accessible.
Mark-and-focus	Cleanse Mark or intercept the high-value hit	Keep multiple threats / switch to a secondary target	A single Mark never rewrites target rules invisibly.
Chakra support engine	Silence/disable the support or defeat it	Use a Basic/reliable-damage win condition while the engine is unavailable	No team requires a Chakra battery because every hero has personal Chakra gain.

Threat archetype	Counter category A	Counter category B	Fairness implementation rule
Boss charge / scripted threat	Interrupt the named charge	Survive it with shield/mitigation or defeat caster before resolution	Boss preview must state resistance and at least two obtainable response types.

26.5 Synergy-strength budget and substitution test

- Use a visible setup → conversion → opponent response chain. A combo is meaningful if it improves time-to-knockout, damage dealt/prevented, survival or action economy—not merely if a tooltip displays +stats.
- Measure a synergy against a functionally equivalent substitute at equal investment. The initial acceptance target is that removing one named partner should not create more than an 8 percentage-point win-rate loss when an accessible substitute restores the same role. This is a test threshold, not a universal industry metric.
- Village/faction bonuses share one total cap: 2 matching tags = +2% ATK/HP, 3 = +4%, 4–5 = +5% maximum. Use only the highest reached tier; do not add tier rewards together. Relationship effects should usually add utility or a one-time small follow-up, with an average normalized effect swing targeted at 3% or less.
- The core kit expresses its role at baseline copy. Duplicate upgrades may improve numbers modestly within shared caps, or grant non-combat customization; they cannot unlock a missing cleanse, an extra mandatory action, the only anti-shield effect, or the only source of a core archetype trigger.
- No automatic global elemental rock-paper-scissors multiplier in MVP. Element/nature tags may affect a specific previewed skill or boss condition, but do not add a universal hidden damage multiplier until all sides of the matrix have accessible answers.
- A hybrid unit must pay an opportunity cost. A healer-buffer that also deals carry-level burst at full uptime is not "flexible"; it erases the reason to choose specialists. Balance that with lower damage, a conditional effect, cooldown trade-off, limited target range or reduced protection uptime.

26.6 One worked damage / synergy example (illustrative, not a balance claim)

Assume an accessible Mark setter applies +10% Vulnerability, an accessible DEF debuffer reduces DEF by 20%, and a finisher hits a target with ATK=1,200, coefficient=1.6 and no crit. Target DEF starts at 1,000. Using Section 24.6, DEF multiplier is 1000/(1000+1000)=0.5 before setup. With 20% DEF reduction, Effective DEF is 800 and multiplier is 1000/1800≈0.5556. If the finisher also has a +20% Attack Bonus, the estimated hit is floor(1200×1.6×1.20×0.5556×1.10)≈1,408 damage. If a raw-stat alternative has 10% more ATK (1,320) but no Mark, no DEF reduction and no Attack Bonus, its matching hit is floor(1320×1.6×0.5)=1,056. Under these assumptions, the prepared combination hits harder despite the alternative's higher raw ATK.

This demonstrates why composition can matter; it does not establish that the prepared team is always superior. The setup sequence can fail if Mark is cleansed, the debuffer is interrupted or the finisher is intercepted. Those response windows are essential. Compare the whole team over repeated matchups, not one damage number, and measure what happens when one enabler is removed.

26.7 Combat rules source-of-truth checklist

Rule area	Authoritative definition	Legacy conflict now treated as historical	Must be logged in combat replay
Damage / healing / Shield	Section 24.6, with Attack Down order, no-MVP-True-Damage rule, Wound cap and Shield timing reconciled in Section 27.	Legacy 100/(100+DEF), flat DEF−PEN, +35% skill cap, "rare True Damage" and formula samples with mismatched attack status.	Base ATK, Attack Down, each bucket, Effective DEF, Crit roll, reduction, Shield absorbed, HP damage and rounding.
Initiative / action economy	Section 25.6 and Section 27.4: one normal action per living unit per round; Speed modifies next-round order only; no meter/advance.	Dynamic turn meter / action advance allowance in older tables.	Round/slot/order, speed snapshot, status skipped, action spent, cooldown and fallback reason.
Personal Chakra / cooldown	Section 25.7 and the synchronized table in Section 20: Basic +25, Technique +15, damage-taken +5 capped +15 per round, takedown +10 once/action, Ultimate costs 100.	Older +10 Technique Chakra and +10 / +20 hit-Chakra drafts.	Chakra before/after, trigger source, cap applied, cooldown remaining.

Rule area	Authoritative definition	Legacy conflict now treated as historical	Must be logged in combat replay
Targeting / Taunt / Intercept	Section 24.5 and 25.10; skill selector, Taunt, squad preset and frontline priority rules.	Any “front-most target” statement that conflicts with the current HP-percentage rule.	Legal candidates, selector, Taunt/intercept decision and tie-break.
Status effects / turn timing	Section 20.7 table, with current effect interactions synchronized in Section 20.8/25.8.	Old generic three-stack DoT and overlapping Stun/Silence DR table.	Application chance, source, duration, start/end tick, cleanse and immunity.
Hard crowd control	Section 25.8; Stun chance tiers and no fractional turns; Silence is soft control.	50%/25% fractional Stun duration and any rule that lets two Stuns before the target action skip two turns.	Base chance, target tier, effective chance, resistance, lost action and tier advancement.
Synergy / tags / follow-up	Sections 24.8-24.10 and 25.5; bounded triggers and +5% ATK/HP maximum tag tier.	Big mono-faction or exact-bond stat locks; recursive follow-ups and clone turns.	Trigger root event, frequency cap, tag threshold, damage bucket and whether chain stopped.
Revive / summon / clone	Section 24.9; clone is a bounded object, not another hero; revive excluded from MVP.	Any extra-turn, recursive spawn or repeat-start-of-battle effect.	Object count, expiry, intercepts remaining, revive eligibility and no-recursion reason.

This checklist is part of the implementation contract. A character cannot introduce a private exception to a core combat rule. New exceptions require a written design reason, a logged decision, counterplay review, and an automated regression test.

26.8 Combat written-specification closure and future implementation gate

The public-source comparison now covers the major team-combat patterns relevant to this design: faction/formation bonuses (AFK Arena/Journey and Mythic Heroes), role coverage and roster transfer convenience (Mobile Legends: Adventure), large teams and faction aura (Idle Heroes), hard action sequences (NIKKE), Speed/Combat Readiness systems (Epic Seven, Summoners War, RAID), shared action resources/weakness break (Honkai: Star Rail), turn-based positioning (Brown Dust 2), card/action resources (Reverse: 1999), and tag/role/previewed-weakness team rules (Cookie Run: Kingdom). Sources have different authority levels, so this is a pattern audit—not an exhaustive account-level or hidden-formula audit of every game.

The combat source-of-truth checklist is a continuing specification-quality requirement, not a gate that blocks broader conceptual research. Undefined combat rules must remain visible in the open-decision register, and future implementation or empirical balance claims remain gated by Section 24.15.

Exit gate	Ready-to-advance requirement	Status in v0.8	What remains before a build can be called balanced
Rules consistency	One authoritative damage formula, one action economy, one Chakra rate, one target policy, one CC rule and one status table.	Written rules reconciled in Sections 24-27; direct damage, status timings and edge cases have a single source of truth.	Implementation and regression still pending; desk closure is not empirical balance.
Roster coverage	Every core combat function has a base-copy provider and at least one accessible substitute.	19 placeholder kits, 10 archetype shells, accessibility rules and normalized stat/coefficient packets are documented.	Final kits and replacements must be implemented and simulated at equivalent investment.
Archetype identity	Each planned team has a win condition, setup, conversion, survival path and fallback.	Ten archetypes have win conditions, weaknesses, two counters and a roster-to-function map.	Simulate every directed matchup; archetype diversity is only a design hypothesis until measured.
F2P access	No mandatory role, answer, or synergy trigger is locked behind paid/limited or duplicate-only progress.	Recorded as a non-negotiable; explicit replacement path required on each kit.	Audit the actual release roster and acquisition paths before launch.
Research evidence	Visible combat systems are source-attributed; unsupported rates/hidden formulas are not copied as facts.	Public-source observations are attributed by source type in Sections 25 and 27; patch-sensitive and community sources are labeled.	Hands-on comparison across current clients and modes remains incomplete; recheck any patch-specific fact before implementation.
Numerical balance	No mandatory meta, permanent stun lock or unanswerable pay advantage within tested role matchups.	Not testable without a simulator or playable build; values remain proposals.	Seeded combat simulation and representative phone playtests; do not claim validation before then.

[FUTURE IMPLEMENTATION SEQUENCE — NOT A CURRENT RESEARCH BLOCKER] Hold implementation/final balance decisions for full modes, equipment, artifacts, pets, awakening/signature power-ups and battle-power rating until combat rules are implemented and the later validation plan has been run. This does not block current conceptual research on idle-RPG modes and other systems. Research them now as theory, record how they depend on combat, and keep unresolved interactions marked [OPEN] or [HYPOTHESIS].

26.9 Version 0.8 change and evidence register (historical; v0.9 changes in Section 27.11)

ID	Integrated change	Reason	Status
C-017	Synchronized active-body tables 12, 14, 28 and 30 plus the older decision register with Sections 24.6, 25.6-25.10.	The v0.7 review found older +35% skill cap, +10 Chakra, 35% Shield and fractional-control descriptions still present in the current body.	Consistency fix completed; originals remain preserved in archival appendices.
C-018	Expanded the public-source combat comparison to Epic Seven, Idle Heroes, Cookie Run: Kingdom and Mobile Legends: Adventure.	Closes the explicit v0.7 benchmark gap while separating official source claims from community observations.	Desk research added; no hands-on client capture or hidden mechanics claimed.
C-019	Added 19 placeholder kits and 10 archetype team shells with accessible replacements.	Roles existed in abstract but were not yet mapped into a coherent, replaceable roster structure.	Design proposal; not a final licensed character kit or empirical tier list.
C-020	Added a directed counter-coverage matrix and a single-source-of-truth rule table.	Prevents undefined targeting/stacking exceptions and premium-only counters from entering new kits unnoticed.	Written coverage complete; must be encoded and regression-tested.
C-021	Added mobile combat research closure gate before modes/power-up add-ons.	Honor project sequencing and avoid covering a weak core system with gear/rarity power.	Settled project workflow; empirical combat balance remains pending prototype.
C-022	Added normalized stat packet and first skill-coefficient budget for a reproducible reference engine.	Convert qualitative role ideas into comparable combat test inputs without pretending values are final.	Specification only; simulator tuning pending.
C-023	Defined deterministic action-resolution sequence, Attack Down timing, post-damage status default, Wound snapshot/cap, Shield Chakra condition, Taunt expiry and follow-up event limits.	Resolve edge cases that otherwise let different characters implement contradictory engine rules.	Written rule set; must be encoded in tests.
C-024	Expanded source audit with official Summoners War Challenge Battle and reconciled incorrect citations in the benchmark table.	Separate public documented mechanic from inference; correct reference attribution.	Desk research; no claim of mode outcomes.
C-025	Updated research registry, roadmap and exit gate so combat spec is drafted but simulator, seed matrix and phone comprehension are next.	Prevent “written” from being mistaken for “balanced” and prevent premature scope expansion.	Project sequence settled.
C-026	Defined Chakra-battery contribution cap and deduplicated incoming Chakra by enemy root action/defender, including multi-hit/follow-up handling.	Prevents ambiguous energy farming and ensures battery kits are bounded rather than mandatory.	Written rule; validate cadence in simulation.

26.10 Sources added in Version 0.8 (historical source notes; see Section 27.10 for v0.9 additions)

These new references support only the described visible/game-facing mechanics. They do not prove balance, commercial success or F2P fairness. Accessed 28 October 2026; time-sensitive features should be rechecked before shipping.

- [R15] Devsisters, “August 26 Update Notice,” CookieRun: Kingdom, 26 August 2026. The update describes boss Weakness previews, composition-specific Fire-tag thresholds, and a Cookie Alliance revision from five teams to three. https://hub.cookierun-kingdom.com/news/updates/august-26-update-notice
- [R16] Google Play editorial, “The different Cookie types in CookieRun: Kingdom.” Used only for the role taxonomy summary (Charge, Defense, Magic, Ranged, Ambush, Bomber, Support, Healing). https://play.google.com/store/apps/editorial?hl=en_US&id=mc_games_editorialevergreen_postinstall_types_of_cookies_in_cookierun_now_fcp
- [R17] MOONTON, “Mobile Legends: Adventure,” Google Play app listing, updated 8 September 2026. Used for the publisher’s description of idle auto-battle, 100+ heroes, seven hero types, team-composition relevance, and level transfer/level sharing. https://play.google.com/store/apps/details?gl=ph&hl=en-gb&id=com.moonton.mobilehero
- [R18] DHGames, “Idle Heroes,” official publisher website. Used for publisher presentation of a large hero roster and skill-specific hero identities; no hidden combat formula inferred. https://ih.dhgames.com/
- [R19] Idle Heroes Wiki, “Tower of Oblivion,” community-maintained reference. Used with low/moderate confidence for a six-hero two-front/four-rear enemy formation and faction aura in that mode; recheck in current client if exact values matter. https://idleheroes.fandom.com/wiki/Tower_of_Oblivion
- [R20] Epic7X, “Combat Readiness Guide” and effect references (community database). Used to illustrate Speed/Combat Readiness manipulation, buffs/debuffs and immunity/resistance interactions; guides and character availability can be patch-sensitive and are not official publisher documentation. https://epic7x.com/combat-readiness-tutorial/ and https://epic7x.com/buff_debuff/immune/

Version 0.8 work record (historical). It added cross-game combat research, 19 placeholder kits, ten archetypes, matchup/counter matrix, and an exit gate. Version 0.9 extends that work with normalized reference stats, initial coefficient budgets, explicit action-resolution state machine, edge-case/status timing audit, source corrections and a refreshed research maturity gate; see Chapter 27.

27. Combat rules closure, normalized stat packet and edge-case audit (Version 0.9)

Purpose and scope. This chapter is the detailed combat reference within the broader game concept, not the project's only active research topic. Version 0.9 standardizes hypothetical reference stats, initial skill-coefficient budgets, action/status timing, edge cases and a future counter-focused regression plan. These values remain proposals and are not balance-validated.

27.1 Research maturity: facts, inference and proposal

[DOCUMENTED PATTERN] Existing games use different action economies and team constraints. CookieRun: Kingdom's official 26 August 2026 update previewed wave-boss weaknesses in the battle lobby, reduced its Cookie Alliance requirement from five teams to three, and used team-condition buffs with 1/3/5 thresholds for one element. This is evidence for previewing encounter rules and for the fact that team-condition effects exist—not evidence that those thresholds are inherently fair. [R15]

[DOCUMENTED PATTERN] Mobile Legends: Adventure's official Google Play listing describes an idle auto-battler with 100+ heroes, seven types, team-composition strategy, and level transfer/level sharing to make other lineups easier to build. This supports the general product lesson that lower investment friction encourages experimentation; it does not prove a particular revenue or F2P outcome. [R17]

[DOCUMENTED PATTERN] The official Summoners War v8.8.5 notice described a Challenge Battle using preset monsters, a pick/ban phase and max-level Summoner Skills. That is a current source example of separating one competitive mode from the player's normal roster investment. This is a design option for later consideration, not a promise to copy that mode. [R21]

[COMMUNITY-ONLY MECHANIC REFERENCE] Epic7X's community guide describes Speed and Combat Readiness manipulation in Epic Seven; the exact interactions vary by unit, effect and patch. Its transferable lesson for this project is only that dynamic turn-meter manipulation can make initiative a full build-around system. The MVP deliberately avoids that complexity by snapshotting Speed at round start and prohibiting action advance. [R20]

[DOCUMENTED PATTERN / OLD SOURCE] Brown Dust 2's older English guideline describes attack/defend phases with skill reservation, attack-order selection and repositioning. It demonstrates the strategic depth positioning can create, but it is a manual tactics game with older guide material, not a direct blueprint for automated idle combat. [R8]

[DESIGN INFERENCE] The practical conclusion is not that these games prove our exact rules. They support a small set of useful design principles: make team conditions visible before the fight; make role identity legible; reduce the cost of trying substitutes; treat tempo manipulation as a high-risk system; and avoid granting large team bonuses that only a narrow roster can activate.

[OPEN EVIDENCE LIMIT] Public sources cannot establish exact balance, monetization sustainability, whether a game is truly F2P-friendly across all account ages, or the win rate of our proposed rules. Those require consistent current-client inspection, telemetry, and our own simulation/playtests. We must not use "popular game does this" as sufficient evidence to adopt a mechanic.

27.2 Normalized reference stats for simulation

The values below are an artificial equal-investment test baseline, not real level-1 stats, a production level cap, or an industry standard. Use them to compare role kits without rarity or progression noise. The first simulator pass must be deterministic under a recorded random seed. Use one primary role multiplier per kit; secondary role tags do not stack stat multipliers.

Baseline stat	Reference value	Rule
Max HP	6,000	Before primary-role multiplier; Max HP is also the denominator for shield and Wound safety caps.
ATK	1,000	Before primary-role multiplier, Attack Down, temporary Attack Bonus and skill coefficient.
DEF	1,000	Before DEF Down and DEF Penetration; use the shared 40% effective-reduction ceiling.
Speed	100	Base initiative value; determine order once at the start of each round.
Crit Chance	10%	Base probability before bonuses; final player Crit Chance is capped at 75% in every MVP mode.

Baseline stat	Reference value	Rule
Crit Damage	1.50×	Baseline final critical multiplier; never exceed 2.0× in MVP.
Starting Chakra	0	At battle start unless an explicitly declared, broadly obtainable kit effect says otherwise; do not use a paid start-Chakra advantage in normalized PvP.
Normal action budget	1 per living unit per round	Follow-up/reaction events have their own strict event caps and never become a second normal turn.

27.3 Primary-role stat multipliers (prototype test values)

These multipliers create distinct stat profiles without making a role label a hard class. All values are provisional. After applying the multiplier, derive actual in-battle stats, then run the universal damage pipeline. A secondary role tag indicates kit function only; it does not apply another row of stats.

Primary role	Reference stats after role multiplier	Balance contract
Guardian / protector	HP ×1.30; ATK ×0.80; DEF ×1.20; Speed ×0.90; Crit 5%	Absorbs or redirects predictable threats; should not also be best damage converter.
Bruiser / interceptor	HP ×1.15; ATK ×0.95; DEF ×1.10; Speed ×1.00; Crit 10%	Trades peak damage for staying power or one bounded intercept.
Single-target finisher	HP ×0.85; ATK ×1.25; DEF ×0.85; Speed ×1.10; Crit 15%	Must have a visible setup or low-HP conversion; cannot one-shot a full-health peer by default.
AoE breaker	HP ×1.00; ATK ×0.95; DEF ×0.95; Speed ×1.00; Crit 10%	Wide pressure with lower per-target coefficient than a dedicated finisher.
Sustained carry	HP ×1.00; ATK ×1.00; DEF ×1.00; Speed ×1.00; Crit 10%	Reliable damage that benefits from a protected multi-round window.
Wound / attrition specialist	HP ×0.95; ATK ×1.05; DEF ×0.95; Speed ×1.00; Crit 5%	Wins through capped delayed damage; not a burst kit by stealth.
Controller / disruptor	HP ×0.95; ATK ×0.85; DEF ×0.95; Speed ×1.10; Crit 5%	Trades damage for control or interrupt; control resistance/access rules remain shared.
Buffer / Chakra support	HP ×0.95; ATK ×0.80; DEF ×0.95; Speed ×1.05; Crit 5%	Enables another function; stacked buffers must not raise a carry beyond defined caps.
Healer / cleanser	HP ×1.05; ATK ×0.80; DEF ×1.00; Speed ×0.95; Crit 5%	Keeps the team active or removes statuses; cannot make attrition unwinnable.
Scout / rear access	HP ×0.90; ATK ×1.05; DEF ×0.90; Speed ×1.10; Crit 12%	Threatens rear line at a stated opportunity cost; should not bypass all protection for free.
Counter / trap / clone	HP ×0.95; ATK ×0.90; DEF ×1.05; Speed ×1.00; Crit 8%	Punishes declared events and uses bounded objects/reactions; no recursive effect chains.

27.4 Skill coefficient budget for first simulation pass

Coefficients are test scaffolding, not final balance. “Coefficient” multiplies the attacker’s ATK after Attack Down, before defensive mitigation. Effects with meaningful control, cleanse, protection or resource utility use less direct damage than pure damage skills. Evaluate both average and 95th-percentile damage across critical/noncritical outcomes.

Action / effect class	Initial coefficient / magnitude	Test range	Required guardrail
Basic single hit	0.85× ATK	0.75–1.00	Reliable fallback action; no extra target or hidden multiplier.
Single-target Technique, damage role	1.15–1.60× ATK	1.00–1.60	Cooldown 2 owner turns by default; if it applies a debuff, that debuff lands after damage unless tagged PRE-DAMAGE.
Single-target Ultimate, damage role	1.50–2.00× ATK	1.50–2.00	Must include the +40% Attack, +25% Skill/Type and +15% Vulnerability cap in the strongest plausible test.
AoE Technique	0.30–0.45× ATK per target	0.30–0.45	Evaluate full-team total damage, not just one-target tooltip number.
AoE Ultimate	0.35–0.50× ATK per target	0.35–0.50	Must not delete an equal-stat team before it has a meaningful response window.
Utility/control damage skill	0.45–0.85× ATK	0.45–0.85	Stronger control, cleanse, interrupt or resource effects pay for some of their power through lower damage.
Single-target heal	1.20–1.80× Healing ATK	1.20–1.80	Apply +25% Healing Bonus ceiling, Healing Suppression and missing-HP cap.
AoE heal	0.45–0.70× Healing ATK per ally	0.45–0.70	Test effective healing rather than gross healing; avoid infinitely resetting battles.
Single-target shield	10–18% recipient Max HP	10–18%	All active shields share 40% Max HP ceiling; repeated shields cannot create Chakra.
Follow-up / reaction hit	0.40–0.70× Basic coefficient	0.40–0.70	No crit, no Chakra, no status/on-hit chain, no reaction recursion; counts inside follow-up caps.

One-shot check for every coefficient pass: run each intended finisher against a full-HP equal-investment non-tank with maximum legal attacker bonuses, mark/vulnerability, and critical hit. The design target is that an ordinary action cannot

remove that target from full HP. A named Execute is the only exception and must require an observable setup/low-HP condition plus at least two accessible answers. Reduce coefficient or setup access before inventing an arbitrary hidden damage clamp.

27.5 Exact action-resolution state machine (authoritative sequence)

This is the order the reference engine must implement. If a new kit cannot be expressed in this sequence, it needs a reviewed, named exception rather than an implicit bypass.

Step	Authoritative rule	Required log / edge-case rule
0. Initialize	Validate each team has up to five living slots, load the deterministic seed and mode rules, initialize status durations/Chakra, and show telegraphs and target policy.	Record roster ID, stat snapshot, seed, side priority, mode rules and opening state.
1. Round start	Apply start-of-round effects in fixed team-side/slot order. Apply Speed changes for this round; do not re-order the queue after it starts.	Record each start effect and current effective Speed.
2. Queue	Sort all living units by effective Speed, descending. Resolve ties by round side-priority (alternates by round) then left-to-right slot. Freeze queue for this round.	Record the whole queue before actions. A KO removes its future slot; it never grants the defeated unit an action.
3. Action-slot start	At the unit's slot start, apply Wound ticks, then other explicitly start-of-slot effects. If a Wound KO occurs, the unit loses the action; Wound grants no takedown Chakra.	Record tick source snapshots, mitigation, shield/HP loss and death reason.
4. Eligibility / control	If already defeated, skip. If Stunned, lose this one normal action. If Silenced, only Basic may be selected. Cooldown/duration timers follow the rules in 27.6.	Log "skipped: defeated", "action denied: Stun", or "skill unavailable: Silence/cooldown"; never silently skip.
5. AI selection	If alive and not Stunned: Ultimate if Chakra=100 and its visible condition/legal target holds; else Technique if cooldown=0 and its condition/legal target holds; else Basic. If no legal ultimate target, fall through to Technique/Basic.	Record why each skill was selected or why AI fell back.
6. Target resolution	Apply skill-authored selector; then Taunt for eligible single-target hits; then one eligible interception. Validate legal target again. In normal Basic, use the selected squad policy, then lowest-current-HP% frontline, then lowest-HP% rear.	Record intended target, final target, redirect/intercept reason and tie-break.
7. Damage / status	Direct hits always land on a legal target in MVP. Compute Attack Down and all damage buckets. Status chance rolls separately. Ordinary on-hit statuses apply after damage; PRE-DAMAGE must be explicitly declared and shown in tooltip/log.	Record each modifier, cap, crit roll, mitigation layer, Shield absorption, HP damage and status chance/result.
8. Follow-up / reaction	Resolve only named event triggers. Max one follow-up per acting unit per round and two per team per round. No follow-up/reflect/lifesteal/reaction chain unless a future separate design decision replaces MVP.	Record trigger source, cap hit, recursion blocked and total follow-ups spent.
9. Chakra / takedown	Basic +25; Technique +15; Ultimate costs 100. Defender +5 per direct enemy attack instance that yields positive post-DEF/post-non-DEF-DR damage before Shield, up to +15 per defender per round. One +10 actor takedown reward per root action that KOs at least one enemy.	Wound, object/clone hits, follow-ups, reaction attacks and Shield generation produce no independent Chakra/takedown event. Shield absorption does not erase Chakra if pre-Shield damage was positive.
10. Slot-end timers	At the end of each recipient's action slot, decrement effects whose duration is measured in recipient slots—even if Stun denied the action. An effect applied after that slot cannot lose duration immediately.	Record old/new duration for each status; remove expired effects in deterministic order.
11. End round / match	Resolve end-of-round effects, victory/defeat and round cap. PvE cap 12 rounds; PvP cap 10 rounds. At PvP cap compare team HP% remaining, then total damage dealt; exact tie is a draw/no rating change.	Record ending condition and all tie-break values; no hidden judge/modifier.

27.6 Status timing and interaction audit

All duration rules use a unit's action slot rather than wall-clock time. "Two target turns" means the next two action slots belonging to that target; it does not mean two global rounds. Effects applied after the target acted start counting at its next slot. Wound ticks at slot start. Speed effects are a separate next-round clock and never reorder the current queue.

Effect	Duration / activation clock	Magnitude / stack rule	Counter / interaction rule
Mark	After triggering hit by default; lasts through next 2 target slots.	+10% Vulnerability; shares total +15% cap.	Cleanse, intercept, decoy, change legal target. PRE-DAMAGE tag is only exception.
Wound	Tick at start of each of target's next 2 slots.	18% source ATK snapshot; total Wound ≤20% target Max HP per slot before mitigation; max 2 sources.	DEF/DR/Shield apply; no Crit, Chakra, takedown Chakra, lifesteal, reflect, on-hit or follow-up.
Attack Down	Applies after hit by default; next 2 target slots.	Strongest active Attack Down only, capped at 40%; applied before Attack Bonus.	Cleanse/wait it out; log effective ATK used.
Defense Down	Applied after hit by default; next 2 target slots.	Default −15% DEF; shares additive 40% cap with DEF Penetration.	Cleanse/eliminate source; show reduction bucket.
Slow	Changes next round's queue only; expires at end of that round.	Default −15% Speed; total Speed change ±20% cap.	Cannot reorder this round; cleanse/Speed-up.
Stun	Denies one next normal action once.	Base default 60%; Tier0/Tier1/Tier2 multipliers ×1/×0.5/×0.25; final status chance clamped 10–90%.	Cannot extend active Stun or lose a second action before acting; immunity after successful Tier-2 Stun until next completed non-Stunned slot, then tier=1.
Silence	Blocks Technique/Ultimate for the next target slot; Basic remains legal.	No Stun tier change. Basic earns +25 Chakra.	Cleanse/wait; does not cancel Charging without Interrupt tag.

Effect	Duration / activation clock	Magnitude / stack rule	Counter / interaction rule
Healing Suppression	Next 2 target slots.	Strongest active effect up to −40% healing received. Shield not reduced unless specified.	Cleanse, protect with Shield/mitigation or remove applier.
Shield	Active through next 2 recipient action slots.	Same-named magnitude does not stack; different sources share 40% Max HP total capacity.	Consumed after DEF and non-DEF DR; Shield does not itself generate Chakra.
Taunt	Expires after 2 eligible enemy single-target redirects or second owner action after application, whichever first.	Eligible single target only; ends if Taunter dies.	AoE, Ignore Taunt and scripted unavoidable attacks bypass; Mark cannot override unless explicitly tagged.
Intercept	At most one eligible hit per effect.	Final recipient is the interceptor; damage pipeline runs once against final recipient.	No double-interception; AoE is not globally redirected.
Clone / decoy	Intercepts one eligible single-target hit or expires at end of following round.	Object/tag, not a hero; no action, heal, Chakra, faction, or formation slot.	AoE/reveal/explicit ignore-decoy tags; no nested objects in MVP.
Interrupt	Only during visible Charging state, when ability has Interrupt tag.	Cancels the specific charged effect; no extra turn denial.	Log charge canceled; Stun DR is unaffected.

27.7 Worked calculation and damage-bucket audit

Example: normalized attacker has base in-battle ATK 1,000 and is under 15% Attack Down; therefore ATK after Attack Down = 850. It uses a 1.4× Technique with +30% Attack Bonus, +15% Skill/Type Bonus, against DEF 1,000 with 15% DEF Down and +10% Vulnerability, and the hit crits for 1.5×. Effective DEF = 1,000 × 0.85 = 850; DEF multiplier = 1,000/(1,000+850) ≈ 0.54054. Damage = floor(850 × 1.4 × 1.30 × 1.15 × 0.54054 × 1.10 × 1.5) = 1,586. At 6,000 HP, that is 26.4% of a full-health reference target before its Shield or non-DEF Damage Reduction. This worked number is for calculation QA, not a claim of final balanced damage.

Bucket audit: Attack Down is a negative stat effect before Attack Bonus; Attack Bonus +40%, Skill/Type +25%, Vulnerability +15%, and Crit are the only ordinary direct-damage offensive multipliers in MVP. DEF Down/Pen is one shared cap; non-DEF reduction is a separate 30% bucket; Shield follows. No hidden Final Damage, true damage, accuracy, evasion or separate flat-penetration bucket exists in MVP.

27.8 Regression and counterplay stress-test pack

Test ID / scenario	Test input	Pass condition
A. Burst pickoff vs protection	Run finisher+Mark against Guardian+Shield; vary side priority and legal focus.	At least one accessible intercept/Shield route and one cleanse/control route meaningfully reduce knockouts; normal finisher cannot one-shot full-HP non-tank.
B. Protect-the-carry vs rear access	Run carry+guardian+healer against scout/rear pressure; test different target policies.	Rear pressure has cost but is real; guard does not make rear immune; protector/carry shell still beats some non-counter teams.
C. Stun-control loop	Attempt many Stuns on one target under deterministic seeds; include Tier 0/1/2, misses and active Stun attempts.	Max one action denied per Stun; no repeated denial before a completed non-Stunned slot; tier and immunity match Table 53 and logs.
D. Silence vs ability reliance	Silence a 100-Chakra target and ensure it can still Basic.	Target still acts, earns Basic Chakra, Ultimate waits; Silence cannot secretly cancel a charging effect.
E. Wound attrition vs cleanse	Apply maximum two Wound sources to high-ATK target; test shields, target turn slots and source death.	Per-slot Wound never exceeds 20% target Max HP before mitigation; Wound gives no Chakra or reactions; cleanse removes one declared source.
F. AoE vs single-target guard	Use five-target AoE into two interceptors and Shields.	AoE target list remains skill-defined; interceptor does not absorb every target's hit; total AoE pressure is measured across all targets.
G. Clone/decoy vs target policy	Create decoy, use Frontline First/Focus Mark/Break Threat and AoE.	Only one eligible single-target hit is redirected; no faction/Chakra/turn is created; expired/invalid decoy is not selected.
H. Chakra-engine disruption	Silence/KO battery; test basic fallback, incoming direct hits and Shield-only hits.	Basic +25, Tech +15, Ult −100; incoming +5 only for positive pre-Shield damage, max +15/round; no recursive resource loop.
I. Speed stack / current queue	Apply Slow and Speed-up before and after the queue is frozen.	Current queue never changes mid-round; modifiers are capped ±20% and affect next-round initiative only.
J. Same-name buff refresh	Apply same named +20% Attack from two sources at different durations.	Magnitude never becomes +40%; strongest active instance controls value and a weaker instance cannot overwrite a stronger one; log refresh/expiry.
K. Tie / defeat ordering	Use equal Speed, same-slot deaths, and simultaneous Wound KOs.	Round-side priority and slot tiebreak deterministic; defeated unit never acts; no double reward from simultaneous KOs.
L. Buff cap abuse	Stack three distinct Attack buffs, Skill/Type buffs, Vulnerability, DEF Down and Penetration.	Each bucket caps exactly; no hidden extra multiplier appears; formula trace matches manual calculation.
M. Trigger recursion	Build test kits that would otherwise cause reflect → follow-up → lifesteal → reaction loops.	Follow-up and reaction ceilings hold; no unbounded event loop; root action grants at most one takedown Chakra reward.
N. Stalled PvP round cap	Force both teams to survive through ten PvP rounds.	Tie-break is remaining team HP% then total damage; exact tie causes draw/no rating change; no mode-specific hidden score.

27.9 Roster synergy fairness and measurable thresholds

The roster must provide role access and different strategic routes, not force a particular canon pair or rarity tier. This is a design rule for a hypothetical Naruto-licensed game; actual Naruto character use requires appropriate licensing. The baseline roster should be testable without paid pulls or duplicates.

Measure	Initial target / rule	Audit method / failure action
Composition coverage	Each five-unit team documents Win Condition, Setup, Conversion, Survival, Fallback.	All 5 fields written for every released kit-composition; no “power score only” assessment.
Archetype viability	Keep at least 8 of 10 current shells viable at normalized stats across their intended PvE/PvP contexts.	Run ≥1,000 seeded outcomes per directed matchup and swap side priority; target ordinary matchup win rates roughly 45–55% among non-specialists, while explicit counter matchups may differ.
Dominance investigation	Any archetype >60% equalized win rate across broad pairings, or <40% without a clearly intended hard matchup, triggers review.	Do not rebalance from tiny samples; include confidence interval, mode and player-skill segment.
Pick-rate alarm	Sustained usage above 35% in broad PvP triggers investigation; >60% is not acceptable without a documented mode/draft reason.	Usage is a signal, not proof; correlate with win rate, roster availability and patch age.
Synergy value	A synergy should improve win rate about 3–7 percentage points over a function-equivalent accessible replacement in its intended matchup.	If it is always optimal regardless of counter/mode, lower effect or add a readable counter window.
Replacement robustness	Removing one named hero and substituting an accessible same-function unit should not reduce the team by >8 percentage points.	If it does, redesign bond trigger or make a second replacement path viable; do not solve via paid duplicate power.
Synergy activation	Intended combos should activate in about 60–85% of genuinely available setup windows after considering enemy disruption.	100% activation indicates low counterplay; below 60% can mean the combo is too conditional or opaque. Log availability and cause of failure.
Counter availability	Every potent archetype has at least two different functional counters, one accessible early.	A counter must be usable by role/function, not only a specific premium/limited character or max-duplicate unit.
Duplicate sensitivity	In normalized PvP, base-copy vs max duplicate effect on combat output target ≤5% after stat normalization.	Measure direct output and win rate; if a specific duplicate unlocks an extra turn, unique required counter or core combo, fail regardless of aggregate stat.
F2P access	A new free account receives all core role functions through starter/early deterministic rewards. One additional coherent archetype should be buildable in the proposed 4–8 week target window.	The 4–8 week value is a project target, not an industry fact; validate with an economy/account simulation later.

27.10 Source notes added in Version 0.9

The following source checks support only the described game-facing mechanics. Official patch notes/store listings are stronger evidence for the mechanics they explicitly describe; community guides can help explain a visible system but can be stale or patch-sensitive. Neither source type validates the proposed Naruto idle RPG numbers. Accessed 28 October 2026.

- [R21] Com2uS, “v8.8.5 Update Notice,” Summoners War: Sky Arena, 30 September 2025. The official notice describes Challenge Battle using preset Monsters, a pick/ban phase and max-level Summoner Skills regardless of account progression. https://summonerswar.com/en/skyarena/news/list/5978
- [R22] Devsisters, “August 26 Update Notice,” CookieRun: Kingdom, 26 August 2026. Official notice describes wave-boss Weakness previews, Cookie Alliance revised from five teams to three, condition-selectable team buffs and Fire-tag thresholds at one/three/five Cookies. https://hub.cookierun-kingdom.com/news/updates/august-26-update-notice
- [R23] MOONTON, “Mobile Legends: Adventure,” Google Play listing, accessed 28 October 2026. Publisher description identifies idle auto-battle, 100+ heroes, seven types, role/team composition for bosses/PvP, and level transfer/level sharing. https://play.google.com/store/apps/details?gl=ph&hl=en-gb&id=com.moonton.mobilehero
- [R24] Epic7X, “Combat Readiness Guide,” community guide. Used only as an example of speed/action-meter interaction in an adjacent turn-based hero collector; not official publisher documentation. https://epic7x.com/combat-readiness-tutorial/
- [R25] Brown Dust 2 English Game Guideline, “Battle” and “Placement,” older community-maintained game guidelines. Describes attack/defend phases, action order and repositioning; use as a historical visible-system reference, not a current patch guarantee. https://browndust2.gitbook.io/guide_en/game-guideline/battle/battle and https://browndust2.gitbook.io/guide_en/game-guideline/battle/placement

27.11 Research closure and next action gate

What is closed at the written-definition level: the MVP team size and slot map; queue freeze/turn order; Chakra income/spending and trigger attribution; target-selection precedence; Taunt/interception and decoy; Attack Down and direct-damage order; healing and Shield; status application timing; Wound snapshot/limits; Stun probability tiers; Silence

vs Interrupt; Speed's next-round-only effect; follow-up/reaction ceilings; defeat/tie order; starter role access; archetype shells; synergy caps; counter categories; and the initial reproducible stat/coefficient packet. These are sufficiently defined to implement a reference engine without inventing rules.

What is not closed: the numerical values may be wrong for fun, pacing or competitive balance; status readability on actual mobile devices is untested; current client behavior in every benchmark game has not been hands-on recorded; the final licensed character roster does not exist; and there are no simulator outcome logs or player data. These are future empirical validation items, not evidence that research must stop and not authorization to start testing now. Until the project owner gives the explicit live-testing instruction, keep working on research/concept completeness and hypothetical scenario review.

FUTURE LIVE-TESTING PHASE — Once the project owner explicitly says live testing is starting, the proposed next work is: (1) implement a deterministic reference engine using this state machine; (2) add automated tests for the scenarios in 27.8; (3) run seeded archetype matchups and substitution tests; (4) retune only from logged results; (5) do phone-size readability tests for target arrows, status timing, Chakra and recap. Until that explicit trigger, do not execute these steps. Current work remains desk research, conceptual specification, source review, and hypothetical reasoning; concept-level research on other game systems can proceed, with combat dependencies clearly marked.

27.12 Version 0.9 change register

ID	Change	Why it matters	Status
C-022	Added normalized baseline stats and first skill-coefficient budgets.	Turns role descriptions into comparable and reproducible simulation inputs.	Provisional inputs; must be retuned with data.
C-023	Defined deterministic action resolution, Attack Down order, post-damage status timing, Wound snapshot/cap, Shield/Chakra condition, Taunt expiry, and follow-up boundaries.	Closes rule interpretation gaps that could let two abilities implement the same system differently.	Written spec; implement and regression-test.
C-024	Corrected benchmark source attributions and added the official Summoners War Challenge Battle source plus CookieRun/MLA/Epic7/Brown Dust context.	Keeps observed game mechanics separate from design inference and removes incorrect citations.	Desk research only; patch snapshots may change.
C-025	Updated coverage ledger and exit gate to identify simulator/regression as the next work rather than another feature group.	Prevents treating a well-written document as empirical balance.	Project sequencing decision; test results pending.

28. Research/concept phase boundary and operating rules (Version 0.9.1)

This section records the project owner's explicit workflow decision and overrides any earlier wording that could be read as authorizing immediate implementation or testing. The project is currently producing a complete, integrated research paper and game concept. Reality changes and balance revisions are expected later, when a build and real testing produce evidence.

28.1 Current phase: research and hypothetical design only

The active task is to research, define, compare, connect and document the game's ideas as thoroughly as possible before implementation. We may reason through hypothetical battles, walk through example turns, compare alternative rules, check arithmetic, look for contradictory rules, trace potential exploits on paper, and examine likely player-facing consequences. These methods can improve clarity and expose logical gaps, but they do not prove that a rule is fun, balanced, readable in practice, technically reliable or commercially successful.

No live testing is to be performed unless and until the project owner explicitly says that live testing is starting. Do not treat a future acceptance criterion, simulation count, tester target or exit gate as a task that should be executed in the present phase.

28.2 What belongs in the current research scope

- Genre identity: what makes an idle RPG an idle RPG; how idle RPGs differ from auto-battlers, turn-based hero collectors, RPG gachas and hybrid games; which genre conventions are essential, optional or design choices.

- Core loop and player cadence: idle/offline progression, claim/login loops, active-session decisions, combat and reward cadence, progression gates, return motivation, and the balance between convenience and meaningful agency.
- Team gameplay theory: roles, composition, win conditions, synergies, counters, targeting, formation, action economy, Chakra/resource rules, statuses, defensive layers, accessibility and hypothetical edge cases.
- Collection and growth: how characters are acquired, developed, substituted, collected and differentiated; how duplicates, levels, rarity and other growth layers could affect choice and fairness.
- Economy and monetization: currency sources/sinks, summon accessibility, pity/guarantees, fragment routes, pacing, F2P viability, pay-to-progress limits, trust and potential exploit or pressure points.
- Conceptual game modes and content roles: what each mode is supposed to teach or reward, what team-building decision it tests, how modes connect to the core loop, and which assumptions depend on unvalidated combat rules.
- Mobile UX, onboarding and readability: what a player needs to understand, what can be automated, what needs clear feedback, what the battle recap should explain, and how to keep the experience legible on a phone.
- Benchmark research: review a suitably broad set of idle RPGs and adjacent hero-collector/turn-based games, distinguish direct genre peers from adjacent references, record source strength and publication date, and extract transferable patterns rather than copying systems blindly.
- System integration and completeness: trace how battle, progression, roster access, economy, modes, retention and monetization affect one another; identify conflicts, missing definitions, trade-offs and decisions that still need owner approval or evidence.

28.3 What is explicitly out of scope until live testing is authorized

- Building or implementing the playable prototype or deterministic combat simulator.
- Running seeded matchups, automated balance batches, removal/substitution tests, telemetry experiments or actual player tests.
- Recruiting participants, measuring live comprehension, reporting live win rates, estimating real retention from this concept, or claiming any numerical balance/readability gate has passed.
- Changing proposed values on the basis of observations that have not occurred, or presenting illustrative calculations as evidence of actual game performance.

It is acceptable to prepare future test plans and define the metrics that should be measured later. The plan is documentation; executing it is deferred. Existing validation chapters and checklists remain preserved for that later phase.

28.4 Required labels: facts, decisions, hypotheses and future tests

- [FACT / SOURCE FINDING] A claim supported by a cited source. Record source date, scope and confidence; live-game details may be patch-sensitive.
- [PROJECT DECISION] A direction explicitly chosen by the project owner. It is not the same as a proven result.
- [PROPOSAL] A recommended design rule or value that has not been accepted as final.
- [HYPOTHESIS] A reasonable theory about likely consequences, player behavior, balance or fun that has not been empirically established.
- [OPEN] A decision, dependency or piece of evidence that remains unresolved.
- [FUTURE VALIDATION REQUIREMENT] A criterion or test to be executed only after the explicit live-testing trigger. It must not be reported as completed merely because it is fully described on paper.

28.5 How hypothetical scenarios should be evaluated

For each important scenario, document the starting state and assumptions, the sequence of rules/actions, the expected result under the written rules, the rule or design principle the scenario examines, possible alternative outcomes, and any unresolved dependency. Use paper calculations where useful. If two written rules conflict, reconcile them in the specification and decision log; if the outcome depends on unknown player behavior or actual balance, keep it marked as a hypothesis for future validation.

A hypothetical scenario can establish that a rule is defined consistently in that example or reveal a logical contradiction. It cannot establish an archetype's actual win rate, the optimal strategy in a live meta, the emotional feel of combat, production performance, or whether players understand the interface.

28.6 Definition of success for the current phase

14. The game concept explains clearly what kind of idle RPG this is and what role idle progress plays alongside active decisions.
15. The major systems and their purpose are documented, including how they connect to the core loop and to each other.
16. Key rules are explicit where they can be specified through research and theory; assumptions, alternatives, dependencies and unresolved questions are visible rather than hidden.
17. External facts are cited and distinguished from project choices, recommendations and hypotheses.
18. The master document retains prior research and historical proposals, while clearly labelling superseded rules instead of silently deleting them.
19. No section describes a proposed value or hypothetical outcome as empirically validated. Future test requirements are ready for later use but remain unexecuted.

28.7 Explicit trigger for the testing phase

The phase changes only when the project owner explicitly instructs that live testing is starting. At that point, review and sequence the already-documented future test plan, confirm the first available implementation and test environment, and record what can actually be measured. Until that instruction appears, maintain the current research/concept-only workflow.

28.8 Immediate research sequence

20. Establish a well-sourced genre map and an operational definition of an idle RPG, including idle rewards, offline gains, active play and auto-battle distinctions.
21. Complete the conceptual core loop from first session through the long-term routine, with the player's decisions and reward timing made explicit.
22. Continue deepening the written combat/team-building theory and review hypothetical battles for rule collisions, counterplay and exploit paths.
23. Research character collection, progression layers, currencies, acquisition and F2P/pay-to-progress constraints as connected systems, not isolated feature lists.
24. Research modes, onboarding, mobile UX, retention and live-operations concepts at the theory level; document which combat or economy assumptions each mode depends on.
25. Run a research-paper integration audit: compare terminology and values across chapters, preserve contradictory historical proposals as history, and maintain one current rule or clearly labelled open decision for each live design question.

29. Holistic research roadmap and system-integration method (Version 0.10.0)

Purpose. The project will not jump randomly between mechanics or try to perfect one subsystem in isolation. First map the complete game concept; then research one connected domain at a time in depth; then revisit earlier domains when new findings reveal dependencies or contradictions. The master remains the single source of truth. This section determines the working process, not an irreversible feature roadmap.

29.1 The two-layer method

Layer A — map the whole game before deep-diving. Every major system receives a purpose, a known dependency list, a maturity label and an owner-facing decision/open-question register. The map is a completeness check; it does not mean every system must be fully specified at once.

Layer B — study one domain holistically. For the selected domain, review credible sources, compare direct genre peers and adjacent games, identify recurring patterns and trade-offs, draft project-specific options, reason through hypothetical cases, and document how the domain exchanges resources, information, power or motivation with other systems. Then record what is established, proposed, uncertain or deferred before moving to the next deep dive.

We will not treat “finished on paper” as “proven in reality.” Paper completeness means terminology is defined, rules do not contradict each other in the reviewed cases, options and trade-offs are visible, and remaining unknowns have been explicitly recorded. Fun, fairness in practice, retention, production performance and balance remain unproven until the project owner later authorizes real testing.

29.2 Whole-game system map

This is the initial map of the game we intend to understand and design. Maturity labels describe the current document, not quality or completion. “Research next” means the next focused desk-research module; it never authorizes implementation or testing.

System domain	What we need to define	Dependencies / current status
Product identity, audience and promise	Define the target player, platform, desired session shape, Naruto fantasy, core value and explicit exclusions.	Foundation for every other domain; revisited when trade-offs conflict.
Idle identity, core loop and player cadence	Define what progresses during absence, what is automated, what active play contributes, when rewards arrive and what makes a return worthwhile.	First deep-dive module; informs combat pacing, rewards, UX, retention and economy.
Combat engine and team-building	Define squads, roles, formations, targeting, action economy, Chakra, statuses, synergies, counters and battle feedback.	Detailed theoretical foundation already exists; remains provisional and receives cross-system review.
Roster, collection and acquisition	Define character identities/roles, recruitment, rarity, duplicates, substitutes, selectors, fragments and roster breadth.	Depends on role coverage and economy; determines whether team choices are accessible.
Progression and build customization	Define leveling, skill growth, equipment, awakening, long-term power curves, reset/ascension if any, and caps/soft caps.	Depends on battle requirements and economy; future balance impact remains unvalidated.
Economy and rewards	Map all currency/resource sources, sinks, rates, scarcity, idle income, upgrade costs, event awards and acquisition pathways.	Must connect idle rewards, progression, collection and monetization; numbers remain proposals.
Modes, encounters and content	Define campaign, bosses, towers/challenges, competitive/cooperative concepts, difficulty gates	Concept research may proceed now; final difficulty

System domain	What we need to define	Dependencies / current status
ladder	and the unique question each mode asks.	tuning depends on unvalidated combat.
Narrative, world and Naruto identity	Define the desired fantasy, story delivery, character presentation, canon boundaries and licensed-IP requirements.	Cross-cutting; commercial use of protected IP requires the appropriate rights.
Onboarding, mobile UX and accessibility	Define the first-session teaching order, information hierarchy, touch targets, battle readability, feedback, automation and low-friction navigation.	Cross-cutting from day one; detailed flow depends on the core loop and system complexity.
F2P fairness and monetization	Define what free players can reliably build, what purchases may offer, pricing/value principles, transparency and competitive-power limits.	Fairness constitution already established as a constraint; economy and commercial sustainability remain hypotheses.
Social, guild and competition	Define solo-first expectations, asynchronous competition, cooperation, guild value, PvP rules and fair reward structures.	Depends on combat, access rules and operating costs; concepts only.
Retention, events and live operations	Define return motivations, event rhythm, catch-up policies, content cadence and sustainable long-term goals.	Must respect the idle promise rather than create compulsory, high-frequency check-ins.
Technical/service integrity and player trust	At concept level, identify account recovery, save/sync, offline-time integrity, anti-cheat boundaries, privacy and reliability requirements.	Document requirements now; implementation architecture and security testing are later work.
Analytics and future validation governance	Define which questions future metrics should answer, how changes would be logged and how decisions are revisited.	Prepare the measurement plan only; do not collect or fabricate data during this phase.

29.3 Recommended research sequence

26. **Pass 0 — Whole-game map (this revision).** Inventory major systems, identify dependencies and establish evidence/decision labels so no important area disappears.
27. **Pass 1 — Idle RPG identity and core loop.** Define the genre promise, active/inactive/return cycle, the player's main decisions, the reward cadence and the boundaries between idle and adjacent genres.
28. **Pass 2 — Combat/team-building fit inside the loop.** Continue refining the existing combat paper against the idle promise: how the player sets up a team, what can progress while absent, and what still rewards active strategic choice.
29. **Pass 3 — Collection and roster access.** Specify how a player gains, replaces and develops shinobi; test hypotheticals for role access, duplicates, limited units and a usable starter account.
30. **Pass 4 — Progression and economy as one connected model.** Map character growth, upgrades, idle reward rates, resources, costs, bottlenecks and acquisition paths together instead of writing unrelated currency lists.
31. **Pass 5 — Content/modes and their purpose.** Define what each mode contributes to the core loop, what team-building decision it asks, what it consumes and what it rewards.
32. **Pass 6 — Onboarding, mobile UX and accessibility.** Turn the concept into a readable learning sequence and interface requirements without relying on testing claims.
33. **Pass 7 — Monetization, social systems, narrative/IP and live operations.** Complete these as connected modules under the already-established F2P, time-respect and licensing constraints.
34. **Pass 8 — Integration and contradiction audit.** Read across the master: harmonize terminology, trace every resource and dependency, label conflicts and preserve old alternatives in the decision log.

The current sequence and its conditions for moving on are in Section 41.1. Keep one primary research question active. Reopen an earlier decision when a new dependency changes its answer and record which later sections inherit the change.

29.4 Standard research packet required for every domain

- Question and scope: define what the system is for and what is outside its scope.
- Terminology: define the terms used by sources and identify where different games use the same label differently.
- Evidence and benchmark review: record direct sources, publication/update dates, source type, exact claim supported, and limitations. Separate direct idle-RPG peers from adjacent games.
- Pattern and trade-off analysis: describe why a mechanic may help, what cost or risk it introduces, and which player behavior it encourages.
- Project options: outline plausible alternatives and their pros/cons before recommending a direction.
- Current project proposal: state a recommended baseline only where there is sufficient reasoning; tag it as a proposal until accepted.
- Hypothetical scenario review: walk through representative ordinary, edge and abuse cases on paper. Calculations may expose contradictions but do not prove player outcomes.
- System links: identify incoming dependencies and outgoing effects on battle, rewards, resources, collection, UX and F2P fairness.
- Decision/open-question register: record chosen direction, alternatives not chosen, reason, confidence, and what evidence would change the decision.
- Paper-completion check: confirm the concept is defined, internally consistent for reviewed cases, source-attributed and honest about unresolved uncertainty.

29.5 Cross-cutting rules that apply to every deep dive

- Team-building remains a central pillar throughout the research; it is not abandoned when the focus shifts to offline rewards, progression or the economy.
- Idle play must respect the player’s absence. Avoid designing the basic value proposition around repeated short-timer check-ins or punishments for ordinary breaks.
- F2P fairness is a constraint across every system, not a chapter to fix at the end. The availability of useful team functions, counters and coherent roster variants must be traced into collection and economy proposals.
- Every numerical value is a project proposal unless a cited source specifically supports it as a benchmark fact. A number copied from another game is not automatically appropriate for this design.
- No actual balance, usability, retention or revenue outcome is to be claimed until real evidence exists. Future test criteria stay documented but unexecuted.
- Naruto-branded commercial use depends on obtaining appropriate rights. Until that exists, examples of named characters are design placeholders and licensing is a material project dependency.

30. Idle RPG identity and core loop: first research foundation (Version 0.10.0)

Scope note. This is the first desk-research foundation, not the final design for every idle reward, mode or economy rate. It uses academic research, a game-design industry article and official storefront/publisher descriptions. Store listings are marketing snapshots, not full technical documentation. Their role is to confirm how a product is presented and which broad features it claims, not to prove hidden formulas or actual balance.

30.1 What does “idle RPG” mean?

There is no single universally enforced taxonomy that cleanly separates every idle game, incremental game, auto-battler, hero collector and RPG hybrid. Academic discussion notes that genre boundaries are blended and change as games combine different kinds of interaction [IR2]. A useful definition should therefore describe the activity the game is designed around, not rely only on the store category or on whether battles can be automated.

For this project, the proposed working definition is: an idle RPG is a role-playing game in which meaningful character/team progression and reward accumulation are designed to continue through automation or elapsed time during periods of little or no player input, while active sessions let the player steer, improve and extend that progress. The RPG identity comes from persistent characters or a party, differentiated abilities/roles, development over time and progression through encounters or an adventure. The idle identity comes from designing for absence as a normal part of the loop, then making the return understandable and worthwhile. This definition is a project-specific operating rule, not a claim that every game sold as an idle RPG uses all of these features.

The strongest source pattern is the repeating cycle of active participation, inactive progression and rewarding return. Villareale and colleagues identified this cycle when analyzing 11 popular idle games; they also note that the amount of active and inactive time differs between games [IR1]. This is stronger support for a recurring design pattern than for one fixed number of daily sessions or a single mandatory offline-reward cap.

30.2 Core identity: essential distinction versus optional conventions

Feature or property	Current interpretation	Implication for our game
Meaningful progress during absence	Core to this project’s idle promise	The account should gain meaningful progress or resources without requiring continuous input. Exact offline cap and reward calculation remain open.
Automation of repetitive combat/work	Strongly expected for this project	Routine battles and reward accumulation should not require constant tapping. Automation alone is not enough to establish an idle identity.
Persistent RPG character/team growth	Core to the RPG side of this project	Characters have distinguishable roles/abilities and can be developed or assembled into teams. A pure resource clicker would not meet the project definition.
Active choices that steer progress	Strong design requirement	The player decides team/formation, improvements, content attempts or strategic direction. The player should not need constant manual execution simply to keep basic progress alive.
Visible, understandable return reward	Core loop requirement	The game explains what accumulated and why, then gives a meaningful next decision. A claim button with no useful follow-on decision would be a weak return loop.
Gacha/summoning	Common in hero-collector variants, not a genre requirement	Potential acquisition route only; our F2P fairness rules govern its eventual design.
PvP, guilds, co-op and competitive seasons	Optional social/content conventions	They can enrich a game but do not make it an idle RPG. Each must justify its costs and fairness risks.

Feature or property	Current interpretation	Implication for our game
Prestige/rebirth/reset and exponential number scaling	Common in some incremental designs, not required for this project's identity	Do not include simply because the game is idle. Evaluate whether a reset fits the ninja fantasy and long-term roster model.
Stamina, VIP, daily login streaks, ads and many currencies	Commercial/retention conventions, not defining mechanics	None should be mistaken for the source of the idle identity. Their costs, player pressure and F2P consequences must be evaluated separately.

30.3 Adjacent labels and how to tell them apart

Label	Primary distinguishing idea	What it means for our research
Idle RPG	Designed-for-absence progression plus persistent RPG growth	The closest primary identity for this project.
Auto-battler / auto-combat RPG	Combat resolves automatically, but absence may not produce meaningful account progress	Auto-battle is a combat mode/feature; by itself it does not prove the game is an idle RPG.
Turn-based hero collector	Collection, distinct character kits and squad decisions are central; progression may be primarily active	Important adjacent benchmark for our battle and collection systems, whether or not the benchmark is idle.
Gacha RPG	Recruitment or character acquisition uses randomized draws or banner mechanisms	Gacha is an acquisition/monetization design, not a genre definition; an idle RPG can exist without it.
Incremental/clicker game	A repeated resource-growth loop and upgrades are the main experience; tapping may be prominent early	Useful for economy and scaling research, but does not automatically supply team RPG depth.
Hybrid idle RPG	Idle income and automated progression coexist with substantial active exploration, tactical encounters or other modes	A valid direction: idle systems need not make every activity passive.

This distinction is practical rather than a rigid industry taxonomy. For each benchmark, record exactly which activity is idle, whether offline progression is meaningful, which decisions require being present, and whether "idle" describes the whole game or only one system.

30.4 What benchmark research currently supports

- [SOURCE FINDING] The 2019 Foundations of Digital Games paper reports a repeated cycle of active participation, inactive progress and return rewards from its analysis of 11 games. It provides a useful conceptual foundation, but its sample spans several idle subgenres and is not a controlled benchmark of modern mobile hero-collector RPGs [IR1].
- [SOURCE FINDING] A 2025 UC Santa Cruz master's thesis reviews genre-boundary ambiguity and studies player engagement through interviews and a short diary study. Its findings are useful as exploratory evidence that idle engagement deserves design study; they do not establish how our target audience will react or guarantee that a proposed loop will be engaging [IR2].
- [INDUSTRY DESIGN REFERENCE] Machinations describes a simple resource loop and a more complex meta loop. In its RPG example, automated combat can produce a common resource while scarce secondary resources and character synergies influence longer-term growth [IR3]. This explains how a simple idle core can support deeper team-building, but its discussion of purchase incentives is not a requirement we should copy. Scarcity should be evaluated under our F2P fairness constitution.
- [OFFICIAL STOREFRONT CLAIM] MOONTON markets Mobile Legends: Adventure around heroes automatically battling and collecting resources while the player idles, alongside gear/evolution, squad deployment, team strategy, and hero level-transfer/sharing features [IR4]. This is a close illustration of an idle loop connected to roster management rather than a standalone clicker.
- [OFFICIAL STOREFRONT CLAIM] DHGames' Idle Heroes listing describes heroes training while the player is away, evolving character abilities, faction-specific hero identities, gear, summons, and a range of PvE/PvP/guild content

[IR5]. This illustrates how idle progression can sit inside a broad hero-collection service, but its paid products and progression requirements are not endorsements for our economy.

- [PUBLISHER DESCRIPTION] Lilith Games' AFK Journey announcement describes idle rewards alongside exploration, quests, and tactical combat features [IR6]. This is an example of an idle-branded hybrid with active content; it does not mean all of those mechanics should be copied into our game.

30.5 Proposed three-loop architecture for the Naruto idle RPG

Loop layer	Working concept	Questions this layer must answer
Core idle loop — when absent	A prepared squad and the account's progression state determine an ongoing reward stream. After elapsed time, the player returns to a transparent summary of earned resources and progress.	Define what progresses, which state controls the reward rate, how time is calculated, whether activity continues through a cap, and how much the player must do to claim rewards.
Active session loop — when present	The player reviews the result, chooses a meaningful next step, improves or rearranges the team, attempts content, receives battle feedback/rewards, and leaves the account set up for future progress.	Make active play valuable through decisions and content access, not through compulsory repetitive tapping or frequent timer collection.
Long-term meta loop — across days/weeks	The player develops a roster, discovers alternative team archetypes, unlocks progression layers, pursues goals and content, and returns to improve a previously blocked strategy.	Link collection, progression, modes, economy, story and social systems to the same core promise; avoid separate currencies and daily chores that exist only to inflate activity.

The sketch below is a conceptual flow, not a complete rules engine:

Prepare squad / set progression state → automated battles or resource accumulation → elapsed time → return summary → spend rewards and review bottlenecks → improve team or attempt new content → leave the account ready for the next idle period.

The team-building connection is crucial: a player should have reasons to think about squad composition and progression before leaving the game, and the return should make those choices legible. If all teams produce the same offline result and composition only matters during a handful of active battles, team-building risks feeling disconnected from the idle promise. That is a design hypothesis to evaluate on paper and later through authorized testing, not a proven outcome.

30.6 Initial design principles for the idle promise

- Meaningful absence: closing the app should not stop all baseline account progress. The exact rate and cap are open; they must be generous enough to make idle progression real rather than a token claim.
- No forced rapid check-in rhythm: do not make the basic economy depend on returning every few hours. Any cap, streak, daily task or time-limited claim needs a clear reason and a player-pressure review.
- Active play adds agency, not mandatory attendance: active sessions should permit strategic choices, new content attempts and goal completion. Continuous manual input should not be required simply to keep the main progression alive.
- Transparent return: show elapsed time, the basis for rewards, the main resources gained, any cap reached, and the most useful next actions. Do not bury meaningful rewards behind multiple claim screens.
- Combat remains strategically relevant: compositions should affect the ability to clear content, overcome a wall and unlock a better reward basis. Do not make team synergies decorative if they have no consequence for progression.
- F2P baseline: all players must be able to obtain meaningful idle income and create a coherent, functional team using ordinary progression routes. Premium spend may improve convenience or broaden choice, but must not be the only route to basic role coverage.
- Predictable, bounded rules: the written concept must state when rewards begin and end, what happens on failure or progression walls, and how duplicate claims/clock anomalies would be handled later. Exact server architecture remains out of scope for this theory pass.
- No genre cargo-culting: no automatic prestige reset, infinite scaling, VIP, battle pass, multi-currency web, or daily streak should be added merely because other idle games use them.

30.7 Offline reward design options — trade-offs, not final decisions

Option	Benefits	Risks / open questions	Current status
Option A — fixed reward rate from the best cleared stage	Simple to explain; makes stage progress and squad strength relevant; easy to estimate in a return summary.	May weaken connection to the squad actually deployed during the absence; a stage-based rate can make one threshold disproportionately valuable.	Candidate baseline for further theory; requires clear wording about what current team/stage state controls it.
Option B — simulate the active team's battles over elapsed time	Strong narrative connection between the prepared squad and idle outcome; can create a clear reason to optimize team composition.	More complex rules; potential to create unexpected battle-state differences and harder-to-explain reward variability.	Keep as an alternative. We do not need to implement or simulate it during research.
Option C — grant a capped offline resource bundle unrelated to combat details	Simple and stable; easiest to understand as a general idle allowance.	Risks making combat, stage advancement and team choices feel disconnected from the main idle income.	Least aligned if the team is meant to be a core identity pillar; not rejected permanently without more comparison.
Option D — hybrid formula tied to progression, with a simple deterministic time calculation	Can keep offline rewards understandable while letting campaign progress and account development increase the rate.	Requires clear definitions for the reward snapshot, progression state changes and the point at which rate updates.	Promising conceptual direction to study further; not a locked formula.

At this stage, do not lock an exact offline cap (for example, 8, 12, 24 or 48 hours), an income rate, or a claim schedule. Those values depend on how often the game expects meaningful active decisions, how progression resources are priced, how long a typical absence should remain useful, and whether offline rewards are tied to stage progression. We should compare those choices as economic and player-experience hypotheses, not choose a number simply because a benchmark uses it.

30.8 Hypothetical first-session-to-return journey

Journey point	Hypothetical intended experience
First encounter	The player learns the premise and sees an automatically resolved battle. Teach the meaning of team slots and the most important result feedback without explaining every future system at once.
First team decision	The player receives enough accessible characters to assemble a coherent starter group. The game shows that role coverage and a simple synergy can matter, without requiring duplicates or a limited summon.
First progression wall	A stage or encounter becomes difficult enough that the player has a reason to inspect the lineup, improve a character or try a different composition. The game must explain the problem instead of presenting only a higher enemy power number.
Leaving the game	The interface explains which progress/rewards will continue while the player is away. The player should not need to keep the app open or leave a battle running just to receive the basic idle benefit.
Return and collection	The game presents accumulated rewards with clear sources and any applicable limits. The player can tell whether the prior squad/progression decisions helped and what resource or tactical bottleneck remains.
Next meaningful choice	The player decides whether to upgrade, adjust the squad, attempt a challenge, recruit from a free path or pursue a longer goal. Not every return must offer a new feature or a mandatory checklist.

This is a paper walkthrough. It does not establish onboarding comprehension, ideal tutorial length, actual return behavior or that rewards feel satisfying.

30.9 Open design register for the next idle-loop pass

Open question	Current hypothesis / next research action
What progresses while absent?	Candidate: baseline resources accrue and are tied to campaign/account progress. Whether stage attempts or other activities simulate is open.

Open question	Current hypothesis / next research action
What state sets the idle rate?	Best cleared stage, current squad, latest successful lineup, chosen idle assignment, account level, or a hybrid. Compare trade-offs before selecting.
Is there a cap on offline gains?	Open. Compare uncapped, fixed cap, milestone-based cap and diminishing return options; check both player freedom and economy stability.
Do rewards require manual collection?	Prefer low-friction collection with one clear summary; exact presentation and any claim flow remain open.
How should active sessions outperform idle-only play?	Candidate: meaningful content, new unlocks, tactical/team adjustments and milestone rewards—not an obligatory stream of manual repetitions.
How does idle progress interact with stamina/attempt limits?	Open; avoid hidden contradictions where a player is promised idle advancement but required resources make all meaningful progress stop immediately.
Can squads be swapped freely between idle and active content?	Open; tie this decision to build identity, preset teams, mode requirements and UX complexity.
What happens during long absence, clock changes or device changes?	Define the player-facing policy now; technical validation and anti-cheat implementation belong to the later implementation phase.
How does idle income avoid pay-to-win dependence?	Must preserve a viable free baseline and accessible team roles. Premium convenience must not become the only way to collect core progress efficiently.
What counts as a coherent core loop?	Open for final concept closure: the loop must connect battle/team choices to visible growth, an intelligible reason to return and a next decision without coercive check-in frequency.

30.10 Interim conclusion and immediate next task

Interim conclusion [RESEARCH SYNTHESIS]: for this project, “idle” should mean that the game is designed to make useful progression during player absence, not merely that battles have an auto button. “RPG” should mean that persistent characters, their growth, and their relationships in a team meaningfully shape the player’s progression. The target is therefore a team-collector idle RPG with auto-resolved baseline combat, meaningful formation/roster decisions and a clear return-reward cycle. The exact offline reward formula, cap, claim flow and the amount of active-play advantage are not settled.

Status update: Section 31 now examines how the idle-reward basis should connect to campaign progression and team-building. The next research pass should remain in this same module and deepen reward cadence, storage/claim rules, and catch-up behavior; do not lock numeric rates or caps merely because a benchmark uses them. The conceptual link is proposed, not empirically validated.

30.11 Sources for the idle-RPG foundation (accessed 28 October 2026 unless publication date is stated)

- [IR1] Villareale, Jennifer; Gray, Robert C.; Furqan, Anushay; Fox, Thomas; and Zhu, Jichen. “Enhancing Social Exergames through Idle Game Design.” Proceedings of the Fourteenth International Conference on the Foundations of Digital Games (FDG ’19), 2019, pp. 1–5. DOI: https://doi.org/10.1145/3337722.3341827. Academic study; 11-game sample across multiple idle subgenres. Used for the active participation → inactive progress → return reward loop, not for hero-collector-specific rates.
- [IR2] Hwang, Daeun. Player Engagement with Idle Games: A Mixed-Methods Exploration with Design Implications. Master’s thesis, University of California, Santa Cruz, June 2025. https://escholarship.org/content/qt0b07v51w/qt0b07v51w_noSplash_5f2385afebb49ef3e458511f62e2af5f.pdf. Academic thesis and exploratory mixed-methods work; use for terminology/engagement context, not as a guarantee of this project’s reception.
- [IR3] Machinations. “How to design idle games.” https://machinations.io/articles/idle-games-and-how-to-design-them. Industry design article; useful for core/meta-loop framing and active/idle balance, but commercially oriented recommendations require independent evaluation under the project’s F2P principles.

- [IR4] MOONTON. “Mobile Legends: Adventure.” Google Play store listing, Philippines. https://play.google.com/store/apps/details?gl=ph&hl=en-gb&id=com.moonton.mobilehero. Official storefront description of idle auto-battle, heroes, resource collection, team strategy and level transfer/sharing; promotional copy rather than technical documentation.
- [IR5] DHGames Limited. “Idle Heroes - Idle Games.” Apple App Store Philippines listing. https://apps.apple.com/ph/app/idle-heroes-idle-games/id1153461915. Official storefront description of training while away, hero identities/growth, equipment, summoning and content modes; exact current feature rules must be verified separately if needed.
- [IR6] Lilith Games. “Introducing AFK Journey – the latest Fantasy RPG published by Farlight Games!” 30 April 2024. https://www.lilith.com/news/7536aa233eec52b424ce5eaa90e81079/?locale=en_US. Publisher description of idle rewards alongside exploration, quests and tactical encounters; used to illustrate a hybrid rather than to infer detailed formulas.

Source-quality note: IR1 is a peer-reviewed conference paper but its game sample is broad, small and not specifically modern hero-collector idle RPGs. IR2 is a university thesis whose results should be read within its study design and limitations. IR3 is a practitioner article. IR4–IR6 are first-party descriptions useful for visible positioning/features but not neutral evaluations. The project’s design choices and hypotheses above are our own synthesis and remain distinct from these sources.

31. Idle rewards, campaign progression, and team-building: an integrated model (Version 0.10.2)

Scope and status. This chapter answers a specific system-integration question: how should a player's squad affect campaign progress, and how should campaign progress affect the rewards earned while the player is away? The source findings below describe other games. The proposed rules for our game are design recommendations, not proof of balance, retention, or player behavior. No simulator or live testing has been performed.

31.1 Research question and dependency chain

The target relationship is a loop, not three separate feature lists:

Build / select a squad → attempt campaign content → composition and upgrades influence whether the squad clears the stage → a new clear grants a one-time reward and may raise the account's idle-reward tier → idle income supplies resources for roster growth and tactical alternatives → the player returns, evaluates the bottleneck, and chooses the next team or upgrade.

This chain should preserve three distinct causal relationships. First, team-building influences the player's ability to earn a new campaign clear. Second, campaign progress can improve the future rate or breadth of idle rewards. Third, those rewards help the player develop the roster. The system should not secretly turn the equipped lineup into a direct hourly-income multiplier; that would make experimentation and roster rotation economically costly.

31.2 What the benchmark evidence supports

[SOURCE FINDING] AFK Journey separates AFK-stage progression from several other activities. A current Prydwen mode guide states that the reached AFK stage affects idle rewards, that stages also unlock many other modes/features, and that an auto-challenge option can continue to the next stage while the team wins. This supports a design in which stage progress is the basis for future idle income and in which a viable team converts its strength into further progression [IR7].

[SOURCE FINDING] AFK Journey team-building guides organize compositions around roles (for example, tank, damage dealers, support, and flexible specialists) and describe how synergy, control, shielding, and anti-healing can be selected for difficult stages. This illustrates that reaching higher stages need not be only a raw-stat check; role choices and matchup tools can matter. These guides also contain season- and roster-specific recommendations, which we should not copy as permanent rules [IR8].

[SOURCE FINDING] NIKKE's beginner guidance reports that campaign clears give immediate leveling resources and increase Outpost Defense level, while Outpost Defense then produces leveling resources over elapsed time. The guide also directs players toward team-building, gear, and other modes when progression stalls. This creates a clear loop between campaign progression, continuing resource generation, roster development, and another campaign attempt [IR9].

[SOURCE FINDING] NIKKE team-building documentation explains that teams must satisfy combat-structure requirements (such as Burst I/II/III access) and gain further value from complementary roles and synergies. Its current story-team guide also recommends accessible substitutes for players who do not own every listed unit, while warning that deeper deficit stages can narrow viable compositions. This is useful both as a positive pattern (team structure creates meaningful decisions) and as a fairness warning (if only a narrow set of rare units can advance the campaign, the idle income advantage compounds roster inequality) [IR10][IR11].

[LIMITATION] These are primarily specialist/community guides rather than complete first-party design specifications. They can document visible gameplay relationships, but they do not establish that the same links or rates are optimal for our project. Exact current game values and roster advice are patch-sensitive.

31.3 The core distinction: team influence versus direct team taxation

Recommended design principle [PROPOSAL]: the player's team should influence idle income indirectly by helping the account earn campaign checkpoints. Once a checkpoint has been cleared and recorded, the resulting income tier belongs to the account's progress, not to the currently equipped team.

This distinction matters. A player should be free to replace a damage dealer with a control unit, test a defensive composition, build a faction-specific squad, or use a favourite character without immediately losing ordinary passive income. A new lineup may be less effective at clearing the next stage, but it should not lower the reward rate already earned by past success.

The campaign therefore has two jobs: it challenges the roster and unlocks future progression, while the idle system provides a reliable resource floor between active attempts. Neither system should completely replace the other. If team choices have no effect on meaningful campaign progress, the team builder becomes decorative. If the idle economy is too tightly chained to perfect campaign clears, the game stops feeling idle-friendly and begins to punish players who lack a narrow set of characters.

31.4 Candidate reward-link models

The following models were compared conceptually. They are alternatives for the design paper, not tested implementations.

Model	How it connects the systems	Useful property	Main risk / assessment
A. Fixed income only	All accounts earn the same base rate, unrelated to campaign depth.	Strong baseline fairness and simple explanation.	Campaign progress can feel disconnected from long-term income. Better as a guaranteed floor than the only model.
B. Best-cleared campaign checkpoint sets future rate	A new permanent clear unlocks a modestly higher idle-reward tier; time then accumulates at that tier.	Team choices help unlock future income without taxing later team swaps.	Can create a compounding lead if rate increases are too large or some stages require rare units. Recommended with a baseline floor and bounded growth.
C. Offline simulation uses the currently equipped team	The current squad is treated as repeatedly fighting new or existing stages while absent; rewards depend on its results.	Can make a pre-departure squad choice feel directly meaningful.	Opaque failure, changing lineups, simulated combat complexity, and a risk that a weak experimental squad harms income. Defer rather than require for the initial concept.
D. Rarity/duplicates directly multiply idle income	Owning a premium unit or more copies increases recurring income independent of campaign clears.	Simple commercial lever.	Directly compounds collection and monetization advantages; conflicts with the project's F2P and team-access principles. Reject for core passive income.
E. Hybrid: universal floor plus checkpoint-based increments	Every account earns a useful baseline; campaign milestones gradually add to the future rate and separately grant first-clear rewards/unlocks.	Links team growth to progress without making absence worthless or requiring a single elite unit.	Requires careful pacing and catch-up design; exact rate curve remains open. Preferred conceptual direction.

31.5 Recommended working model for our Naruto idle RPG

Status: [PROPOSAL — not yet owner-approved as final, and not empirically validated]. Adopt Model E as the current design direction: a universal idle-income floor plus gradual, account-level increases tied to campaign checkpoints.

A. Universal baseline. Every account earns useful core progression resources while away, even when the player is stuck at a campaign stage. The floor should not require a specific premium character, rare duplicate, paid subscription, repeated hourly check-in, or successful unattended fight. "Useful" is a design goal whose exact resource quantities and rates remain open.

B. Campaign-based rate progression. A new highest-cleared checkpoint can increase the rate at which future idle rewards accrue. The increase should be gradual or placed at understandable milestone bands, not a sequence of steep jumps. The

reward basis should be displayed clearly (for example: “income tier based on Chapter X, Stage Y”). The exact growth curve, checkpoint spacing and cap are unresolved.

C. Keep first-clear rewards separate. Clearing a new stage can grant its one-time reward immediately. That reward is distinct from the ongoing hourly output. Avoid designing the best basic resources as a one-time prize only, or as a mandatory rare-unit drop: the player still needs an ordinary source of upgrade materials after a campaign wall.

D. Use team composition to earn progression, not to tax ordinary absence. The selected team affects battle performance and therefore whether a new checkpoint is reached. Once a checkpoint has been cleared, idle income uses the account’s recorded progression tier, independent of the current formation. Changing teams must not reduce the previously earned rate.

E. Active play creates agency, not attendance pressure. While present, the player can inspect enemy details, adjust formation, try a counter archetype, upgrade units, and use an in-session auto-challenge option. Offline rewards continue at the recorded tier. Automatic offline advancement through never-cleared stages is not required for the initial concept; it remains an alternative to examine later because it introduces hidden-battle and explainability questions.

F. Preserve counter and role access. A player should have accessible ways to assemble the basic functions needed to make campaign progress: damage, survival, support/sustain, and relevant counter tools. A stage should not effectively require one limited shinobi when an accessible role-equivalent or alternate team strategy could achieve the same tactical purpose. This connects directly to the existing roster-access and F2P fairness rules in Sections 24–27.

G. Keep ordinary idle output separate from rarity and duplicates. A rare unit may offer a different tactical solution, and duplicates may improve character strength modestly under the current fairness proposal, but rarity or duplicates should not directly multiply account-wide idle income. Otherwise roster advantage would generate resources that widen the same roster advantage again.

H. Explain what happens at a wall. If the next campaign stage cannot be cleared, the already-earned idle tier stays active. The player can accrue resources, develop characters, review the stage’s tactical demands, or earn supplementary rewards from accessible modes. A stalled stage must pause further checkpoint-based rate improvements, not erase the base income or invalidate past clears.

31.6 Accrual accounting and claim rules: decisions we must define

Even a good conceptual reward model can produce confusing incentives if we do not state when rates apply. Recommended rule [PROPOSAL]: rewards already earned for elapsed time must not be repriced upward merely because a player clears a higher stage before tapping “claim.” The rate active during each accrual interval determines the value of that interval. Newly cleared checkpoints improve future accrual only. This removes a stockpiling/claim-timing exploit and makes the reward summary explainable.

The simplest conceptual calculation is: accumulated idle rewards equal the sum of (eligible elapsed time in each income tier × that tier’s published rate), plus any explicitly earned non-paid modifiers. The exact implementation is future technical work; no server code or simulator is being built now.

The amount of time that can be stored, whether over-cap time is discarded or partially retained, how temporary events modify rates, and whether “instant rewards” use the current or historical income tier remain open. Do not import a benchmark’s 12-hour or 24-hour cap merely because it is familiar. A cap should be chosen in relation to the intended return cadence and should be reviewed for pressure to check in repeatedly.

31.7 Hypothetical scenario review

These scenarios test whether the proposed rules are internally understandable. They do not demonstrate real balance or player satisfaction.

Scenario 1 — The player changes their team after a successful campaign push. The player clears a checkpoint with a burst team, then switches to a control-heavy team to build familiarity. Expected result: the recorded idle-reward tier remains unchanged. The new squad may be worse or better at the next stage, but cannot erase prior progression income. Principle: account progress, not currently equipped team, determines the base rate.

Scenario 2 — The player hits a hard campaign wall before going offline. They have not cleared the next checkpoint. Expected result: baseline rewards continue at the last recorded tier while they are away. On return, they have some

resources to upgrade or experiment with a counter composition. Principle: a progression wall slows the next rate increase but does not turn idle progression off.

Scenario 3 — A player lacks the featured limited damage dealer. Their available roster can cover basic roles but has a different win condition, such as control plus sustained damage. Expected result: the player may need a different tactical route or more development, but no rule directly requires the featured character to unlock an idle multiplier. Principle: limited units add choices rather than owning access to basic campaign income.

Scenario 4 — A high-progression account earns much more than a newer account. Expected result: higher campaign checkpoints can pay a higher rate, but growth is gradual and has defined milestones; the baseline remains useful and the roster is not directly multiplying idle output. Principle: progression should feel earned without making the gap compound without bound. The actual curve remains an open question.

Scenario 5 — The player leaves for longer than the storage cap. Expected result: the system follows its clearly documented cap/over-cap rule, but the player is not required to make several short daily visits to preserve most of the baseline value. Principle: the cap is a cadence/experience decision, not a disguised attendance tax. The exact rule remains open.

Scenario 6 — The player clears a higher stage but delays claiming the accumulated bank. Expected result: only the future elapsed interval uses the new rate; previously accrued time keeps its original value. Principle: claiming order must not create an exploitable retroactive rate increase.

Scenario 7 — The player experiments with a low-power but interesting team and loses. Expected result: the loss consumes no idle income, does not reset the best checkpoint and does not reduce the account's current reward rate. The player can freely return to a stronger team. Principle: optional experimentation should be low-friction and not financially punish curiosity.

Scenario 8 — A campaign chapter unlocks a new mode with targeted materials. Expected result: the new mode adds a distinct way to pursue those materials, while baseline idle rewards still cover ordinary development. Principle: mode unlocks broaden choices rather than becoming mandatory payment/attendance gates for the basic roster.

31.8 Main design risks and safeguards

Risk: campaign-income snowball. Clearing more stages increases income, which makes the roster stronger, which makes the next clear easier. Safeguard [PROPOSAL]: preserve a baseline floor; keep checkpoint increments bounded; ensure rewards do not rise exponentially at every stage; support alternate accessible compositions and non-campaign ways to make incremental progress. Whether a specific curve succeeds cannot be known from theory alone.

Risk: rare-character bottleneck. If a unit is uniquely able to clear a stage, its owner receives both the clear reward and a better future income stream. Safeguard [PROPOSAL]: every essential role and common counter category has accessible alternatives; do not attach idle-output multipliers to rarity/dupes; design stage mechanics to reward different viable tactical answers, not just a single damage check.

Risk: idle income becomes irrelevant. If stage progress does not change income or unlock anything meaningful, offline rewards and campaign feel disconnected. Safeguard [PROPOSAL]: give campaign checkpoints a visible but modest effect on future income, separate first-clear rewards, and mode unlocks.

Risk: team choice becomes economically constrained. If income directly depends on the currently assigned squad, players will avoid experimenting or may be afraid to change teams before going away. Safeguard [PROPOSAL]: use the best-cleared account checkpoint for base idle income; keep squad configuration relevant to battle attempts rather than the passive rate itself.

Risk: claim timer creates compulsory check-ins. A short cap can make players feel that absence wastes value. Safeguard [PROPOSAL]: defer exact cap decisions; compare the experience across realistic absence windows and prefer a generous, transparent rule over a narrow timer designed solely to increase visits.

Risk: the reward model hides causality. If the player cannot tell why income changed, the relationship between campaign progress and idle rewards feels arbitrary. Safeguard [PROPOSAL]: the return screen names elapsed time, income tier, resources accrued, cap status if any, and which campaign checkpoint last changed the rate.

31.9 Provisional decision register

Question	Current working direction	Status / reason
What sets the recurring income tier?	Highest recorded campaign checkpoint, with gradual/bounded increments.	[PROPOSAL] Best link found so far between campaign and idle income; exact curve is open.
Does the current equipped squad change income?	No direct effect on the base rate after a checkpoint is cleared.	[PROPOSAL] Protects roster experimentation and avoids an income penalty for changing teams.
What if the player is stuck?	Baseline idle income continues at the current tier.	[PROPOSAL] Keeps idle progression meaningful at a campaign wall.
Can premium rarity or duplicates multiply idle income?	No direct roster-wide income multiplier from rarity or duplicates.	[PROPOSAL] Aligns with F2P/pay-to-progress constraints.
Does a new clear reprice banked time?	No; new tiers affect future accrual only.	[PROPOSAL] Avoids claim-timing exploit and makes the math transparent.
How long can rewards be stored?	Not decided in this round.	[OPEN] Requires separate cadence, cap, and return-behavior research.
Does offline time auto-push uncleared stages?	Not required for the initial concept; in-session auto-challenge may be considered.	[OPEN] Needs a distinct comparison of complexity, transparency and idle identity.
Exact resources and income rates?	Not decided.	[OPEN] Must be modeled with progression costs later; do not copy a competitor's number.

31.10 Interim conclusion and next research step

Interim conclusion [RESEARCH SYNTHESIS]: the best current direction is to link team-building to campaign success and link campaign checkpoints to future idle income, while making the income itself an account-level benefit that does not vanish when the team changes. Provide a universal idle floor, modest checkpoint-based increases, separate first-clear rewards, accessible team counters, and transparent accrual rules. This model captures the progression loop seen in current benchmark guides without copying their exact rates or creating a direct premium-roster income multiplier.

Not settled: the reward cap, storage/over-cap behavior, exact income curve, resources included in the idle bank, catch-up rules for new/returning players, stage-milestone spacing, and whether unattended auto-challenge should ever advance beyond cleared stages. These remain visible open questions.

Status: completed by Section 32. The storage-cap, claim-cadence and catch-up comparison is now documented. The model remains provisional and unvalidated; Section 32 records current benchmark caveats and hypothetical scenarios. The next research task is to define which resource types should accrue while idle and which should come only from active modes, one-time clears, events or other sources.

31.11 Sources for campaign / idle-reward / team-building integration (accessed 28 October 2026)

- [IR7] Prydwen Institute. "AFK Journey Game Modes." Current community guide accessed 28 October 2026. https://www.prydwen.gg/afk-journey/guides/game-modes. Used for the described relationship between reached AFK stage, idle reward output, mode unlocks, and in-session auto-challenge. The page also contains PTR/current-mode details that should not be treated as durable rules.
- [IR8] Prydwen Institute. "AFK Progression Teams." Current community guide accessed 28 October 2026. https://www.prydwen.gg/afk-journey/guides/afk-progression-teams. Used as a visible example of role coverage, synergies, control, sustain and flexible team construction for AFK progression. Current roster/team examples are patch- and season-sensitive.

- [IR9] Nikke.gg. “Nikke Beginner’s Guide.” Page dated 30 April 2024; current page/search index reviewed 28 October 2026. https://nikke.gg/beginners-guide/. Used for the broad claim that campaign stages give immediate leveling resources and raise Outpost Defense, which produces passive resources; also for the campaign → idle income → squad development progression loop. Community guide, not a full official specification.
- [IR10] Nikke.gg. “Team Building Guide.” Current guide accessed 28 October 2026. https://nikke.gg/team-building-guide/. Used for the visible team-structure requirement and synergy/role rationale. The page displays an original publication date of 12 November 2023; exact current implementation details should be rechecked when they matter.
- [IR11] Prydwen Institute. “Meta Teams (Story) — NIKKE.” Page last updated 4 July 2026; accessed 28 October 2026. https://www.prydwen.gg/nikke/guides/meta-teams/. Used to illustrate accessible substitutes and the fact that increasingly difficult stages may narrow viable team options; exact meta lineups are not adopted as project rules.

Source-quality note. IR7–IR8 and IR11 are specialist/community guides with useful visible-mechanic descriptions but are patch-sensitive and not neutral evaluations. IR9–IR10 are community guides; they support only the claims stated above. No benchmark source validates the proposed reward curve, universal floor, cap, accrual accounting or F2P thresholds for this project. Those remain proposals/open decisions.

32. Idle reward storage, claim cadence, and catch-up policy (Version 0.10.3)

Purpose and status. This chapter studies what happens between player sessions: how long idle rewards can accumulate, whether the rate changes after storage fills, how collection is presented, and how the game supports a player returning after a longer absence. The external examples below are source findings with specific confidence limits. The proposed model is a theory-stage recommendation only. No code, simulator, seeded run or live/player test was performed.

32.1 Four concepts that must not be conflated

- Accrual rate: the resources generated per unit of elapsed time at the account's current reward tier.
- Storage window / cap: how much elapsed time can be represented in the ordinary idle bank before rewards stop or change rate. It is not necessarily a cap on inventory or on resources already claimed.
- Claim cadence: the frequency at which the interface nudges or requires the player to collect. A game can keep accumulating rewards without demanding that the player open the app at a specific time; alternatively, a short hard cap can make frequent check-ins economically optimal.
- Catch-up support: a separate system for a player who has missed longer-term account progression or events. It is not the same as ordinary idle earnings, and it should not simply multiply the offline bank by every day absent.

Design implication: a generous storage window, a readable claim flow, and a fair returner package solve different problems. Combining them into a single "login reward" makes it difficult to understand what the player is earning and why.

32.2 What the benchmark evidence currently supports

[SOURCE FINDING] AFK Arena is described in a recent AFK Global search-index result as having a 12-hour base AFK reward window, with Fast Rewards granting two hours of AFK income per use. The same result describes a daily free use and additional uses paid in diamonds. The corresponding page open in this research pass resolved to an older, differently dated guide, so the exact current details are treated as a source-version discrepancy rather than fully verified current behavior [IR12].

[SOURCE FINDING WITH CONFLICTING SECONDARY DATA] Prydwen's NIKKE Outpost guide says Outpost Defense storage fills after 12 hours and links Outpost level to story-stage clears [IR13]. A separate third-party NIKKE calculator currently describes a 24-hour capacity with rewards continuing at 15% of normal rate after capacity [IR14]. These two secondary sources disagree on present storage behavior. The important research conclusion is not that one number is correct; it is that patch-sensitive cap values must not be copied from a single guide without in-client verification later.

[SOURCE FINDING] Prydwen's AFK Journey returnee guide, updated 10 January 2026, says players returning after an absence may receive season-dependent rewards such as resonance boosts or additional pulls [IR15]. This is an example of return support being a separate re-entry mechanism rather than only a larger offline bank. Exact availability and reward contents vary over time.

[SOURCE FINDING] An official NIKKE event page for December 2025–January 2026 defined an eligible "returning player" for a specific invitation-code campaign as one inactive for at least 14 consecutive days who had triggered the Welcome Back Mission [IR16]. This documents one event's eligibility rule, not a universal rule for all events or a permanent threshold. Third-party October 2026 patch notes also describe a Welcome Back redesign with a longer inactivity threshold; treat any such threshold as patch-dependent until confirmed in the live client [IR17].

[SYNTHESIS] The relevant transferable patterns are: caps can differ by game or change over time; fast-reward purchases can convert premium currency into time; VIP-linked longer caps or higher idle income can create compounding economic advantages; and returner rewards can be handled separately from normal idle accrual. None of these sources establishes the best cap, overflow rate, cadence or catch-up threshold for our project.

32.3 Storage-cap models and their trade-offs

Model	How it works	Player / economy consequence	Research assessment

Model	How it works	Player / economy consequence	Research assessment
A. Short hard cap (e.g., 8–12 h)	Full-rate accrual stops when the bank fills.	Creates clear urgency to claim; sleeping, school, work or a long day away can cause lost value and may encourage multiple daily check-ins.	Simple to explain, but likely conflicts with a low-pressure idle promise.
B. One-day hard cap (e.g., 24 h)	Full-rate accrual continues for about one normal day, then stops.	Reduces twice-daily pressure, but weekends, travel or missed days still discard all extra time.	A better baseline than a short cap, yet still attendance-sensitive.
C. Full-rate window plus slow overflow	Rewards accrue at full rate for an initial window, then accrue more slowly for a limited overflow window.	A missed collection loses less value without allowing unlimited full-rate resource hoarding.	Preferred provisional model for this concept; exact durations and overflow rate remain open.
D. Long full-rate bank (e.g., several days)	The same ordinary rate is retained over a much longer absence.	Very forgiving, but can produce large resource injections and widen the gap between progression tiers if the economy is not designed for it.	Worth comparing; should not be chosen without reviewing resource sinks and pacing.
E. Unlimited full-rate accumulation	No meaningful cap on offline time.	Maximum convenience but permits potentially very large stockpiles and can make progression pacing difficult to govern.	Not recommended as the first working model for a live-service hero collector.

32.4 Claim cadence: reward the return, do not tax the absence

The claim screen may be a satisfying return moment, but the player should not need to visit at narrow intervals to avoid a severe penalty. This matters especially for a mobile game whose audience includes students, workers, and players who cannot reliably log in at the same time every day.

- Use a single “Collect All” action for the ordinary idle bank, with a clear per-resource breakdown and a visible explanation of whether the bank is full or in overflow.
- Do not use a daily attendance streak as the basis for ordinary idle income. Missing a day should not reset the player’s reward tier or erase previously accumulated resources.
- Treat notification timing as player-controlled convenience, not a sequence of escalating “your rewards are being wasted” alerts. Never make push notifications the only way to preserve value.
- Keep ordinary reward collection independent of current squad selection. Experimenting with teams should not require remembering to equip a supposed “best farming” squad before leaving the app.
- If instant “Fast Rewards” are eventually offered, disclose the conversion (for example, a specified amount of elapsed-time income). Do not make premium spending increase the base offline cap or provide a large permanent idle-rate multiplier that compounds into competitive power.

Design interpretation: a claim button is acceptable; an attendance obligation disguised as a claim button is not. Whether a free daily instant-reward use belongs in the game is an economy decision to study later, not part of this chapter’s settled rules.

32.5 Catch-up after a longer absence is a separate system

Ordinary idle rewards compensate for time away within the storage policy. Catch-up support helps a returning account re-enter a game that may have added chapters, characters, currencies, modes and progression layers since the player left. A large offline bank alone cannot reliably solve that problem because it grants generic resources, not necessarily the specific knowledge, roster access or mode access needed to resume play.

Candidate returner support package — proposal, not final design:

- A short “what changed” briefing that highlights major feature unlocks and important system changes since the last session, without overwhelming the player with every patch note.
- A bounded package of common development materials or catch-up tokens that can be invested into the roster the player actually wants to use, not a compulsory featured character.
- A guided but optional re-entry route: campaign recommendation, a suggested counter team, and a visible explanation of the highest-value next upgrade. It should not force a single meta composition.
- A clear separation between resources already earned through normal idle accrual and additional returner support. Each component should have its own eligibility and expiry rules, explained before collection.
- Eligibility threshold, package size, cooldown and frequency remain open. Benchmarks have used different event-specific inactivity thresholds; this evidence does not justify selecting 7, 14, 21 or 28 days as an industry standard.

Fairness guardrail: returning players may receive help overcoming accumulated content and roster gaps, but not a permanently exclusive competitive advantage that active players can never earn through ordinary play. Any special catch-up unit, selector or resource should have a fair long-term route or be non-essential to core team function.

32.6 Recommended provisional model for the Naruto idle RPG

[PROPOSAL — WORKING BASELINE, NOT A FINAL BALANCE DECISION] Use a full-rate idle window of 24 hours, followed by a limited 48-hour overflow window at 25% of the normal rate. After 72 elapsed hours, ordinary idle-bank accrual stops until the bank is collected. These numbers are a deliberately explicit working hypothesis for comparing alternatives on paper; they are not an industry standard, a user-approved final setting, or a proven optimal value.

1. The best-cleared campaign checkpoint determines the idle rate. Team composition affects the player’s ability to clear new stages but does not directly multiply the idle bank.
2. Accrual is segmented when the reward tier changes. Time earned before a new campaign clear uses the prior tier’s rate; the new rate begins only from the actual clear timestamp. Claiming late cannot retroactively reprice old accrual at a higher rate.
3. Full rate runs for the first 24 hours. For the next 48 hours, accrual runs at 25% of the normal rate. This makes a three-day absence less punishing while limiting the resource injection compared with three full-rate days.
4. At the end of the overflow window, ordinary accrual pauses rather than generating unbounded resources. A longer-term returning-player package is a separate mechanic governed by its own eligibility rules.
5. The claim interface shows total elapsed time, full-rate time, overflow time, resources earned at each rate tier, and whether accrual has paused. Use clear labels rather than hidden calculations.
6. Do not put limited-event currency, exclusive ranking rewards, or rare character drops into the ordinary idle bank by default. Define the reward basket separately in the next resource-economy research pass.

Cross-system finding (v0.10.11): Under this unchanged storage proposal, claims separated by 72 and 96 hours yield only three full-rate days of materials per week. Section 40.9 connects that loss to counter preparation and compares possible remedies; the cap is not changed in this revision.

Why this is the provisional recommendation: a one-day full-rate bank supports a normal daily rhythm; a low-rate overflow window reduces the penalty for a disrupted schedule or a longer weekend; and a bounded bank avoids creating an unlimited accumulation strategy. These are reasoned design trade-offs, not measured player outcomes.

32.7 Hypothetical scenarios under the provisional model

The arithmetic below is illustrative only. It checks what the written rule means; it is not a simulator run or evidence of actual game balance.

Absence / situation	Accrual under proposal	Expected experience	Rule being checked
8 hours away	8 hours at 100% rate.	The player receives the full normal return reward.	Ordinary short absences are unaffected.
27 hours away	24 h at 100% + 3 h at 25% = 24.75 full-rate-equivalent hours.	Only three hours are in overflow; missing a routine daily login does not lose all subsequent value.	Overflow is transparent and mathematically simple.

Absence / situation	Accrual under proposal	Expected experience	Rule being checked
60 hours away	24 h at 100% + 36 h at 25% = 33 full-rate-equivalent hours.	A long absence still grants meaningful resources but less than 60 full-rate hours.	The overflow dampens accumulation.
Five days away	Ordinary bank reaches its 72-hour elapsed-time limit; only 36 full-rate-equivalent hours are banked.	The player may still be eligible for separate returner support if its later-defined threshold is met.	Long absence is handled by two systems, not unlimited idle banking.
Higher campaign stage cleared after returning	Previously elapsed time remains priced at the earlier rate; new rate starts at the clear timestamp.	The player cannot delay a claim to reprice earlier hours at the higher tier.	No retroactive reward-rate exploit.
Player changes the active team before leaving	No rate change unless a campaign checkpoint was actually cleared.	The player can experiment with roster options without a farming-team tax.	Account progression, not current squad, determines idle rate.
Player spends on convenience	The bank cap and ordinary idle rate remain the same for all accounts.	A purchase may save time, but not create a compounding higher passive-income rate.	Protects the pay-to-progress principle.
Player repeatedly closes/reopens the app	Elapsed-time accounting continues from recorded timestamps; reopening does not reset the overflow allowance.	No benefit from artificially cycling the session.	The later implementation must define reliable timekeeping.
Player returns after a very long break	Ordinary idle bank remains bounded; returner support is checked separately.	The player receives a clearer path back without a multi-week pile of unbounded resources.	Catch-up is distinct from raw absence duration.

32.8 Exploit, fairness and readability safeguards

- No retroactive income repricing after a campaign clear, rate upgrade or reward-rule change. Record the current tier and the time at which the new tier begins.
- No pay-exclusive increase to the ordinary storage window or base idle rate in the working model. If paid convenience features are later considered, compare their indirect progression effect against the project's F2P fairness rules.
- Do not make rare units, duplicate copies, or the current meta team the only way to improve the idle income tier. The rate follows earned account progression, and essential campaign functions must have accessible alternatives.
- One absence should not be artificially split by app reopens to exploit a new overflow window. The eventual implementation specification needs a trusted elapsed-time source and explicit handling of clock changes, server downtime and duplicate claim requests.
- Do not let a returning-player reward be repeatedly farmed by rotating accounts or by repeatedly crossing the same inactivity threshold. Eligibility cadence and account-level limits must be explicit if that system is adopted.
- Avoid punishing active players to help returning players. Returner rewards should speed re-entry, not create permanent exclusive power. If returner rewards include characters or selectors, core-role functionality must remain accessible to the full player base by other routes.
- Do not claim this model has solved retention, player wellbeing, economy inflation or monetization. These are hypotheses that require later empirical and economic validation.

32.9 Current decision and open-question register

Item	Current status	What remains open
Separate ordinary idle bank from returner catch-up	Recommended design principle.	Exact content and eligibility rules for returner support.
Do not tie idle rate to current squad	Consistent with Section 31 working model.	No change unless later research finds a compelling reason; any change must address roster-rotation costs.

Item	Current status	What remains open
24 h full-rate + 48 h at 25% overflow	Provisional working model for theory comparisons.	Exact cap, overflow percentage, rate curve and resource basket must be reconciled with progression costs and resource sinks.
Long absence support	Separate returner system recommended for study.	Threshold, frequency, reward composition, expiry, and fair routes for active players.
Claim UI and notifications	Recommend one-tap collection and user-controlled reminders.	Exact summary screen, notification defaults, and accessibility behavior.
Paid instant rewards	No paid cap / base-rate bonus in the working model.	Whether a modest, transparent instant-time conversion belongs in the broader economy.
Economic sustainability and user response	Open hypothesis.	Requires later economy modeling and, only after explicit authorization, implementation/live testing.

32.10 Interim conclusion and next research task

Interim conclusion [RESEARCH SYNTHESIS]: the best current theoretical direction is a bounded full-rate bank with a transparent low-rate overflow, plus a distinct catch-up path for genuinely long absences. This reduces pressure for repeated daily check-ins without turning offline time into unlimited full-rate accumulation. Campaign progression continues to determine the idle-rate tier; the currently equipped squad, paid status and claim timing do not retroactively alter accrued rewards.

Status: completed by Section 33. The resource basket and source-allocation map now identify which resource families are proposed for idle accrual, first-clear rewards, repeatable active modes, recurring objectives, events, social play or specialized systems. The map intentionally does not set exact quantities or costs. Next research task: study resource sinks, upgrade costs, bottlenecks and progression pacing so the proposed income sources connect to a coherent economy without false precision.

32.11 Sources for storage caps, claim cadence and catch-up (accessed 28 October 2026)

- [IR12] AFK Global, “AFK Arena beginner guide.” Search index says 12-hour rewards and two-hour Fast Rewards, but the opened URL resolved to an older guide; source-version discrepancy flagged. https://www.afk.global/afk-arena/guides/beginners-guide
- [IR13] Prydwen Institute, “Outpost output — NIKKE.” Community guide describes a 12-hour storage cap; patch-sensitive and not independently verified in-client. https://www.prydwen.gg/nikke/guides/outpost-output
- [IR14] Nikke Explorer, “Outpost Defense Calculator & Case Yields.” Indexed page says 24-hour capacity and 15% overflow, conflicting with IR13; use as evidence of disagreement, not a confirmed fact. https://nikke.exynan.my.id/outpost-defense
- [IR15] Prydwen Institute, “AFK Journey Returnee Guide,” updated 10 January 2026. Describes season-dependent returner rewards such as resonance boosts or extra pulls; community source. https://www.prydwen.gg/afk-journey/guides/returnee-guide
- [IR16] Level Infinite / NIKKE, official “Re:White Memory” event page (31 December 2025–29 January 2026): one campaign used 14 consecutive inactive days plus a Welcome Back Mission. This is event-specific, not universal. https://nikke-en.com/events/whitememory/?share_code=TDEHZNSKZWAD
- [IR17] Nikkers.cc, “Update on October 8 — NIKKE Patch Notes.” Third-party summary of a Welcome Back redesign; verify the live implementation and eligibility before production decisions. https://nikkers.cc/updates/2026-10-08

Source-confidence note. Most sources are secondary and patch-sensitive. The AFK Arena search result conflicts with the opened page, and NIKKE guides disagree on capacity. These sources inform design trade-offs; they do not establish a confirmed current cap or prove what is optimal for our game.

33. Idle resource basket and source-allocation map (Version 0.10.4)

Purpose and status. This chapter decides, at the conceptual level, which resource families belong in the ordinary idle-reward bank and which should be earned through other activities. It does not set final quantities, drop rates, progression costs, or monthly income totals. Benchmark observations are explicitly sourced; the resource-source table is a project proposal. No prototype, simulator, seeded run, or player test was performed.

33.1 The question: what should the idle bank actually contain?

An idle RPG must provide meaningful progression while the player is away, but “idle game” does not mean every reward and every currency should be generated passively. If the bank contains every resource needed for summons, rare character access, skill upgrades, premium gear, competitive ranking and event completion, active modes risk becoming optional chores with no distinct purpose. If the bank only provides tiny amounts of ordinary currency, the idle promise becomes hollow and progression can feel like a series of paywalls.

The design task is therefore to assign each resource a primary source, allowed secondary sources, and a guardrail. We must also distinguish a resource’s source from its use: “skill material” is not a sufficient label unless we know which skills it upgrades, how often the player needs it, and whether missing the associated mode blocks an essential role or counter.

33.2 What current benchmark sources support

[SOURCE FINDING] AFK Arena’s guide describes AFK rewards as including recurring resources such as Gold, Hero EXP and Hero’s Essence whose rate rises with campaign progress, while campaign quests and chapter completion pay separate rewards including diamonds, hero/scroll rewards, gear, and feature unlocks [IR18, IR19]. This supports separating the repeated idle basket from one-time stage milestones. Exact rates and details are version-sensitive and the guide is community-maintained.

[SOURCE FINDING WITH SOURCE-DATE CAVEAT] The NIKKE.gg beginner guide says campaign clears grant immediate leveling resources and increase Outpost Defense level, while Outpost Defense produces recurring resources including Credits, Battle Data, Core Dust, Commander EXP and equipment [IR20]. The search result and opened page showed inconsistent update dates, so this is used as a structural observation only. It does not verify every current yield or cap.

[SOURCE FINDING] AFK Journey’s official “Talk with Designers: New Season Preview” describes a season update in which one-time AFK-stage rewards were moved to a Season Handbook, equipment rewards were changed from random drops to Select Equipment Chests, and some stage-progression materials were moved earlier in the season [IR21]. This is a useful documented example of separating one-time milestone rewards from recurring/season systems and using selection to reduce gear-drop frustration; it is not evidence that our game should copy that exact arrangement.

[SOURCE FINDING] The official Google Play listing for Idle Heroes markets passive hero training while away alongside hero summoning/training, gear, heroic quests, towers, arena and guild boss battles [IR22]. The listing shows that idle progression and active/social content coexist, but it does not disclose enough economy detail to assign exact resource sources.

[SYNTHESIS] These examples support a layered resource economy: a predictable idle floor for recurring development; immediate one-time campaign milestones; mode-specific rewards for activities that need their own purpose; and events/objectives that provide additional, more deliberate sources. Sources do not establish exact percentages or the optimal basket for our project.

33.3 Resource-source principles for our project

1. Give every resource family a declared primary source and explainable secondary sources. Avoid “this can drop anywhere” as the only economy rule; diffuse sourcing makes bottlenecks harder to understand and tune.
2. Use idle accrual for predictable, repeatable development inputs. Its purpose is to make useful progress during absence, not to replace every active activity.
3. Use campaign first-clears for milestone rewards, introductions to new mechanics, and one-time account advancement. Do not count a first-clear reward as a sustainable recurring income source.

4. Use repeatable active modes for targeted and specialized materials. Keep an alternative route or a modest baseline for anything required to maintain the basic function of a team, so one locked mode or difficult boss cannot freeze all account development.
5. Use daily/weekly objectives to bundle flexible recurring rewards, but avoid making every important resource depend on a long chain of daily check-ins. Weekly completion and cumulative milestones can smooth missed days.
6. Use events for thematic rewards, optional acceleration, cosmetics and selected progression choices. Do not put an indispensable role, mandatory counter, or permanently exclusive core-upgrade requirement behind a brief event window.
7. Keep premium summon economics auditable. Summon currencies, banner tickets and pity progress should be tracked explicitly rather than hidden in a random idle loot bag. The income budget must be reconciled with the project's existing illustrative pull and selector targets before exact rates are set.
8. Separate energy/entry systems from the ordinary idle bank. If a mode uses energy, its regeneration, cap, free refills and event grants are a separate system, not another ambiguous component of the general offline timer.
9. Treat rank and social participation fairly. PvP and guild modes may provide their own tokens and prestige, but basic team functionality should not require top-percentile placement or membership in a high-ranking guild.

33.4 Proposed resource-source matrix

All entries in this table are recommendations for our concept, not claims that every benchmark game uses the same allocation. "Idle bank" means the ordinary offline accumulation system defined in Section 32, with the working cap/overflow model still provisional.

Resource family	Proposed idle-bank treatment	Primary complementary sources	Main guardrail / reason
Basic currency (working name: Ryo)	Include as a core idle resource.	Campaign clears, daily/weekly objectives, general shops, selected mode rewards.	Must remain useful over long progression; watch inflation and avoid making one source the only route.
Shinobi EXP / training data	Include as a core idle resource.	Campaign and milestone rewards, objectives, limited event bundles.	Allows a returning player to strengthen a core squad without mandatory farming sessions.
Common promotion / level-cap essence	Include a baseline flow; tune separately from EXP.	Repeatable training or tower modes, weekly deterministic shop/choice, achievements.	Prevent the first upgrade bottleneck from freezing an account that has hit a campaign wall.
Basic skill-up materials	At most a modest baseline if needed for core progression; not every advanced skill item.	Repeatable training mode, stage milestones, weekly shop and selected events.	Advanced skill optimization can reward active play, but basic role function cannot depend on rare drops.
Basic equipment enhancement material	Include a low, predictable supply if equipment is a universal growth layer.	Campaign milestones, equipment mode, weekly objectives and shops.	Make ordinary gear growth possible; keep high-end upgrades and set optimization from being completely passive.
Equipment pieces	May include common/basic pieces or clear-choice starter gear; do not rely on random idle drops for important slots.	Campaign first-clear chests, targeted equipment activity, crafting/selection systems.	Use selector choices at key milestones to reduce bad-luck dead ends; advanced gear should have understandable routes.
Character-specific shards / duplicates	Do not make random copies a standard idle-bank output.	Recruitment, focused fragment routes, achievements, faction/tower milestones, shops and events.	A duplicate lottery in passive rewards is hard to plan around and can widen roster gaps; duplicates must not unlock mandatory core functions.
Universal character fragments / selector currency	Not a default hourly drop; consider a small, explicit predictable stream only if the acquisition budget needs it.	Weekly milestones, long-term achievements, deterministic shops and broad-access events.	Preserve player agency by allowing targeting. Reconcile income with the existing illustrative 100-fragment selector and weekly-fragment targets.

Resource family	Proposed idle-bank treatment	Primary complementary sources	Main guardrail / reason
Standard summon currency / tickets	Not part of the ordinary idle basket by default.	Campaign first-clears, recurring missions, achievements, weekly milestones, events and accessible shops.	Keep monthly F2P income and pity progress auditable. Missed optional dailies should not silently destroy the whole guarantee plan.
Featured / limited-banner tickets or currency	Do not generate as an opaque idle-bank drop.	Banner-specific missions, event milestones, compensation and any explicitly budgeted free allocation.	Clearly show counter carry-over, expiration and guarantee rules; do not make a core counter or essential role limited-only.
Mode-specific tokens (boss, tower, gear dungeon)	Keep out of ordinary idle accrual.	The named mode, weekly milestones, participation floor and possibly a limited exchange shop.	The mode needs a reason to exist. Essential base-team upgrades still need an accessible alternate route.
PvP currency / seasonal points	Keep out of ordinary idle accrual.	Battle participation, ranked season, milestone tiers and periodic reset rewards.	Rewards can scale with effort/rank, but the majority of essential growth cannot be locked to the highest rank brackets.
Guild / alliance currency	Keep out of ordinary idle accrual.	Guild boss participation, cooperative milestones, support or contribution tasks.	No basic progression should require a perfect guild; provide a fair baseline for casual or newer guilds.
Event-specific currency	Do not accrue automatically outside its own event unless explicitly designed as a grace/catch-up rule.	Event stages, cumulative event missions, participation milestones and event shop exchanges.	Provide cumulative milestones and reruns/alternate access for important long-term resources; cosmetic exclusives are safer than power exclusives.
Energy / mode-entry tickets	Separate regeneration/bank from ordinary idle rewards.	Timed regeneration, occasional mission/weekly rewards, event grants and free refill rules.	State caps and overflow plainly; do not stack multiple hidden timers or force frequent check-ins.
Returner/catch-up tokens	Separate returner system; not part of normal idle bank.	Explicit returning-player track, evergreen catch-up missions or account-progression grants.	Help with obsolete content and roster breadth without creating repeated inactivity exploits or permanent power advantage.
Cosmetics / profile identity	No need for passive bank inclusion.	Achievements, event shops, optional purchases, collection milestones and community rewards.	A safe monetization and engagement lane because it does not need to alter combat strength.

33.5 Three candidate idle-basket models

Model	Idle basket	Advantages	Risks / assessment
A. Broad universal basket	Nearly all currencies, materials, tickets and some specialized mode rewards.	Very simple claim experience; large visible reward pile.	Blurs purpose of modes, hides economy balance in offline accrual, enables stockpiling of limited resources. Not preferred.
B. Core-growth basket	Basic currency, character EXP and a predictable baseline of ordinary progression materials; perhaps low-tier gear/gear materials.	Strong idle identity while leaving player choices and mode purposes intact.	Preferred working model. The exact core basket and rates must be matched to costs and the progression curve.
C. Minimal basket	Only basic currency and EXP; almost all other materials require active play.	Very clear separation between passive and active play.	Risks creating hard bottlenecks and makes “idle” feel hollow if critical upgrades require frequent grinding. Keep only if other sources are generous and flexible.

33.6 Recommended working allocation for the Naruto idle RPG

[PROPOSAL — THEORETICAL, NOT OWNER-APPROVED AS FINAL] Start from Model B: a core-growth idle basket, with a tightly limited list of resource families. Idle rewards should be predictable enough that players can estimate normal progress, but they should not complete every character collection, high-end equipment set or mode-specific upgrade path by themselves.

- Core idle bank candidates: Ryo/basic currency, Shinobi EXP, and one common progression essence. Consider low-tier equipment enhancement materials only after the resource-cost map shows they are necessary to keep ordinary growth moving.
- Campaign clear rewards: first-clear premium currency or standard tickets in an explicit budget, starter or selector resources, equipment choice chests at meaningful checkpoints, and feature-unlock rewards. First-clear payments must be counted once rather than treated as recurring income.
- Repeatable active-mode rewards: specialized skill materials, boss/tower currencies, selected gear/crafting ingredients and tactical resources that give the mode a distinct purpose. If any such item becomes a mandatory baseline bottleneck, add an accessible side route or guaranteed weekly allowance.
- Recurring objective rewards: standard summon income, universal fragments, broad choice chests and modest flexible material bundles. Prefer weekly/cumulative completion to an economy where missing a single day permanently breaks the featured guarantee budget.
- Event reward layer: event currency, themed cosmetics, limited story rewards, optional acceleration and clearly described selectors. If an event contains a core progression material, provide enough time and a cumulative path for ordinary players; do not assume everyone logs in at the event reset.
- Competitive and social layer: PvP and guild tokens mainly fund mode-specific cosmetics, seasonal prestige, specialized sidegrades and supplemental resources. A lower-rank or casual guild reward floor should still support participation without becoming the single source for required character power.
- Summoning layer: keep premium currency and ticket flows visible in a dedicated acquisition budget. Earlier sections contain more than one illustrative recurring-pull target (for example, around 45–50 pulls plus fragment income in one passage and 60 pull-equivalents per 28 days in another); this is an internal inconsistency to reconcile in the later economy budget, not a value to silently choose here. The 70-pull featured guarantee and 100-fragment selector are also proposals, not verified industry standards.

33.7 Hypothetical scenarios: does the allocation stay coherent?

These are paper scenarios to expose logical gaps. They are not simulated outcomes or measured player results.

Scenario	Expected behavior under the proposal	Failure condition to watch for
New account with a small roster	Idle bank supports basic leveling; early campaign rewards provide starter gear, clear milestones and early recruitment resources; the tutorial reveals one active mode at a time.	The player cannot level a coherent starter team without repeatedly farming a mode that requires a different team.
Player hits a campaign wall	Basic income continues; active play allows alternative tactics and supplementary materials; campaign clear rewards stop until a new stage is won.	All useful upgrades require progress beyond the wall, creating a circular deadlock.
Player returns after several days	The core basket yields resources within the cap/overflow policy; catch-up is a separate, transparent track; special event currencies are not silently duplicated.	The bank is large but the player lacks the specific resource needed to use the new systems or understand where to go.
F2P player targets a featured shinobi	Visible free summon routes and guarantee counter show the expected time-to-guarantee; income is audited across all recurring and one-time sources.	The guarantee depends on impossible daily attendance, elite PvP rank, or an expiring event that most players cannot complete.

Scenario	Expected behavior under the proposal	Failure condition to watch for
Player experiments with an alternative squad	Current squad choice does not change idle basket rates; the account can reallocate training resources without permanently losing investment.	Idle income encourages equipping one “farming team” and discourages team-building experimentation.
Player skips a limited event	Missed cosmetics may remain exclusive; core-role and essential upgrade access returns through evergreen, future-event or alternate sources where appropriate.	The missed event permanently removes the only accessible counter or fundamental upgrade.
Low-rank PvP or casual guild member	Participation and milestone rewards provide some mode-specific value without becoming the sole source of critical progression.	Low-rank players accumulate a permanent power deficit that cannot be closed through PvE or long-term progression.
Player stockpiles resources before a new character release	Ordinary resources can be saved; limited tickets and pity follow clear carry-over/expiry rules; event currency expires only if warned before earning/spending.	A hidden cap or surprise expiration invalidates long-term planning, or the resource bank trivializes new content progression.
Temporary balance change raises upgrade costs	The research spec requires communicated compensation / grandfather rules before any later implementation change; historical costs are versioned.	Previously saved balances suddenly lose practical value with no explanation or adjustment path.

33.8 Resource sourcing, sinks and free-to-play fairness

- No single source should be the only route to a fundamental combat role, essential counter or baseline character improvement. The source map should identify a primary route and at least one accessible alternative for critical resources.
- Idle rewards must be valuable but not sufficient to unlock every strategic option without active decisions. Active modes should grant distinctive, understandable rewards—not simply require a manual battle for the exact same generic currency the idle bank supplies.
- First-clear rewards should emphasize discovery, unlocking and one-time acceleration. Recurring needs must have recurring sources so a player who reaches a wall is not permanently stuck after spending the initial chest.
- Use deterministic choice rewards at selected milestones when random drops could leave a slot, role or essential upgrade unavailable for too long. Randomness may create excitement, but it should not be the only route to a required solution.
- Publish an economy map: each resource should list source(s), sink(s), storage cap if any, expiry if any, tradability, purchase routes, F2P route, and whether the item affects competitive power.
- Do not create a paid-only “idle basket multiplier” as the default monetization lever. If time-saving products are later studied, price them transparently and examine their cumulative effect on roster power and the project’s fairness goals.
- Avoid making daily missions the only way to earn the core free summon budget. Use a mix of weekly cumulative milestones, campaign progression, achievements and event routes so one missed day is not equivalent to losing an entire week of value.
- Do not overproduce currency just because it is passive. Every idle resource needs a known purpose and a corresponding sink, and the long-run supply must later be compared with upgrade costs before numeric rates can be finalized.

33.9 Current allocation register

Question / decision	Working status	What must happen before finalization
Ordinary idle basket	Core-growth basket preferred: basic currency + Shinobi EXP + one common progression essence; low-tier gear materials optional.	Map all upgrade sinks and costs; compare early-, mid- and late-game requirements on paper.
Basic equipment	Starter/common gear may come from campaign milestones and possibly the basic idle layer; premium/high-end sets need distinct routes.	Define gear slot progression, replacement frequency, and sources for bad-luck protection.

Question / decision	Working status	What must happen before finalization
Skill/promotion materials	Baseline access plus specialized active-mode sources; no mandatory single-mode gate.	Map character skill-up costs and determine which materials are essential vs optimization-only.
Summoning income	Dedicated source budget; no default opaque idle-ticket drops.	Resolve conflicting illustrative monthly income targets and align with banner release cadence, 70-pull guarantee and 100-fragment selector proposals.
Character fragments	Targetable predictable sources rather than a random idle-copy lottery.	Connect the fragment economy to role coverage, roster breadth and duplicate power rules.
Event currencies	Event participation / cumulative milestones; limited to the event by default.	Decide expiry, reruns, catch-up and whether each item is cosmetic, acceleration or essential progression.
Mode tokens	Mode-specific by default, with accessible fallback for essential progression.	Define each mode's unique reward identity in the later mode research round.
Exact quantities and rates	Open; no number is set by this chapter.	Complete sinks/cost/pacing research first. Future implementation or empirical tests require the owner's explicit authorization.

33.10 Interim conclusion and next research task

Interim conclusion [RESEARCH SYNTHESIS]: the recommended idle bank is a deliberately limited core-growth basket, not a copy of the whole reward economy. Campaign first-clears deliver one-time milestone rewards; repeatable active modes supply specialized materials and tactical goals; daily/weekly objectives and evergreen achievements contribute predictable acquisition income; events provide thematic rewards and optional acceleration; PvP and guild systems provide participation/seasonal rewards without monopolizing essential progression. All allocation rules remain proposals until their sinks, costs and progression effects are reconciled.

Status: completed at the conceptual-framework level in Section 34 (Version 0.10.5). Exact costs, rates and pacing are still open. Next research task: build a paper cost-versus-income budget for core character progression and equipment, then reconcile the result with summoning, fragments and release cadence. No rates should be finalized until the resource costs, dependable income sources and target progression windows are compared.

33.11 Sources for resource-source allocation (accessed 29 October 2026)

- [IR18] AFK Global, "AFK Arena beginner guide." Community guide; indexed summary describes campaign-scaled Gold, Hero EXP and Hero's Essence. The opened page shows an older publication date, so use for structural background only. https://www.afk.global/afk-arena/guides/beginners-guide
- [IR19] AFK Global, "Campaign chapter rewards and unlocks." Community guide separating one-time campaign rewards/unlocks from recurring AFK income; patch-sensitive. https://www.afk.global/afk-arena/guides/campaign-chapter-rewards
- [IR20] Nikke.gg, "Nikke Beginner's Guide." Community guide linking campaign clears, Outpost Defense and development resources; search/opened dates conflict, so treat as structural evidence only. https://nikke.gg/beginners-guide/
- [IR21] Farlight Games, "Talk with Designers: New Season Preview." Official source describing one season's shift of one-time stage rewards to a Season Handbook and random gear rewards to selectable chests; season-specific. https://afkjourney.farlightgames.com/news/a0ae0b8ad8c35da97805079b672b7287
- [IR22] DHGAMES, "Idle Heroes," official Google Play listing updated August 2026. Markets idle training alongside quests, towers, arena, gear and guild battles; does not specify exact economy rules. https://play.google.com/store/apps/details?hl=en_US&id=com.droidhang.ad

Source-confidence note. IR21 is a first-party designer statement about one season change; IR22 is a publisher store listing, not detailed economy documentation. IR18-IR20 are community sources with the date/version limitations stated above. The source-allocation matrix is our proposal, not a claim about hidden benchmark economies. Exact quantities and yields remain unresearched and unvalidated.

34. Resource sinks, upgrade-cost structure and progression pacing (Version 0.10.5)

Purpose and status. This chapter connects the Section 33 resource-source map to how the game spends resources. It establishes a first-pass sink register, describes how costs should behave across early, mid and late progression, and defines the accounting required before exact values can be chosen. All Naruto Idle RPG mechanics in this chapter are [PROPOSAL] or [RESEARCH SYNTHESIS], not owner-approved final rules. Benchmark numbers are examples from named

games and must not be copied as our economy. No implementation, economy simulator, seeded battle run or player test was performed.

34.1 Research questions and evidence rules

- A resource source (Section 33) answers “where does it enter the account?” A resource sink answers “what consumes it, and for what player-facing benefit?” A stock is the amount currently held. A recurring income rate must be distinguished from one-time first-clear or milestone rewards.
- Every resource family needs a primary purpose, allowed secondary uses, a dependable acquisition path, a restriction / expiry rule where relevant, and a guardrail against becoming either worthless or an unavoidable hard gate.
- “Mandatory” means the resource is required for baseline team functionality or ordinary progression. “Optimization” means it improves efficiency or options but is not required to keep a viable team functioning. The same resource may serve both, but its baseline and advanced uses must be separated clearly.
- Keep three evidence layers separate: source-derived benchmark observations; project-specific design proposals; and unresolved values that require later owner decisions or validation. No benchmark source establishes the optimal cost curve for our game.

34.2 What the benchmark evidence supports

AFK Arena — multiple resource tracks and milestone costs. A community-maintained guide published 21 October 2026 reports that hero leveling consumes Gold and Hero EXP regularly, with Hero’s Essence charged at selected breakpoints tied to skill upgrades. Its example cost for one hero from level 1 to 100 is 1,953,229 Gold, 1,088,586 Hero EXP and 900 Essence; it reports still steeper EXP costs later. The same guide describes focusing level investment on five heroes and using the Resonating Crystal to share levels with additional heroes. This is useful structural evidence of separate cost tracks, milestone gates and shared-level infrastructure, not an economy target for Naruto Idle RPG. The page is community-maintained, not an official publisher specification, and current values remain patch-sensitive [IR23].

AFK Arena — reversible early investment. The same guide describes free hero resets before campaign stage 4-36 and a diamond cost after that point. The transferable lesson is not the exact stage or fee; it is that early experimentation can be made low-risk, then the game can teach players to make more considered investments after they have learned the systems. A refund or respec policy for our game remains an open design choice [IR23].

AFK Journey — revise reward supply when the cost structure changes. In a first-party seasonal designer communication, Farlight Games described moving some upgrade materials earlier in a season, replacing random equipment drops with Select Equipment Chests, and increasing hero-upgrade material drops after the new feature raised development costs. This is direct evidence that a cost-system change should trigger a review of material supply and reward certainty. It is one season-specific revision, not proof that the same schedule or chest system fits our game [IR21].

General economy-design references. Unity’s economy guide defines sources as ways resources enter the game and sinks as ways they are spent, and emphasizes considering their combined effect on progression. A Game Developers Conference session on sink design discusses sink trade-offs and spreadsheet-based analysis. These sources support keeping an explicit accounting model; they do not provide the correct values for our title [IR24, IR25].

34.3 Proposed design principles for the Naruto idle RPG

- Design costs and income together. Do not define a resource’s idle or active supply before listing the upgrades that consume it; do not define costs without checking whether dependable income can cover the intended progression path.
- Protect the functional baseline. The first useful version of a character’s role, core counter or synergy should work without a high-rank skill, rare random material, limited-event item or expensive duplicate. Advanced investments may improve reliability, efficiency, durability or flexibility, but should not sell access to a required tactical function.
- Make the major costs legible. If a promotion, skill unlock or equipment tier requires a large payment, signal the threshold and the benefit before the player commits. Avoid hidden cost jumps and undisclosed expiration.
- Avoid redundant taxes. A single basic currency may be shared by several systems, but do not layer a fee onto every upgrade by default. Each added cost must have a clear economic purpose and be included in the time-to-afford calculation.

- Minimize regret during early learning. Evaluate a low-cost reset, refund or transferable investment for baseline levels and replaceable starter gear. This is a proposal for discussion, not a locked refund rule.
- Let active modes specialize rewards without monopolizing essentials. A boss, tower, PvP tier, guild or short event may improve access to specialized or advanced materials; if a material becomes required for basic team function, provide a dependable alternate route or a baseline allowance.
- Keep the idle promise honest. Core idle income should support progress during a campaign wall or ordinary absence. It need not replace every active mode or produce every premium, summoning, competitive and high-end optimization resource.

34.4 First-pass resource-sink register

The register below assigns a primary proposed sink to each family already present in Section 33. It is a working map of intended purpose, not a finalized price list. Where the current design does not yet specify an upgrade mechanism, the row explicitly leaves that rule open.

Resource family	Primary proposed sink	Required guardrail / open rule
1. Ryo / basic currency	Routine level-up fees and possibly basic skill / gear operations.	Keep it broadly useful across progression. Avoid taxing every system with Ryo; compare aggregate demand against dependable income.
2. Shinobi EXP	Character levels; potentially account-level training only if that system is later defined.	Keep the main purpose narrow and visible. Track how costs shift from building the starter team to building a second or third viable team.
3. Common promotion / level-cap essence	Discrete advancement thresholds or level-cap breaks.	Use predictable, communicated gates. A core account should not freeze because all reliable essence income comes from a single mode or random drop.
4. Basic skill-up materials	Skill-rank improvements beyond a character's functional baseline.	Keep role-defining mechanics available at base rank. Higher ranks may improve strength or efficiency, not unlock the only practical counter to a common strategy.
5. Equipment enhancement materials	Enhancing owned equipment through defined tiers.	Avoid failure-based destruction, hidden odds or breakage in the baseline proposal. Evaluate partial refund, transfer or salvage when an item is replaced.
6. Equipment pieces / gear	Filling equipment slots and progressing into higher tiers or sets.	Use milestone rewards or choice chests for critical early slots. Do not let a long run of random drops deny a necessary slot or make early enhancement spending feel wasted.
7. Character-specific shards / duplicates	Character acquisition or advancement tiers, if duplicate-based advancement is retained.	Duplicates must not be the only route to essential role functionality. Decide advancement value, fodder rules and duplicate conversion together; exact duplicate costs remain open.
8. Universal fragments / selector currency	Targeted recruitment or a fragment selector.	Preserve player agency and roster breadth. Align the accumulation threshold with a paper budget before treating any selector size or weekly income as settled.
9. Standard summon currency / tickets	Standard recruitment only, by default.	Keep summon income auditable. Avoid unrelated shop sinks unless explicitly budgeted, and never mix one-time campaign rewards into a recurring monthly-income claim.
10. Featured / limited-banner currency	Recruitment on the named limited or featured banner.	Publish expiry and guarantee carry-over clearly. Do not let a limited banner become the only route to an essential role or counter.
11. Mode-specific tokens	Specialized items in the mode's own exchange shop (boss, tower, gear activity, etc.).	Give the mode a unique reason to play. If an item becomes a baseline progression requirement, create an accessible alternate source or guaranteed periodic allowance.
12. PvP currency / seasonal points	Competitive cosmetics, prestige, sidegrades or supplemental resources.	Participation should have value at lower ranks. Avoid making top-percentile rank the sole route to durable combat power.

Resource family	Primary proposed sink	Required guardrail / open rule
13. Guild / alliance currency	Cooperative rewards, guild shop and social progression.	Do not require a perfect or high-ranking guild for basic growth. Include a meaningful participation floor for new and casual members.
14. Event-specific currency	Event-shop rewards, thematic cosmetics, selected progression choices and optional acceleration.	State expiry before earning/spending. Essential materials need cumulative milestones, reruns or alternate access; missed events must not permanently delete a core build option.
15. Energy / mode-entry tickets	Entry into the named repeatable activity or limited attempt.	Keep regeneration, storage cap and free refills in a separate, transparent system. Avoid stacked hidden timers and frequent check-ins as a substitute for progression design.
16. Returner / catch-up tokens	Older progression materials, broad resource bundles or catch-up steps.	Use a bounded catch-up route rather than a permanent exclusive power tier. Do not reward repeated inactivity loops or make new active players feel foolish for playing.
17. Cosmetics / profile identity	Skins, effects, titles, profile frames and non-combat customization.	Treat this as a safe optional sink for collection and monetization. Cosmetic spending should not alter a normalized PvP outcome.

34.5 Character-progression cost architecture

These are proposed cost behaviors, not formulas or values. The categories should be costed independently and then evaluated together, because a single hero may simultaneously require currency, EXP, promotion essence, skill materials and equipment investment.

Upgrade area	Candidate cost structure	Pacing and fairness rule
Character levels	Ryo + Shinobi EXP on ordinary level steps; promotion/level-cap essence at explicit milestones if retained.	Track total cost by level band and account stage. If an account hits a campaign wall, baseline income should still move at least one meaningful growth track. Avoid making all resources rise at the same rate by default.
Skill ranks	Skill materials, with Ryo only if the fee has a defined purpose.	Functional baseline kit first; incremental improvement later. Show the next rank and its cost. Review whether high-rank costs create specialization choices or simply stop all progress.
Promotion / level caps	A distinct threshold payment and any agreed character advancement requirement.	A threshold should correspond to a legible benefit. Do not hide role access inside a duplicate tier. Compare milestone reward timing with the new cost so the player is not asked to pay before a reasonable source exists.
Second team / replacement hero	Shared progression infrastructure or recoverable investment should be considered before a completely separate full-cost track.	Roster experimentation matters to counters and modes. Evaluate refunds, transfer, shared-level systems or cheaper re-entry paths as alternatives; none is locked yet.

34.6 Equipment-cost architecture and replacement regret

- Separate obtaining an item from improving it. Campaign milestones can provide starter / common items; targeted gear activities and selectors can handle later needs. Random drops may add variety, but should not be the only practical path to a required early slot.
- Do not assume every enhancement should be invested fully. If early gear is expected to be replaced often, keep early upgrade cost low, make enhancement transfer possible, or return part of the investment through salvage. Pick one clear policy rather than mixing several opaque refund rules.
- Keep high-end set completion and stat refinement as optional optimization until the rest of the equipment economy is specified. Do not set an equipment-tier count, per-slot price, rarity curve, enhancement success chance, or salvage ratio in this research pass.

- If a future gear system uses random rolls or randomized crafting, separately assess the cost of a failed attempt, bad-luck protection, and whether the player can see expected effort. Failure that destroys scarce materials can multiply the effective sink; it should not be introduced by default.

34.7 Early, mid and late progression pacing

Progression band	Player-facing goal	Proposed cost and supply behavior	Failure signal
Early game — functional team	Level the starter core; learn roles; fill basic equipment slots; unlock and understand the main upgrade tracks.	A dependable flow of basic currency and EXP, a clear path to the first essence / skill thresholds, milestone gear choices, and low-regret learning investments.	A single rare material, duplicate requirement or random equipment slot prevents a functional team.
Midgame — specialization and alternatives	Improve skill ranks, refine equipment, build a second role option or counter, and participate in specialized modes.	Costs can create choices between upgrading the current squad and broadening the roster. Advanced materials may be specialized, but core upgrades need a dependable path.	Every newly obtained hero competes with the entire main team for full-cost resources, so testing substitutes becomes irrational.
Late game — optimization and breadth	Long-horizon investment, advanced gear choices, marginal skill improvements, sidegrades and collection goals.	Use deeper sinks for optional optimization, while maintaining visible recurring progress and viable counters. Keep high-end costs transparent and distinguish a prestige goal from a core requirement.	Escalating costs consume every source, idle income becomes meaningless, or vertical investment outpaces the available counter / roster design.

Stage definition. “Early”, “mid” and “late” should eventually be anchored to campaign checkpoints and system unlocks, not arbitrary calendar days alone. Account age is useful for cohort analysis later, but it is not a reliable substitute for actual player progression.

34.8 Paper affordability model before any numeric balance pass

For each resource r and planning window W, calculate dependable supply separately from one-off rewards:

RecurringSupply_r(W) = Idle_r(W) + RepeatableActive_r(W) + RecurringObjectives_r(W) + ScheduledEventBaseline_r(W)

One-time campaign first-clears and unique milestones should be recorded in a separate column, not silently included in recurring monthly income. Random drops should be shown as variable income, not the only dependable source of a required resource.

A first-pass affordability estimate for a multi-resource goal G is:

TimeToAfford(G) = max over required resources r of max(0, Cost_r(G) - Balance_r) / ReliableIncomePerDay_r

This is an accounting tool, not a simulator result. It assumes a defined target player profile and reliable income rates; if a required resource has no dependable path, the estimate is effectively unbounded and that dependency must be reviewed. When several upgrades compete for the same resource, the budget must include their shared demand rather than treating all income as freely available to every goal.

- Budget windows to prepare in the next pass: seven days for early upgrade milestones; 28 days for a recurring progression and recruitment budget; and one campaign checkpoint interval to distinguish recurring supply from first-clear rewards. These are proposed accounting windows, not finalized player-session expectations.
- Prepare at least three theoretical profiles: idle-focused; regular active; and highly engaged. The comparison asks whether each profile can make useful progress, what optional acceleration active play provides, and whether extra engagement creates an unfair competitive gap. No player outcomes should be claimed from paper calculations.

34.9 Hypothetical edge cases and required responses

Paper scenario	Expected behavior under the proposal	Failure condition to watch for
Campaign wall for several days	Idle baseline still adds useful core resources even though stage-based rate increases stop. A player can make a modest level / gear / skill improvement.	If passive rewards do not support any meaningful action, the wall becomes a hard stall; if they bypass every active mode, specialized modes lose purpose.

Paper scenario	Expected behavior under the proposal	Failure condition to watch for
New counter character is acquired	The account can try its baseline role without first fully rebuilding every resource track. Optional upgrades improve reliability over time.	A counter exists on paper but remains unusable for a long time because all EXP / currency / skill materials are trapped in the current lineup.
Early gear is replaced	The policy (low early cost, partial salvage or transfer) is clear before investment.	Resources are repeatedly burned on near-disposable equipment with no visible warning or recovery path.
Only one game mode supplies an essential material	A guaranteed baseline or secondary path is specified before the mode becomes a required gate.	A locked mode, difficult boss or competitive rank freezes account development.
An upgrade system gets more expensive after revision	Review costs and supply together; communicate the revision and evaluate compensation / grandfathering if prior balances lose value.	A cost increase is shipped conceptually without asking whether the corresponding resource income also needs to rise.
Player misses an event or several daily tasks	Permanent core access remains available through milestones, reruns or alternate routes; optional event cosmetics may remain exclusive if clearly disclosed.	A brief absence permanently destroys a featured guarantee plan or removes the only accessible counter.

These are paper scenarios used to expose logical gaps. They are not simulation outputs, live observations or measured player results.

34.10 Current proposal register and open decisions

Decision area	Current status	What remains to be decided
Resource sink assignments	Provisional register in 34.4	Review against the actual named upgrade systems before final costs.
Character level-cost curves	Open	Define per-resource cost bands and expected time-to-afford for early / mid / late checkpoints.
Skill material costs and rank effects	Open	Separate baseline functionality from optimization; specify the material families and which ranks unlock major effects.
Promotion / cap gates	Open	Choose what is required at a threshold and how players access its material outside one restrictive mode.
Equipment slot / tier progression	Open	Define slot count, item lifespan, milestone selectors, enhancement transfer / salvage policy and high-end optional sinks.
Reset / respec / investment recovery	Open	Compare full refund, partial refund, transferable investment and shared-level infrastructure. AFK Arena is a benchmark example only, not our chosen rule.
Recurring income and target windows	Open	Quantify idle, repeatable-active, objective and event baseline supply only after the cost register is reconciled.
Summon / fragment budget	Open; prior illustrative targets conflict	Keep the historical 45–50 recurring pulls per 28 days and 60 pull-equivalents per 28 days examples visibly unresolved; reconcile them with banner cadence, 70-pull guarantee and 100-fragment selector proposals in the dedicated economy pass.
Event / mode token sinks	Open	Mark each item essential, supplemental, optional or cosmetic; define expiry, reruns and alternate access.
Cost-change compensation policy	Open	Specify how historical balances and announced costs would be handled before any future live economy change.

34.11 Interim conclusion and next research task

Interim conclusion [RESEARCH SYNTHESIS]: the current direction is a narrow set of recurring core-growth sinks, specialized but accessible secondary sinks, clear milestone costs and an explicit separation between baseline functionality and optional optimization. Costs, reliable income and progression windows must be co-designed and tracked in a paper budget. The AFK Arena example shows why several resources and milestone costs can be tightly

coupled; AFK Journey's documented seasonal changes show that increased costs should trigger a review of drop rates and reward certainty. These are design lessons, not a prescription to copy either game.

Status: the first paper cost-versus-income framework and roster-progression integration are now recorded in Section 35 (Version 0.10.6). No actual game prices or rates are locked. Next research task: reconcile the core-progression budget with standard summons, universal fragments, featured-banner cadence and the unresolved recurring-pull targets; then extend the same budget to active-mode, event and endgame sinks.

34.12 Sources for resource sinks, upgrade costs and progression pacing (accessed 29 October 2026)

- [IR23] AFK Global, "Beginner guide to the first weeks, feature unlocks and early spending." Community-maintained AFK Arena guide, published 21 October 2026 in the opened page. Used for a current example of campaign-scaled AFK resources, separate Gold / Hero EXP / Hero's Essence costs, selected essence breakpoints, five-hero leveling / Resonating Crystal, and early reset rules. Not an official publisher specification; exact in-game values are patch-sensitive. https://www.afk.global/afk-arena/guides/beginners-guide
- [IR21] Farlight Games, "Talk with Designers: New Season Preview." First-party AFK Journey designer communication describing season-specific changes to upgrade-material timing, Select Equipment Chests and increased hero-upgrade material drops after development costs rose. Supports the principle of revisiting supply when costs change, not a recommendation to copy the exact seasonal system. https://afkjourney.farlightgames.com/news/a0ae0b8ad8c35da97805079b672b7287
- [IR24] Unity, "Building an in-game economy." Unity Gaming Services guide describing resources, sources, sinks and the need to consider their combined influence on progression. General design reference, not idle-RPG-specific balance data. Accessed 29 October 2026. https://unity.com/how-to/building-game-economy-guide-part-2
- [IR25] Game Developers Conference, "Economic Balancing and Improved Monetization Through Clever Sink Design," Vili Lehdonvirta, Free to Play Design & Business Summit, GDC 2014. Session overview describes currency and item sinks, trade-offs and spreadsheet-based analysis. Older general design reference; used for method only. https://www.gdcvault.com/play/1021008/Economic-Balancing-and-Improved-Monetization

Source-confidence note. IR23 is a recent specialist/community guide, not official publisher data. IR21 is first-party but documents a particular season revision. IR24 is a general Unity economy guide; IR25 is an older GDC session description. The sink register and progression framework above are our project-specific proposals and inferences. Exact costs, rates, target durations, refund policy and fairness outcomes remain unvalidated.

35. Cost-versus-income model and roster-progression integration (Version 0.10.6)

Purpose and status. Section 34 defined the first-pass sinks and cost behaviors; this chapter connects those costs to dependable resource income and to a practical roster-progression design. The benchmark figures below describe other games, not this project. Proposed implementation patterns and example values are labelled as hypotheses. No final game prices, rates, wait times, or roster-sync rules are approved by this chapter; no simulator or live test has been performed.

35.1 New research questions

- For each major upgrade, how many separate resources must be present at once, and which resource is expected to be the limiting factor?
- How much of a player's dependable daily or weekly income should go toward essential team development, and how much should remain available for choice and specialization?
- Can a free-to-play player use a newly acquired tank, healer, debuffer or counter without paying the full cost of rebuilding a character from level one?
- How should the economy change from first-session onboarding to midgame roster-building and endgame optimization without making the resource curve feel like arbitrary exponential inflation?

35.2 Benchmark findings and evidence limits

Finding A [SOURCE FINDING; specialist guide]. AFK Global's leveling guide, published 21 October 2026, describes AFK Arena as charging Gold and Hero EXP on every level, with Hero's Essence concentrated at specific level breakpoints that also unlock skill ranks. Its published totals make the relative cost growth visible: one hero from levels 1-100 is listed at about 1.95 million Gold, 1.09 million EXP and 900 Essence; the 200-240 band is listed at about 21.80 million Gold, 76.46 million EXP and 70,000 Essence. The page is a current specialist/community guide rather than a publisher-owned economy specification, so use those values as a documented example, not as a universal benchmark [ECO1].

Design implication [INFERENCE]. The limiting resource can change by progression band: EXP may dominate ordinary growth while a threshold material gates a skill unlock. A single "upgrade power" score is therefore insufficient; the paper budget must forecast each resource separately and identify the slowest requirement.

Finding B [SOURCE FINDING; official publisher communication for a specific season]. AFK Journey's Talk with Designers: New Season Preview describes replacing random AFK-stage equipment drops with Select Equipment Chests, moving several one-time stage rewards into a seasonal handbook, making some progression materials available earlier, and increasing hero-upgrade material drop rates because the updated development systems would increase hero-development costs [ECO2]. This records one seasonal change, not an evergreen rule for all versions.

Design implication [INFERENCE]. When a new system increases the cost of development, evaluate its supply routes and timing in the same design pass. Choice rewards can reduce bad-luck frustration for important equipment slots, while one-time milestones should not be counted as sustainable recurring income.

Finding C [SOURCE FINDING; general developer guidance]. Unity's economy guidance defines the task as balancing sources and sinks, taking player engagement differences into account, and keeping early progression sufficiently smooth. It also identifies starting balances, periodic rewards and progression costs as variables to test after a working implementation exists [ECO3, ECO4].

Project boundary. Unity's later testing advice is not authorization to run tests now. For this project, use accounting, paper examples and hypothetical player journeys only until the owner explicitly changes the research/concept phase.

35.3 Cost-versus-income accounting method

Calculate dependable income separately for each resource. Let D(r) be dependable daily income for resource r: ordinary idle accrual + a conservative allocation from repeatable active play + daily rewards + weekly rewards divided by seven. Keep first-clear rewards, limited events, one-time compensation, lucky drops and paid purchases in separate columns; they may accelerate progress, but should not be necessary for the baseline affordability promise.

For a proposed purchase, let C(r) be its cost, B(r) the player's current balance, and S(r) the dependable daily surplus available after previously committed mandatory spending. If S(r) is positive, an initial time-to-afford estimate is T(r) = max(0, C(r) - B(r)) / S(r). When the upgrade requires multiple resources simultaneously, its minimum estimated wait is controlled by the slowest required resource: T(upgrade) = max over all required r of T(r). This estimate assumes future income is stable and does not yet model cost increases or spending interruptions; those must be added in later paper journeys.

Accounting field	How to calculate / classify it	Why it matters
Dependable recurring income	Idle income + conservative repeatable activity income + daily rewards + weekly rewards / 7.	Defines the baseline without relying on one-time rewards or lucky drops.
Variable / milestone income	Campaign first-clears, random bonus drops, short events, compensation, purchases.	Tracked separately so a one-time spike is not mistaken for repeatable progress.
Committed spend	Expected costs already allocated to baseline team/functionality upgrades.	Prevents the same daily resource from being promised to several simultaneous systems.
Daily surplus	Dependable recurring income minus committed baseline spend reserved elsewhere.	If zero or negative, the proposed plan cannot fund the purchase without changing priorities or adding an accessible source.
Bottleneck resource	The required resource with the longest time-to-afford estimate.	Shows which cost or source should be adjusted if the intended pacing is missed.

Accounting field	How to calculate / classify it	Why it matters
Reserve / optionality	Resources deliberately left uncommitted for experiments, replacements and future choices.	Avoids an economy that is technically balanced but leaves players unable to respond to a new counter need.

35.4 Illustrative calculation (not a game balance value)

Purely hypothetical example. An upgrade costs 300 Ryo and 600 Shinobi EXP. The player already has 100 Ryo and 150 EXP. After accounting for other baseline commitments, the model estimates a dependable surplus of 80 Ryo per day and 300 EXP per day.

Resource	Cost	Current balance	Amount still needed	Dependable daily surplus	Estimated time
Ryo	300	100	200	80 / day	2.5 days
Shinobi EXP	600	150	450	300 / day	1.5 days

Because both resources are required, the upgrade is estimated to take 2.5 days, not 2.5 + 1.5 = 4 days. Ryo is the bottleneck. If the desired pacing were faster, legitimate options would include lowering that upgrade's Ryo fee, raising a dependable Ryo source, providing a clearly signposted milestone reward, or removing a redundant Ryo tax elsewhere. Adding another simultaneous rare-material requirement would normally make the wait worse. These figures only demonstrate the calculation; they are not recommended prices or income rates for the future game.

35.5 Proposed future-game integration: shared baseline plus individual specialization

The main game-design question is how to make roster experimentation affordable without removing all meaningful character investment. AFK Arena's Resonating Crystal illustrates a shared-level approach: a small set of trained heroes establishes a level that can be shared with additional roster members [ECO1]. This project should not copy its exact rules, roster count, unlock stage or cost. The relevant transferable idea is that the game can separate the cost of maintaining a baseline roster level from the cost of specializing individual characters.

Model	What the player pays for	Main advantage	Main risk
A. Fully individual leveling	Ryo and EXP for every character's levels.	Strong individual resource sinks and a clear personal investment path.	High replacement and experimentation cost; a newly acquired counter may be unusable without extensive repeated leveling.
B. Fully shared level	One account/team-wide level investment applies to most or all eligible characters.	Fast roster experimentation and little duplicated leveling spend.	May remove too much individual growth and requires other progression layers to carry the investment fantasy.
C. Hybrid shared baseline + personal specialization [PREFERRED TO STUDY]	A shared training baseline covers ordinary levels after a clear unlock; individual skills, promotion milestones and equipment provide character-specific investment.	New or underused characters become easier to field while strategic specialization retains meaningful costs.	Needs careful handling of promotion caps, gear gaps, and the power difference between baseline and optimized characters.

Working proposal, not owner-approved final: investigate Model C as the leading concept. A future "Shinobi Training Hall" or equivalent account training system could raise a shared baseline through Ryo and Shinobi EXP. Eligible recruited shinobi would reach that baseline once the feature unlocks, while personal skill ranks, equipment, promotion and advanced mastery remain individual investments. Do not copy AFK Arena's exact pentagram/slot mechanic by default. Before selecting this model, compare it on paper against fully individual leveling and the shared-level alternative.

- Keep basic role functionality at the base kit: a character should not need late skill ranks merely to provide the only practical cleanse, interrupt, protection or other essential counter available to the player.
- Use character-specific materials to improve performance, reliability, flexibility or synergy, rather than to sell access to the character's basic job in the team.

- When a training baseline unlocks, explain exactly which stats it provides and what remains individual. Do not make the same level benefit get charged both in the shared training track and again in each character's level ledger.
- If individual levels are retained for the whole game, consider a recoverable early investment or a low-cost baseline-sync option for bench characters. This is an alternative to Model C, not an additional mandatory system.

35.6 Proposed upgrade layers and practical sinks

Layer	Main cost role	Integration rule for the future game
Training / ordinary levels	Ryo + Shinobi EXP; shared baseline is the preferred model to evaluate.	Budget as the routine growth sink. Do not make every other feature charge the same resources by default.
Promotion / advancement thresholds	Common essence and any explicitly designed advancement materials.	Use visible milestones with clear benefits. Check that the required resource is accessible before the player reaches the gate.
Skill progression	Basic skill materials for upgrades beyond the baseline kit; advanced materials for specialization only after further design.	Keep role-defining functionality at baseline. Tell the player which skill effect changes before spending.
Equipment acquisition	Campaign milestone rewards, targeted mode rewards, craft/selector routes where later specified.	Use selection or predictable milestones for important slots; do not leave essential equipment solely to repeated random drops.
Equipment enhancement	Enhancement materials for stat/quality improvement.	Separate item acquisition from item enhancement. Evaluate transfer, refund or salvage so early replacement does not destroy most invested value.
Advanced optimization	High-tier equipment sets, advanced skill investment, sidegrade options and mastery systems if approved.	Can be a longer-term sink, but must not be required for baseline role coverage or core campaign participation.

Equipment policy recommendation [PROPOSAL]. Prefer deterministic enhancement costs rather than failure-based destruction in the baseline version. If gear is expected to be replaced, choose one transparent protection rule—low early costs, transfer of enhancement, or partial salvage. Do not choose a refund percentage, gear-tier count, upgrade success rate or enhancement price until equipment lifespan and supply have been mapped.

35.7 Candidate pacing envelopes for later paper review

The windows below are working hypotheses for affordability modelling, not industry facts, launch promises or approved final balance settings. Use them as test questions for the paper budget; revise them if the game loop, session design or resource yields point elsewhere.

Player journey / purchase	Candidate planning envelope	Pass condition to investigate
First-session starter team	Several clearly taught upgrades affordable from starting grants within the tutorial / first session.	The player can learn team roles and attempt a coherent first squad without grinding or needing a lucky pull.
Routine core-team growth	A meaningful baseline upgrade roughly every 1-2 daily cycles for a regular active F2P player; an idle-focused player blocked by campaign should still have a meaningful core-growth step within about 2-3 days.	At least one relevant progression track can move from dependable sources; the slowest resource is identified and no paid source is required.
Make an owned alternate/counter character fieldable	Investigate a roughly 3-7 day regular-play planning band for baseline readiness, separate from the time needed to obtain that character.	Players can prepare a counter or alternate role without dismantling the primary team or waiting for an exclusive short event.
Endgame optimization / niche sidegrade	Multi-day to week-scale or longer goals may be acceptable where the reward is incremental and optional.	The grind adds mastery or customization, not the only access to an essential role, cleanse, protection or counter.

Reason for using ranges rather than exact prices: Unity's guidance supports making player-engagement differences explicit and evaluating source/sink curves across the journey [ECO3, ECO4], but neither Unity nor the benchmark games establish universal wait-time targets for this project. These bands are design hypotheses to compare on paper, and they must not be described as measured player behaviour.

35.8 Resource safeguards and edge-case audit

- Resource dominance: if one resource controls almost every upgrade, lowering its income can stop the entire game. Keep the dominant bottleneck visible and give it dependable sources; avoid hidden multi-resource tolls at every step.
- Parallel commitments: do not promise the same EXP or Ryo stream to leveling, gear enhancement, skill ranks and shop purchases simultaneously. Mark core commitments and optional spending separately in the budget.
- One-time reward illusion: starter grants and first-clears may make early affordability look healthy. Recalculate after those rewards end to confirm the recurring economy can sustain repeat needs.
- New roster acquisition: simulate on paper a player pulling a tank or cleanse counter after the main team is already highly invested. The player should be able to equip and field the counter without an unbounded duplicated leveling bill.
- Equipment replacement: consider a player replacing a starter weapon soon after spending enhancement materials. Apply the declared transfer/salvage policy consistently and avoid surprise loss of most investment.
- Low engagement / campaign wall: use the guaranteed idle floor and accessible recurring routes, not repeated high-rank PvP, perfect guild activity, daily streak perfection or paid refills, to sustain basic growth.
- Cost increase after a feature change: any new upgrade tier must trigger a paired review of supply, milestones and affordability. Preserve historical costs in the decision log and decide explicitly whether compensation or transition protection is needed.
- F2P and payer comparison: optional spending may shorten waits or broaden convenience, but it must not convert an essential role or counter from an attainable target into a pay-only requirement.

35.9 Current proposal register and open decisions

Question	Current status	Required next step
Use a shared roster baseline?	Hybrid shared baseline + individual specialization is preferred to study; not a locked project decision.	Compare the three models through hypothetical journeys: starter account, new counter acquisition, low-engagement return and endgame roster expansion.
What is the required income per day?	Open; no income rate selected.	Populate resource-by-resource dependable income using the 17-family source map and distinguish guaranteed, variable and one-time sources.
What is the intended wait time for upgrades?	Candidate planning windows in Section 35.7 are hypotheses only.	Check them through paper accounting after the shared baseline / individual level model is selected.
How should gear investment be protected?	Transfer, refund or salvage are alternatives; no percentage or exact rule selected.	Define gear lifespan, replaceability, selector routes and whether gear carries character identity or can move freely.
How does this affect summons and fragments?	Not yet reconciled; prior conflicting recurring pull targets remain unresolved.	Build a separate acquisition ledger and then reconcile the combined weekly/monthly F2P budget, selector thresholds, banner cadence and guarantee carry-over.
When can economy tests run?	Not authorized in current phase.	Only proceed after explicit owner approval; future implementation/testing gates remain separate from this paper model.

35.10 Interim conclusion and next research task

Interim conclusion [RESEARCH SYNTHESIS]. The game should budget progression in time-to-afford, not only in raw cost numbers. Recurring dependable income, one-time milestones and variable rewards must be kept separate. Every multi-resource upgrade must be checked against its slowest required resource. The leading roster concept to study is a shared

training baseline with character-specific specialization, because it may reduce duplicated leveling costs and make counters easier to field; this remains a proposal until compared to alternatives. Gear should use visible costs and a clear investment-protection policy. Candidate timing windows remain hypotheses, not validated balance.

Next research task: reconcile this core-progression model with standard summons, universal fragments, featured-banner cadence, the proposed 70-pull guarantee, 100-fragment selector, and conflicting illustrative F2P pull-income targets. Then connect the combined budget to repeatable mode rewards, events and endgame sinks. Do not finalize prices or rates before the combined budget is internally consistent. Continue research and hypothetical paper journeys only; no prototype, simulator, seeded run or player test without explicit authorization.

35.11 Sources for cost-versus-income and roster-progression integration (accessed 29 October 2026)

[ECO1] AFK Global. “Hero leveling costs, level caps and the Resonating Crystal.” Published 21 October 2026 on a specialist/community guide site. Used for the guide’s current displayed example of multi-resource leveling costs, breakpoint-based Essence costs and shared-level roster concept; not treated as official publisher documentation. https://www.afk.global/afk-arena/guides/leveling

[ECO2] Farlight Games / AFK Journey. “Talk with Designers: New Season Preview.” First-party designer communication for a specific seasonal update. Describes Select Equipment Chests replacing random stage equipment drops, moving some milestone/material rewards earlier, and raising hero-upgrade material drop rates because development costs were increasing. These details are specific to the described change. https://afkjourney.farlightgames.com/news/a0ae0b8ad8c35da97805079b672b7287

[ECO3] Unity. “Building an in-game economy.” General developer guidance on identifying resource sources and sinks and balancing their effects on progression. https://unity.com/how-to/building-game-economy-guide-part-2

[ECO4] Unity. “Designing a balanced in-game economy.” General developer guidance on engagement profiles, smooth early progression, player journeys and later validation of cost/reward variables. https://unity.com/how-to/design-balanced-in-game-economy-guide-part-3

Evidence boundary. [ECO1] is a specialist guide and includes game-specific current values; [ECO2] is first-party but describes one season’s planned changes; [ECO3-ECO4] are general design guidance. The shared-training proposal, affordability equations as applied to this project, candidate target windows and all future-game integration recommendations are project inferences/proposals, not benchmark facts.

36. Summon guarantees, universal fragments and F2P roster-acquisition budget (Version 0.10.7)

Purpose and status. This chapter reconciles the project’s earlier 45-50 pull-equivalent target, the illustrative 60 pull-equivalents per 28-day gross budget, the proposed 70-pull featured guarantee, the 100-fragment standard-pool selector, and the tentative 4-6 week new-hero cadence. It adds explicit accounting categories and source-grounded benchmark comparisons. All Naruto Idle RPG values remain proposals unless labelled as arithmetic from an existing proposal. No prototype, simulator, seeded run or player test was performed; the Library master was not modified.

36.1 Research questions and accounting units

- Can the proposed featured guarantee be reached from recurring, featured-eligible income on a cadence consistent with new-hero releases?
- Are standard recruitment pulls, featured-banner pulls and universal fragments separate budgets, fungible currencies, or overlapping ways to spend the same income? This must be explicit before income can be described as “pulls per month.”
- Does the 100-fragment selector give enough predictable roster breadth without creating more characters than players can reasonably prepare for battle?
- Which guarantee, counter carry-over, expiry and pool-entry rules keep recruitment transparent and reduce bad-luck dead ends?

Three accounting fields are required even if the eventual game uses a single shared currency: (1) general/standard recruitment value; (2) featured-eligible pull value, including whether any general currency can be redirected to a featured banner; and (3) universal-fragment income. Each reward source must be counted once and tagged as dependable recurring, repeatable but effort-dependent, variable/seasonal, or one-time/progression-triggered. “Pull-equivalent” is a planning unit, not proof that a player can spend that value on any banner.

36.2 Reconciliation of the two earlier pull targets

Internal arithmetic [PROJECT-DOCUMENT AUDIT]. Section 21.2’s existing 60-unit budget is made from 8 login/attendance + 14 daily activity accumulated through weekly completion + 12 weekly objectives + 10 campaign/tower milestones + 10 monthly event participation + 6 guild/social/rotating activity. The first three rows total 34; adding guild/social activity gives 40; adding the monthly event row gives 50; adding the campaign/tower milestone row gives 60. The document itself describes campaign/tower rewards as partly one-time or progression-driven. They must not be counted as a dependable recurring floor once the player has exhausted relevant milestones.

Budget layer	Existing target arithmetic	Interpretation for planning
Routine objectives	8 + 14 + 12 = 34	Routine-source subtotal, conditional on ordinary completion; not a universal guaranteed reward independent of activity.
Repeatable social / rotating activity	34 + 6 = 40	Adds effort-dependent income; test accessibility for players who do not rank highly or belong to top guilds.
Monthly event rewards	40 + 10 = 50	A plausible upper recurring/periodic target if an event is available and its cumulative participation path is accessible. Event schedules and completion rates remain unverified.
Campaign/tower milestones	50 + 10 = 60	Gross period value that includes progression-triggered or one-time rewards. Do not use as the long-run recurring floor.

Recommended interpretation [PROPOSAL]. Preserve both existing figures while assigning them different meanings: treat 45-50 as the provisional recurring/periodic acquisition target to examine after rewards are source-tagged, and treat 60 as a gross progression-window target that can include milestone value. This resolves the arithmetic ambiguity without silently pretending every source recurs. It does not yet solve the banner-eligibility question: the amount of that budget usable toward featured pity remains open.

Important caveat. The 34 routine subtotal, 40 with social activity, and 50 with events are arithmetic from the current project’s illustrative rows; they are not verified game income, measured completion rates or promises. If the event line is not offered in every 28-day window, or some objectives require attendance beyond the target profile, the dependable realized total can be lower.

36.3 Current benchmark findings: guarantees, carry-over and selection

Finding A [FACT — official publisher help page, accessed 29 October 2026]. HoYoverse’s Wish guarantee article states that a Genshin Impact Character Event Wish guarantees a 5-star character within 90 wishes; the first 5-star has a 50% chance to be the promoted character, and a non-promotional result makes the next 5-star guaranteed promotional. The two Character Event Wish banners share pity and guarantee status, and progress carries between them. The page also distinguishes weapon-banner rules from character-banner rules [SUM1]. This illustrates why rarity pity and featured-character guarantee must be described separately and why counter scope must be explicit.

Finding B [FACT — official publisher board rules, accessed 29 October 2026]. Neverness to Everness (NTE) states that its Limited Boards share an S-class character pity counter, Points Gift progress and Board Modification progress. After 70 consecutive rolls without the featured S-class, the board moves to a modified/rate-up state; if the character has still not been obtained, roll 90 automatically grants the featured character [SUM2]. This is a useful wording warning: “rate-up after 70” is not the same as “featured character guaranteed by 70.” The project’s 70-pull proposal is the latter, more generous mechanic and must be written unambiguously.

Finding C [SOURCE FINDING — current specialist/community guide, accessed 29 October 2026]. AFK Global's AFK Arena summoning guide, published 21 October 2026 in the accessible page, reports a guaranteed Elite-tier hero at each 100 Tavern-token milestone and describes a Hero Choice Pack that redirects eligible Elite results to the player's chosen hero, subject to monthly chances and progression unlocks [SUM3]. The milestone reward is not itself a promise of a specific featured hero, and the guide is not publisher-owned. Use it as a structural example of combining a general-pool milestone with a separate targeting mechanism, not as proof that the same rates or access rules are ideal here.

Synthesis [INFERENCE]. Recruitment systems can offer (a) rarity guarantees, (b) featured guarantees, (c) cumulative milestone rewards, and (d) deterministic selector paths. They solve different player problems. For this project, deterministic featured access and a universal-fragment standard selector are complementary rather than interchangeable, provided their budgets are independently tracked and neither is the sole route to an essential role or counter.

36.4 What a 70-pull featured guarantee actually costs in time

For a hard guarantee at 70 featured-eligible pulls, the maximum saving time before one featured copy is affordable is: DaysTo70 = 70 × 28 / FeaturedEligiblePullsPer28Days. This is a deterministic affordability calculation, not the expected number of pulls if the featured character can arrive earlier. It also assumes the player does not spend this budget on another banner family.

Featured-eligible pulls per 28 days	Days to save 70 pulls	Meaning
45	43.6 days	Near a six-week target cadence only if essentially all of this income can be used on featured recruitment.
48	40.8 days	Matches the earlier example; still assumes all counted pulls are featured-eligible.
50	39.2 days	About 5.6 weeks at this full featured-eligible rate.
60	32.7 days	Optimistic gross-rate illustration; the current 60-source table includes milestone value and is not a sustainable floor.

Desired one-featured-guarantee cadence	Featured-eligible pulls required per 28 days	Budget implication
Every 28 days (4 weeks)	70	Requires 70 dependable featured-eligible pulls per 28 days; current 45-50 total target cannot support this by itself.
Every 35 days (5 weeks)	56	Requires more than the 45-50 recurring gross target unless extra featured-eligible sources are added.
Every 42 days (6 weeks)	46.7	At least about 47 featured-eligible pulls per 28 days are needed if starting with no saved pity and no other carryover surplus.

Sensitivity note [ARITHMETIC]. The "39-44 days" result is a best-case for a 45-50 monthly budget only when every counted pull can be spent on that featured-banner family. If only half of a 50-pull gross budget is featured-eligible (25 per 28 days), saving 70 takes 78.4 days. Therefore total monthly summon-equivalents and featured pity income must never be presented as the same measure until the currency and source allocation is fixed.

Recommended reading of the existing 4-6 week release proposal [PROPOSAL]. The 70-pull guarantee does not imply an F2P player should obtain every newly released hero. A four-week release cadence requires a much larger featured budget than the current target; a six-week cadence is closer to 45-50 only if most or all of that recurring target is featured-eligible. Keep release cadence a hypothesis until the banner budget is resolved. Ensure gameplay content remains counterable with existing, standard-pool or deterministic-access alternatives.

36.5 Universal fragments and targeted standard-pool access

Existing project proposal: 15-20 universal hero fragments per week; 100 fragments select one eligible standard-pool hero. Arithmetic: 100 ÷ 20 = 5 weeks (35 days); 100 ÷ 15 ≈ 6.67 weeks (46.7 days). The corresponding 28-day yield is 60-80 fragments if those weekly sources are consistently available and completed. These are planning outputs from project proposals, not measured game values.

Design implication [INFERENCE]. If the featured budget were also sufficient for one targeted copy every roughly 39-44 days, the current fragment target could provide a second targeted standard-pool acquisition every roughly 35-47 days. That may support roster breadth, but it could also cause acquisition to outpace preparation costs, skill materials or equipment. The v0.10.6 shared-training baseline proposal is therefore linked to, not independent from, the acquisition budget. Model acquisition and fieldability together.

- Selector scope: keep the current 100-fragment selector scoped to eligible standard-pool heroes unless a separately budgeted rule is approved. Do not quietly include current limited/featured heroes in this pool.
- Source resilience: spread the 15-20 weekly target across accessible repeatable activities, cumulative weekly objectives and deterministic shop/achievement paths. No essential part of the amount should depend solely on top-ranked PvP or a high-ranking guild.
- Duplicate handling: duplicates may contribute to fragments as a variable bonus, but do not count uncertain duplicate conversion toward the dependable weekly fragment floor. A base copy must still contain the hero's core tactical identity and essential counter function.
- Expiry and stockpiling: define whether fragments persist indefinitely; the working fairness direction is that universal fragments and earned selector progress should not expire unexpectedly. Featured pity/spark progress is separately proposed to carry across banners in its family and not expire.

36.6 Recommended acquisition architecture for the future game

Working architecture [PROPOSAL FOR OWNER REVIEW; NOT FINAL]. Use a three-ledger recruitment economy while maintaining clear UI explanations even if the implementation eventually uses a fungible currency:

Ledger	What it measures	Working policy / guardrail
A. Standard recruitment	Summons or equivalent value usable in the general pool; progress to general-pool milestone rewards.	Maintain useful general roster growth. State which sources and currencies qualify. Do not count these toward featured pity unless they can actually be spent on that featured banner.
B. Featured-eligible budget	Pulls actually usable on the featured banner family and the 70-pull featured guarantee counter.	The 70-pull proposal guarantees the named featured copy by the 70th eligible pull, with no 50/50. Carry counter and featured-guarantee status across banners of the same family/reruns; display progress and rules. Keep this an auditable budget, not random idle-ticket drops.
C. Universal fragments	Fragments toward a selector for an eligible standard-pool hero.	Working proposal: 100 fragments per selector and 15-20 per week across several accessible sources. Separate its income and spending from both pull ledgers.
D. One-time / variable supplements	New-player grants, campaign/tower milestones, event-specific rewards, compensation and activity-dependent rewards.	Track separately. Do not use first-clears as recurring income. Mark seasonal event timing and completion requirements. Count a source once only.

Banner UI requirement [PROPOSAL]. Show the exact featured character, rates by rarity and the featured result, remaining pulls to the hard featured guarantee, which banners share that counter, what resets it, whether ticket/value sources qualify, expiry/carry-over of tokens, and which selector pools exclude limited heroes. "Pity" is not precise enough without stating whether it guarantees any high-rarity unit or the featured character itself.

Monetization tie-in [EXISTING PROJECT PRINCIPLE]. Spending may improve convenience, cosmetics or roster breadth earlier, but must not sell the only functional cleanser, guard, disruptor, anti-control tool or other counter to a common strategy. Limited-first-release characters require a deterministic later access route or an accessible functional equivalent. Duplicates must not be required to unlock the base kit or a mandatory synergy.

[SOURCE-ALLOCATION FOLLOW-UP ADDED IN v0.10.8] Section 37 now proposes how recruitment sources can be assigned to standard-pool access, featured-guarantee spending and universal fragments while keeping the number of currencies low. This is a new proposal for owner review, not an approved change to the earlier source-allocation map.

36.7 Paper acquisition profiles and failure cases

Paper profile / scenario	Question to calculate	Pass condition for the concept
Routine F2P with complete weekly objectives	How much featured-eligible value and how many standard pulls are earned over 28 and 42 days, excluding first-clear rewards?	Guarantee timing is calculated from featured-eligible value only; standard and featured progress are not conflated.
Player skips some daily tasks	Does weekly catch-up preserve most planned recruitment income?	A missed day does not permanently destroy guarantee access; the lower profile's cadence is shown honestly.
Player spends on standard recruitment	How much longer until a featured target if the player divides fungible currency or uses standard tickets?	UI makes the opportunity cost visible; no claim that a full featured guarantee is still funded on the original schedule.
Campaign/tower first-clear month	How much does progression milestone income inflate the apparent monthly total?	The first-clear bump is labelled one-time/progression-driven and is not projected into future months.
Unlucky featured-banner player	Can the player still reach the 70th featured-eligible pull and is the counter preserved across a rerun?	The featured copy is guaranteed by the stated threshold, with no hidden 50/50 and no unexpected counter reset.
New featured hero fills a missing counter role	Is there an existing or deterministic alternative if the player does not target the banner?	No boss or expected PvP archetype requires that limited hero as the only functional answer.
Fragment income at lower/upper target	How long to accumulate 100 fragments, and can the player prepare the selected hero?	Selector timing and readiness costs are budgeted separately; the selector is not assumed to provide free equipment/skill materials.
High duplicate luck	Do extra copies inflate fragments or power beyond the baseline acquisition model?	Duplicates are a variable upside, not needed for viability or assumed in the dependable free budget.

All profiles above are paper-accounting scenarios for the research phase only. They are not a simulator run, seeded pull result, measured completion rate, probability estimate or player test. Where a quantity is unknown, record a variable rather than inserting a guessed rate.

36.8 Proposed reconciliation and unresolved decisions

Question / topic	Working status after this research	What remains before finalization
45-50 vs 60 pull-equivalents	Proposed interpretation: 45-50 is a recurring/periodic gross target to examine; 60 is a gross window including progression milestones. Not yet a final value.	Rebuild the income ledger source by source and assess routine, social, event and new-player cohorts. Preserve the old values in the historical decision record.
Featured pull entitlement	Open. Existing "standard/featured" terminology is too broad to calculate featured guarantee cadence.	Specify standard-only, featured-eligible and fungible sources; calculate a 28-day and 42-day featured budget before promising time to target.
70-pull featured guarantee	Existing project proposal remains: no 50/50 failure step, guaranteed featured by 70 eligible pulls, carryover within the same featured banner family, no expiry for spark progress.	Owner approval remains open; benchmark comparison does not validate its economics. Test the cost only in paper budgeting until live-test phase is authorized.
Banner release cadence	4-6 weeks remains a hypothesis. A 70-pull guarantee every 28 days requires 70 featured-eligible pulls per 28 days; every 42 days requires about 46.7.	Choose whether the game expects players to target every release or selectively target new units. Ensure counters and roles are not limited-only.
100-fragment selector / 15-20 per week	Existing proposal gives a selector every about 35-46.7 days if earned consistently.	Check it together with the featured budget and preparation costs. Decide selector pool-entry timing, expiry and duplicate conversion.

Question / topic	Working status after this research	What remains before finalization
Standard vs featured pool structure	Separate accounting ledgers recommended. Final currency architecture remains open.	Compare shared spendable currency versus separate ticket sources on player agency, budget clarity and how easy it is to ensure predictable featured access.

36.9 Interim conclusion and next research task

Interim conclusion [RESEARCH SYNTHESIS]. The two historical pull targets can be reconciled only at the gross-source level: in the current 60-row example, excluding 10 campaign/tower milestone value yields 50; source availability and completion still vary. However, neither 45-50 nor 60 tells us how many pulls can actually be spent on featured recruitment. The next budget needs separate standard, featured-eligible, universal-fragment and one-time/variable ledgers. With 45-50 featured-eligible pulls per 28 days, accumulating a 70-pull featured guarantee takes about 39-44 days; with only part of the income eligible, it takes longer. A 100-fragment selector at 15-20 weekly fragments takes about 35-47 days. These deterministic calculations are useful constraints, not tested balance results.

Follow-on research task after Section 37: assign conservative, evidence-aware quantities to the source ledger; audit whether the current routine 34-pull-equivalent subtotal is dependable; quantify how much weekly/periodic/event value can truly support featured recruitment; then re-run the paper readiness arithmetic with low, central and high-source-availability cases. Section 37 supplies the first source-by-source allocation proposal but does not finalize rates. Do not run a prototype, simulator, seeded pull or player test without explicit owner authorization.

36.10 Sources for summon and fragment economy research (accessed 29 October 2026)

[SUM1] HoYoverse Help Center. “How does the Wish guarantee system work?” Updated 17 September 2025. Official explanation of Character Event Wish 90-wish 5-star guarantee, first featured 50/50, next featured guarantee after a non-featured result, and shared pity/guarantee between Character Event Wish banners. Used as a structural example of clear counter scope and carry-over; its thresholds are not adopted for this project. https://support.hoyoverse.com/hc/en-us/articles/50333940684953-How-does-the-Wish-guarantee-system-work

[SUM2] Perfect World / Hotta Studio. “Alluring Shadows” Limited Board - NTE. Official board rules dated 21 August 2026. States that Limited Boards share an S-class character pity counter, Points Gift progress and Board Modification progress; rate-up modification follows 70 rolls without the featured S-class and the 90th roll guarantees it if still unobtained. This demonstrates that an intermediate rate-up threshold differs from a hard featured guarantee. https://nte.perfectworld.com/en/article/news/gameevent/20260821/263691.html

[SUM3] AFK Global. “Summon rates, the wishlist and every summon pool in AFK Arena.” Specialist/community guide published 21 October 2026 in the opened result. Describes Tavern summoning and a guaranteed Elite-tier reward at 100 summon-token milestones, plus targeted pool/choice mechanics. This is not a publisher-owned economy spec; exact game rules should be confirmed in the live client before production use. https://www.afk.global/afk-arena/guides/summoning

Evidence boundary. [SUM1] and [SUM2] are official publisher descriptions for the named games and their stated banner families/dates. [SUM3] is a current specialist guide, not an official publisher source. The proposed 45-50 recurring gross target, 60 gross-window interpretation, 70-pull no-coin-flip guarantee, 100-fragment selector, source-ledger structure and proposed cadence are project proposals or arithmetic, not external facts or validated game balance.

37. Acquisition source ledger, recruitment routing and battle-ready character budget (Version 0.10.8)

Purpose and status. This chapter translates Sections 33, 35 and 36 into a source-by-source proposal: which activities support flexible recruitment, which contribute to the 70-pull featured guarantee when spent on a featured banner, which provide universal fragments, and how a recruited character becomes ready for tactical use. It preserves the earlier 45-50 and gross 60 targets as unapproved proposals and does not assign final source quantities. Library synchronization remains pending; this is the working master only. No prototype, simulator, seeded run, or player test was performed.

37.1 Key finding from current benchmark review

The practical lesson is to separate how a player earns a reward from how the player spends it, and to state eligibility rules explicitly. Current NIKKE patch notes (retrieved through a specialist patch-notes mirror on 29 October 2026) describe separate Ordinary Recruit and Advanced Recruit vouchers; Special Recruit eligibility limited to Gems or Advanced Recruit Vouchers; a 200 Gold Mileage Ticket exchange for the featured character; mileage that persists after the banner ends; free pulls that do not earn mileage; and a free SSR plus selector options through event and login routes. The same patch notes distinguish per-banner recruitment milestone counts and specify which pull types count. This is a useful source-accounting example, but because the accessible detailed page is a specialist mirror rather than the publisher-hosted patch page, its precise current mechanics should be checked against the live client before any production use.

The official HoYoverse Help Center describes Primogems as earnable from missions, events and in-game mail while Genesis Crystals are the purchased currency that can be converted to Primogems. A separate official guide explains that Character Event Wish banners share pity/guarantee status: the exact rules are visible and scoped to a banner family, but the named game uses a 90-wish five-star guarantee and 50/50 mechanism that this project is not adopting. These sources support clarity and explicit routing rules, not the idea that our thresholds should match theirs.

AFK Journey's official designer update (5 September 2025) describes moving one-time AFK-stage rewards into a seasonal handbook, replacing random equipment rewards with Select Equipment Chests, increasing hero-upgrade material drops when development costs rise, and returning prior Rate Up heroes. It also describes an event path through which two crossover heroes could be acquired, with an advancement reward for one. These are specific changes/announcements from that season, not evergreen current rules. The transferable lesson is to adjust costs, sources and reward certainty as one system, and to give players predictable ways to revisit important characters.

37.2 Source reliability classes

Class	Meaning for the economy ledger	How it may be counted
A. Dependable recurring floor	Available in ordinary cycles without leaderboard rank, a high-tier guild or a limited event; weekly cumulative paths let players recover missed daily tasks.	May support the conservative 28-day baseline after exact availability and completion requirements are budgeted.
B. Repeatable but effort-dependent	Requires active mode participation, daily/weekly objectives, guild contribution or a repeatable shop exchange.	Count separately from the passive floor; estimate availability for a normal player and a low-activity player.
C. Scheduled / seasonal variable	Events, rotating modes, returner campaigns, anniversary grants or event-store bonuses.	May improve the expected month but is not part of the guaranteed floor unless cadence and minimum reward are specified.
D. One-time / progression-triggered	First clears, new-player onboarding, account achievements, one-time tower milestones or first-time selector rewards.	Count once, by cohort and account progress. Never roll forward as recurring monthly income.
E. Paid	Purchased currency, paid passes or bundles.	Exclude from F2P budgets. Evaluate only as separate acceleration/convenience supply; never define essential counter access by paid income.

A source should be tagged for both reliability and eligibility. For example, a fixed event that occurs monthly can be scheduled but still participation-dependent; a free recruitment attempt may have zero pity/mileage value if the game rules exclude it. The ledger therefore needs at least: source, cadence, user action, audience/eligibility, resource granted, standard/featured/fragment eligibility, recurrence class, and whether the amount is already counted elsewhere.

37.3 Recommended activity-to-reward allocation

Activity / source	Primary proposed reward role	Recruitment / featured treatment	Guardrail and ledger tag
Login calendar + ordinary daily/weekly missions	Core repeatable Recruitment Seals; small predictable fragment contribution where the weekly plan needs it.	Recruitment Seals are flexible: if spent on standard recruitment they do not count toward featured pity; if spent on the featured family they do. Tag as Class A or B after completion burden is checked.	Use weekly cumulative completion and catch-up rather than punishing one missed day. Count once only.

Activity / source	Primary proposed reward role	Recruitment / featured treatment	Guardrail and ledger tag
Repeatable PvE / training / tower participation	Targeted universal fragments and mode-specific growth items; modest flexible Recruitment Seals only where needed to give the activity a recruiting identity.	Any seals awarded can be routed by player choice; do not claim the same source funds both standard pulls and featured pity in the same budget.	Baseline fragment access must not require clearing the wall that the selected counter is intended to answer. Tag as Class B.
Campaign and tower first-clears	Starter selector / role-coverage grant, flexible seals, broad choice chests and one-time account milestone rewards.	Use to teach pools and pity rules, but exclude completely from the recurring 28-day floor.	Tag Class D; separate onboarding cohort from mature-account budget.
Rotating events and event story/challenge tracks	Bonus Recruitment Seals, fragments, choice rewards and selected character trial access.	If an event gives featured-eligible seals, count them as Class C unless an equivalent recurring minimum is guaranteed. No unique essential counter may be event-only.	Prefer cumulative milestones, clear expiry rules, reruns or a permanent alternative.
PvP participation / Arena	A modest fragment or utility reward for participation; seasonal prestige and top-rank rewards are cosmetic-first.	Do not fund the conservative featured guarantee primarily from high-rank placement. A participation seal is Class B; rank-only bonuses are excluded from baseline.	Losses should still count for participation; avoid reward structures that compound rank into permanent power.
Guild / alliance activity	Small participation-based fragment or Recruitment Seal contribution plus mode-specific shop value.	Do not make featured pity dependent on being in a top guild. Count only what ordinary/new guild members can reliably access.	Provide meaningful lower-participation access; Class B, with rank-based extras excluded from floor.
Achievements and account milestones	One-time universal-fragment bundles, role selectors and flexible seals.	Useful to smooth the first roster; never counted as recurring. A selector should make up a genuine role gap, not force a random DPS.	Class D; avoid designing achievements that only high spenders or endgame accounts can reach.
Returner / catch-up track	Bounded flexible seals, universal fragments and core-training resources to reduce the obsolete-content gap.	Do not include in the ordinary active F2P baseline. If returning-player rewards contain an eligible featured seal, show that distinction transparently.	Should accelerate re-entry, not create permanent exclusive power or reward repeated inactivity loops.
Shop exchange	Convert mode-specific tokens into a small, clearly budgeted source of fragments, standard/utility items or flexible seals.	Every exchange has an opportunity cost. Track token generation and shop stock limits separately; do not count the token reward and the purchased item as two summon rewards.	Keep core selection available without a high-rank currency; document stock reset and rollover rules.
Paid purchase / pass	Convenience, cosmetic value and optional time saving; any recruitment value is an explicitly bounded paid layer.	Paid currency is excluded from F2P source sums. Paid/free split and pity eligibility must be visible; purchases must not unlock the only functional role/counter.	Separate monetization model from F2P balance math. Never call paid supply “free monthly income”.

37.4 Recommended low-currency architecture to compare

Working proposal [NOT OWNER-APPROVED]: use one flexible, earnable Recruitment Seal for standard and featured recruitment, plus one persistent Universal Fragment resource. The 70-pull counter is progression state, not a third currency. Every seal can be spent once; if used on standard recruitment it does not advance the featured counter, and if used on a featured banner in the shared featured family it advances that family’s counter. Special event grant labels can describe which banner/pool a seal is eligible for, but avoid creating a new named ticket for every banner unless a clear player-experience need justifies it. This is an economy architecture proposal, not a change to the older resource-source map yet.

Model	Advantage	Risk / cost	Working assessment
A. One flexible Recruitment Seal + Universal Fragments	Few persistent currencies; player chooses standard breadth or featured target; every free seal is auditable.	Players who spend on standard summons delay featured pity; UI must make that trade-off very clear.	Preferred candidate for further paper comparison because the project explicitly values currency simplicity and player agency.
B. Separate standard tickets + featured-only tickets + fragments	Clear source targeting; featured income can be guaranteed independently if supplied.	More currency families, higher chance of confusion, and risk that featured tickets feel scarce or forced toward a schedule.	Viable fallback if the budget shows the flexible model cannot protect predictable featured access.

Model	Advantage	Risk / cost	Working assessment
C. Shared premium currency but separate summon tickets for every banner	Fine-grained event control and source attribution.	Highest complexity; may create expiry/currency conversion friction and too many rules to explain.	Not preferred unless there is compelling gameplay or business evidence.

Important distinction: “separate ledgers” do not require separate currencies. For each source, record total seals earned and then record player spending by banner family; do not assign the same seal to both the standard-pull total and featured-pity total. The displayed 70-pull guarantee means 70 actual eligible featured pulls, not 70 earned seals regardless of where the player spent them. If a banner-specific event excludes free pulls from its reward milestones, explain that separate rule in the event UI instead of assuming all attempts count.

37.5 Worked source reconciliation using existing project values

The current Section 21.2 gross example is 34 routine login/objective pull-equivalents + 6 social/rotating + 10 event + 10 campaign/tower = 60. Section 36 interpreted the first three groups (34 + 6 + 10 = 50) as a provisional periodic gross target and the last 10 as one-time/progression-triggered. Section 37 further separates reliability: 34 is the largest candidate recurring subtotal, but it is not yet proven to be a dependable floor; 6 is participation/rotation-dependent; 10 is event-calendar/participation-dependent; and 10 is one-time. Keep all source values as existing illustrations until the underlying source rows and completion rules are audited.

Paper case	Counted free seals per 28 days	Days to earn 70 seals if all are saved for featured	Interpretation
Routine subtotal only	34	57.6	Illustrative lower-source case; exceeds an 8-week acquisition-plus-readiness window if readiness adds several days.
Routine + social/rotating	40	49.0	Conditional on repeating and completing these sources.
Routine + social/rotating + event	50	39.2	Current provisional 50 total; depends on event availability/participation and assumes every seal is saved for featured use.
Gross including milestones	60	32.7	Optimistic one-time-window illustration only; invalid as a repeatable monthly guarantee claim.

The arithmetic makes the pressure visible: a 42-day guarantee cadence requires about 46.7 featured pulls per 28 days. The routine 34 subtotal alone provides about 34 × 42 / 28 = 51 pull-equivalents over 42 days, short of 70 by roughly 19. The 50-pull provisional window can theoretically reach 70 in 39.2 days only if the social/event rows recur on schedule and the player saves all of those seals for the featured family. Do not promise every-six-week featured access until this floor is rebuilt from source evidence and the player-spending trade-off is explicit.

37.6 Acquisition and battle-readiness must be budgeted together

Acquiring a character is not the same as making that character useful. This project already says a base copy must perform its intended role immediately, duplicates cannot unlock an essential counter, and an alternative archetype should become buildable within roughly 4–8 weeks of ordinary targeted play. Treat that existing target as an end-to-end horizon from the start of a targeted plan to fielding a practical alternative—not an additional 4–8 weeks after the character is acquired. This interpretation should be confirmed in the future economy decision log.

Readiness layer	Proposed rule	Resource / system implication
Base tactical function	The newly acquired base copy has the complete role-defining mechanic, including its essential counter functionality where intended.	Do not gate the only cleanse, anti-control, interruption or protection answer behind duplicates, high-rank awakening or signature equipment.
Battle-ready baseline	Candidate target: an existing-account player can prepare an owned/selected alternative for practical use within about 7 days, assuming the player has the common training and equipment baseline. This is a hypothesis, not a promise.	Use the proposed shared training baseline, reusable generic gear or safe transfer/salvage rules; ensure baseline skill function does not require rare specialized drops.

Readiness layer	Proposed rule	Resource / system implication
Target-to-field-ready horizon	Use the existing 4–8 week target from the start of focused roster-building. For a standard-pool selector at 15–20 fragments/week, 100 fragments take 35–46.7 days; adding at most 7 days for baseline preparation gives about 42–53.7 days (6.0–7.7 weeks).	This route fits the existing target arithmetically if the weekly fragment sources are accessible and the preparation ceiling is maintained.
Featured-target route	If all 45–50 pull-equivalents per 28 days are truly spendable recurring seals, 70 pulls take 39.2–43.6 days; with up to 7 days readiness, the total is about 46.2–50.6 days (6.6–7.2 weeks).	If only the routine 34 subtotal is dependable, 70 pulls take about 57.6 days; plus 7 days readiness becomes about 64.6 days (9.2 weeks), beyond the 8-week target. Source reliability or cadence must change, or the promise must be narrowed.
Advanced optimization	Higher skill ranks, advanced gear sets and duplicate-based sidegrades may take longer after the base character is field-ready.	Keep optimization separate from basic team functionality. Define respec/refund or transfer rules before pricing high-cost upgrades.

Field-ready does not mean fully optimized or competitive at every rank. A later metric should define minimum role function, training level relative to the account's active roster, a basic gear baseline and the required core skill rank. Do not assign a numeric power ratio yet; that depends on the unresolved leveling/gear system and is for later paper modelling.

37.7 UI / system integration for the future game

- Recruitment screen: show the seal balance, actual cost per pull, banner-family pity count, remaining eligible pulls, current selection, carry-over scope, and the difference between standard and featured outcomes.
- Reward screen: tag every reward as recurring, activity-dependent, seasonal/variable, or one-time. Show whether it contributes to the shared Recruitment Seal balance, fragments, or a banner-specific event milestone.
- Source ledger in the design/admin tools: one row per source with cadence, expected eligible amount, low-activity availability, account progression gating, expiry, source reliability and whether the same reward is already counted in the 28-day budget.
- Roster preparation panel: show the actual missing resources to field a new hero; allow users to see common training / equipment transfer options before spending hard-to-replace resources on an older character.
- Test-drive / preview: time-limited playable preview or non-ranked trial could let players evaluate a featured character's tactical role before committing seals. It must not substitute for a permanent route to an essential counter.
- Respec and replacement safety: investigate a free early-game skill reset and predictable equipment transfer/refund/salvage so experimenting with an acquired counter does not permanently destroy a player's prior investment. This follows existing project fairness principles; exact refund rates remain open.

37.8 Failure cases and paper acceptance checks

Case	Question to answer on paper	Provisional acceptance condition
Player saves all seals for a featured target	Is the 70th featured-eligible pull affordable under the conservative recurring floor?	State an honest max-saving cadence; no use of one-time or paid rewards to disguise it.
Player spends some seals on standard recruitment	How much does this delay featured pity?	The UI makes the trade-off visible; fragments/selector preserve deterministic roster breadth.
Player misses several dailies	Can weekly cumulative rewards recover the baseline?	One missed day does not permanently break the free acquisition plan.
Player skips or cannot finish an event	Does the conservative featured floor or essential counter access collapse?	The core guarantee budget excludes the event unless it is genuinely guaranteed; alternatives remain.
New account receives weak random pulls	Can it still form the initial role-complete team?	Starter roster or selector covers the missing core functions without luck.
Player selects a standard hero with fragments	Can the hero be field-ready inside the overall 4–8 week target?	100-fragment path plus readiness costs stays within the stated horizon in the intended source-availability case.
Player gets a new hero but lacks gear / skill materials	Does acquisition convert into a usable counter, or just an unusable roster icon?	Base kit works; common training and gear source is accessible; specialized drops are not a baseline lock.

Case	Question to answer on paper	Provisional acceptance condition
Free attempt contributes to a banner event but not pity (or vice versa)	Are separate counters and eligibility rules clear?	Each system explicitly lists whether free pulls count. The same pull is never double-counted in rewards or budget.
High-rank guild / PvP dependence	Can a new, low-rank or solo player earn the core fragment floor?	Participation-based baseline is available; rank adds prestige rather than mandatory counter access.

These are conceptual acceptance checks for later review. They have not been executed as a simulator, prototype, seeded run, playtest or production QA.

37.9 Interim decision register and next task

Question / design decision	Working status	What remains open
Source reliability classes	Working proposal: dependable recurring, effort-dependent repeatable, variable/seasonal, one-time/progression, paid.	Audit each Section 21.2 source and exact cadence/completion requirement; classify conservatively.
Recruitment Seal currency model	Candidate preferred for further paper comparison: one flexible earnable seal for standard or featured use, plus universal fragments; pity is a counter, not a currency.	Compare with separate standard/featured ticket model. Owner approval not yet given.
70-pull guarantee cadence	A full 50 seals/28d saved to featured equals 39.2 days; 34/28d equals 57.6 days. Both are arithmetic from current examples.	Build a dependable source floor and decide whether every-six-week access is a goal or only a best-case player choice.
100 universal fragments	Existing 15–20/week proposal; selector in 35–46.7 days. This combines with an at-most-7-day fieldability hypothesis to fit the 4–8 week overall target.	Check targetable pool eligibility, catch-up routes, duplication conversion and actual common training costs.
Battle-ready threshold	New proposal: owned/selected alternative can be prepared within about 7 days, as an intermediate target that helps keep total alternate-build time within 4–8 weeks.	Define readiness concretely after levelling, gear, skill and reset options are reconciled.
Event / social / PvP / guild income	Supplemental and conditional; excluded from the guaranteed floor until a dependable minimum is documented.	Budget the expected and conservative scenarios separately, without assuming every player performs every activity.

Interim conclusion [RESEARCH SYNTHESIS]. The next usable economy spec should minimize resource-family count while making the accounting strict. A flexible Recruitment Seal plus universal fragments is a promising candidate because it preserves player agency; however, the 70-pull guarantee is only meaningful if eligible featured spends, carry-over scope and source cadence are exact. Under the existing numbers, 34 routine pull-equivalents/28 days is insufficient for a 6-week featured guarantee (needs about 46.7/28 days), and the 50 gross recurring/periodic target reaches the guarantee in 39.2 days only if all seals are eligible, available, and saved for that purpose. The 100-fragment selector at 15–20/week can be paired with a proposed ≤7-day baseline preparation path to place a practical alternative within about 6–8 weeks. This is a proposed budget architecture and arithmetic, not validated game balance.

Task status update (recorded in v0.10.9): completed conceptually in Section 38. The source rows have been stress-tested with conservative, ordinary and high-engagement assumptions over 28- and 42-day windows. Exact rewards and overlap still require a canonical source table before any numeric value can be finalized. Next research task: integrate the acquisition budget with recurring-mode, event and endgame sinks; define a canonical source-ID ledger; and build end-to-end paper journeys for new and mature accounts. Do not run a prototype, simulator, seeded pull or player test without explicit owner authorization. Follow-up completed in Section 39 (v0.10.10): mapped mode-specific reward roles, compared flexible and split ticket architectures, and built conditional 42-day new/mature journeys. Next: reconcile the unique source-ID ledger with core character/equipment costs.

Completion note (v0.10.9): The source-level audit and player-profile budget exercise described above is now recorded in Section 38. The figures remain paper assumptions, not observed earnings or final game rates.

37.10 Sources for acquisition-source and readiness research (accessed 29 October 2026)

[SRC1] HoYoverse Help Center. “What are the two currencies in Genshin Impact?” Updated 16 September 2025. Official description of Genesis Crystals as purchase-acquired currency convertible to Primogems and Primogems as also earned through missions, events and in-game mail. Used to distinguish paid currency from earnable currency and to support

source-ledger transparency, not to adopt the product's currency design wholesale. https://support.hoyoverse.com/hc/en-us/articles/50333919936153-What-are-the-two-currencies-in-Genshin-Impact

[SRC2] HoYoverse Help Center. "How does the Wish guarantee system work?" Updated 17 September 2025. Official description of the Genshin Impact Character Event Wish 90-wish five-star guarantee, initial 50% featured chance, next featured guarantee after a non-featured result, and shared pity/guarantee between two Character Event Wish banners. Used as an example of clearly scoped counter carry-over; exact values are not adopted. https://support.hoyoverse.com/hc/en-us/articles/50333940684953-How-does-the-Wish-guarantee-system-work

[SRC3] NIKKE.gg. "October 29 Patch Notes." Updated 26 October 2026. Current patch-note mirror describing separate ordinary/advanced recruitment voucher eligibility, 200 Gold Mileage Tickets for a featured character, mileage persistence after banner end, free pulls not earning mileage, free SSR/selector routes through event and login sources, and per-banner event-count eligibility. This is a specialist mirror rather than the publisher-hosted rule page; verify exact live-client rules before relying on it for production. https://nikke.gg/october-29-patch-notes/

[SRC4] Farlight Games / AFK Journey. "Talk with Designers: New Season Preview." Published 5 September 2025. Official seasonal designer communication describing one-time stage rewards moved to a Season Handbook, random gear drops replaced with select chests, upgrade-material supply increased in response to higher development costs, returning Rate Up heroes and a specific event path for free crossover characters. The post refers to planned/version-specific changes and is not an evergreen current economy spec. https://afkjourney.farlightgames.com/news/a0ae0b8ad8c35da97805079b672b7287

Evidence boundary. SRC1, SRC2 and SRC4 are official publisher/help-center sources for the stated product rules and dated update. SRC3 mirrors detailed patch notes and is labelled lower confidence than a directly opened publisher page. All Naruto-specific source assignments, the flexible Recruitment Seal proposal, the source reliability classes, the ≤7-day fieldability hypothesis and calculations using 34/40/50/60 seals are design inference or arithmetic from existing project proposals, not benchmark facts or validated balance.

38. Acquisition source reliability audit and player-profile budgets (Version 0.10.9)

Purpose and status. This chapter audits the illustrative source rows from Section 21.2 and the allocation proposal in Section 37. It builds conservative, ordinary and high-engagement 28-day and 42-day accounting cases; measures how spending on standard recruitment delays a 70-pull featured guarantee under the flexible Recruitment Seal proposal; and checks the 100-fragment selector against a proposed battle-readiness target. All scenario values are arithmetic assumptions derived from existing project proposals, not verified game rewards or player-behavior data. No prototype, simulator, seeded summon, or live/player test was performed. The Library master remains untouched; v0.10.9 is the working copy only.

38.1 Findings from current benchmark review

The evidence supports separating recurring activity rewards, variable or rank-dependent income, one-time progression grants, and paid currency rather than presenting them as one stable monthly total. AFK Arena specialist guides updated 21 October 2026 separate recurring daily/weekly/arena sources from large one-time chapter, Peaks of Time and player-level rewards; their exact amounts are third-party guide data and are not directly portable to Naruto Idle RPG [ACQ1]. A second current AFK Arena guide distinguishes ordinary summons, wishlist targeting, a Tavern-token milestone and named-hero targeting systems, showing that random collection and deterministic targeting can serve different purposes [ACQ2].

Official HoYoverse help describes earnable Primogems as coming from daily commissions, exploration, quests, limited-time events, achievements and promotional codes, while distinguishing them from purchase-acquired Genesis Crystals [ACQ3]. Its official guarantee guidance documents separate pity counters by banner type and explains carry-over within the Character Event Wish family [ACQ4]. This is evidence for clear source labeling and explicit counter scope, not a recommendation to copy its specific 90-wish/featured-chance model.

Current NIKKE patch notes reproduced by a specialist mirror distinguish ordinary from advanced recruitment voucher eligibility, explain which pulls earn Gold Mileage, and note that unused Gold Mileage persists after a Special Recruit ends; the same update lists free attempts and login/event acquisition routes [ACQ5]. Because this is a patch-note mirror rather than a directly opened publisher rules page, treat it as a lower-confidence, date-specific example and re-check live rules before any production decision. AFK Journey official designer communication describes moving one-time stage rewards to a Season Handbook, replacing random equipment drops with select chests, and making new upgrade resources available earlier in a season [ACQ6]. Together, these sources reinforce the need to separate source cadence and reward eligibility.

38.2 Audit of the existing source rows

The existing Section 21.2 table totals 60 pull-equivalents by adding 8 login/attendance + 14 daily activity + 12 weekly objectives + 6 guild/social/rotating + 10 monthly event + 10 campaign/tower milestones. For this audit, the first three rows are treated as completion-dependent recurring candidates; social/rotating income is optional effort-dependent; events are variable by calendar and participation; and campaign/tower milestones are one-time or progression-driven. The 10 campaign/tower units are therefore excluded from the mature-account recurring scenarios.

Critical accounting risk: the row “daily activity, accumulated through weekly completion” may overlap with “weekly objectives.” The current document does not specify unique reward IDs, objective thresholds or whether these rows represent separate rewards. The scenario sums below temporarily assume they are non-overlapping, solely so the sensitivity can be calculated. If the source definitions overlap, subtract the duplicate value before treating any total as an income estimate. This is a blocking open question for final rates.

Source row	Existing illustration per 28 days	Scenario allocation: conservative / ordinary / high	Reliability and audit note
Login / attendance	8	5 / 7 / 8	Candidate recurring source. Make claim windows forgiving; amount and attendance thresholds are proposals.
Daily activity, accumulated through weekly completion	14	7 / 11 / 14	Effort-dependent. Verify whether this overlaps the weekly-objective row; do not count the same reward twice.
Weekly objectives	12	6 / 10 / 12	Effort-dependent. Define completion burden, catch-up route and guaranteed accessible minimum.

Source row	Existing illustration per 28 days	Scenario allocation: conservative / ordinary / high	Reliability and audit note
Guild / social / rotating activity	6	0 / 3 / 6	Conditional Class B. Conservative case excludes it; no top-rank or top-guild requirement for baseline power.
Monthly event participation milestones	10	0 / 6 / 10	Variable Class C. Ordinary/high cases assume an event is scheduled and its cumulative route is available; not an evergreen floor.
Campaign / tower milestones	10	0 / 0 / 0 recurring	Class D, one-time/progression-triggered. May accelerate an eligible new account once, but must not be projected into later months.

38.3 Conservative, ordinary and high-engagement accounting cases

The profiles are stress cases, not measured personas. Conservative assumes partial completion of basic tasks and no social/event value; ordinary assumes most routine rewards, some social participation and partial event completion; high-engagement assumes every illustrated recurring source, the social/rotating row and full event milestones. The milestone row remains separate in every case. Fractional 42-day totals are averages from multiplying a 28-day window by 1.5; actual event timing and week boundaries may not distribute rewards linearly.

Paper profile	Rows included	28-day recurring / periodic total	42-day linearized total	Milestone treatment
Conservative	5 + 7 + 6 + 0 + 0	18	27	Up to +10 once, only if relevant new-account milestones are available and actually cleared
Ordinary	7 + 11 + 10 + 3 + 6	37	55.5	Up to +10 once, not part of the mature-account cadence
High-engagement	8 + 14 + 12 + 6 + 10	50	75	Up to +10 once; do not treat milestone value as recurring
Routine rows at full listed value, excluding social and events	8 + 14 + 12	34	51	Not a verified dependable floor; assumes all routine rows are separate and fully completed

Interpretation: the old 45-50 target is not a conservative recurring floor. Under the scenario assumptions here, it is closest to the ordinary-to-high gross periodic band, and the high case depends on full social/event participation. The 34 routine subtotal is still conditional because the task thresholds and row overlap are unresolved. The 60 total remains a first-progression-window example, not a mature recurring budget.

38.4 Featured guarantee sensitivity under the flexible Recruitment Seal proposal

For an account starting at zero pity, use the simple accounting formula: days to guarantee = 70 × 28 ÷ (pull-equivalents per 28 days × share of seals spent on the featured banner). This calculation assumes all earned pull-equivalents use the flexible seal, the same average source availability persists, and only seals spent on the featured banner count toward its 70-pull counter. It excludes one-time milestone rewards, paid purchases, pre-existing pity and stochastic drop outcomes. Standard summons purchased with these seals do not advance featured pity under the proposed rules.

Profile	Save 100% for featured	Spend 25% on standard (75% remains)	Spend 50% on standard (50% remains)
Conservative: 18 / 28 days	108.9 days	145.2 days	217.8 days
Ordinary: 37 / 28 days	53.0 days	70.6 days	106.0 days
High-engagement: 50 / 28 days	39.2 days	52.3 days	78.4 days

For context, a guaranteed featured copy every 42 days needs 46.7 featured-eligible pulls per 28 days; every 56 days needs 35. The ordinary profile reaches a 70-pull guarantee in about 53 days only if it saves all seals, but spending 25% on standard recruitment stretches the estimate to about 71 days. The high case reaches 70 pulls in about 39 days if all source assumptions hold, but spending 25% on standard recruitment stretches it to about 52 days. The conservative case cannot support an 8-week featured guarantee even when it saves every seal. These are not promises to acquire every new hero; content must remain answerable without collecting every release.

38.5 Fragment-selector and counter-readiness sensitivity

Section 21.2 currently proposes 100 Universal Fragments for a selectable standard-pool hero and 15-20 fragments per week from multiple sources. To expose the effect of source reliability, this audit uses 10 / 15 / 20 fragments per week as conservative / ordinary / high scenarios. A proposed maximum 7-day preparation period is added after the selector is reached. “Battle-ready” means able to serve the needed team function at the account’s current baseline with accessible common training, usable baseline equipment and the role-critical portion of the kit; it does not mean endgame-optimized or guaranteed to win a specific matchup.

Profile	Fragments/week	28-day fragments	42-day fragments	100-fragment selector time	Selector + up to 7 days readiness
Conservative	10	40	60	70 days	77 days (~11.0 weeks)
Ordinary	15	60	90	46.7 days	53.7 days (~7.7 weeks)
High-engagement	20	80	120	35 days	42 days (6 weeks)

Payout-timing correction (v0.10.11): The selector times above are continuous-rate estimates. If 15 fragments arrive in weekly batches, a zero-balance account reaches 100 on its seventh payout, which is day 49 in the aligned week-end example. Adding seven preparation days reaches day 56. Exact availability must use payout dates and eligibility, not weekly income divided by seven alone.

A useful mathematical requirement follows: to acquire 100 fragments within 49 days and reserve up to 7 more days for baseline preparation, the player needs about 14.3 fragments per week on average. Rounded to a design target, a dependable 15-fragment weekly floor is the minimum candidate for a guaranteed 8-week selector-to-field-ready path. If the 15-20 target can only be reached by doing several optional or rank-gated modes, it is an expected yield, not a dependable floor. The conservative 10-per-week case takes about 77 days including readiness and therefore fails the 8-week target; that failure is visible rather than hidden.

38.6 New-account versus mature-account handling

New-account cohort: preserve an onboarding role-coverage selector or equivalent guaranteed starter grant so the first coherent squad does not depend on random pulls. One-time campaign/tower grants can help the first account journey and may reduce the first guarantee savings time if the rewards are truly available and featured-eligible, but must be tracked in a separate first-clear ledger. Never multiply them into later 28-day budgets.

Mature-account cohort: use recurring/periodic source availability only. A cleared campaign or tower milestone is not paid again unless a documented reset or repeatable reward explicitly says so. The ordinary case estimates 37 pull-equivalents per 28 days before accounting overlap; this is enough for a 70-pull featured guarantee in about 53 days only when all units are saved and every modeled source recurs as assumed. It does not support both frequent standard recruiting and a 6-week featured guarantee at the same time without more eligible income.

Counter readiness: when a newly recruited or selector-obtained character fills a role gap, use the proposed shared training baseline, a basic gear route or transfer/refund option, and an accessible role-critical skill upgrade to reduce the preparation delay. Do not require a duplicate, rare mode-only material or a limited-event equipment drop for the character’s essential function. The ≤7-day readiness allowance remains a hypothesis to be checked against the full cost-and-income model.

38.7 Source-ledger requirements for the future game

Before final numeric economy design, each reward source needs a unique ledger row with: source ID; activity name; cadence and reset; reward item and amount; whether the reward is pull-equivalent, fragment or material; completion threshold; account/progression eligibility; conservative availability class; whether it is one-time, repeating, seasonal, rank-dependent or paid; which banner/pity counters it qualifies for; expiry and carry-over rules; source-overlap ID to prevent double counting; and expected yield over 28 and 42 days. Where the source is a shop exchange, record both the currency spent and the value acquired so that the same flow is not counted as two income sources.

Recommended UI integration [PROPOSAL]: display recurring, activity-dependent, event, milestone and paid rewards distinctly; show the Recruitment Seal balance and the explicit opportunity cost of standard spending; show featured-eligible pulls remaining and carry-over scope; show Universal Fragment balance and selector eligibility; and provide a “what prevents this shinobi from being field-ready?” checklist. Weekly cumulative tracks and catch-up should prevent a single missed day from permanently breaking the target path.

38.8 Interim design recommendations and decision register

- [PROPOSAL] Keep the single flexible Recruitment Seal plus Universal Fragments as the leading low-currency model to compare, not a final owner-approved choice. A seal spent on standard recruitment cannot simultaneously count toward featured pity. If predictable featured access must remain independent of standard spending, compare a split-ticket model and record its extra currency complexity.
- [PROPOSAL] Treat 45-50 pull-equivalents as a gross periodic target for now, not the conservative guaranteed floor. Treat the gross 60 as including one-time/progression value. Do not finalize either target until daily and weekly rows have unique non-overlapping reward IDs and completion thresholds.
- [HYPOTHESIS] If a featured guarantee within 8 weeks is an explicit target for a player who saves all seals, budget at least 35 eligible pulls per 28 days; a 6-week target needs about 46.7. An ordinary player who spends some seals on standard pulls will need more total income or a longer guarantee horizon.
- [PROPOSAL] If an essential role alternative must be targetable and field-ready within 8 weeks, preserve a dependable 15 Universal Fragments per week from evergreen, accessible sources and hold the baseline readiness route to at most 7 days. Optional or rank-based modes can supply additional fragments but should not be required to meet this floor.
- [PROJECT PRINCIPLE] Campaign first-clears, beginner grants, events, returner tracks and paid currency must remain separate from the mature recurring floor. No limited unit may be the only functional answer to a common strategic problem.
- [OPEN] Confirm whether the 14 daily-activity row and 12 weekly-objectives row overlap. Confirm event schedules, source completion burden, which seal types are eligible for featured pity, fragment pool eligibility and whether the 15-fragment weekly floor can be met without mandatory high-rank or multi-mode participation.

38.9 Paper acceptance checks (not executed)

- Source accounting: every budget reward has one unique source ID; daily and weekly sources are proven non-overlapping before being summed.
- Conservative accounting: event, top-rank, paid and first-clear rewards are excluded from the mature-account dependable floor unless their recurring eligibility is documented.
- Featured accounting: any presented guarantee-time estimate uses featured-eligible seals only, after subtracting the player's standard-spending share and existing pity progress.
- Selector accounting: a 100-fragment selector's time is shown at 10 / 15 / 20 fragments per week and is combined with a separate readiness allowance.
- New-account accounting: onboarding grants and first-clears are a separate first-progression budget; they do not recur in the mature profile.
- Strategic access: an essential counter remains accessible through starter roster, standard selector or deterministic evergreen route without an exclusive duplicate, top-rank requirement, or one-time-only event.

These are paper checks only. None was run in a prototype, simulator, seeded pull model or live playtest.

38.10 Interim conclusion and next research task

Interim conclusion [RESEARCH SYNTHESIS]. The audit demonstrates that the previous budget should be treated as a conditional range, not one recurring number: 18 / 37 / 50 pull-equivalents per 28 days under conservative / ordinary / high assumptions, respectively, before one-time milestone rewards and pending overlap correction. Saving all seals for featured recruitment yields a 70-pull threshold in approximately 109 / 53 / 39 days; standard spending materially lengthens that timeline. A 100-fragment selector at 10 / 15 / 20 fragments per week takes 70 / 46.7 / 35 days, plus the proposed 7-day readiness allowance. The 15-fragment weekly floor is a candidate requirement if the 8-week role-counter target is a real design promise; it is not yet validated as affordable.

Next research task: reconcile the acquisition ledgers with repeatable-mode, event and endgame resource sinks; resolve whether daily/weekly rewards overlap; and model whole 42-day journeys for new and mature accounts using both flexible-seal and split-ticket architectures. Then connect the resulting acquisition schedule to the core character/equipment cost model. Continue paper research only; live testing, simulation or prototype work requires explicit owner authorization.

38.11 Sources for acquisition source reliability and paper budgets (accessed 29 October 2026)

[ACQ1] AFK Global. “Where to get diamonds and the best ways to spend.” Updated 21 October 2026. Specialist/community guide listing current steady quest/arena sources separately from large one-time chapter, Peaks of Time and player-level rewards. Used as structural reference only; amounts are game-specific and should be rechecked against the client. https://www.afk.global/afk-arena/guides/diamonds-and-store

[ACQ2] AFK Global. “Summon rates, the wishlist and every pool.” Updated 21 October 2026. Specialist/community guide differentiating general-pool summons, wishlist targeting, Tavern-token milestone and named-hero targeting. Used to illustrate distinct random and deterministic acquisition routes; not a value target for our game. https://www.afk.global/afk-arena/guides/summoning

[ACQ3] HoYoverse Help Center. “How can I get more Primogems?” Updated 17 September 2025. Official list of daily commissions, exploration, quests, limited events, achievements and codes as earnable sources. https://support.hoyoverse.com/hc/en-us/articles/50333936115097-How-can-I-get-more-Primogems

[ACQ4] HoYoverse Help Center. “How does the pity system work for different banners and reruns?” Updated 23 September 2025. Official description of independent banner-type counters and Character Event Warp guarantee behavior. https://support.hoyoverse.com/hc/en-us/articles/50913710696217-How-does-the-pity-system-work-for-different-banners-and-reruns

[ACQ5] NIKKE.gg. “October 29 Patch Notes.” Patch-specific specialist mirror indexed 27 October 2026 and describing the 29 October 2026 update. Discusses ordinary/advanced voucher eligibility, Gold Mileage eligibility/carry-over, free special-recruit attempts, login selector box and event/mission rewards. Lower confidence than official publisher documentation; verify against live client before production use. https://nikke.gg/october-29-patch-notes/

[ACQ6] Farlight Games / AFK Journey. “Talk with Designers: New Season Preview.” Official developer communication dated 5 September 2025 describing one-time stage rewards moved to a handbook, select equipment chests replacing random equipment drops, and progression materials made available earlier in a season. Specific to the update described. https://afkjourney.farlightgames.com/news/a0ae0b8ad8c35da97805079b672b7287

Evidence boundary. ACQ3, ACQ4 and ACQ6 are official publisher/help-center sources for the specific stated mechanics. ACQ1 and ACQ2 are specialist/community guides with current visible update dates but are not publisher specifications. ACQ5 is a specialist patch-note mirror and is explicitly lower confidence. Scenario allocations, 18/37/50 profile totals, featured savings times, fragment scenarios, the 15-per-week floor and 7-day readiness allowance are project arithmetic/proposals/hypotheses, not external facts or measured game balance.

39. Mode reward architecture, recruitment-ticket alternatives and full account journeys (Version 0.10.10)

Purpose and status. This chapter connects the recruitment-source audit in Section 38 with the existing 17-family resource map, resource sinks and proposed core-progression budget. It compares a flexible Recruitment Seal with separate standard/featured tickets and maps rewards to campaign, idle income, repeatable PvE, guild, PvP, event, seasonal and endgame modes. It includes paper-only new-account and mature-account journeys over 42 days. All allocations and scenario numbers are theoretical unless explicitly identified as a source-supported benchmark observation. No prototype, simulator, seeded recruitment run, or player test was conducted. The Library remains unsynchronized; this is the working master only.

39.1 What the benchmark evidence supports

[SOURCE FINDING] HoYoverse's official help center lists daily commissions, exploration, story/world quests, limited events, achievements and promo codes as Primogem sources. A separate official article explains that banner-family pity can carry across Character Event Wish banners, while each banner family has its own rules. This supports clear source accounting and visible counter scope; it does not mean our game should copy the 90-wish threshold or 50/50 rule [MODE1-MODE2].

[SOURCE FINDING] AFK Journey's official designer update describes moving one-time seasonal stage rewards into a Season Handbook, replacing random equipment drops with selectable equipment chests, granting class-specific trial materials with a universal fallback, and adjusting upgrade-material drops in response to changed development costs. These are specific changes described for that version, not evergreen assumptions. They support the design principle that reward certainty, resource supply and upgrade costs should be reviewed together [MODE3].

[SOURCE FINDING WITH DATE / SOURCE CAVEAT] A specialist NIKKE beginner guide updated 27 August 2026 describes campaign clears both granting immediate development resources and raising the Outpost Defense passive-income level. It also describes the Synchro Device as allowing additional units to share a baseline based on the account's top five levels, reducing the need to individually level every roster unit. Use this as a structural example only; detailed current values should be checked against the live client [MODE4].

[SOURCE FINDING WITH SOURCE CAVEAT] AFK Arena's specialist guide updated 21 October 2026 distinguishes the general Tavern pool from targeted/advanced pools, describes wishlist effects, and reports milestone rewards at every 100 eligible Tavern-token summons. This illustrates how a game can combine broad random recruitment, targeting and deterministic milestones while keeping their eligibility separate. It is not a publisher-owned economy specification [MODE5].

[SOURCE FINDING WITH PATCH-SENSITIVITY CAVEAT] The AFK Journey 1.8.1 notes mirrored by AFK Global describe guild-only seasonal progression and rewards, a daily-cleared/swept Union Campaign, and guild resources used for guild features. Because the accessible detailed notes are a specialist mirror and mechanics are patch-sensitive, use these only as a current structural observation, not as a claim that every season uses the same system [MODE6].

Synthesis. Strong recurring economies distinguish at least four things: (1) repeatable income, (2) account-progression or first-clear grants, (3) mode-specific rewards, and (4) seasonal or participation-dependent bonuses. A mode should have a clear reward purpose. One-time first-clear rewards can accelerate a new account but cannot be counted in the mature-account monthly floor. Seasonal rewards may be valuable without being permanent sources of core power. Targeting or deterministic milestones can reduce acquisition uncertainty, but the exact generosity must be budgeted rather than inferred from a benchmark.

39.2 Proposed source-to-mode reward allocation

The following is the recommended architecture to model, not a finalized source or quantity list. The canonical ledger still needs unique source IDs and verification that daily and weekly reward rows do not overlap. Do not sum a row until its cadence, audience, reset and eligibility have been specified.

Mode / source class	New-account purpose	Mature-account purpose	Accounting rule / guardrail

Mode / source class	New-account purpose	Mature-account purpose	Accounting rule / guardrail
Campaign and first-clears	Guaranteed starter role/coverage selection; one-time recruitment and fragment milestones; feature unlocks.	New chapters may give one-time expansion rewards, but old cleared stages do not count as recurring income.	Separate from mature recurring budget; first-clear rewards cannot be recycled into a monthly floor.
Ordinary idle bank	Core development resources accrue while away; rate can rise at documented campaign/account checkpoints.	Keeps baseline leveling/progression moving even during a campaign wall.	Do not silently place rare characters, featured pity tickets or all specialized materials in the idle basket.
Daily and weekly objectives	Small predictable Recruitment Seal amounts; cumulative weekly milestones; part of the Universal Fragment floor.	Main dependable recurring source if requirements remain accessible and rewards are non-overlapping.	A missed day must not invalidate the whole weekly reward. Verify the existing 14 daily / 12 weekly accounting rows before totaling.
Repeatable progression PvE / tower / training	Early guaranteed materials, feature tutorials and modest roster-targeting fragments.	Specialized skill, promotion, gear or boss materials; an accessible fallback for role-critical upgrades.	Make the mode useful, but never the only path to an essential role or counter. Avoid requiring every mode to reach a basic summon budget.
Guild and social modes	Tutorial/support access, low-pressure participation rewards and social bonuses.	Guild-specific currency, co-op progress, prestige and supplemental materials.	No ideal-guild requirement for the dependable recruitment/fragment floor. Top guild placement must not gate essential counters.
PvP / ranked	Beginner-protected milestones and participation rewards; no need to win at high rank.	Seasonal prestige, cosmetics and modest general income; participation floor for broad rewards.	Keep power-critical sources out of top-rank-only rewards. Rankings may reward performance without defining roster access.
Events and seasonal modes	Bonus seals/fragments, choice chests and thematic/cosmetic rewards; help explain the event's mechanics.	Variable bonus income, seasonal materials and optional acceleration; old story may return through an evergreen archive.	Do not count event currency as a fixed core floor unless calendar and participation assumptions are explicit. State expiry and rerun/catch-up rules.
Endgame boss / raid / challenge modes	Unlock later; early players should not be expected to earn endgame-specific resources.	Higher-tier specialization materials and longer-term optimization, plus cumulative participation rewards.	Difficult or ranked rewards may accelerate optimization, but must not be the only way to obtain baseline counter access.
Returning-player catch-up	Not applicable until return conditions are met.	Bounded catch-up for missed progression and outdated content.	One-time/conditional; not counted as a recurring mature-account source. Do not create a reward exploit for repeated short absences.
Paid offers	Optional convenience, cosmetics or accelerated acquisition only.	Optional time-saving and cosmetic purchases.	Exclude paid rewards from all free-to-play income scenarios. No paid-only mandatory role, counter or essential functionality.

39.3 Flexible Recruitment Seal versus separate tickets

Neither architecture can be selected by simplicity alone. The decision depends on whether player choice or a predictable banner budget is the priority, and whether our total free acquisition budget can sustain both standard roster growth and a featured guarantee.

Architecture	How it works	Advantages	Risks / conditions
A. One flexible Recruitment Seal + Universal Fragments (current leading proposal)	Player chooses standard or featured recruitment. Only actual eligible featured spends advance the 70-pull featured counter. Fragments target a standard-pool character.	Few persistent currencies, clear choice, easier inventory/UI, flexible use when a player needs broad roster growth.	Players can spend seals on standard and delay the featured guarantee; must show exact opportunity cost and never promise a calendar guarantee that assumes every seal is saved.
B. Separate Standard tickets + Featured tickets + Universal Fragments	Sources allocate a fixed amount to each pool; featured-only tickets advance the featured counter, standard tickets cannot be diverted.	Easier to ring-fence a banner budget and forecast featured access if the ticket amounts are genuinely dependable.	Currency clutter, stranded balances, forced allocation and low standard income if too much is routed to featured tickets. A "guaranteed schedule" is dishonest if its sources depend on event completion or one-time clears.

[RECOMMENDATION — WORKING PROPOSAL, NOT OWNER-APPROVED] Keep Architecture A as the leading model for the next paper-budget round because the project has repeatedly preferred fewer currency families and player agency.

Preserve the Universal Fragment route as a deterministic alternative. Do not promise that every player can obtain every featured release on a fixed schedule under the present budget.

If the owner later decides a featured character must be guaranteed within a fixed number of weeks regardless of spending choices, reconsider Architecture B only after increasing or explicitly reallocating the total free budget. Do not try to promise both high standard-banner volume and a fixed featured schedule from the same limited seal pool without showing the arithmetic.

39.4 Whole-account 42-day paper journeys

Use the Section 38 recurring scenarios as assumptions: 18 / 37 / 50 pull-equivalents per 28 days for conservative / ordinary / high-engagement profiles. Linearizing to 42 days gives 27 / 55.5 / 75 recurring pull-equivalents. A new account may earn up to the existing illustrative 10 one-time campaign/tower milestone equivalents during the window only if the relevant milestones are actually reached. Those rewards must not be carried into the mature-account recurring budget.

Journey / profile	42-day acquisition income	Featured-eligible spends if flexible seals are saved	Design interpretation
New account — conservative	27 recurring; up to +10 one-time if earned	27 if all recurring seals go to featured	Starter role/coverage selection is the protection against an unlucky first 42 days; do not promise the featured guarantee.
New account — ordinary	55.5 recurring; up to +10 one-time if earned	55.5 if all recurring seals go to featured; at most 65.5 if all 10 milestones are earned and eligible	Even the full illustrative milestone bonus remains below 70. The starter selector and targetable fragments must keep team function independent of banner luck.
Mature account — conservative	27 recurring; no exhausted campaign first-clears	27 in 42 days	The 70-pull guarantee needs about 109 days if all seals are saved; a 25% standard-spend share stretches it to about 145 days.
Mature account — ordinary	55.5 recurring; no first-clear bonus	55.5 in 42 days	About 53 days to 70 if every seal is saved; about 71 days if 25% of seals go to standard recruitment.
Mature account — high engagement	75 recurring only under the high-case assumptions	75 in 42 days	About 39 days if all are saved, but this depends on variable social/event participation and is not a dependable floor.

Fragment route. At the proposed 15 Universal Fragments per week, an account earns 90 over 42 days and reaches 100 in about 46.7 days; the additional ≤7-day readiness allowance produces a theoretical 53.7-day path. This assumes all 15 weekly fragments come from accessible, non-overlapping evergreen sources. A starter selector is still needed for the first functional team; the 100-fragment selector is a later targeting route, not the onboarding solution.

Separate-ticket sensitivity. If only 60% of the recurring pull-equivalent budget were routed to featured tickets, the 18 / 37 / 50 profiles would receive 10.8 / 22.2 / 30 featured tickets per 28 days, reaching 70 in roughly 181 / 89 / 65 days. At an 80% featured allocation, they would receive 14.4 / 29.6 / 40 per 28 days, reaching 70 in roughly 136 / 66 / 49 days. These percentages are sensitivity examples, not proposed allocations. They show that split tickets improve predictability only if enough total income is available; they do not create value by themselves.

39.5 Endgame and seasonal integration

- [PROPOSAL] Keep permanent roster access and seasonal progression distinct. Endgame modes can grant advanced skill/gear materials and optional optimization; season-only currencies may reset only when clearly communicated and when the reset does not revoke a permanent character, essential counter or core progression function.
- [PROPOSAL] Provide reward floors based on participation or cumulative milestones. Top rank and guild strength may provide prestige, cosmetics or faster optional optimization, but not the only practical route to core counter access.
- [PROPOSAL] Let the campaign improve future idle income, but do not make one-time campaign/tower rewards a recurring character-access budget. As accounts mature, recurring PvE, objectives and predictable evergreen acquisition must replace exhausted first-clears.
- [PROPOSAL] Use events to create variety and optional acceleration. If an event contributes to the published free-acquisition target, define its calendar, participation requirement, eligibility and fallback route. A temporary banner must not be the sole path to a commonly needed counter.

- [PROPOSAL] Mode reward identity: campaign = milestone/unlock; idle bank = baseline growth; training/tower = specialized progression; guild = cooperative resources; PvP = ranked prestige plus participation floor; events = themed bonus/selection; endgame = optimization. Cross-mode shop exchanges can add flexibility, but every exchange rate is an explicit sink/source and must be counted once.

39.6 Canonical source ledger requirements

Before any value becomes a final income promise, one canonical ledger row is required for every reward source. Minimum fields: Source ID; mode; activity/reset cadence; eligibility/unlock; participation threshold; exact reward and amount; currency family; standard/featured/fragment eligibility; recurring/variable/seasonal/one-time/paid classification; expected low-activity availability; expiry/carry-over; account stage; and duplicate-count check. Give each reward instance one source ID, not a second ID because it appears in a different UI summary.

Use two separate accounting outputs: (1) “total earned” by resource and source; (2) “eligible featured pulls actually spent” by banner family. Never label all earned Recruitment Seals as featured-pity progress. Report a range or uncertainty when eligibility or completion rate is not known. Do not sum a source row twice across daily and weekly objectives.

[UI PROPOSAL] In the recruitment screen, show the current featured-family pity count, exact number of eligible spends remaining, carry-over scope, seal balance and banner option cost. Show the forecast “at current earned rate” as a range rather than a promise, and explain that spending seals on the standard pool delays the featured guarantee. In mode reward screens, tag each item as core progression, roster access, mode specialization or optional optimization.

39.7 Design recommendations and open questions

- [PROPOSAL] Keep one flexible Recruitment Seal plus persistent Universal Fragments as the leading low-currency architecture. Do not introduce separate ticket families solely to make the UI look like another game.
- [PROPOSAL] Do not make a fixed “one featured character every six/eight weeks” promise yet. With 37 pull-equivalents per 28 days in the ordinary Section 38 case, a 70-pull guarantee takes about 53 days only when all seals are saved for featured use; spending 25% on standard recruitment stretches it to about 71 days.
- [PROPOSAL] Maintain a new-account starter selector that fills a missing core role or counter. The role must work at base copy, and this starter grant is accounted as one-time onboarding rather than recurring mature-account income.
- [HYPOTHESIS] Preserve a 15 Universal Fragments/week evergreen target and a ≤7-day preparation path for an owned counter, subject to source-ID audit and core-level/gear cost compatibility. These are candidate fairness targets, not tested results.
- [OPEN] Confirm whether the 14 daily-activity and 12 weekly-objective rows overlap; define what 18/37/50 profile sources actually include; decide what the featured banner release cadence should be; decide whether featured pity ever uses restricted sources; decide fragment pool eligibility and duplicate conversion; define endgame mode resets/catch-up; and set acceptable activity burden.
- [OPEN] Decide whether the player-facing contract is “70 eligible featured pulls guarantees one featured copy” only, or whether there will also be a minimum calendar cadence. A pull-count guarantee is auditable; a calendar guarantee requires a dependable source budget that remains sufficient after standard spending.

39.8 Paper acceptance checks (not run)

- No duplicate source ID is counted twice, including daily tasks also summarized in weekly completion.
- New-account first-clear rewards and onboarding grants are separated from the mature-account recurring floor.
- The ordinary mature-account 42-day example reproduces 55.5 recurring pull-equivalents before optional milestone/event sources.
- Featured pity increases only from eligible featured-banner spends; standard spending does not secretly advance it.
- The fragment selector route remains available through accessible, evergreen sources without mandatory PvP rank, guild entry or a short event window.
- Every active/endgame mode has a reward identity; no single mode is the only practical route to a core role or counter.
- Every seasonal reset, expiry and banner counter carry-over rule is visible before a player spends resources.

These are conceptual paper checks. They have not been executed against code, a simulator, seeded recruitment model, prototype or live players.

39.9 Interim conclusion and next research task

Interim conclusion [RESEARCH SYNTHESIS]. A mode's rewards should have a primary identity while remaining connected to the whole economy. First-clear and event bonuses may help new accounts but are not a mature-account floor. The flexible Recruitment Seal remains the leading low-currency proposal, but it only preserves a featured guarantee if actual featured-eligible spends reach 70; seals spent in standard recruitment must not be counted as pity. Under existing Section 38 assumptions, the ordinary account reaches 70 in about 53 days if it saves every seal and about 71 days if it spends 25% on standard. Split tickets make allocation predictable only by fixing the distribution; at the same total free income, a fixed featured allocation can leave too few standard summons. The 100-fragment target at 15/week creates a deterministic standard-pool route in about 46.7 days plus the proposed ≤7-day readiness period, if the fragment floor is reliable. All values remain paper hypotheses, not tested game balance.

Next research task: construct the canonical source-ID ledger with no overlap, fill the exact cadence/eligibility fields for daily, weekly, mode, social, event and milestone rewards, then run paper-only 56-day journeys against the core character/equipment cost model. Only after that should the project consider locking any free summon, fragment or calendar-access target. No prototype, simulator, seeded run or player test without explicit owner authorization.

39.10 Sources for mode rewards and ticket architecture (accessed 29 October 2026)

[MODE1] HoYoverse Help Center. "How can I get more Primogems?" Updated 17 September 2025. Official list of earnable currency sources: daily commissions, exploration, story/world quests, limited events, achievements and promotional codes. https://support.hoyoverse.com/hc/en-us/articles/50333936115097-How-can-I-get-more-Primogems

[MODE2] HoYoverse Help Center. "How does the pity system work for different banners and reruns?" Updated 23 September 2025. Official description of banner-family counter scope, separate standard/light-cone counters and carry-over between Character Event Wish banners. Rules are specific to the named game and are not adopted as Naruto RPG rates. https://support.hoyoverse.com/hc/en-us/articles/50913710696217-How-does-the-pity-system-work-for-different-banners-and-reruns

[MODE3] Farlight Games / AFK Journey. "Talk with Designers: New Season Preview." Published 5 September 2025. Official designer communication describing one-time AFK-stage rewards moved to a handbook, Select Equipment Chests, class-specific trial materials with a universal fallback, progression-material timing and supply changes when development costs rise. Version-specific, not an evergreen economy rule. https://afkjourney.farlightgames.com/news/a0ae0b8ad8c35da97805079b672b7287

[MODE4] NIKKE.gg. "NIKKE Beginner's Guide." Updated 27 August 2026. Specialist guide describing campaign progress raising Outpost Defense income, immediate campaign resources, tower-specific resources and the Synchro Device's top-five-level baseline for other units. Structural comparison only; recheck detailed values in the live client. https://nikke.gg/beginners-guide/

[MODE5] AFK Global. "Summon rates, the wishlist and every summon pool in AFK Arena." Updated 21 October 2026. Specialist/community guide describing distinct general and targeted summoning pools, wishlist targeting and a guaranteed reward at 100 eligible Tavern-token summons. Not a publisher-owned economy specification. https://www.afk.global/afk-arena/guides/summoning

[MODE6] AFK Global. "AFK Journey 1.8.1 Patch Notes." Dated 22 September 2026. Specialist mirror of patch notes describing current guild modes, seasonal resources and progression limited to guild-mode effects. Patch-sensitive and lower confidence than a direct official source. https://www.afk.global/afk-journey/patch-notes/1-8-1

Evidence boundary. MODE1-MODE3 are publisher/help-center sources for the specific mechanics described. MODE4-MODE6 are specialist/community sources and should not be treated as official source specifications. The source-to-mode allocation, flexible-versus-split ticket comparison, 42-day player journeys, 18/37/50 scenario values, 70-pull guarantee and 15-fragment weekly target are project proposals or arithmetic from existing assumptions, not observed results or a verified benchmark standard.

Task progression (v0.10.11): Section 40 now records the canonical draft source register and 56-day paper review requested in Section 39.9. Acquisition accounting is explicit for the candidate core schedule; material quantities, equipment coverage and preparation costs remain open. The next research packet must price that baseline package and resolve the idle-claim constraint before any access deadline is promised.

Status update (v0.10.10): This chapter completes the mode reward / ticket architecture and account-stage journey research task left open in Section 38. The next work is the canonical source-ID ledger and paper reconciliation with character/equipment costs.

40. Canonical reward ledger and 56 day readiness review

Version 0.10.11 | 30 October 2026. This chapter turns the source requirements in Section 39 into a canonical draft register and a dated, deterministic 56-day accounting example. The main finding is that obtaining a counter and preparing it are separate commitments: the fragment route can reach its seventh weekly payout within the window, while the current idle cap can leave too little material surplus for preparation.

Status and scope. [PROJECT-DOCUMENT AUDIT] marks findings about the existing master. [PROPOSAL] marks new design choices for owner review. [ILLUSTRATIVE ARITHMETIC] marks explicit hypothetical quantities and calculations. New reward amounts, resets, unlock conditions and readiness packages are not approved settings or measured player behavior. This pass used desk research and closed-form bookkeeping only; no game prototype, gameplay simulation, seeded summon model or live test was run.

40.1 Evidence used in this pass

[SOURCE FINDING] HoYoverse states that Encounter Points can substitute for Daily Commissions when claiming the same daily commission rewards [LED1]. The useful distinction is between an activity that earns eligibility and the reward entitlement it satisfies. Our inference: alternate tasks should not create a second income row when they redeem the same entitlement.

[SOURCE FINDING] Unity documents exchanges with a named ID, explicitly specified resources received, and explicitly specified resources spent [LED2]. Its resource documentation distinguishes balances, owned items and purchases [LED3]. These support separate grant, conversion and spend records. They are accounting examples, not a recommendation of an implementation service or a claim about game generosity.

[SOURCE FINDING WITH ACCESS LIMIT] Indexed text from AFK Journey's official New Season Preview describes moving one-time stage rewards to a handbook and replacing random equipment drops with selectable chests [LED4]. The opened page exposed a script shell, so this pass relies only on the visible official indexed excerpt. Our inference is limited to clearer reward timing and equipment choice; it does not establish current rates or an affordability target.

40.2 Canonical identity and the counting rule

[PROPOSAL] Maintain one register of reward definitions and a separate record of each earned occurrence. A definition ID identifies the rule; an occurrence combines that ID with the account, cycle or content milestone, and reward tier. A weekly reward needs a new occurrence each week, not a new definition. A display in the daily screen, weekly screen or mailbox does not create a new reward.

Required field	Meaning
Definition and occurrence	Stable source ID; account + reset cycle or milestone + reward tier. Record the original entitlement when a reward is displayed elsewhere.
Eligibility and timing	Unlock, eligible player group, qualifying activity, earn time, claim time, reset, expiry and any catch-up rule.
Reward vector	Every currency or item and its exact quantity; a bundle is one grant containing several resources. Mark unknown quantities OPEN.
Accounting class	Core recurring, optional repeatable, event/seasonal, one-time, paid, conversion, refund, or progress-only.
Banner and overlap rules	Standard/featured eligibility; which spending advances pity; linked alternative claim routes; first-clear scope; duplicate-display reference.
Cost link and status	Related cost/sink ID; amount paid in a conversion; proposal/decision status and evidence. Unknown eligibility excludes a source from a dependable budget.

A single action can legitimately complete two distinct published reward entitlements. That is not double-counting. Counting one payout twice because it appears in two screens is. Likewise, earning a mode token and exchanging it for an

item creates a token grant plus a conversion; it does not mint a second independent free-income allowance. Do not combine unlike resource units into one value without an explicit conversion rule.

40.3 What the old daily and weekly totals actually establish

[PROJECT-DOCUMENT AUDIT] Section 21.2 lists 8 attendance + 14 daily-activity + 12 weekly-objective pull-equivalents per 28 days. Sections 38-39 explicitly say that the daily and weekly definitions may overlap. The supplied master contains no entitlement IDs or payout specification capable of proving that the rows are independent. This uncertainty cannot be repaired by asserting that the old total was correct.

[ILLUSTRATIVE ARITHMETIC] If O is the overlapping portion of the 14 and 12 rows, unique routine income is 8 + 14 + 12 - O = 34 - O. With 0 <= O <= 12, that is 22-34 per 28 days, or 44-68 over 56 days if the same pattern repeats. This is a sensitivity bound on the old figures, not a measured income range. The ordinary 37 and high 50 cases also include conditional social/event value and are not routine floors.

[PROPOSAL] Replace the ambiguous 14-plus-12 structure in the next design candidate with one explicitly funded weekly activity entitlement. Preserve 26 per four weeks as a provisional allowance to compare with the previous upper bound. This creates a new, auditable proposal; it does not retrospectively certify or add to the old 14 and 12 rows.

40.4 Quantified core reward proposal

[PROPOSAL] For this paper candidate, recruitment, fragments and the weekly core route unlock when the introductory squad tutorial is complete. A core visit consists of collecting available idle resources and either reviewing/changing the squad or attempting an unlocked solo encounter. It requires no win, purchase, summon, resource spend, guild membership or PvP rank. Two visits on distinct reset days complete the weekly route. Both can occur during ordinary core play; they do not require separate modes.

Candidate reset rule: days turn at 00:00 UTC (08:00 in Manila); weeks turn Monday at the same time. Activity quantities repeat in a four-week cycle, 6 / 7 / 6 / 7 seals, with its first production epoch still to be published. Earned payouts auto-credit and do not expire. Uncompleted weekly eligibility expires at reset; no retroactive full-week payout is assumed. This deliberately exposes the missed-week limitation for future catch-up work. The 56-day example starts at the beginning of cycle week 1, with tutorial completed and no opening seals, fragments or pity.

Source ID	Eligibility and reset	Exact candidate reward	Counting rule
CORE01	First core visit in each week.	2 flexible seals + 10 Universal Fragments per week.	One weekly occurrence; credited immediately.
CORE02	Second core visit on a different reset day in that week.	6 / 7 / 6 / 7 seals by cycle week + 5 fragments every completed week.	Replaces the old daily/weekly pair. Do not add 14 + 12 again.
PROG01	Daily task or alternate core activity.	Eligibility progress only; 0 currency and 0 fragments.	Feeds the same CORE02 entitlement; never an income row.
START01	Introductory role-coverage milestone, once per account.	1 choice of an eligible starter that fills a missing core function.	A one-time character grant; not a seal, fragment or pity grant.
FIRST01	Introductory tutorial completion, once per account.	5 flexible seals.	First half of the old illustrative 10 milestone allowance.
FIRST02	First story-chapter completion, once per account.	5 flexible seals.	Second half of that allowance. Included only when earned; no recurring reset.

At full candidate completion, CORE01 + CORE02 yields 34 seals and 60 fragments per 28 days; 68 seals and 120 fragments over 56 days. The fragment split, 10 on first visit plus 5 on second, is a new proposal, not a recovered old decision. One seal buys one eligible pull in this candidate. Only a seal actually spent in the featured family advances the proposed 70-pull counter. Fragments target the eligible standard pool and do not advance featured pity.

Reliability is conditional on access and completion. A player returning once each week receives 2 seals and 10 fragments weekly under this proposal, not the full two-visit reward. The core route is available without optional modes, but no

completion or calendar guarantee is claimed for every player. Restricted promotional tickets, if later added, require their own eligibility records.

40.5 Remaining source register and resource coverage

[PROPOSAL] The following IDs complete the source-category draft for Sections 33 and 39. OPEN is an unresolved design input, not a zero reward. Zero in the counted-budget column means no dependable numeric credit is taken until that input is specified. Subtypes such as a particular boss, event or achievement need child definitions before release; this is not a claim that every future content reward has already been priced.

ID and class	Rule and eligibility	Reward and budget treatment
IDLE01 Core recurring	After idle unlock; disjoint accrual intervals at the applicable campaign tier. Existing 24h full + 48h at 25% proposal; then pause.	Ryo, EXP, common essence and basic enhancement vector; exact rates OPEN. No default seals or fragments.
MAT01 Core recurring	One weekly baseline material entitlement after training unlock. Qualify through CORE02 OR three eligible solo training attempts; both routes share one claim.	Material vector M_r OPEN. Accessible alternate route; no numeric income credited until quantities and unlock timing are selected.
GEAR01 Core recurring	One baseline equipment choice per week after the gear tutorial; claim through core play. No win/rank requirement.	1 selectable basic piece, PROPOSAL. Slot/tier rules OPEN. One piece is not assumed to fill an entire loadout.
PVE01 Optional repeatable	Per published solo mode and reward tier; entry limits, unlock and reset OPEN.	Specialized materials or mode tokens. Core acquisition budget counts 0; random drops excluded from readiness floor.
SOC01 Optional repeatable	Guild/co-op participation; per-mode eligibility and reset OPEN.	Guild tokens and optional materials; 0 core-budget credit. No dependence on ideal guilds.
PVP01 Optional or seasonal	Participation and ranked rewards have separate tier definitions; season schedule OPEN.	PvP tokens, cosmetics and supplemental rewards; 0 core-budget credit. Top-rank income cannot fund essential access.
EVT01 Event or seasonal	Event ID, start/end, milestones, expiry and fallback must be published; currently OPEN.	Event tokens and optional grants. No event assumed in the base 56-day journeys.
END01 Optional optimization	Endgame unlock, boss/reset and tier-specific prerequisites OPEN.	Advanced materials or prestige; 0 baseline-readiness credit.
ACH01 One-time	One claim per published achievement ID; prerequisite OPEN.	Reward vector OPEN; never multiplied into a recurring floor.
RET01 Conditional catch-up	Return eligibility and cooldown OPEN; separate from idle accrual.	Catch-up vector OPEN; 0 ordinary recurring credit.
ENTRY01 Entry entitlement	Per-mode energy regeneration/cap or free entry rules OPEN.	Entries fund mode access; not seals or idle materials. Do not count replenishment as acquired loot.
SHOP01 Conversion	Published offer ID, purchase limit and reset; spend and receive vectors both required.	Subtract tokens/currency spent; add output once. Not a second source of gross free value.
RECRUIT01 Conversion	Spend one eligible seal; standard and featured outcomes tracked separately.	Character result may be random. No lucky result or duplicate-conversion income assumed.
SELECT01 Conversion	Spend 100 persistent Universal Fragments for one eligible standard-pool choice.	Deduct 100; grant 1 selected hero. Eligibility list OPEN; preserve enough broadly accessible counters.

ID and class	Rule and eligibility	Reward and budget treatment
SALVAGE01 Refund or conversion	Published gear replacement/refund policy; exact rule OPEN.	Recover only allowed prior investment, with a recorded input. Never count both a transfer and a refund of the same value.
PAID01 Paid	Offer-specific entitlement and limits; no assumed purchase.	Excluded from every F2P scenario.
COMP01 Exceptional grant	Specific service/compensation notice and claim eligibility.	Excluded from dependable income; do not finance recurring targets with compensation.

Coverage of the 17 earlier resource families: Ryo/EXP/essence/enhancement connect to IDLE01 and MAT01; basic skills to MAT01/PVE01; gear to GEAR01 and first-clear definitions when specified; characters/duplicates to START01, RECRUIT01 and SELECT01; fragments to CORE01/CORE02; standard/featured spending to RECRUIT01; mode, guild, PvP and event tokens to their named rows; energy to ENTRY01; returner resources to RET01; cosmetics to their event, achievement, PvP or paid entitlement. Standard and featured are spending destinations under the flexible model, not two duplicated balances. Old 17-family coverage does not require 17 permanent currencies.

40.6 Dated 56 day acquisition schedule

[ILLUSTRATIVE ARITHMETIC] The table uses eight complete weeks, both core entitlements earned every week, no optional sources, no opening resources, and no missed reset. To avoid assuming early collection, it records each week's earned rewards at its end. These dates are a conservative bookkeeping schedule for this example; actual earlier qualifying visits may credit rewards earlier. An account joining midweek needs its own partial-week schedule.

Week	Day	Seals earned	Seals total	Fragments earned total	Selector status
1	7	8	8	15	Not yet
2	14	9	17	30	Not yet
3	21	8	25	45	Not yet
4	28	9	34	60	Not yet
5	35	8	42	75	Not yet
6	42	9	51	90	Not yet
7	49	8	59	105	Select 1; 5 remain
8	56	9	68	120	1 selected; 20 remain

At day 49, 105 fragments permit one SELECT01 conversion, leaving 5; day 56 leaves 20 after that same single conversion. The table's fragment total is cumulative gross earned, not the wallet after spending. At zero opening pity, 68 seals still cannot guarantee a featured copy. This candidate therefore does not support an eight-week featured promise from routine income alone.

Payout-aware correction: 100 / 15 weeks is an average-rate estimate, not a claim date. With weekly batches, ceil(100 / 15) = 7 payouts. At 10 per weekly payout, ten payouts are needed; at 20, five. Reserve seven preparation days only after the character is available unless materials were accumulated beforehand. Pre-saving can reduce the post-acquisition wait, but must appear as an actual opening reserve in the preparation budget.

40.7 New accounts and spending choices

56 day case	Seals and featured allocation	Fragment route and interpretation

56 day case	Seals and featured allocation	Fragment route and interpretation
Mature, all core rewards, save seals	68 earned; up to 68 eligible featured spends. Short by 2 for a 70-pull threshold.	Selector on seventh payout. An essential counter must not depend on a featured result.
New account, both first-clears earned	68 recurring + 10 one-time = 78. Up to 70 can fund one guarantee, leaving 8 unspent.	One-time START01 covers the first squad. A later standard selector is separate. The +10 is conditional on completing both milestones.
Mature, 25% spent on standard	17 standard spends + 51 featured spends = 68. Featured threshold short by 19.	Same fragments if both core visits completed. Standard pulls do not secretly add featured pity.
New account, 25% policy with integer rounding	Reserve 20 of 78 for standard; 58 for featured. Featured threshold short by 12.	The approximate 25% split is explicit; no fractional ticket or additional income is invented.
Split tickets at equal total income	Illustrative 80% earmark: 54 featured + 14 standard = 68, rounding featured down. Short by 16.	A split makes allocation fixed; it does not create more income. This is a window-level sensitivity, not a new exact payout schedule.
One visit every week	16 seals and 80 fragments in 56 days.	No 100-fragment selector in eight weeks. The full route is conditional, so a catch-up alternative remains a design question.
First visit every week; second visit missed in weeks 3 and 6	68 - 6 - 7 = 55 seals; 120 - 5 - 5 = 110 fragments.	By day 49, 105 - 10 = 95 fragments; selector arrives with week 8. A seven-day preparation allowance then extends to day 63.

The optional-mode envelope stays separate. Repeating the old social 6 plus event 10 assumptions twice would add up to 32 seals over 56 days, yielding 100 with the 68 core candidate. This is only a conditional illustration: it requires two eligible event windows and the full assumed social participation. The base cases above count neither. Do not use the 100 figure as a dependable floor.

A calendar featured guarantee is a separate product commitment. Even with no standard spending, 70 pulls in 56 days needs 35 eligible pulls per 28 days, versus the candidate 34. If 25% is spent on standard, at least 94 total integer seals in the window are needed to leave at least 70 after an explicit allocation. Increasing income solely to meet that promise requires a new whole-economy budget; it is not adopted here.

40.8 Acquisition is only the first part of a counter journey

[PROPOSAL FOR THE NEXT COSTED PACKAGE] Define baseline readiness as a recruited shinobi providing its intended role at the account's current shared training baseline, with a usable equipment loadout and its functional base kit. It does not mean maximized stats or a guaranteed matchup win. The shared baseline must be available by the time the route is advertised; any personal promotion cap that blocks it is a real cost, not an exception hidden from the ledger.

Period	New account	Mature account
Introduction and week 1	START01 provides missing role coverage; starter level/gear grants must be specified separately. FIRST01/02 pay only on their actual milestones.	Use an already functional squad. Opening seals/fragments are zero in this conservative example; older first-clears pay nothing.
Weeks 1 to 6	Earn core rewards while building the shared baseline. Do not charge the same common levels again for every bench hero.	Continue baseline growth and reserve common materials for the intended counter. Optional events, guilds and PvP are unnecessary for this acquisition path.
Seventh payout at day 49	105 fragments earned; spend 100 on the eligible standard counter.	Same selector timing when all core conditions are met.
Days 49 to 56	Use only actual material reserves and residual income after starter/core commitments; include any missing gear pieces.	Use only residual income after ongoing core growth; no assumption that the original squad is dismantled for gear.

Period	New account	Mature account
Day 56 review	Acquisition fits the paper window; preparation remains conditional on the resource and gear checks below.	The same distinction applies. Shared levels alone do not prove sufficient gear or material access.

Gear is a separate constraint. The proposed GEAR01 can cover one missing basic piece in a week once its unlock and tier rules are satisfied. If the counter needs two missing pieces, or that week's choice was already spent, a one-piece allowance alone cannot establish seven-day readiness. Reusing unequipped spare gear, loaner gear, class-shared gear or another deterministic source are alternatives to compare; none is silently assumed to exist. Equipping the counter by weakening the only functioning squad is not counted as a successful breadth result.

Record the campaign checkpoint and required baseline when preparation begins. If the account advances during those seven days, recompute any affected requirement and segment the income rate at the change. A fixed-checkpoint example cannot establish readiness against an unspecified moving level or gear target.

40.9 Idle storage changes the material budget

[PROJECT-DOCUMENT AUDIT AND ARITHMETIC] Section 32 currently proposes 24 hours at full rate, the next 48 hours at 25%, then no further accrual until collection. At a fixed campaign rate, a claim after h hours contains e(h) = min(h, 24) + 0.25 × min(max(h - 24, 0), 48) full-rate hours. Divide by 24 to obtain full-rate-day equivalents. This quantity is not a new currency.

Collection schedule over seven days	Full-rate-day equivalents	Share of daily collection
Seven claims, each 24 hours apart	7.00	100%
Four claims, gaps 24 / 48 / 48 / 48 hours	4.75	67.9%
Two claims, gaps 72 / 96 hours	3.00	42.9%
One claim after 168 hours	1.50	21.4%

Two core visits can satisfy the acquisition track but collect only three full-rate days of idle materials under the proposed cap. A preparation claim based on seven full-rate days would overstate that player's idle supply by 7 / 3, about 2.33 times. This is a deterministic consequence of the written cap, not evidence about how often real players log in. Campaign-rate changes must be segmented as Section 32 requires; no new rate is applied retroactively.

[DESIGN INFERENCE] The project must decide which collection schedule its readiness promise covers. Possible remedies are an accessible weekly material allowance, a smaller baseline preparation cost, or a longer full-rate storage window. Requiring every optional mode to repair the deficit would contradict the purpose of the core route. Altering the storage cap is not free: its extra resource supply has to be reviewed against all sinks. No cap change is selected here.

40.10 Reconcile the cost vector before claiming readiness

For every resource r, use the dated balance: available_r(t) = opening reserve_r + eligible grants_r(up to t) + conversion outputs_r - conversion inputs_r - committed core spending_r - optional spending_r. Earliest readiness is the first date on which all required resource costs, gear pieces, level/cap prerequisites and unlocks are satisfied together. No balance can go negative, and one reserve cannot finance two simultaneous goals. A shared-level cost paid by the account is not paid again by the newly selected shinobi unless the design explicitly requires a separate benefit.

Cost ID	Baseline requirement	Input that remains open
COST-LEVEL	Ryo/EXP for account baseline growth. Incremental hero level cost can be zero only under the shared-baseline candidate.	Level bands, unlock and treatment of personal promotion caps.

Cost ID	Baseline requirement	Input that remains open
COST-PROMO	Essence or cap payment only if the target needs a separate promotion.	Whether shared training also supplies that cap; otherwise exact cost and accessible supply.
COST-KIT	Role-defining cleanse/control/protection works at base copy and kit.	Extra ranks are optional performance investment; any required rank must be explicitly justified and costed.
COST-GEAR	List every missing usable item and all required enhancement resources.	Slot count, usable tier, selector coverage, equipment transfer/loan policy and price.
COST-CORE	Reserve the ongoing growth commitments of the existing squad.	Stage-specific spending schedule; cannot allocate the same Ryo/EXP to the counter twice.
COST-SELECT	100 fragments in the current candidate.	Eligible pool and substitutes; does not pay any equipment or training bill.

[ILLUSTRATIVE ARITHMETIC ONLY] The following normalized example shows how the method detects a failure. For each idle resource, one U_r equals that resource's own output from 24 full-rate hours at a fixed checkpoint. U_Ryo and U_gear are different units and cannot be exchanged or added. Assume zero preparation reserve at acquisition, no non-idle material grant, seven days available, three U_Ryo committed to core growth, one U_gear committed elsewhere, and a counter package costing two U_Ryo plus two U_gear. These are demonstration values, not selected game prices. Assume shared levels/caps and an accessible needed gear piece solely for this example; removing either assumption adds a separate unmet requirement.

Resource	Counter cost	Daily claims gross / core / free	Two weekly claims gross / core / free
Ryo units	2 U_Ryo	7 - 3 = 4 available; sufficient	3 - 3 = 0 available; short by 2
Enhancement units	2 U_gear	7 - 1 = 6 available; sufficient	3 - 1 = 2 available; sufficient
Incremental EXP, essence and kit materials	0 in this specific shared-baseline example	Only valid under the stated cap/kit assumptions	Any extra required cost must be added; zero is not a general rule
Missing gear piece	1 accessible piece assumed	GEAR01 must be unspent and able to supply the required slot/tier	If it cannot, the resource arithmetic alone does not establish readiness

The daily-collection example can fund its specified package; the two-visit example cannot because Ryo has no residual supply. For that demonstration, a dependable MAT01 grant of at least two U_Ryo during the preparation week would close the Ryo gap. In general, the required accessible supplement is max(0, counter cost_r + committed core cost_r - opening reserve_r - collected income_r). This calculates a design requirement; it does not establish a final material grant. If one choice chest must cover several scarce resource types, the same chest cannot be assumed to satisfy all of them.

At the same illustrative commitment and cost, two weekly collections under a 48-hour full-rate window followed by 24 hours at 25% would yield 4.5 U_Ryo and leave only 1.5, still short. A 72-hour full-rate window followed by 48 hours at 25% would yield 6.25 for the 72/96-hour gaps and leave 3.25. Those are sensitivity examples only. They explain why changing one timer requires a complete source/cost comparison rather than a blanket claim that a longer cap fixes the economy.

40.11 Findings and the next research packet

[RESEARCH SYNTHESIS] This pass specifies distinct core entitlements, preserves one-time and optional income separately, and replaces fractional average dates with a concrete eight-week schedule. Under the candidate full core route, 68 routine seals do not fund the 70-pull featured threshold in 56 days. A standard-pool selector becomes available at the seventh weekly payout, but a seven-day preparation target has no spare week and still requires material and gear coverage. Missing a weekly condition or collecting idle materials less often can extend the journey.

Question	Status after this pass
Are the old 14 and 12 payouts independent?	Still historically unknown. New candidate CORE02 replaces their combined allowance; it does not prove the old rows were additive.
Does a core acquisition route avoid mandatory optional modes?	Structurally yes in the new proposal; its reward amounts, two-visit condition and completion burden remain unapproved.
Is an eight-week featured promise supported?	No under the core 68-seal example. It remains a separate, unfunded product commitment.
Is a standard counter acquired and ready by day 56?	Acquisition reaches day 49 with full weekly completion. Readiness is conditional; the sparse-collection cost example fails.
Is the full numeric source ledger finished?	No. The canonical draft defines IDs and quantified acquisition entitlements; material rates, content-specific child rewards and costs remain OPEN.
What has actually been checked?	Deterministic totals, conservation, payout counts and idle-cap arithmetic. No gameplay effectiveness, player effort, retention or live balance has been measured.

[RECOMMENDATION, NOT OWNER-APPROVED] Continue with flexible seals plus persistent standard-pool targeting as the leading comparison model. Keep base counter functionality independent of featured acquisition. Compare an accessible weekly material package with a longer full-rate bank and lower preparation costs before selecting any one remedy. A small bounded catch-up rule should be considered separately because the current proposed core track still loses uncompleted weekly eligibility.

Next research task: define one explicit baseline counter-readiness package at an early and a mature campaign checkpoint. Specify the shared-training unlock and personal-cap treatment, usable gear slots/tier and transfer policy, common material sources, and exact core commitments. Price each required resource, then compare the unchanged bank plus MAT01 against longer storage alternatives for two-, four- and seven-visit schedules. Include a missed week, a midweek new account, one missing gear slot and a campaign wall. Keep first-clears, events, paid income and luck out of the mature dependable case. Reconcile 56-day access only after every required source and cost has a defined value; all work remains on paper.

40.12 Source register and evidence limits

Sources accessed 30 October 2026. The official pages below support the narrow structural observations in Section 40.1. No benchmark source supplies the proposed 2/6/7-seal grants, 10/5-fragment split, two-visit condition, material costs or wait-time promises. Those belong to this project's hypothetical design.

[LED1] HoYoverse Help Center. How do Encounter Points work? Updated 17 September 2025. Official, directly opened. Used only for alternative completion paths redeeming commission rewards.

https://support.hoyoverse.com/hc/en-us/articles/50333944750105-How-do-Encounter-Points-work

[LED2] Unity documentation. Add a virtual purchase to your game. Official, directly opened. Used for explicit transaction identity, input costs and output rewards.

https://docs.unity.com/en-us/economy/add-virtual-purchase

[LED3] Unity documentation. Introduction to resources in Economy. Official, directly opened. Used for the distinction among balances, inventory and conversions; no service selection is made.

https://docs.unity.com/en-us/economy/item-types

[LED4] Farlight Games. Talk with Designers New Season Preview. Official indexed excerpt; live page body was not readable in this pass. Used only for handbook reward timing and selectable equipment, not current numeric rates.

https://afkjourney.farlightgames.com/news/a0ae0b8ad8c35da97805079b672b7287

Historical cross-references: Sections 21.2, 32.6, 33.4, 34.4-34.8, 35.3-35.8 and 38-39 supply the earlier project assumptions. Their proposed numbers remain historical unless explicitly revised above. Preserved benchmark citations and appendices have not all been re-audited by this pass.

Handoff to v0.10.12. Section 41 specifies the baseline package requested in 40.11 and compares its cost with the paid preparation example above. Section 41.1 gives the current research order.

41. Research priorities and baseline counter readiness

31 October 2026 | Version 0.10.12. First decide what it takes to field a recruited cleanser. Which levels and gear does it inherit? Can the original five return with their gear intact? Those answers connect the combat rules in Sections 24–27 to the budgets in Sections 35–40.

The table sets our research order. New training, equipment and reward rules are proposals. The numerical cases are paper budgets.

41.1 Research order and conditions for moving on

Priority and scope	What the next module needs from this one
1 Field an owned counter	Specify training, caps, base skills, gear and the cost of switching. Sections 41.3–41.6 deliver the early and mature cases for budgeting.
2 Fund shared squad growth	Price a complete upgrade at each checkpoint. Show dated balances at two, four and seven visits, including a campaign wall. Record the first unaffordable input and the smallest remedy that covers it. Next primary module.
3 Time access to essential roles	Map starter and selector eligibility to Section 26. Trace 56 days for new, mature, late-joining and returning accounts. Date two distinct answers to each threat, one broadly obtainable before its gate.
4 Design campaign and optional encounters	Write the campaign teaching sequence, its story beats, one boss and the optional modes. For each, specify its tactical question, two answer categories, unlock, cost and reward ID. Send any new mandatory cost back to priority 2.
5 Teach the loop on mobile	Trace a first session and a return visit from claim through squad choice to the battle recap. Specify the exact screen where a player learns the threat, a legal response, a missing resource and its accessible source.
6 Define social and competitive play	Separate solo access, co-op participation and ranked rewards. Specify normalization, entry, matchmaking and what a player who skips guild or PvP retains. Check the role-access budget again.
7 Plan events and paid progression	Map event timing, absences, catch-up, long-term goals, paid offers and disclosures to source IDs. Compare free and paying paths through the same required encounter.
8 Audit the whole concept	Trace a returning player from saved state through claim, recruitment, battle and recovery. Reconcile duplicate definitions and version changes. Record open production, service integrity and future measurement questions.

At every handoff, check the 29.2 inventory for product identity, Naruto-inspired narrative and rights, accessibility, solo viability, payment fairness and save integrity. Advance when the next module has an explicit rule, dated price or documented shortfall to work with. Questions requiring player evidence belong in the Section 28 future validation register.

41.2 Why roster readiness comes first

Section 40 traced a standard selector to the seventh weekly payout. One weekly gear choice may still leave a second slot empty. Before increasing income, decide which investments belong to the squad and which a player chooses for one shinobi.

[SOURCE FINDING] Farlight Games describes shared levels and equipment in AFK Journey [READY1]. The Mythic Heroes guide asks players who swap in a new hero to check its runes and artifacts [READY2]. These peers separate the level a recruit receives from the equipment a player must assemble.

[SOURCE FINDING] Guild Wars 2 offers a narrower comparison from an adjacent genre. Its Legendary Armory gives eligible characters access to account-owned equipment, with explicit restrictions on eligibility and copies [READY3]. We use this distinction when separating an account permission from a physical item that can occupy only one active slot.

Progression model	Effect on the squad decision
Individual levels and gear	A new counter needs a second level budget and an equipment route. This preserves personal investment but adds a repeat cost each time the player changes composition.
Share all levels and equipment	Replacing a teammate is simple. Sharing every advanced build also reduces the scope for a player to specialize a particular shinobi.
Share the earned baseline	Use Section 35's hybrid model. Fund squad training once and issue basic gear at the earned band. Owned equipment and advanced skill ranks give a player something specific to build toward. Leading candidate.

41.3 The minimum package for an owned shinobi

[PROPOSAL] A shinobi is fieldable when the player owns one eligible copy, can place it in a legal formation, receives the squad's earned level and promotion cap, has its defining skill available, and can fill both basic equipment slots without taking equipment from another member. Its effectiveness against a particular encounter is a separate question for the combat and content review.

Requirement	Candidate rule and accounting treatment
Unlock and timing	The introductory squad tutorial activates shared training and standard equipment before paying START01 or opening ordinary recruitment. A pre-unlock tutorial unit uses its scripted tutorial loadout. The roster shows the unlock condition for any early grant.
COST-LEVEL and COST-PROMO	All owned deployable shinobi inherit the Hall's purchased level up to its earned account cap. The squad pays Ryo and EXP for levels and any common essence required for Hall promotions once. Taking a new copy to that same point costs 0 Ryo, 0 EXP and 0 essence. The next unearned cap still costs the squad resources.
COST-KIT	The first copy gets its defining skill under the usual targeting, cooldown, Chakra and status rules. It costs 0 skill materials and 0 duplicates to access that function. Optional ranks may improve its numbers.
COST-GEAR	Two basic slots, tool and protective gear. Each owned unit receives a permanent standard-issue loadout in its earned Hall band. Base enhancement is +0. These items are derived equipment, with 0 Ryo and 0 enhancement material due at recruitment. The account can field all five at once.
Transfer and reward credit	Standard gear has no sale, salvage or conversion value. A personal item replaces the standard item in its slot and retains its enhancement when moved. It occupies one active slot. A saved preset holds a reference. Removing that item restores standard gear. GEAR01 still pays one actual choice per eligible week.
COST-SELECT and COST-CORE	SELECT01 costs 100 fragments for an eligible standard-pool hero. START01 is a distinct one-time role choice. The weekly core-growth budget stays on the account ledger. Hall access and issued gear are persistent permissions and create no recurring wallet grants.

The Hall band determines the standard gear floor for all eligible roles. A stronger personal item contributes only the replacement's stats in that slot. Standard issue is accounted for once inside the reference loadout. Use the Section 27 role multipliers for comparison. Map the combat stats of each earned Hall band separately when pricing growth. A higher earned band updates the baseline for the entire roster at the time it is purchased.

41.4 Early and mature account budgets

The following two checkpoints are illustrative inputs. "Level 20" and "level 80" give the bookkeeper a fixed reference. Actual campaign bands, stat curves and currency denominations are inputs for priority 2.

Condition	Early checkpoint E	Mature checkpoint M

Condition	Early checkpoint E	Mature checkpoint M
Opening state	Tutorial complete. Hall level and cap 20. Band I earned. The account holds its unused START01 choice and can select K-09 for cleanse.	Hall level and cap 80. Band II earned. The account owns five functional members and can spend 100 fragments on an eligible K-09.
Result when K-09 joins	Level 20, cap 20, base cleanse, Band I tool and protection at +0. No reserve or weekly gear choice is required.	Level 80, cap 80, the same base cleanse and both Band II pieces at +0. A spent GEAR01 or two empty personal slots do not delay fielding.
Incremental preparation	0 Ryo, EXP, common essence, skill material or enhancement material. 0 physical gear claims. The starter choice is consumed once.	The same zero preparation vector. The 100-fragment conversion is charged once. The existing five keep their levels, gear and presets.
Weekly squad commitments	Reserve 3 Ryo, 2 EXP, 1 common essence and 1 enhancement in resource-specific 24h units.	Reserve 3 Ryo, 3 EXP, 2 common essence and 1 enhancement in the same units.

Each resource unit equals 24 hours of its own full-rate output at that checkpoint. Assume zero reserves, fixed ordinary idle rates and payment after the last collection. With gaps of 72 and 96 hours the player collects 3 units of each resource. After the illustrative commitments, E retains 0 Ryo, 1 EXP, 2 essence and 2 enhancement units. M retains 0, 0, 1 and 2. The next shared upgrade competes for the Ryo and EXP already spent.

A role example makes the distinction concrete. Against K-06 Wound, an owned K-09 can take the flex slot and use its base cleanse on an eligible ally at a legal activation. Target selection for the paper case is one allied Wound source, removed on K-09's action before the ally's next tick. The normal cooldown and turn order still matter. A second route is to focus K-06 with a damage dealer such as K-03. The encounter designer must specify when each route is available. Equipping the cleanser alone cannot settle which route wins.

41.5 Compare supply remedies against the same bill

Section 40.10's comparison bill charges the recruit 2 Ryo units and 2 enhancement units, with one usable physical piece already available. Reserve 3 Ryo and 1 enhancement unit for ongoing squad growth. The table gives the extra Ryo still needed at the end of seven days. The new fielding proposal removes the personal preparation bill. Its account commitments still apply.

Seven-day comparison	2 visits	4 visits	7 visits
Idle units under the 24h full + 48h at 25% bank	3.00	4.75	7.00
Old bill under that bank: Ryo shortfall	2.00	0.25	0.00
Old bill plus a funded MAT01 grant of 2 Ryo units: Ryo shortfall	0.00	0.00	0.00
Old bill with 48h full + 24h at 25%: Ryo shortfall	0.50	0.00	0.00
Old bill with 72h full + 48h at 25%: Ryo shortfall	0.00	0.00	0.00
Proposed standard issue under the original bank: additional Ryo required for fielding	0.00	0.00	0.00

Visit gaps in hours are 72/96 for two visits, 24/48/48/48 for four, and seven gaps of 24 for daily visits. The two longer banks yield 4.50 and 6.25 full-rate days respectively at two visits, and 7.00 at both higher frequencies. Under the old bill, the Ryo gap is the positive part of 3 + 2 minus collected Ryo and any timely grant. The 2-visit player also has exactly 2 enhancement units after the core commitment. Count MAT01 once, only when its entitlement has been earned before the bill is due.

[RECOMMENDATION] Carry the shared baseline into the next budget and retain Section 32's storage rule as the comparison case. Standard issue removes the cost triggered by switching characters. A longer bank boosts every idle resource, while MAT01 can target one short resource. Decide those two policies from the shared growth budget. Under the charged alternative, a missing physical piece can still block fielding even when both currency balances are sufficient.

41.6 Recruitment dates and failure cases

Paper case	Outcome under the proposed baseline
Aligned account, both visits	The seventh weekly posting brings cumulative fragments to 105 on day 49. Spend 100 on an eligible standard counter. Five remain, and the counter can enter the next legal lineup that visit. Day 56 ends at 20 after the single conversion.
Miss a whole week	Miss week 3 entirely. Six credited weeks yield 90 by day 49. The seventh credited week yields 105 by day 56. Fielding can occur at that posting if the player chooses the counter.
Miss second visits in weeks 3 and 6	Section 40.7 gives 95 fragments by day 49 and 110 by day 56. The standard-issue proposal permits fielding on day 56. With only one visit in every week, the account has 80 fragments at day 56 and still needs another acquisition route or later payouts.
Join on Wednesday	Tutorial completes on day 1. Earn 10 on that visit, then return each Monday and Thursday from the next week. Five complete weeks bring the total to 85. Monday day 41 brings it to 95, and Thursday day 44 to 100. Immediate credit follows CORE01/02. Day 49 is specific to Section 40's week-end posting convention.
Two missing gear pieces	With zero physical spares and GEAR01 spent, both standard slots still populate. Removing K-09 restores the original five with every previously equipped item. The old one-piece-per-week physical route cannot fill two empty slots in that same entitlement.
Blocked campaign	At a fixed checkpoint, apply only the already-earned Hall band and idle rate. If the account lacks a cleanser before a required wall, identify an earlier reachable substitute such as focus on the applier. If neither route is accessible, move the access point or redesign the gate.

The mature full-core schedule still produces 68 seals in 56 days against the proposed 70-pull featured threshold. The standard-issue policy affects the cost after acquisition. It cannot turn a featured pull into a guaranteed result or put an ineligible hero in the selector. The proposed standard list includes K-09 for these cases. Priority 3 must map the rest of the pool against the actual campaign order.

41.7 Handoff to shared growth pricing

Priority 1 is specified for paper budgeting. Track ownership, fieldability at the earned baseline and effectiveness in an encounter separately. The first two follow from the roster and equipment rules. The third requires evidence about the matchup.

Priority 2 will price one meaningful Hall upgrade at E and one at M. For each, state the last cleared campaign checkpoint and exact IDLE01 output per hour in Ryo, EXP, common essence and enhancement materials. Specify the upgrade's before-and-after level, cap and stat benefit, each resource cost, opening wallet, payment date and any distinct eligible MAT01 entitlement. Use the two-, four- and seven-visit calendars from 41.5 and a stalled account. Record the day the slowest requirement is met and the resource displaced by that payment.

The handoff is complete when every required resource has a published route and dated balance, or a named shortfall with a concrete alternative. Recheck the candidate 1–2 daily-cycle and 2–3 day stalled-account pacing questions in 35.7 against actual visit dates. If two visits per week cannot meet a shorter calendar target, distinguish time affordable from the next opportunity to collect and spend. Reopen priority 1 if a standard item is too weak for the advertised encounter, a shared cap hides a personal fee, or equipping the counter consumes another teammate's gear.

41.8 Sources and scope

Consulted 31 October 2026. Section 41.2 uses these sources for design comparisons. All project prices and calendar outcomes follow from the assumptions above.

[READY1] Farlight Games. AFK Journey. Developer-authored App Store description of shared levels and equipment. Direct idle RPG peer. https://apps.apple.com/us/app/afk-journey/id1628970855

[READY2] IGG. Mythic Heroes Day 1 Guide. Publisher guidance on a new recruit's runes and artifacts. Direct idle RPG peer. https://www.mythicheroes.com/7day-guide-day1/

[READY3] ArenaNet. Legendary Armory. Guild Wars 2 support explains account ownership, character eligibility and copy limits. MMO equipment analogy. https://help.guildwars2.com/hc/en-us/articles/4403435663635-Legendary-Armory

42. What the next shared Hall upgrade costs

Priority 2 | 31 October 2026. At a fixed campaign wall, the proposed early upgrade is affordable at the first or second collection under all three schedules. The mature promotion needs more EXP. A player visiting twice must wait until day 7 even though the full week produces enough currency. We recommend carrying this slower path into encounter planning before promising a deadline.

42.1 Two purchases and their prices

Design proposals for paper comparison. E and M are placeholder IDs for the last cleared campaign stage. Both accounts start at day 0 just after an empty claim, with zero balances. Neither clears another stage during the week. The Hall upgrade is part of the weekly core commitment in Section 41.4. Counting it again as an extra purchase would overstate the bill.

Input	Early stage E	Mature stage M
Shared purchase	Hall level 20 to 21. Cap 20 to 25.	Hall level 80 to 81. Cap 80 to 85.
Immediate benefit	All eligible shinobi inherit level 21. For this example base HP, ATK and DEF rise by 2% of their level-20 values.	All eligible shinobi inherit level 81. Illustrative base HP, ATK and DEF rise by 1% of their level-80 values.
IDLE01 hourly income	10 Ryo, 20 EXP, 2 common essence, 1 enhancement material.	40 Ryo, 60 EXP, 4 common essence, 2 enhancement materials.
Price of this purchase	300 Ryo, 720 EXP, 48 essence, 0 enhancement.	1,920 Ryo, 3,600 EXP, 192 essence, 0 enhancement.
Other core bill due on day 7	420 Ryo, 240 EXP, 0 essence, 24 enhancement.	960 Ryo, 720 EXP, 0 essence, 48 enhancement.

Each purchase opens the next cap and pays for the first level above the old cap. Levels after 21 or 81 require separate future payments. Role multipliers, Speed, skill access and the gear band stay at their prior values for this step. The listed percentage describes the illustrative base-stat benefit before role multipliers. Encounter effectiveness remains open.

The two bills sum to the Section 41.4 weekly commitments. In E they total 720 Ryo, 960 EXP, 48 essence and 24 enhancement. In M they total 2,880 Ryo, 4,320 EXP, 192 essence and 48 enhancement. MAT01 earns zero budget credit in the base case. Recruitment seals and fragments stay in their own ledgers.

42.2 When the player can collect and pay

A day is 24 hours after the empty opening claim. Credit 24 hours at full rate and the next 48 hours at one quarter rate, then pause until collection. Pay the Hall bill at the first visit with enough of every input. The other core bill has a day-7 due date and has priority over optional spending. The examples assume it can wait until that date.

Visits	Claim days	Cumulative full-rate days at each claim	First Hall payment E / M
2	3, 7	1.50, 3.00	Day 3 / day 7
4	1, 3, 5, 7	1.00, 2.25, 3.50, 4.75	Day 3 / day 5
7	1 through 7	1, 2, 3, 4, 5, 6, 7	Day 2 / day 3

Divide each price by its own full-rate daily output. E needs 1.25 days of Ryo, 1.50 of EXP and 1.00 of essence. M needs 2.00, 2.50 and 2.00 respectively. EXP is the last requirement in both cases. At four visits M has only 3,240 EXP on day 3, which leaves a deficit of 360. The next visit on day 5 covers it. For daily E the bank already holds enough at hour 36 across the first two accrual intervals, but the spend occurs at hour 48 when the player returns.

42.3 Dated wallets and competing uses

Amounts below follow the order Ryo / EXP / essence / enhancement. The purchase-day wallet is after the Hall payment and before the separate day-7 bill. The final wallet deducts both bills exactly once.

Case	Pay day	After Hall purchase	After both bills on day 7
E · 2 visits	3	60 / 0 / 24 / 36	0 / 480 / 96 / 48
E · 4 visits	3	240 / 360 / 60 / 54	420 / 1,320 / 180 / 90
E · 7 visits	2	180 / 240 / 48 / 48	960 / 2,400 / 288 / 144
M · 2 visits	7	960 / 720 / 96 / 144	0 / 0 / 96 / 96
M · 4 visits	5	1,440 / 1,440 / 144 / 168	1,680 / 2,520 / 264 / 180
M · 7 visits	3	960 / 720 / 96 / 144	3,840 / 5,760 / 480 / 288

The mature two-visit player reaches day 7 with 2,880 Ryo and 4,320 EXP before spending. Paying for the Hall leaves 960 Ryo and 720 EXP. Those balances are exactly what the other core bill consumes. An optional purchase of 100 Ryo before the bills are settled creates a 100-Ryo deficit. A newly owned cleanser can still use the already-earned Hall level under Section 41.3 during that wait.

42.4 What would justify a change

For a day-3 mature upgrade at two visits, the first claim yields 1,440 Ryo, 2,160 EXP, 144 essence and 72 enhancement. The smallest timely supplement for this bill is 480 Ryo, 1,440 EXP and 48 essence together. Funding only the EXP shortage leaves two blockers. A MAT01 award tied to the second weekly visit arrives on day 7 in this calendar and cannot solve the day-3 shortage. The alternate training route would need its own proven unlock and claim time.

A storage change has a different effect. With 48 hours full and the following 24 hours at one quarter rate, the first 72-hour claim yields 54 full-rate hours, or 3,240 mature EXP. It still misses the price by 360 EXP. With 72 hours full the first claim yields 4,320 EXP and covers all three inputs on day 3. That rule raises income across the idle basket on every qualifying long absence. A lower Hall price or a targeted timely grant has a narrower effect.

Pacing decision for the next paper pass. Retain the original bank as the comparison case and use the observed payment dates from this ledger. The early daily case meets the 1–2 day planning window. The mature daily promotion takes three days, and the mature stalled two-visit case exceeds the 2–3 day window. These examples warrant distinct targets for routine levels and cap promotions. A four-day gap between visits can itself prevent a three-day payment even with sufficient resources in storage.

42.5 Handoff to essential role access

Priority 2 now has a priced reference at E and M, six dated wallets and an explicit shortfall against the accelerated target. Priority 3 should place the first Wound encounter on the same calendar as its eligible cleanser and a second response such as focusing the applier. For each route, write down the first ownership date, inherited Hall level, gear entitlement and first legal opportunity to respond. If a required encounter expects level 81 before day 7 on the mature two-visit path, reopen its timing, the price or the supply route. The campaign stage IDs, rates, prices and stat increases in this chapter remain illustrative.

Basis. Section 32.6 supplies the storage candidate, Section 35.7 the pacing questions, Section 40 the source identities, and Sections 41.4–41.7 the commitments, calendars and handoff. All new quantities are project design inputs. The arithmetic establishes payment dates under those assumptions.

43. Essential-role access I: first mandatory Wound encounter

Priority 3A | 31 October 2026 | Version 0.10.16. Research result: the first required Wound fight should be answerable by a guaranteed, accessible focus-fire team at the Hall level already earned; choosing a cleanser provides a second tactical

route. An owned cleanser is immediately fieldable by Section 41, but an unowned cleanser cannot be assumed before the seventh complete weekly core payout. This is a working paper recommendation, not an observed combat or retention result.

Evidence labels in this chapter: [PROJECT DECISION] is already recorded direction; [WORKING PROPOSAL] is a selected but unapproved game rule; [CALCULATED] follows from stated assumptions; [ILLUSTRATIVE COMBAT] uses invented post-mitigation values to expose dependencies; [OPEN / FUTURE VALIDATION] remains unproven. Do not treat a designed answer as a verified win. No prototype, simulator, seeded run, player study or live testing was performed.

43.1 Constraint chain and what must be true before a mandatory gate

Dependency	Existing authoritative baseline	New question / requirement
Role coverage	Sections 24-27: K-06 Wound, K-09 cleanse, K-03 finisher, K-12 Mark, K-19 reliable striker; 5-unit round-based auto combat.	Name which of these a new account reliably owns, not just which exist in the catalog.
Acquisition	Section 40: CORE01 + CORE02 = 15 Universal Fragments per fully completed week; SELECT01 = 100; START01 = one starter choice.	The earliest conservative fully aligned selector is week 7 / day 49, not an opening-day entitlement.
Immediate fielding	Section 41: owned deployable units inherit earned Hall level and standard-issue 2-slot gear; first-copy role skill works at 0 extra cost.	Record ownership and first legal combat activation separately from preparation cost.
Shared-growth deadline	Section 42: E can buy level 21 on day 2/3; M two-visit account buys 81 on day 7.	First required Wound gate cannot demand a discretionary level 21 or level 81 before it can be earned.
Content access	Exact stage at which a normal account reaches E or M is not yet dated.	Use E/M as state anchors, not an invented promise that tutorial immediately grants Hall 20/80.

[WORKING PROPOSAL] Separate two clocks. A 56-day acquisition calendar begins when the tutorial/recruitment systems are available. Encounter time E0 begins when the account legitimately reaches early campaign checkpoint E (Hall 20, cap 20, Band I); the calendar date when E0 occurs remains OPEN. This prevents treating a hypothetical E checkpoint as day 0 for every actual new account. The first Wound gate W-E unlocks at E0 without a real-time expiry, and it must be solvable at Hall 20. The day-3 upgrade to 21 is help, not an admission fee.

43.2 Guaranteed starter access: close the roster-entitlement gap

[PROJECT-DOCUMENT AUDIT] Section 26.3 lists a balanced K-01/K-03/K-09/K-12/K-19 squad but describes a roster shell, not an unambiguous entitlement. Section 40 START01 promises one eligible starter selection. Treating all five roster shells as automatically owned would double-count the starter grant or assume unseen acquisition sources.

Entitlement	Selected candidate	Cost, trade-off and role
START00 — introductory core (NEW; once per account)	Grant exactly K-01 Guardian, K-03 Finisher, K-12 Marker and K-19 Reliable Striker at the tutorial grant.	One grant of four owned role shells, not four seal pulls, four pity advances, or an additional weekly reward. Basic survivability, mark, target conversion and fallback damage are deterministic.
START01 — fifth slot choice (existing ID clarified)	Allow K-09 Cleanser OR K-10 Healer/Protector; the choice is displayed before the first required Wound stage.	One unit only. K-09 exchanges sustain for debuff removal; K-10 exchanges immediate cleanse for heal/shield. No currency or duplicate required.
SELECT01 — later flexible upgrade	Keep K-09 in the eligible 100-fragment standard selector (Section 41). K-10 eligibility also proposed for parity.	A K-10 starter can later choose K-09; selector conversion consumes 100 fragments exactly once. No featured pull requirement.
Migration / existing accounts	Any older account that missed START00 must receive its four core role entitlements or functionally equivalent verified substitutes before W-E becomes mandatory.	Migration semantics, duplicate-grant compensation and save integrity are OPEN for Priority 8. Never revoke existing owned units.

[RECOMMENDATION] Adopt START00 + one START01 flex as the candidate. Do not silently give an already-drafted five-unit sample team plus another START01 hero. The focus route uses K-03 + K-19 and does not require K-09 or a successful Mark application. K-12 Mark increases tactical interest when it lands; it cannot be a mandatory proc. K-01 provides protection. If

acquiring four tutorial heroes creates too little roster aspiration, vary their narrative identity and optional builds rather than removing a core counter.

43.3 W-E encounter contract and honest reachability

Design parameter	Paper contract at W-E	Why this restriction exists
Entry and progression	Normal solo campaign at checkpoint E; Hall level 20/cap 20 and earned Band I are sufficient; no entry energy, guild, PvP or banner condition.	A required lesson may test choices, not whether a casual player paid for level 21 or joined a social mode.
Wound source	One advertised K-06-class Wound applier on the front row, legally reachable by standard single-target Basic and the K-03 finisher; no Taunt, intercept, protected rear, instant revive or hidden shield before its first answer window.	Keeps direct focus a real route without adding a hidden second roster check.
Threat visibility	Show Wound intent, expected target window, 2 target-slot duration, 18% source ATK snapshot, ordinary mitigation and 20% max-HP pre-mitigation cap.	The fight teaches how Wound operates without a mystery damage rule.
Cleanse answer	K-09 at first copy can use its Technique to remove one declared Wound source from a legal allied target when available. Initial Technique cooldown is ready; AI condition: use if at least one ally has Wound.	Makes the advertised role true without duplicates or an Ultimate requirement. General Technique cooldown remains 2 full own turns after use.
Damage answer	K-03 plus K-19 can attack the front-row source before the source gets a second effective application; no hit must rely on a successful K-12 Mark roll.	Makes a guaranteed-owned path even when START01 selected K-10.
Timing / persistence	Wound tick is at victim action-slot start; one action per living unit per round. Defeating the applier prevents future casts but does NOT retroactively remove already attached Wounds in an unfinished battle.	Preserves effect identity and avoids falsely describing focus as cleansing. Existing Wound may still hurt a surviving ally.
Reward identity	Do not invent a new W-E reward amount in this packet. Content completion must later receive a distinct once-only FIRST-stage reward ID, not another claim of CORE01/CORE02 or FIRST02.	Priority 4 will price the stage; no duplicated 5-seal chapter reward assumed.

[OPEN] The status-application chance, enemy HP/ATK/DEF, unit Speed values, opening positions and attack coefficients are not yet approved for W-E. Consequently these rules specify legal response windows, not win rates or a guaranteed victory. A miss on Wound application must not make progression impossible; tutorial UI can preview the threat even if it does not apply on the first attempt.

43.4 Paper-only first-round action example: cleanse versus focus

[ILLUSTRATIVE COMBAT] Hold the same legal round-1 Speed queue for both routes: enemy K-06 at 120, allied K-12 at 115, chosen support K-09 or K-10 at 110, K-01 at 108, K-03 at 105, and K-19 at 90. Speeds are fabricated order markers, not canon stats; equal-speed side-priority is not needed. Assume K-06 successfully applies one Wound to K-19 before K-19 acts and W-E provides no guard/interception. The queue is frozen at round start as Section 27 requires.

Action slot	If START01 selected K-09: cleanse	If START01 selected K-10: focus
K-06 / Speed 120	Wound applied to K-19 (status success assumed for the example).	Same application; no counter occurs before it.
K-12 / 115	May Mark K-06; an unsuccessful Mark roll does not invalidate the path.	May Mark K-06; no damage threshold relies on it.
Support / 110	K-09 Technique checks visible Wound and removes one source before K-19 starts its slot. No Ultimate or duplicate.	K-10 may heal or shield under its real authored condition, but cannot claim cleanse.
K-01 / 108; K-03 / 105	Guardian takes its legal action; K-03 follows standard target and conversion rules.	K-03 damages the accessible K-06; current Mark may help but is not necessary for the paper threshold.

K-19 / 90	No Wound tick if cleared before this slot; K-19 may take its action if alive.	Wound tick happens FIRST. If alive afterward, K-19 then attacks the same legal focus target.
Next window	K-09 Technique is on cooldown for the next 2 own turns. A second Wound source or quick reapplication is meaningful counterplay.	If K-06 falls, no further source casts; an already attached Wound still needs mitigation, expiry or cleanse while battle continues.

[CALCULATED ILLUSTRATION, NOT BATTLE VALIDATION] Choose source attack snapshot 800, victim maximum HP 950, victim DEF 500 and no non-DEF reduction/shield. One Wound tick pre-mitigation is min(0.18 × 800, 0.20 × 950) = min(144, 190) = 144; final damage floor(144 × 1000/(1000+500)) = 96. If K-19 began with 950 HP, one isolated tick leaves 854 HP and it can act (all other incoming damage excluded). For a hypothetical unshielded K-06 with 1,000 HP, assume K-03 deals 680 and K-19 deals 360 post-mitigation without Mark: 680+360=1,040, so source falls after K-19 acts and before round 2. These damage figures were assigned for a conditional example; the document has not proved real base-kit damage reaches them.

Critical counterexample: if K-03 plus K-19 total only 850 against the same 1,000 HP source, the paper focus route fails to remove it in that window. If K-19 takes other damage before its slot and the Wound KO occurs at slot start, its 360 never happens. If a tank or Taunt prevents focus, the route is not legal. The stage must be designed against these cases, not declared fair because a paper average looks good.

43.5 Eight-week ownership, fieldability and first action by account type

Date convention: day 0 is a reference account-start or return instant, not a verified campaign milestone. Aligned full-core payout cases conservatively post at the end of completed weeks; actual qualifying visits can award earlier. Week 1–8 total fragments at day 7/14/21/28/35/42/49/56 are 15/30/45/60/75/90/105/120. SELECT01 consumes 100 once; 68 recurring seals in 56 days still miss the 70-featured-pull guarantee by 2. Existing Section 40 quantities and Section 41 midweek example are reused, not re-awarded.

Account (all F2P)	**Guaranteed W-E response on first legal W-E attempt**	**Optional K-09 acquisition if not START01**	**Hall / equipment and caveat**
New account; aligned week start	START00 owns K-03 + K-19 at tutorial; K-09 available immediately only when START01 chosen. W-E becomes legal at E0, date of reaching E OPEN.	Full weekly route earns 105 fragments day 49; SELECT01 then fields K-09 immediately, 5 fragments left. Day 56 120 gross / 20 after one selection.	At E0 inherited level 20, Band I +0 standard gear. Relative to a separate zero-wallet E calendar, Hall 21 is affordable day 3 on two visits, day 3 on four, or day 2 daily; it is not W-E admission.
Mature aligned account; Hall 80, 0 starting fragments	Existing START00-equivalent K-03/K-19 must remain accessible; if already owns K-09 it can cleanse day 0. Otherwise use the focus route at the fixed Wound gate.	SELECT01 at the seventh full weekly payout, day 49. No assumed featured guarantee, optional drop or premium route.	Earned Band II and Hall 80 at day 0; under fixed M rates the two-visit account pays to 81 on day 7, not before. A K-09 owned earlier inherits 80 immediately and requires zero personal preparation.
Late join Wednesday; tutorial on day 1	At tutorial day 1, START00 focus units and START01 option. No forced seven-week wait for first counter lesson when E is later reached.	10 fragments on joining Wednesday; Mon/Thu visits thereafter. Existing Section 41.6 schedule: 85 after 5 complete weeks; 95 Mon day 41; 100 Thu day 44 -> select on day 44.	First legal W-E is when E achieved, not necessarily day 1. K-09 joins with whatever Hall level/band is already earned and its normal first action order.
Returning mature account; 4 missed weeks, 30 persistent fragments at return day 0	Saved or grandfathered START00-type focus access remains. A previously owned K-09 fields on the return visit; absent ownership, focus remains the default.	No retroactive missed CORE grants. Five subsequent complete weekly payouts add 75 to opening 30, reaching 105 at return+day 35; SELECT01 then leaves 5. With 0 opening fragments the full path is return+day 49.	Keep saved Hall level and equipment state. Idle bank still has 24h full + 48h quarter rate then pauses; do not assert it funded upgrades throughout a four-week absence.

[CALCULATED SENSITIVITY] If weeks 3 and 6 miss CORE02 but CORE01 is still earned, week 7 is at 95 fragments and week 8 at 110, so the counter selector moves from day 49 to day 56. With only CORE01 once a week and zero initial fragments, day 56 reaches 80 and day 70 is the tenth payout reaching 100. A returning account with 30 opening fragments and only

one visit per week reaches 100 at its seventh weekly payout, return+day 49. No number here is a promise that an actual player has reached Hall 20 or Hall 80 by these calendar dates.

43.6 Chosen policy versus alternatives: costs and reversal conditions

Option	Decision and benefit	Trade-off / reason to revisit
A. Mandatory K-09 START01 choice	Reject. Makes a guaranteed cleanser but turns player choice into a fake choice.	Revisit only if every alternative is inaccessible despite reasonable E tuning; prefer fixing the encounter first.
B. START00 focus core + START01 K-09/K-10	LEADING WORKING PROPOSAL. Two structurally different pathways at the E gate with no random acquisition or personal re-leveling.	Costs four guaranteed starter units and demands deliberate early encounter tuning; K-09 versus K-10 choice must stay meaningful in later fights.
C. Make every starter a cleanser	Reject for now. Fast, but makes debuff mitigation less of a roster decision.	Could re-examine if small-screen auto-AI cannot reliably execute targeted cleanses.
D. Delay all Wound gates until day 49	Reject. Calendar lock substitutes collection time for tactical learning; late/returning accounts wait longer.	A later advanced Wound challenge can use selector timing if other combat answers remain available.
E. Increase idle bank to 72h full or grant level 81 early	Reject as first response. Changes the whole resource economy to solve one content-access problem.	Reopen only when a broad shared-growth affordability failure remains after moving or scaling the encounter.
F. Premium/featured-only anti-Wound	Reject under the established F2P fairness principle.	No acceptable mandatory paid gate.

Decision reversal ladder: (1) verify legal target access and K-03/K-19 first-copy damage at Hall 20/Band I using the authored stat packet; (2) if weak, reduce W-E enemy HP or guard pressure without stripping Wound of meaning; (3) if counter still fails, move W-E to a reachable earlier-granted substitute or give an explicit accessible anti-Wound tool before the stage; (4) only then reconsider starter allocation; (5) do not distort recurring seals, featured pity or full idle storage as a localized fix. The next pass must record every rule changed, side effect on other threats, and the rejected alternative.

43.7 Cross-system acceptance contract and unresolved research

Paper acceptance check	Result now	Remaining work before claim of validity
Two response categories at W-E	DESIGNED: guaranteed focus (K-03/K-19) and optional cleanse (K-09 through START01).	Not proved: real coefficients, start-of-slot survival, status chance, AI sequencing and enemy HP.
Affordability / owned-unit switch	SPECIFIED: Section 41 baseline has zero personal Ryo/EXP/gear prep; no Hall21/81 admission fee.	No signed-off stage-to-Hall timing; fielding is not proof of victory.
RNG/premium independence	SPECIFIED: starter focus grant independent of gacha; standard selector after 100 fragments; no featured guarantee dependency.	Future content must not sneak in a required K-16 rear attacker or mandatory K-12 Mark proc.
Late/returner access	CALCULATED under published CORE01/02 assumptions: midweek selector day44; 30-fragment returner selector return+35 at two visits/week.	Publish weekly cadence and reset epoch, grandfather starter claims, preserve old player state.
Wound persistence and cleanse behavior	WORKING PROPOSAL: defeating applier does not cleanse existing ticks; K-09 Technique removes one source; starts available.	Update combat source-of-truth in Section 27 when kit details are finalized; audit status source-death and revive interactions.
Mobile explanation	DRAFT: show target, status tick timing, accessible response and after-action recap.	Priority 5 must specify exact player screens and accessibility treatment; no user comprehension test has run.

Explicit unresolved questions: the stage and reward IDs for W-E; complete START01 eligibility; tutorial grants that avoid choice regret; actual Hall-20 damage/defense packet and Speed ordering; whether the first mandatory Wound encounter introduces one or two sources; status application roll and telegraph; migration fairness; event/idle claim dates for an actual novice; low-end mobile legibility; and all other mandatory threat gates. Section 28 keeps experimental checks for a future authorized phase.

43.8 Handoff: finish Priority 3 before opening Priority 4

[NEXT RESEARCH — PRIORITY 3B] Build the full mandatory-threat access matrix: control/interrupt, burst protection, guarded rear pressure, sustain/healing suppression, shield/decoy and Wound with second-source pressure. For each earliest required gate, name (a) its obtainable first-copy baseline answer, (b) a distinct alternative answer, (c) the date/visit at which both can be owned and fielded for aligned, midweek, returning and one-visit accounts, (d) legal targeting and first action, (e) a Hall-level and reward-ID floor, and (f) the smallest content/access correction when either answer fails. Do not assume K-06 or K-09 access alone proves all archetype counter coverage. If a future mode requires an unpriced new material, send it to Priority 2 rather than silently adding supply.

[PERFORMANCE RECOMMENDATION FOR NEXT PASS: EXTRA HIGH] Priority 3B reconciles multiple enemy archetypes, fifteen-plus roster substitutes, tutorial entitlements, reset calendars, shared growth, and legal combat targeting across four account types. It needs cross-chapter contradiction checking, not a single bounded comparison. After that matrix has an explicit rule or documented shortfall for every required threat, advance to Priority 4 campaign and optional encounter design. Project stays in theory/research mode only.

43.9 External comparison, dated evidence and limits

[ACCESS 31 October 2026; PEER1] IGG, Mythic Heroes official Day 1 Guide. Describes common 2-front/3-back composition and warns that swapped heroes may need rune/artifact checks. It motivates our avoidance of a hidden new-hero preparation tax, but does not establish any fair Wound timing. https://www.mythicheroes.com/7day-guide-day1/

[ACCESS 31 October 2026; PEER2] Prydwen, AFK Journey Characters Info, third-party current guide. Describes Resonating Hall and shared class equipment as ways to reduce hero-swap friction. This is a guide snapshot, not a verified developer specification or evidence of our proposed economy rates. https://www.prydwen.gg/afk-journey/guides/characters-info/

[ACCESS 31 October 2026; PEER3] Roblox Creator Hub, Onboarding. Documents teaching core-loop essentials quickly and improving first-time player experience via later experimentation. This is adjacent developer guidance, not a Naruto/idle RPG metric or permission to run current playtests. https://create.roblox.com/docs/production/game-design/onboarding

44. Essential-role access II: full mandatory-threat coverage

Priority 3B | 31 October 2026 | Version 0.10.17. Research scope: make every mandatory early threat answerable without a lucky summon, premium purchase, new-hero level tax or calendar-locked selector. This paper design checks ownership, five-slot team legality, target reachability, action order, shared-Hall preparation, and entitlement timing. It does NOT establish actual damage thresholds, win rates, source probabilities, fun or usable mobile UX. No prototype, seeded simulation, gameplay implementation or live user test was performed.

Evidence language: [DOCUMENT AUDIT] quotes/reconciles existing Sections 24-27 and 40-43; [WORKING PROPOSAL] selects a reversible rule; [CALCULATED] is arithmetic from named assumptions; [CONDITIONAL] is an if/then encounter requirement rather than a measured success; [OPEN] needs the next paper pass or later authorized testing.

44.1 Diagnosis: kit coverage is not owner access

[DOCUMENT AUDIT] Section 26.2 lists K-04 for AoE, K-08 for Interrupt, K-16 for rear access and K-17 for Healing Suppression. Section 43 guarantees the first four introductory characters K-01/K-03/K-12/K-19 and the fifth-slot K-09 OR K-10 choice. The existence of other K-IDs in a hypothetical roster does not prove that any player owns them. SELECT01 costs 100 Universal Fragments; the full weekly core acquisition path reaches this on the seventh completed weekly payout, while a once-weekly CORE01-only player does not reach it by day 56. Therefore demanding K-04/K-08/K-17 effects in required early fights without separate guaranteed access creates an unacknowledged roster gate.

[WORKING PROPOSAL] Separate first encounter pedagogy from long-term roster expansion. Keep START00 (four characters) and START01 (exactly one K-09/K-10). Add exactly THREE one-time, pre-gate story recruit entitlements, not a free character per enemy type. Grant a functional first copy of K-08, K-04 and K-17 before their respective introductory mandatory threats. This yields eight distinct owned characters after all three story grants, not eight active battle slots; a legal team still fields at most five. K-16 and K-18 remain optional standard-pool/other earnable specialization, never mandatory to clear the proposed early gates.

44.2 One-time grants and the no-double-counting contract

Reward source / trigger	Authoritative candidate entitlement	Ledger and save rules
STORY01 / at first charge-preview checkpoint, before entry	One K-08 Ability Disruptor, at first-copy Interrupt Technique; no duplicate required.	Pre-entry claim auto-resolves before battle validation. No seals, fragments, pity, consumable trial or chapter-clear duplicate. Unique account source. If already owned, credit character ownership once, no extra copy or substitute currency until a separately approved policy.
STORY02 / at first decoy-or-protected-rear checkpoint, before entry	One K-04 AoE Formation Breaker, with a legal Rear Row Technique at base copy.	Same once-account entitlement. A row-targeting skill is not redirected by single-target Taunt; damage subject to the normal per-target budget.
STORY03 / at first sustain checkpoint, before entry	One K-17 Anti-Sustain Debuffer, whose base Technique applies Healing Suppression after a legal direct hit.	Same once-account entitlement. First-copy anti-heal is a real effect, not unlocked behind skill materials, fragments, duplicate ranks or an Ultimate.
Reward category	Separate STORYxx from START00, START01, CORE01/02, SELECT01, PVE01/FIRSTxx and RECRUIT01.	Exactly one transaction identity per account and source. Neither a stage-first-clear nor the same chapter reward can count the character a second time; no implicit +3 recruitment pulls.
Returning / mature / already past checkpoint	On first eligible return, reconcile unclaimed STORYxx for every qualifying completed checkpoint before any newly mandatory encounter.	Idempotent grant on migration: owned→no duplicate, unowned→one first copy, grant state persists; previously obtained optional copies preserved. No requirement to replay cleared content.
Equipment / progression price	Each story shinobi inherits earned Hall level/cap and two standard-issue gear slots.	Incremental hero preparation vector = 0 Ryo, 0 EXP, 0 essence, 0 personal gear resources, 0 duplicate and 0 skill materials; this DOES NOT waive a separately priced Hall promotion.

Choice trade-off: eight guaranteed characters reduce the urgency of early recruitment, but keep ordinary five-slot team building viable. The alternative of only offering randomized summons or a weekly selector would preserve short-run gacha tension at the expense of mandatory fairness. Loan units would be less expensive in roster grants but create expiring permissions and save/retry ambiguities. Three persistent story copies are the leading paper solution; revisit after roster-aspiration and onboarding evidence, not by reintroducing pay-only gates.

44.3 Mandatory-threat access matrix: eight introductions

[WORKING PROPOSAL] Gate IDs are ordering anchors, NOT confirmed campaign stage numbers or reward quantities. All eight belong to a low-stat early instructional band with legal minimum Hall level 20, cap 20, Band I, and base first-copy skills. Later harder repetitions may demand Hall 21 or more only after Priority 2 establishes a genuinely payable path, including once-weekly players. Each gate has at least two different functional answer categories. Neither must imply a free win, but both must be legal at their stated first action and attainable without random recruitment.

Ordered gate / mandatory threat	Path A: owned response, legal window	Path B: different functional response; design constraint
W-E / one front-row Wound source	START00 K-03 + K-19 focus accessible front applier before another effective application; no Mark proc assumed.	START01 K-09 cleanses applied Wound OR K-10 pre-emptive Shield/mitigation on chosen target; both choices must have a meaningful threat response. Source death does not erase attached Wound.

Ordered gate / mandatory threat	Path A: owned response, legal window	Path B: different functional response; design constraint
B-E / front-row burst and Mark	K-01 Taunt eligible single-target burst (protect threatened K-03 or K-19), then convert with basic damage.	K-03 + K-19 focus the fragile burst source before the next charged/telegraphed high-damage attempt. If START01=K-09, cleansing Mark helps; if K-10, its Shield helps; neither variant is mandatory.
C-E / single-target Stun or Silence	K-01 redirect an explicitly Taunt-eligible control hit and keep at least one striker functional; global Stun DR prevents indefinite denial.	K-03 + K-19 eliminate the accessible control source within bounded rounds; K-09 cleanse if chosen, or K-10 sustain if chosen. No AoE multi-Stun or untargetable caster in introductory version.
T-E / single-target visible Charging	STORY01 K-08 Technique interrupts legal Charging target after cast is telegraphed and before resolution; it cannot act retroactively.	K-01 Taunt the declared single-target charged hit, then K-03/K-19 defeat exposed caster during recovery. First version is not unavoidable unmitigated AoE and must leave K-01 alive enough to protect or force a meaningful survival trade.
D-E / one decoy or bounded Shield	STORY02 K-04 rear-row or row AoE pressures real threats while decoy only intercepts eligible single-target hits; row AoE is not redirected.	K-19 Basic consumes a one-hit decoy or the player waits its one-following-round expiry, then K-03 focuses the exposed source. Any shield must obey 40% max-HP active cap and cannot be refreshed into an indefinite 12-round stalemate.
R-E / one protected rear-row threat	STORY02 K-04 explicitly targets Rear Row with reduced per-target AoE value even while Taunt/one interception shields the front.	K-03 + K-19 legally clear a defeat-able front guard, then attack the revealed rear enemy. Never pair an invincible guard with an untargetable rear objective; both paths must fit PvE 12-round cap.
S-E / healer or sustain engine	STORY03 K-17 first-copy Technique applies bounded Healing Suppression (up to 40%) to a legal, exposed healer; after-hit application cannot affect a heal already resolved.	K-03/K-19 burst or sustained focus removes the healer/support before net recovery overwhelms output. First healer must be reachable: front row or unguarded, not an uncounterable protected rear plus full refresh.
WW-E / two Wound sources, clearly previewed	STORY01 K-08 interrupts one visibly Charging source; K-03/K-19 focus a second reachable source; legal queue must give K-08 a true opportunity.	Defensive route: START01 K-09 removes one source on a legal ally OR K-10 uses a pre-tick Shield; K-01 protects the priority victim and damage team removes sources. Both openings must obey shared two-source/20%-MaxHP per-slot Wound cap and allow survival even when K-09 clears only one source.

Definition of two answers: Path A and Path B must change a different tactical lever (prevent/interrupt, protect/mitigate, target/bypass, or outlast/eliminate). Two characters dealing the same front-target Basic do NOT count as two counters. On every gate, document two legal five-character squads or two legal tactic/formation plans with the accessible owners listed above; no future skill may turn a baseline counter into an Ultimate-, dupe-, pay- or featured-only action.

44.4 First-copy combat/AI amendments needed to make the matrix honest

Owner / available effect	Minimum base-copy specification (reversible)	Timing/AI failure to avoid
K-01, START00	Taunt eligible single-target hostile actions; no interception of row/all-target AoE by default. Show target and expiry.	Cannot claim it prevents Wound or Stun if enemy skill is untauntable. It may act after the first hit; encounter must survive that opening.
K-03 + K-19, START00	Non-critical, no-Mark direct damage exists at first copy; legal front targeting and focus preset.	Do not count a chance-based Mark, Critical, unearned Ultimate or K-19 action after a KO as guaranteed damage.

Owner / available effect	Minimum base-copy specification (reversible)	Timing/AI failure to avoid
K-04, STORY02	Technique: legal Rear Row AoE when ≥1 living rear enemy; no Ignore-Taunt wording needed because row AoE is not single-target. Ordinary lower per-target coefficient.	If no legal rear target, Technique must fall back to Basic instead of wasting a turn; targeting cannot unexpectedly switch to front.
K-08, STORY01	Technique: Interrupt only a visible Charging target with a legal single-target selector; ready from first copy; cancel specific charge without also applying an unpriced Stun.	Give player an actual slot between charge start and resolution. If Ultimate is available, its condition must not override needed charged-response Technique or its authored Ultimate must also legally Interrupt.
K-09, START01 or SELECT01	Technique: clear exactly one named Wound source on one legal affected ally; may be expanded to other debuffs only with an explicit target/priority rule.	One cleanse cannot be counted as clearing two Wounds. At its 2-full-own-turn cooldown, back-to-back applications can matter.
K-10, START01	Technique: bounded Shield to legal threatened ally, ready on first copy; Ultimate may heal once 100 personal Chakra is earned.	A turn-1 heal is not assumed. Shield exists before victim tick only if K-10 acts first; cannot exceed total 40% max-HP cap.
K-17, STORY03	Technique: applies Healing Suppression at base copy, after legal direct hit; capped at 40%, duration next two target slots.	If healer already healed that round, retroactive anti-heal is invalid. If Taunt redirects the skill, mark the target as unreachable.
K-12, START00	Mark +10% vulnerability as setup, not part of the minimum no-RNG success budget.	Failed Mark must not invalidate every first-gate answer. Focus Mark preset cannot override Taunt.

[RULE RECONCILIATION] Section 24.4 currently selects Ultimate before Technique when its authored condition and target are valid. A new “interrupt priority” hidden branch would violate that rule. Specify K-08 Ultimate’s target/condition so it is not eligible while a charge requires an interrupt, or give it the legal Interrupt tag at an appropriate cost. For K-09, choose a deterministic affected ally selector (earliest fatal tick, then lowest current HP%, then slot order); for K-10, choose a visibly threatened ally selector. These are working targets, not finalized skill cards. If target condition selection requires foresight beyond telegraphed action state, revert to a simpler visible condition.

44.5 Two clocks: grant time versus 56-day recruitment time

For every account, define G_i as the first timestamp when mandatory checkpoint i becomes enterable; define C_i as its story grant transaction timestamp. Policy: C_i ≤ G_i and the player cannot press Enter before successful grant reconciliation. There is no assumption that G_i occurs on a fixed account day. A player cannot reach a gate without visiting, but when they do, its story counter is present on that visit. START00/01 remain their own tutorial entitlements. This is a gate-relative guarantee, not a promised calendar date.

Account and visit pattern	Guaranteed gate access / first response	Optional selector timeline / growth constraint
Aligned new F2P; ≥2 qualifying weekly visits	START00 + START01 at tutorial; STORY01/02/03 on each pre-gate visit. First-copy counter fields that visit at earned Hall level; no incremental hero materials.	15 fragments per fully completed week; 105 on seventh payout/day49; selector optional to early gates. Hall21 E: first affordable day3 with 2 visits in the fixed comparison.
Late Wednesday new F2P; Mon/Thu thereafter	Same entitlement at tutorial and each actually reached gate, including midweek first eligibility. No 7-week lock on story tools.	Section 43 late-join case reaches 100 fragments on day44 under its stated cadence, but exact future gate arrival dates remain unspecified.
Mature F2P; Hall80/own roster already saved	On login, reconcile already-passed story checkpoints idempotently. If next required stage is pending, all prerequisite STORYxx are granted before entering.	Prior owned units and equipment preserved. No current-stage retroactive level credit. With 2 visits/week M Hall81 paper payment day7, but this is not a new-counter preparation fee.

Account and visit pattern	Guaranteed gate access / first response	Optional selector timeline / growth constraint
Returning F2P; four-week absence	No missed grants from CORE01/02; STORYxx from previously cleared checkpoint is reclaimed once with no expiry or re-clear. Current-owned first-copy counters inherit saved Hall.	With 30 fragments saved and 2 visits/week, standard selector reaches 100 after five full credited weeks/return+day35; without saved fragments, return+day49.
One-visit/week F2P, starting zero fragments	All pre-gate STORYxx still granted on the visit when the gate is reached. No CORE02 requirement, premium chest or special weekly claim to unlock counter.	CORE01 earns 10 fragments per weekly visit: 80 by day56; 100 only on tenth weekly payout/day70. Section 42 shared-growth weekly bill is NOT sustainably covered at checkpoint E by the 1.5-bank-day weekly collection; see Section 44.6.

44.6 Priority 2 reopened: one-visit shared-growth deficit

[CALCULATED, FIXED CHECKPOINT ONLY] Section 40.9 gives one claim after 168h = 36 full-rate equivalent hours = 1.50 full-rate days. At E: hourly 10 Ryo, 20 EXP, 2 essence, 1 enhancement -> weekly collected 360 Ryo, 720 EXP, 72 essence, 36 enhancement. At M: 40 Ryo, 60 EXP, 4 essence, 2 enhancement -> 1,440 Ryo, 2,160 EXP, 144 essence, 72 enhancement. Section 42 E total week bill is 720 Ryo / 960 EXP / 48 essence / 24 enhancement. Section 42 M total is 2,880 / 4,320 / 192 / 48. One visit cannot fund both bills in week 1 at E or M.

Fixed checkpoint / one claim on day7	Collection versus currently stated weekly spending	Result / policy consequence
E, full Hall21 + other core bill	Gross 360/720/72/36; full bill 720/960/48/24 (Ryo/EXP/essence/enhancement).	Short 360 Ryo and 240 EXP; sufficient essence and enhancement. Even “other core bill” alone needs 420 Ryo and exceeds 360 by 60 Ryo.
M, full Hall81 + other core bill	Gross 1,440/2,160/144/72; full bill 2,880/4,320/192/48.	Short 1,440 Ryo, 2,160 EXP and 48 essence; sufficient enhancement.
E, weekly other-core-only recurrence	Other bill 420 Ryo, 240 EXP, 24 enhancement vs gross 360 Ryo, 720 EXP, 36 enhancement.	Ryo loses 60 per week from zero; impossible to pay that *weekly obligation* indefinitely without grants, a reduced optional bill or a broader bank. No later compulsory Hall21 may rely on an uncorrected one-visit schedule.
M, other-core-only, Hall deferred	After recurring 960 Ryo and 720 EXP each weekly claim, residual is 480 Ryo, 1,440 EXP, 144 essence, 24 enhancement/week.	Hall81 requires 1,920 Ryo, 3,600 EXP and 192 essence; earliest theoretical fund after four claimed weeks (day28), if no other spending and checkpoints/rates fixed. Not the two-visit day7 case.

Illustrative decision hierarchy: do not extend the idle bank to 72h full automatically, and do not silently add one-visit compensating income. First distinguish OPTIONAL ongoing weekly growth from a mandatory charge: a player must never enter debt or lose earned level because they did not buy an optional weekly package. If recurring core activity is mandatory for campaign pace, its E one-visit Ryo floor is at least 420 weekly Ryo versus 360 collected, so Section 42 pricing/schedule requires a fresh paper budget. A proposed small recurring catch-up allowance must have its own source ID, exact resource vector, eligibility and sink audit. Keep advanced Hall21+ gates OUT of mandatory content until this is reconciled.

44.7 Failure audit: do not confuse legal availability with demonstrated success

Adversarial case	Why a claimed counter might fail	Required correction before content lock
First burst kills K-19 before its slot	Total K-03+K-19 focus paper damage cannot include K-19 after KO.	Reduce first burst, delay lethal cast, or guarantee alternative K-01 redirect/Shield timing and survivor strike.
Taunt or intercept protects a Wound/burst source	Focus Mark preset cannot override legal Taunt; K-03 cannot hit illegal target.	Place first source in accessible row or allow K-04 only after STORY02; otherwise move gate.

Adversarial case	Why a claimed counter might fail	Required correction before content lock
Charged attack resolves same enemy slot	K-08 acts too late even with an Interrupt-tag Technique.	Two-stage charge must visibly persist across at least one K-08 legal action; otherwise count only defensive route, not interruption.
K-08 has 100 Chakra and an eligible non-Interrupt Ultimate	Default AI chooses Ultimate over Technique and fails to stop the charge.	Authored Ultimate condition excludes this state OR carries a properly costed Interrupt; no hidden priority exception.
K-17 acts after heal and anti-heal applies post-hit	Anti-heal arrives too late to stop first heal; shield remains unaffected.	Create a second heal window after K-17 acts, keep single-target legal and ensure focus alternative fits cap.
K-04 Technique is only Front Row despite label	Rear threat remains unreachable under Taunt; row targeting does not magically bypass legal selector.	Specify Rear Row in first-copy kit; test zero-rear-target fallthrough.
Both Wound sources apply immediately before K-08 or K-10 acts	“Interrupt” and “shield-before-tick” may be legally impossible.	Start a clearly signaled round-1 Charging window and schedule round-2 resolution after a valid player response.
Single victim retains 2 Wound sources after one K-09 cleanse	One active source still ticks at its victim slot; Wound is not automatically erased when caster dies.	Use remaining-source + single-target damage survival inequality and no assumed free second cleanse.
Decoy/Shield continuously refreshes	Waiting does not expose target; 12-round PvE cap becomes hidden wall.	Limit refill opportunity, allow K-04 legal AoE pressure or create vulnerable cast/recovery windows.
Returning account past all story checkpoints but missing characters	Story unit is not in account despite mandatory progression.	Grant before next entry idempotently; do not delete premium or preexisting copies.

Minimum mathematical authoring contract for a focus route: source HP at relevant action window ≤ sum of guaranteed non-critical post-mitigation damage from still-living, legally targeting owned actors before the next effective cast. If the inequality is false, we do not claim the focus route. For defensive routes, compute each scheduled pre-action Wound tick or charged hit after DEF, non-DEF reduction and Shield in the exact Section 27 order; victim HP must remain positive at the action window that is necessary to answer. These are analytic inequalities to be filled with future skill/stat cards, NOT simulated outcomes.

44.8 Trade-offs, dependency map and gate to Priority 4

Selected working policy	Reason / cost	Reopen when...
Three deterministic STORY grants + optional selector	Guarantees specific tactics before first required introduction; reduces early gacha scarcity and gives player eight owned heroes.	Acquisition identity, team learning, save migration or roster variety becomes measurably harmful; any replacement still must be deterministic and free.
First mandatory threat intros at earned Hall20/Band I	Prevents calendar-based compulsory promotions while shared-growth schedule remains unproven for one visit/week.	P2 sets an affordable, dated growth floor for every supported cadence and P4 authored enemy stats meet those floors.
K-04 row attack, K-08 Interrupt, K-17 anti-heal at base copy	Prevents counters being locked behind dupes or Ultimates; costs distinct specialization pressure and kit coefficient budget.	Their exact skill cards violate action economy, first-copy role budget or 5-slot competition, requiring a different common counter.
Two functional response categories with telegraph + target contract	Maintains strategy in auto combat; costs UI, telegraph authoring and battle recap complexity.	One answer is illegal because of Taunt, Speed order, cooldown, status odds or target protection; fix first gate rather than adding premium item.
No extra seals/fragments for story copies	Preserves Section 40 numeric seal/fragment floors, sourced once.	P4 adds any separate chapter first-clear rewards: those must be counted in a new source ledger entry, not folded into STORYxx.

[PRIORITY EXIT STATUS] The ACCESS part of Priority 3 now has a selected theoretical structure for eight early threats: source identity, first-copy owner, legal response categories, pre-gate claim condition and Hall20/Band I fielding. It is NOT quantitatively closed: Section 27 does not yet provide final first-copy skill/Speed/enemy damage packets; P2 has an

uncovered one-visit weekly Ryo obligation; campaign stage identities and reward amounts await P4. Do not label every gate “tested” or “balanced.”

[NEXT RESEARCH] Before full Priority 4 encounter authoring, reconcile Priority 2C/3C together: (1) solve one-visit and missed-week material/promotion affordability without double-counting sources; (2) write explicit first-copy skill cards and an exact paper speed/target/HP packet for W-E, T-E, R-E and S-E; (3) evaluate no-RNG, worst-action-order, starter-choice and defender-KO cases manually under Section 27. Only then select campaign stage order and stage-specific reward IDs. Recommended next performance level: EXTRA HIGH, because a one-cadence economy correction can change the progression gates and every proposed counter action window.

44.9 Evidence, outside comparisons, limitations

[SOURCE, 31 October 2026; ACCESS4] Riot Games, “Quick Gameplay Thoughts: May 14” (14 May 2021). Developer explains counterplay as an actionable response and distinguishes strategic and tactical answers. Analogical design guidance, not a quantitative rule for an idle RPG. https://www.leagueoflegends.com/en-us/news/dev/quick-gameplay-thoughts-may-14/

[SOURCE, 31 October 2026; ACCESS5] Riot Games, “Clarity in League” (12 March 2021). Developer describes information hierarchy, meaningful visual telegraphs and reducing visual noise. Used for an auto-battle telegraph/recap design principle, not proof of our UI effectiveness. https://www.leagueoflegends.com/en-us/news/dev/clarity-in-league/

[SOURCE, 31 October 2026; ACCESS6] Android Developers, “Authentication & Onboarding” (current guidance; accessed 31 October 2026). Recommends contextual rather than overloaded upfront instruction. Analogical support for awarding/teaching roles near their first relevant battle. https://developer.android.com/design/ui/mobile/guides/patterns/onboarding

[SOURCE, 31 October 2026; ACCESS7] IGG, official Mythic Heroes Day 1 Guide. Notes balanced 2-front/3-back formations and checking new recruits’ gear. Supports the concern about hidden preparation costs, not the proposed number of free recruits. https://www.mythicheroes.com/7day-guide-day1/

Source caution: None of the sources prescribe our STORY01/02/03 entitlements, eight-gate order, exact 20-level floor, 8-character tutorial roster or expected fight outcomes. Those are internal working design proposals. Commercial use of Naruto IP still requires appropriate rights/licensing; this conceptual research grants no such rights.

45. Economy and four-encounter paper reconciliation

Joint Priority 2C + 3C | 31 October 2026 | Version 0.10.18 | Research only. This chapter resolves the meaning of the one-visit weekly spending deficit, proposes eight first-copy skill cards and performs four transparent no-crit/no-proc encounter calculations. It is not a combat simulator, a playable implementation, a seeded battle run or an empirical balancing claim. Section 27 remains the authority for combat clocks and formulas; Section 42 remains the priced fixed-stage example.

Evidence key: [DOCUMENT AUDIT] = previously stated rules; [PROPOSAL] = chosen reversible rule; [CALCULATED] = arithmetic at precisely described assumptions; [CONDITIONAL] = true only if authors reproduce those inputs; [OPEN] = needs subsequent research or authorized gameplay testing.

45.1 Why the one-visit result is a design-contract problem, not a currency paradox

[DOCUMENT AUDIT] The existing bank credits the first 24 hours at 100%, the next 48 hours at 25%, and then stops. A first collection 168 hours after an empty bank receives 24 + 48 × 0.25 = 36 equivalent hours = 1.50 full-rate days. Section 42 bundled one Hall promotion and another core-spending package as a hypothetical day-7 commitment. Nowhere was that second package priced as an unavoidable game-entry fee. Treating the *illustrative full budget* as an involuntary debt creates an invented wall.

Fixed checkpoint / first weekly claim	Gross income (Ryo / EXP / essence / enh.)	Bills and genuinely payable choices
Early E, day 7	360 / 720 / 72 / 36	Hall 20→21 = 300 / 720 / 48 / 0: PAYABLE day 7. Remainder 60 / 0 / 24 / 36. Other 420 / 240 / 0 / 24 is not payable simultaneously.
Mature M, day 7	1,440 / 2,160 / 144 / 72	Hall 80→81 = 1,920 / 3,600 / 192 / 0: not payable. Other core = 960 / 720 / 0 / 48: optional, payable; remainder 480 / 1,440 / 144 / 24.

Fixed checkpoint / first weekly claim	Gross income (Ryo / EXP / essence / enh.)	Bills and genuinely payable choices
Mature M, day 14 after optional week-1 spend	1,920 / 3,600 / 288 / 96 before Hall on second claim	Hall 80→81 now exactly payable for Ryo/EXP; after it: 0 / 0 / 96 / 96. The week-2 optional package is deferred.
Early E, day 14 after Hall-first day 7	420 / 720 / 96 / 72 before optional spend	Optional 420 / 240 / 0 / 24 now payable; after it: 0 / 480 / 96 / 48. No new income or special reward was assumed.

[PROPOSAL, DECISION 2C-01] Interpret the Section 42 “other core bill” as a voluntary spending basket, not a recurring maintenance fee, debt, expiring quest, loss of earned power, or a requirement to unlock a mandatory encounter. Hall upgrades are voluntary account investments and permanently owned; the next campaign gate is separately authored against a *published attainable baseline*. Players may bank, spend, or defer any optional purchase. No penalties for returning after missed weeks; idle-storage earnings stay exactly as currently modeled.

[CALCULATED] For one claim per week, E Hall21 is payable at day 7 under an empty-wallet start. M Hall81 is payable by day 14 whether the player saves everything in week 1 or buys only the optional M other-core package in week 1, then reserves the full second claim for Hall. Day 14 is a first *visit* payment, not a real-time claim before logging in. The older two-, four-, and seven-visit budgets from Section 42 are unchanged. Day-7 compulsory simultaneous payment at E or M remains mathematically impossible under these inputs; do not imply otherwise.

[GUARDRAIL] Do not claim this proves equitable progression speed. Daily players collect 7 equivalent days/week while once-weekly players collect 1.5; that is a material opportunity gap. The working design accepts different *speed* while protecting base counter access and avoiding mandatory calendar gates. Whether this gap is enjoyable or economically sustainable is for later economy studies. We are not adding a compensatory daily-login quest, seal, fragment, debt waiver or hidden “return boost” to these ledgers.

45.2 Alternative solutions and actual opportunity costs

Option	Benefit	Cost / verdict
A. Remove compulsory recurring bill (CHOSEN)	E Hall on day 7; M Hall on day 14; no bank inflation or new reward ID.	Players choosing one weekly claim progress materially slower. Mandatory content must never assume paid optional packages.
B. Keep bill compulsory; issue catch-up grants	Can buy both day-7 bills.	Needs separate weekly source: E +360 Ryo/+240 EXP; M +1,440 Ryo/+2,160 EXP/+48 essence. Creates a large new recurring source that needs eligibility, sinks, anti-stacking and save audits.
C. Extend full-rate bank to 72 h	Raises once-weekly supply; helps two-visit promotion timing.	One weekly claim then yields 72 equivalent hours, enough for E/M complete day-7 bills (M EXP = 4,320 exactly); doubles the present 36h equivalent to 72h. Raises all long-gap currency income, not just a single gate.
D. Discount promotions only	Advances Hall, not optional recurring packages.	Would require E/M revised Ryo, EXP and essence prices and alter every higher Hall pricing comparison; do not change one price in isolation.
E. Put Hall21/81 behind early mandatory gates	Easy to make levels feel necessary.	REJECT until the gate-relative earned-level requirement is actually affordable for the supported visit patterns.

Correction to the comparison above: a 72-hour full-rate bank yields 72h for a seven-day gap, double the current 36h; exact week-one E/M bills become affordable only because E=3 daily-Ryo units and M=3 daily-Ryo/EXP units. An additional long-term 25%-rate tail, if ever added, would be a *separate* proposal; neither option is silently selected. A fixed Hall20/Band I required encounter remains the standing design constraint.

45.3 Baseline proxy and first-copy skill cards: one authoritative action economy

[PROPOSAL] For these four *isolated arithmetic packets only*, use Section 27 normalized reference stats: HP 6,000, ATK 1,000, DEF 1,000, Speed 100, then one primary-role multiplier per K-ID. This is NOT a derived Hall20 stat curve. Basic coefficient 0.85, no Crit, no unearned Mark, no faction/bond bonus, no vulnerability, no Attack Bonus, no Shield unless separately declared. Full-precision calculations are floored ONCE per hit. Most enemy stats below are purpose-authored illustrative inputs and are not automatically equivalent to level-20 opponents.

Owned first-copy kit (stat proxy)	Technique / predicate at 0 Chakra; 2 subsequent own-turn cooldown	AI, targeting and exact failure rule

Owned first-copy kit (stat proxy)	Technique / predicate at 0 Chakra; 2 subsequent own-turn cooldown	AI, targeting and exact failure rule
K-01 Guardian: HP 7,800 / ATK 800 / DEF 1,200 / Spd 90	Taunt: 0 damage, eligible single-target hostile actions, up to 2 redirects or owner's second subsequent slot (existing expiry rule).	Free first copy, applies during owner round-1 slot; cannot redirect AoE or retroactively redirect a prior hit.
K-03 Finisher: HP 5,100 / ATK 1,250 / DEF 850 / Spd 110	Focused Technique: 1.40 × ATK, single legal front-row enemy with published threat tag when present.	Technique before Basic when eligible; no hidden Mark, Crit, full-health Execute or guaranteed on-hit status.
K-19 Flexible Striker: HP 6,000 / ATK 1,000 / DEF 1,000 / Spd 100	Reliable Technique: 1.30 × ATK, legal front-row focused enemy, no debuff.	Normal Basic 0.85 if Technique unavailable. Chosen proxy role = sustained carry, not an extra stat multiplier.
K-09 Cleanser: HP 6,300 / ATK 800 / DEF 1,000 / Spd 95	Technique: remove exactly one Wound source from one affected ally; 0 damage.	Only if ≥1 ally has Wound. Prioritize next fatal tick, then earliest tick, lowest HP%, slot order. Never counts as two-source cleanse.
K-10 Healer/Protector: HP 6,300 / ATK 800 / DEF 1,000 / Spd 95	Technique: Shield 12% of target Max HP, 2 recipient slots, under 40% combined cap.	Only for visible threatened ally; earliest threatening tick, then lowest HP%, slot. Never pretend its Ultimate heal is available at starting Chakra 0.
K-08 Disruptor: HP 5,700 / ATK 850 / DEF 950 / Spd 110	Technique: 0.55 × ATK to one legal visible Charging enemy; INTERRUPT tag cancels that specific charge.	Only available in Charging state. Skill's Interrupt is deterministic *only when legal*, not a debuff proc. Ultimate is ineligible while a Charging target needs interrupt unless its authored kit also includes Interrupt.
K-04 AoE Breaker: HP 6,000 / ATK 950 / DEF 950 / Spd 100	Technique: 0.40 × ATK to each living rear-row enemy, no extra debuff.	Only if rear enemy exists; row targeting bypasses single-target Taunt, does not bypass DEF/shield; if no rear target, Basic fallback.
K-17 Anti-Sustain: HP 5,700 / ATK 850 / DEF 950 / Spd 110	Technique: 0.55 × ATK to one legal exposed healer; 90% chance to add 40% Healing Suppression, next 2 target slots.	Applied AFTER direct damage. If 10% status failure occurs, no anti-heal credit. First S-E healer self-heals (same recipient); cannot claim effect against healing of a different ally.

[STATUS-RULE CONFLICT RESOLUTION] Section 24 caps player status application at 90%; therefore K-17 Healing Suppression cannot be declared a guaranteed no-RNG route. At the failed-status branch, the first S-E must still allow the pre-owned K-03/K-19 focus plan to succeed. Similarly K-08 Interrupt is a discrete named cancellation against visible Charging rather than a Stun/status roll; if a future definition makes it probabilistic, revisit the entire T-E route. K-09 cleanse and K-10 Shield are deterministic legal skills, not chance-applied debuffs.

45.4 W-E packet: Wound still matters after the source is defeated

[WORKING ENCOUNTER INPUT] One front-row enemy W-Applier: 1,850 HP, 600 DEF, 900 ATK, Speed 118. Round-1 single-target attack applies Wound to K-19 after its 0.45× direct hit. A second weak non-Taunting enemy remains alive so the battle does not automatically end upon W-Applier KO. Assume it does not threaten the calculations below. START00 owns K-03 and K-19; either K-09 or K-10 can be chosen through START01. First-copy Techniques have zero cooldown at battle start.

Step / condition	Paper calculation	Finding
Enemy acts first, Spd118	Direct to K-19 = floor(900×0.45×1000/(1000+1000)) = 202. Wound snapshot = 900×18% = 162 pre-DEF.	Two future K-19 slots each take floor(162×0.50)=81, unless cleansed/shielded. No tick occurs at application.
K-03 and K-19 focus, Spd110/100	K-03 = floor(1,250×1.40×1000/1600)=1,093; K-19 = floor(1,000×1.30×1000/1600)=812. Total = 1,905.	1,905 ≥ 1,850: source is defeated before its round-2 reapplication, if BOTH actors live, legally target it and choose Technique.
Optional K-09 response, Spd95	K-09 acts after K-19 in round 1 and removes W-Applier's named Wound source from K-19.	Cleansing cancels the two pending 81-HP ticks; killing the source alone does NOT.
Optional K-10 response, Spd95	K-10 grants a 12%×6,000 = 720 shield to K-19 in round 1, before its next slot.	The next 81-HP tick is absorbed; the following one is too if Shield duration still covers K-19's next two slots.

[ADVERSARIAL BRANCH] If K-19 is KO'd before its action or the enemy is protected by Taunt/intercept, only K-03's 1,093 damage remains and the turn-1 focus certificate FAILS. Remedy the specific encounter (enemy first-turn damage, target

legality, source HP, or exposed position); do not “solve” it with an assumed Crit, duplicate, premium unit or free second action. Wound remains meaningful in the focus route because it persists after source death while the second enemy remains.

45.5 T-E packet: distinguish canceling a charge from surviving it

[WORKING ENCOUNTER INPUT] Exposed charger: Speed 118, 800 DEF, 3,600 ATK, HP not fixed by this defensive timing packet. Round-1 action enters clearly visible Charging; no hit occurs on charge entry. Its round-2 action would deal a 1.60× single-target hit, originally aimed at K-03; the attack is Taunt-eligible. If interrupted in round 1, the declared charge is canceled and the enemy can next take its ordinary action according to its authored cooldown; no extra Stun or removed slot is granted. K-08 is granted by STORY01 before entry.

Path	Timing and arithmetic	Constraint
A: Interrupt	Round1 charger (118) enters Charging → K-08 (110) Technique hits for floor(850×0.55×1000/1800)=259, then cancels that charged effect.	Legally targeted, deterministic Interrupt; no assumption of damage KO or new Stun. AI must not burn Technique on a non-Charging target.
B: Guardian protection	Round1 K-01 (90) applies Taunt after charge entry. Round2 enemy (118) fires eligible hit toward K-03 but Taunt redirects to K-01. Guardian takes floor(3,600×1.60×1000/2200)=2,618.	Guardian survives at 7,800−2,618=5,182 HP (ignoring other damage), while absent redirect K-03 would take floor(5,760×1000/1850)=3,113.
Failure examples	If charge resolves at the same round-1 slot, no K-08 response window; if attack ignores Taunt, K-01 cannot be listed as Path B.	Move resolution to round2, change attack tags, or design another affordable defensive route; do not invent hidden player priority.

[OPEN] The caster’s HP, Basic damage, exact post-interrupt cooldown and target-selection UI need authored cards before this is a complete 12-round encounter. This packet validates the *response window and a survival inequality* under its inputs, not a win-rate or full fight outcome.

45.6 R-E packet: a real rear-row action advantage, not imaginary bypass

[WORKING ENCOUNTER INPUT] Front guard: 1,500 HP, 600 DEF, no active Taunt/Intercept in the numeric line, and rear caster: 1,200 HP, 600 DEF, Speed 85. The rear caster can act at its slot unless KO’d. Both are enemy targets; K-03 and K-19 begin with Technique ready, K-04 is owned through STORY02. The guarded variant with Taunt or Intercept is NOT numerically certified by this no-guard line.

Sequence	Arithmetic	Outcome
Round1 front clearing	K-03 Technique 1,093 + K-19 Technique 812 = 1,905 vs front guard 1,500 HP.	Guard defeated during round1. No special rear targeting was used by either single-target kit.
Round1 true rear access	K-04 Technique rear row = floor(950×0.40×1000/1600)=237 to caster (1,200→963).	Because it is ROW damage, frontline single-target priority does not redirect it.
Round2 with K-04 prior hit	K-03 Basic = floor(1,250×0.85×1000/1600)=664; K-19 Basic = floor(1,000×0.85×1000/1600)=531; combined =1,195.	963 ≤1,195: rear caster KO before its Speed85 slot in round2.
Round2 WITHOUT K-04	Rear HP 1,200 − 1,195 = 5.	Caster survives to act once in round2; next legal Basic can remove it in round3 if allies survive.

[AUDIT] K-04 provides an earlier pressure window rather than a magical guard bypass. The actual first R-E with active Taunt/interception still needs a specific legal selector and front-row HP/mitigation packet. A permanently invulnerable guard, a fully protected caster and a 12-round timer would falsify the fallback path. If row-only attacks never reach the required backline in the authored fight, the STORY02 grant has no instructional purpose.

45.7 S-E packet: 90% anti-heal, 100% accessible fallback

[WORKING ENCOUNTER INPUT] A front-row exposed self-healer has 2,400 HP, 600 DEF, 850 Healing ATK, Speed 90. It self-heals for 1.20× Healing ATK at the end of each own normal slot if alive; it has no additional damage or Shield in this deliberately isolated packet. K-03/K-19 have the same first-copy Techniques and Speed 110/100; K-17 (STORY03) has

Speed110 and a first-copy 0.55× hit plus 90%-chance Healing Suppression applied after the hit. Tie order among K-17 and K-03 does not change this instance because both act before healer, and application is post-hit.

Branch	Calculation across action slots	Consequence
K-17 status SUCCEEDS (90% branch)	K-17 direct ≈ floor(850×0.55×1000/1600)=292. K-03+K-19 deal 1,905. Healer HP before heal = 2,400−292−1,905=203. Heal 1,020×0.60=612 → 815. Round2 K-03+K-19 Basics = 664+531=1,195.	Healer is removed in round2 before its next self-heal. This branch is NOT a guaranteed event.
K-17 status FAILS (10% branch)	Same 292 hit and 1,905 focus damage; first heal is 1,020, so remaining HP 1,223. Round2 Basics 1,195 leave 28; healer self-heals 1,020 → 1,048. Round3 Basics remove it if both survive.	The anti-heal specialist still supplies direct damage, but the status cannot be counted as certain.
No K-17 owned / focus-only	Round1: 2,400−1,905+1,020=1,515. Round2: 1,515−1,195+1,020=1,340. Round3: 1,340−1,195+1,020=1,165. Round4 Basic/Technique opportunity removes it before another heal (if both survive).	Guaranteed role-access fallback, under the packet assumptions and without assuming K-17 or a successful status. The K-03 Technique is in fact ready by round4.

[RULE CHECK] K-17's anti-heal affects the unit RECEIVING healing, so the first S-E healer must self-heal. Anti-heal applied to an enemy healer would not normally reduce that healer's healing of *another ally*. If a later boss heals allies, K-17 needs a legal recipient target or a different counter. This is an explicit constraint, not a hidden anti-heal aura. No shield, damage-over-time, enemy attack, Crit, susceptibility bonus or Mark was credited. The no-K-17 fallback requires K-03 and K-19 to survive through round4; future enemy-damage packets must test that.

45.8 Universal adversarial gates and remaining contradictions

Check / account	Status after paper reconciliation	Why not done
One-visit E and M economy	RESOLVED AS DESIGN CONTRACT: other-core basket optional; E Hall21 day7; M Hall81 day14. Two-/four-/seven-visit figures intact.	Quantities remain hypothetical; once-weekly cash growth is slower and weekend absences still cap income.
W-E guaranteed focus	CONDITIONAL PASS: 1,905 noncrit focused damage against legal 1,850-HP source; both survive their action slots.	Enemy non-source adds, true Hall20 stat mapping, and target legality for the actual encounter are not authored.
T-E timing	CONDITIONAL PASS: charge in round1, interrupter acts in middle, Taunt survives charged hit next round.	Enemy HP, damage to whole squad, full charge cooldown and actual encounter completion not priced.
R-E rear pressure	CONDITIONAL PASS: K-04 rear hit causes earlier round-2 KO; no-K-04 focus can finish after guard.	The sample guard has no active Taunt/Intercept; a true protected-guard encounter needs a separate packet.
S-E anti-heal	CONDITIONAL PASS: 90% proc shortens fight; non-proc and non-owner accessible focus branch still works.	If healer's ally is recipient, target/caster logic changes; if enemy attacks, survival must be demonstrated.
K-09 vs K-10 choice	DESIGNED: cleanse one Wound source vs Shield before next slot.	Both paths and auto-target priorities need UI confirmation and future gameplay validation.
Status/AI invariants	DESIGNED: Skills conditional; no special Ultimate-over-Tech override, no guaranteed 90%-chance debuff.	Normal Ultimates for named kits still need complete cards before authoring later boss behavior.
Hall progression bridge	OPEN/BLOCKER: reference combat numbers ≠ actual Hall20/Band I stat package.	Must map Hall stats, enemy stat curves, and gear band values before claiming any mandatory content is tuned.
Two-source Wound, burst/control, clones	ACCESS MATRIX ONLY in Section44.	Do not extrapolate four paper packets into balanced coverage for all eight threats.

[AUDIT RULES] Preserve the written global 12-round PvE cap; one normal action per living unit per round; 2-front/3-rear placement; named Taunt/Intercept legality; target-slot Wound timing; a two-full-subsequent-own-turn Technique cooldown; Status chance cap 90%; normal personal Chakra; shared defense/bonus caps. No paper scenario may quietly use a different turn order or apply a status before its written activation step. Any newly authored enemy trait must be previewed and logged.

45.9 Decision record, research sequence and next performance level

ID / selected working rule	Rejected alternative / cost	Reopen condition and dependency
2C-01 Optional weekly core basket; no debt/decay, no forced day7 total payment.	Compulsory basket + catch-up grant or global bank inflation; both require new currency source and sink audits.	If new campaign rules REQUIRE routine optional spend for first-gate completion, revise gate or create separately sourced accessible budget first.
3C-01 First-copy Techniques as named in §45.3; normal CD and Chakra unchanged.	Free automatic double actions or 100% healing-debuff application; conflict with §27 and existing status cap.	If skill budget, relative class strength or future seeded outcomes break counter diversity; reopen actual coefficients with explicit cost.
3C-02 Four conditional fight packets at reference stats; not Hall20 certification.	Marking every early threat “tested” from role presence or a noncrit damage sum.	When Hall20→normalized stat conversion, enemy loadouts, full ability cards and actual encounter stage ordering are authored.
3C-03 Publish visible enemy intent and legal response before entry; inspect recaps.	Untelegraphed unavoidable charge, protected enemy that cannot be targeted, chance debuff as only path.	If first-party UX design cannot display correct target/tick/charge state on small phones, simplify the fight rather than silently buff premium counter kits.

[HANDOFF] Section 45 closes the narrow P2C ambiguity (optional vs compulsory bill) and provides four reproducible reference-paper response windows, but DOES NOT close Priority 3 quantitatively. Priority 3D should author the missing Hall20/Band I to reference-stat bridge, first required enemy attacks, both START01 choices, true protected R-E guard, and the remaining B-E/C-E/D-E/WW-E packets. Then, and only then, begin Priority 4A stage order, boss narrative, costs/reward IDs and instructional UI. Avoid spinning up a simulator or running live tests before explicit owner authorization.

[NEXT RESEARCH PERFORMANCE: EXTRA HIGH] A Hall-to-combat bridge and four missing threat packets could contradict the shared economic floor, skill accessibility and turn-order/target rules; this requires cross-chapter reconciliation rather than a narrow mechanical check. Revisit to HIGH if the next packet is limited to one numerical gate and its ledger.

45.10 External evidence: design analogies only (31 October 2026)

[SOURCE E18-A] Riot Games, “Quick Gameplay Thoughts: May 14” (14 May 2021): counterplay is a meaningful available response, not necessarily immunity. https://www.leagueoflegends.com/en-us/news/dev/quick-gameplay-thoughts-may-14/

[SOURCE E18-B] Riot Games, “Clarity in League” (12 March 2021): design clear ability communication and visible information hierarchy. https://www.leagueoflegends.com/en-us/news/dev/clarity-in-league/

[SOURCE E18-C] Microsoft Xbox Accessibility Guideline 109 (accessed 31 October 2026): show progress prerequisites, actionable objectives and revisitable contextual tutorials. https://learn.microsoft.com/en-us/gaming/accessibility/xbox-accessibility-guidelines/109

[SOURCE E18-D] Apple Human Interface Guidelines, “Designing for games” (accessed 31 October 2026): teach through play rather than requiring separate upfront study. https://developer.apple.com/design/human-interface-guidelines/designing-for-games

46. Priority 3D: Hall-level combat bridge and four missing threat packets

31 October 2026 | v0.10.19 | Pre-development research only. Method: static arithmetic and exact-rule walkthroughs, not executed combat simulation, seeded matches, implementation or live observation. This section is a reversible research proposal. Read Sections 23, 24–27, 41–45 before reusing its numbers. The complete project remains in theory/concept design.

46.1 Evidence boundaries and the real design question

[DOCUMENT AUDIT] Section 27 gives normalized HP 6,000 / ATK 1,000 / DEF 1,000 / Speed 100 before one primary-role multiplier; Section 45 proves selected arithmetic only at that reference packet. Section 42 prices Hall20→21 (+2% to base

HP, ATK and DEF) and Hall80→81 (+1% to its level-80 values) but does not state the level-22 through level-80 stat function. Section 41 includes standard issue once in the combat baseline. Sections 43–44 guarantee four starter characters, one K-09/K-10 choice and three story recruits before their relevant gates. Never convert one of these into a claim that the complete high-level growth curve or real encounters have been tested.

[DESIGN QUESTION] Can we declare Hall20/Band I unambiguously, produce reproducible checks for B-E/C-E/D-E/WW-E, and identify where a first-copy, five-slot, noncritical squad is actually vulnerable? Existence of a counter is not equivalent to mandatory-win certainty.

46.2 A narrow, exact Hall20 reference bridge—without inventing the future curve

Variable / checkpoint	Chosen paper rule	Why / what remains open
Hall20 / Band I	Base in-battle reference HP=6,000, ATK=1,000, DEF=1,000 before exactly ONE Section 27 primary-role multiplier; Speed=100 before role multiplier; Crit baseline per role.	Two standard-issue gear slots are INCLUDED in these reference totals; never add another hidden gear bonus.
New first copy	Inherits currently earned Hall level/cap and Band I standard gear; base defining Technique available without duplicates or payment.	Use role profile only once; replacing another unit does not strip its equipment.
Hall20→21	Increase each kit's HP, ATK and DEF base reference by 2%, then apply its one role multiplier. For a basic unmodified role: 6,120 HP, 1,020 ATK, 1,020 DEF.	Speed, Crit, named skill coefficient, two-slot rules and Wound timing unchanged. Matches Section 42 immediate-benefit assumption.
Hall80→81	Retain Section 42's 1% relative upgrade benefit, but do NOT assert absolute Hall80 HP/ATK/DEF until levels 22–80 are priced and mapped.	No fabricated 20–80 exponential curve; high-level enemy/gear design remains separately open.
Enemy difficulty	Author each early enemy at fixed Hall20 reference-facing HP/ATK/DEF with visible traits and a legal target. If later enemy scaling is used, price and review it separately.	No invisible level sync, enemy power auto-scaling or arena normalization inside mandatory campaign.
Rounding	Keep precision through role/level multipliers; floor once per final stat shown, and once per completed damage hit; no intermediate flooring in the damage equation.	Use the same ordering at all checkpoints; explicit integer stat snapshots are required before future testing.

[CALCULATED CHECK] Hall20 K-03 focused Technique vs a 600-DEF target: floor(1,250×1.40×1000/1600)=1,093; K-19: floor(1,000×1.30×1000/1600)=812; combined 1,905. At Hall21, with 2%-higher player ATK and the SAME 600-DEF enemy, 1,115+828=1,943. If enemy HP 1,850 and DEF 600 also rise 2%, they become HP 1,887 and DEF 612; player output becomes 1,107+822=1,929, still above 1,887. These are two explicit enemy-policy cases, not a universal scaling theorem.

[NONOBVIOUS SCALING RISK] Under the written Section 24 DEF multiplier 1000/(1000+DEF), multiplying both sides' HP/ATK/DEF equally does NOT conserve time-to-KO. Against baseline DEF 1,000 the multiplier is 0.50; doubling target DEF to 2,000 changes it to 0.333. Doubling attacker ATK then gives damage ×1.333 while the defender's HP doubles; fractional HP removed per hit falls to two-thirds of its previous value. If higher-tier battles should retain similar pacing, vary skill budgets, enemy DEF/HP ratios or role mechanics deliberately and re-audit the 12-round cap. Do not silently change Section 24's universal defense formula.

46.3 Packet conventions and team-validity audit

[SHARED ASSUMPTIONS] All four packets use Hall20/Band I base copy, 0 starting Chakra, no Crit, Mark, buff, bonus gear, RNG damage, duplicate skills, unearned faction effects or hidden extra action. Source roles and first-copy Techniques are as in Section 45. One normal action per living shinobi per round. Speed order is sampled at each round start; same-Speed ties follow the documented side/slot rule. The player owns K-01, K-03, K-12 and K-19 and either K-09 or K-10; the story grants K-08 prior to T-E/WW-E, K-04 prior to D-E/R-E and K-17 prior to S-E. A five-member squad is named per packet; 8 owned ≠ 8 deployed.

[RULE LIMIT] A Technique used in round 1 has its next two subsequent owner action slots on cooldown, so rounds 2 and 3 use Basic unless another legal action is stated. An Ultimate at 100 Chakra has priority when eligible, but the paper

calculations stop before any undeclared Ultimate is necessary. Wound ticks at the affected unit's slot start. A clone absorbs at most ONE eligible single-target hit, contributes no action, and is not a sixth enemy combatant.

46.4 B-E: telegraphed burst—eliminate source or protect a target

[ENCOUNTER INPUT, NEW PROPOSAL] One exposed front-row burst attacker, HP 1,850, DEF 600, Speed 118, ATK 7,600. Round 1 it visibly designates K-03 for a Round-2 single-target Burst, but deals no damage. If alive at its round-2 slot, it casts 1.30× ATK, Taunt-eligible, with no hidden pre-damage Mark or vulnerability. The large attack is deliberately lethal to K-03 at base copy; surrounding enemies, if later added, need separate budgets. A one-round telegraph is the accessible response window.

Response / legal squad	Written action and arithmetic	Conditional outcome
A / K-01,K-03,K-19,K-12,K-09 or K-10	Round1 K-03 and K-19 Techniques hit exposed front source for 1,093+812=1,905 ≥1,850.	Source defeated before r2 Burst; requires both actors alive, able to target the source and not diverted by Taunt.
B / same five with K-01 Taunt	If source survives r1 (e.g., a different team choice), K-01 applies Taunt at Speed90 r1. In r2 redirected Burst vs K-01 DEF1,200 = floor(7,600×1.30×1000/2,200)=4,490.	K-01 survives at 7,800−4,490=3,310 HP, ignoring other attacks; without legal redirect K-03 would receive floor(9,880×1000/1,850)=5,340>5,100 HP.
START01 check	If K-10 chosen, shield on K-03=12%×5,100=612; alone it absorbs 612 of 5,340, leaving 4,728 HP damage and 372 HP. K-09 choice does not grant a shield.	Both starter choices retain K-01 route; do NOT demand K-10. If any prior chip exceeds 372, K-10-alone survival fails and must not be sold as reliable.

[DESIGN COST] This is a tutorial-style one-enemy decision window, not a certified full encounter. Add normal enemy damage or a guard before the first Burst only after writing a new survival and targeting audit. Keep charged-attack T-E distinct: B-E teaches survival versus fast elimination, T-E teaches a named Interrupt action.

46.5 C-E: soft control—redirect or defeat before application

[ENCOUNTER INPUT, NEW PROPOSAL] One exposed front-row Seal Controller, HP 1,500, DEF 600, ATK 900, Speed 118. Round 1 it announces a Taunt-eligible single-target Silence targeting K-03 but deals no damage. Round 2, if alive, it deals 0.30× ATK to its final legal target and attempts a visible 80%-chance Silence lasting one target action slot; if the status fails, damage still applies. Introductory C-E uses Silence rather than Stun so players learn the Basic fallback without a stolen normal action; a later Stun tutorial must honor diminishing returns and accessible answers.

Response / legal squad	Written action and arithmetic	Conditional outcome
A / K-01,K-03,K-19,K-12,START01	Round1 K-03/K-19 Techniques produce 1,905 vs controller 1,500 HP.	Controller eliminated before its round-2 Silence; no status roll required.
B / same squad with defensive choice	Round1 K-01 uses Taunt. Illustrative player chooses Basic on K-03 and K-19: 664+531=1,195, leaving controller 305 HP. Round2 control redirects to K-01, damage=floor(900×0.30×1000/2200)=122; Silence can fail (20%) or affect K-01.	K-03 remains free to act round2, and an available 1.40× Technique defeats the 305-HP controller before later casts (if K-03 was Basic round1 and thus has no cooldown).
No invisible immunity	Silence applied to K-01 restricts its Technique/Ultimate but still permits Basic on its next action. No probability is credited as guaranteed protection.	If enemy Silence ignores Taunt, changes target after Taunt, or fires in round1, Path B fails and must be reauthored.

[LIMITATION] The "choose Basic" branch illustrates a reachable composition/AI scenario; our actual automatic selector prefers a legal Technique when ready. Consequently the alternate full round-1 scripted Basic line is NOT a guaranteed player-selectable behavior under current auto-AI. The strict alternate available to ordinary auto-battle is K-01 Taunt plus surviving Silence when focus damage is obstructed by an authored legal condition. A future design must either create a previewed legal Technique target condition or permit a published tactic preset without granting manual mid-battle control. Record this as a REAL AI design gap, not a passed automatic-play branch.

46.6 D-E: two bounded decoys—row pressure versus exhausting intercepts

[ENCOUNTER INPUT, NEW PROPOSAL] Rear-row source HP 1,450, DEF 600, Speed 85; two one-hit decoys exist at battle start, within the Section 24 maximum of two. Each decoy intercepts one eligible single-target hit, cannot stop row/AoE,

and neither acts nor reapplies itself. The source uses a weak, separately logged Basic and no status; its exact damage is outside this access-window comparison. The player must own K-04 via STORY02 before entering.

Response / legal squad	Written action and arithmetic	Conditional outcome
A / K-01,K-03,K-19,K-04,START01	R1 K-03 & K-19 each use a single-target Technique; the two decoys consume those hits (0 damage to source). K-04 row Technique hits real rear source for floor(950×0.40×1000/1600)=237; source at 1,213. R2 K-03 Basic=664, K-19 Basic=531, K-04 Basic=504 once rear target legal.	After the first two R2 Basics source has 18 HP; K-04 Basic defeats it before its Speed85 R2 slot. K-04 is not assumed to bypass DEF or shield.
B / K-01,K-03,K-19,K-12,START01	Same two initial intercepts consume r1 K-03/K-19 Techniques. Decoys gone. R2 K-03+K-19 Basics=1,195, leaving source 255 HP. R3 next legal Basic finishes if units survive and target remains exposed.	Accessible no-K-04 fallback without luck; source receives at least one more action window. Any r2 re-summon or undying extra decoys invalidate this packet.

[AI CONTRACT] Decoys may absorb eligible single-target attacks even if the attacker selects the real source; they do NOT create a hidden additional row or force a special K-04 target. The K-04 Basic in r2 is legal only when no front-row enemy remains. If the encounter later adds a front guard, the access-window proof must be rerun; do not re-label this packet as the protected R-E test from Section 45.

46.7 WW-E: two Wound sources—kill both, or mitigate while killing

[ENCOUNTER INPUT, NEW PROPOSAL] Two exposed front-row appliers, A: HP1,500/DEF600/ATK900/Speed118 and B: HP1,500/DEF600/ATK850/Speed117. Each uses a round-1 0.45× direct hit plus a named two-target-slot Wound, to K-19 and K-03 respectively. They then use Basic in r2 while the Wound Techniques cool down, with no extra source or healer. The Wound snapshot rate is 18% ATK before DEF. Both players are guaranteed. A second enemy cannot immediately turn into an untelegraphed AoE Wound engine.

Response / squad	Written action and arithmetic	Conditional outcome
A / K-01,K-03,K-19,K-12,START01	R1 A→K-19 direct floor(900×0.45×1000/2000)=202, 2 pending Wound ticks floor(162×1000/2000)=81 each. B→K-03 direct floor(850×0.45×1000/1850)=206, pending ticks floor(153×1000/1850)=82. K-03+K-19 focus A for 1,905 ≥1,500.	A defeated r1, BUT its Wound on K-19 persists; B also remains. No action-free cleansing is assumed.
Round2 focus continuation	B speed117 acts before K-03/19. K-03 takes its 82 Wound tick at slot start; K-19 takes its 81 tick. K-03 Basic=664, K-19 Basic=531, K-01 Basic=425; total 1,620≥B HP1,500.	B defeated during r2 under legal focus, if K-01 is able to Basic and all three remain alive. Need account for B's r2 Basic and any Taunt before certifying total party HP.
B / same five; K-09 selection	If START01=K-09, it can cleanse exactly ONE named Wound source from ONE ally on its r1 Speed95 slot. E.g., cleanse B→K-03; A→K-19 remains for ticks. Damage squad still handles sources.	Cannot call this "two-source cleanse." Choosing K-09 costs the K-10 shielding alternative.
B / same five; K-10 selection	If START01=K-10, shield one threatened ally by 12% max HP after the r1 attacks and before their next r2 slot. K-03 shield=612; K-19 shield=720.	Shield absorbs later ticks under its written lifetime, but no second simultaneous shield from the same Technique. Does not remove Wound.

[LIMITATIONS] Source B's exact r2 Basic target and damage are not yet authored. The Wound snapshots are individually under the 20% recipient Max-HP per-slot safety cap; with two sources on DIFFERENT victims the shared cap is not stressed. A later design placing both Wounds on the SAME victim must rerun the combined cap before calling the fight fair. WW-E is a timing demonstration, not a survival certificate for all rounds or both starter choices.

46.8 Counter-argument and true unresolved contradictions

Claim we can support	What would invalidate it / next research burden
A measurable Hall20/Band I reference now exists.	Future gear stat values or rarity bonuses that double-count standard gear; later-level curve that causes 12-round timeout or invalidates base kit roles.

Claim we can support	What would invalidate it / next research burden
B-E has two distinct response categories, focus and guard.	A hidden front protector, source action before the round-1 window, non-Taunt burst, or earlier damage preventing Guardian survival.
C-E demonstrates redirectable soft control and non-guaranteed debuff.	Automatic action priority cannot select the illustrated slower plan; a legal auto-AI fallback needs an authored trigger and full protection test.
D-E AoE addresses bounded clone interception without being a premium monopoly.	New decoys are generated every round, row target becomes invulnerable, or caster has strong damage not affordable during r3 fallback.
WW-E has an accessible two-source focus line and partial mitigation option for both START01 choices.	Combined Wounds exceed source cap; a victim dies before it can act; B's r2 hit removes K-03/K-19/K-01 before the necessary attacks.
Earlier W-E/T-E/R-E/S-E packets remain reference checks.	The source B-E/C-E/D-E/WW-E packets do not repair the still-open T-E whole-fight HP, R-E true Taunt guard, or S-E enemy survival gaps.

[IMPORTANT SELF-CRITIQUE] Priority 3D is NOT finished merely because every threat now has one arithmetic table. These packets deliberately isolate response windows; enemy formation, other attacks, end-to-end 12-round path, complete Ultimate cards, starting gear and earned-level economics must eventually be unified. Do not treat "conditional access-window passes" as empirical win rates, verified balance or automatic-gameplay guarantees. Do not run scripted battles/seeded matches before the owner expressly opens the implementation/validation phase.

46.9 Decision register, order of work and next performance recommendation

ID / proposal selected	Rejected alternative and trade-off	Reopen when
3D-01 Hall20/Band I equals Section 27 reference INCLUDING standard gear; Hall21 +2% HP/ATK/DEF.	Invent full level20–81 exponential curve immediately. Faster document completion, but creates unchecked defense creep and false cost certainty.	Hall costs and gear bands for 22–80 are designed, or a previewed skill requires a new stat.
3D-02 First C-E teaches Silence rather than Stun.	Immediate hard Stun tutorial. More dramatic but hides loss of normal action and makes early counter dependence less clear.	A Stun packet is independently reviewed under global DR and counter availability.
3D-03 Two decoys may absorb two hits, then expire; no round-by-round refill in first D-E.	Unlimited replacement decoys. Makes basic focus potentially unable to progress within 12 rounds.	A later clone specialist has explicit accessible AoE/cooldown counter and survival budget.
3D-04 Two-source Wound distributed across two allies for first WW-E.	Stack both on one target immediately. More challenging but needs a combined cap/tick/survival packet first.	Paper tests establish realistic worst-case two-source stacking and START01 fairness.

[NEXT RESEARCH] Priority 3E: rather than claiming full readiness, (1) author one authoritative visible auto-AI condition for C-E and five-slot presets; (2) add exact enemy Basic damage/targets and non-source adds across all eight early encounters; (3) complete true R-E protected-guard/Intercept and T-E full-fight cases; (4) write starting Hall20→21 and mature 80→81 stat *and* enemy-envelope risk checks, without inventing prices for missing levels. Only after all named blockers have a passing paper certificate or a deliberately revised encounter should Priority 4 begin its content order. Recommended performance: EXTRA HIGH, because changing one AI selector, stat curve or first-copy counter affects several already-authored encounters and fairness guarantees.

46.10 Outside source comparison and evidence limits

[SOURCE E19-A; accessed 31 October 2026] Unity, "Designing a balanced in-game economy: How-to guide," stresses mapping progression sources/sinks against player journeys and reviewing early friction. Used as a planning analogy only; not a source of our numerical coefficients. https://unity.com/how-to/design-balanced-in-game-economy-guide-part-3

[SOURCE E19-B; accessed 31 October 2026] Farlight Games, AFK Journey official App Store listing: shared levels and equipment enable trying new formations. Supports the product-value rationale for inheritable growth, not the Hall20 stat packet. https://apps.apple.com/ph/app/afk-journey/id1628970855

47. Priority 3E: auto-combat integrity, Wound timing corrections and full opponent windows

31 October 2026 | v0.10.20 | Pre-development desk research only. Scope: reconcile Sections 24–27 and 43–46; author explicit auto-action conditions, correct same-round Wound timing, complete two provisional two-round encounter paths, and identify what eight early enemy cards actually specify. All numeric encounters are original proposals using the Hall20/Band I reference; arithmetic checks are not executed battle simulations, playtest evidence or win-rate measurements.

Evidence labels: [PRIOR RULE] preserved written mechanic; [CORRECTION] an inconsistency in earlier example; [CHOSEN PROPOSAL] reversible design value; [ARITHMETIC] exact floor calculations under declared assumptions; [CONDITIONAL] only true for the written lineup, enemy card and turn order; [BLOCKER] cannot yet call the mode complete. No Naruto IP rights are assumed.

47.1 What failed the audit—and what must not be silently changed

Issue from v0.10.19 or earlier	Diagnosis under the authoritative rules	New handling
C-E “choose Basic” branch	A legal ready Technique is selected ahead of Basic (§24.4); neither a tactic preset nor a preferred playstyle can suppress that priority.	Withdraw the voluntary-Basic round-1 example. Preserve Ultimate → eligible Technique → Basic. Redesign the control lesson, not the engine.
C-E Silence lands in round 2	K-03 and K-19 normally KO the 1,500-HP caster in R1 (1,905 combined). If it somehow survives, their R1 Techniques are on cooldown in R2–3, so R2 Silence often denies no eligible ability.	Treat current C-E as a weak teaching case even though the damage and target timing are legal. Require an actual opportunity-cost proof before naming it a Silence tutorial.
W-E/WW-E “two future ticks”	Wound resolves at the victim’s action-slot START; if the source acts earlier in round 1, the first tick is in round 1.	Correct both encounters. K-09 (Spd95) and K-10 (Spd95) do not prevent K-19’s (Spd100) first tick unless action order or prevention is changed.
R-E claimed earlier KO via K-04	Previous examples omitted actions from other deployed teammates; their Basics may remove the protected rear enemy equally quickly.	Price every deployed slot. Attribute K-04 guaranteed rear pressure while the guard lives, NOT a universal one-round advantage.
T-E checked only one hit	Source HP and post-interrupt behavior were unknown, so “interrupt” was not a full-fight outcome.	Author a bounded one-source enemy card and verify both legal finish sequences under noncritical damage.
Level80/81 scaling claimed by implication	No absolute Hall80 HP/ATK/DEF or enemy tier envelope exists in the master.	Use only 1% symbolic comparisons. Do not infer the Hall22–80 curve, higher-level KO thresholds, or mandatory gates.

47.2 The strict visible auto-AI contract: keep action priority and make intent testable

[CHOSEN PROPOSAL, 3E-AI-01] A skill may expose ONE locally evaluated activation condition and a legal target selector; different skills do not coordinate through an invisible conditional tree. At each owner action slot: if Stunned, lose normal action; if Silenced, only Basic is legal; otherwise choose a legal Ultimate at 100 Chakra, else a ready Technique with its explicit condition true, else Basic. Targeting presets alter legal target order only. A scripted paper comparison cannot instruct the player to use Basic while a valid higher-priority action exists. This tightens the interpretation of §24.4, not the underlying action hierarchy.

Kit / action	One visible Technique condition	When false; guardrail
K-01 Taunt	A living enemy has a previewed forthcoming Taunt-eligible single-target hostile action aimed at an ally other than K-01 before K-01’s next action slot.	Use Basic. The preview must identify target, eligibility and firing window; no retroactive protection. Do not pretend Taunt can absorb AoE.
K-03 / K-19 front Techniques	A living, legally targetable frontline enemy is present. Authored threat tags rank eligible targets; no presumed Mark/crit.	Use Basic if the Technique has no legal target or remains on cooldown. A tactic preset cannot suppress the Technique just to lengthen a tutorial.
K-08 Interrupt	At least one living enemy is visibly in Charging state AND within the authored Technique’s legal single-target range.	Use Basic otherwise. At 100 Chakra, K-08’s future Ultimate must be ineligible while an interruptible charger is present unless it also interrupts; otherwise Ultimate priority can break the advertised counter.
K-04 Rear Row	At least one living rear-row enemy exists, regardless of frontline Taunt.	Use Basic if no legal rear target. Row damage is not a redirected single-target attack; ordinary DEF/shields still apply.

Kit / action	One visible Technique condition	When false; guardrail
K-09 Cleanse	At least one ally currently has a named, removable Wound instance.	Use Basic if no Wound. If multiple allies qualify, next earliest unmitigated lethal tick, then earliest tick, lowest HP%, slot order; cannot cancel ticks already processed.
K-10 Shield	At least one ally is threatened by a visible incoming effect before that ally's next action slot.	Use Basic if no qualifying threat. Shield only starts after K-10 acts; it does not retroactively absorb damage. Later Ultimate still needs its own authored predicate.
Player tactic presets	Frontline First, Focus Mark and Break Threat reorder only targets that each selected action can legally reach.	No manual mid-battle Basic override, extra action, hidden bypass or guaranteed status success. The preview and recap show which tactic was active.

[OPEN] We have not yet authored every kit's Ultimate. In a 0-starting-Chakra tutorial, no Ultimate is initially legal, so the certificates below can close narrow opening windows. To extend them beyond early rounds, each Ultimate must have its own legality predicate and must not silently override a required defensive Technique. K-01's targeting predicate is a proposal to validate against all later enemy telegraphs.

47.3 Correct the timing ledger: applied Wound may tick immediately in the same round

[PRIOR RULE, §24.2 / §24.6] A living unit begins its slot by processing Wound before taking its normal action. A source can apply Wound on its earlier initiative slot; "two target slots" means the recipient's next TWO slots—including the current round if their slot has not happened. A source's defeat never removes existing Wound. Shield absorbs Wound after mitigation; Cleanse removes only remaining instances before later ticks.

Packet	Previously understated timing	Corrected Hall20 arithmetic
W-E: source Spd118 → K-19 Spd100 → K-09/K-10 Spd95	R1 application, followed by K-19 turn before support can act.	Direct 202 to K-19, then first Wound tick 81 at K-19 R1 slot: 6,000−202−81=5,717 before K-19 attacks. K-09 can remove only the remaining second tick (81). K-10 can shield before that second tick only.
WW-E: sources Spd118/117, K-03 Spd110, K-19 Spd100	Both first ticks occur R1—not R2—because each source applies before the corresponding victim acts.	K-03: 5,100−206−82=4,812 before its R1 action. K-19: 6,000−202−81=5,717 before its R1 action. A later no-cleanse R2 tick subtracts a further 82/81.
K-09 vs two sources	Removing one Wound does not erase an already-processed tick or a second source on another ally.	When support Spd95 acts R1, the first ticks have happened. Cleanse prioritizes the earliest next lethal/eligible tick and removes exactly one remaining source. K-10 shields only one legal target for its next two recipient slots.

[CORRECTION RECORD] The two-tick avoidance language in Sections 43.4 and 45.4 and the delayed-first-tick phrasing in §46.7 are now superseded only for those examples. All their enemy input stats remain historical. This does not alter the global Wound formula or its 20%-Max-HP pre-mitigation cap; it corrects when that existing formula is invoked. These earlier contradictions must remain visible so future research can trace them.

47.4 Exact proposed enemy attack cards: one source of truth per introductory packet

[DESIGN BOUNDARY] These eight cards define only the introductory threat's immediate enemy actions, target rules, sources and adds. They do not define a complete campaign stage number or reward. Listed optional adds are not secretly assumed in the earlier isolated calculation; introducing an add requires a new total-party survival packet. All new Basic damage uses the Section 24.6 floor and target's own DEF with no crit/buff unless stated.

Threat	Enemy / adds and actions	AI and additional integrity condition
W-E	Wound source: HP1,850, DEF600, ATK900, Spd118; R1 0.45× to K-19 plus two-slot Wound (18% snapshot). Candidate non-source raider: HP3,000, DEF600, ATK400, Spd80; 0.55× Basic against K-01 front-left =100.	First Wound tick is K-19 R1. Raider persists as a second enemy only after it passes its own survival/target check; its presence keeps already-applied Wound mechanically relevant after source KO.
B-E	Exposed front burst source: HP1,850, DEF600, ATK7,600, Spd118; R1 telegraph K-03, R2 Taunt-eligible 1.30× burst if alive; no adds.	K-03+K-19 noncrit Techniques deal 1,905 and normally KO it R1; Guardian redirect is a contingent defense if focus is unavailable, not a guaranteed second observed fight path.

Threat	Enemy / adds and actions	AI and additional integrity condition
C-E	Current controller: HP1,500, DEF600, ATK900, Spd118; R1 announces R2 0.30× damage + 80% Silence targeting K-03. No adds currently authorized.	FAIL AS SILENCE TEACHING: the 1,500-HP caster normally dies in R1 to 1,905 core damage; if it survives, R2 Silence may deny no available skill. Existing voluntary-Basic alternate is invalid. Keep C-E as a redesign blocker rather than inventing a new AI.
T-E	NEW coherent one-source charger: HP3,450, DEF800, ATK3,600, Spd118; R1 Charging, R2 either 1.60× Taunt-eligible hit toward K-03 or, if interrupted R1, 0.85× Basic. Charge is one-use in this intro; no adds.	K-08 Interrupt cancels only that charge; no extra Stun. After the interrupted R2 Basic the same source does not instantly queue another charged attack; otherwise the first-copy Technique cooldown would fail.
D-E	Rear source: HP1,450, DEF600, proposed ATK600, Spd85; R1/R2 0.60× Basic against K-01 front-left =163. Exactly two one-hit decoys at setup. No refills/additional attackers.	Decoys intercept eligible single-target hits once each; never intercept row/AoE. This adds enemy Basic damage omitted from the §46 response-window example.
R-E	Front guard REVISION: HP2,700, DEF600, ATK600, Spd92; 0.60× Basic to front-left K-01 =163. Rear caster: HP1,200, DEF600, ATK650, Spd85; 0.70× declared single hit toward K-03, Taunt-eligible, redirects to K-01 =206. No adds.	Guard starts Taunt and a single eligible Interception protection for rear-directed single hits while alive. K-04 rear row is immune to that redirection ONLY because it is not single-target; damage still uses DEF.
S-E	Exposed self-healer: HP2,400, DEF600, ATK850, Spd90; 0.30× Basic to K-01 front-left =115 plus its already-authored end-of-own-slot 1.20× ATK self-heal (before anti-heal). No adds.	Self-heal triggers once per own normal slot, not an additional normal action. K-17 90% Healing Suppression still has a failure branch; focus-only survival must also be checked.
WW-E	A: HP1,500 DEF600 ATK900 Spd118; B: HP1,500 DEF600 ATK850 Spd117. Both R1 direct0.45× plus Wound: A targets K-19, B targets K-03. If B survives R1, R2 Basic0.85× against K-01 front-left =328.	First Wound ticks are R1. When A is focused down R1, B's R2 Basic is resolved before player R2 slots. No hidden third source, AoE, instant refresh, or free Technique in R2.

[AUDIT NOTE] Packet R-E changes guard HP from the earlier illustrative 1,500/no-guard case to a genuinely surviving guard with active protection. This is a deliberate new proposal, not a retroactive correction to the arithmetic of the older 1,500-HP case. W-E's additional raider and S-E's weak Basic similarly extend earlier isolated packets and require rechecking any broad "all threats passed" claim.

47.5 Full T-E early fight: interruption versus surviving the telegraphed hit

[CHOSEN PROPOSAL] Starting player: Hall20/Band I, 0 Chakra, no Crit, no proc, no Mark, standard gear once. Formation front-left K-01, second front slot any eligible ally (put K-03 there for the declared charge), rear slots the other three. Full 5-member squad A = K-01/K-03/K-19/K-08/K-09. Five-member squad B = K-01/K-03/K-19/K-12/K-10. Each unit takes its normal slot; contribution from K-12 is set to zero for a conservative lower bound, not to a fictitious "skip action". If K-12 acts and adds legal nonnegative damage, that can only shorten this specific source-only fight. K-10's R1 shield is a legal action because K-03 is visibly threatened by a future charged hit.

Legally authored path	Exact no-crit two-round ledger	Verdict / trade-off
A: K-08 interrupt	R1 enemy Spd118 begins Charging. K-08 Spd110 Tech: floor(850×0.55×1000/1800)=259, INTERRUPT. K-03 Tech=972; K-19 Tech=722; K-09 Basic=377; K-01 Basic=377 because charge already canceled. Total=2,707; HP3,450→743. R2 enemy Basic to K-01=floor(3,600×0.85×1000/2200)=1,390; K-01 HP7,800→6,410. K-08 Basic=401; K-03 Basic=590: HP743−401−590≤0 before K-19 acts.	CONDITIONAL PASS: the Interrupt plan neutralizes the higher-risk burst but not the enemy's normal R2 action. Distinguishes interruption from an extra Stun. No starting Ultimate or duplicate needed.
B: K-01 Taunt + K-10 Shield	R1 Charging → K-03 Tech972 + K-19 Tech722 =1,694; source remains 1,756 HP (excluding nonnegative K-12 output). K-10 Shields visibly threatened K-03 for 612; K-01 Taunts the R2 attack. R2 charge redirects onto K-01: floor(3,600×1.60×1000/2200)=2,618; HP7,800→5,182. K-03 Basic590 + K-19 Basic472 + K-10 Basic377 + K-01 Basic377=1,816, enough to KO the remaining 1,756 in round 2.	CONDITIONAL PASS: works even without interrupt, provided Taunt lasts until enemy R2 and final action contributions stay legal. Shield on K-03 may be unused because Taunt redirected damage; that is a real opportunity cost, not "double protection".

Legally authored path	Exact no-crit two-round ledger	Verdict / trade-off
B variant: K-09 instead of K-10	With K-09 no active Wound, R1 Basic377; R1 source HP drops to ≤1,379. R2 K-03 Basic590 + K-19 Basic472 + K-09 Basic377=1,439, enough before K-01's own R2 Basic. Burst still lands on K-01 first for 2,618.	START01 choice is not a paid or random gate. A legal early attacker may end the fight sooner; this does not invalidate the protective route.

[FAIL CONDITIONS] An added front guard that draws away legal K-03/K-19 attacks; charged hit tagged Ignore Taunt; charging that resolves within the same slot; an unauthorized K-08 Ultimate that overrides its interrupt at 100 Chakra; a new second cast before K-08 recovers; or any unseen damage above K-01's remaining health voids these certificates. The full-fight result is ONLY for the explicitly authored single-source tutorial, not a future boss with adds.

47.6 Protected rear R-E: protection is real, but the value of AoE is conditional

[CHOSEN PROPOSAL] Enemy front guard HP2,700/DEF600/Spd92, live Taunt plus one bounded Interception, and rear caster HP1,200/DEF600/Spd85. Player squad with K-04 and K-10 = K-01/K-03/K-19/K-04/K-10. Player source-order for same-Speed K-19 and K-04 is recorded as K-19 then K-04. K-03 and K-19 must first target the front guard. Enemy caster's single-target intent toward K-03 is visible before K-01's R1 slot; K-01 Taunt is therefore legal.

Round / step	Arithmetic and legality	Finding
R1 source and target pressure	K-03 Tech1,093 + K-19 Tech812 vs front guard: HP2,700→795. K-04 Rear Row Tech floor(950×0.40×1000/1600)=237 despite guard Taunt/Intercept: rear HP1,200→963. K-10 shields K-03; K-01 Taunts the announced caster hit. Guard's earlier 0.60 Basic to K-01 =163; rear caster's declared 0.70 hit redirects to K-01 for 206. K-01 HP7,800→7,431.	Row attack hits while guard remains alive. The Interception has no eligible row attack to redirect, so it does not silently consume an extra row hit.
R2 K-04 team	K-03 Basic vs front guard=664, leaving 131. K-19 Basic vs guard=531, KO. K-04 Basic vs now-exposed rear caster=504; rear HP963→459. K-10 Basic vs rear=425; HP459→34. K-01 Basic vs rear=425; KO before rear Speed85.	CONDITIONAL PASS for K-04/K-10: both enemies removed in round 2; guard's R2 slot and rear caster's R2 slot are prevented by their respective KOs.
Conservative no-K-04 five-slot line	Squad K-01/K-03/K-19/K-12/K-10. Exclude K-12's unknown positive contribution as a FLOOR, not a manual skip. Guard HP795 after R1 core damage. R2 K-03 Basic664 leaves 131, K-19 Basic531 KOs guard. K-10 Basic425 + K-01 Basic425 against rear HP1,200 leaves 350 unless K-12 adds enough. Rear caster then deals another 206 to K-01; in R3 K-03 Basic664 can finish the 350.	CONDITIONAL LOWER BOUND: ≤3 rounds if K-12 does no harmful retargeting. NOT a proven one-round advantage for K-04, because an authored K-12 action ≥350 rear damage could finish the no-K-04 version in R2.
K-09 starter-choice sensitivity	K-09 has no Wound to cleanse here, so its R1 Basic adds 425 to the guard, leaving 370 after core damage. R2 K-03 Basic can KO guard immediately; K-19 Basic531 + K-09 Basic425 + K-01 Basic425 already exceed 1,200 rear HP.	With K-09, even the no-K-04 team can finish in R2 before crediting K-12. K-04's "always one round faster" proposition is FALSE for the current enemy HP. Its guaranteed contribution is early rear damage under protection.

[INTERCEPTION VALIDITY] The guard's one-hit Interception must be tested against any future optional single-target rear-access attacker (for example K-16) separately; no current first-copy named core unit has been credited with such a hit. An intercepting guard cannot intercept its own incoming hit twice. A row/AoE action never becomes "eligible" solely because the front guard has an Interception icon. Enemy protection remains active and UI-visible; it is not magic invulnerability.

[STRATEGIC IMPLICATION] We should not buff K-04 merely to force a contrived round advantage in an easy introductory fight. Instead its reason to exist is guaranteed rear access while the front remains protected, multiple rear targets, or later guard HP/shield budgets for which early rear chip changes the outcome. A future candidate encounter with a second rear target must be audited against actual five-slot skill coefficients before approval.

47.7 Survival envelope and Hall 20→21 / Hall 80→81 boundary checks

[ARITHMETIC, HALL20] The eight authored introductory packets are intentionally low-damage except for the telegraphed burst. Example direct-damage upper examples at no Crit: T-E burst to protected K-01 =2,618 leaving 5,182; interrupted T-E R2 Basic =1,390 leaving 6,410; R-E both R1 enemy Basics against K-01 =163+206=369 leaving 7,431; WW-E B's R2 Basic against K-01 = floor(850×0.85×1000/2200)=328. None of these comparisons secretly adds enemy attacks that the card

does not authorize. They do NOT replace a full battle survival model when added enemies, shield decay or statuses are introduced.

Comparison	Exact result or symbolic invariant	What it does NOT prove
Hall20→21, player vs same T-E enemy	Guardian Hall21 HP7,800×1.02=7,956; DEF1,200×1.02=1,224. Same R2 charged damage floor(5,760×1000/2,224)=2,589; survivor HP5,367.	Does not guarantee a higher-level boss is unchanged; enemy HP/DEF/ATK policy must be declared independently.
Hall21 same W-E source	K-03 ATK1,275 and K-19 ATK1,020 against unchanged DEF600: floor(1,275×1.40×1000/1,600)=1,115 and floor(1,020×1.30×1000/1,600)=828; sum1,943.	Does not imply Hall22 or Hall80 values; this is only a +2% first-step stat example.
Hall80→81 same enemy, symbolic	If player ATK rises 1% and enemy DEF stays fixed, direct noncrit pre-floor damage rises 1%, except separately capped effects. If player HP/DEF rise 1%, incoming noncrit raw attack against that DEF is no larger than before (assuming enemy ATK fixed).	Absolute level-80 HP/DEF/ATK and stage difficulty remain unknown. Floors, shields, max effects and boss resistances can change thresholds.
Hall80→81 mirrored enemy +1% ATK/DEF/HP	For uncapped DEF e, damage ratio =1.01×(1000+e)/(1000+1.01e); fractional enemy HP removed ratio /1.01 = (1000+e)/(1000+1.01e). With e=1000 that fraction is 2000/2010≈0.9950 of prior fraction.	Equal percentage scaling does NOT exactly conserve time-to-KO. Do not fabricate the missing 22–80 Hall function or infer normalized PvP equivalence.

[ECONOMY DEPENDENCY] Section 45's one-visit decision remains: at E, Hall21 payable on day 7; at M, Hall81 payable on day 14 under the precise paper wallets. Mandatory W-E through WW-E introductory lesson floors remain Hall20/Band I; no new gate may require the optional "other core" basket, Hall81 or paid resources. A single symbolic mature-level percentage is not permission to choose a high-level encounter gate.

47.8 Closure table: what actually passed, what remains blocked

Research object	Classification after v0.10.20	Next evidence required
Automatic action priority and tactic boundaries	SPECIFIED: no voluntary Basic override; one visible skill activation predicate; full 5-slot action accounting mandated.	Full Ultimate conditions, multi-charger priority tie rules and ability-availability UI in future design.
Same-round Wound	CORRECTED: K-19/K-03 first ticks in R1 under authored Speed order.	Apply same tick timing to every older paper packet and any future two-source stack/cap example.
T-E one-source charged lesson	CONDITIONAL FULL TWO-ROUND PAPER PASS for both interrupt and redirect, plus both START01 choices.	Any additional enemy, changed charge cooldown or future staged version needs a new damage/target ledger.
R-E K-04 protected rear	CONDITIONAL TWO-ROUND PASS for the K-04/K-10 five; no-K-04 route can clear ≤3 rounds under lower-bound actions.	Author K-12 full first-copy skill and test starter K-09/K-10 branches; K-04 speed advantage not universal.
C-E actual control teaching	BLOCKED: round-2 Silence currently lands during cooldown; voluntary Basic scripted route invalid.	Choose meaningful status timing or a separate training variant while preserving the Ultimate→Technique→Basic rule and early accessibility.
W-E, D-E, S-E, WW-E added enemy damage	EXPLICIT ENEMY CARDS proposed; direct-damage lower-risk envelopes written.	Full owner-turn ledgers for the newly added W-E raider and all no-crit first-copy lineups, including actual K-12 skills and status failures.
Hall20/Hall21 and mature Hall80/81	Hall20 to reference mapped, Hall21 +2% exact; M 1% step checked symbolically.	Author prices/stat formulas for Hall22–80 and enemy stat envelopes before higher-tier campaign readiness.
Priority 4 campaign and PvP balance	NOT AUTHORIZED AS COMPLETED; only concept planning allowed now.	Do not claim full difficulty progression, paid/F2P win-rate parity, simulated matches, or tested UX; keep all as future validation gates.

47.9 Working decisions, rejected approaches and explicit next research order

Decision ID / provisional selection	Alternative rejected / reason	Reopen when
3E-01: Keep existing AI priority; visible one-predicate activation for each Technique.	Manual mid-battle cast, hidden tactic-based Basic override, unconditional taunt; all change idle identity, introduce extra UI or distort documented conditions.	An authored Ultimate/counter conflict shows a particular kit has no legal automatic answer.
3E-02: Correct same-round Wound ticks across W-E/WW-E and cleansing/shield descriptions.	Credit K-09/K-10 for preventing an earlier slot they act after; violates status timing and misstates kit value.	Only if the owner explicitly chooses a new global Wound tick timing, with all statuses, balance and UX re-reviewed.
3E-03: T-E source-only 3,450-HP two-round tutorial; active source uses Basic after a canceled charge.	Treat interrupt as a stun/KO; assume zero other enemy actions; or loop permanent charges faster than K-08 cooldown.	Extra enemy sources, charge recast, a different eligible target or later-level stat scaling.

Decision ID / provisional selection	Alternative rejected / reason	Reopen when
3E-04: R-E 2,700-HP guard with active Taunt/Intercept and bounded rear caster.	Invulnerable guard, row interception, or universal “K-04 always 1 round faster” claim.	Full K-12 kit, both START01 paths and extra enemy composition are authored; then tune HP for distinct, non-forced strategic value.
3E-05: Leave C-E control tutorial openly blocked, not fake a Basic choice or guarantee a status proc.	Change tactic presets into action overrides without UI/economy audit; invent 100% Stun/Silence immunity as one premium kit.	A separate, legal five-unit control packet demonstrates a meaningful lost ability/action and two available responses.

[NEXT RESEARCH, PRIORITY 3F] First, resolve the C-E status-teaching failure with a legal first-copy auto-combat packet and a sensible controlled status that actually affects an available action; specify K-12’s real first-copy Technique and its target condition so no fifth slot is omitted. Second, extend W-E/WW-E to multi-round status and HP survival, including one-visit and both starter choices. Third, revisit the R-E row-pressure benefit for two rear targets and deliberate enemy guard budgets; do not artificially buff K-04 to win one spreadsheet. Fourth, create an explicit Hall-level envelope policy before assigning any later mandatory gate. Only after these dependencies are specified may Priority 4 author a real stage and reward-ID sequence.

[NEXT RESEARCH PERFORMANCE: EXTRA HIGH] The control tutorial, K-12 fifth-slot targeting, status order, shared-level stat envelope and two-choice survivability overlap across multiple earlier chapters; treating them independently would reintroduce the contradictions this pass exposed. No simulator, seeded matches or playable code until the owner authorizes production/validation.

47.10 Outside design references and limitations

[SOURCE E20-A, accessed 31 October 2026] Riot Games, “Clarity in League” (12 March 2021), explains that gameplay-relevant events should have legible information hierarchy, clear visual communication and minimal excess noise. Used to justify previewed target, next firing slot, Taunt eligibility and visible post-battle reasons—not our numeric thresholds. https://www.leagueoflegends.com/en-us/news/dev/clarity-in-league/

[SOURCE E20-B, accessed 31 October 2026] Microsoft Xbox Accessibility Guideline 109, “Objective clarity,” recommends reviewable goals/prerequisites and actionable next steps. Used to require clear first-copy counter and gate-requirement UI—not to assert that any proposed encounter is accessible in practice. https://learn.microsoft.com/en-us/gaming/accessibility/xbox-accessibility-guidelines/109

[EVIDENCE LIMIT] The owner has not authorized production, executable combat simulation, balance runs, or playtesting. These numbers are transparent static arithmetic under reversible encounter contracts. Sources are external design analogies; all character coefficients, levels, target rules and enemy HP/ATK are original project proposals.

48. Priority 3F: meaningful control, five-slot scouting, status survival and enemy envelopes

1 November 2026 | Version 0.10.21 | RESEARCH ON PAPER ONLY. This is a narrow reconciliation packet, not the next campaign chapter. It reads Sections 20, 24–27 and 43–47 as its authorities. It changes only the declared first-copy K-12 card and the deliberately open C-E/WW-E/R-E encounter candidates, and leaves the account-wide Hall system, existing damage equation, source archive and 56-day reward ledger intact.

Evidence labels: [PRIOR RULE] current authoritative specification; [SELECTED PROPOSAL] reversible working design; [EXACT ARITHMETIC] declared no-crit/no-proc static calculation; [CONDITIONAL] holds only under the named action order, RNG branch and enemy card; [OPEN] not established. Scripted combat simulation, seeded matches, live playtests and implementation remain explicitly unapproved.

48.1 Formal dependencies, invariants and why the previous control lesson failed

[PRIOR RULE] Initiative is frozen at each round start; the ordinary action priority is legal Ultimate → eligible Technique → Basic, except a Silenced shinobi must Basic. Two-turn Technique cooldown counts the next two owner slots after casting. Silence lasts the next recipient action slot and does not deny Basic, Chakra or all movement. Player-applied status chances cap at 90%; enemy 80% Silence is an authored encounter probability, not an inevitable outcome. Wound ticks before its victim acts, potentially in the application round. Victory may end the encounter before a surviving character receives another tick.

[REJECTED] Delaying an original 1,500-HP controller to round 2 required a source that the starter team normally KOs in round 1, and the victim's Technique was already on cooldown. Telling the auto-AI to choose Basic to preserve a later Technique violates the game identity. We instead require the controller to act BEFORE a readied first Technique, and make its target rule visible before battle.

48.2 K-12 Scout / Mark Setter — first-copy card now closed for opening encounters

Field	Chosen first-copy specification	Failure / scope boundary
Hall20/Band-I proxy	HP 5,700; ATK 850; DEF 950; Speed 115; 0 starting Chakra. No stats double-counted from free standard gear.	Normalized role proxy, NOT proof of full growth by level or completed kit at Hall80.
Basic	Single legal target following standard Taunt/preset/front-lowest-HP% order, coefficient 0.85× ATK; +25 Chakra.	No true damage, guaranteed reveal, Ignore-Taunt, or extra action.
Technique: Scout Mark	One visible condition: any living legally targetable FRONTLINE enemy exists. Select the frontline enemy with the lowest HP% (slot order tie), then apply direct 0.55× ATK and separately roll 80% chance for ordinary Mark AFTER damage. Two subsequent K-12 own-slot cooldowns. +15 Chakra.	Mark is +10% Vulnerability for target's next two action slots in the shared +15% bucket. An 80% chance is NOT guaranteed. No Mark on the attacking hit itself; no rear access.
Technique false / Silence	If no legal front enemy, Technique unavailable: Basic falls back against a legal rear target. While Silenced, Basic is compulsory even if Technique is ready.	No manual Basic choice while Technique is legal; no invented advanced fallback AI.
Ultimate	No Ultimate is ready at 0 Chakra. Future 100-Chakra Ultimate target, payload, condition and conflicts remain OPEN.	Do not extrapolate these opening-certificates to long fights with unpriced Ultimates.
Strategic role	Early Mark source, faster than K-03; substitute accessible focus targeting when Mark fails; countered by cleanser/guard/decoy.	K-12 does not become a second burst finisher or mandatory rear-row assassin.

[EXACT ARITHMETIC] Against DEF600, K-12 Technique = floor(850 × 0.55 × 1000/1600) = 292; Basic = floor(850 × 0.85 × 1000/1600) = 451. A successful post-hit Mark raises future eligible attacks by ×1.10 until expiry; the 292 applying hit is not boosted. This is a new specification. Prior Section 47 zero-credit comparisons were conservative lower bounds, not evidence K-12 has no action.

48.3 C-E revised: Silence now denies an available first action, and formation is a real counter

[SELECTED PROPOSAL, C-E-v2 supersedes §47.4 only for the Silence lesson] Single exposed front controller: HP3,000 / DEF600 / ATK900 / Speed118. Round 1 immediately uses a 0.30× noncritical direct hit aimed at the unit in the visible FRONT-RIGHT player slot, followed AFTER damage by an 80% roll to apply Silence for that victim's next action slot. Round 2, IF alive, uses a 0.85× Basic that specifically targets K-01 Guardian by visible role tag; no second Silence, extra enemy, invisible pre-round stun or Crit. Round 3+ unpriced and not needed in these limited two-round lines. The target indicator is shown on the opponent preview before team placement.

All five guaranteed player units: K-01/K-03/K-19/K-12/START01, with START01 either K-09 or K-10. Both supports have no eligible Wound/shield threat at their R1 slot; they use Basic. K-01 has no future eligible attack against another ally after its R1 slot (R2 targets K-01), so uses Basic as well. Mark chance is evaluated in two branches; no damage crit or Ultimate is claimed.

Plan and actual full team	Round 1 no-Crit ledger (DEF600)	Round 2 / conclusion
A. EXPOSE K-03: front-left K-01, front-right K-03; rear K-12, K-19, START01. Enemy Silence succeeds.	Enemy goes first: K-03 takes floor(900×0.30×1000/1850)=145 HP damage; Silence locks K-03 to Basic. K-12 Technique292; K-03 Basic664; K-19 Technique812; support Basic425; K-01 Basic425. Total = 2,618; controller remains 382 HP.	R2 controller acts before players, Basics K-01 for floor(900×0.85×1000/2200)=347. K-12 Speed115 Basic451 KOs remaining382 before other player slots. K-03 lost one eligible Technique and enemy gained one extra attack.
A with K-12 Mark SUCCESS (80% independent)	K-12 damage292, then K-03 Basic730 + K-19 Technique893 + support Basic467 + K-01 Basic467 = 2,849. Controller stays alive on 151 HP.	R2 enemy Basic still hits K-01. K-12 Basic ≥451 immediately KOs source; status good luck did not erase the teaching distinction.
A if enemy Silence FAILS (20%)	K-12 damage292; K-03 ready Technique1,093; K-19 Technique812; support Basic425; K-01 Basic425 = 3,047 without Mark, ≥3,000.	Controller KOs in R1. This is the acknowledged RNG branch; mission success must not depend on the debuff applying.
B. REPOSITION: front-left K-19, front-right K-01; rear K-03, K-12, START01. Enemy Silence succeeds on K-01.	Direct to Guardian floor(900×0.30×1000/2200)=122. K-12 Technique292 + K-03 Technique1,093 + K-19 Technique812 + support Basic425 + K-01 Silenced Basic425 = 3,047.	Controller KOs in R1 even if K-12 Mark fails. K-01 loses no eligible Technique: its only legal attack at this slot was Basic. Distinct answer is team placement, NOT status immunity.

[DESIGN INTERPRETATION] Two early answers: rearrange the formation to put a unit without a needed round-1 Technique in the targeted slot, or accept the lost K-03 Technique and win after the additional exposed opponent Basic. Both are available before any random recruitment. This is a meaningful opportunity cost without forcing an 80% proc, requiring K-09 to cleanse Silence (it currently cleanses only Wound), or inventing a stealth action selector. A mandatory tutorial may show a fixed status-application demonstration but its required success conditions must hold if the Silence roll fails in actual campaign play.

[CONDITIONAL PASS] C-E-v2 is a coherent paper lesson only if its target-by-slot telegraph is fully visible, the 5-unit attack order is retained and the enemy does not gain another defender, status or higher DEF. The encounter still needs player-facing teaching copy and future authorized usability checks. R1 full-party damage is 2,618–2,849 after a successful Silence, versus at least 3,047 when the threatened skill is protected. Exactly one additional enemy Basic, not indefinite control.

48.4 W-E with the previously disclosed raider: every slot and surviving Wound instance

[PRIOR INPUT] Wound source HP1,850, DEF600, ATK900, Speed118 opens with 0.45× direct to K-19 and two-slot 18%-ATK Wound. Second enemy raider HP3,000/DEF600/ATK400/Speed80 Basics 0.55× against K-01. Use K-01 front-left, K-03 front-right, K-12/K-19/support behind. R1 K-12 Mark292 and K-03 Technique1,093 and K-19 Technique812 kill source before support, even if Mark fails; total 2,197. Raider is a separate surviving enemy. K-01 Taunt predicate false because raider already targets K-01, so its R1 Basic425 is legal against raider.

Round	Five-slot action and opposing actions	Survivability with either START01 selection

Round	Five-slot action and opposing actions	Survivability with either START01 selection
R1	Source acts first: K-19 direct202. K-12 Technique292 + K-03 Technique1,093 + K-19 Technique812 kill source (1,850 HP); K-19 has already ticked81 at its slot start, leaving 5,717 HP. K-09 cleanses the remaining Wound OR K-10 shields K-19 for720 after that tick. K-01 Basic425 leaves raider2,575; raider Basic100 hits K-01 →7,700.	The source's defeat does not dispel its Wound; no support is fast enough to prevent the first K-19 tick. Both starter choices are legal and cost no special gear.
R2	K-12 Basic451, K-03 Basic664, K-19 Basic531, K-09/K-10 Basic425, K-01 Basic425: 2,496 vs raider, HP2,575→79. Raider survives to act, Basics K-01 for100 →7,600.	K-09 removes the second Wound: K-19 stays5,717. K-10 shield absorbs second81: K-19 HP stays5,717, with shield639 before lifetime expiry. Without either intervention K-19 would fall to5,636.
R3	K-12 Speed115 Basic451 KOs raider HP79; fight ends before slower ally/enemy slots. No unearned third Wound tick occurs.	K-01 ends at7,600; K-03, K-12 and K-09/K-10 take no HP damage under this declared enemy card. Claimed no-Crit 3-round win conditional on exactly this two-enemy script.

[SENSITIVITY] Mark success cannot hurt either line, but the guaranteed no-Mark lower bound was used. If a new enemy Taunt/decoy displaces K-12/K-03/K-19 from the source on R1, or a new enemy attacks before support, reprice the whole ledger instead of describing this packet as automatically balanced.

48.5 WW-E two Wound sources: deliberately reveal the second tick without making cleansing compulsory

[SELECTED PROPOSAL, WW-E-v2] Keep A: HP1,500/DEF600/ATK900/Speed118 and B: DEF600/ATK850/Speed117 and all previously published actions, but change B HP from 1,500 to 1,900 ONLY for this transparent two-source follow-up. The old 1,500 B often dies before K-19's R2 slot, so its projected second Wound tick was not experienced. This 400-HP increase makes both victim R2 slots occur without new damage or unavailable characters; it is a new proposed encounter value, not a correction to global Wound.

Action window / source	Exact worst-first no-Mark calculation	K-09 branch versus K-10 branch
R1 hostile opens	A: K-19 direct202 + first tick81, leaving K-19=5,717. B: K-03 direct206 + first tick82, leaving K-03=4,812. K-12 Mark Technique292 to A, K-03 Technique1,093 to A, K-19 Technique812 to A: combined2,197, A defeated. K-01 Basic425 to B: HP1,900→1,475.	K-09 cleanses one remaining source, prioritizing the earlier R2 victim slot K-03; cannot remove Wound from BOTH allies. K-10 instead shields K-03 for612 after the already-processed R1 tick.
R2 before first player slot	B Speed117 Basic to K-01 = floor(850×0.85×1000/2200)=328, leaving K-01=7,472. K-12 Speed115 Basic451 → B1,024 HP.	K-09 has removed K-03 next tick; K-10 shield will absorb K-03 next82 tick. In either branch K-19 still carries A's second Wound81.

Action window / source	Exact worst-first no-Mark calculation	K-09 branch versus K-10 branch
R2 finishing slots	K-03 Speed110 Basic664 → B360 HP; K-19 Speed100 first takes second Wound81, leaving5,636 HP, then Basic531 defeats B. Battle ends before support/K-01 R2 actions.	K-03 ends4,812 HP with either cleanse or shield; K-19 ends5,636 HP; K-01 7,472. Every original R1 Wound is counted, and both second ticks matter. K-10 shield retains530 after absorbing82 until expiry.

[CAP AND FAIRNESS] A's 162 raw snapshot against K-19 (20% Max HP =1,200) and B's 153 raw against K-03 (20% =1,020) are individually below per-slot pre-mitigation cap. They affect DIFFERENT victims; a double stack on one victim is still OPEN. With B HP1,900, the no-Mark case can finish by K-19 R2 and K-09/K-10 do not add offensive damage to the win. With B HP1,500, the encounter ends sooner and K-19's unused future Wound tick must NOT be charged after victory. Both are valid encounter designs but teach different things.

48.6 Protected rear R-E with TWO rear enemies: real area utility, not a fabricated round advantage

[SELECTED PROPOSAL] Keep the v0.10.20 front guard HP2,700/DEF600/ATK600/Speed92 (0.60× to K-01) with one Interception and visible Taunt. Replace the lone rear caster with TWO independent casters, each HP1,200/DEF600/ATK650, Speed85 and84, each a single 0.70× cast against K-03 if alive, both Taunt-eligible. No healing, shields, unannounced adds or reinforcements. If K-01 uses R1 Taunt at Speed90, its two eligible redirects are spent on the two later caster hits (206 each); front guard Basic163 already hits K-01. No row attack is redirected.

Five-slot lineup	R1, with correctly ordered targets	R2–R3 / what is actually proved
Row team: K-01,K-03,K-19,K-04,K-10	K-03+K-19 Techniques=1,905 on guard, leaves795. K-04 Technique=237 to EACH rear caster, both1,200→963 while guard lives. K-10 shields K-03 against the previewed caster target. Guard Basic163 K-01; K-01 Taunts both later hits206+206; K-01 ends7,225.	R2 K-03 Basic664 leaves guard131; K-19 Basic531 KOs guard; K-04 Basic504 to caster1→459; K-10 Basic425→34; K-01 Basic425 KOs caster1. Caster2 remains963, R2 attack245 on K-03 is absorbed by the R1 shield. R3 K-03 Basic664 and K-19 Basic531 KO caster2: three rounds.
Mark team: K-01,K-03,K-19,K-12,K-10, Mark FAIL	K-12 Technique292 + K-03 Technique1,093 + K-19 Technique812=2,197 guard damage, guard remains503. Shield/Taunt and enemy hits as above.	R2 K-12 Basic451→guard52; K-03 Basic664 KOs guard. K-19 Basic531, K-10 Basic425, K-01 Basic425 KO caster1; caster2 acts into surviving K-10 shield. R3 K-12 Basic451 + K-03 Basic664 + K-19 Basic531 KO caster2. Also three rounds.
Mark team, Mark SUCCESS	R1: K-12=292, K-03=1,203 and K-19=893 → guard HP312; no R1 guard KO.	R2 K-12 marked Basic496 KOs guard. K-03 Basic664, K-19 Basic531, K-10 Basic425 KO caster1; K-01 Basic425 chips caster2 to775. R3 K-12 Basic451 + K-03 Basic664 finish caster2. Also three rounds.

Five-slot lineup	R1, with correctly ordered targets	R2–R3 / what is actually proved
Starter sensitivity K-09 instead of K-10	K-09 has no Wound here, so R1 Basic425 hits the guard; with successful Mark the guard can die R1, changing whether its Speed92 Basic occurs. K-01 still Taunts the two slower rear-caster hits.	No shield is available for caster2 R2; its 245 damage strikes K-03 (HP5,100→4,855) if that enemy survives. Under the stated targets and no-Crit attacks, the source-only rear encounter remains solvable without treating K-04 as required.

[DESIGN CHOICE] The row attacker delivers 474 total rear damage before the guard is removed and preserves its advertised rear-row identity. K-12 improves concentrated frontline damage, and a successful Mark can change when the guard dies. Both builds plausibly need three rounds here; inventing an alleged one-round AoE advantage would be false. Keep a later multi-target or guarded-sustain encounter as a possible K-04 payoff, only after all five kit cards and enemy timing are priced.

48.7 Higher-level enemy-envelope policy—requirements, not invented character growth

[SELECTED POLICY] Hall20/Band I and the Hall20→21 +2% HP/ATK/DEF bridge remain the only absolute early-account calibration. Hall80→81 has only the previously declared symbolic +1% step; Hall22–80, equipment tiers, talent ranks, stage unlocks, boss resistance tables and reward amounts have NO authored full curves. They may not be silently inferred from two endpoint examples.

Future mandatory encounter constraint	Proposed paper acceptance test	Reason to reopen
Expected roster and earned gear	Specify exactly which characters, Hall band and standard equipment are guaranteed by a supported schedule, including a weekly visitor and both support choices.	Would require an unowned, paid, duplicate-locked, or historically missed choice.
Complete enemy card	Author every enemy HP/ATK/DEF/Speed, target rule, telegraph, skill cooldown and add/guard behavior, at the actual gate level.	A new source or passive changes the number of attacks before response.
Action-economy integrity	Use one frozen initiative queue per round, all five actions, legal skill activation, cooldowns, K-12 Mark hit/proc branches and status-clock order.	Unseen Basic override, extra action, target shortcut or omitted ally input is needed.
Accessible two-answer proof	For each mandatory threat, write one deterministic accessible answer not dependent on positive status luck; alternative may rely on a distinct accessible mechanic (formation/protection/interrupt).	Only one high-rarity unit, lucky debuff, or hidden lineup survives.

Future mandatory encounter constraint	Proposed paper acceptance test	Reason to reopen
Survival and combat pacing	Report exact damage and HP at each recipient slot, plus 12-round timeout exposure. Proposed tuning warning: an untelegraphed normal enemy action should not erase a full-health equal-tier core unit; a lethal telegraphed attack needs two response categories.	One hit unpredictably kills the only counter, or success requires unrealistically short/long fights.
Stat-scaling bands	For any level beyond 21, publish actual player and enemy stat snapshots and compare ratio of damage dealt to enemy HP and enemy damage to vulnerable ally HP. Do not use equal-% scaling as proof of constant turns-to-KO.	DEF growth, shields, Wound caps or enemy HP create unexpected 12-round stalls.
Economic gate independence	Publish the dated cheapest viable Hall/gear/resource pathway for one-, two- and seven-visit weeks; no compulsory optional “other-core” bundle.	A mandatory fight is available before its obtainable counter can be fielded.

[DECISION NOTE] Reject a single universal +X% enemy-growth formula now: the current cap and defense multiplier do not preserve fractional HP removed under equal stat growth (see §47.7). Reject arbitrary permanent damage caps and generic 50%-of-HP gates until whole-fight evidence exists. A stage-specific envelope with audited numbers permits intentional short tutorials, defensive bosses, and later higher-HP attrition while retaining accessibility. This postpones detailed balance rather than claiming it has been validated.

48.8 Contradictions, reversals, dependencies, next research recommendation

ID / selected proposal	Why chosen / rejected alternative	Evidence that would reverse it
3F-01 K-12 first copy: 0.55× Mark 80%, Speed115, visible frontline predicate	Closes omitted fifth slot and avoids guaranteed +10% vulnerability. Reject automatic rear-bypass and 100% Mark.	Subsequent first-copy kit audit or 5-slot log reveals impossible targeting or excessive core burst.
3F-02 C-E-v2 Speed118 R1 front-right 80% Silence; HP3,000	Silence meaningfully prevents a ready K-03 Technique; slot placement offers a deterministic answer. Reject the old R2 cooldown Silencing, manual Basic suppression and forced status outcome.	If its status roll fails too often to teach the mechanic; revisit with a separate optional predictable tutorial preview, not mandatory gacha.
3F-03 W-E raider validated in R1–R3 with both starters	Preserves persistent Wound and modest opposing follow-up damage. Reject unmodeled raider deletion.	Additional enemies, longer Wound, new auto skill conditions or shield expiration modify ledger.
3F-04 WW-E B HP1,900 instead of1,500 for a two-second-tick lesson	Makes K-03 and K-19 R2 Wound slots occur. Reject charging future ticks after final KO.	If the added 400 HP makes the tutorial unintuitive or creates difficulty for a valid alternate squad.

ID / selected proposal	Why chosen / rejected alternative	Evidence that would reverse it
3F-05 R-E two rear casters; no artificial K-04 speed win	Confirms row pressure distinct from single-target Mark without granting free KO. Reject K-04 buff for a preselected spreadsheet result.	New rear enemy healing/shield or alternative tactic creates a real observed timing gap; rerun all five slots.
3F-06 Higher tier gate envelope, no Hall22–80 curve yet	Blocks false high-level readiness and protects economic/role fairness. Reject untested exponential scaling.	When equipment bands, Hall costs, and actual stage roster/stat assignments exist for every intervening gate.

[RESEARCH HONESTY] Section 48 resolves selected PAPER contradictions, not all Priority 3 deficiencies. K-07/K-08 other control branches, first-copy Ultimates, healer/anti-heal and clone full-round enemy ledgers, simultaneous multiple Wounds on the same target, arena stat normalization and long-term Hall22–80 remain open. Numbers should never be represented as measured win rates or player comprehension.

[NEXT PASS — PRIORITY 3G] Author the minimal first-copy Ultimate contracts and 100-Chakra interactions for essential roles; bring the rest of the eight early encounter enemy cards to five-slot, two-starter, no-positive-proc static ledgers; audit attack/status expiry and optional guard/decoy variants; reconcile the resulting counter accessibility with the 56-day entitlement model. Only after the Priority 3 exit checklist is satisfied may Priority 4 write an actual mandatory stage order. PERFORMANCE: EXTRA HIGH because every Ultimate, enemy target and economy gate can invalidate an otherwise legal first-copy counter. No executable simulator, seeded match, playable UI or live playtesting without explicit owner authorization.

48.9 External comparison and evidence limit

[SOURCE E21-A, checked 1 November 2026] Riot Games, “Clarity in League,” 12 March 2021. Relevant external principle: players need to recognize consequential attack/control cues and their priority before responding. This supports our previewed targeted front-right slot and hit-by-hit recap, not any numerical HP or status probability. https://www.leagueoflegends.com/en-us/news/dev/clarity-in-league/

[SOURCE E21-B, checked 1 November 2026] Microsoft, Xbox Accessibility Guideline 109 (objective clarity). It recommends replayable tutorials and accessible explanations, not merely static controls screens. Relevant to a replayable Silence formation lesson and showing target/alternative counter. Not evidence that our actual UI is usable yet. https://learn.microsoft.com/en-us/gaming/accessibility/xbox-accessibility-guidelines/109

49. Priority 3G: first-copy Ultimates, 100-Chakra integrity and eight-threat readiness audit

1 November 2026 | Live master v0.10.22 | DESK RESEARCH ONLY. This chapter reconciles §§24–27 and 41–48. New character coefficients and target predicates are reversible proposals; calculations are static arithmetic, not an executable battle simulation, seeded matchup, player test, proof of enjoyment, or final Hall progression balance. Naruto IP commercial rights are not assumed.

49.1 Research objective and evidence classification

[PRIOR AUTHORITATIVE RULE] Five deployed characters, a frozen Speed queue, one normal action per living unit per round, personal Chakra 0–100, Basic +25, Technique +15, Ultimate −100 and no Ultimate Chakra generation, +5 incoming direct-hit Chakra per qualifying enemy root action (maximum +15 per defender/round), +10 one root-action takedown, two subsequent own-slot Technique cooldowns, status clocks, Target/Taunt/Intercept sequence, DEF formula, and Ultimate→eligible Technique→Basic auto priority. PvE ends at the 12-round cap. These do not change in this revision.

[RESEARCH QUESTION] Does introducing first-copy Ultimates create an inaccessible mandatory counter, secretly displace the defensive Technique at 100 Chakra, produce illegal damage at full HP, or invalidate our earlier 0-Chakra introductory encounters? The test is rule compatibility and conditional, reproducible arithmetic—not measured balance.

49.2 Chakra availability: exact timing, without an early-Ultimate assumption

Owner sequence from 0 Chakra	Chakra after owner action (no incoming/takedown)	Earliest Ultimate slot	Qualification
Basic, Basic, Basic, Basic	25, 50, 75, 100	Owner slot 5	The Ultimate requires 100 at selection time; earning 100 on slot 4 never grants an extra slot-4 action.
Technique, Basic, Basic, Technique, Basic	15, 40, 65, 80, 100	Owner slot 6	Tech is unavailable for the next two owner slots; this is the standard introductory damage-kit pattern.
Basic + incoming + KO bonuses	Each qualifying incoming root contributes +5, at most +15 per round; root-action KO contributes +10	Potentially earlier than the no-bonus rows	An Ult cannot be certified by assuming those bonuses; log the particular incoming hits and actual KOs.
Silence or Stun at 100	Silence: legal Basic, +25 clamped at 100; Stun: no normal action	After control expires at next eligible owner slot	Never expend 100 Chakra on an illegal Ultimate, and never grant deferred extra actions.

[SELECTED PRECISE INTERPRETATION] Chakra is clamped to [0,100] after each income event. Ultimate legality is checked at the owner action slot before the selected action. If 100 Chakra is stored but no legal Ultimate target/visible condition exists, continue to a legal Technique or Basic and retain the stored 100 (no overflow or automatic discharge). An Ultimate costs 100 on actual resolution; if its target dies before resolution, validate and reselect a legal target before paying, or fall back if none. This resolves an otherwise unspecified full-meter edge case; it does not change income quantities.

[ADVERSARIAL NOTE] A zero-starting-Chakra opening does not imply that an Ultimate is impossible in EVERY three-round fight: exceptional repeated direct-hit income and credited takedowns could advance a slot-3 use. The specific Hall20 tutorial calculations in §§45–48 claim no Ultimate because their declared action and incoming-damage ledgers do not generate 100 Chakra before their ending slots. Future encounter designers must account for all resource triggers, not silently assume they never occur.

49.3 Minimal first-copy Ultimate contracts for nine relevant shinobi

[WORKING PROPOSAL, NOT FINAL ROSTER LOCK] The contracts below complete only the initial Hall20/Band-I functional set. All are accessible at the first owned copy: no skill materials, duplicates, character awakening, premium banner, skill-rank payment or extra combat slot. Damage uses §24.6, is direct unless stated, may Crit only where ordinary direct damage allows, and does not create extra attacks or Chakra. Conditions are displayed as single visible skill eligibility statements; selection and ties are explicitly stated. All targets must be legal after Taunt/Intercept.

Kit / proposed Ultimate	100-Chakra activation + target selection	Payload and counter-integrity constraint
K-01 Guardian — Shelter Oath	A living enemy is previewed to make an eligible single-target offensive action. Self-targeting protection.	Refresh (do not stack) the existing 2-redirect Taunt, and Shield self for 10% own max HP for 2 recipient slots (Hall20 = 780). Maintains the Taunt protection role if it supersedes the Technique. AoE and Ignore-Taunt remain counters.
K-03 Finisher — Execution Window	A legal target is visibly at or below 30% HP. Single target; target lowest HP% among eligible enemies, obeying Taunt.	Named Execute: 1.85× ATK direct hit against eligible ≤30%-HP target only; if none, fall through to Technique/Basic while retaining Chakra. Threshold appears in preview and log. Recovery/heal, Shield, Taunt/Intercept and target removal are response categories.
K-19 Reliable Striker — Final Strike	At least one legal offensive target exists. Standard single-target priority.	1.50× ATK direct damage; no Mark, free follow-up, ignore DEF or extra action. Replaces rather than supplements Basic/Technique.
K-12 Scout — Clear Opening	At least one legal frontline enemy exists; lowest HP% frontline, slot-tie break.	1.50× ATK direct damage; NO guaranteed Mark or rear bypass. The separate 80% Mark remains in its Technique. If frontline absent, fall through to normal legal Basic.

Kit / proposed Ultimate	100-Chakra activation + target selection	Payload and counter-integrity constraint
K-08 Disruptor — Severing Counter	At least one living enemy is visibly Charging AND a legal single-target selection exists; earliest resolving charge then lowest HP%.	0.80× ATK direct hit plus deterministic Interrupt to that one named charge. No additional Stun or action theft. If no Charger, Ultimate is ineligible: Technique if legal, else Basic, retaining 100 Chakra.
K-04 AoE Breaker — Rear Cascade	At least one living rear-row enemy is legal for a rear-row skill.	0.45× ATK direct damage to each legal living rear-row enemy, subject to each target DEF and Shield. No global Taunt ignore for single-target follow-ups; no random target leakage.
K-09 Cleanser — Restoring Thread	At least one ally has Wound OR an ally is at or below 80% HP; prioritize next fatal Wound tick, then lowest HP%, then slot.	Exactly ONE named Wound source removed from that ONE ally (if present), then heal same ally 1.35× Healing ATK, limited by missing HP. Does not cleanse multiple targets/sources, Silence, Stun or healing suppression.
K-10 Protector — Recuperation Seal	At least one ally is at or below 80% HP; select lowest current HP%, slot tie.	Heal same ally 1.50× Healing ATK and apply a 12%-Max-HP Shield to that ally for 2 recipient slots; standard same-named Shield and 40% combined cap. If nobody injured, the Ultimate is ineligible and the existing visible-threat Shield Technique may be selected instead.
K-17 Anti-sustain — Recovery Break	At least one reachable enemy with a visible Healer tag exists; lowest HP% tagged legal target, slot tie.	0.80× ATK direct hit; AFTER damage, 90% chance of −40% Healing Suppression for next 2 target slots. Not guaranteed; if proc fails, accessible focus damage must still answer early sustain.

[NON-DESIGNATED KITS] K-02, K-05, K-06, K-07, K-11, K-13, K-14, K-15, K-16 and K-18 do NOT gain invented Ultimate values in this pass. Their eventual skills require separate authoring and full interaction reviews. The initial nine cards do not imply a completed 19-hero roster.

49.4 Ultimate burst guardrail and exact Hall20 examples

[ARITHMETIC, UNBUFFED] Against DEF600, K-03 Execute = floor(1,250×1.85×1,000/1,600)=1,445 (only when target ≤30% HP); K-19 Ultimate = floor(1,000×1.50×1,000/1,600)=937; K-12 Ultimate = floor(850×1.50×1,000/1,600)=796; K-04 Ultimate hits EACH rear DEF600 target for floor(950×0.45×1,000/1,600)=267; K-17 Ultimate vs DEF600 =floor(850×0.80×1,000/1,600)=425. K-08 interrupt Ultimate vs DEF800 = floor(850×0.80×1,000/1,800)=377. K-09 unmodified gross heal =floor(800×1.35)=1,080; K-10 gross heal =1,200, then missing-HP cap. These are reference examples, not independent DPS promises or first-round skills.

[FULL-HP ONE-SHOT AUDIT] An unrestricted 1.70× K-03 Ultimate against a full-HP Hall20 K-03 (HP5,100/DEF850) with +40% Attack Bonus, +25% Skill Bonus, +15% Vulnerability, ×2 Crit and 40% effective DEF reduction would hit for floor(1,250×1.70×1.40×1.25×1.15×2×1000/(1000+850×0.60))=5,664, violating §27.4's proposed no-full-health-one-shot guardrail. We reject unconditional 1.70×. The chosen 1.85× is ONLY executable against an already ≤30%-HP target, never a full-health target. Its threshold and counters must be taught. We are not claiming normalized PvP balance from this sample.

[ULTIMATE OPPORTUNITY COST] K-03 stores Chakra if no execute target exists, which can feel sluggish; K-08 stores it if no charger exists. This is intentional for counter reliability and visible skill identity, but must be re-evaluated in long fights and PvP: ultimate animation/value cannot be priced only from a short tutorial. Alternative—always-fire full-damage Ultimate—was rejected because it weakens timing identity and can override necessary Techniques.

49.5 Audit of the eight first mandatory threat packets at the new Ultimate boundary

Encounter	Previously established accessible response	3G status after adding Ultimate cards	Unresolved robustness work

Encounter	Previously established accessible response	3G status after adding Ultimate cards	Unresolved robustness work
W-E: one Wound source + raider	START00 K-03/K-19 focus; K-09 remove one source or K-10 mitigate later tick. §48.4 detailed to round 3.	CONDITIONAL opening pass unchanged: neither defensive support Ultimate can be assumed at initial Chakra 0. K-19 may retain a tick after source KO.	Need explicit Wound status chance/any first-hit crit and all hidden damage tags; later casts and 100-Chakra alternative battle.
B-E: exposed burst	K-01 Taunt redirects an eligible hit; K-03/K-19 focus source. §46.4 initial window.	CONDITIONAL: K-01 future Ultimate preserves Taunt rather than overriding it at 100. No stronger tank assumed in opening.	Full five-slot, both-support, all enemy attacks and after-Taunt expiry ledger remains unclosed.
C-E-v2: front-right Silence	Move K-01 to targeted slot; K-03 retains R1 Technique. §48.3 includes 80% hit/fail branches.	CONDITIONAL: first-round 0-Chakra result unchanged; if future 100-Chakra K-03 has no ≤30% Execute target, it falls back legally.	Controller status-proc expiry and extended 12-round reapplication with two-starter lines not yet evaluated.
T-E: telegraphed charger	STORY01 K-08 Interrupt or START00 K-01 Taunt; §47.5 full 2-round single-source paper.	CONDITIONAL: at 100 Chakra K-08 Ultimate ALSO interrupts; K-01 Ultimate ALSO Taunts if chosen. Priority conflict closed on written kit.	Multiple simultaneous chargers, no legal target for prioritized cast, or added front guard require fresh full card.
D-E: two decoys	STORY02 K-04 row damage or expiring one-hit decoys then focus. §46.6 access window.	PARTIAL: K-04 Ultimate row attack is never intercepted as single-target. No extra summon turns.	Full enemy direct attacks, status clocks, decoy expiration if K-04 unavailable, both support choices need complete roster-ledger accounting.
R-E: front guard + two rear casters	STORY02 K-04 early rear damage or accessible defeat-the-guard; §48.6 both need 3 rounds in modeled lines.	CONDITIONAL: K-04 Ultimate keeps rear-row identity; cannot create an imagined round-1 Ultimate.	Guard Intercept against optional K-16; higher-HP guards; any new Taunt and future Ult retarget must be separately priced.
S-E: exposed healer	STORY03 K-17 90% anti-heal or START00 focus K-03/K-19; §45.7 isolated recovery.	PARTIAL: at 100 K-17 Ultimate preserves 90% suppression, NOT 100% certainty. Intro focus still must pass failed-proc branch.	Enemy Basic damage, healing timing after actions, surviving player HP, both START01 choices and longer sustain/cooldowns are missing.
WW-E: two Wound sources	Focus A then B; K-09 remove only one; K-10 shield one. §48.5 opponent cards through round2.	CONDITIONAL: neither K-09 nor K-10 Ultimate is credited in opening. No cost/ownership change.	Both Wounds on SAME victim; cap distribution when binding; added third enemy or overlapping control need more work.

[BOUNDARY] An accessible response window is not a certified mandatory stage. The current list lacks complete 12-round, full-enemy/full-player hostile ledgers for B-E, D-E and S-E. Some prior tables use explicitly conservative omitted positive ally damage; those are bounds, not legal “skip this unit” manual action instructions. Do not claim eight fully validated fights or choose a final campaign gate number from this table.

49.6 Same-target two-source Wound: close allocation ambiguity without inventing extra ticks

[PRIOR RULE] Wound is 18% applier ATK snapshot, active for next 2 victim action slots including the same round if the application preceded that slot. Up to TWO sources; total Wound pre-mitigation per victim slot is capped at 20% victim Max HP. No Wound Chakra, Crit, lifesteal, reactions or takedown. §48.5 puts one source on each of two allies, so it does not establish allocation if both attach to one ally.

[SELECTED CAP-ALLOCATION PROPOSAL, 3G-W-01] For each victim slot, calculate each surviving Wound instance’s pre-mitigation raw snapshot. Let sum S and C=0.20×victim MaxHP; multiply ALL simultaneously active raw instance values by common factor min(1,C/S) when S>0. Apply the victim’s DEF and non-DEF reduction to each scaled source, floor only once per source at the end of its damage pipeline, then total the HP/Shield absorption. This is unbiased by application

order and retains per-source identity so a ONE-source cleanse works. Two sources do not become a merged uncleanseable status. A third source is illegal under the existing cap.

Example (K-19 HP6,000, DEF1,000)	Source A and B	Slot outcome
Ordinary WW-E same victim: ATK900/850	Raw Wounds =162+153=315, under 20%-HP cap 1,200. After DEF: 81+76=157.	If both apply before K-19 R1 slot, both tick for 157 in that same R1 slot; source death afterward does not erase remaining next slot.
Direct hit and R1 tick boundary	Enemy 0.45× direct hits =202+191=393; Wound tick157.	K-19 6,000−393−157=5,450 before its own R1 normal action, absent Shield or other damage.
R2 partial intervention	K-09 later R1 removes A only: remaining B R2 tick76. K-10 later R1 shields K-19 for 720, potentially absorbing both R2 ticks157.	K-09 path K-19 reaches5,374 after R2 tick; K-10 path remains5,450 with at most563 Shield (assuming no intervening direct hits). Neither rewinds R1.
Binding cap illustration (NOT a recommended first encounter)	Two 4,000-ATK sources yield 720+720=1,440 raw, over cap1,200. Shared scale=1,200/1,440=5/6; each becomes600 raw; after DEF1,000 each deals300.	Total post-DEF Wound =600 for that victim slot (before Shields); cleanse one would leave the other 300. The 4,000 ATK inputs are for the arithmetic of the cap only, not a proposed onboarding enemy.

[COUNTERARGUMENT] An oldest-source-first cap is easier to implement but can make the second applier deal zero damage in a capped packet, concealing its contribution and making cleanse priority unintuitive. Common scaling adds an extra log field and must be represented plainly: “two Wounds; 1,200 raw cap shared proportionally.” Reopen if player comprehension, desync/integer precision, or later defensive effects require a simpler aggregate policy. This is a reversible written rule until engine implementation and UI validation are authorized.

49.7 Acquisition and pay fairness: no new cost to unlock a necessary Ultimate

[ACCESS CROSS-CHECK] START00 K-01/K-03/K-12/K-19 is a one-time tutorial core; START01 gives exactly one K-09 or K-10 choice; STORY01/02/03 provide K-08/K-04/K-17 at their first relevant gates and before entry. Every corresponding Ultimate above is included in the first owned copy, uses only earned personal Chakra, and inherits the same Hall and standard gear. No new STORY receipt, fragments, seal, equipment upgrade, skill material, awakening, duplicate or passive-purchase condition has been added. The seven-week 100-fragment optional selector remains a separate acquisition route, not an early Ult unlock.

[FAIRNESS STRESS GATE] For every future mandatory encounter, compare (i) K-09 starter, (ii) K-10 starter, and (iii) ordinary account without any OPTIONAL selector purchase. Any new stage requiring K-09’s or K-10’s specific Ultimate rather than their base role, or requiring a character not granted before the gate, is a documented failure. Do not claim all future event/guild/ranked Ult synergies are free-to-play normalized until mode-specific investment rules are written.

49.8 Decisions, rejected alternatives, reversal conditions and handoff

Decision ID	Chosen working policy and rejected alternative	Reopen trigger
3G-01	At 100 Chakra choose ONLY a legal predicate-matched Ultimate. Otherwise Technique→Basic; 100 remains stored. Reject mandatory unconditional auto-cast.	Long fights routinely leave core characters at full meter without meaningful, understandable decisions.
3G-02	Nine first-copy Ultimate cards, no dupe/material gate. Reject blanket 19-Ultimate authoring before role and enemy interactions are reviewed.	Future character-release plan needs a new mandatory role or alternate roster coverage.
3G-03	K-01 Ultimate retains Taunt; K-08 Ultimate retains Interrupt; K-10 long-term heal/shield does not supplant first-turn Shield at zero Chakra. Reject priority exception that silently favors Techniques.	A valid 100-Chakra defense case still causes the AI to skip the only legal counter.
3G-04	K-03 named ≤30% Execute; no unconditional 1.70× full-health hit. Reject hidden damage clamping.	Broader maximum legal buff/DEF calculations or future normalized roles violate no-one-shot rule.

Decision ID	Chosen working policy and rejected alternative	Reopen trigger
3G-05	Two Wounds on one victim share the per-slot cap proportionally before mitigation. Reject application-order bias.	Source-specific defensive mitigation or intelligibility tests justify aggregate or different rounding.
3G-06	Eight encounter table is an audit of written counter integrity, NOT completed whole-fight balance. Reject premature Priority 4 stage order.	Each threat has its full legal five-action, both-starter, no-positive-proc enemy/ally status and round ledger.

[NEXT RESEARCH — PRIORITY 3H: END-TO-END ENCOUNTER CERTIFICATES] First close B-E, D-E and S-E with full enemy cards and every living five-slot action across the fight, no omitted player turns; extend T-E/C-E to charged/status retry where applicable; apply new same-target Wound cap and full-meter AI conditions in the longest plausible encounters. Distinguish truly distinct counter categories and show failure/reversal branches. Defer Hall22–80 and Priority 4 campaign stage numbering until the first mandatory threat set has an honest gate-ready paper certificate. Recommended performance: EXTRA HIGH because a single Ultimate/cooldown/status decision may alter all eight access proofs. No implementation, executable simulator, seeded match, external player trial or real-world balance validation until the project owner explicitly authorizes it.

49.9 Relevant external design comparisons—limited evidence

[EXT-E22-1 | checked 1 November 2026] Riot Games, Teamfight Tactics patch 15.1 notes (2025), publishes role-specific Mana-generation changes and targeting tie criteria. Useful analogy: resource and target priorities belong in visible role/rule contracts. This is NOT evidence that our +25/+15/100 values are optimal. https://teamfighttactics.leagueoflegends.com/en-sg/news/game-updates/teamfight-tactics-patch-15-1-notes-2025/

[EXT-E22-2 | checked 1 November 2026] Riot Games, Teamfight Tactics patch 14.15 notes (2024), describes Mana overflow/carryover behavior and visual inspectability of PvE opponents. This highlights why our new 100-Chakra retention must be explicit; it does not dictate our independent design. https://teamfighttactics.leagueoflegends.com/en-us/news/game-updates/teamfight-tactics-patch-14-15-notes/

[EXT-E22-3 | checked 1 November 2026] Microsoft Xbox Accessibility Guideline 109, objective clarity, stresses that players can review the actions needed for progression and revisit instructional material. Supports a replayable “why didn’t my Ultimate fire?” recap and visible gates, not any battle coefficient or gameplay usability claim. https://learn.microsoft.com/en-us/gaming/accessibility/xbox-accessibility-guidelines/109

50. Priority 3H: complete burst, decoy and sustain encounter certificates

1 November 2026 | Live master v0.10.23 | Paper research only. Extends Sections 27 and 43–49 without altering the authoritative combat engine. Every stated action and integer hit below is a hand-authored deterministic ledger, not a programmed simulator, randomized trial, actual playtest, measured win rate, or certified fun/balance outcome. Replacement enemy cards are new proposals and do not erase older historical examples.

50.1 Why three previous "accessible" encounters were not complete

[AUDIT FINDING] The previous B-E enemy had only 1,850 HP, so the core K-03+K-19 Techniques alone delivered 1,905 and killed it in round 1; its telegraphed round-2 burst was not actually experienced. The previous S-E healer had only 2,400 HP; after K-12, K-03, K-19, the ordinary starter-support Basic and K-01 are included, its required first heal may never occur. The previous D-E access example did not count all normal Basics and prematurely assumed Techniques on an enemy with no frontline. Thus all three previously described counter *windows* were narrower than end-to-end fight certificates. This is a research correction, not retroactive balancing proof.

[INVARIANTS, UNCHANGED] START00 always grants K-01/K-03/K-12/K-19. START01 grants exactly K-09 OR K-10; STORY02 unlocks K-04 and STORY03 unlocks K-17 before their teaching gates. Hall20/Band I normalized proxies only. Starting personal Chakra 0; one normal action per living unit per round; frozen descending Speed order with round-side ties and slot breaks; only legal Ultimate→Technique→Basic; Technique +15, Basic +25, Ultimate −100; 2 subsequent own-slot

Technique cooldown; no criticals, gear bonuses, beneficial Mark rolls, summons or additional enemies unless declared. The battle ends as soon as the final enemy falls, so later queue entries are labeled ENDED, not incorrectly logged as voluntary skips.

[ATTACK RULE] In the following tables direct damage for attacker ATK A, coefficient c and defender DEF D is floor(A×c×1000/(1000+D)), with no other buckets. Against enemy DEF600: K-12 Technique292/Basic451, K-03 Technique1,093/Basic664, K-19 Technique812/Basic531, K-01 Basic425, K-09 or K-10 Basic425, K-04 row Technique237 and Basic504, and K-17 Technique292/Basic451. K-01 has 7,800 HP and 1,200 DEF. Each card explicitly states when K-01 Taunt or K-10 Shield has a legal upcoming threat; no manual Basic override is permitted.

ID	Affected historical card	Revised candidate, not an implicit original-rule amendment
B-E-v2	Old B-E: HP1,850 exposed charger, Speed118, 7,600 ATK	Set source HP to 2,500 (DEF600; Speed118; ATK7,600 unchanged). R1 telegraph R2 1.30× single-target burst aimed at K-03, Taunt-eligible; R2 resolves once if alive. No other attacks/adds.
D-E-v1 audited	Old D-E: HP1,450 rear source, two one-hit decoys, Speed85, ATK600	Retain HP1,450, DEF600, Speed85, two setup one-hit decoys (no re-summon); each surviving owner slot enemy Basic0.60× against K-01. No frontline enemy. Reject this as evidence of K-04 superiority.
S-E-v2	Old S-E: HP2,400, DEF600, ATK850, Speed90, self-heal1.20×	Replace with HP4,700, DEF600, ATK850, Speed89 (avoids unresolved Speed90 side-tie); Basic0.30× aimed at K-01 then self-heal1.20× ATK at end of own slot if alive, once/slot. No adds/shield or second action.

50.2 Burst B-E-v2: real protection window and early knockout are distinct

[PAPER CERTIFICATE INPUT] Enemy B-E-v2 source 2,500 HP/600 DEF/7,600 ATK/Speed118. R1 only previews R2 burst at K-03; R2 1.30×, eligible for K-01 Taunt. K-12 Scout Mark 80% roll is assigned its adverse FAILURE result. Both valid squads use front K-01/K-03 and rear K-12/K-19/START01. K-01 Speed90, K-03 Speed110, K-12 Speed115, K-19 Speed100, support Speed95. Enemy begins R1 at Speed118, R2 attack before player slots.

Step	START01 K-09: accessible focus	START01 K-10: protection then focus
R1 enemy Speed118	Telegraphs R2 K-03 burst; no damage.	Same telegraph; no damage.
R1 K-12 Speed115	Technique292, Mark FAIL; source2,500→2,208; Chakra15.	Same: 2,500→2,208; Chakra15.
R1 K-03 Speed110	Technique1,093; source2,208→1,115; Chakra15.	Same: 2,208→1,115; Chakra15.
R1 K-19 Speed100	Technique812; source1,115→303; Chakra15.	Same: 1,115→303; Chakra15.
R1 support Speed95	K-09: no Wound, Basic425; source303→KO, combat ENDS.	K-10: R2 threat to K-03 precedes K-03 next slot, so Shield K-03 for 612; Chakra15; source remains303.
R1 K-01 Speed90	ENDED: no enemy alive; no artificial Taunt/Basic action.	Visible R2 single-target hostile action is aimed at K-03; Taunt legal. K-01 Technique Taunt, Chakra15. Source303 remains alive.
R2 enemy Speed118	ENDED.	R2 burst redirects onto K-01. Damage floor(7,600×1.30×1000/2,200)=4,490; K-01 HP7,800→3,310. K-03 Shield612 remains unused (Taunt succeeded). K-01 earns +5 incoming direct-hit Chakra.
R2 first ally Speed115	ENDED.	K-12 Technique on cooldown; Basic451, source303→KO, combat ENDS. R2 others no action.
Verdict	One-round deterministic focus KO even with failed Mark, without K-09 cleanse, paid kit or Ultimate.	Two-round defensive KO; a dangerous 4,490 hit is survived by the guaranteed Guardian, no duplicate or lucky status.

[FAILURE / HONEST COST] In this exact auto-AI contract K-09's own R1 Basic causes the early KO; K-09 is NOT actively "choosing focus" manually. K-10's K-03 shield is paid as an action but is not consumed because the later K-01 Taunt redirects the burst. This redundant protection is a real cost of separate local predicates. An Ignore-Taunt tag or enemy front guard invalidates the defensive certificate; a faster follow-up burst after K-01 falls would require another response. The two paths share core damage but differ between finish-before-cast and survive-and-finish; they should not be portrayed as completely independent damage systems.

50.3 Decoy D-E: the original level is solvable, but it is not a K-04 showcase

[PAPER CERTIFICATE INPUT] Rear source 1,450 HP/DEF600/ATK600/Speed85. Two setup decoys each intercept the next eligible single-target hit once. They are not frontline enemies or independent action slots. K-03/K-19 Techniques require a frontline target and therefore fall back to Basic. K-12 Technique also needs a frontline target, so its Basic is compulsory here. K-04 Technique is row damage, unaffected by single-target decoys. Source Basic floor(600×0.60×1000/2,200)=163 to K-01 after K-01 R1 slot. K-10 has NO legal shield here: source strikes K-01 only after K-01 has completed its current slot, not before its next immediate R1 action. K-01 Taunt is also ineligible when enemy targets K-01 itself. Both supports therefore use Basic; no manual overrides.

Round / slot	With STORY02 K-04: K-01/K-03/K-19/K-04/START01	Without K-04: K-01/K-03/K-19/K-12/START01
R1 initial objects	Two one-hit decoys.	Same two decoys.
R1 first eligible attacker	K-03 Speed110 Basic hits decoy1. +25 Chakra, no HP source damage.	K-12 Speed115 Basic hits decoy1. +25 Chakra, no HP source damage.
R1 next attacker	K-19 Speed100 Basic hits decoy2. +25 Chakra, no HP source damage.	K-03 Speed110 Basic hits decoy2. +25 Chakra, no HP source damage.
R1 remaining attackers	K-04 Speed100 row Technique237 source→1,213 (+15 Chakra); support Speed95 Basic425→788; K-01 Speed90 Basic425→363.	K-19 Speed100 Basic531 source→919; support Speed95 Basic425→494; K-01 Speed90 Basic425→69.
R1 source Speed85	Basic163 to K-01: HP7,800→7,637; K-01 earns +5 direct-hit Chakra.	Same Basic163; K-01 HP7,637 and +5 incoming Chakra.
R2 earliest eligible allied attack	K-03 Speed110 Basic664 source363→KO; combat ENDS before K-19/K-04/support/K-01.	K-12 Speed115 Basic451 source69→KO; combat ENDS before K-03/K-19/support/K-01.
Certificate result	Two rounds; K-04 bypasses the decoys but does not shorten this completed fight.	Two rounds; guaranteed core consumes the decoys and finishes the source, no positive proc.

[DECISION 3H-D01] Keep D-E as a teaching encounter about bounded decoy interception and target eligibility, NOT a mandatory proof that the rewarded K-04 is faster. The useful early K-04 payoff is instead documented in the separate protected-rear multi-caster R-E case (§48.6), where row pressure reaches both rear targets. Rejected: artificially inflating K-04's coefficient or pretending a source-only fight has more enemies. The old §46.6 single-action comparison is superseded by this full five-member accounting for D-E only. Revisit if decoy objects gain renewed charges, a front guard, or independent actions; all would require a new complete certificate.

50.4 Sustain S-E-v2: full five-member focus and both anti-heal outcomes

[PAPER CERTIFICATE INPUT] New self-healer HP4,700 / DEF600 / ATK850 / Speed89. Once per living own normal slot it Basic-attacks K-01 for floor(850×0.30×1000/2,200)=115 then self-heals floor(850×1.20)=1,020 gross, reduced to 612 if K-17's 90% Healing Suppression SUCCESS and 1,020 if FAIL. No attacking/healing add or passive trigger. K-12 Mark is assigned FAIL. K-01 Taunt is ineligible (the hostile target is already K-01). K-10 Shield is ineligible (the K-01-targeted attack happens after K-01's R1 slot, and no ally is hit before their immediate next R1 slot); both support choices use ordinary Basic425. Player tie order sets K-03 before K-17 at Speed110, with both preceding K-19 Speed100, START01 Speed95, K-01 Speed90, enemy Speed89. Enemy R1 and R2 actions are fully priced.

Branch / round	Whole five-player legal action ledger; enemy actions	Enemy and K-01 HP after round

Branch / round	Whole five-player legal action ledger; enemy actions	Enemy and K-01 HP after round
Focus-only, R1 (K-12 Mark FAIL)	K-12 Tech292; K-03 Tech1,093; K-19 Tech812; START01 Basic425; K-01 Basic425 =3,047. Healer Basic115 to K-01; then heal1,020.	Healer4,700−3,047+1,020=2,673. K-01:7,800→7,685.
Focus-only, R2	K-12 Basic451; K-03 Basic664; K-19 Basic531; START01 Basic425; K-01 Basic425=2,496. Healer survives at177, then Basic115 and heal1,020.	Healer2,673−2,496+1,020=1,197. K-01:7,685→7,570.
Focus-only, R3	K-12 Basic451; K-03 Basic664; K-19 Basic531. Healer1,197→746→82→KO. START01 and K-01 do NOT act because combat ended.	WIN R3 before healer Speed89. K-01 HP7,570.
K-17 SUPPRESSION SUCCESS, R1	K-03 Tech1,093; K-17 Tech292 with 90% suppression SUCCESS; K-19 Tech812; START01 Basic425; K-01 Basic425=3,047. Healer Basic115 then heals612.	Healer4,700−3,047+612=2,265. K-01 HP7,685.
K-17 SUCCESS, R2	K-03 Basic664; K-17 Basic451; K-19 Basic531; START01 Basic425; K-01 Basic425=2,496. Healer2,265→1,601→1,150→619→194→KO before own Speed89 slot.	WIN R2. K-01 HP7,685, no R2 healer Basic.
K-17 SUPPRESSION FAIL, R1	Same 3,047 direct team damage; failed status does not lower the heal. Enemy Basic115 to K-01 and heals1,020.	Healer2,673. K-01 HP7,685.
K-17 FAIL, R2	Same 2,496 allied Basics: healer2,673→177; then enemy Basic115 and full heal1,020.	Healer1,197; K-01 HP7,570.
K-17 FAIL, R3	K-03 Basic664; K-17 Basic451; K-19 Basic531. Healer1,197→533→82→KO; later allies ENDED.	WIN R3 before enemy Speed89. K-01 HP7,570.

[DEDUCTIONS] With either START01 choice, all three declared branches preserve K-01 above 7,500 HP and remain under the 12-round cap. K-17's meaningful value is one less enemy Basic/heal cycle IF its 90% debuff succeeds: R2 clear rather than R3; failed status does not cause a mandatory loss. This is conditional arithmetic at the specified new 4,700-HP enemy, NOT a guaranteed 90% win-rate statement. The old 2,400-HP healer example remains historical but is not a valid introductory healing lesson after all five player actions are counted.

[AI / CHAKRA AUDIT] In S-E-v2, R1 damage characters use Techniques (+15), R2/R3 Basics (+25), and no character reaches 100 Chakra before battle end. Each enemy Basic yields K-01 +5 incoming direct-hit Chakra (maximum twice across R1/R2); K-01 uses Basic +25 each completed turn. K-10 and K-09 both use Basic because their Technique predicates are false here. Replacing the healer's self-heal with an ally heal, adding a shield, allowing a second free heal per slot or increasing enemy Speed back to 90 changes the record and requires a new gate review.

50.5 Initiative, cooldown, resource and worst-case certificate register

Requirement	Audit result for B-E-v2 / D-E / S-E-v2	Unresolved expansion or reversal
Five-slot actions	All legal living slots before battle end are listed. No dropped positive support/Guardian Basics; ENDED explicitly replaces future slots after last enemy KO.	Other five introductory threats retain separate §48–49 papers and are not re-certified by this chapter.
Negative proc branch	K-12 Mark fails in B/S; K-17 suppression FAIL included; K-09 vs K-10 both supported. Decoys and Taunt/Shield have declared deterministic rules.	Crits and 80%/90% status rolls are not probabilistic balance measurements. New enemy control RNG requires fresh branches.
Cooldowns	First-copy Techniques in R1; on two subsequent own slots R2 and R3, damage kits use Basics; no illegally refreshed Technique or manual Basic override.	R4 technique readiness and longer Ult interaction require a new complete long-fight example.

Requirement	Audit result for B-E-v2 / D-E / S-E-v2	Unresolved expansion or reversal
Chakra	B ends R1/R2; D ends R2; S ends R2/R3. Starting 0; completed action increments and K-01 incoming bonuses cannot make a 100-Chakra Ultimate fire in these particular certificates.	Cannot generalize from short fights to all 12-round boss or PvP encounters.
Survival and damage	B K-01 may take 4,490 and survives at3,310; D takes163 and survives at7,637; S takes 115 per surviving healer slot and remains≥7,570.	Other allies are not attacked by these specific cards; adding targeting or attack AoE invalidates certificate.
Accessible acquisition	B uses guaranteed START00 plus either START01, D no K-04 and K-04 alternatives, S uses START00 focus or STORY03 K-17 granted before its teaching gate. No seals or duplicate prerequisite.	Story timing and Hall/gear affordability remain assumptions requiring eventual campaign-stage budgets.
Encounter intent	B-E-v2 makes protective redirect observable only in K-10 branch; D demonstrates two consumable decoys but not faster AoE; S-E-v2 enables meaningful healer cast and status-failure fallback.	Not final enemy design, narrative quality, interface clarity or enjoyable encounter pacing.

50.6 Integrity: no silent changes to global rules, no invented production evidence

[CONTRADICTION LOG] Original B-E/S-E became short no-mechanic fights when the fifth ally and starter support acted. Revised *enemy* HP for the actual proposed lessons instead of nerfing accessible core skills, changing Chakra, forcing a voluntary Basic, or making a lucky Mark/Healing Suppression mandatory. D-E stayed numerically unchanged and was reclassified as a decoy tutorial with equally fast alternative routes. No all-eight-encounters-pass claim is made.

[SCOPE LIMIT] The three revised cards prove conditional, reproducible damage/HP/action consistency only. They do not establish actual Hall20 stat mapping from the complete economy, true gear tiers beyond the reference, conversion from normalized stats to account power, randomness across repeated battles, player comprehension, fairness across paid investment, server saving, or real gameplay success. For these entries status=DESK CERTIFICATE, not GAME VALIDATION.

50.7 Choices, rejected alternatives and precise next handoff

Decision	Selected working rule and trade-off	Reopen only if
3H-B01	B-E-v2 2,500-HP burst target, visible R2 cast; K-09 core focus ends R1, K-10/K-01 protection ends R2. Reject 1,850-HP source as a protection lesson.	New front guard, Ignore-Taunt, R1 chip damage, or a different intended support availability changes the two routes.
3H-D01	Keep existing D-E two one-hit decoys and 1,450-HP rear source as decoy rules lesson; K-04 is not asserted faster than K-12. Reject fake forced AoE win.	Two rear threats, decoy refreshes, different intercept priorities, or different auto predicates are actually authored.
3H-S01	New S-E-v2 self-healer 4,700 HP, Speed89: no-K-17 focus wins R3; K-17 success R2, miss R3. Reject 2,400-HP “healer encounter” and permanent 100% anti-heal.	Healing recipients change, healer Speed rises, new shields/control appear, or complete roster/gear inputs change.
3H-I01	Use complete pre-KO five-unit ledgers and enemy actions with worst-status branches. No skipped legal action to manufacture weakness.	A new authored kit changes a predicate or the encounter acquires additional enemies/effects.
3H-U01	Do not declare Priority 3 finished. Next audit must reconcile all eight threat certificates to one coherent chronological starter acquisition/campaign plan; no Hall22–80 curve is fabricated.	Owner authorizes new project scope or future validated balance data permits replacing provisional numbers.

[NEXT RESEARCH — PRIORITY 3I: EIGHT-THREAT CERTIFICATE RECONCILIATION AND GATE DEPENDENCIES] Bring W-E, C-E, T-E, R-E and WW-E under the same R1-to-KO ledger standard with both START01 choices, no positive status assumptions, full enemy action counters, cooldowns, defensive timing and all five living actions. Reconcile conditions for K-04 and K-08 story entitlement before their mandatory tests. Then write a provisional gate ordering and cheapest

obtainable Hall/gear floor *without* inventing the level22–80 curve; only afterwards begin Priority4A authored campaign chapters, bosses and reward sources. Recommended thinking level: EXTRA HIGH because the remaining encounters cross-check control timing, wound caps, interrupts, row protection, roster acquisition, idle economy and survival together. Still no executable simulation, implementation, seeded tests, player testing or actual balance validation.

[SOURCE BASIS] Internal canonical rules §§24.2–24.6, 27.2–27.7, 41–42 and 43–49 in this master. This is an internal contradiction/arithmetical audit with no new external factual claim; earlier cited primary-source design comparisons remain referenced in their original chapters. All enemy HP/Speed replacements are project proposals, not source-derived facts.

51. Priority 3I: eight-threat reconciliation, full opening ledgers and pre-gate access

1 November 2026 | Live master v0.10.24 | PAPER RESEARCH ONLY. This chapter consolidates W-E, C-E-v2, T-E, R-E-v2 and WW-E-v2 to the same five-slot and enemy-action closure standard as Section 50, then sequences all eight proposed mechanic gates against guaranteed roster entitlements. It does not create a campaign chapter, measured balance result, executable simulator, revised Hall22-80 economy, or player-test claim.

EVIDENCE: [PRIOR] = already written elsewhere; [ARITHMETIC] = transparent hand-accounted integer outcome at declared stat cards; [WORKING DECISION] = reversible choice; [CONDITIONAL] = validity requires exactly its stated enemies, targets, statuses and owned units; [OPEN] = not established. Historical earlier paper illustrations are retained but superseded for the specific numeric contracts listed here.

51.1 Cross-packet invariants: one authoritative opening state

[PRIOR] Hall20/Band-I stat proxy is NOT the complete level-1-to-20 progression curve. START00 gives K-01, K-03, K-12 and K-19; START01 grants exactly one K-09 OR K-10. K-08/04/17 are one-time persistent story first copies before their own teaching gates. The legal squad size stays five. Standard issue is included once in these stats and must not be added again. No seal, fragment, featured copy, duplicate, premium item or personal gear-fee is required for these opening responses.

[PRIOR] K-01 HP7,800 DEF1,200 ATK800 Spd90; K-03 HP5,100 DEF850 ATK1,250 Spd110; K-12 HP5,700 DEF950 ATK850 Spd115; K-19 HP6,000 DEF1,000 ATK1,000 Spd100; K-09/10 HP6,300 DEF1,000 ATK800 Spd95; K-08 HP5,700 DEF950 ATK850 Spd110; K-04 HP6,000 DEF950 ATK950 Spd100. All begin at 0 Chakra. Basic=0.85x ATK, +25 Chakra; eligible Technique=+15 Chakra and unavailable on the next two owner slots; Ultimate requires 100 available before its slot. Frozen Speed queue, normal Ultimate->eligible Technique->Basic, enemy and ally actions stop at last-enemy KO. All hits below are floor(ATK*coefficient*1000/(1000+defender DEF)), with no Crit, positive Mark, unlisted proc, additional enemy or unsupported action.

[REVIEW RULE] Every certificate must document at least five slots while the fight is alive or explicitly say ENDED after final KO. Enemy actions and status ticks happen at the specified slots. Favorable K-12 Mark is assumed to FAIL; enemy Silence and Wound application are examined at their adverse successful-application branch. Silence failure is a separate shorter branch, not a required win condition. Death/ownership/status chances are not turned into fabricated win rates.

51.2 W-E — Wound source and surviving raider: both START01 choices

[ENEMY] W source front HP1,850/DEF600/ATK900/Spd118 applies 0.45x to K-19 followed by 18%-ATK two-slot Wound in R1; raider front HP3,000/DEF600/ATK400/Spd80 uses 0.55x Basic against K-01 in each living owner slot. Both teams are K-01/K-03/K-12/K-19 plus either K-09 or K-10. K-12 Mark FAIL, Wound SUCCESS (adverse), no other enemy effects.

Window	All living action slots and enemy outcomes	Starter-choice difference / HP state
R1 hostile and first three allies	Source direct202 to K-19, then Wound pre-K-19 slot81; K-12 Technique292 + K-03 Technique1,093 + K-19 Technique812 =2,197, KOs source. K-19 HP6,000-202-81=5,717.	The enemy source can die but its already-applied Wound continues. K-12/K-03/K-19 each end R1 at15 Chakra.
R1 support / Guardian / raider	K-09 cleanses named Wound OR K-10 shields K-19 for720; K-01 Basic425 against raider3,000->2,575; raider Basic100 to K-01.	K-01 HP7,700; supports +15 Chakra; K-01 +25 Basic and +5 incoming direct hit. No extra action after source KO.
R2 entire team and surviving enemy	K-12 Basic451; K-03 Basic664; K-19 Basic531; support Basic425; K-01 Basic425 =2,496 to raider2,575->79. Raider Basic100 K-01.	K-09: removed second Wound81 before R2. K-10: Shield absorbs81 at K-19 R2 slot; both leave K-19 at5,717; K-01 ends7,600.
R3 first ally	K-12 Spd115 Basic451 KOs raider79; battle ENDED before K-03/K-19/support/K-01 or enemy actions.	No invented third Wound tick; both legal five-unit rosters clear in three rounds.

[CONDITIONAL] If Wound fails, K-09 may take a legal Basic against the raider and the fight can end earlier; the adverse successful Wound ledger above is the more demanding teaching branch. A new enemy Guard/Intercept, extra pre-support

hit, repeated Wound application or source respawn invalidates this certificate. The two actual answers are early elimination of the applier plus either one-source cleanse or timed shield mitigation; the same foundational attack package contributes to both.

51.3 C-E-v2 — Silence must deny a ready Technique, not an ability already on cooldown

[ENEMY] Front controller HP3,000/DEF600/ATK900/Spd118. R1 single hit0.30x to the clearly previewed FRONT-RIGHT occupant, then 80%-chance Silence lasting the target's next owner action. R2 if alive Basic0.85x aimed at K-01. There is no third-round enemy action in the actual two-round ledger. K-12 Mark FAIL. Same owned START00 plus either START01; support has no eligible Wound/Shield and therefore Basics425.

Route / branch	Exact living-slot math	Enemy / squad result
Keep K-03 front-right; Silence SUCCESS	Enemy R1 direct145 to K-03 then Silence. K-12 Tech292; K-03 Silenced Basic664 (not Technique); K-19 Tech812; support Basic425; K-01 Basic425 ->2,618.	Controller HP3,000->382 after R1; K-03 HP4,955. R2 controller Basic347 hits K-01 HP7,800->7,453; K-12 Basic451 KOs controller before other R2 allies act.
Move K-01 front-right; Silence SUCCESS	Enemy R1 direct122 to K-01; Silence restricts K-01 to Basic, already its legal offensive action. K-12 Tech292 + K-03 Tech1,093 + K-19 Tech812 + support Basic425 + K-01 Basic425 =3,047.	Controller KO in R1; K-01 HP7,678. Formation, not premium Silence immunity, preserves K-03 Technique.
K-03 stays front-right; Silence FAILS	Enemy R1 direct145 to K-03, but K-03 uses legal Tech1,093. Full R1 team contributes3,047 without Mark.	Controller KO in R1. Tutorial must teach anticipation even though 20% RNG branches skip the actual restriction.

[WORKING DECISION] Keep the R1 status probability 80% and require a conspicuous target-by-slot preview. A single run cannot guarantee that Silence will demonstrate; later UX design should provide a non-random pre-battle explanation/replay rather than secretly turn the live status into 100%. The follow-up controller Basic does not arrive in the reposition or failed-Silence branches. No manual Basic override was introduced.

51.4 T-E — charger: K-08 interrupts or guaranteed Guardian survives

[ENEMY] Single front charger HP3,450/DEF800/ATK3,600/Spd118. R1 enters visible Charging with no hit. R2 either 1.60x Taunt-eligible hit toward K-03 if not interrupted, or 0.85x Basic aimed K-01 if canceled. No add, repeated R2 charge or third action before KO. K-12 Mark FAIL when present. Same account at Hall20/Band I; K-08 already granted by STORY01 before the teaching gate.

Owned five-unit team	Round 1 exact actions and source HP	Round 2 and conditional result
K-01/K-03/K-19/K-08/K-09	K-08 interrupt Tech259; K-03 Tech972; K-19 Tech722; K-09 Basic377; K-01 Basic377 after charge canceled. Total2,707; HP3,450->743.	Enemy canceled-charge Basic1,390 to K-01 HP7,800->6,410. K-08 Basic401 then K-03 Basic590 KOs source; other R2 actors ENDED.
Same K-08 team, K-10 replaces K-09	After K-08 cancellation, K-10 no longer has a pending charged threat to shield, so legal Basic377; same R1 total2,707.	Same enemy Basic1,390 and early R2 KO. K-10 does not automatically consume a redundant defensive Technique.
K-01/K-03/K-19/K-12/K-10	K-12 Tech259; K-03 Tech972; K-19 Tech722 =1,953. K-10 shields intended K-03 for612, K-01 Taunts future charged hit. Enemy HP1,497.	Enemy charged hit2,618 redirects to K-01 HP5,182. K-12 Basic401 ->1,096; K-03 Basic590 ->506; K-19 Basic472 ->34; K-10 Basic377 KOs; K-01 ENDED.
Same no-K-08 team, K-09 replaces K-10	K-12 Tech259 + K-03 Tech972 + K-19 Tech722 + K-09 Basic377 =2,330, source HP1,120. K-01 Taunts.	R2 charged hit2,618 to K-01 HP5,182. K-12 Basic401 ->719; K-03 Basic590 ->129; K-19 Basic472 KOs; K-09/K-01 ENDED.

[COST / INVALIDATORS] K-08 consumes a five-slot place rather than being a sixth unit. Interruption cancels only the named cast, never Stuns or removes R2 Basic. Guardian protection takes 2,618 HP damage and can leave K-10's shield unused. An Ignore-Taunt tag, same-slot charge resolution, an extra enemy, unavailable STORY01 grant or an extra charged

recast requires a new authored certificate. Source and kit first-copy Ultimates cannot activate from the declared 0-Chakra opening.

51.5 R-E-v2 — front guard plus two rear casters: audit four legal squads

[ENEMY] Front guard HP2,700/DEF600/ATK600/Spd92, active Taunt and one single-target rear Interception, uses Basic0.60x on K-01 (163). Two rear casters each HP1,200/DEF600/ATK650/Spd85 and84, each Basic0.70x aimed K-03 (245 versus K-03), Taunt-eligible and redirected to K-01 for206 while K-01 Taunt charges remain. No reinforcements. STORY02 K-04 is granted before an encounter that asks for row attack. Compare K-04 versus K-12 with both START01 choices; K-12 Mark FAIL.

Team	R1 full living slots	R2 / R3, enemy damage and actual result
K-01/K-03/K-19/K-04/K-10	K-03 Tech1,093 + K-19 Tech812 -> guard795; K-04 row Tech237 to BOTH rear casters ->963 each; K-10 Shield K-03 for612; guard Basic163 to K-01; K-01 Taunt; two casters hit K-01 for206+206.	R2 K-03 Basic664 -> guard131; K-19 Basic531 KOs guard; K-04 Basic504 -> caster1 459; K-10 Basic425 ->34; K-01 Basic425 KOs caster1. Caster2 hits K-03 for245, absorbed by K-10 Shield. R3 K-03 Basic664 + K-19 Basic531 KOs caster2. WIN R3.
K-01/K-03/K-19/K-04/K-09	Same front initial1,905 and K-04 row237 each; K-09 Basic425 -> guard370; guard Basic163 K-01; K-01 Taunt; two caster redirects206+206 to K-01.	R2 K-03 Basic664 KOs guard; K-19 Basic531 -> caster1 432; K-04 Basic504 KOs caster1; K-09 Basic425 -> caster2 538; K-01 Basic425 -> caster2 113. Caster2 hits K-03 for245. R3 K-03 Basic664 KOs caster2. WIN R3.
K-01/K-03/K-19/K-12/K-10	K-12 Tech292 + K-03 Tech1,093 + K-19 Tech812 -> guard503; K-10 Shield K-03 for612; guard Basic163 K-01; K-01 Taunt; two caster redirects206+206 to K-01.	R2 K-12 Basic451 -> guard52; K-03 Basic664 KOs guard; K-19 Basic531 -> caster1 669; K-10 Basic425 ->244; K-01 Basic425 KOs caster1. Caster2 attacks K-03 for245 absorbed by shield. R3 K-12 Basic451 + K-03 Basic664 + K-19 Basic531 KOs caster2. WIN R3.
K-01/K-03/K-19/K-12/K-09	R1 K-12 Tech292 + K-03 Tech1,093 + K-19 Tech812 + K-09 Basic425 -> guard78; guard Basic163 K-01; K-01 Taunt; rear hits206+206 K-01.	R2 K-12 Basic451 KOs guard; K-03 Basic664 -> caster1 536; K-19 Basic531 -> caster1 5; K-09 Basic425 KOs caster1; K-01 Basic425 -> caster2 775. Caster2 hits K-03 for245. R3 K-12 Basic451 -> caster2 324; K-03 Basic664 KOs caster2. WIN R3.

[RECHECK] In all four lines, K-01 R1 receives 163+206+206=575 direct damage and ends at7,225 HP. In K-10 lines K-03 absorbs the later R2 caster2 attack into its 612 shield; in K-09 lines K-03 takes 245 HP damage and remains4,855. R2 K-01 uses Basic because its Taunt Technique is still on own-slot cooldown; it never refreshes Taunt for a free third redirection. No row attack is intercepted. The outcome is three rounds in all four chosen no-Mark, no-Crit branches: K-04 provides early TWO-target rear access but NOT a guaranteed speed advantage. Enemy first-caster choice defaults to caster1 on lowest slot to make its order reproducible.

51.6 WW-E-v2 — two Wound sources and a real second tick

[ENEMY] A front HP1,500/DEF600/ATK900/Spd118 directly hits K-19 for202 and applies a two-slot 18%-ATK Wound; B front HP1,900/DEF600/ATK850/Spd117 directly hits K-03 for206 and applies a two-slot 18%-ATK Wound. B if alive R2 Basics0.85x at K-01 for328. K-12 Mark FAIL. Both START01 choices use the same guaranteed four-core team; one-source-only cleanse remains a hard limit.

Window	Exact five-unit and enemy ledger	K-09 / K-10 impact

Window	Exact five-unit and enemy ledger	K-09 / K-10 impact
R1 two hostile sources	A hits K-19 202; K-19 first Wound81 before K-19 R1 slot. B hits K-03 206; K-03 first Wound82 before K-03 R1 slot. K-12 Tech292 + K-03 Tech1,093 + K-19 Tech812 target A; A KOs. K-01 Basic425 to B1,900->1,475.	K-03 HP5,100-206-82=4,812; K-19 HP6,000-202-81=5,717. K-09 cleanses ONE Wound on earlier next-acting K-03; K-10 shields K-03 for612; neither can prevent R1 already-paid ticks.
R2 before K-19 slot	B Speed117 Basic328 to K-01 HP7,800->7,472. K-12 Basic451 -> B1,024; K-03 next-slot Wound removed (K-09) or shield absorbs82 (K-10), then K-03 Basic664 -> B360.	K-03 remains4,812 HP in both routes. Both K-12/K-03 Techniques are on cooldown.
R2 completion	K-19 takes second tick81 from surviving A Wound, HP5,717->5,636, then Basic531 KOs B360. Fight ENDED before START01 or K-01 R2 turns.	B has no R3 cast; K-01 ends7,472. K-10 shield remains530 immediately after K-03 tick. K-09 cannot cleanse both victims. No third Wound tick.

[SCOPE] The §49 same-target simultaneous dual-Wound proportional cap is an authoritative candidate for *another* fight, not exercised here: A and B Wound DIFFERENT units. Their raw snapshots162 (K-19 20%-HP cap1,200) and153 (K-03 cap1,020) are individually below the pre-DEF cap. If both attack the same victim, recompute the §49 allocation, victim selection and cleanse priority; do not transplant this ledger.

51.7 Eight-threat consolidation: admissible responses, grant order and proof ceiling

Provisional relative gate (not a stage #)	First-copy access on entry	Two distinct advertised actions	Desk evidence / remaining constraint
G1 W-E: named Wound source	START00 + START01 choice	Eliminate applier; cleanse ONE Wound or shield next tick.	§51.2 three rounds, adverse applied Wound; no Hall21 purchase.
G2 B-E-v2: telegraphed burst	START00 + either START01	Finish before cast; Guardian redirect + survivable burst.	§50.2 R1/R2; K-10 branch pays redundant shield.
G3 C-E-v2: Silence	START00 + either START01	Reposition front-right sacrificial/low-dependency role; accept restriction and finish R2.	§51.3 80% apply and 20% fail branches; genuine UI preview needed.
G4 T-E: charging source	STORY01 grants K-08 before entry; START00/01 remain	Interrupt charge; Guardian Taunt/optional Shield survives.	§51.4 R2 KO; interrupted source still Basics.
G5 D-E: two decoys	START00/01 already sufficient; STORY02 K-04 granted before G5 so both response categories are owned	Consume two one-hit decoys; rear-row attack bypasses them.	§50.3 2-round both; no guaranteed K-04 speed win.
G6 R-E-v2: guard + rear casters	STORY02 already granted before G5; K-04 remains owned at G6; core K-12 always owned	Row pressure both rear; eliminate guard and reach rear later.	§51.5 3-round all four no-proc branches.
G7 WW-E-v2: dual Wound	START00 + both possible START01 choices	Eliminate two sources while surviving ticks; one-source cleanse or Shield.	§51.6 R2 KO; R1/R2 ticks counted; not simultaneous same-target.
G8 S-E-v2: self-healing enemy	STORY03 grants K-17 before entry; START00 fallback remains	Focus repeated direct damage; optional Healing Suppression shortens fight.	§50.4 R2 on suppression success, R3 failure/focus.

[PROVISIONAL ORDER] W-E, B-E, C-E, T-E, D-E, R-E, WW-E, S-E minimizes mechanic stacking: damage-over-time before two sources; burst protection before a telegraphed charge; single-choice control before row/guard complexity; anti-healing after sustained Basics are familiar. Since G5 now offers row attacks as an alternative to spending attacks on decoys, STORY02 must be granted before G5, not deferred until G6. The relative labels G1-G8 are research anchors ONLY, not approved chapter names, unlock stages, server timestamps or reward IDs. They can be separated by ordinary content; nothing forces eight battles consecutively at exactly level20.

[GRANT TRANSACTION] Before enabling G4, issue STORY01/K-08 exactly once; before G5 issue STORY02/K-04 (so two counter categories are already accessible for D-E and it remains usable at G6); before G8 issue STORY03/K-17. Before *every* entry, reconcile its required grant with inventory; if already owned, do not duplicate or award seals/fragments; if offline, grant deterministically upon return at first eligible visit. The button remains blocked only until the OWNED first copy is confirmed, never until a weekly selector, purchase, rank, guild membership or banner. STORY timing is gate-relative C_i <= G_i, not account-day-specific.

51.8 First-account and returner feasibility: distinguish proven from merely assumed

Account scenario	Guaranteed availability at each gate	Hall/gear and calendar inference	Open risk
Brand-new account completing tutorials	START00 four + one START01; STORY01/02/03 arrive before designated G4/G5/G8.	Fieldability at earned Hall20/Band I uses inherited levels, defining first-copy abilities and two standard slots. Actual number of hours/days from fresh account to Hall20 is NOT priced.	No calendar guarantee for G1-G8; tutorial investment may take time.
Mature account at earned level/band	Same once-only entitlements and legal five-unit rosters; no duplicate prerequisites.	Actual stats might differ from Hall20 proxy. A high-Hall fight needs a new envelope/gear budget, not copied Hall20 enemy HP.	No 22-80 curve or equivalent late gate certified.
Midweek or twice-weekly visitor	If account reaches gate on a visit, required story copy is granted immediately; no requirement to finish CORE02 for a role.	Section42 wallets remain valid at chosen fixed E/M; Hall21 only optional. Exact claim opportunity differs by visitation.	Stage travel time and tutorial completion unpriced.
Once-weekly or returning player	Persistent story grants do not expire; missing earlier grant resolves at next eligible gate visit.	Existing 24h full +48h quarter bank means 36 equivalent hours per weekly claim; E Hall21 optional affordable day7 from E checkpoint, M Hall81 day14 in that model.	Early Hall0-20 costs, calendar wall, inactive-user UI and catch-up still open.

[CHEAPEST LEGAL FLOOR vs PROVEN ATTAINABILITY] The lowest *published combat certificate* remains already-earned Hall20/cap20/Band I, first owned copy of each required free character, its defining base skill, and its included standard gear. No personal gear purchase, 100-fragment selector, featured guarantee, Hall21 promotion or optional weekly growth basket is a legal mandatory admission fee. This is NOT yet a fully priced newcomer journey to Hall20: the master only prices Hall20->21, Hall80->81 and selected fixed-stage earnings. To claim genuine F2P affordability from account creation, author the Hall0/1->20 upgrade ledger, checkpoint unlock rules and tutorial resource grants in a dedicated next pass.

[COUNTING INTEGRITY] START00/START01, STORY01/02/03, CORE01/02, SELECT01, IDLE01 and ordinary first-clears keep their prior unique-source identity; do NOT use a story grant as a 34-seal or 60-fragment core reward, and do NOT award three story units on every campaign reset. 56-day full-core fragments can reach a 100-fragment selector on the seventh fully completed weekly payout; the three essential story grants do not depend on that schedule. These are separate inventories and clocks.

51.9 Decisions, rejected alternatives, falsifiers and precise next handoff

Decision / current selection	Alternative not chosen and reason	Reopen condition
3I-01: Use actual living-slot ledgers and negative status branches.	Do not omit K-12, skip the support Basic or grant a post-KO action to manufacture threat.	Any first-copy kit/stat/action priority revision requires full packet reprice.
3I-02: Keep C-E-v2 front-right telegraphed control.	Do not guarantee Silence or teach by forcing an illegal Basic.	Unclear mobile target preview, changed control clock or player misunderstanding.
3I-03: Retain T-E guardian route independent of K-08.	Do not make interruption mandatory or the canceled charge a free Stun.	New Ignore-Taunt action, extra charger, faster recast, or KO to Guardian.

Decision / current selection	Alternative not chosen and reason	Reopen condition
3I-04: Keep protected rear as trade-off, not premium payoff.	Do not buff K-04 for spreadsheet round advantage or deny K-12 its existing early hit.	A later genuinely multi-target shield/healer produces a meaningful differentiated answer.
3I-05: Propose gate-relative STORY01/02/03 and G1-G8 teaching sequence.	Reject fixed-account-day mandatory gates, random recruitment, eight simultaneous grants and fake stage IDs.	Complete Hall1-20 and onboarding budgets show rewards/access arrive too late, or later mode reuses roles as pay gate.
3I-06: Mark Priority3 encounter rule coherence as conditional desk closure only.	Reject saying all-eight passed live testing or calling Hall20 attained automatically by day7.	A real historical gap, status counter contradiction or future engine/UX evidence falsifies a paper case.

[NEXT RESEARCH — PRIORITY 3J: FIRST ACCOUNT TO HALL20 + STORY-GRANT AFFORDABILITY] Price the currently missing Hall1->20/cap sequence and initial resource/reward ledger; compare first seven visits for once-weekly, twice-weekly and daily schedules, plus a return/midweek case. Determine which earliest story milestone can legally offer Hall20/Band I and both START01 alternatives, and whether any first-copy switch incurs a fee. If the budget cannot get a supported player to E0 without hidden rewards, reduce or move the E0 gate BEFORE authoring a full campaign. Then Priority 4A may define actual stage IDs, narrative goals, boss arcs and unique reward-source identities, with all eight conditional certificates still subject to future prototype simulation and live UX/balance evidence. RECOMMENDED LEVEL: EXTRA HIGH because a revised onboarding grant or Hall promotion can alter idle supply, roster promises, combat stat envelopes and first mandatory gate placement. No game code, executable simulator, seeded matches or playtesting until owner authorization.

[EVIDENCE AND SCOPE] This chapter uses the internal dated source of truth: §§24-27 combat mechanics, §§40-42 reward/account price candidates, §§43-49 first-copy entitlements and enemy cards, and §50 the three recent desk encounter certificates. No external game or source was used to claim these authored enemy stats or probabilities. Previous sourced comparisons remain intact in their own sections. Actual Naruto intellectual-property rights for commercial release remain unassumed.

52. Priority 3J: new-account Hall 1 to 20 affordability and tutorial source ledger

1 November 2026 | v0.10.25 | PRE-DEVELOPMENT RESEARCH ONLY. This packet closes a missing first-account currency route at the existing Hall20/Band I reference. It does not approve actual stages, stat growth, playtest performance, or a finished onboarding flow. All newly named grants and prices below are [WORKING PROPOSAL]; arithmetic derived from those proposals is [CALCULATED]. Existing §§24–27, 32, 40–42 and 43–51 remain the comparison baseline.

52.1 Problem and design constraints

[AUDIT] Previous manuscripts priced the step Hall20→21 and fixed-stage income but never stated what Hall1→20 costs or which first-account rewards finance it. Therefore the eight at-Hall20 encounter certificates could not prove a new free account would reach that baseline. The missing bridge cannot be repaired by assuming unspecified first-clear rewards, calendar streaks, random summons, a mandatory Hall21 payment, or unpriced returner compensation.

[PROJECT CONSTRAINTS] Keep the four START00 shinobi and one START01 K-09/K-10 choice; Hall20/cap20 and Band I baseline; two included standard gear slots per owned shinobi; 0 first-copy personal preparation costs; Section 32 bank (24h full +48h at 25%, then stop); Section 42 E rate (10 Ryo/20 EXP/2 essence/1 enhancement per hour) once E0 is cleared; and gate-relative STORY01/02/03 grants. Tutorial Ryo/EXP may not count toward seals, fragments, pity, GEAR01, or any other award.

52.2 Four level bands: exact 19 shared Hall purchases

[WORKING DECISION 3J-01] The Training Hall is created at level 1 with initial cap 20. Levels 2–20 are 19 shared, permanent account purchases funded only by Ryo and Shinobi EXP. No promotion essence, enhancement, copies, skill books, weekly

claim, or personal gear fee is charged before Hall20. The 20→21 step remains a distinct 300-Ryo, 720-EXP, 48-essence upgrade, which raises the cap to 25 under §42. This is a candidate novice schedule, not a demonstrated long-term curve.

Hall step	Training band	Ryo	EXP	Cumulative Ryo / EXP
1 → 2	A / introductory	8	12	8 / 12
2 → 3	A / introductory	8	12	16 / 24
3 → 4	A / introductory	8	12	24 / 36
4 → 5	A / introductory	8	12	32 / 48
5 → 6	B / formation	12	18	44 / 66
6 → 7	B / formation	12	18	56 / 84
7 → 8	B / formation	12	18	68 / 102
8 → 9	B / formation	12	18	80 / 120
9 → 10	B / formation	12	18	92 / 138
10 → 11	C / targeting	16	24	108 / 162
11 → 12	C / targeting	16	24	124 / 186
12 → 13	C / targeting	16	24	140 / 210
13 → 14	C / targeting	16	24	156 / 234
14 → 15	C / targeting	16	24	172 / 258
15 → 16	D / final training	20	30	192 / 288
16 → 17	D / final training	20	30	212 / 318
17 → 18	D / final training	20	30	232 / 348
18 → 19	D / final training	20	30	252 / 378
19 → 20	D / final training	20	30	272 / 408

[CALCULATED] Cost bands: Hall1→5 is 4×(8/12)=32 Ryo/48 EXP; 5→10 is 5×(12/18)=60/90; 10→15 is 5×(16/24)=80/120; 15→20 is 5×(20/30)=100/150. Grand total = 272 Ryo and 408 EXP, with zero essence/enhancement. Any later cap-at-5/10/15 mechanic would contradict this first-account baseline and must be explicitly re-priced, not smuggled in.

52.3 Four one-time onboarding grants, no fabricated first-clears

[WORKING DECISION 3J-02] Choose four durable, once-per-account, noncombat training checkpoints ONB01–ONB04. Each checkpoint visibly grants its listed Ryo and EXP, then teaches and enables purchasing its matching Hall band. They are available without PvP, guilds, advertisements, gacha, additional equipment or timed objectives. They can be resumed after logout. This is a deliberately guided first-session bridge; it does NOT imply the live user will enjoy four consecutive upgrades. A later user-study could replace parts with verified, affordable combat clears at earlier levels.

Unique ID	Guaranteed checkpoint	Grant Ryo / EXP	Band spend Ryo / EXP	Wallet Ryo / EXP	Hall result
ONB01	Open Training Hall; account created / START00 delivered	40/60	32/48	8/12	Hall 5
ONB02	Choose START01 flex; learn formation	56/84	60/90	4/6	Hall 10
ONB03	Review targeting preview; confirm five-slot roster	80/120	80/120	4/6	Hall 15
ONB04	Inspect Wound preview; complete preparation lesson	104/156	100/150	8/12	Hall 20

[CALCULATED] Total one-time ONB01–04 grant = 280 Ryo +420 EXP. Required Hall1→20 purchases =272 Ryo +408 EXP. After a clean first tutorial, player retains 8 Ryo and12 EXP. No idle claim is needed for the baseline. These four grant IDs are ADDITIONAL explicit inputs, not evidence that a previous reward table already financed early growth; earlier versions correctly marked this income missing.

[TRANSACTION POLICY] Reserve ONB identities once per account, with idempotent grant transactions and durable completion. On interruption, reopen the oldest unfinished step, show its resources and legal next upgrade, and never re-grant an already credited step. Unlock unrestricted optional currency sinks and the shop only once ONB04 is complete, preventing a newcomer from spending the tutorial budget elsewhere before Hall20. If the same hero is owned from recruitment, START00/START01/ STORY must reconcile the first copy without duplicate free pulls. This is a progression-safety constraint, not a license for misleading or hidden currencies.

52.4 E0 reward checkpoint and first Wound gate placement

[WORKING DECISION 3J-03] At the completion of ONB01, publish tutorial checkpoint E0 as an earned noncombat campaign/tutorial clearance and unlock ordinary IDLE01 at the already-defined early E hourly vector: (10 Ryo,20 EXP,2

essence,1 enhancement). E0 is the first eligible idle-rate floor. It requires neither winning a below-Hall20 fight nor earning a random shinobi. The account may continue ONB02–04 immediately without waiting for resource accrual. The first mandatory G1 Wound battle is offered only after ONB04 has credited Hall20/cap20, Band I standard issue, START00 and one START01 choice. E0 is NOT an approved narrative stage number or a disguised win against G1.

[FAIRNESS / ALTERNATIVE] Giving the E early fixed rate at ONB01 intentionally front-loads training value; it can reduce the importance of early idle accumulation and makes tutorial completion more directed. Reject the alternative of forcing a one-day/weekly idle claim before G1 merely to demonstrate idle mechanics: that creates a clock wall without a tactical purpose. Also reject requiring combat victories at Hall5/10/15 until actual sub-20 role stat packets and those fights are authored. Change this policy if onboarding evidence shows unengaging upgrades; the smallest safe revision would preserve the guaranteed resource budget while changing how its milestones are presented.

52.5 Idle-bank formula and first seven visits: four fully specified calendars

[BASELINE] Define H(Δ)=min(Δ,24)+0.25×min(max(Δ−24,0),48), with Δ hours since the last claim. For gaps >=72h, H=36 equivalent full-rate hours, not 72h. No claim time before E0 receives credit. The following tables begin just after ONB04 at tutorial day0, assuming E0 was activated earlier that same uninterrupted session, that the account has the stated 8/12 currency remainder, no event/first clear/weekly material rewards are credited, no later idle rate is unlocked, and no optional purchases occur. A calendar claim is available ONLY when the player returns and presses Collect.

Daily visits · visit	Day	Earned hours	Cumulative hours	Wallet Ryo	EXP	Ess.	Enh.
1	1	24.0	24.0	248	492	48	24
2	2	24.0	48.0	488	972	96	48
3	3	24.0	72.0	728	1452	144	72
4	4	24.0	96.0	968	1932	192	96
5	5	24.0	120.0	1208	2412	240	120
6	6	24.0	144.0	1448	2892	288	144
7	7	24.0	168.0	1688	3372	336	168
Two visits/week (72h/96h gaps) · visit	**Day**	**Earned hours**	**Cumulative hours**	**Wallet Ryo**	**EXP**	**Ess.**	**Enh.**
1	3	36.0	36.0	368	732	72	36
2	7	36.0	72.0	728	1452	144	72
3	10	36.0	108.0	1088	2172	216	108
4	14	36.0	144.0	1448	2892	288	144
5	17	36.0	180.0	1808	3612	360	180
6	21	36.0	216.0	2168	4332	432	216
7	24	36.0	252.0	2528	5052	504	252
Once-weekly visits · visit	**Day**	**Earned hours**	**Cumulative hours**	**Wallet Ryo**	**EXP**	**Ess.**	**Enh.**
1	7	36.0	36.0	368	732	72	36
2	14	36.0	72.0	728	1452	144	72
3	21	36.0	108.0	1088	2172	216	108
4	28	36.0	144.0	1448	2892	288	144
5	35	36.0	180.0	1808	3612	360	180
6	42	36.0	216.0	2168	4332	432	216
7	49	36.0	252.0	2528	5052	504	252
Midweek start (48h then 96h/72h) · visit	**Day**	**Earned hours**	**Cumulative hours**	**Wallet Ryo**	**EXP**	**Ess.**	**Enh.**
1	2	30.0	30.0	308	612	60	30
2	6	36.0	66.0	668	1332	132	66
3	9	36.0	102.0	1028	2052	204	102
4	13	36.0	138.0	1388	2772	276	138

Daily visits · visit	Day	Earned hours	Cumulative hours	Wallet Ryo	EXP	Ess.	Enh.
5	16	36.0	174.0	1748	3492	348	174
6	20	36.0	210.0	2108	4212	420	210
7	23	36.0	246.0	2468	4932	492	246

Account visit pattern	First claim day	7th claim day	Full-rate-equivalent hours	Wallet after 7 visits Ryo/EXP/Ess/Enh	Earliest optional Hall21
Daily visits	1	7	168.0	1688/3372/336/168	day 2
Two visits/week (72h/96h gaps)	3	24	252.0	2528/5052/504/252	day 3
Once-weekly visits	7	49	252.0	2528/5052/504/252	day 7
Midweek start (48h then 96h/72h)	2	23	246.0	2468/4932/492/246	day 6

[CALCULATED DAY-7 SAME-CALENDAR CHECK] Daily (7 claims) earns168 full-rate-equivalent hours by day7: wallet1,688 Ryo/3,372 EXP/336 essence/168 enhancement. Twice-weekly (days3 and7) earns72h: wallet728/1,452/144/72. Once-weekly (day7 only) earns36h: wallet368/732/72/36. This is a substantial speed gap, not equal progression. Seven *visits* of weekly and twice-weekly schedules both collect252 equivalent hours but represent very different elapsed calendar time (49 vs24 days); never compare those rows as equal-duration outcomes.

[CALCULATED] Under no optional spending, Hall20→21 is first affordable at collected visits day2 (daily), day3 (twice-weekly), day7 (weekly), and day6 (midweek). The midweek day2 wallet is 308 Ryo/612 EXP/60 essence: Ryo and essence pass, but EXP fails. On day6, the next 96-hour-gap claim adds 360 Ryo/720 EXP/72 essence, so all three requirements pass. Check the slowest required input, not only the Ryo threshold.

[COMPARABILITY] Section42's zero-wallet early E cases were designed after an empty claim; this new ledger begins with the proven 8-Ryo/12-EXP tutorial remainder. Therefore post-tutorial wallets differ by exactly8/12 from that historical opening, but the earliest purchased Hall21 visit is unchanged for the three common schedules. The illustrative §42 other-core basket (420 Ryo/240 EXP/24 enhancement) remains OPTIONAL, not a debt or gate; charging it would require an explicit separate spending-path calculation.

52.6 Returning, interrupted and incomplete-training cases

Case	Guaranteed owner state and claim	Funding result	Remaining risk
Stops after ONB02 at Hall10	ONB01–02 claimed once. ONB03/04 remain persistent; E0 idle already unlocked at ONB01.	Resume ONB03/04 and train through20 from guaranteed later grants; idle may add funds but is not required.	No actual combat below Hall20 is assumed; restore resumable tutorial UI.
Hall20 account away 14 days	Same four core plus starter support; no repeated ONB or START grant. At return, 14-day bank is capped at36h.	Earn360 Ryo/720 EXP/72 essence/36 enhancement + tutorial leftover8/12; can buy optional Hall21 then.	No retroactive 14-day income, return bonus or reset of missed rewards.
Hall20 account away 28 days	Persistent owned roster, Hall level and gear unchanged; ONB and STORY grants stay unique.	Same single bank cap36h, regardless of28 days idle. Upon return, claim and resume progression.	Returner-specific RET01 still unpriced, not silently credited.
Player joins midweek	ONB grants and starter choice independent of weekly reset. E0 unlock at ONB01; tutorials can finish same session.	First claims after2/6/9 days show distinct cash clocks; no CORE02 qualification needed for G1.	Weekly seal/fragment schedule remains separately governed by §40.
G4/G5/G8 unlocked after a long absence	STORY01/02/03 are issued once and before corresponding fight, upon eligible return.	Zero cost for owned first-copy deployment; Hall/band at time of fight must be audited separately.	Do not count duplicate summon or fragments if story unit already owned.

52.7 Gate certificate, accounting controls, exclusions and falsifiers

[CONDITIONAL EARLIEST ACCESS] Given each ONB step is claimable without combat and has no timer, a player can reach the Hall20/cap20/Band-I paper baseline and enter the first G1 Wound lesson within the same completed introductory visit (day0) under this proposal. G2–G8 still have only relative gate order; nothing here assigns their actual chapter numbers, completion dates, enemy stat growth, resource costs, new story reward vectors, or player success. START01 K-09 and K-10 remain mutually exclusive one-time choices, but BOTH first-encounter response paths have the existing §51 certificates. Switching owned shinobi has 0 personal preparation cost; included standard-issue gear is not a recurring claim.

[ACCOUNTING INVARIANTS] ONB01/02/03/04 are each first-account Ryo/EXP grants only. START00, START01, FIRST01, FIRST02, STORY01/02/03, CORE01/02, IDLE01, GEAR01, MAT01, SELECT01 and RET01 retain distinct source/transaction identities. ONB grants do not produce fragments, seals, gems, skill materials or duplicate hero copies. One grant receipt + one spend per level step; retry is idempotent; spending cannot produce a negative balance. Tutorial E0 tier never retroactively increases pre-unlock time; no superposed duplicate IDLE01 accrual. A weekly reset never replays ONB or STORY. This model still needs eventual backend-save integrity design.

[STOP CONDITIONS / REOPEN] Reopen if any ONB step requires a Hall level higher than its already-funded previous band; the shop opens before ONB04, exposing new-player spending regret; first Wound requires a paid personal gear/counter; actual tutorial includes below-Hall20 fights; E0 does not count as a legitimate best-cleared checkpoint under later narrative design; ONB grants collide with pre-existing FIRST01/FIRST02 resources; or an actual user cannot follow four noncombat purchase cycles. Under any of these, author the revised price+source ledger and rerun all four calendars BEFORE approving campaign entry stages.

[EXPLICIT LIMIT] The 1→20 currency route is now closed under chosen paper assumptions, not the level1–19 combat stat curve. Later chapters still need the per-level HP/ATK/DEF/Speed map, tutorial UX and low-level opponent models if any such encounters are introduced, the exact story-stage ID/rate graph after E0, optional-spend trajectories, real time-to-upgrade usability, long-run F2P catch-up, and empirical onboarding satisfaction. This study does not justify saying early progression is fun, tested or optimally paced.

52.8 Decision record and next research level

Candidate	Disposition and justification	Cost / sensitivity / reversal
Wait for idle to fund Hall20	REJECT: calendar wall before first meaningful battle, with 1.5 full-rate-day weekly collections.	May lengthen tutorial; no guaranteed same-session G1.
Unpriced first-clear windfall	REJECT: no sub20 combat or first-clear ledger exists; circular funding.	Would conceal a mandatory resource prerequisite.
Lower required first gate below Hall20	RESERVE if four training grants fail UX; avoids full-band tutorial but requires fresh stats and eight desk ledgers.	A larger combat rewrite than funding one shared baseline.
Four bound noncombat grants and cap20	SELECT working baseline. Exactly priced, resumable, independent of attendance/pulls.	More guided early flow; grants deliberately front-load 280/420 and shop lock reduces freedom.
E0 tutorial idle unlock	SELECT early fixed tier at ONB01; unlocks idle promise before training is finished.	Pre-combat income exists; must validate this does not distort long-run supply.

[NEXT — PRIORITY 4A: CAMPAIGN CONTENT GRAPH AND REWARD-ID LEDGER] With the first-account Hall20 funding bridge explicitly priced, prepare a *paper* chapter-and-stage skeleton that maps G1–G8 into actual campaign checkpoints, story order, unique first-clear and repeatable reward IDs, idle-tier transitions, unlock prerequisites and solo-mode boundaries. Before finalizing a low-level fight or any Hall21+ mandatory gate, return to §52.7 exclusions and author its new stat/affordability certificate. RECOMMENDED RESEARCH LEVEL: EXTRA HIGH because even a narrative stage reward can affect the global economy and roster grant clocks. Preserve all unresolved design/validation cases. No executable prototype, simulation, or player testing is authorized.

[EVIDENCE] Calculations are internal design arithmetic grounded in §§32, 40–42 and 43–51 (master saved 1 November 2026). No external game claims validate these authored amounts, and no external data proves optimal onboarding. This packet introduces a proposed numeric training cost/grant schedule and explicitly supersedes §51.8's statement that Hall1→20 was unpriced; prior historical paragraphs stay in the archive as dated research.

1 November 2026 | v0.10.26 | RESEARCH / WORKING DESIGN. This packet converts the relative G1–G8 teaching order from §51 into a single traversable campaign candidate without changing combat coefficients, earnable Hall20/Band-I access or the previously priced onboarding economy. [PROPOSAL] identifies chosen candidate rules; [CALCULATED] identifies arithmetic and ordering deductions; [OPEN] identifies values excluded from numeric F2P claims. No playable content, simulation, remote service, or player test exists.

53. Priority 4A: campaign stage graph, unique reward identities and idle-tier boundaries

53.1 Scope, upstream authority, and explicit non-goals

[UPSTREAM AUTHORITY] §23 continuity and project constraints; §§24–27 normal round/auto/target/stat rules; §32 24h full + 48h quarter-rate idle bank; §40 unique reward definitions and FIRST01/FIRST02/CORE01/CORE02; §41 no gear-tax on owned characters; §42 fixed early E idle vector (10 Ryo/20 EXP/2 essence/1 enhancement per full-rate hour); §§43–51 eight conditional Hall20/Band-I enemy certificates and gate-relative story grants; §52 ONB01–04 total 280 Ryo/420 EXP funding 272 Ryo/408 EXP Hall1→20. The HALL-20 fights are paper reference packets, not real measured difficulty or approved battles.

[OBJECTIVE] Produce precise stage IDs, node type, prerequisite edge, legal roster-grant timing, unique first-clear identities, first-chapter reward boundary, replay rules, and rate-change conditions. The graph is a candidate teaching arc, NOT a complete narrative script, art plan, chapter duration, later Hall progression curve, or guarantee a new account clears eight fights in one session.

[SELECTED SCOPE] Three short chapters, 15 linear nodes: 8 previously examined mandatory battle nodes and 7 zero-combat narrative/preparation nodes. The graph deliberately adds NO unpriced mandatory fight before or between G1–G8. It supports stage replay after a first clear and allows players to stop between any two nodes; neither optional PvP/guild involvement nor extra recruitment unlocks a mandatory edge. All eight reference fights retain an already-earned Hall20/cap20/Band-I admissible baseline. Do not call an extra battle balanced by copying Hall20 numbers without a full new certificate.

53.2 Exact campaign node graph and first-clear identity

[CANONICAL GRAPH] Account created → START00 four guaranteed first copies → START01 K-09 OR K-10 → four ONB tutorial steps (E0 after ONB01; Hall20 after ONB04; FIRST01 on tutorial completion) → C01-01 (G1) → … → C03-04 (G8). Every row below is a proposed immutable identity. Predecessor means a FIRST CLEAR prerequisite, never a demand for repeated farming or a specific paid shinobi. SCN-* nodes are noncombat scene completion records; FC-* nodes are battle first-clear records. None of these completion IDs themselves pay an unpriced currency.

Stage ID	Node / mechanic	Required prior state	Unique completion and grants
C01-01	G1 • Wound (battle)	ONB04 + Hall20/Band I	FC-C01-01 • first-clear progress; no extra priced bundle
C01-02	Wound debrief (scene)	C01-01	SCN-C01-02 • story progress only
C01-03	G2 • Burst (battle)	C01-02	FC-C01-03 • first-clear progress
C01-04	Formation briefing (scene)	C01-03	SCN-C01-04 • targeting preview; no spendables
C01-05	G3 • Silence (battle); Chapter 1 complete	C01-04	FC-C01-05 + FIRST02 (5 seals, once only)
C02-01	Disruption recruit (scene)	C01-05	SCN-C02-01 + STORY01/K-08 once
C02-02	G4 • Charged attack (battle)	C02-01; K-08 grant reconciled	FC-C02-02 • first-clear progress
C02-03	Area-recruit briefing (scene)	C02-02	SCN-C02-03 + STORY02/K-04 once
C02-04	G5 • Decoys (battle)	C02-03; K-04 grant reconciled	FC-C02-04 • first-clear progress
C02-05	Backline tactics (scene)	C02-04	SCN-C02-05 • no spendables
C02-06	G6 • Protected rear (battle)	C02-05	FC-C02-06 • first-clear progress
C03-01	Attrition forecast (scene)	C02-06	SCN-C03-01 • status-preview lesson
C03-02	G7 • Multiple Wounds (battle)	C03-01	FC-C03-02 • first-clear progress

Stage ID	Node / mechanic	Required prior state	Unique completion and grants
C03-03	Healing-control recruit (scene)	C03-02	SCN-C03-03 + STORY03/K-17 once
C03-04	G8 • Sustain (battle); Chapter 3 capstone	C03-03; K-17 grant reconciled	FC-C03-04 • first-clear progress; no priced boss bonus

[GRAPH INVARIANT] Exactly 8 battle cards (G1–G8), 7 noncombat nodes, 14 directed predecessor edges. Chapter 1 ends at C01-05, Chapter 2 at C02-06, and Chapter 3 at C03-04. STORY01 arrives before G4, STORY02 arrives before G5, and STORY03 before G8. G1 has no preceding campaign battle. Completion state of any scene or battle is persistent; replay never traverses a first-clear grant a second time.

53.3 Reward-definition crosswalk: no double credits, no invisible money

Source definition	Trigger / category	Exact grant or current budget treatment	Unique counting rule
ONB01–04	Four tutorial checkpoints	280 Ryo + 420 EXP total; spend 272/408 on Hall1→20	Once per account; never from FC IDs.
E0 / IDLE01	E0 at ONB01; claim accumulated idle intervals	Early E rate 10 Ryo/20 EXP/2 essence/1 enhancement per effective hour; no higher rate from G1–G8	One time interval claimed once; no retroactive tier changes.
START00/START01	Introductory roster ownership	Four fixed first copies; ONE K-09 or K-10 choice	First copy reconciled, not a seal or fragment.
FIRST01	ONB04 tutorial-completion event	5 flexible Recruitment Seals	Exactly one entitlement, not paid again at G1.
FIRST02	C01-05 Chapter 1 completion	5 flexible Recruitment Seals	One entitlement, NOT a second FC-C01-05 payout.
STORY01/02/03	C02-01/C02-03/C03-03 scenes	K-08/K-04/K-17 first copies, respectively	If already owned, resolve entitlement, give no duplicate, seals or shards.
FC-C* / SCN-C*	8 battle or 7 scene first-completion events	Persistent progress/unlock only; incremental material reward OPEN and 0 assumed in all current budgets	No recurring calendar or replay credit; later material grant requires a new published revision.
CORE01/CORE02	Distinct qualifying core visits per week	Existing 2+6/7/6/7 seals and 10+5 fragments by earned week	Stage victory may qualify a visit; same weekly entitlement not minted per victory.
GEAR01/MAT01	Separate weekly eligible entitlements	Standard basic piece policy / material vector partially OPEN	Never automatically paid again by C nodes.
PVE01/PVP01/SOC01/EVT01	Future optional repeatables	Amounts, eligibility and reset OPEN	All excluded from mandatory F2P path.

[NUMERICAL CHECK] The one-time introductory seal candidate is FIRST01 5 + FIRST02 5 = 10 flexible seals if both milestones are completed. No other fixed seal is granted by C01–C03 stages in this packet. These ten seals are separate from CORE01/02 but cannot be counted until their own conditions occur. If a player completes both milestones in the first session, that is still not evidence they completed CORE01, CORE02, or a featured guarantee. The 34 seals/60 fragments per 28-day full-core candidate remains exactly the same (subject to existing weekly eligibility).

[REWARD HONESTY] A first-clear record is not itself a paid currency grant. Stage rewards not specified here are OPEN, not assumed 0 in the envisioned finished game; they carry zero credit in these paper budgets. The next research must select meaningful nonessential first-clear value and deduct its opportunity cost against Hall21+, first-chapter seals, equipment, and repeatable mode rewards. Do not silently invent eight stages × X Ryo, claim a free featured guarantee, or assume every cleared node increases idle earnings.

53.4 Rate graph, unlock gates and what cannot be extrapolated

[RATE POLICY — SELECTED] E0 at ONB01 is the only new-account idle-tier unlock supported by a priced rate in the research to date. G1–G8 do not advance IDLE01 automatically, and nothing in this chapter creates E1, E2 or an M checkpoint. The §42 mature M reference rate is an independently defined scenario, not a reward for clearing C03-04. If later design introduces an idle-tier step at a boss/stage, its exact four-resource hourly vector, effective timestamp, prerequisite Hall affordability, and all four claim calendars must be published before that stage becomes an authoritative earning tier.

[RATE INTEGRITY] On any future tier change, segmented accrual applies the rate in force during each eligible interval; a newly cleared node cannot retroactively upgrade already elapsed offline time. No replay reopens previous intervals; no offline accrual before E0. Tier ownership is monotonic, but the amount claimable is capped by §32 storage rules at each interval. Rate progress and actual earning are separate bookkeeping objects.

[MODE BOUNDARIES] CORE01/02 eligibility opens after the completed tutorial as in §40. GEAR01 and MAT01 keep their own existing published prerequisites, not implied bonuses for Chapter 1. Chapter 1 completion may expose a *non-rewarding practice/replay entry* in the navigation as a WORKING UX proposal; it does not fund seals, PvP, guild participation, ranked matchmaking, energy purchases or a new currency. Separate mode-cost and reward ledgers must be written before launching a repeatable trial, boss farm or other solo activity.

53.5 Eight-encounter placement audit and affordability certificate

Campaign gates	Counter possession and accessible solution	Earned level / equipment	Certificate boundary
C01-01,03,05: G1–G3	START00 + either START01; focus, guardian/formation or cleanse/shield	Already-earned Hall20, standard Band I, 0 personal fee	Existing §51/§50 paper packets; no Hall21 spend.
C02-02: G4	STORY01/K-08 before entry; guardian alternative remains	Same Hall20/Band I; included first-copy kit	§51 charged attack two-response packet.
C02-04,06: G5/G6	STORY02/K-04 before G5; decoy consumption / focus alternative	Same Hall20/Band I	§50–51 both routes work; no forced faster K-04 win.
C03-02: G7	START00 and chosen support; eliminate both wounders / mitigate	Same Hall20/Band I	§51 multi-Wound negative branch calculated.
C03-04: G8	STORY03/K-17 before entry; core focus works if debuff fails	Same Hall20/Band I	§50 sustain alternate 2/3-round packet.

[STRICT CLAIM] The graph is reachable without *additional* mandatory growth after ONB04 under the chosen Hall20 reference and each past paper packet. This is an affordability/legality argument only. It does NOT prove that 15 nodes can be completed in a short first session, that all eight battles are balanced with a permanent flat stat baseline, or that the same player appreciates zero intermediate numeric first-clear rewards. We intentionally do not invent Hall22–80 character scaling, staged enemy inflation, a boss HP multiplier, or a grind requirement to address those unknowns.

[FAILURE CASES] (i) If C01-01 is accessible before Hall20, the first encounter lacks a valid baseline: block until ONB04. (ii) If STORY02 grants K-04 at C02-06, G5's advertised area counter is unowned: grant at C02-03. (iii) If a first-clear reward and FIRST02 are two separate displayed envelopes of the same 5 seals, prevent the second monetary grant. (iv) If a G8 story grant arrives after battle entry, defer entry only long enough to confirm ownership, not until a weekly reset. (v) If C02-06 difficulty silently scales with narrative chapter number, the §51 Hall20 certificate no longer applies; request new enemy ledger before publish.

53.6 Navigation, comeback state, repeat rules, and alternate player paths

[ENTRY/UI CONTRACT] Show current chapter, single next node, threat preview for each battle, two accessible suggested answers, current Hall and standard equipment, and missing requirement with its *guaranteed* source. Show a completion stamp separately from claimable reward. On story nodes with free recruitment, present the granted character and an immediate team-swap option; do not require consuming extra resources. Replay buttons appear only after a node is cleared and do not award core currency or weekly rewards twice.

[RESUME CONTRACT] Persist the highest cleared node, every individual FC/SCN occurrence, START choice, STORY grant receipts, FIRST01/FIRST02 receipts, current Hall/personal gear and last IDLE01 claim. An interrupted clear resolves server-side once, then may be viewed again without duplicating a reward. The next stage is visible after all required grants and predecessor clear records are committed. A later patch may reassign a displayed story label but must not recycle the immutable ID for a different entitlement.

[SEGMENTS] Daily, twice-weekly, weekly and returning players see the same mandatory graph and grant order, because G1–G8 contain no calendar-locked prerequisite. A midweek player can still complete ONB01–04 outside the weekly reset. A lapsed account retains prior first clears and unused START/STORY entitlements; their status is resolved on return. A premium buyer may own STORY units early, but does not gain a second first copy on the grant scene or skip encounter mechanics. This is paper accessibility, not an estimated playtime.

53.7 Decision register: rejected alternatives, costs and reversal conditions

Decision and selected candidate	Rejected alternative and reason	Trade-off / condition to reopen
4A-01: 15 nodes; 8 mandatory certified battles + 7 noncombat scenes.	Eight battles consecutively lacks context; filler encounters would require unpriced combat certificates.	May feel text-heavy; revisit with mobile UX research and story pacing evidence.
4A-02: 3 chapters, FIRST02 only at C01-05.	Pay 5 seals at every chapter end: would invent +10 seals and change guarantee clocks.	Chapters 2–3 need meaningful but priced closure rewards later.
4A-03: fixed E0 rate throughout this teaching arc.	Automatic idle-rate increases every boss: no priced income tier or affordability check.	Flat rate may damp progression; reopen with complete future tier-vector model.
4A-04: gate-relative story grants at separate scene nodes.	Random gacha/selector before necessary counter: luck/time wall.	Extra owned units may reduce gacha desire; compensate with differentiated roster breadth, not paywall.
4A-05: stage FC identities progress-only for numeric budgets.	Invent a currency chest before sinks and economy are priced.	Weak reward salience; prioritize FIRST CLEAR reward pricing in 4B/4C.
4A-06: G8 acts as mechanics capstone only.	Invent a fully new boss/raid layer and call it campaign-ready.	Needs authored boss telegraphs, narrative stakes, UI accessibility and full survival audit.

53.8 External references, integrity warning and next handoff

[SOURCE ANALOGY, not a claim about our game] Farlight Games, “Talk with Designers: New Season Preview”, indexed official developer account (accessed 1 November 2026), describes moving some AFK-stage first-clear rewards to a handbook and making equipment rewards more selectable. Design implication: separate an earned encounter clear from the *presentation and transaction* of its one-time reward. We do not infer its numeric rewards or copy its season structure. https://afkjourney.farlightgames.com/news/a0ae0b8ad8c35da97805079b672b7287

[SOURCE — CURRENT SERVICE STATUS] Unity official Economy documentation, accessed 1 November 2026, states Economy stopped accepting new project sign-ups on 8 September 2026; Unity now points new projects toward Cloud Code, Cloud Save and Remote Config for server-authoritative economy functions. Do NOT treat historical §40 Unity Economy examples as an endorsed selectable backend for this unbuilt game. This is a technology/status note only, not an engine or service decision. https://docs.unity.com/en-us/economy

[SOURCE — ACCOUNTING PATTERN] Microsoft PlayFab official developer documentation, “Idempotent transactions and retry patterns in Economy v2” (updated 15 April 2026; accessed 1 November 2026), recommends deterministic quest-plus-player transaction identity for server-side quest grants. This supports a stable first-clear occurrence key, not a claim our project uses PlayFab. https://learn.microsoft.com/en-us/gaming/playfab/economy-monetization/economy-v2/tutorials/idempotent-transactions-and-retries

[LIMITS] These sources inform presentation and transaction integrity only. All stage IDs, chapter partitions, grant placement, tentative story names, failure cases and budget treatment above are ORIGINAL PROJECT PROPOSALS. No production game engine, backend, content asset, actual player account, simulation or human test was used. Nothing in this section grants rights to commercial Naruto IP.

[NEXT — PRIORITY 4B: CAPSTONE BOSS AND OPTIONAL SOLO MODES] Author G8 as a teaching capstone with legitimate telegraph, multi-round fail branches and distinct accessible answers; define one optional solo mode with one narrowly defined reward purpose and transparent entry/reset costs. First check whether the fifteen-node arc is too many taps and whether zero interim material payout is sufficiently motivating; then price any added first-clear payout against §§40–42 before it is published. Do NOT introduce a second source of weekly seals, paywall counter, unmodeled Hall21+ mandatory fight or new idle tier without a full recalculation. RECOMMENDED PERFORMANCE LEVEL: EXTRA HIGH because one boss or repeatable resource changes combat, content pacing and economy integrity together.

54. Priority 4B: G8 capstone boss and zero-inflation optional solo mastery

1 November 2026 | v0.10.27 | [PROJECT PROPOSAL / DESK CALCULATION]. Authoritative upstream: §24.4–24.7 and §27 action order, §45/§50 five-member damage packets, §49 first-copy Ultimate boundaries, §50 S-E-v2 healer, §52 Hall20/Band I, §53 immutable campaign nodes and economic source IDs. This study does NOT claim live gameplay, engagement, measured win rate, art completion, backend selection or Naruto intellectual-property license.

54.1 Research question, success criteria, and alternatives

[QUESTION] Can C03-04/G8 feel like a recognizable campaign capstone without requiring an unowned premium counter, charging another Hall upgrade, inventing a permanent resource faucet, or invalidating the eight-threat teaching sequence? Can one optional solo activity teach a repeatable skill without becoming a daily obligation?

[DECISION] Use one mechanically modest, high-clarity capstone variant of the already certified self-healer (S-E-v2), not an extra mandatory stage and not an HP-inflated raid. Preserve the combat engine. Add exactly one conditional, non-recursive shield to the boss at the end of its SECOND living normal action, and publish its condition at battle entry. This allows a real late-fight phase for ordinary focus squads while a successfully applied anti-heal plan can end the fight before that phase. Call it S-E-v3 / G8-BC01; keep S-E-v2 in §50 as historical certified teaching reference, not automatically a current G8 victory ledger.

[REJECTED A] Two bosses or a multi-target medic-and-guard combination: changes legal target pools, loss branches and difficulty beyond the previously certified five-member context. [REJECTED B] Full boss immunity to healing suppression: would make freshly granted K-17 irrelevant; a boss-resistance tag also needs an explicit accessibility review. [REJECTED C] Hard-turn enrage, unavoidable paid counter or increased Hall21 entrance: unbudgeted difficulty and potentially compulsory daily resource collection. [REJECTED D] Recycle S-E-v2 verbatim and merely call it a boss: provides no distinctly readable capstone escalation.

54.2 New G8-BC01 enemy contract: telegraph, triggers and mechanical budget

Field	S-E-v3 / G8-BC01 authored contract	Why this restriction matters
Identity	Medical Supply Commander, an ORIGINAL working enemy role; no canon Naruto assignment. ONE exposed FRONT enemy only.	Narrative identity can change later without recycling FC-C03-04.
Stats	Max HP 4,700; DEF 600; ATK 850; Speed 89; Hall20/Band-I teaching reference.	No unpriced stats or new frontline guard.
Normal enemy slot	Basic coefficient 0.30 to K-01: floor(850×0.30×1000/(1000+1200))=115 damage; then heal SELF for floor(850×1.20)=1,020 gross, or 612 under 40% Healing Suppression. Heal is capped by missing HP.	Preserves S-E-v2 known order and counter value; no added enemy action.
NEW Ward	Once only, after the boss completes its SECOND living normal slot (not at R1): if HP after that slot's heal is ≤2,350 (50% max), gain a 1,000-point shield, expiring after player R3 if not consumed. No HP restored by shield.	Legal 1,000 < 40% of 4,700 = 1,880 shield cap; later focus can break it.
Telegraph / UI	At entry: "Medical recovery after each attack. If still active after its second turn at half HP or less: Emergency Ward 1,000." After R1, show forecast; on R2 trigger, log shield separately from heal.	Visible and nonrandom; distinguish HP recovery from shield.
Counter access	Core K-01/K-03/K-12/K-19 plus K-09 OR K-10 can focus. STORY03 grants K-17 before C03-04. K-17 applies Healing Suppression at 90%, including explicit FAIL branch.	Two accessible tactical families; no 100%-proc or premium requirement.
Unchanged rules	0 starting Chakra; auto Ultimate→eligible Technique→Basic; 2 subsequent own-slot Technique cooldowns; one normal action each; max 12 PvE rounds; defeat immediately ends the battle.	No stealth manual Basic overrides, free boss turns or round-clock shortcuts.

[TELEGRAPH CONTEXT] Emergency Ward is a new named boss passive, not a new global shield category, no follow-up normal action and no second heal. Its check occurs at the end of the living boss's R2 action only: after direct damage, after healing, and before end-of-slot cleanup. It has an explicit one-time flag, a 1,000 shield budget and expiry. If the boss dies before its R2 slot, no Ward occurs. If the boss heals above the threshold at its R2 slot, no Ward occurs and this one-time opportunity is spent. The shield does not refresh, carry to later attempts, or convert into reward.

54.3 Complete adverse-proc five-shinobi paper certificate, including both starter choices

[FIXED REFERENCE] START00 is K-01/K-03/K-12/K-19; START01 is K-09 or K-10. FOCUS squad has those five. ANTIHEAL squad swaps K-12 for the owned STORY03 K-17, preserving K-01/K-03/K-19/START01. K-12 Mark is forced to FAIL for the adverse reference. K-17 Healing Suppression is evaluated SUCCESS and FAIL independently. No crit, extra gear, faction bonus, duplicate or Ultimate. Enemy DEF600 makes the existing paper hit packets: K-12 Tech292 / Basic451; K-03 Tech1093 / Basic664; K-19 Tech812 / Basic531; K-01 Basic425; support Basic425; K-17 Tech292 / Basic451. K-10 Shield and K-01 Taunt do not activate against the slow enemy already targeting K-01, and K-09 has no Wound to cleanse. Both support picks legally Basic for425; the numbers below apply to each.

Branch / round	Exact legal action/HP and shield sequence	Boss and Guardian final state
Core focus R1	K-12 Technique292 (Mark FAIL); K-03 Technique1093; K-19 Technique812; START01 Basic425; K-01 Basic425: total 3,047. Boss Basic115 then heals1,020. Ward cannot trigger in R1.	Boss 4,700−3,047+1,020=2,673 HP, shield 0. K-01 7,800−115=7,685 HP.
Core focus R2	K-12 Basic451; K-03 Basic664; K-19 Basic531; support Basic425; K-01 Basic425: 2,496. Boss 2,673→177; then Basic115 and heal1,020→1,197. End of SECOND living boss slot and ≤2,350: 1,000 Ward ONCE.	Boss HP1,197 + shield1,000. K-01 HP7,570.
Core focus R3	K-12 Basic451 consumes 451 shield (shield549); K-03 Basic664 consumes remaining549 shield and deals115 HP (boss1,082); K-19 Basic531→551 HP; support Basic425→126; K-01 Basic425→KO. Boss R3 does not occur.	WIN R3. K-01 remains7,570. Both START choices identical at this reference.
K-17 SUPPRESSION SUCCESS R1	K-03 Technique1,093; K-17 Technique292 and 90%-chance status SUCCESS; K-19 Technique812; support Basic425; K-01 Basic425: 3,047. Boss Basic115 and heals612. R1 Ward disallowed.	Boss HP4,700−3,047+612=2,265. Shield0; K-01 HP7,685.
K-17 SUCCESS R2	K-03 Basic664; K-17 Basic451; K-19 Basic531; support Basic425; K-01 Basic425: 2,496; boss2,265→KO (on K-01 slot). Its R2 turn never occurs, so no shield or attack.	WIN R2. K-01 HP7,685. No forced proc assumption for alternative core path.
K-17 SUPPRESSION FAIL R1–R2	Same legal damage sequence as K-17 SUCCESS, but enemy R1 heal1,020→2,673 and R2 after2,496 damage is177 HP. Boss R2 attacks115, heals1,020→1,197, gains one-time Ward1,000.	End R2 boss HP1,197 + shield1,000; K-01 HP7,570.
K-17 FAIL R3	K-03 Basic664 consumes shield (remaining336); K-17 Basic451 removes336 shield and deals115 HP (boss1,082); K-19 Basic531→551; support Basic425→126; K-01 Basic425→KO. Boss never acts R3.	WIN R3; K-01 HP7,570. Both START choices remain legal and no proc is needed for a core victory.

[ARITHMETIC AUDIT] The Ward adds a visible 1,000 points of absorbable damage ONLY in the two R3-clear branches. Focus R3 total available damage is 451+664+531+425+425=2,496, strictly greater than 1,197+1,000=2,197, leaving 299 damage headroom. The boss still dies in R3, but unlike S-E-v2 all five living player actions are needed in the unfavorable Mark branch; this is conditional and will not always hold with different squad AI or balance patches. Boss heals 1,020 both R1/R2 on those branches; its R2 shield does not heal its HP.

[CHAKRA / SURVIVABILITY] Each player starts at 0; R1 attacking Techniques +15 and Basics +25; R2/R3 Basics +25, with K-01 gaining +5 on each received direct hit in the R3-clear branch. No unit reaches 100 Chakra before victory; no Ultimate cast is claimed. K-01 takes exactly 115 damage per living boss slot (twice in R3-clear branches; once with successful suppression), remaining at 7,570 or 7,685, respectively; other allies are not attacked by this deliberately bounded card. Two-turn cooldowns keep R2/R3 as Basics; a future longer boss must reopen this and the R4 Ultimate budget.

[CERTIFICATE CLASS] These are manual deterministic paper paths for declared cases, not a general proof that players will survive every roll or gear state, not a combat simulator, and not empirical win rates. The proposed extra shield is legal under the declared shield cap and creates a distinct second-phase action; its fairness against all accessible legal rosters remains a FUTURE TEST. If the enemy becomes rear-row, obtains a heal target ally, cleanses suppression, attacks other players, or has a different Speed, these certificates expire.

54.4 Boss presentation and chapter closure: strong information, low extra friction

[NARRATIVE WORKING BEAT] C03-03 recruits K-17 while revealing that the Medical Supply Commander can repeatedly restore itself. On C03-04 entry the enemy card shows (1) guaranteed slot heal, (2) optional one-time late shield, (3) K-17's 90%-chance suppression and (4) ordinary focus damage as a reliable substitute. R1 shows the enemy restoring HP; R2 shows either an early knockout or a shield and its remaining absorb value. Victory explains exactly which effect changed outcome. This is a system narrative, not a finished Naruto script or a claim to licensed characters.

[MOBILE UX SPEC] At preflight, show "Restores up to 1,020 HP after acting" and "Emergency Ward: 1,000 shield after its second turn if at or below 50% HP." The boss healthbar must separate green HP and a distinct outlined shield segment; the battle recap shows gross heal, suppression-reduced heal, direct damage absorbed and source of the Ward. One entry preview, no intermediate unskippable confirmation or extra tap to claim a cosmetic. Avoid timed QTE, manual interrupt and hidden percentage thresholds. Introduce chapter-specific audio/animation later and test legibility on lower-end devices; no art or UX test exists.

[FAILURE UX] If the player loses, show the two accessible responses: (A) focus damage and (B) K-17 suppression plus fallback if it fails. Display current earned Hall20/Band-I and character entitlements. Do not suggest paying, pulling featured characters or buying Hall21 to solve this Hall20 reference. A player may change formation or re-enter immediately at no stamina cost. Boss is one C03-04 node, not a new chapter node; graph still has 15 nodes and exactly 8 mandatory combats.

54.5 The first optional solo mode: Tactical Archive TA-01

[SELECTED OPTIONAL MODE] One pilot solo activity named Tactical Archive / Formation Forecast, TA-01. It unlocks on FIRST clear of C01-05 (after Chapter1) and is explicitly optional. Players view the already documented C01-05/G3 controller preview: a Speed118 foe targets the FRONT-RIGHT formation occupant with 80%-chance Silence. They choose which available introductory ally to put in the target slot, see a deterministic preview of the *targeting identity*, then may watch/replay the established G3 encounter. The mastery question is whether they protected K-03's ready R1 Technique from being the designated Silence target. The result is judged by target-slot/formation legality, NOT by whether the 80% Silence roll happens to fail in a particular fight.

Contract item	Selected rule	Implication / dependency
Unlock and entry	After FC-C01-05. Solo and offline-compatible for preview; save grants and best-formation receipt persist when connected.	No guild, PvP, subscription, roster luck or calendar participation.
Eligible roster	START00 and either START01; may use later owned units, but the certificate only requires K-01 and K-03 plus legal slots.	No newly minted demo character; input belongs to real owned account.
Goal	Show that a valid formation places K-01 at FRONT-RIGHT, preserving K-03's ready R1 Technique. Provide G3 preview and optional battle watch.	Checks tactical choice; cannot be accidentally "passed" because Silence fails its 80% roll.
Attempts and reset	Unlimited attempts, zero energy/Ryo/EXP cost. No daily/weekly reset, tickets, score-rank ladder or reward multiplier.	Repeat for learning, not compulsory engagement or currency farming.
One-time benefit	TA01-COS01: one nontradable cosmetic dossier/profile emblem, exactly once after successful target-formation verification. Replay pays zero currency and zero duplicate cosmetic.	One cosmetic reward purpose only; not a sixth team preset or new power stat.
CORE and IDLE exclusions	TA-01 completion does not grant CORE01/CORE02 qualification, idle hours, higher IDLE01 tier, FC campaign first clear, STORY grant or FIRST02.	Does not double-credit existing weekly seals/fragments or affect progression arithmetic.
Account and recovery	Store TA-01 mastery receipt and TA01-COS01 entitlement as two persistent records tied to one immutable account+reward key; retries return the same award.	Idempotent design intent, backend not selected and no service implemented.

[DISTINCTIVENESS] Campaign replay asks “Can I win the encounter again?” Tactical Archive asks “Can I predict and change the unit targeted before an 80% control roll?” The optional lesson is useful even when no currency is paid, but intrinsic learning and one cosmetic may be insufficient for long-term retention. Do NOT silently treat this as a full-fledged tower, wave trial or guild boss. Only TA-01 is proposed now; do not pre-author a huge catalogue or promise repeatable economic rewards.

54.6 Campaign first-clear reward salience, strict source-cost audit

Reward ID / event	Proposed credited value	Economic and transaction status
FIRST01 at ONB04	5 flexible Recruitment Seals, once.	UNCHANGED §53; no new seals.
FIRST02 at FC-C01-05	5 flexible Recruitment Seals, once.	UNCHANGED §53; no per-fight repeat.
STORY01/02/03	K-08 / K-04 / K-17 guaranteed first copies at scenes.	UNCHANGED; no duplicate seal/fragment conversion.
C02-06 chapter closure	LORE-C02-01: persistent nontradable tactics-record entry, no numeric resources.	NEW working noncurrency entitlement; zero progression power.
FC-C03-04 chapter closure	COS-C03-01: one nonstat “Field Medic Breaker” profile title; narrative boss dossier, once.	NEW working cosmetic; no unlock rate, pity, equipment or currency changes.
TA01-COS01	One solo Tactical Archive profile emblem after verified formation result.	NEW one-time cosmetic; independent of mandatory campaign.
All other FC/SCN nodes	Persistent progress only; 0 EXTRA material ASSUMED in this paper budget, actual finished-game reward OPEN.	Not a claim future final game should have empty rewards.
Recurring trials / extra IDLE tier	NONE newly authored. No extra seals, fragments, Ryo, EXP, enhancement, essence, resets or premium tickets.	Existing 34 seals/60 fragments per full 28d core budget unchanged; existing fixed E0 rate unchanged.

[OPPORTUNITY-COST COMPARISON] Rejected for this pass: award +80 Ryo and +120 EXP at C03-04. At E0 rates (10 Ryo and 20 EXP per full-rate hour), those two numbers separately correspond to 8 and 6 equivalent full-rate hours, respectively; pricing them would change Hall21+ wallet calendars and may affect a later promotion gate. An arbitrary +5 seals per later chapter adds 10 total seals and materially changes the §40 acquisition path. Cosmetic/lore-only rewards avoid those immediate currency effects but could feel underwhelming; keep player-value assessment OPEN. No monetary value of artwork or production cost has been estimated.

[TRANSACTION INTEGRITY] FC-C03-04 is the chapter-win event; COS-C03-01 is a reward definition credited once, not an extra FC event. TA01-COS01 has a unique per-account occurrence key and cannot be exchanged for a roster duplicate. A disconnect between victory and reward acknowledgment should show a pending claim or safely reconcile the existing receipt—not award a fresh second copy. Nontradable cosmetics can be previewed without an open economy backend decision. Actual server implementation remains out of scope.

54.7 Pacing, player agency, boss originality and limitations

[15-NODE FRICTION AUDIT] §53 requires eight battle completions and seven scene completions. A worst-case forced-tap count cannot be claimed without screen flows; however seven mandatory scene completions risk becoming seven pointless confirmations. PROPOSED navigation rule: scenes are short, skippable on replay, permit immediate next-node navigation, and combine grant acknowledgment with team-switch preview on STORY scenes. STORY scenes still commit their immutable ownership grant before G4/G5/G8 entry. During eventual usability testing, compare minimum input actions and time-to-first-battle against a merged-scene alternative; choose a lighter presentation only if reward and grant integrity survive.

[BOSS IDENTITY] An ordinary self-healer plus one late Ward is a restrained introductory medical-supply boss, not a mature raid. It teaches recovery, suppression odds, shields and five-member target pressure, but uses only one enemy and does not use Ultimate Jutsu. For now that restraint is deliberate: adding an army of minions, multiple phases or an untelegraphed wipe move would create an entirely different encounter certificate. The artistic/narrative identity, distinctive Jutsu, environmental storytelling, and animation readability are OPEN and later require original-IP or licensed Naruto choices.

[NO FALSE PROOF] Three deterministic ledger branches at a Hall20 proxy do not establish expected win rates, satisfaction, retention, F2P commercial viability, performance on low-tier devices, or success across all placements. The 90% success branch is not “90% faster wins” in a real game. All other stage packets remain independently conditional, and any changed combat status, ability, enemy targeting or Hall budget triggers revision. No prototype, executable simulator, human playtest, network backend or live monetization exists.

54.8 Decision register, rejected alternatives, dependent systems and falsifiers

Decision	Reason and cost	Reversal condition
4B-01: G8-BC01 one boss with 1,000 Ward on qualifying R2 slot.	Makes a visible late defensive phase while preserving affordable Hall20/five-slot paths. Adds boss-specific passive/UI burden.	If alternate legal team lacks damage to clear or UI cannot explain delayed Ward, reduce/retime Ward and recertify all branches.
4B-02: keep 90% K-17 suppression and core focus fallback.	Specialist meaningfully shortens favorable case; no luck/paywall. Failure branch requires full five-unit engagement.	If a later build cannot clear without successful status, adjust encounter rather than raising status to a guaranteed hidden proc.
4B-03: TA-01 no recurring material rewards.	Provides strategic prediction with minimal economy disturbance. Intrinsic engagement and one cosmetic may have low retention.	Price one bounded resource only after its purpose, sinks, frequency and F2P parity can be calculated against §40/§42.
4B-04: cosmetic/dossier Chapter2–3 closure.	Avoids new seals or idle-tier inflation. Subjective value uncertain and art costs unpriced.	If mockups/usability show unrewarding progress, evaluate one-time material budgets and rerun full acquisition calendars.
4B-05: no new stage or level gate.	Preserves 15-node graph and Hall20 access, ensures single boss reward event. May make early progression too flat.	If authored story pacing requires extra fights, independently price grants and certify each new enemy/level.
4B-06: battle recap + predictable formation lab.	Teaches controllable decisions without confusing failed Silence roll with tactical success. Requires explicit preview and unique mastery receipt.	If formation predictability or targeting rules change, revise TA-01 before launch.

54.9 Sources, inference limits and next research handoff

[PEER / COMMUNITY COMPARISON] Prydwen, AFK Journey Dream Realm guide (accessed 1 Nov 2026), describes repeatable daily rotating boss damage/rank rewards. We treat it as a third-party mode description, not a numeric template. The potential for repeated boss play is interesting but a ranking-driven material faucet is explicitly rejected for early TA-01. https://www.prydwen.gg/afk-journey/guides/dream-realm

[FIRST-PARTY COMPARISON] Supergiant Games, Hades Welcome to Hell Update Patch Notes (developer, historical), documents boss changes under Extreme Measures, Pact modifiers and bounty limits. Design inference: optional challenge variants can change an encounter’s tactical contract and reward eligibility together; this does not establish a specific balancing number for our idle RPG. https://www.supergiantgames.com/blog/hades-welcome-to-hell-update-patch-notes/

[SERVICE DESIGN ANALOGY] Unity Cloud Code documentation for game state management (accessed 1 Nov 2026) explains persistent server-authoritative state and verification of legal actions. This supports separating mastery receipt from reward credit; it is NOT an engine choice. Unity Economy stopped new project signups on 8 Sept 2026; older Economy citations remain conceptual, not a backend recommendation. https://docs.unity.com/en-us/cloud-code/3.0.0/game-state-management ; https://docs.unity.com/en-us/economy

[EVIDENCE LABELS] All boss names, stats, 1,000 shield, the precise R2 trigger, five-member calculations, TA-01 challenge rules and cosmetics are original [PROJECT PROPOSALS] with [CALCULATED] arithmetic. The cited games/services are external examples only. An additional system-wide economy/balance proof and player evidence are [OPEN].

[NEXT — PRIORITY 4C: CAMPAIGN PACING AND FIRST-CLEAR VALUE] Study whether 15 nodes / seven scenes create excessive friction, decide how to present optional skip/recap and accessible tactical cues, and run a 28–56-day

source/sink comparison for at least two candidate Chapter2/3 first-clear reward packages (cosmetic-only versus bounded materials). Keep first-play Hall20, fixed E0, FIRST01/FIRST02, weekly seal/fragment floor, starter grants and the G8 boss certificate as baseline. Only adopt a material change if an explicit dated wallet and all-source collision audit justify it. RECOMMENDED THINKING LEVEL: EXTRA HIGH because UI pacing, stage satisfaction, progression incentives and economic fairness must be reconciled together. Research/concept only.

55. Priority 4C: opening campaign pacing and bounded first-clear value

1 November 2026 | v0.10.28 | [PAPER RESEARCH / WORKING PROPOSALS]. This packet audits §53's 15-node graph and §54's first-clear value gap using §40 core entitlements, §42 early income, §52 Hall1→20 onboarding, FIRST01/FIRST02, STORY01–03, G8-BC01 and TA-01 as binding comparison assumptions. It does not establish real first-session duration, player satisfaction, retention, working UI, an implemented ledger, or a commercial Naruto license. New quantitative rewards are proposed, never retroactively attributed to earlier chapters.

55.1 Decision question, hard constraints and acceptance tests

[QUESTION] Can eight mechanical lessons and seven noncombat scene records feel continuous on a small phone, while Chapters 2 and 3 have meaningful, nonpaywalled closure value? Does adding a modest one-time material grant undermine early Hall affordability, summon economics, repeatable mode purpose, or idle rewards?

[BASELINE] Keep exactly 15 durable content IDs (8 FC battles + 7 SCN scenes) and 14 predecessor edges; Hall20/Band-I first gate; E0 at ONB01 and fixed hourly rate 10 Ryo/20 EXP/2 essence/1 enhancement; 24h full + 48h quarter-rate bank; FIRST01 and FIRST02 exactly five seals each; STORY01/02/03 guaranteed first copies; 34 CORE seals and 60 CORE fragments per fully completed four-week cycle; TA-01 zero recurring resources. G8 boss stats and adverse branches remain untouched. No chapter completion buys an Ultimate, exclusive counter, gear baseline, or extra fight.

[PAPER ACCEPTANCE TESTS] (1) Each original FC/SCN and receipt ID remains unique; (2) a skipped story scene does not skip its grant or unlock; (3) no purchase/mode/wait requirement is added; (4) full and partial 28/56-day wallets are traceable; (5) the chosen grant cannot independently force Hall21 at day1 when only one daily claim has occurred; (6) no new seals/fragments, idle-tier acceleration or recurring income are implied; (7) minimum required scene-dismiss taps fall relative to the explicit blocking-modal comparison; (8) any new rewards remain labeled proposals until real user feedback exists.

55.2 Graph-preserving presentation: seven scene records, not seven forced pop-ups

[WORKING DECISION 4C-01] Preserve every §53 stage identity for save and reward integrity, but stop treating a SCN as a mandatory full-screen interruption. Present noncombat scenes as inline narrative strips on the same campaign route, with one optional Expand/Read scene control, persistent View Again in the dossier, and a normal Next Battle control. When a player proceeds or chooses Skip Story, commit the SCN receipt and any associated STORY grant exactly once BEFORE enabling the dependent battle. A skip bypasses presentation, NEVER ownership, tutorial readiness, or the predecessor record. Never show an additional claim modal for an already-auto-credited first copy. The player may pause at any node.

[FORMAL TAP-COUNT COMPARISON] Count only EXTRA taps caused by scene-specific blocking prompts, not battle Start/Continue, team changes, or device controls. A hypothetical blocking flow requiring OPEN then DISMISS on each of 7 scenes adds 7×2=14 extra scene-specific taps. The selected inline/auto-credited treatment targets 0 EXTRA mandatory modal-dismiss taps; ordinary Next Battle remains necessary regardless. These are interface-design counts, not measured player taps, durations, or completion rates. Optional Read Scene and Swap Shinobi add voluntary inputs; first-session actual taps remain OPEN until a clickable mockup is assessed.

Node ID / content	Nonblocking scene contract	Persistent grant/recap invariant
C01-02 · Wound debrief	Inline status aftermath; optional expanded log; keep Next G2 visible	SCN-C01-02; no currency
C01-04 · formation briefing	Inline illustration of who Silence will target; optional Try Formation	SCN-C01-04; formation choice remains player-controlled
C02-01 · disruption recruit	Auto-credit K-08, show optional equipment/team-swap drawer	SCN-C02-01 + STORY01 before G4
C02-03 · area recruit	Auto-credit K-04; optional AoE-target preview	SCN-C02-03 + STORY02 before G5
C02-05 · rear tactic	Inline frontline/row target graphic; nonmandatory replay	SCN-C02-05 before G6

Node ID / content	Nonblocking scene contract	Persistent grant/recap invariant
C03-01 · attrition briefing	Inline two-source Wound timing preview; status tooltip	SCN-C03-01 before G7
C03-03 · healing recruit	Auto-credit K-17; optional skill preview and swap drawer	SCN-C03-03 + STORY03 before G8

[UI GUARANTEES] Before a fight, show named threat, two currently owned/available answer categories, earned Hall/baseline gear, forecasted target and status timing, plus a noncolor-only icon and short explanatory text. If a counter is missing due to a pending grant transaction, reconcile that same grant and unlock; do not sell a pull. After battle, a concise Why This Worked recap is expandable and can be reopened from the permanent tactics dossier. Story skip, reduced motion, larger text, contrast, readable tap areas, and a screen-reader-friendly narration transcript must not change SCN or FC reward eligibility.

[TRADE-OFF] Inline presentation lowers interruption but makes plot beats easier to miss. Keep Chapter1/2/3 optional recap cards and a chapter closure moment that combines story, cosmetic and any material entitlement in ONE receipt. Revisit if future usability evidence shows players cannot remember why a character joined or which mechanic their team answered. No time-to-first-battle or emotional-engagement claim is made.

55.3 First-clear reward alternatives and selected bounded material candidate

Candidate / event	Incremental economic grant (beyond existing §54 rewards)	Strength and cost	Decision
A · existing Chapter2 dossier, Chapter3 title	0 Ryo; 0 EXP; 0 essence; 0 gear; 0 seals; 0 fragments	Zero inflation and clear identity; early victories may feel materially empty	Retain as reference
B · Chapter3 measured boss bonus	ONE 80 Ryo + 120 EXP at FC-C03-04, atop existing COS-C03-01 title; Chapter2 still dossier	Visible postboss accelerator; no new recurring faucet or entitlement before the boss	SELECT provisional
C · equal bonuses at Chapter2 AND Chapter3	80 Ryo +120 EXP each: total 160 Ryo +240 EXP	Stronger immediate chest, but can advance Hall21 to the first daily claim if both cleared very early	Reject at this baseline

[WORKING DECISION 4C-02] Select B as the narrow candidate: new reward definition FIRST03 grants 80 Ryo and 120 Shinobi EXP exactly once after FC-C03-04 / Chapter3 boss victory, in addition to the existing nonstat COS-C03-01 title. FIRST03 has no essence, enhancement, Recruitment Seal, Universal Fragment, pity, class gear, ticket, standard shinobi copy, or IDLE01 change. Existing Chapter2 LORE-C02-01 stays informational. Present FIRST03 in the Chapter3 victory recap, not a new popup. A player may bank or later spend its resources freely after ONB04. It cannot be claimed from replay, TA-01, a failed attempt, a premium purchase, or another calendar reset.

[UPSTREAM RATIONALE] Section 54 explicitly kept +80 Ryo / +120 EXP open as a candidate at C03-04. This pass prices and compares it without pretending an earlier first-clear paid it. The amount is equal to 8 E0-equivalent full-rate hours of Ryo or 6 hours of EXP, but NOT an exchangeable 8-hour idle time coupon; it is two fixed resource balances. The reward pays only part of the optional Hall20→21 bill (300 Ryo, 720 EXP, 48 essence) and no essence by itself. A status-only Chapter2 reward may still feel weak; we are declining to compensate with a second unpriced chest.

55.4 Dated 28- and 56-day idle-wallet comparison: three reward architectures

[SCENARIO ASSUMPTIONS] ONB01–04 are completed at tutorial day0; opening wallet is 8 Ryo /12 EXP /0 essence /0 enhancement. E0 is activated at day0; no Hall21 purchase, shop spend, new idle tier, GEAR01/MAT01, event, guild/PvP material or extra rewards are included. To test maximum early effect of candidate B and C, assume their chapter-clear receipts BOTH become eligible and are paid at day0—a deliberately optimistic timing stress case, not a claim the user can actually complete the entire 15-node arc immediately. If G8 is cleared later than a listed snapshot, its +80/+120 is absent until the actual grant. Idle accrual uses $H(\Delta)=\min(\Delta,24)+0.25\times\min(\max(\Delta-24,0),48)$; each claim after a ≥72h gap collects 36 E0-equivalent hours. Day28/56 balances are taken only after that day's scheduled collection when one exists.

Snapshot / account	Claims / eff. h	A: Ryo / EXP	B: Ryo / EXP	C: Ryo / EXP	Essence / Enh. (all)
28 · Daily	28 / 672	6728 / 13452	6808 / 13572	6888 / 13692	1344 / 672
28 · Twice weekly (days 3/7 pattern)	8 / 288	2888 / 5772	2968 / 5892	3048 / 6012	576 / 288

Snapshot / account	Claims / eff. h	A: Ryo / EXP	B: Ryo / EXP	C: Ryo / EXP	Essence / Enh. (all)
28 · Once weekly	4 / 144	1448 / 2892	1528 / 3012	1608 / 3132	288 / 144
28 · Midweek start (day 2 then 4/3 gaps)	8 / 282	2828 / 5652	2908 / 5772	2988 / 5892	564 / 282
56 · Daily	56 / 1344	13448 / 26892	13528 / 27012	13608 / 27132	2688 / 1344
56 · Twice weekly (days 3/7 pattern)	16 / 576	5768 / 11532	5848 / 11652	5928 / 11772	1152 / 576
56 · Once weekly	8 / 288	2888 / 5772	2968 / 5892	3048 / 6012	576 / 288
56 · Midweek start (day 2 then 4/3 gaps)	16 / 570	5708 / 11412	5788 / 11532	5868 / 11652	1140 / 570

[CALCULATED] In the no-spend snapshots, B is exactly +80 Ryo and +120 EXP relative to A at ANY point after FIRST03 is earned; C is +160/+240 if both hypothetical receipts are earned. Essence and enhancement are identical across the alternatives and do NOT change with the one-time bonus. As a fraction of A's Ryo wallet B contributes ~5.52% on a once-weekly day28 case (80/1448), ~2.77% on that account's day56 case (80/2888), ~1.19% on a daily day28 case (80/6728), and ~0.59% on daily day56 (80/13448). This is a one-time delta, not 80 Ryo per chapter replay or per week.

[ACQUISITION CROSS-CHECK] FIRST01+FIRST02 remain a separate 10-seal maximum for completed tutorial + Chapter1. At full CORE01/CORE02 participation, the 28-day gross is 34 recurring seals/60 fragments + 10 one-time introductory seals = 44 seals/60 fragments; over 56 days it is 68 recurring +10 one-time=78 seals/120 fragments. A once-weekly visitor earns only the first-visit CORE share: 8 recurring seals/40 fragments in four weeks, 16 seals/80 fragments in eight weeks, plus any actually completed first-clear seals. FIRST03 adds ZERO seals/fragments under all scenarios. The 70 featured-pull guarantee applies only to seals actually spent in the eligible featured banner; the 68 recurring seals alone still do not meet 70. Banner choices and calendar eligibility remain separate from held balances.

55.5 Hall21 timing, source collision, player paths and smallest safe change

[FIXED UPGRADE CONTRACT] The optional Hall20→21 step still costs 300 Ryo, 720 EXP and 48 essence; it raises the Hall cap to25 under §42. Starting at 8/12/0, one daily claim on day1 adds 240 Ryo/480 EXP/48 essence and the second adds the same on day2. Model both hypothetical chapter reward receipts as occurring at day0 solely for stress testing; actual G8 access is gated behind the earlier 14 nodes and could occur much later.

Route / day 1 wallet Ryo/EXP/essence	Hall21 affordable day1?	First affordable scheduled visit (no other spends)	Reason
A / 248 / 492 / 48	NO	Daily day2; twice-weekly day3; weekly day7	Ryo short52 and EXP short228 on daily day1
B / 328 / 612 / 48	NO	Daily day2; twice-weekly day3; weekly day7	EXP short108 on daily day1, despite adequate Ryo
C / 408 / 732 / 48	YES under day0 clear assumption	Daily day1; twice-weekly day3; weekly day7	Both gifts together cross 300R/720EXP/48ess floor

[DECISION] B does not move the earliest optional Hall21 purchase forward for these three visit schedules in the maximum-early-receipt stress case, but its resources remain real spendable progress and may affect other unpriced sinks. C would move one daily player's Hall21 purchase to day1 and is rejected pending a proper later-level growth curve. This does NOT imply B is permanently balance-neutral; future stages or equipment prices could make the extra 80/120 cross a different threshold. Preserve the 0 additional Hall21+ mandatory gate policy until full growth pricing exists.

[TRANSACTION IDENTITY] An encounter win is the persistent FC-C03-04 occurrence. FIRST03 is a reward-definition entry whose unique payout key is (account, FC-C03-04, FIRST03-v1), independent of COS-C03-01 title and LORE-C02-01 dossier. On a retry or reconnect, read the prior receipt; grant missing components once; never mint new seals/fragments from a cosmetic claim. The UI must distinguish a successful first clear, an already-received reward and a pending reconciliation. No service provider is selected; authoritative persistence is a design requirement, not an implemented security guarantee.

55.6 First-session navigation and reward presentation contract

Moment	Mandatory information/action	Optional, always revisitable	Forbidden shortcut
ONB04 → G1	Confirm Hall20, standard Band I, 5 owned shinobi, G1 Wound preview	More detail on Wound timings and team swap	No forced idle wait or premium summon
G1/G2/G3 aftermath	One compact win/lesson panel + Next; FIRST02 shows with G3 closure once	Expanded battle log, cutscene/story transcript	No extra claim modal paying FIRST02 twice
STORY01/STORY02/STORY03	Auto-credit permanent first copy BEFORE eligible battle	Character details and team preset switch	Skip Story cannot discard shinobi
G8 victory	One consolidated panel: boss outcome, COS-C03-01, FIRST03 80R/120EXP	Read boss dossier, share text summary, open inventory	No duplicate reward on replay or reconnect
Return after absence	Claim once, show saved next graph node and earned reward receipts	View prior scenes/recaps, optional Tactical Archive	No lost first clear, forced recap or fabricated offline money

[PLAYER-AGENCY RULE] New players can read lore at their own pace; mandatory mechanics are communicated in concise battle previews and demonstrated during fights. Skip Story is reversible from the dossier. A first-time grant scene may visually be skipped but its ownership receipt must still be committed before the battle button becomes active. Reduced motion never hides a threat wind-up, warning icon, duration, magnitude, target legality, or a reward entitlement. System subtitles and text equivalents remain available for all mechanics-critical feedback. Avoid irreversible choices except the disclosed START01 starter choice.

[FUTURE EMPIRICAL GATES — NOT TESTED] Instrument number of voluntary scene expansions, screen taps from ONB04 to first G1 entry, number of blocked/unreadable threat previews, frequency of skip/recap use, delayed/failing grant reconciliation, first-clear-to-next-battle continuation, and whether a player can name a counter category after the recap. Evaluate A/B reward preference using mockups without altering unrelated stats; do not infer engagement simply from reward size. No performance threshold has passed in this research.

55.7 Working decisions, rejected options, limits and falsifiers

Decision	Trade-off and rejected alternative	Condition to reopen
4C-01: Keep 15 logical nodes, inline seven SCN presentations	Preserves receipts without seven modal gates; fewer forced taps may reduce dramatic storytelling	If mocked flows hide grants or story motivation, add selective chapter beats but never duplicate payouts
4C-02: Select FIRST03 once at G8: 80 Ryo/120 EXP	Adds tangible boss reward with small persistent wallet change; rejects two equal bonuses for day1 Hall21 risk	If cost graph / later level curve creates new wall or bonus proves negligible, reprice AFTER publishing sink schedule
4C-03: Chapter2 dossier and Chapter3 title persist	Avoids extra seals, premium counters and repeatable farming; reward salience may remain weak	If user tests show low clarity/value, alter cosmetics/presentation before adding currency
4C-04: Core/idle/recruitment payouts unchanged	Maintains initial F2P comparability; no new rate progression within 15-node arc	Reopen only with segmented idle rate and 28/56-day re-budget
4C-05: One consolidated reward receipt at each chapter closure	Minimizes friction; requires clear itemized entitlement and failure recovery	If players mistake pending for missing rewards, improve recovery UI and logging
4C-06: no estimated minutes or pass-rate claims	Avoids false precision; decisions are limited to paper screen/tap contracts	Replace with observed usability/time data once production/user testing explicitly authorized

[CONDITIONALITY] We have not measured which reward architecture feels satisfying, whether seven inlined scenes are memorable, whether 15 nodes overstay welcome, whether a tutorial and 8 Hall20 fights can actually be enjoyed together, or whether G8-BC01 remains balanced after real player inputs. Material FIRST03 is a proposal, not a statement that past historical stage rewards now included it. Historical §54 rejects granting it without a full comparison; this chapter supplies that comparison and selects a conservative option. Supersession is explicit, not erasure.

55.8 Research sources and handoff

[SOURCE — APPLE OFFICIAL] Apple Developer, Human Interface Guidelines: Onboarding and Designing for Games, consulted 1 November 2026. Apple recommends contextual, interactive learning, fast optional onboarding, usable controls and readable instructions. We use this to question forced scene interruptions, not to infer exact retention/tap savings or mandate an Apple-specific platform choice. https://developer.apple.com/design/human-interface-guidelines/onboarding ; https://developer.apple.com/design/human-interface-guidelines/designing-for-games/

[SOURCE — ROBLOX OFFICIAL] Roblox Creator Hub, Onboarding, consulted 1 November 2026. Treat first-time play as a funnel where each step may lose participants and require instrumentation. This is a process analogy; no Roblox-specific retention statistic is imported. https://create.roblox.com/docs/production/game-design/onboarding

[ORIGINAL PROJECT DATA] All 15 node names, one-time FIRST03 receipt, 80R/120EXP, dated wallets, Hall21 affordability, mobile flow and acceptance gates are original paper design proposals/calculations from §§40–42, 52–54. The sources do not validate any numeric value. No game implementation, simulator, real playtest, purchase service or licensing grant was created.

[NEXT — PRIORITY 4D: FIRST-SESSION EXPERIENCE AND CAMPAIGN NARRATIVE INTEGRATION] Develop a state-by-state opening UX script and chapter story beat matrix in the actual 15-node graph, showing what is taught vs merely told, how character grants are represented narratively, where players can choose/pause/skip, and how G8 closes a coherent theme. Also reconcile day0 pay timing versus likely progression pacing and validate whether optional Hall21 income remains intelligible without adding pressure. No new modes, power currencies, unpriced battles or assumed Naruto IP rights. RECOMMENDED LEVEL: EXTRA HIGH, because UX presentation, acquisition receipts, story semantics and tactical onboarding need joint consistency. If narrative authorship stays symbolic and no new mechanics/costs are introduced, the narrower subsequent writing pass may be HIGH.

56. Priority 4D: first-session player agency, original narrative spine and durable progression

56.1 Research scope, authority, outcomes and alternative designs

[SCOPE / EVIDENCE] 1 November 2026. Continue §§23, 26, 40–42, 46–55. This is a PAPER design of first-time-player experience (FTUE) and narrative semantics—not a playable prototype, proof of retention, implemented saving, authored Naruto canon scenario or rights clearance. Keep the 15 campaign node IDs, eight Hall20/Band-I conditional battle packets, seven scene receipts, four ONB funding events, both START01 options, three story recruits, FIRST01/FIRST02/FIRST03, TA-01 and the E0/CORE budgets unchanged. The aim is not more content: it is to give players a coherent reason to make the choices the existing mechanics already support.

[QUESTION] Can a completely new account see the stakes, fund its baseline team, exercise one meaningful tactical choice before the first Wound fight, and recover from any interruption—with no mandatory pull, waiting clock, separate grant, irreversible hidden default, or seven forced story dialogs? The following is an executable-by-humans UX/narrative contract for future mockups; numeric success and session-time targets must await authorized testing.

Candidate	Advantages	Costs / failure modes	Decision
A. Cinematic-first 15-node introduction	Stronger prewritten story exposition	Delays mechanics, adds forced taps, difficult to resume	REJECT for default flow
B. Context-first, single supply-route crisis	One continuous motive; first meaningful formation/support choice early; inline skippable beats	Less cinematic exposition; needs well-written concise cards	SELECT working proposal
C. Mechanics-only auto-skip narrative	Fastest surface path	Why each encounter matters and why new recruits join becomes opaque	Retain only as player-selected compact presentation

56.2 Narrative spine: The Broken Supply Route (original-IP-safe working wrapper)

[WORKING NARRATIVE PROPOSAL 4D-01] The 15 nodes describe a crisis around interrupted medical supplies, misleading communications and a defended field clinic. A dispatch team investigates injuries (G1), protects a convoy (G2), and discovers attempts to suppress warnings (G3). In Chapter 2 they learn that the attackers use a charged strike, misdirection and protected rear assets (G4–G6); they gain the necessary specialists through on-route alliances, not lucky recruitment. In Chapter 3 the team faces simultaneous attrition threats (G7) and reaches the Medical Supply Commander who preserves the clinic defenses through self-recovery and a telegraphed Ward (G8-BC01). The moral through-line is coordinated aid under incomplete information—not a claim about Naruto canon, a new faction system, or a branching dialogue mechanic.

[NARRATIVE RIGHTS BOUNDARY] All locations, names, dialogue, allegiances and the Medical Supply Commander are original placeholders. Do not assign a Naruto canon figure, Hidden Village, signature jutsu, clan relationship or authorized audio/visual asset until an explicit IP decision and rights review. The combat IDs K-01…K-19 and the 15 event IDs remain stable when story names are later rewritten. Original-IP fallback must support exactly the same counter and reward entitlements.

[CAUSE-AND-EFFECT] Each scene must answer a question raised by the preceding fight, foreshadow the next threat, or show why a recruit joins. No additional battle or dialogue gate is required merely to justify a grant. The seven SCN receipts preserve graph order even when their text is collapsed into the screen already in view. A cosmetic title and tactics dossier can represent the resolution without becoming gameplay power.

56.3 First-session interaction sequence and two equivalent training paths

[WORKING DECISION 4D-02] Offer GUIDED and COMPACT onboarding presentation. Both run exactly the four ONB transactions, buy the same 19 Hall levels for 272 Ryo/408 EXP, credit 280 Ryo/420 EXP, open E0 at ONB01, retain 8 Ryo/12 EXP, grant FIRST01 5 seals at ONB04, reconcile START00 ownership, require one explicit START01 support selection and provide Band-I equipment. Neither path can omit an entitlement, change the numerical economy, skip a mandatory character choice, or issue a full-currency reward without its unique receipt. COMPACT means fewer explanatory screens—not artificially faster resource accrual or automatic battle wins.

[BATCH-TRANSACTION NOTE] The presentation may display a single “Train to level X” control for each funded ONB band instead of 19 separate level taps. This control still commits the documented sequence of individual level costs as ONE atomic aggregate transaction only if the band is fully funded and eligibility is satisfied; its receipt discloses before/after Hall, exact Ryo/EXP debits and restored balance. Do not create a second reward ID or new discount. No implementation is authorized; atomicity is a future service requirement.

Step / first-screen surface	Player decision, visible information	Durable receipt / next eligibility
F0 • initial play / minimal start	Show story problem in one 1–2 sentence panel. “Start Guided” or “Start Compact”; all story text can be reopened. No forced ad, offer, summons, or notification request.	Local choice is a UI preference; no money. F1 is available.
F1 • starter team preview	Show K-01, K-03, K-12, K-19 as the four fixed recruits. Show frontline/rear slot and the simple claim: every slot has a job.	START00 first-copy ownership credited only once, no seals or duplicate shards.
F2 • Hall training / first band	ONB01 40R/60EXP → buy Hall1→5 for32R/48EXP. Label the shared Hall, not four personal upgrades. E0 idle rate unlocked here.	ONB01 and E0 durable; 8R/12EXP remains; no claim wait.
F3 • subsequent training bands	ONB02 56R/84EXP → Hall5→10 (60R/90EXP), ONB03 80R/120EXP → Hall10→15 (80R/120EXP), ONB04 104R/156EXP → Hall15→20 (100R/150EXP). Each grant carries the current wallet forward.	After ONB04: Hall20, 8R/12EXP; FIRST01 5 seals once; standard Band-I tools included.
F4 • support decision	Contrast K-09 Cleanser (remove one Wound) vs K-10 Protector (shield/recover) with a two-line trade-off and actual Wound timing. Ask explicit selection; allow Back/More Details.	Exactly one START01 first copy; choice not auto-made by timeout or compact mode.
F5 • first actual tactical choice	Show G1 Wound preview: who applies Wound, when victim takes a tick, which available solutions exist. Allow one meaningful formation/target preset change, or knowingly keep default.	Save chosen five-slot squad/preset; no extra resources or battle-modifying tutorial buff.

F6 • C01-01 G1 battle	Run existing G1 conditional Hall20/Band-I rules; no demo stats, paid counter, precharged Ultimate or mandatory technique override. Offer replay/recap on loss.	FC-C01-01 only after committed victory; no additional numeric grant.
F7 • debrief / next action	A short result card identifies source, counter, and next threat; inline SCN-C01-02 automatically becomes reviewable. Next G2 is prominently visible.	SCN-C01-02 occurs once. G2 shown; past script may be reopened.
F8 • optional return/idle example	Only after first meaningful combat, unobtrusively explain that E0 has been earning idle resources. Show Collect only if eligible; never require collection or a purchase to continue.	IDLE01 claim credits unique elapsed interval; no retroactive E0 or duplicate grant.
F9 • future chapter closures	G3 shows single FIRST02 + progress recap; G8 shows G8 dossier, COS-C03-01 and FIRST03 80R/120EXP in one receipt.	Single transaction per entitlement; revisits and scene skips give no additional grant.

[HANDOFF / CAVEAT] The ONB grant values above are those of §52, not a new grant. The per-band wallet is cumulative: after ONB01 spend the balance is 8/12; ONB02 brings it to64/96 before costing60/90, leaving4/6; ONB03 leaves4/6; ONB04 leaves8/12. The condensed display must never misleadingly show an unearned resource after a band. START00 ownership before the Hall is a presentation rule; the acquisition registry remains idempotent.

56.4 Canonical stage-to-story beat matrix: all 15 nodes, no hidden rewards

Stage / existing gate	Story beat; player-relevant question	Player learns or can choose	Receipt / immutable restriction
C01-01 G1 Wound	A medic reports delayed harm from one attacker. Can the team stop the source or sustain the wound?	Preview Wound tick timing; show K-09 vs K-10 available approaches and target preset.	FC-C01-01; zero extra priced currencies.
C01-02 debrief	An injured courier survives, revealing an ambush toward the convoy.	Optional wound recap; Next G2 always visible.	SCN-C01-02, no spendables.
C01-03 G2 Burst	A convoy is threatened by a telegraphed burst assault.	Protect a frontline ally or defeat the exposed attacker; show intent.	FC-C01-03; original B-E conditional card.
C01-04 formation briefing	A scrambled dispatch warns that the next enemy silences a predictable slot.	Optional formation-preview widget; player may reposition.	SCN-C01-04, no mandatory modal.
C01-05 G3 Silence / Ch1 end	Restore communications by protecting the shinobi whose Technique matters.	Understand Silence versus cooldown; recap and 5-seal FIRST02 once.	FC-C01-05 + FIRST02 once.
C02-01 K-08 alliance	A disruptor joins to prevent a catastrophic charged strike.	See K-08 first-copy interrupt; optional team-switch drawer.	SCN-C02-01 + STORY01 first copy before G4.
C02-02 G4 Charged	An enemy announces a charged attack on the supply route.	Interrupt through K-08 or protect with existing guardian.	FC-C02-02, no new resource price.
C02-03 K-04 alliance	A formation breaker joins after misleading attackers scatter the party.	Preview row targeting vs one-hit decoys.	SCN-C02-03 + STORY02 first copy before G5.
C02-04 G5 Decoys	False targets block straightforward single-target attacks.	Use K-04 row pressure or exhaust the decoy with guaranteed tools.	FC-C02-04; no false K-04 speed promise.
C02-05 backline tactic	Courier traces reveal a protected command unit behind guards.	Optional targeting illustration; explain that rear access is not equal to faster victory.	SCN-C02-05, no spendables.
C02-06 G6 Rear	A defended controller shelters behind the front.	Choose row pressure or legal frontline elimination.	FC-C02-06; three-round conditional paper card.
C03-01 attrition warning	Medical stocks are contaminated from two independent sources.	Preview simultaneous Wounds and the shared per-slot cap.	SCN-C03-01, zero reward currency.
C03-02 G7 Multi-Wound	The team must halt two Wound attackers without relying on rare cleansers.	Plan source elimination or mitigation; disclose Wound timing.	FC-C03-02, existing conditional packet.
C03-03 K-17 alliance	A field specialist identifies the clinic defender's self-recovery.	Auto-credit K-17; show anti-heal is probabilistic and optional.	SCN-C03-03 + STORY03 first copy before G8.
C03-04 G8 Capstone	The Medical Supply Commander self-heals and may activate a one-time Ward. Secure the route.	Preview healing, 50% HP threshold and visible Ward; choose suppression or core focus.	FC-C03-04 + cosmetic COS-C03-01 + FIRST03 80R/120EXP once.

[NARRATIVE QUALITY CHECK] The scenes must avoid asserting a new protagonist, fixed Naruto-era chronology, officially licensed location, voice cast or additional plot ending. Each scene carries one plot function and one mechanic-related statement; lore is expandable, not mandatory. SCN completion means the underlying information and any recruit have been durably offered/credited, not that every line of dialogue was read.

56.5 Player agency, reveal budget, accessibility and loss explanation

[WORKING DECISION 4D-03] Allow three independent preferences: (a) Guided/Compact mechanical teaching; (b) Read/Skip narrative dialogue; (c) reduced-motion/low-effects battle presentation. None alters enemy stats, payout

identities, eligible recruits, summon guarantees or mandatory combat difficulty. “Skip” for a narrative SCN advances the same receipt and credits its character before the next gate; “Compact” tutorial still processes ONB01–ONB04 safely. Do not invent an instant “Skip all battles” choice.

[REVEAL BUDGET] Show only the required fact for the present decision. At START01: Wound removal vs protective recovery. At G1: which enemy applies Wound, target legality and damage timing. At C01-04: who will be Silenced and how formation alters the threat. At G4: visible charging window. At G8: self-heal/one-time Ward and suppression’s chance. Advanced Chakra timing, relative SPD, detailed coefficients and complete status taxonomy remain available through tap-for-details or the replayable dossier. No numeric claim about reduced cognitive load or reading time is yet supported.

Surface / moment	Mandatory readable content	Optional expansion / guardrail
Starter choice	K-09: removes one Wound; K-10: shield and recovery. Both paths can clear the opening reference.	Exact stats, Wound tick clock, heal/shield card. No timer/default selection.
Enemy preview	Threat type, imminent target/row, legality of two accessible answers, boss Ward threshold where relevant.	Specific values and turn timing. Never imply guaranteed status application.
After battle	One sentence on why it ended, key counter used or missed, next available action.	Full initiative, Wound ticks, damage and cooldown traces; replay without currency.
Chapter reward	Show each unique grant with its source ID, credited or pending state, and current inventory.	Read previous receipts. Never label a failed/pending credit “collected”.
Accessibility	Text labels alongside icons/colors; adjustable text and subtitle display before any cinematic; tap-safe targets.	Reduce motion, skip animations but not tactical events, replay tutorials; verify target sizes on real devices later.

[EXTERNAL DESIGN CONTEXT] Apple Game Onboarding advocates interactive, progressive teaching, agency and optional tutorials; Roblox Creator Hub recommends context-specific teaching and observes FTUE funnels; Microsoft accessibility guidance recommends on-demand tutorials and subtitle controls; Android lifecycle guidance warns that activity interruption can lose state. These are *transferable practices*, not proof of our particular funnel speed or outcome. See §56.10 and source URLs.

56.6 Durable state, idempotent receipts and interruption/recovery contract

[STATE CONTRACT 4D-04] Treat UI presentation as a projection of an authoritative durable account snapshot: ownership entitlements (START00, START01, STORY01/02/03), ONB01–04 credit and debits, Hall level, E0 accrual anchor, formation/preset, battle first-clear flags, SCN receipts, FIRST01/02/03, cosmetic receipts, and optionally a current narrative scroll position. In a production design, all grant/debit/unlock mutations must be atomic or idempotent with a unique transaction key. Do not assume any particular backend or call this implemented.

Interruption / alternate path	Expected restore or duplicate behavior	Check / unresolved implementation question
Close between ONB grant and Hall purchase	Grant receipt persists; offer the specific funded band at next launch. No second grant.	If aggregate purchase partially failed, recover via transaction ledger, not optimistic UI.
Close after ONB04 but before support choice	Hall20, Band-I and FIRST01 survive. START01 remains unselected; show comparison on return.	Never silently choose K-09/K-10 from inactivity.
Skip STORY01/02/03 dialogue	Credit character once, record SCN and grant outcome, allow immediate legal next battle.	If already owned, credit an entitlement receipt only—not a free duplicate/shard.
Disconnect after story first-copy credit before recap	On return, show ownership and a one-time informational recap; no duplicate copy.	Does UI distinguish information shown from currency credited?
Close after boss win before FIRST03 display	Victory and FIRST03 are pending/committed atomically or in a recoverable workflow.	Persist an unclaimed/pending receipt; never fake a second 80R/120EXP.
Replay battle or story later	May view result and text; no first-clear, idle-rate, seal or character duplicate.	If first-clear result incomplete, reconcile server/account state before enabling claim.
Return after long idle absence	Show collected/uncollected idle bank per original 24h full + 48h quarter-rate cap. Next node persists.	No recurring story reward or implied 72h full-rate payout.
Low-end device dies during battle	Restore from last committed battle entry or valid completed result; no invented win/loss/reward.	Future engine/server must specify authoritative battle outcome protocol.

56.7 Paper funnel, qualitative exit gates and presentation counterfactuals

[MEASUREMENT REGISTER, NOT RESULTS] Later authorized studies should measure: first launch → START00 → ONB01–04 credited → START01 explicit choice → first meaningful formation/target decision → G1 battle submitted → G1 resolved → first recap seen → G3/FIRST02 → G8/FIRST03; revisit route after pause; how often optional story and battle details are opened. For each step record completion and abandonment with player consent and appropriate privacy controls. No rates, time-to-battle promises, conversion lifts, or statistical success are claimed in this document.

Question / failure test	Paper acceptance rule	What would reverse the proposal
Does a player understand why to choose K-09 or K-10?	Both choices describe accurate role and Wound timing, and neither is recommended as uniquely viable.	If comprehension fails, shorten comparison / add in-battle illustration, not mandatory copy.
Does compact onboarding secretly change progression?	Same ONB receipts, Hall20, 8R/12EXP remainder, FIRST01, E0, START00 and one START01.	If even one currency/ownership state differs, fail compact path.
Can a scene skip block a counter recruit?	STORY01/02/03 credited before eligible combat gate regardless of dialog visibility.	If grant arrives after battle unlock or gets lost, fail graph.
Can an abandoned session resume safely?	No extra grant, no vanished Hall purchase and no lost first-clear or title.	If any half-committed state cannot be reconciled, redesign transaction boundaries.
Does G8 feel like a chapter conclusion without a new reward faucet?	Boss reveal, legible self-heal/Ward counter, one receipt for title + 80R/120EXP; no new seals.	If later qualitative tests show no satisfaction, improve boss/story presentation then re-evaluate bounded rewards.
Does opening play teach tactics rather than menu compliance?	Before G1 the player is shown a meaningful, reversible formation/target choice and sees its battle impact in recap.	If choices have no observable effect, rewrite encounter/preview instead of more exposition.

56.8 Decision record, conflicts and unresolved trade-offs

Decision retained or newly proposed	Rejected alternative and concrete cost	Revisit trigger
4D-01: original “Broken Supply Route” narrative wrapper	Immediate Naruto canon cast/locale assumes rights and narrative period; original wrapper could feel less familiar.	Approved IP direction and narrative world bible.
4D-02: Guided and Compact ONB presentation share exact ledger	Skip all funding or inject free Hall20 state would erase source-cost trail; 19 forced level taps add friction.	A real mockup shows batch control confusing or guided route unengaging.
4D-03: support choice remains explicit, no hidden default	Timed/default K-09 choice could permanently alter roster without player agency.	If players abandon choice, improve comparison/deferral but preserve one-time entitlement semantics.
4D-04: scene skip awards its prior-defined recruit and receipt	Forcing reading to acquire K-08/04/17 turns counter into interface barrier.	Only legal/safety-critical notification must be separately acknowledged.
4D-05: minimal contextual story; optional transcripts	Full cinematic interrupts play, while no story erases motives and social context.	Future authorized tests show comprehension or emotional investment is poor.
4D-06: no engine/backend selected; save invariants are requirements	Assuming a cloud service, account identity or seamless offline merge would overstate implementation.	Technical architecture research chooses transaction, sync and conflict-resolution policy.

[RECONCILIATION NOTES] 1. §53 C03-04 originally marked boss payout “OPEN”; §55 selected FIRST03 once and supersedes that historical provisional zero in later paper budgets. 2. §52 credited FIRST01 at ONB04 and E0 at ONB01; neither is duplicated by narrative presentation. 3. §55 preserves seven SCN event identities; this chapter changes only presentation and authors original story semantics, not the stage order. 4. The boss G8 remains the 4,700-HP Medical Supply Commander with one conditional shield: story presentation cannot alter its timing. 5. TA-01 remains optional after C01-05 and does not become a required tutorial or daily quest.

56.9 Next research handoff: from script to mobile UX information architecture

[NEXT RESEARCH — PRIORITY 5A] With the canonical opening sequence and stage-semantic matrix written, design a navigation and state-information architecture for a small-screen phone: precise screen routes (Home/Training Hall/Squad/Campaign/Battle/Recap/Recruitment/Idle/Archive), notification priorities, persistent CTA hierarchy, account recovery and safe tutorial/popup interruptions. Trace two new-account paths (Guided/Compact), one returning player, and the two START01 choices without new power grants or alternate encounter outcomes. Test the *paper* route for every named reward and tool tip; flag unreadable or ambiguous flows. RECOMMENDED LEVEL: EXTRA HIGH, because UI

structure must reconcile transaction integrity, character choice, resource ownership and combat previews across multiple chapters. If scope narrows to typography and layout after this map, HIGH is sufficient.

56.10 Sources, evidence limits and preservation protocol

[FTUE1 — APPLE OFFICIAL, 1 NOV 2026] Interactive, progressive, optional game onboarding. https://developer.apple.com/app-store/onboarding-for-games/ ; https://developer.apple.com/design/human-interface-guidelines/onboarding

[FTUE2 — ROBLOX OFFICIAL, 1 NOV 2026] Contextual prompts and funnel analysis; no numeric rates imported. https://create.roblox.com/docs/production/game-design/onboarding ; https://create.roblox.com/docs/production/game-design/onboarding-techniques

[FTUE3 — MICROSOFT OFFICIAL, 1 NOV 2026] On-demand tutorials and subtitles; no compliance claim. https://learn.microsoft.com/en-us/gaming/game-publishing/concepts/metadata-accessibility

[FTUE4 — ANDROID OFFICIAL, 1 NOV 2026] Activity interruption motivates recoverable state; no backend chosen. https://developer.android.com/guide/components/activities/activity-lifecycle.html

57. Priority 5A: mobile navigation, information access and interrupted-session integrity

2 November 2026 | Version 0.10.30 | RESEARCH/CONCEPT ONLY. This packet is a logical UI and state-ownership architecture, NOT a designed screen mockup, clickable prototype, backend, telemetry deployment, or observed usability finding. Existing §§24–27 combat and §§40, 52–56 economy/campaign rules remain authoritative.

57.1 Problem, design principles and source boundaries

[QUESTION] On a compact phone, can a first-time or returning player always answer four questions—What should I do next? What will it cost? Why did I lose? Did my reward actually post?—without memorizing a large menu tree or risking duplicate grants? The answer must preserve the existing tactical and financial contracts.

[NON-NEGOTIABLE DEPENDENCIES] Starter entitlements START00 and explicit START01; training grants ONB01–04, aggregate band upgrade and Hall20/Band-I state; E0 idle accrual at ONB01; tutorial FIRST01; G1–G8/SCN graph; chapter FIRST02; G8 FIRST03 (80 Ryo/120 EXP); STORY01/02/03 grant before their gates; optional TA-01; 34 seals/60 fragments per full 28-day core; existing banner/selector disclosure; no new currencies or payout events. UI projection cannot edit these source rules.

[EVIDENCE] Official Android UX recommends 3–5 top-level destinations on compact screens, ≥48dp touch targets, and adaptive navigation for larger form factors. W3C WCAG 2.2 establishes a distinct web target-size minimum (24 CSS pixels subject to exceptions)—not a substitute for Android phone touch guidance. Android official state-restoration documentation distinguishes transient UI state from durable application data. All exact click paths and screen labels below are our own proposals, not externally validated benchmarks. See §57.11.

57.2 Top-level architecture: choose four persistent destinations

Alternative	Benefit	Cost / provisional decision
A. Four primary destinations: Home, Squad, Campaign, Recruit	Fast reach to recurring core loop; each label corresponds to a player intention	SELECTED. Idle is a Home action; Hall lives in Squad; Tactical Archive lives under Campaign; settings/details are contextual or profile routes.
B. Five including “Modes” / “More”	Broader future scalability for multiple optional modes	DEFER. Adds a permanent empty or weakly defined slot while the only designed optional solo mode is TA-01.
C. Everything in one scrollable Home dashboard	Potentially fewer top-level route changes	REJECT. Too many unlike actions; unclear hierarchy and no stable Squad/Campaign mental model.
D. Three destinations with Recruit under Home	Less tab competition	NOT SELECTED. Recruitment odds, currency use and guarantee warrant a distinct predictable destination.

[WORKING DECISION UX-01] For compact portrait phones, expose exactly four labeled bottom destinations, never icon-only: Home, Squad, Campaign, Recruit. The bottom bar persists on these four root pages, but focused Battle, Preflight, Recap, modal confirmation and sensitive transaction states replace or cover it. Hide only when doing so prevents an invalid route during an unresolved action. A tablet/landscape layout can later use a navigation rail without changing route identities. This does not select an engine or Android UI framework.

[RETURN MECHANISM] Every non-root screen has a labeled Back/Close and returns to the correct parent. Deep links reconstruct a parent route and entitlement guard; no push message or UI shortcut may bypass an unlock. Retain a “Resume mission” contextual action in Home, rather than creating a fifth permanent tab.

57.3 Screen inventory and route ownership

Route ID and parent	Primary player question / on-screen essentials	Authoritative information and allowed action

Route ID and parent	Primary player question / on-screen essentials	Authoritative information and allowed action
HOME / root	What is next? Show one current objective, claim-ready idle summary, limited wallet preview, resume/return status.	Derive from Hall, mission graph, pending receipt and claim state; route to next eligible action, never generate rewards.
SQUAD / root	Who is on the five slots? Formation (2 front/3 rear), roles, selected preset, missing counter warning.	Roster ownership, equipped Band-I gear, saved formation; mutations only via explicit Save.
HALL / SQUAD or HOME contextual	What upgrade is affordable, what will be consumed, and what will change?	Published Hall cost, owned wallet, before/after level; one transaction keyed by entitlement and selected band.
CAMPAIGN / root	Where am I in the 15-node graph? Threat, prerequisite, reward receipt and next legal node.	FC/SCN IDs, grant receipts, graph edges; story skip cannot withhold STORY character.
PREFLIGHT / CAMPAIGN	Who acts, what threat should I answer, and which two legal routes are available?	Enemy preview, target legality, current squad and accessible counters; editing squad returns here.
BATTLE / PREFLIGHT	What happened this round? Readable intent, target, statuses, HP/shields/Chakra and current action.	Resolved combat state; 1x/2x and low-effect settings are presentation only; cannot edit damage rolls.
RECAP / BATTLE or CAMPAIGN	Why did I win/lose? Damage/healing/control, first-death timing, one legal suggestion, receipts.	Battle result + transaction receipts; no implied reward until confirmed; retry retains same difficulty.
RECRUIT / root	What pool, price, odds, guarantee and spend source apply before purchase?	Live eligible banner definition, wallet, pity/selector ledger; explicit confirm before immutable purchase.
IDLE / HOME	What accrued, at which rate, since when, and what remained uncollected?	E0 or earned later rate, eligible elapsed time and claim receipt; never client-calculated credit.
ARCHIVE / CAMPAIGN	Can I practice a known lesson without stamina, recurring resources or ranked pressure?	TA-01 unlock, cosmetic first-clear receipt and replayable result; no recurring payout.
DOSSIER / context	What does a status/skill/role/counter actually mean?	Authoritative kit/version and status timing; accessible from squad, preflight, battle pause and recap.
WALLET / contextual	What is each balance for and where is an eligible source?	Currency balances and canonical source IDs, cooldowns and eligibility—not invented “earn now” shortcuts.
SETTINGS / profile	Adjust motion, effects, sound, captions, text preference, notification opt-in.	Presentation preferences only; never creates alternate entitlement or combat calculations.

[RULE] “Primary CTA” is contextual rather than universally “Battle”: new accounts see Complete Training / Choose Support; later accounts see Continue Mission; returning accounts may see Review Result or Claim Idle. One dominant action is allowed per surface. A shop promotion cannot displace a blocked essential counter, training payment, mission continuation, or pending-result recovery.

57.4 Logical journeys: new accounts, returners and starter-choice parity

Profile and trigger	Minimal legal route (routes, not fabricated measured taps)	Invariant / stop condition
Guided first session	HOME/ONBOARD → START00 → ONB01–04 via HALL → START01 K-09 or K-10 → SQUAD/PREFLIGHT → G1 BATTLE → RECAP	Hall20, Band-I, FIRST01 exactly once; same total Ryo/EXP as compact. Start choice explicit.
Compact first session	HOME/ONBOARD → START00 → four validated batch-band receipts → START01 explicit choice → PREFLIGHT → G1 → RECAP	Only explanations and repeated clicks compress, not grant IDs, battle count or cost.
Returning after >72h	HOME with rate-aware accumulated claim and “bank capped” notice → IDLE → Claim receipt → Resume Campaign/Archive	No unbounded backfill, no double claim, no requirement to watch idle animation or purchase.

Profile and trigger	Minimal legal route (routes, not fabricated measured taps)	Invariant / stop condition
Interrupted after G8 victory	HOME pending resolution → fetch authoritative G8 result and FIRST03 + cosmetic receipt → RECAP	If victory committed, show same reward exactly once. If unresolved, do not infer either win or loss.
Recruiting after weekly reward	HOME/WALLET → RECRUIT → banner/selector detail → visible terms and confirmation → receipt → SQUAD	Keep free/paid balance types, pull eligibility and pity visible; no claim that all seals guarantee banner within 56 days.
G4/G5/G8 story skip	CAMPAIGN → SCN Skip/Read → check STORY01/02/03 posted → PREFLIGHT	K-08 before G4, K-04 before G5, K-17 before G8; do not require scene viewing.

[PATH LIMIT] These are designed route transitions, not observed interaction times. Bottom-tab roots should normally be reachable in one top-level selection; the optional Hall and Idle pages are one contextual route from Home when relevant. We do not certify “two taps” without a rendered prototype or motor-accessibility review.

57.5 Notification and information priority: reveal only what matters

Priority	Example and placement	User action / suppression
P0 — unresolved integrity	G8 result still reconciling, ONB or recruit transaction pending; persistent nonblocking status card	Check status; do not offer a second submission or falsely claim success.
P1 — one time critical decision	START01 Choose K-09/K-10; first legal formation adjustment before G1	Requires explicit choice/Save; never auto-select due to timer or skip.
P2 — actionable gameplay	Preflight enemy threat, Wound next-action tick, K-08 charge window, K-17 status chance	Inline preview and optional details; match actual automatic AI timing.
P3 — routine value	Idle ready/capped, available Hall purchase, CORE objective completed	Home card; no full-screen auto-popup on every return.
P4 — optional discovery	Tactical Archive, lore, cosmetic title, feature announcement	Section label and opt-in; never obscure tactical warning or receipt.
P5 — commercial content	Shop / pass / cosmetic offer	Separate from required navigation; no deceptive “Fix defeat” purchase prompt.

[LANGUAGE RULE] A loss statement must identify a grounded cause (“Silence denied K-03’s ready Technique”) and a legal response (“Move the predicted target / focus the controller”) only when the battle log supports it. Avoid “Power too low” as an unverified general diagnosis or redirecting a free player straight to Recruit. Distinguish “available,” “owned,” “equipped,” and “fieldable.”

57.6 Action-state contract, purchasing and reward receipts

User-visible action	Before submission / pending	Authoritative outcome / failure recovery
ONB training band	Exact grants, Ryo/EXP debit, before/after Hall; prevent shop diversion	One posted band transaction or no mutation; retry checks same ONB receipt.
START01 support choice	Two first-copy kits and trade-off; explicit select and confirm	K-09 OR K-10 receipt, not both; if interrupted before commit, show choice again without guess.
IDLE claim	Eligible hours, rate steps, bank cap and estimate; distinguish estimate from confirmed balance	Post unique claim interval; display exact credited vector; if unknown, refresh state not re-issue award.
Campaign first-clear	Preview FIRST02/FIRST03 where eligible, show one-time and replay distinction	Victory record → receipt; replay shows already claimed; G8 material+title remain separate receipt classes.
STORY recruit from scene	Before G4/G5/G8 verify entitlement independent of dialogue read flag	Post STORY01/02/03 once even on Skip; reenter preview only when ownership confirmed.
Summon or selector purchase	Current wallet, banner eligibility, odds, 70-pull guarantee scope, fragment selector terms, currency order	Confirm price, unique purchase key, durable pull/pity/account result; network error never means free repeat.
Saved formation	Slots and targeting preset preview, show who leaves team, no destructive gear transfer	Save explicit new formation; invalid five-slot or targeting rule blocks battle start.

User-visible action	Before submission / pending	Authoritative outcome / failure recovery
Battle result	If interrupted after submission, display “Checking result” not auto-retry	Reconcile authoritative battle id and reward receipt; render replay/recap if committed; pending cannot grant.

[TRUST BOUNDARY UX-02] Never derive owned currency or first-clear rewards from an animation, local countdown or toast. The UI may show provisional estimates, but only an authoritative transaction receipt changes the displayed confirmed wallet. Receipts expose source ID, debit/credit, before/after balance when available, and whether already claimed. Back, retry and reconnect must not replay the grant.

[INFORMATION SEPARATION] Display user-earned CORE01/02 objectively; FIRST01/02/03 as one-time; STORY01/02/03 as noncurrency first copies; TA-01 emblem once; E0 as time-accrual. Do not combine these into a misleading “daily prize” total. A locked mode is labeled “Not unlocked” with its actual condition, not “Claimable.”

57.7 Compact combat information and accessibility envelope

Element	Default concise display	On-demand explanation / restriction
Top status / action order	Five ally portraits, five enemy silhouettes, clear active unit and readable next threat	Show Speed explanation and resolved order; never conceal a charge or Silence target behind particle effects.
HP, shield and Chakra	Separate HP, shield and personal Chakra meters with numeric inspect	Color + icon + text, never hue alone; Ultimate 100 cost and legality visible.
Wound / Silence / Stun / Heal suppression	Distinct icon + duration/next trigger glyph, stack count and affected target	Explain same-round next-slot tick, status chance and current cleansability.
Formation and target eligibility	2 front / 3 rear representation, selected preset and predicted enemy target	Inspect legality and who is protected; warn if an auto-Technique target is illegal.
Battle recap	Cause of win/loss, first decisive enemy action, damage/heal/control contributions	Detailed event log and one accessible counter hint when supported by recorded state.
Touch / text / motion	Proposed ≥48dp targets with ≥8dp separation for critical phone actions; text-safe scaling and reduced motion	W3C web 24 CSS-pixel minimum is a different criterion. Values here are proposed Android-inspired screen acceptance constraints.
Audio / interruptions	Text/visual warnings independent of sound, optional captions and no forced animation to claim	Device interruption restores from durable snapshot. Reduced effects does not remove actionable telegraph.

[RESPONSIVE SCOPE] Portrait-compact is the design reference; tablet and landscape may use a rail and wider recap. The state and route IDs must not change. Exact pixel font sizes, safe areas, thumb-reach compliance, color contrast, 30fps low-effects performance and screen-reader focus order need prototype/device validation; no compliance is claimed here.

57.8 Conflict and edge-case paper audit

Case	Proposed UI behavior	Unresolved proof obligation
Player leaves mid ONB03 grant/purchase	Reopen Hall with posted receipt and net balance, or safely resubmit same idempotent key	Atomic transactions and exact crash recovery require technical design.
Story dialog skipped at K-08 grant	Confirm owned K-08 before G4 preflight, regardless of read flag	Scene UI and ownership storage cannot race.
G8 victory posts while app closes	Home pending status resolves same battle ID, shows FIRST03 once and title once	Server authority, battle persistence and offline policy unchosen.
Idle bank cap reached	Explain 24h full +48h quarter + stop, show elapsed/uncredited time honestly	Exact UI display for time zones/clock drift still unspecified.
No legal Technique due to Silence	Battle shows Basic fallback and denied Technique reason in action/recap	Critical for 3E/3F control teaching; no hidden manual override.
Recruitment intent with insufficient eligible seal	Show source eligibility and actual pity route before a purchase choice	Banner-specific free/paid order and live terms need later product/review audit.
Player with small screen or larger text	Allow scrolling/details; keep action buttons and preflight threats visible; no clipping	Physical-device text scaling and TalkBack not tested.
Returning player sees 5 alerts	Display highest priority actionable item plus collapsed others; no serial modals	Funnel/user research must determine which notice is salient.

57.9 Decision register, trade-offs, falsifiers and exclusions

[SELECTED PROPOSALS] UX-01 four persistent roots; contextual Hall/Idle/Archive/Dossier; battle-focused overlays; one dominant actionable next step; priority P0–P5; explicit transactional receipts; preflight→battle→recap with causally grounded explanations; two presentation paths with identical progression. These are design decisions for future testing, not coded features.

[REJECTED] A fifth empty Modes tab; forced separate UI modal per SCN node; automatic START01 timeout choice; Home as a monetization billboard; treating a wallet animation as credit authority; claiming every defeat means insufficient power; assigning imaginary eligible sources; adding a new currency to make a menu feel busy.

[COSTS / TRADE-OFFS] Four roots push future social/endgame modes into a later navigation review; Home contextual shortcuts increase conditional UI complexity; exposing receipts requires a deliberate data schema; preflight explanations consume limited screen space; safely hiding commercial prompts may reduce early monetization impressions. These costs are accepted provisionally to preserve player agency, fairness and trust.

[FALSIFIERS / REOPEN IF] An authorized small-device study shows a substantial share of players cannot find Recruitment/Hall/Archive; first-session players misinterpret the support choice; returners mistake capped accrual for lost rewards; effect icons remain unreadable at enlarged text; preflight cannot show legal counters without deep nested navigation; critical actions exceed touch target bounds; root nav changes break uninterrupted battles. No observed data exists yet.

57.10 Next research handoff and performance-level choice

[NEXT RESEARCH — PRIORITY 5B] Specify screen-by-screen minimum content and hierarchy on a small portrait phone: Home, Squad/Hall, Campaign/Preflight, Battle/Recap, Recruit/Wallet and claim flows. Write the exact short player-facing copy for FIRST01/02/03, START01 and idle overflow without changing transaction amounts. Test static paper layouts against 48dp target envelopes and large text, with explicit missing-space notes. Then trace a first session and returning session, comparing whether players can reach the next meaningful action and understand why. Performance recommendation: HIGH for bounded visual hierarchy/copy once §57 route architecture is stable; use EXTRA HIGH if downstream state, purchase or combat semantics must be reconciled. No production, prototype, simulator or live user test until owner authorizes.

57.11 External references and limits

[UXA1 — ANDROID OFFICIAL, accessed 2 November 2026] Layouts and navigation patterns: compact navigation bar commonly 3–5 destinations, adaptive rail for larger layouts; descriptive guidance, not a Naruto game requirement. https://developer.android.com/design/ui/mobile/guides/layout-and-content/layout-and-nav-patterns

[UXA2 — ANDROID OFFICIAL, accessed 2 November 2026] Mobile app accessibility: recommend ≥48dp touch targets. https://developer.android.com/guide/topics/ui/accessibility/views/apps-views ; https://support.google.com/accessibility/android/answer/7101858?hl=en

[UXA3 — W3C OFFICIAL, accessed 2 November 2026] WCAG 2.2 criterion 2.5.8 has web 24×24 CSS-pixel minimum with exceptions; do not confuse this with native Android 48dp guidance. https://www.w3.org/TR/wcag/

[UXA4 — ANDROID OFFICIAL, accessed 2 November 2026] Save UI states: transient screen state is not persistent game ownership/wallet data. This is an architecture analogy only; no technology is selected. https://developer.android.com/topic/libraries/architecture/saving-states

[UXA5 — APPLE OFFICIAL, accessed 2 November 2026] Design for games encourages teaching through play and effective defaults; transfer principle only. https://developer.apple.com/design/human-interface-guidelines/designing-for-games

58. Priority 5B: compact-phone screen hierarchy, exact microcopy and accessible states

2 November 2026 | Version 0.10.31 | RESEARCH/CONCEPT ONLY. This packet is a paper UI/content specification. No visual mockup, running UI, physical device measurement, screen-reader inspection, usability session or backend implementation was performed. Canonical §57 routes and §§24–27, 40, 49–56 authority remain unchanged.

58.1 Decision scope and invariant checklist

[DESIGN QUESTION] What is the smallest amount of truthful information each critical phone screen can show before the player acts, while still exposing a tactical response, real source/cost, and confirmed reward state? The current objective is legibility and state honesty, not maximizing the number of metrics on screen.

[UNCHANGED CONTRACT] Four roots: Home, Squad, Campaign and Recruit. Child pages: Hall, Idle, Preflight, Battle, Recap, Wallet, Details and Tactical Archive. START00 four recruits; ONB01–04 to Hall20/Band I; START01 explicit K-09 or K-10; FIRST01 five seals; FIRST02 five seals; FIRST03 80 Ryo and 120 EXP; STORY01/02/03 K-08/K-04/K-17; G1–G8; TA-01 cosmetic emblem. Current CORE01/02 and E0 are unchanged. Pity and acquisition visibility remains controlled by §§36–40.

[INFORMATION PRIORITY] First: an unresolved integrity state or irreversible choice; second: next legal action with cost/eligibility; third: immediate tactical reason; fourth: secondary details. Never use hidden tooltips as the sole carrier of a dangerous enemy intent, source shortfall or irreversible selection.

58.2 Reference small-screen geometry and scaling stress case

Use 360×640 dp as an ILLUSTRATIVE portrait design canvas, not an assumed minimum supported phone. For paper placement only, reserve approximate 24dp status, 56dp app header, 80dp root navigation and 24dp gesture/navigation inset, yielding 456dp of flexible content. Actual status/navigation insets and device sizes vary; these allocations are not measured. Battle, Preflight and Recap may replace the root bar to return space to priority information. Do not hard-code screen-height assumptions.

Area	Compact working hierarchy	Large-text / failure behavior
Header (approx. 56dp)	Screen title, labeled Back/Close where needed, one contextual utility; no rotating promotion	Allow title to wrap or truncate with accessible full label; do not shrink below legibility
Primary task area	One next action or danger summary in the first scroll region; lead with verb and actual prerequisite	The dominant action can move lower in a scroll surface, never underneath an overlay
Secondary detail	Expandable cost breakdown, noncritical metrics and receipts; disclosure remains one labeled action away	Content can grow vertically; avoid fixed-height text boxes and horizontal scrolling of prose
Bottom destination bar (roots only)	Home / Squad / Campaign / Recruit: icon plus text labels, distinct selected state	Each destination retains a 48dp or greater touch envelope; no two-row compression
Touch / activation envelope	Working Android target at least 48×48dp and about 8dp between neighboring hit regions	Visual icons may be smaller than their active hit area; do not overlap regions
Stress profiles	Illustrative 360×640 and constrained 320×568; 100% and 200% font-scale paper passes	At enlarged scale, show core task and important warnings in the vertical scroll, with accessible non-drag alternatives

[EXTERNAL GUIDANCE] Android recommends ≥48dp touch targets and commonly a 3–5 destination compact navigation bar. WCAG 2.2 SC 2.5.8 uses a distinct web 24×24 CSS-pixel threshold with exceptions. Treat 48dp as our conservative native-phone reference; neither guideline constitutes a tested pass for our unrendered UI.

58.3 Screen-by-screen minimum visible information: strict content contracts

Screen / route	Always visible before decision	Optional detail / one dominant action
Home	Current objective with verb, next eligibility, current pending result if any; idle-bank status if actionable	Show Continue training / Choose support / Review result / Continue mission or Claim idle as context demands; wallet breakdown can expand
Squad / Hall	Five slots and 2-front/3-rear roles; current Hall level and attainable next cost; distinguish owned vs fieldable	Inspect abilities/gear; Train only after presenting Ryo+EXP+essence and before/after states; warn of missing input

Screen / route	Always visible before decision	Optional detail / one dominant action
Campaign / mission list	Current node, cleared/locked status and nearest next legal fight; campaign chapter context	Continue mission; expand dossier/earlier scenes; a locked node states a verifiable condition
Preflight	Enemy intent, timing (e.g., Wound at next victim action), legal target rules, accessible responses and team formation	Review counters/Adjust squad first; Begin Battle only after valid formation and pending receipt checks
Battle	Active actor; readable ally/enemy HP and shield; current personal Chakra; charged intents; next important status tick	Pause/Details; auto-actions remain authoritative; avoid claiming manual skill override
Recap	Victory/defeat; real causal event; exact reward receipt status; one legal next action	Timeline, damage/healing/control breakdown; do not assert a loss cause not supported by the event log
Recruit / banner	Banner type, useable seal quantity, pool eligibility, rate disclosure and applicable pity family	Recruit with explicit cost and acknowledgement; until committed, show Pending rather than new ownership
Wallet / reward claim	Confirmed balance; distinct one-time/weekly/idle source labels and claim state	Show source ID, debits/credits, before/after when available; retry reads the same receipt, never duplicates grant
Tactical Archive	Lesson goal, no-stamina/free-repeat rule, one-time cosmetic reward state	Practice or Review Explanation; no recurring seals/materials

[EXCLUSION] Do not push the Store ahead of a legal counter or a pending-reward recovery. Keep cosmetic monetization outside the primary task hierarchy. A visible “Claim” control cannot represent an owned balance until confirmation.

58.4 Exact player-facing copy and transaction states

Copy below is the DEFAULT ENGLISH WORKING SCRIPT. Braced labels such as {X} are runtime substitutions from authoritative state and must never be replaced with made-up numbers. Confirmed statements appear ONLY after a posted source-ID receipt.

Source / state	Exact short copy	When allowed / safe action
START01 unchosen	Choose your starter support. K-09 removes one active Wound on its eligible turn. K-10 protects allies but does not remove Wound.	Two selectable cards, explicit Confirm choice, no timeout, no automatic default
START01 pending	Saving your choice… Please do not choose again.	Lock second submit, offer Check status; reconstruct result on return
START01 confirmed	{K-09 or K-10} joined your squad. Your choice is saved.	Only after confirmed ownership/choice receipt
FIRST01 confirmed	Training complete. 5 Recruitment Seals added.	Only after FIRST01 receipt, exactly once
FIRST02 confirmed	Chapter 1 complete. 5 Recruitment Seals added.	Only after FIRST02 receipt, exactly once
FIRST03 confirmed	Chapter 3 complete. 80 Ryo and 120 Shinobi EXP added.	Only after FIRST03 receipt; cosmetic title shown separately
Idle active	Idle rewards saved: {elapsed}. Full rate for the first 24h; quarter rate for the next 48h.	Elapsed and collectible estimate must be computed from actual saved accrual
Idle bank full	Idle bank full. Rewards stopped accumulating after 72h. Claim to restart accrual.	Show only when full; never promise retroactive gains beyond the cap
Idle pending	Checking your idle reward claim…	No repeat grant; reconcile identical receipt ID
Idle posted	Idle rewards collected: {confirmed resource breakdown}.	Post-confirmation; record collection interval/source
Missing Hall resource	Cannot train yet. Need {missing amount}{resource}. Available source: {eligible source}.	Only name a source this account can currently reach; otherwise say “No eligible source is available yet”
Battle loss diagnosed	K-03’s ready Technique was blocked by Silence. You can change the predicted target or focus the controller.	Only when log establishes readiness, denied action and valid answers
Unknown battle loss	Battle lost. Review the action timeline before changing your team.	No speculative “power too low” or paid purchase recommendation
Pending battle receipt	Checking your saved battle result… Your rewards are not yet confirmed.	Reopen pending receipt and preserve battle outcome without resubmitting the grant
Already claimed	Already collected. Your balance includes this reward.	After duplicate-safe verified receipt lookup

58.5 Preflight, battle and recap: causal explanation without overload

Moment	Initial concise information	More detail if requested / failure case

Moment	Initial concise information	More detail if requested / failure case
Before G1 Wound	“Enemy plans Wound. Wound damages its target at the target’s next action—even in the same round.”	Explain K-09 action timing and source-focus route; preview must not imply cleanse prevents an earlier tick
Before G3 Silence	“Enemy targets this formation slot with Silence before K-03 acts.”	Show predicted target and that Silence can deny a ready Technique; show formation adjustment when legal
Before G4 Charge	“Enemy is charging an attack. K-08 can interrupt; K-01 can absorb a redirected hit.”	Show action window and visible target/Guard legality; do not promise universal survival
Before G5 Decoys	“Decoys intercept eligible single-target hits. A row attack can bypass them.”	No claim that K-04 always wins in fewer rounds
Before G8 Healing	“Boss heals after acting and may raise a one-time Ward. Focus or reduce healing.”	K-17 suppression is probabilistic; keep damage-focused route accessible
During battle	Show action actor, predicted target where applicable, current status effect and next tick/charge	Tap effects for source, duration, cap and actual timing; never rely on red/green alone
After battle	One supported causal sentence plus 1–2 permitted actions; reward status with source ID	If a cause is not proven by log, show neutral recap; show numerical breakdown only on expansion

[HONESTY LIMIT] “Next turn” is prohibited as Wound shorthand when the victim’s next action is still in the current round. Use “next action” or exact timing. The client must not infer K-03 readiness merely from a status icon; it needs the resolved combat event log.

58.6 Two paper journeys and 200% text stress test

Persona / moment	Next action discoverability	Stop / recovery invariant
Guided new account	Home → training (ONB01–04) → explicit START01 support → Squad/Preflight G1. Expandable coaching explains each step.	No First01 seals before confirmation; one support kit, Hall20/Band I; do not suppress required choice
Compact new account	Same required receipts and START01, shortened explanations and grouped presentation; G1 remains first tactical check.	Shorter copy ≠ different rewards, free levels or fewer meaningful choices
Daily return	Home distinguishes ready idle claim, confirmed last receipt and current mission; player chooses next meaningful action.	Never reset current stage or generate a new first-clear reward
Once-weekly return	Home notes bank cap of 72h and shows actually collected capped resources; no guilt or fake “lost all rewards” alert.	Progress and inventory persist; returner sees the same eligible source and truthful capped accrual
Connection lost mid-claim	Home/Recap shows “Checking…” and one Check status option; the original receipt is recovered.	No second grant ID, no phantom balance, no infinite loading as a claimed success
Large-text / 320×568 stress	Primary task and risk copy wrap in a scrollable column; utility cards become vertical; no two-column dense data.	At 200% text scale, no clipping/overlap, no overlapping touch regions, readable buttons; actual pass requires device render

58.7 Negative-state contracts, trade-offs and acceptance criteria

State / candidate	Selected response and cost	What would falsify it
Small screen vs all battle metrics	Show only active actor, critical intent, HP/shield, Chakra and actionable status first; deeper metrics on demand. Costs extra inspection taps.	A player cannot anticipate damage/status with basic view, or misses the most urgent intent
One CTA vs multiple claims	One dominant CTA, secondary claim still reachable if permitted. Costs conditional prioritization logic.	Users cannot locate Claim Idle, START01 or Review Result at a sensible point
Exact receipt copy vs polished celebration	Show receipt-confirmed amounts first; animations/cosmetics may follow. Costs spectacle-first sequencing.	Players treat unposted rewards as owned or lose trust after reconnection
Large text vs dense squad card	Switch cards to vertical stacking and natural-height content; prefer scroll over tiny text. Costs visible-at-once density.	Content clips, CTA becomes unreachable, icons lack text alternatives, touch regions overlap
Paper compliance vs real accessibility	Record these as design constraints, not WCAG/Android compliance claims. Costs additional future QA and physical testing.	Any real device/screen-reader audit fails a target or source/intent explanation

PAPER ACCEPTANCE GATES (not observed outcomes): each root has one meaningful primary task and a correct parent route; all first-clear rewards have exact source IDs and explicit pending/posted states; every first-gate Preflight describes a legal accessible answer without relying on a premium hero; user never must infer a Wound tick from “next turn”; every mandatory touch control has planned ≥48dp hit area; no important information depends on color alone; 320×568 at 200% text admits scrolling rather than clipped or hidden controls; both Guided and Compact remain receipt-equivalent.

58.8 Decisions, rejected alternatives and next research handoff

[WORKING DECISION 5B-01] Use one-CTA-per-surface hierarchy, cause-first preflight/recap, truthful transaction microcopy, source-specific collection states and a vertical large-text fallback. Keep four roots and all combat/economy numbers unchanged. A complete visual design has NOT been produced.

[REJECTED] Fixed-height cards that truncate warnings at enlarged text; icon-only status indicators; “One tap to win” or “Power too low” diagnoses; a full-screen reward celebration before confirmation; conflating once-weekly bank saturation with loss of existing inventory; automatically choosing K-09/K-10; giving out unopened first-clear rewards to make a UI animation look consistent.

[UNRESOLVED] Exact visual hierarchy after localisation, 320dp phones with display cutouts, screen-reader focus/announcements, right-to-left support if needed, device-specific font scale, keyboard and gesture accessibility, error focus restoration, reduced-motion interactions and actual click/touch burden. No empirical claim can be made until an authorized prototype and device checks.

[NEXT RESEARCH — PRIORITY 5C] Before progressing to social/PvP interface designs, audit failure/recovery and accessibility across the critical journeys: pending claim, app restart mid-recruitment, first-time combat loss, 72-hour bank, enlarged text and screen-reader narration. Produce a risk-ranked scenario matrix and minimal error-state copy, reconcile any contradictions with §57/58. PERFORMANCE LEVEL: HIGH, because this is a focused accessibility and state-communication audit; raise to EXTRA HIGH only if an account-state or combat semantics contradiction forces a system redesign. Remain in theory/research; do not implement or run usability tests.

58.9 Official sources, consulted 2 November 2026

[5B1] Google, Android Accessibility: touch target size ≥48dp; recommends target spacing and large accessible regions. https://support.google.com/accessibility/android/answer/7101858?hl=en

[5B2] Google, Android Developers: navigation bars are appropriate for three to five primary destinations on compact devices. https://developer.android.com/design/ui/mobile/guides/layout-and-content/layout-and-nav-patterns

[5B3] W3C, WCAG 2.2 SC 2.5.8: 24×24 CSS px minimum for web pointer targets subject to exceptions. This is NOT a native Android 48dp equivalence claim. https://www.w3.org/TR/WCAG22/

[5B4] Google, Android Developers: make views accessible through text descriptions and touch target sizing. https://developer.android.com/guide/topics/ui/accessibility/views/apps-views

[PROJECT CONTEXT] §§23, 26–27, 40, 49–57 of this master; sources here support UI accessibility concepts only, not the proposed game percentages, currencies, copy or engagement outcomes.

59. Priority 5C: accessible recovery, transaction uncertainty and first-loss integrity

2 November 2026 | Version 0.10.32 | CONCEPT RESEARCH ONLY. This chapter defines proposed accessibility and recovery behavior. No code, simulated backend, physical-device inspection, screen-reader testing, purchase, user research or security verification was performed. Existing §§23–27, 40, 49–58 are authoritative unless a change is explicitly proposed below.

59.1 Research question, sources and strict non-regression boundary

[DESIGN QUESTION] Can every critical action be understood and safely resumed after a network failure, app closure, large-text reflow or assistive-technology interruption—without inventing combat causes, spending a resource twice or announcing an unconfirmed reward?

[UNCHANGED PROJECT CONTRACT] Four navigation roots (Home, Squad, Campaign, Recruit); Hall 20/Band I onboarding through ONB01–04; START00 roster and START01 deliberate K-09/K-10 choice; FIRST01/FIRST02 five seals each; FIRST03 80 Ryo plus 120 EXP once; STORY01/02/03 K-08/K-04/K-17; G1–G8 and TA-01; IDLE01 24h full rate + 48h quarter rate, then accrual pauses; CORE and banner/fragment rules unchanged. No new currencies, attempt fees, shop shortcuts, combat mechanics or numerical rewards in this pass.

[SOURCE FINDING] W3C WCAG 2.2 guidance distinguishes status messages that should be announced without stealing focus from meaning-changing navigation; its focus-order guidance requires a logical sequence. Android advises meaningful unique labels for interactive controls, with decorative elements excluded, and offers accessibility announcements through native semantics. Google Play Billing requires verified PURCHASED status before paid entitlement and a separate acknowledgement/consumption path. These are comparisons and future engineering requirements, NOT proof that this proposed game conforms to WCAG, Android accessibility, or billing policy.

59.2 Risk-ranked paper scenarios: player outcome and invariant

Priority / interruption	Visible state and next safe action	Required invariant / failure gate
P0 · Claim response lost (FIRST01–03 or weekly)	Show “Checking saved claim…” and one “Check status” action; after confirmed receipt show exact source and new balance.	Same account/source grants once; unknown response is not a failure. Never display an unposted resource as owned.
P0 · Recruit app closes after submission	On reopen show “Checking recruitment result…” and same request; show shinobi and pity change only from completed receipt.	Repeated device request does not consume another seal, grant a second shinobi or advance pity twice.
P0 · Support choice closes during confirmation	Restore selected K-09/K-10 preview and committed START01 result; if unknown, block conflicting new selection pending reconciliation.	Exactly one START01 hero; no automatic selection; same entitlement on Guided/Compact and scene skip.
P0 · Paid purchase pending or uncertain	Separate platform pending from account grant: “Payment pending. No items delivered yet.” Link to status, not “Buy again”.	Verify purchase with store/backend before granting; no duplicate delivery on callbacks or relaunch.
P0 · Hall purchase response lost	Show prior confirmed Hall and current pending attempt; resolve transaction receipt before allowing another promotion.	No double Ryo/EXP charge; no level jump on UI optimistic guess.
P1 · First combat loss and recap reload	Restore confirmed loss and event-log-based diagnosis, or “Battle details unavailable” plus basic formation guidance.	Never invent which character acted, was stunned or targeted if record absent; replay cannot duplicate first-clear.
P1 · Idle claim at or after 72h bank cap	Display accrued payable amount, 24h + 48h at 25%, and stopped accrual beyond 72h; offer Claim or status check.	Time windows never double claimed. Do not imply stored currency disappears after 72h.
P1 · Return mid-story or before recruit	Resume at the authoritative stage/choice; offer Review Story and Continue; confirm grants independently of scene playback.	Skipping/read-state cannot suppress STORY grants or re-run FIRST receipts.
P1 · TalkBack/Switch Access loses focus	Restore focus to meaningful pane title/status then triggering control, or safe nearest parent if removed.	No focus trap or jump behind a modal; irreversible choice retains confirm/cancel.
P2 · 320dp / 200% text / landscape interruption	Reflow details as vertical scroll; preserve confirmation controls and readable warning text.	No clipped status, overlap, essential control off-screen with no scroll, or color-only meaning.

59.3 Transaction state machine and receipt identity

[PROPOSAL 5C-01] Formalize source-specific state phases: NOT_SUBMITTED → SUBMITTED_UNKNOWN → CONFIRMED_POSTED or CONFIRMED_REJECTED, with ALREADY_POSTED as a resolved replay of an existing result. A local animation or HTTP timeout cannot move the account to CONFIRMED_POSTED or CONFIRMED_REJECTED. The system of record and receipt type must be selected in future architecture; do not pretend a backend now exists.

[IDEMPOTENCY DESIGN] One-time source entitlements should have an account-specific source uniqueness constraint (e.g., account + FIRST03). A recruitment/spend request needs a stable client-generated request identity reconciled with a server-recorded result and atomic costs/rewards/pity changes. Do not use the recruitment animation, screen transition or

current balance alone as the receipt. Retried status checks re-use the original identity; a deliberate new summon is a distinct confirmed action.

[BOUNDARY] For optional paid purchases, the provider purchase token and PURCHASED/PENDING state are separate from internal source IDs. Verification and acknowledgement/consumption belong to a future platform integration. Do not instruct the user to repurchase while status is unknown. No vendor or backend is selected by this research.

Action / receipt key	Uncertain state (allowed)	Settled response / prohibited behavior
FIRST01 tutorial; FIRST02 chapter; FIRST03 G8	Re-check by account + unique source; show “Checking saved reward…”	Confirmed exact amount; otherwise nonfinal pending. Prohibit animation-based grants and duplicate source awards.
START01 support choice; STORY01/02/03 recruits	Choice pending or saved; identify claimed source before presenting choice again	Grant precisely one selected starter; persistent story shinobi ownership despite skip or replay.
IDLE01 collection period	Pending claim ID and server-stamped interval; keep collect action guarded until known	Post only eligible elapsed seconds; preserve last claimed timestamp; no double-count overlapping intervals.
Hall upgrade	Pending atomic debit/level change; show confirmed prior level until resolution	Both debits and advancement commit together or neither; no duplicate promotion.
Recruitment Seal / featured or standard pity	Original request submitted, result unknown; only status retry	One seal cost and one pity/result outcome per request; no new pull without fresh player intent.
Google Play optional purchase	Provider pending/verification/acknowledgement distinct from in-game result	No goods on PENDING; verify PURCHASED, then grant once and acknowledge/consume as appropriate.

59.4 Accessibility semantics, focus and announcements

[PROPOSAL 5C-02] Every actionable element exposes accessible name, role, value/state and next consequence; a decorative portrait/artifact has no redundant accessibility focus. Screen-reader reading order should mirror causal order: screen identity → important warning → current value/source → available action → secondary detail. Never use a status icon, color change, vibration or animation as the only message.

[ANNOUNCEMENT POLICY] Announce pending → confirmed/rejected changes politely once per semantic change, not on every spinner tick or Chakra frame. Do not move focus for a transient confirmation toast. For explicit dialogs/irreversible starter selection, give the dialog a title and move focus into it; after dismissal return focus to the initiating element or its nearest surviving parent. Only urgent actionable blocking errors warrant intrusive announcement. Avoid auto-expiring critical status messages.

Surface	Speech/reading priority	Focus and input fallback
Home / return state	“Home. Resume Chapter {n}, stage {id}.” Then pending claims, next legal objective and idle summary.	Restore root/last meaningful task, not Store; status announcements do not steal focus.
K-09/K-10 selection	“Choose one permanent starter support: K-09 Cleanser / K-10 Protector.” Describe distinct roles and confirmation.	Two labeled options and separate Confirm; no timed choice, swipe-only control or default silent acceptance.
Squad and formation	Read unit ID, role, row/slot, HP and what changing slot affects, including Taunt/targets.	Provide non-drag action such as Choose position/Move to slot; support Switch Access sequence.
Preflight G1–G8	Name specific telegraph and one accessible counter, including Wound “next action” rather than “next turn”.	Text alternatives for highlighted enemies and actionable Review Formation; detailed rules on demand.
Battle live	Periodic digest of whose action, critical status/telegraph and result, not every damage animation.	Pause/reduced motion and accessible battle event log; avoid constantly resetting focus.
Recap / first loss	Outcome → observed causal event → actionable legal counter → relevant resource/source.	One clear Try Again/Return control; allow replay of text explanation without relaunching battle.

Claim or Recruit receipt	State as Pending / Confirmed / Rejected, exact source ID or plain-language label, and associated resources.	“Check status” operable with assistive tech; no hidden success behind animation.
Wallet / idle full bank	Accrued balance, capped collection behavior, elapsed eligible period, named resource sources.	Currency amounts readable in full; labels and values available without horizontal scrolling.

59.5 Exact recovery microcopy: separate uncertainty from failure

All braces below indicate authoritative runtime substitutions, not invented values. Example English only; localization and voice output require later work. Never use these phrases to fake a server result.

Trigger	Working visible / spoken copy	Safe action / restriction
Claim submitted, response unknown	“Checking your saved reward. Do not claim again yet.”	Check status; may leave screen; do not show grant as confirmed.
Existing grant found	“Already claimed: {source label}. Your current balance is {balance}.”	View receipt; do not credit again.
Confirmed reward posted	“{source label} confirmed: +{amount} {resource}.”	Continue; speech announcement once, no focus theft.
Confirmed nonexecution	“This action was not completed. Your balance is unchanged.”	Retry only if system has confirmed nonexecution and unchanged debit.
Recruit status unknown	“Recruitment result not confirmed. We are checking the original request.”	Check status; disable second paid/seal spend until resolution.
Store payment pending	“Payment is pending. No purchased items have been added.”	Review status via platform; no duplicate purchase prompt.
First defeat	“Battle lost. {Observed event}. Try {accessible response}.”	Use event log for both placeholders; otherwise show neutral “Review formation”.
Idle storage full	“Your idle bank stopped adding rewards after 72 hours. Previously earned rewards are still available to claim.”	Claim saved amounts; no suggestion accrued balance was erased.
Disconnected while inspecting	“Connection interrupted. Your last confirmed progress is saved as shown.”	Only if confirmation exists; otherwise “Progress is being checked.”
Screen reader state	“Reward pending.” → “Reward confirmed.”	Announce changes once, independent of button label.

59.6 Edge-case walkthroughs and recoverability proofs still owed

PAPER WALKTHROUGH A — Unknown FIRST03: boss defeat posts a unique combat result; the FIRST03 reward request is sent; connection ends before the answer arrives. On resume, an account/source lookup either finds the receipt (show 80 Ryo and 120 EXP as confirmed) or returns an explicit nonposted status. A timeout alone gives neither answer. Retrying the same claim must never create a second FIRST03 payment. If system cannot reach authority, retain pending and show last confirmed wallet, not a speculative new balance.

PAPER WALKTHROUGH B — Recruit interrupted: a player confirms exactly one draw using a seal; animation stops before shinobi reveal. The app resumes with the original request identifier. If the draw committed, reveal that single existing shinobi and updated pity, without charging another seal. If it did not commit, expose only a confirmed nonexecution retry. An indeterminate response must not unlock an unguarded “Summon again” action.

PAPER WALKTHROUGH C — G1 first loss with TalkBack: Preflight announces Wound’s next-action timing; after defeat, Recap gives the result, exact causal line if present, and an accessible focus/formation answer. Returning to Preflight preserves selected formation and predictable focus. If the event record is unavailable, a factual generic tip is allowed; fabricated source-target or damage explanations are forbidden.

PAPER WALKTHROUGH D — 72-hour return: the player visits after more than 72 hours. The collection screen distinguishes 24 hours at full rate, 48 hours at quarter rate, and zero beyond 72; it displays the *actual server-derived resource amount*, not a blanket promise based on this example. Claim must use one unique collection interval; dismissing the screen, relaunching or using large text never burns a collection or grants it twice.

PAPER WALKTHROUGH E — 320dp/200% text: a pending-reward pane expands vertically; the title, status, resource figures, and Check status action remain reachable in reading order. An interrupted modal returns focus to a meaningful control. If the user has reduced motion enabled, essential outcome text remains unchanged. This is a design exercise, not a visual or TalkBack pass.

59.7 Decisions, rejected alternatives, cross-system effects

Decision and reason	Trade-off / dependent section	Reopen trigger
5C-D1: Source-ID receipts and ambiguous-outcome state; prevent duplicate or lost rewards.	Needs future authoritative receipt store/transactions, §§40, 52–58. Adds status UI and reconciliation complexity.	Any reward source cannot be uniquely identified or atomic internal debit/post cannot be specified.
5C-D2: Separate paid purchase provider verification from in-game earned grants.	Backend/payment integration later; purchase tokens and acknowledgements are not ordinary reward IDs.	Store policy changes or proposed architecture cannot reconcile refunds and pending purchases.
5C-D3: Accessible names, logical order and polite state announcements; avoid focus theft.	Adds semantic design, native input testing, translation and screen-reader review; §§57–58.	Essential information hidden in art, focus trap, duplicate speech or 200% text clipping.
5C-D4: First-loss recap only diagnoses recorded causes; prevent false training.	Requires full trusted combat log or noncausal fallback; §§24–27, 47–51.	Replay missing, event log not authoritative, or accessible answer not actually owned.
5C-D5: No new daily/weekly material payments or compulsory logins.	Retains FIRST/CORE/IDLE source budget and economy integrity; §§40, 52–55.	Recovery system requires a newly minted source or consumes resources to check existing state.

[REJECTED] Optimistic “success” reward toast before receipt; interpreting a transport timeout as confirmed failure; disabling all navigation until online for a noncritical pending claim; unguarded retries; automatic second summon after restart; color-only status changes; live-announcing every combat frame; using placeholder combat logs as a cause-of-loss diagnosis; assuming a guest account can be recovered without verified credentials or saved state.

[UNRESOLVED] Exact backend/account identity and offline persistence; what server time source defines idle interval, eventual consistency and receipt retrieval SLA; cross-device conflicts; Play purchase refund/revocation; guest-account transfer; loss of device; consent and data minimization for event logs; translations, TalkBack/Switch Access/keyboard/voice navigation, 320dp layouts, text at 200%, localization truncation, high-contrast and motion settings; observed accessibility/usability measures. These require future implementation and tests authorized by the owner.

59.8 Paper acceptance gates and next research

[PAPER GATE] Every named critical action has an immutable entitlement/source identity, an explicit unknown state, a confirmed terminal outcome and an accessible resolution path; negative state never silently invents a second cost or first-clear; each primary surface has one meaningful announced role/name, logical navigation order, non-color status and non-drag fallback where relevant. These are acceptance *requirements*, not measured passes.

[FUTURE VALIDATION — NOT PERFORMED] On a real build: reproduce network loss immediately before and after commit, duplicate callbacks and reboots; assert invariant rewards/pity/upgrade/wallet; use TalkBack and Switch Access through START01, G1 loss, FIRST03 claim and 72h return; use 320dp and 200% text without clipped warnings; test reduced motion, offline return and locale expansion; ensure announcements and focus persist. Record observed failure counts rather than claiming zero now.

[NEXT RESEARCH — PRIORITY 5D] Investigate localization, low-end device constraints, and a consolidated mobile UX exit gate before social/ranked UI. Distinguish text expansion and right-to-left support from verified layout; set minimum performance/asset budgets only as proposals, compare offline/online transitions with the recovery contract, and enumerate required future device tests. RECOMMENDED LEVEL: HIGH, because this is a bounded presentation/performance scope anchored by §§57–59. Escalate to EXTRA HIGH if network/service selection or combat semantics require an architectural redesign. Remain in concept research.

59.9 External sources and provenance

[5C1] W3C WAI, Understanding WCAG 2.2 SC 4.1.3 Status Messages (accessed 2 Nov 2026). Guidance for accessible status announcements without shifting focus. Web criterion used as comparative practice, not a native game compliance certificate. https://www.w3.org/WAI/WCAG22/Understanding/status-messages.html

[5C2] W3C WAI, Understanding WCAG 2.2 SC 2.4.3 Focus Order (accessed 2 Nov 2026). Logical focus sequence preserving meaning. https://www.w3.org/WAI/WCAG22/Understanding/focus-order.html

[5C3] Google Android Developers, Make apps more accessible (Views), accessed 2 Nov 2026: accessible names, unique descriptions, and suppressing decorative focus. https://developer.android.com/guide/topics/ui/accessibility/views/apps-views

[5C4] Google Android Developers, Jetpack Compose accessibility semantics, accessed 2 Nov 2026: polite live-region usage, limited assertive urgency, pane titles. Not a selection of Compose as engine/UI toolkit. https://developer.android.com/develop/ui/compose/accessibility/semantics

[5C5] Google Android Developers, Play Billing security, accessed 2 Nov 2026: verify purchase tokens and PURCHASED state; pending payments have no entitlement, server-side acknowledgement recommended. This does not select a payment implementation. https://developer.android.com/google/play/billing/security

[5C6] Google Android Developers, Integrate Play Billing Library, accessed 2 Nov 2026: distinguish payment pending, verification, grant and acknowledgement/consumption. https://developer.android.com/google/play/billing/integrate

[INTERNAL BASE] Master §§23, 24–27, 40, 47–58, Appendix E. All game-specific IDs, amounts and proposals originate from this master and current analysis; the external sources do not verify them.

Appendix A. Historical report preserved from the Library: Executive Summary

Archive status: Full saved report text, converted to Word formatting. Its in-report citation markers (for example 【17†L173-L181】) are legacy tool references and may not resolve when viewed outside the original research result. Use Section 15's source URLs for currently accessible references, and re-verify patch-sensitive claims. Archived statements are not automatically approved design decisions.

Executive Summary

Anime-themed idle RPGs blend **incremental progression** with collectible heroes and turn-based combat. Players form teams of characters (e.g. Naruto ninjas), engage in **auto-battles** or simple turn-based fights, and earn resources continuously – even when offline【17†L173-L181】【65†L119-L122】. The core loop typically cycles through **accrue idle rewards → upgrade heroes/gear → auto-battle stages → repeat (occasionally prestiging)**. Progress is driven by leveling characters, unlocking gear, and resetting (ascension) for permanent bonuses, fostering a satisfying sense of growth【40†L114-L122】【59†L52-L60】. Collection ("gacha") mechanics supply a steady stream of new characters of varying rarity, encouraging team-building and strategic synergies【50†L152-L160】.

A Naruto-themed idle RPG must include robust **fighting mechanics** (emphasizing ninja skills), a layered **progression system** (levels, gear tiers, awakening, prestige), and a multi-currency **economy** with sinks to control inflation. Monetization would center on gacha draws (banner rates ~1% for top-tier heroes as seen in similar games), battle passes, and optional ads, while ensuring fairness (e.g. pity systems). Retention strategies include daily login rewards, timed events (festivals, story chapters), guilds/clans, and push notifications. Balance hinges on smooth power curves and catch-up mechanics so new players aren't hopelessly behind. The UX/UI should cater to "idle" play: clear feedback of offline gains, simple skill activation, and convenient auto-battle options. Analytics should track DAU/MAU, retention (D1/D7), ARPU, progression speed, etc. Anti-grind designs (e.g. automated leveling) and anti-pay-to-win safeguards (frequent free draws, generous events) are essential. Finally, using the Naruto IP legally requires licensing from rights holders (Kishimoto/Shueisha/Pierrot/Bandai Namco)【31†L32-L37】, adhering to brand guidelines (no unauthorized characters, etc.).

The following report dives deeply into each area – core loop, combat, progression, collection, economy, monetization, retention, balancing, UX, analytics, anti-grind/anti-P2W, and legal – with examples and recommendations, supported by industry sources. A comparative table of 6–10 anime-style idle RPGs (e.g. AFK Arena, Idle Heroes, Saint Seiya, ML Adventure) highlights differences in loop, monetization, gacha rates, retention features, pacing and session length.

Core Gameplay Loop

At the heart of an idle RPG is a **low-pressure loop** of gradual progress【17†L186-L189】. A typical cycle is: **(1) Team Setup → (2) Auto-Battle/Idle Combat → (3) Collect Rewards → (4) Upgrade Progress → (repeat)**. Players recruit and arrange a team of characters, then start battles (often auto-resolved) against waves of enemies. Victory yields experience, gold, loot and character shards【40†L114-L122】【65†L119-L122】. These resources are spent on leveling heroes, equipping gear, enhancing skills, and unlocking new abilities. After upgrading, the player returns to battle more powerful enemies on higher stages. The process runs continuously – notably **offline**, as most idle RPGs grant rewards even while the player is away【65†L119-L122】【59†L52-L60】. This "offline accumulation" is a defining feature: for example, IdleOn advertises that "your heroes keep fighting even when you're gone"【59†L52-L60】, and AFK Arena explicitly promises auto-battles and offline rewards【65†L119-L122】.

Once players hit major thresholds (e.g. max level or a key boss), they perform a **prestige/ascension** reset: they lose short-term progress (characters revert to level 1) but gain permanent boosts (e.g. a multiplier on future gains). These prestige loops spark long-term engagement by allowing exponential growth over multiple runs【17†L149-L158】【40†L125-L133】. The core loop thus has two layers: a short loop of continuous upgrades, and a long loop of periodic resets for permanent advancement.

The following flowchart depicts this loop: an idle rewards accumulation leads to upgrades, which enable auto-battles, yielding more rewards. Boss defeats trigger optional resets that feed back into upgrades.

【42†L9-L12】【40†L125-L133】

```
flowchart TD
  A[Idle Resource Accumulation\n(offline gains, quest rewards)] --> B[Claim & Spend Resources]
  B --> C[Upgrade Heroes/Gear & Team Setup]
  C --> D[Auto-Battle Begins]
  D --> E[Gain XP, Gold, Loot]
  E --> F{Boss Defeated?}
  F -- No --> A
  F -- Yes --> G[Boss Rewards & Milestone]
  G --> H[Optional Prestige/Ascension]
  H --> I[Permanent Bonuses (XP multipliers, etc.)]
  I --> A
  G --> A
```

Examples from existing games: IdleOn features “unlimited idle gains” and allows hiring defeated foes as passive damage dealers【59†L52-L60】【40†L125-L133】. AFK Arena’s description highlights 1,000 free hero summons and “auto-battle your way to victory and earn rewards even while offline”【65†L119-L122】. These illustrate how games market the loop of continuous progress with minimal player input.

Recommended parameters: Victory rewards should scale so players see steady progress over ~seconds/minutes of play. Boss or map milestones might occur every few hours of cumulative progress. A prestige reset (with reasonable timing) every **1–3 days** of idle play can keep long-term motivation – more frequently leads to grind, less makes progression dull. Upgrade costs typically escalate (double/triple per level), enforcing cadence. For Naruto-themed flavor, prestige could be called “Rebirth” and grant, say, a +5% XP or currency bonus per reset.

Combat Mechanics

Combat in an idle turn-based RPG is usually **automated** or semi-automated to support the “idle” premise. In each stage or battle, characters perform actions in sequence without direct player input (beyond initial setup or occasional skill triggers). Key mechanics include:

1. **Turn-based Auto-Battle:** Characters take turns attacking based on speed or initiative, using basic attacks and skills automatically. The player typically cannot control actions mid-battle, although some games allow manual activation of ultimate skills. AFK Arena, for example, is known for its completely auto-resolved combat (players choose heroes and formation only)【65†L119-L122】.
2. **Skill Timing / Ultimates:** To retain engagement, games often give players a button to trigger each character’s ultimate or special skill. Skill cooldowns might be visible, requiring the player to time activations (e.g. pop Naruto’s Rasengan when available). Some idle RPGs bypass this and automate all skills (simplest design), but adding one or two manual abilities per combat can add a layer of strategy.
3. **AI Difficulty Scaling:** Enemies grow stronger each stage. Boss enemies use stronger AI patterns. A modern Gacha game often normalizes stats for PvP (as in Naruto Mobile’s equalized PvP), but for idle PvE content, enemy stats just scale with level. We recommend **graduated difficulty** so early stages breeze through, mid-game requires strategy, and late-game feels challenging. Stage bosses (every 10–20 stages) should have unique mechanics or resistances.
4. **Auto-battle features:** Many titles include a 2x/4x speed toggle, “sweep” functions (instant clear of stages once maxed), or an “AFK Mode” showing expected rewards if left idle. AFK Arena has a “AFK Combat Rating Boost” that auto-battles and even gains rewards while offline【65†L119-L122】. Ensure clear UI for these (a toggle or an “Auto” indicator).

Example: IDLE RPG (Steam, 2026) describes “classic turn-based battles” where you “hire defeated enemies as assistants” and use prestige points【40†L114-L122】【40†L125-L133】. IdleOn even lets you unlock passive worker roles (e.g. “miners”) to simulate continuous battles【59†L52-L60】. While Naruto RPG won’t literally hire defeated ninjas, analogous features could be “Assistants” or “Shadow clones” that continue fighting or gathering.

Pros/Cons:
- *Pros:* Auto-battle minimizes grind and keeps users engaged in upgrades rather than twitch play. It's accessible to casual players and suits mobile "AFK" style.
- *Cons:* Too much automation can feel passive. Always-on auto may bore players who enjoy interaction. Balancing purely idle content is harder, since strategy only lies in pre-battle preparation. Adding limited interactive skills or formations adds strategic depth. Also, fully automated pacing means tuning rewards carefully to avoid either stagnation or pay-to-win rush.

Implementation Notes: Use simple turn queues and states (attack, use skill if ready). Provide visual feedback (damage numbers, hit animations) to make progression satisfying. Keep performance in mind: avoid heavy 3D when hundreds of fights occur (use stylized or 2D effects if needed). Bug reports from trailers indicate performance issues (lag) can sour impressions【34†L423-L433】【65†L119-L122】 – ensure smooth animations and loading.

Recommended parameters: Base attack speed such that a full team round takes ~3–5 seconds at normal speed. Ultimates recharge every ~20–60 seconds. Boss fights might last ~30–60 seconds. Auto-battle speed toggle (2× or 4×). Allow offline progress for at least 8–12 hours (IdleOn uses *unlimited*, but typical games cap 8h–12h). Skill damage and AI aggression should scale so that small increments in hero power noticeably shorten battle time, maintaining dopamine.

Progression Systems

Progression in an idle RPG is multi-dimensional. Common elements include:

1. **Hero Levels:** Characters gain XP from battles and level up, increasing base stats. Early levels (1–50) might be fast, later levels (50–100+) slower, requiring more XP per level (common exponential growth).
2. **Equipment/Gear:** Heroes equip weapons, armor, accessories, etc. Gear has its own levels or star ranks. Upgrading gear (with resources) boosts hero stats. Often gear tiers increase over time (Tier 1 → Tier 2 → etc). For Naruto, gear could be "ninja tools" like kunai, scrolls, chakra enhancers with quality tiers (Bronze, Silver, Gold, etc).
3. **Hero Rarity & Awakening:** Gacha characters come in rarities (e.g. 1★ common to 5★ legend). Higher rarity starts with better base stats. Awakening (or "limit break") is a secondary progression: using special materials or duplicate shards to unlock character passives or higher potential. For instance, Idle Heroes has "transcendence" (E1–E7) for top-tier heroes; AFK Arena has Lightbearer / Mauler factions, not explicit awaken but equips. In Naruto terms, "Awakening" could be unlocking a character's next form or hidden potential (e.g. Kakashi's second Sharingan).
4. **Ascension/Prestige:** At key points (e.g. after stage 1-100, or when progression plateaus), players can reset progress. This is often called "Prestige" (Idle RPG), "Transcendence" (Idle Heroes), or "Rebirth." The reset strips heroes back to level 1 (sometimes keeping one copy if dupes were used), but awards permanent resource bonuses or stat multipliers【17†L149-L158】【40†L125-L133】. This dramatically accelerates subsequent runs. The design here must balance short-term sacrifice for long-term gain.
5. **Meta-Progression:** Over multiple ascensions, players unlock higher "new game plus" tiers. Possibly a second currency for meta-upgrades (like "prestige points" invested in global boosts【40†L125-L133】). Example: IDLE RPG (Steam) lets players invest Prestige Points into parameters【40†L125-L133】.
6. **Currencies/Resources:** Besides XP/Gold, include upgrade currencies like Shards (for gacha/awakening), Upgrade Ore (for gear), and premium gems. Multiple currencies create choice and complexity.

Examples: Idle Heroes differentiates between *items* like Scrolls/Orbs (for summoning) and *materials* like Rune Shards or Upgrade Materials. AFK Arena uses "Hero Scrolls" and "Diamonds." Generally, progression is gated by activity: stages, events, and elite challenges yield scarce upgrade materials.

Pros/Cons:
- *Pros:* Layered progression (level → gear → awaken → prestige) provides many growth avenues and prevents stagnation. Reset mechanics (prestige) keep players engaged long-term and mitigate power creep.
- *Cons:* Can overwhelm new players with too many systems. Requires careful pacing so players hit each feature at the right time. Risk of pay-to-win if late-game gear/upgrades are too scarce without spending.

Implementation Notes: Plan stat growth curves so no single parameter dominates. Show clear requirements for next milestone (e.g. "Reach level 50 to awaken"). Use daily logins and events to grant small amounts of premium resources

(shards, gems) to support progression. Consider "skip" content (sweep tickets or auto-clear) once a stage is repeatedly cleared, to avoid boredom.

Recommended parameters (example values):
- Hero Level Cap: Increase gradually – e.g. unlock up to level 50 within first week's play, then cap at 50 until heroes are transcended or new content released. Maximum level could be 100+ as game matures.
- Gear Upgrades: Tiered, with each tier requiring exponentially more gold. Tier progression could unlock every ~50 hero levels.
- Prestige Reset: Offer voluntary prestige after boss levels (e.g. every 100 levels). Provide a flat 10–20% permanent bonus to XP or gold per prestige, stacking multiplicatively.
- Awakening/Ascension Materials: Tied to events or high-level challenges – e.g. 3–5 duplicates or special shards needed per star upgrade.

Collection Mechanics

Heroes/Characters: The core collectible in a gacha RPG are the heroes. Characters vary by **rarity** (e.g. 1★–5★), type/class (attacker, defender, healer), faction or element, and have unique skillsets. Designing a **large roster** of diverse heroes encourages players to experiment with teams. For a Naruto theme, include all fan-favorites across the timeline (Naruto, Sasuke, Sakura, etc) at various incarnations, each classified by village or role (Striker, Tank, Support, etc).

1. **Rarity Distribution:** Typical rarity pools might be: 5★ (legendary) ~0.5–1.5% pull rate, 4★ (epic) ~5–10%, 3★ (rare/common) ~80%【50†L199-L203】. This skew means players collect many low-tier heroes and few top-tier. To avoid frustration, include a pity or guaranteed system (e.g. every 40 summons guarantees a 5★).
2. **Fusions/Evolution:** Allow combining characters or shards to create higher-tier ones. For example, Idle Heroes lets players merge lower-star heroes into a higher-star version. Possibly implement a "fusion" system (e.g. combining 5 of the same rank to ascend). This uses up duplicates or shards and prevents hoarding too many weak heroes.
3. **Team Composition and Synergies:** A key strategic layer: some heroes boost others of same faction/village. E.g. having a Uchiha or Konoha ninja team gives passive buffs. Many idle RPGs use faction synergy: AFK Arena has 7 factions, Idle Heroes has multiple tiers and sets. Naruto idle could use *Konoha vs. Akatsuki* synergies or *Taijutsu vs. Genjutsu* types. Encouraging balanced team-building (tank/frontline, DPS, support) ensures players try different combinations.
4. **Recommended design:** Start with small teams (3–5 units) that scale to larger (6–9) as game grows. Each hero has 2–4 skills (normal+2 ultimates). Provide a "skill-up" or mastery system so players invest duplicates to improve skill damage or reduce cooldown.

Examples: Mobile Legends Adventure (a known hero-collector RPG) uses a 5-hero formation with synergies; Idle Heroes uses 6-hero teams with formation bonuses. The Naruto blueprint could allow, say, 4 ninjas (2 front, 2 back) plus 1 support. Synergies: e.g. full *Naruto Team* (Naruto, Sasuke, Sakura, Kakashi) gets a massive team buff.

Pros/Cons:
- *Pros:* Collection drives engagement and monetization. Evolution/fusion gives purpose to extra heroes. Synergies reward thoughtful composition.
- *Cons:* Collectible fatigue (too many heroes) can occur. Overpowered synergies may force specific team meta (hard to balance all heroes). Rarity drops being too rare can frustrate free players.

Implementation Notes: Implement an in-game codex of heroes, categorize by rarity. Provide multiple sources for characters: standard banners, event banners, loyalty rewards. Ensure even 1★ heroes have eventual utility (e.g. feeding into fusion or as awaken fodder). Example: Idle Heroes uses a "Summon Tokens" currency for generic draws, plus faction-specific summons for targeted pulls.

Recommended parameters:
- Base hero pool: Start ~30 characters, expanding monthly by 5–10.
- Rarity pull rates: ~0.7–1.0% for 5★ (legendary), ~7–10% for 4★【50†L199-L203】.
- Team size: unlock 3 slots at start, expand to 5 by mid-game.
- Synergy bonus: define clear set bonuses (e.g. 3* Konoha ninjas = +10% all stats). Test to avoid one synergy dominating.

Economy and Resource Flows

Idle RPGs use multiple currencies and resources. A balanced economy **channels player engagement** and controls inflation. Key components:

1. **Currencies:** Usually a free *soft currency* (e.g. Gold, Coins) earned in every battle, and a *premium currency* (Gems, Crystals) usually bought with real money or granted sparingly (logins, events). Additional currencies include *summon tokens* (for gacha pulls), *stamina/energy* (to limit attempts), and *special mats* (forge ores, awaken fragments, etc).
2. **Income Sources:** Main sources are stage rewards (idle/autobattle yields soft currency and items), daily quests, weekly events, login bonuses, and ads. Most games award a flat amount per hour offline (AFK Arena grants rewards per offline minute【65†L119-L122】). *Resource refreshes*: energy refills or extra daily currency if players watch ads.
3. **Spending Sinks:** Upgrades create sinks to prevent inflation. Examples: hero leveling costs (growing exponentially), equipment crafting/upgrading, skill upgrades, and event shops (players trade tokens for items). Idle Heroes players advise **saving resources for events** rather than blowing them on minor upgrades【56†L62-L70】, indicating event currencies are major sinks. Sinks also include limited-time gacha spins.
4. **Inflation Control:** To avoid runaway wealth, costs ramp up exponentially. New currency types (e.g. Legendary Shards) can be introduced with level expansions. Impose level caps or diminishing returns. Events and bosses should require ever-increasing amounts of resources, matching exponential income growth.

Examples: The Gacha Game wiki notes many games "feature several in-game currencies with intricate conversion methods, obscuring the value of non-premium currencies"【50†L199-L203】. For instance, one might have Coins (infinite earnings), Medals (rare currency from PvP), and Crystals (premium shop money). Idle Heroes uses gear upgrade stones, rune stones, and scrolls as intermediate currencies.

Recommended resources flow: Ensure *soft* currency (e.g. Gold) is plentiful from early play (so upgrades feel satisfying), but require *hard* currencies (gems/shards) for top-tier pulls and awakening, which come slower. Example split: each stage clear gives 100–1000 Gold and 1 XP orb. Daily quests or events might give 50–200 premium Gems per week. Limit refund (catch-up) to major resets (e.g. bonus currency on prestige).

Pros/Cons:
- *Pros:* Multi-currency lets designers fine-tune motivation. Sinks give players goals to spend.
- *Cons:* Too many currencies confuse players. Poor balancing can either stall progression (if costs too steep) or trivialize currency (if too easy).

Implementation Notes: Track currency supply in analytics. If players amass gold quickly, raise costs or introduce a new sink (e.g. gear upgrades). If demand outstrips supply (players stuck), add more giveaways or reduce costs. Offer ways to convert surplus (sell common shards for gold). Visualize flows for players in UI (e.g. dashboards showing how to earn each currency).

Monetization

Most anime idle RPGs are free-to-play with **gacha monetization**【50†L152-L160】. This typically includes:

1. **Gacha (Loot Boxes):** The primary draw is character banners. Common models: *"10-pull"* with a chance for high-rarity heroes. Rates ~1% for top-tier【50†L199-L203】. Some games use multiple banners (e.g. "featured Naruto characters" vs general pool). Consider a guaranteed pity: e.g. every 30–50 pulls guarantees a ★5 hero.
2. **Shops and Direct Purchases:** In-game shop sells premium currency packs, resource bundles, and exclusive heroes. Often there's a "limited shop" with rotating deals. Bundle examples: "Naruto Starter Pack" giving shards and currency at a discount. Pricing tiers from ~$1.99 up to ~$99.99. Ensure value packs give a clear edge in progress, but also have smaller packs for casual spenders.
3. **Battle Passes:** Season or monthly passes (free + premium track) are common. Rewards: minor heroes, shards, currency. Encourages regular play. Example: a "Ninja Pass" that over 30 days grants enough currency for 10 summons plus some gear. *Battle passes* are accepted by players when they give predictable rewards.

4. **Ads:** Optional ads can accelerate progress. E.g. watch an ad to double offline gold for the next hour, or get a free summon ticket. This engages non-spenders and can drive retention without mandatory paywalls.

5. **Fairness/Anti-P2W:** Balance must be mindful. As reviews note, many gacha games are perceived as pay-to-win【65†L243-L247】. To mitigate, include generous free aspects: regular free summons (AFK Arena offers daily 10 free pulls【65†L119-L122】), pity on banners, and events that guarantee rare heroes if objectives met. Purchasable currency should only **accelerate** progression, not gate core content. Value packaging (e.g. "best value" bundles that promise big items) can feel fair if clearly marked. Implement spending limits or exclusive cosmetics (i.e. allow whales to gain vanity or small boosts, but avoid exclusive unbeatable heroes).

6. **Ads/Rewards balance:** Since user instructions said "non-pushy", ensure ads are **rewarded** (player opts-in for a meaningful bonus). No forced ads.

7. **Example Mechanics:** Naruto Mobile (Tencent) reportedly uses an aggressive VIP system (though not cited), which may not suit all markets【32†L67-L75】. Instead, a Naruto idle RPG could adapt a soft VIP: small bonuses for spending but not gating content (e.g. every $ spent gives loyalty points for freebies). Character sales or skins could complement gacha (exclusive costumes).

8. **Pros/Cons:**

    1. *Pros:* Gacha draws and shop items directly monetize. Battle passes and ads increase ARPU (average revenue per user) and retention respectively.

    2. *Cons:* High prices or poor drop rates cause backlash. As the reddit points out, pricing balanced for one region may offend another【32†L68-L77】. Ensuring perceived fairness is critical.

9. **Implementation Notes:** Clearly communicate drop rates (many regions now require it). Offer small refunds (soft currency) for duplicate characters. Track whales vs. regular spenders for balance. For Naruto, consider tie-ins (e.g. selling "chakra refills" which restore stamina, a common Pokémon-like model) and check legal restrictions for loot boxes (some regions regulate gacha).

Retention & Engagement

Key retention drivers in idle games include **daily login rewards**, **events**, and **social features**. Specific tactics:

1. **Daily/Monthly Logins:** Reward players for each consecutive day played (with bonuses increasing over a 7- or 30-day cycle). Common rewards: premium currency, summon tickets, or exclusive characters after long streaks. This forms a basic habit loop.

2. **Time-Limited Events:** Frequent events (weekly/monthly) keep content fresh. Examples: *Raid/World Boss events* (players team up to slay a boss), *gacha festivals* (increase drop rates on certain heroes for a week), *challenge towers* (progressively harder stages with leaderboard). Such events often award rare materials or even limited characters. Idle Heroes has many periodic events where certain currencies can be spent for exclusive gear【56†L63-L72】. Naruto idle could run events like "Village Battle Event" or new story arcs when anime episodes release.

3. **Meta-Progression Systems:** Implement systems like guilds/clans (collectively earn guild points or unlock guild shop items), daily achievements, and a sense of overall progression (e.g. an "Adventure Rank" that unlocks new features as it increases). AFK Arena, for instance, has player levels and guild buildings that grant buffs.

4. **Notifications & Reminders:** Idle games often use push notifications for "your offline rewards are full" or "energy refilled" to bring players back. Use these sparingly and with valuable info (not spam).

5. **Community/Social:** Incorporate friend lists, friend bonuses (e.g. borrow friend heroes in battle, or send gifts), chat and social media integration. Given Naruto's fanbase, social sharing (e.g. "My Uzumaki team just defeated the boss!") could help organic growth.

6. **Metagame Progress:** A "Campaign map" that visually shows progress through the Naruto story (e.g. Konoha exams, Chunin Exams, Great Ninja War chapters) provides a narrative long-term engagement. Completing chapters could unlock new game modes or characters.

Examples: Idle RPG Wikipedia notes such games often see "very high player retention" due to steady progression and frequent feedback【17†L186-L189】. The Gacha wiki observes that limited banners and events drive retention and marketing (limited-time availability creates urgency)【50†L189-L198】. Naruto idle should follow these patterns: e.g. have "character banners" for seasonal village festivals, and special login weeks after big anime releases.

Recommended cadence:
- Daily quests (small tasks) reset every day (e.g. do 10 stages, win 3 PvPs).
- Weekly events with cumulative rewards (e.g. "Complete 100 stages for an SSR shard").
- Monthly story event unlocking new chapters and an exclusive hero.
- Seasonal events tied to holidays or Naruto anniversaries.

Balancing & Pacing

A well-paced idle RPG avoids steep power spikes or stalls:

1. **Power Curve:** Early game should feel fast and rewarding. Mid-game slows down to lengthen engagement. Late-game (post-prestige) opens new strata. Use **soft caps** (e.g. slowdown of stat growth) to make progression feel meaningful. After each prestige, reset power curve so player feels a new ramp. Common trick: *diminishing returns* on stats at high levels, making additional levels less impactful, then cured by ascending.
2. **Soft Caps:** Impose level caps or gear tiers. For example, hero level cap might be 50 until certain upgrade is purchased. Or cap gear rarity at Bronze for a while, then Silver gear unlocks later. These ensure players pivot to other tasks (like ascending or new characters) rather than brute-forcing one hero.
3. **Hard Caps:** A true max-level or gear cap can exist (e.g. max level 100). If so, ensure there is next progression path (promote characters, new mode) to keep playing. Idle games usually have effectively no permanent cap, only prestige resets.
4. **Catch-up Mechanics:** Implement ways for new or returning players to reach near-parity. Examples: "paywall dust" – free summons if you skip for a week, or friendship summons. Pity systems (guaranteed rare after X pulls) also help mid-spenders. Many idle RPGs (AFK Arena, Idle Heroes) give an initial hero pool (clearing initial campaign grants most heroes【65†L114-L122】) so newbies start with a variety.
5. **Fairness:** As noted, avoid "whale traps" late-game. Ensure free players can still access endgame content, maybe at slower pace. Provide occasional tier boosts (like "new player missions" or bonus XP boosters for first week). The reddit guide for Idle Heroes stresses *saving resources for events*, implying events are intentionally generous catch-up opportunities【56†L62-L70】.

Implementation Notes: Regularly review progression speed via analytics: track how many days on average to reach key milestones (first SSR, first prestige). Adjust drop rates or earnings if the average strays too far from intended. For Naruto idle, consider giving a small buff (e.g. a set % XP boost) to players who haven't logged in in 7 days, as a friendly catch-up.

UX/UI Considerations for Idle Play

Design UI for *glanceability* and minimal input:

1. **Progress Indicators:** Clearly display offline gains accrued, next rewards threshold, and idle counters. An "Idle Reward" button (like AFK Arena's "Claim Offline Rewards") should summarize time played and multiplier from equips/buffs.
2. **Auto-Controls:** Provide one-touch auto or manual battle mode. Allow players to queue stages. Visible toggles for speed. If interactive skills exist, place big icons/buttons. Show skill cooldown timers graphically.
3. **Consolidated Upgrades:** Bulk upgrade options (e.g. "Level Up 10 Times" or "Max Level Up to 1-star") reduce tedium. A single hero can have all upgrade options on one screen.
4. **Minimal Clicks:** Minimize unnecessary navigation. Idle players log in sporadically; reward immediate gratification. For example, after battle or login, pop a summary of gained loot and a prominent "Upgrade" suggestion.
5. **Performance:** As one player review noted, poor optimization (lag) severely hurts user experience【34†L423-L433】【65†L119-L122】. Use lightweight graphics for battles. A Naruto idle should still feel lively (use flashy jutsus in moderation), but mobile-optimized.

6. **Idle-specific UI:** Show "Offline Progress" bar. Indicate when game is stopped earning (e.g. cap reached). Mobile push notifications should mention key info ("Offline reward full!").
7. **Inspirations:** Many idle games use a **main hub screen** with buttons for Story, Arena, Guild, Shop, Team, Events, etc. The idle RPG's main UI should show currency balances (soft money, gems) at top, quick access to upgrade menus. Use Naruto-themed art for icons (Kunai icon for challenges, Ramen bowl for stamina, etc).

Analytics & KPIs to Track

To ensure design goals are met and to balance live operations:

1. **Retention Metrics (D1/D7/D30):** Essential for all mobile games. Idle RPGs aim high: consistent login rewards and passive progression can boost retention【17†L186-L189】. Track how many return after day 1, 7, 30; target improvements via events or UI tweaks.
2. **Engagement Metrics:** Daily Active Users (DAU), session length, sessions per day. Idle games often have short sessions (players check rewards/upgrades 2–3 times a day) so average session might be 3–10 minutes. Monitor if session length drifts too short (unengaged) or too long (complexity or grind too high).
3. **Progression Velocity:** Track how many stages or levels players clear per day/week. If average progress stalls (power gap to new content), adjust difficulty or drop rates.
4. **Monetization Metrics:** ARPU/ARPPU, conversion rates (percentage who spend), average purchase. Track gacha spending and pitfall (e.g. high spending concentrated among few whales【69†L1-L9】). If whales dominate, consider introducing more mid-tier spend incentives.
5. **Item Flow:** Track currency inflow/outflow and inventory. If economy is unbalanced (players hoard too much currency, or can't spend enough), tune costs or sources.
6. **Feature Usage:** Measure which features/events players engage with. For example, do most players participate in daily dungeons? Is PvP used? This guides which new content to build.
7. **Churn Triggers:** Use funnels to see where players quit. E.g. if new players repeatedly fail a boss or run out of energy, they may leave. Check where big drop-offs occur (UI confusion? insufficient tutorials?).

No direct sources here, but these are industry-standard KPIs. Analytics tools (Firebase, GameAnalytics, etc.) should be integrated.

Anti-Grind and Anti-Pay-to-Win Patterns

Anti-Grind: Idle design inherently reduces grind by automating battles. Further anti-grind strategies:

1. **Automatically Collecting:** Let players queue up many fights at once (e.g. "Simulate 100 battles"). Reward players for logging in (collecting 8h of offline gold in one tap).
2. **Catch-up Systems:** Offer generous initial boosts: newbies might get a "first-time double XP" potion, or "new player bundle". Weekly milestone rewards for maintaining login can soften the grind.
3. **Event Funneling:** Schedule events that reward common resources to meet upgrade demands (as Idle Heroes guide emphasized, events supply the hard-to-get Awakening Crystals, suggesting the game expects players to wait for events【56†L62-L70】).
4. **Reduce Repetition:** Include auto-skip cinematics, auto-complete lower difficulty levels.

Anti-Pay-to-Win: To avoid P2W backlash (as AFK Arena reviews note)【65†L243-L247】:

1. **Balance Pay vs F2P:** Ensure top-tier heroes can be earned without paying (via pity, shop, or events). Provide marginal differences rather than absolute power gaps (e.g. paying might get a unique skin or slight % boost, not a totally OP hero).
2. **Transparency:** Clearly show drop rates (regulations often require it), and assure players that persistence or skill (in team building) can compete with spenders.
3. **Whale Protection:** Consider "hard cap per character", meaning even spending more can only get so many resources per cycle, to avoid infinite strengthening.

4. **Inclusive Rewards:** Every player should occasionally feel rewarded by play, not just spend. This includes free daily/weekly summons and rewarding event attendance (log-in five days to get a mythic shard, etc).

Implementation Notes: Continually survey player sentiment. If many call the game P2W or too grindy (as reddit threads on idle games often do【12†L25-L33】), recalibrate. For example, reduce cost of an upgrade or increase freebies. Balance patches should address any hero or combo that trivializes content.

Legal/IP Considerations

Using the Naruto license means obtaining a formal agreement with IP holders (Masashi Kishimoto, Shueisha, TV Tokyo, Pierrot, and Bandai Namco Entertainment, as credited on the official site【31†L32-L37】). Key points:

1. **Licensing:** Acquire rights for character likenesses, story elements, music, etc. This likely involves negotiations with Shueisha/Pierrot and Bandai Namco (who appear on Naruto Mobile's credits【31†L32-L37】). Budget for high licensing fees.
2. **Content Restrictions:** Follow Naruto's brand guidelines. Characters must act "in character" (e.g. Naruto can't have a gunslinging modern move out of nowhere). Avoid infringing crossovers (no, say, Dragon Ball cameos).
3. **Age Rating:** Naruto contains ninja violence but is generally family-friendly. The game should reflect the same tone (likely a "Teen" or "PG-13" style).
4. **Legal Disclaimer:** The game should include copyright attributions (as seen in [31]). Use official art assets or commissions approved by the licensor.
5. **Regional Sensitivities:** The reddit discussion pointed out that pricing acceptable in China may offend Western users【32†L68-L77】. Similar caution with content (Naruto is popular globally, but certain story arcs might not be suitable for younger audiences without context).

In summary: **Plan on an expensive license** and ongoing compliance (e.g. approvals for new characters, storylines). Emulate successful licensed games (Bandai Namco has experience with Naruto games) by contacting rightsholders early.

Comparative Table of Representative Anime Idle RPGs

Game: AFK Arena**** - **Core Loop:** Team collect → auto-battle → earn rewards → upgrades【65†L119-L122】 - **Monetization:** Premium bundles, battle pass, event packs - **Gacha Rate (top-tier):** ~1.0% legend, 5–10% epic【50†L199-L203】 - **Retention Features:** Daily login, 10 free summons/day【65†L111-L118】, guild tech, events - **Progression Speed:** Moderate; first prestige ≈1 week - **Typical Session Length:** ~5–10 min (very idle-focused)

Game: Idle Heroes**** - **Core Loop:** Similar idle loop with offline gains - **Monetization:** Premium summon scrolls, VIP packs, events - **Gacha Rate (top-tier):** ~1.0% for Transcend 5★【56†L93-L100】 - **Retention Features:** Daily chests, clan bosses, month-long events【56†L62-L70】 - **Progression Speed:** Slow midgame; major breaks at events - **Typical Session Length:** ~5–15 min

Game: Mobile Legends: Adventure**** - **Core Loop:** Collect heroes from ML universe → auto-battle - **Monetization:** Hero summons, gems shop, BP - **Gacha Rate (top-tier):** ~0.3–1.5% for legend - **Retention Features:** Daily & weekly tasks, guild war, 72-hour battle royale event - **Progression Speed:** Fast early; slows after 2nd reset - **Typical Session Length:** ~5–20 min

Game: Saint Seiya Awakening**** - **Core Loop:** Auto-turn combat with awaken/upgrade heroes - **Monetization:** Character draws, gear packs, tickets - **Gacha Rate (top-tier):** ~1.5% for top SS - **Retention Features:** Time-limited banners, daily sign-ins, VIP system - **Progression Speed:** Moderate; punctuated by chapter releases - **Typical Session Length:** ~10–20 min

Game: One Punch Man: The Strongest**** - **Core Loop:** Idle stamina battles → collect and upgrade Saitama-esque heroes - **Monetization:** Character draw tickets, coin packs, crates - **Gacha Rate (top-tier):** ~0.8% for SSR hero - **Retention Features:** Daily login, dim sum events, alliances - **Progression Speed:** Fast (1st prestige ~3–5 days) - **Typical Session Length:** ~5–15 min

Game: Honkai: Star Rail** (anime-style RPG) **-** Core Loop:** Manual/auto turn battles, gear and XP → reset+Skyfold - **Monetization:** Gacha banners, light cones, supply packs - **Gacha Rate (top-tier):** ~0.6% 5★ character【67†】 - **Retention Features:** Daily commissions, resonators, logins - **Progression Speed:** Steady (no idle, so slower effective) - **Typical Session Length:** ~10–30 min

(Gacha rates are illustrative. Features like "idle" vs "manual" affect session length.)

Sources: Gameplay and feature descriptions are based on official game sites and storefronts【40†L114-L122】【59†L52-L60】【65†L119-L122】, Wikipedia on idle/incremental games【17†L169-L178】【17†L186-L189】, and gacha mechanics【50†L152-L160】【50†L199-L203】. Community insights (player guides, reviews) inform best practices【56†L62-L70】【65†L119-L122】. Licensing information from the Naruto official site【31†L32-L37】. All referenced content is up-to-date as of late 2026.

Appendix B. Historical report preserved from the Library: Executive Summary (1)

Archive status: Full saved report text, converted to Word formatting. In-report citations may be tool-local; treat claims and numbers as historical research leads pending verification.

Executive Summary

This report analyzes the design of anime-style idle/RPGs (e.g. *AFK Arena*, *Idle Heroes*, *Mythic Heroes*, etc.) with the goal of creating a **Naruto-themed idle game** that is *truly F2P-friendly* and rewards synergy. We compare core mechanics across these games, derive concrete formulas/tables for progression and summoning, and propose team-synergy rules (with Naruto-themed examples). We pay special attention to **summoning/pity systems**, **duplicate handling**, **awakenings**, **economy/currency flows**, and **F2P vs P2W monetization**. To sustain free players, we recommend generous pity/compensation and robust offline/idle gains, while still retaining engaging pay options (cosmetics, convenience). Key findings include: an event-driven "1000 summons" model (as in AFK Arena's recent *Companions* update【27†L105-L113】【14†L64-L69】), clear hard-pity thresholds (guaranteed rare hero after *N* pulls【77†L273-L282】), extensive daily rewards/events【77†L248-L250】【58†L172-L180】, and explicit role synergies (e.g. ninja-village and team-7 buffs) that make team building meaningful. We also outline alternative monetization models (e.g. low-cost subscriptions, battle passes, extra currency sales) balanced with strong free rewards. Finally, we include mermaid diagrams showing system interactions and a player-progression flowchart, plus a "design log" of open questions and proposed A/B tests.

****Aspect: Core Loop**** - ****AFK Arena (Lilith Games): Idle offline gain + auto-battles ⇒ Campaign progression ⇒ gather rewards【74†L9-L12】【19†L61-L69】 - ****Mythic Heroes (IGG): Similar: idle & AFK battles in Greek/Norse myth settings【78†L138-L142】 - ****Idle Heroes (DroidHang): Idle training ("Idle Master") and auto-combat【19†L61-L69】 - ****Naruto Idle (Concept): Idle/offline training plus turn-based battles; village quests and boss raids

****Aspect: Summoning/Gacha**** - ****AFK Arena (Lilith Games): *Legendarium* (universal banner): 10-summon costs diamonds; occasional events give free pulls (e.g. "1000 summons" login)【27†L105-L113】【14†L64-L69】. Hero shards fuse (5→1★, 10→2★ etc.) for awakening. - ****Mythic Heroes (IGG): Mythic "Stargazer Summon": Hero shards + gold for pulls. No built-in pity noted, but community "fair SSR rates"【78†L232-L240】. Rares (SSR/UR) drop ~1–2%. Excess shards exchange for banner currency. - ****Idle Heroes (DroidHang): Heroic Scrolls from event/shop or crystals; no official pity. 5★ heroes can transcendence-ascend after 5 dupes (Spirit). Transcendence "rebirth" gives stat boost. CR shards convert to gold or other shards. - ****Naruto Idle (Concept): Summon ninja characters by banner (e.g. *Village Summon*, *Legendary*). Hard pity at ~100 draws for top rarity (mirroring common gacha standards【77†L273-L282】). Duplicates → spirit shards or automatic limit-break.

****Aspect: Pity/Guarantee**** - ****AFK Arena (Lilith Games): **None in base game**, but massive login events effectively act as pity (AFK's "10/day" for 1000)【27†L105-L113】. Could add: hard pity = SSR after ~80–100 pulls, soft pity gradually increasing odds【77†L273-L282】. - ****Mythic Heroes (IGG): No official hard pity reported; player feedback suggests summoning feels "fair for SSR" at start【78†L232-L240】. To ensure fairness: implement hard pity (e.g. 90+1 system). -

****Idle Heroes (DroidHang)**:** No hard pity (long-time players report low event rewards relative to level【58†L114-L122】). Many players suggest adding a pity or increased free pulls to aid late-game. - ****Naruto Idle (Concept)**:** Use **hard pity** (e.g. guarantee an S-rank ninja every 100 pulls, with a "spark" currency on each pull for a bonus)【77†L273-L282】. Offer daily/weekly free summons (e.g. 2 free 10x pulls/week or accumulating scroll tokens) to ensure F2P progress.

****Aspect**:** Duplicates/Upgrade**** - ****AFK Arena (Lilith Games)**:** Excess 5★ copies → faction bonuses; extra low-rarity auto-sell to tokens. Heroic traces (ether) for breakthrough. 5 copies → 1★; 10→2★; etc. - ****Mythic Heroes (IGG)**:** Commonly: 5 duplicate = 1-star upgrade (limit break); extra = artifact/special currency. - ****Idle Heroes (DroidHang)**:** 5 duplicate = Transcendence (new Hero "reborn" at 1★, more powerful). Extra rank-up stones convert to gold or training XP. - ****Naruto Idle (Concept)**:** Proposed: **Shard System**: e.g. each SSR hero needs 5 copies to awaken +1★. Additional copies give "S-Rank upgrade stones" to permanently improve stats. Duplicate SR/SSR can convert to "Spirit points" for power-ups. Nurturing a hero could also require generic "training points" earned in quests.

****Aspect**:** Progression**** - ****AFK Arena (Lilith Games)**:** Hero levels 1–160; gear (Lightbearer Crest, etc. up to +20). Power spikes at 30/60/80☆. Campaign stage XP + gold; faction trials unlock. 7 factions【74†L9-L12】 give formation buffs. Prestige through faction 'Majesty' resets. - ****Mythic Heroes (IGG)**:** Levels up to a cap (e.g. 100); gear tiers with limited upgrade mats; Ascension (limit-break) at level cap using Hero-specific tokens. Many side systems (pet companions, trinkets). - ****Idle Heroes (DroidHang)**:** Level cap 200; gear Enhancements require gold and shards. Transcendence resets level (prestige) for bonus. Frequent soft-caps (Progress resets like "Outer Ring 2000th floor") force new mechanics. - ****Naruto Idle (Concept)**: Levels & Gear:** Experience from battles (XP can scale ~100×L^1.2) and daily missions. Level cap (e.g. 100) then "Promotion" needing special materials. **Skill Upgrades:** Karma scrolls (rare currency from dailies) raise ninja skills. **Prestige/Ascension:** "Inner Peace" system: rebirth at cap resets level to 1 but grants permanent % stat bonus.

****Aspect**:** Economy & Currency**** - ****AFK Arena (Lilith Games)**:** Multiple currencies: Diamonds (premium), Gold (levels), Hero Shards/faction coins (faction trials, Hero union), Ether/Traces (breakthrough), feathers (cosmetics). Inflation managed by sinks (gear costs, Ascension)【19†L61-L69】【77†L244-L252】. - ****Mythic Heroes (IGG)**:** Premium gems, Gold, Hero Soulstones. Events yield "Relics" or "Gold Pieces" that buy limited equipment. Market/shop rotates. Many paid bundles for currency. Late-game inflation noted in reviews. - ****Idle Heroes (DroidHang)**:** Premium "Crystals"; common Gold/Glory; Hero Soulstone shards; Guild coins, Arena Medals. High gold inflation – sellers for gold (e.g. purchase gold with cash) and gold boosters exist. Subscription doubles idle gold【70†L1-L4】【58†L172-L180】. - ****Naruto Idle (Concept)**: Currencies:** *Premium*: Ryo (wind-style currency for summons, sold cheaply in small packs). *Regular*: Coins (level-up gear), Scrolls (skill-ups), Chakra (rare hero token used in limited shops). *Special*: Stamps (exchange for specific hero shards in Shop). **Sinks:** leveling gear/skills, upgrading village HQ (buffs), purchasing NPC tutor boosts. Control inflation via **multiple currencies** requiring conversion (coins → scrolls, etc.) so no single sink.

****Aspect**:** Monetization**** - ****AFK Arena (Lilith Games)**:** Gacha-centric. Small convenience subs (e.g. AFK Arena's $2.99/month for passive gold bonus)【19†L73-L79】. Cosplay skins. Events encourage spending (e.g. gem packs for summons). Classic AFK criticized as Pay-to-Win【27†L239-L247】【58†L114-L122】, but new "Companions" shifted to F2P focus (1000 summons)【27†L105-L113】【14†L64-L69】. - ****Mythic Heroes (IGG)**:** Standard gacha IAP: gem packs, hero bundles, cosmetic skins. "Starter pack" deals. No heavy subscription model noted; likely whales only competitive in PvP. Late-game dead zone made whales reluctant【28†L281-L289】. - ****Idle Heroes (DroidHang)**:** Summon scroll packs, gold packs, XP boosts, premium sets. $0.99 weekly "Golden Dragon Avatar" sub for +100% idle gold【19†L73-L79】. Many players say heavy spending *accelerates* progression but is not strictly required【58†L172-L180】. - ****Naruto Idle (Concept)**: F2P-First Model:** Emphasize small purchases & cosmetics. Options: low-cost *"Daily Nindo"* ($1/wk for bonus stamina and modest free summons), *Battle Pass* (rewards including rare shards), and **cosmetic skins** only via cash or event achievements. Avoid paywalls on core progression. Maximize free gems/runes via logins & achievements. Consider *"gacha filler"* (e.g. cheap single-summon tickets) and generous pity (hard cap) to reduce frustration【77†L273-L282】【58†L172-L180】.

****Aspect**:** Retention**** - ****AFK Arena (Lilith Games)**: Daily login** (gifts for 1–30 days, culminating in rare hero shards【77†L248-L250】). Recurring events (Labyrinth, Shadow Faction Wars) and timelines. Season battles (PvP ladder resets) to reward active play. Special story chapters and co-op challenges. - ****Mythic Heroes (IGG)**:** Weekly/monthly events tied to holidays (Greek festival, etc.). Login calendar grants month-long rewards (coins, gems). Arena/Guild boss rewards. "Treasure trials" mini-game. - ****Idle Heroes (DroidHang)**:** Daily Quests (Skirmish, Campaign, Arena). Weekly events (e.g. Ura/Valor Compass). Guild Wars, Arena with seasonal resets. Cumulative login (avatars and scrolls). Many complaint

(in 2021) that late-game daily rewards didn't scale with level【58†L114-L122】. - ****Naruto Idle (Concept)**:** **Daily/Weekly**: Login streak bonus (increasing Ryo, summon tokens). Weekly Ninja Exam (challenge tower) with leaderboard. Special *Village Festivals* (limited-time raids giving gear/rares). **Battle Pass** track (free and premium tiers with hero shards). **Login/Idle**: Offline training yields "Will/Spirit" tokens over time, incentivizing daily collection. Avoid grind walls by letting players accumulate offline rewards even when not powerful.

****Aspect**:** PvP & Matchmaking**** - ****AFK Arena (Lilith Games)**:** Ranked Arena with "Combat Rating" matchmaking. Shuffled rosters (ban phase) to avoid stale meta. Guild Wars (real-time co-op boss battles). PvP rewards are mostly cosmetic/leaderboard. - ****Mythic Heroes (IGG)**:** Arena with staged tiers; uneven match-ups at high rank. Friendly duels. Limited-time "God Arena" vs themed teams. - ****Idle Heroes (DroidHang)**:** Random Arena rank with tickets; players advance through leagues. Guild vs guild real-time battles. Many players find higher tiers dominated by whales. Soft reset (Season return currency). - ****Naruto Idle (Concept)**: Balanced PvP:** Use hidden MMR/battle rating to match fairly, not absolute power. Seasonal ladder with rank rewards. Team-ban drafting to force strategy over brute force. Cosmetic rewards (titles, skins). For F2P catch-up: "Weekly Rival" boss battles that let mid-tier players challenge top-tier NPC teams for gems (like *challenge tower* rewards).

****Aspect**:** Team Roles/Synergy**** - ****AFK Arena (Lilith Games)**:** Heroes have roles (Tank, Warrior, Ranger, Mage, Support) and factions (Lightbearer, etc.) with faction-buffs (e.g. +ATK if ≥3 same faction). Formations (offense/defense frontlines) matter. Synergy examples: e.g. two support heroes can amplify healing. - ****Mythic Heroes (IGG)**:** Character classes (e.g. Guardian, Berserker, Mage). Some synergy combos (e.g. all-Greek heroes get special bonus on certain stage). Requires mix of roles (at least 1 frontline tank, backline DPS). - ****Idle Heroes (DroidHang)**:** Classes (Fighter, Priest, Mage, Ranger, Support). Each with passive group buffs (e.g. all Fighters +crit chance). Synergy: matching 3 of a class gives team bonus. Role needed: one healer/priest recommended per team. - ****Naruto Idle (Concept)**: Naruto Roles:** e.g. *Atacker* (Naruto, Sasuke – AOE damage), *Tank* (Kakashi, Guy – block/damage soak), *Healer/Support* (Tsunade/Sakura heal or buff, Shikamaru crowd-control). **Synergy Rules:**

1. *Village Bonds:* Naruto characters from same village (Konoha, Suna, etc.) grant each other bonuses (+5% ATK/DEF for 2, +10% for 3).

2. *Team Combos:* Specific trios (e.g. Team 7 = Naruto+Sasuke+Sakura) unlock extra effects like a one-time *Team Attack* (massive damage).

3. *Clan Buffs:* Uchiha clan heroes together boost Critical Rate, etc.

4. *Chakra Types:* Matching chakra-nature (2+ Fire ninjas = +fire damage).

5. **Role Synergy:** Ensure at least one Tank, one Healer, two DPS. E.g.: Naruto (Shadow Clone passive spawns at 50% HP, plus rubidy synergy), Sasuke (crit buff for Uchiha ally), Sakura (group heal+red buff), Kakashi (slash all enemies on death), Gaara (ground ATK to stun).
   Example Teams:

6. *Balanced:* Naruto, Sakura, Kakashi, Rock Lee, Hinata (Healer/Buff). Roles cover DPS, heal, defense, speed buff.

7. *Susanoo Uchiha Team:* Itachi, Sasuke, Obito, Kakashi, Tobi – all Uchiha; they gain a big crit/ATK buff【74†L9-L12】.

8. *Medical Squad:* Sakura, Shizune, Tsunade, Kabuto, Healing Jutsu: powerful healing synergy (e.g. combined ultimate).

9. *Madara Clan:* Madara, Hashirama, Sasuke, Itachi, Minato – unstoppable but balanced with fewer buffs. Counters: Teams need edges (e.g. Susanoo Uchiha vs *sharingan-proof* characters like Pain/Ten) to avoid rock-paper-scissors breakdown. |

Formulas & Tables (Examples). To concretize progression, we propose typical formulas: e.g. *XP to next level* = $100 \times L^{1.2}$, *Gold for next level* = $50 \times L^{1.5}$ (escalating cost). *Summon currency:* 10 Ryo (premium) for 1 summon; pity triggers guaranteed SSR at 90 summons (hard pity)【77†L273-L282】. Heroes gain +10% stat per Promotion (ascension). *Duplicate payout:* each extra SSR beyond 5 → 1 full awaken star; beyond that, add 10% to spark gauge. For example tables:

Level (L): 1→2 - **XP needed:** 100 - **Gold cost:** 50 - **Cumulative Gold:** 50

Level (L): 2→3 - **XP needed:** 120 - **Gold cost:** 71 - **Cumulative Gold:** 121

Level (L): 3→4 - **XP needed:** 139 - **Gold cost:** 98 - **Cumulative Gold:** 219

Level (L): ... - **XP needed:** ... - **Gold cost:** ... - **Cumulative Gold:** ...

Level (L): 99→100 - **XP needed:** ~6300 - **Gold cost:** ~5000 - **Cumulative Gold:** ...

(Numbers illustrative; actual tuning may differ.)

Summons: 1–80 - **SSR Chance:** 1% each - **Guaranteed SSR?:** No - **Notes:** Standard rate

Summons: 81–89 - **SSR Chance:** +0.5% each - **Guaranteed SSR?:** Soft pity boost - **Notes:** Up to ~5%

Summons: 90**** - **SSR Chance:** N/A - **Guaranteed SSR?: Yes** - **Notes:** 100% SSR at 90

Currency Flow: e.g., 10 minions defeated = 1 Hero XP stone; daily quest = 1000 Gold + 5 Chakra Ryo. *Inflation control:* use multiple sinks (hero gear costs gold and shards; skill upgrades use rare scrolls; prestige resets use a new currency).

Balancing F2P vs P2W (Monetization Models): We outline several options:

1. *Cosmetics-First:* Sell skins and optional cosmetics (headbands, costumes) only. All combat power obtainable via play. Pro: Purely skill-based/strategy. Con: Fewer direct revenue if players ignore cosmetics.

2. *Small Bundles & Subscriptions:* Like Idle Heroes, offer a cheap weekly sub ($0.99–$2.99) giving +50–100% idle gold and daily bonuses【19†L73-L79】. Include small packs (e.g. “Daily Ryo Bundle” for $0.99 that yields 500 Ryo usable on summons). Pro: steady revenue, low entry. Con: Could slow F2P progress if poorly tuned.

3. *Battle Pass:* Offer a season pass (free+premium track). Premium track (e.g. $5/month) unlocks extra shards, currencies. Keep most essential items in free track (e.g. a 4★ hero shard at tier 10, full SSR hero at tier 50). Pro: predictable revenue and rewards; Con: needs continuous content.

4. *Limited Deals:* Occasional time-limited package (e.g. $1 for 100 Ryo, $5 for 550 Ryo). Pro: common tactic, players expect it. Con: Must ensure freebies are comparable; risk bad PR if too expensive.

Pros/Cons Table (F2P-Friendly vs P2W):

Model: Hard Pity (guarantee)**【77†L273-L282】** - **Pros:** Reduces frustration; F2P knows maximum spend. Good marketing. - **Cons:** Less chance-driven revenue; whales may feel capped.

Model: Soft Pity (increasing odds)**** - **Pros:** Encourages players to keep pulling; less guaranteed whales. - **Cons:** Less predictable F2P outcome; still feels RNG.

Model: All-F2P events (like AFK 1000 summons)**【27†L105-L113】** - **Pros:** Great goodwill; attracts players; kills P2W narrative. - **Cons:** Potential revenue loss if whales reduce spend.

Model: Small Subscription**【19†L73-L79】** - **Pros:** Reliable monthly income; minimal annoyance. - **Cons:** Could upset hardcore F2P if power too strong (keep buff mild).

Model: Gacha Packs**** - **Pros:** Target whales; easy to implement. - **Cons:** Widely seen as P2W; must be optional, not mandatory.

Model: In-Game Ads/Videos**** - **Pros:** Can reward gems for watching ads. - **Cons:** Affects immersion; low revenue.

Team Building & Synergy: A well-crafted team system is crucial. Roles (Tank/DPS/Healer/Support) must complement each other, and thematic “hero synergies” should reward creative composition. In Naruto terms, we envision:

1. **Role Definitions:**
    1. *Tanks:* High defense/HP (e.g. *Gaara*, *Rock Lee*, *Kakashi when shielded*).
    2. *DPS (Attackers):* High damage (e.g. *Naruto*, *Sasuke*, *Madara*).
    3. *Healers/Support:* Provide healing or buffs (e.g. *Sakura* heals, *Shikamaru* stuns crowd).
    4. *Debuffers:* (e.g. *Kurenai* genjutsu reduces enemy accuracy, *Haku* slows).
2. **Synergy Rules:** Use *qualifiers* (faction/village, signature combos): see table above. For example, any **three Konoha ninjas** might grant a flat +10% ATK buff to the team. A team with **Naruto & Sasuke & Sakura** (Team 7) gains a special "Resonance" effect: once per battle unleash a group buff or all-target nuke. Villages act like factions; using *Team 8* (Hinata+Kiba+Shino) could give a speed/stealth bonus.
3. **Counters and Balance:** Make sure no one team is "always best." For instance, Uchiha-dominant teams might struggle against *Yin-Yang Discord* counters (Hinata/Neji type healers). A team full of attackers but no healer should be vulnerable to attrition in long battles.
4. **Example Compositions:**
    1. **Balanced Ninja Team:** *Naruto (Atk)*, *Sakura (Heal/Support)*, *Kakashi (Tank/DPS)*, *Rock Lee (DPS/Rage)*, *Hinata (Debuff)*. Covers offense, defense, heal, control. This team would synergize off-season (all Konoha for +stat buff) and use Shuriken/Clone combos.
    2. **Sharingan Squad:** *Itachi, Sasuke, Kakashi, Obito, Kiba.* Three Uchiha grants big crit buff; Kakashi's "Double Sharingan" skill triggers extra bonuses. Countered by clones/misdirection (Naruto/Shino).
    3. **All-Out Attack:** *Madara, Hashirama, Sasuke, Kabuto, Deidara.* Nearly all heavy hitters. High base damage but very fragile. They'd smash non-tanks but struggle if a healer like Sakura is present (she can outlast them).
    4. **Healing Circle:** *Sakura, Shizune, Tsunade, Tenten, Ino.* Focus on healing and buffing each other; little direct damage. Surprising niche: they can out-sustain tougher teams, but fail if enemy one-shots a healer early.

Teams should *feel* like real ninja squads, not random collections. By giving clear bonuses for Naruto-specific combos (e.g. "Team 7" or "Hokage Council"), we encourage players to experiment with themed squads. We also recommend a UI that highlights known synergies (e.g. tooltip: "Naruto+Sasuke: Ultimate Duo – +5% crit").

Analytics & Telemetry Needs: To tune F2P balance, the game should track engagement vs spend (segmented by free vs paying users), pull rates, pity outcomes, event conversion rates, and Gini coefficient of power distribution. A/B tests could vary pity thresholds or daily resource caps. For example, test whether offering 5 vs 10 daily free summons increases long-term retention. Track which team comps players use and win with most, to adjust power levels or buff synergies.

Design Log & Open Questions:

1. *Pity Level:* Hard pity at 90? Soft vs hard? Experiment: some players get guaranteed SSR at 80, others at 100, measure retention.
2. *Duplicate Use:* Should 5 or 6 copies awaken a hero? Try both and survey satisfaction.

3. *Offline Gain:* How much idle XP/coins should accrue per minute? Too high kills progression; too low bores player. Start at 1x speed (normal mode), upgradable by prestige bonuses.

4. *Energy Stamina:* Will we have a spending-limited energy (like 10 tries/campaign per day)? If yes, compensate free by offering a small amount of "free energy refill" coupons via quests. Could A/B test with vs without purchase refill.

5. *Sub vs Battle Pass:* Try launching a $1/wk sub (bonus gold) vs a $4-month battle pass; which gives steadier revenue?

6. *Team Synergy Discovery:* We might not reveal all synergies upfront—players could experiment or learn on wiki. But ensure at least 5–10 well-known combos (e.g. core Naruto teams) for early game enjoyment.

7. *Balancing:* Will Naruto characters have destiny ties (like Naruto & Kurama) or just regular synergy? Perhaps special "bonded" abilities if using exactly the original Team 7. If too complex, maybe skip Kurama integration.

Mermaid Diagrams: Below are illustrative system flows:

```
flowchart LR
  Summoning["Summoning (Ryo → Hero)"] --> Heroes["Heroes Unlocked"]
  Heroes --> Team["Team Composition"]
  Team --> Combat["Battles/Quests"]
  Combat --> Rewards["Rewards (Gold, Shards, XP)"]
  Rewards --> Currency["Currencies Earned"]
  Currency --> Summoning
  Currency --> Upgrades["Level/Gear Upgrades"]
  Upgrades --> Team
  Combat --> Banners["Event Progression"]
  Banners --> Currency
  style Rewards fill:#e6ffe6
  style Upgrades fill:#e6f7ff
```

```
flowchart TD
  subgraph Player Journey
   A[Login / Collect Idle Gains] --> B[Choose Stage / Battle]
   B --> C{Battle Results}
   C --> |Win| D[Earn XP/Gold/Shards]
   C --> |Lose| E[Streak Bonus?]
   D --> F[Upgrade Heroes / Summon Heroes]
   F --> G[Stronger Team → Return to B]
   E --> B
   D --> H[Check Events / Daily Tasks]
   H --> I[Get Extra Rewards → D]
   G --> I
  end
```

Conclusion: By synthesizing mechanics from top idle RPGs and biasing toward player-friendly design, our Naruto idle game can offer **deep strategy and satisfaction without punishing F2P players**. Generous summoning (event pity, free pulls) and transparent progression (no pay-locked content) will build goodwill【27†L105-L113】【58†L172-L180】. At the same time, synergistic teams and a rich anime setting (Naruto clans, jutsus) provide the strategy and fan-service to keep both free and paying players engaged. The recommendations above (formulas, tables, designs) should serve as a concrete blueprint for development and tuning, ensuring the final game feels both authentic to Naruto and fair to all players.

Sources: Authoritative sources on idle/RPG mechanics, F2P/gacha design, and the games themselves were used. Key references include official descriptions for *AFK Arena*【74†L9-L12】【14†L64-L69】, *Mythic Heroes*【78†L138-L142】【

78†L232-L240】, *Idle Heroes*【19†L61-L69】【58†L172-L180】, as well as gacha system analyses【77†L273-L282】【77†L248-L250】 and player reviews.【27†L105-L113】【58†L114-L122】. Full links are listed below.

Source URLs:
- AFK Arena – App Store description (Lilith Games)【14†L64-L69】
- AFK Arena – Google Play page【27†L105-L113】
- AFK Arena Official Site (Lilith Games)【74†L9-L12】
- Idle Heroes – App Store description & reviews【19†L61-L69】【58†L172-L180】
- Mythic Heroes – Google Play page (IGG)【78†L138-L142】【78†L232-L240】
- *Gacha game* – Wikipedia (pity system, daily login)【77†L273-L282】【77†L248-L250】

Appendix C. Historical report preserved from the Library: Executive Summary (2)

Archive status: Full saved report text, converted to Word formatting. In-report citations may be tool-local; treat claims and numbers as historical research leads pending verification.

Executive Summary

Anime-style idle RPGs (like AFK Arena, Idle Heroes, etc.) rely on **team composition and synergy** to create depth. Teams typically balance archetypes (tank, physical DPS, magical DPS, healer, buffer, debuffer), and leverage explicit synergy mechanics (faction bonuses, combo triggers, leader buffs, positioning) to reward smart builds. We catalog patterns from top idle RPGs (e.g. Idle Heroes, AFK Arena, Legends of Glory) and show how these inform a Naruto-themed design. Key points: idle RPG heroes fall into classic RPG roles, and synergy often comes from matching factions or enabling combo chains【81†L62-L69】【86†L153-L156】. Cumulative duplicate systems (shards/awakening) plus pity timers shape progression (e.g. 10 daily summons “for a total exceeding 1,000” in AFK Arena【95†L67-L70】). A balanced F2P economy uses generous daily rewards and catch-up mechanics (offline training, free summons, “resonator” or similar) so free players progress steadily without being forced to pay. We outline design options (stat growth formulas, buff-stacking rules, resource requirements) with tables and diagrams. Finally, **Naruto-specific team templates** illustrate these principles (e.g. combining chakra-users, medical-nin, and taijutsu fighters into cohesive squads). Unresolved issues (ideal buff stacking rules, exact F2P/pity balance) and alternatives (e.g. optional battle passes, cosmetic currencies) are identified.

Roles/Archetypes in Idle RPGs

Idle RPGs generally divide heroes into familiar roles (with synergy-defined archetypes):
- **Tanks/Frontline Defenders:** High HP/defense units that absorb damage and protect allies. (E.g. Gaara-like “Shield” heroes or armored samurai.)
- **Physical DPS/Bruisers:** Melee attackers dealing sustained physical damage (e.g. Naruto as taijutsu DPS).
- **Magic/Ninjutsu DPS:** Ranged or area attackers using chakra or magic (e.g. Sasuke’s lightning jutsu).
- **Healers/Clerics:** Restore team HP and remove debuffs (e.g. Sakura/Tsunade-type medical nin).
- **Buffers:** Grant positive buffs (attack%, crit%, speed) to allies (e.g. Kakashi boosting chakra regen).
- **Debuffers/Controllers:** Inflict negative statuses (stun, slow, defense-down) on enemies (e.g. Shikamaru’s trap to immobilize foes).
- **Utility/Support:** Other roles like “summoners” (e.g. Kabuto summons), disablers, or hybrid fighters.

Each role interlocks: tanks soak damage so squishies survive, healers sustain the tanks, buffers amplify DPS, and debuffers weaken bosses. For example, Idle Heroes advertises “more than 400 Heroes in different factions with specific skills,” encouraging players to cover roles【81†L62-L69】. The chosen roles must complement each other: a team with only DPS and no healer or tank will suffer in sustained fights, while too many healers and no damage leads to slow clears. Good idle-RPG design explicitly defines these roles, often in guild raids or arena where balanced team-building is key.

Synergy Mechanics

Idle RPGs enrich team-building with explicit synergies:

1. **Faction/Alignment Bonuses:** Placing heroes from the same faction or theme yields buffs. For example, *Legends of Glory* notes each hero has "exclusive skills and faction synergies" and urges players to "build the strongest team" by combining them【86†L153-L156】. In a Naruto game, alliances like *Konohagakure clan* or *Akatsuki team* could grant collective bonuses when mixed.
2. **Leader Skills & Tags:** Some heroes provide passive buffs to teams (e.g. "Konoha Leader: +10% attack to all Konoha ninja"). These tag-based skills reward specific compositions.
3. **Combo Chains/Triggers:** Abilities can trigger follow-up actions in others. For instance, one hero's taunt can force enemies onto a tank, enabling high-DPS allies to focus them. Or a stunlock from one hero lets a second hero land critical hits.
4. **Buff Stacking:** Multiple heroes may add the same buff (e.g. two allies each granting +15% attack). The stacking rule (additive vs multiplicative) matters. A simple formula is additive stacking: *Total attack buff* = BaseAttack × (1 + 0.15 + 0.15) = Base×1.30 if two heroes each give +15%. Alternatively, some games multiply buffs for diminishing returns (final attack = Base×1.15×1.15). Design choice: additive stacking keeps synergies linear, while multiplicative stacking penalizes heavy stacking.
5. **Positional Strategy:** Many idle RPGs allow front/back row placement. Example: a front-line tank (Naruto in Sage Mode, Gaara) stands in front, while fragile DPS (Sasuke, Shikamaru) stay back. Some games have row bonuses (e.g. "front row heroes get +X% defense"). This adds strategic depth: e.g. *Tsunade is placed back to heal, while Hatake Kakashi stands front to engage*.
6. **Tag/Genre Synergy Examples:** Although less common, one could include Naruto-themed combos. For instance, a buff like "If Naruto and Sasuke are both on team, grant +20% group attack for 6 seconds" (hypothetical). These special combos encourage collecting related heroes.

In sum, strong synergy ties units together. As one example, Legends of Glory's marketing explicitly advises summoning and team-building to exploit faction synergies【86†L153-L156】. We adopt a similar approach: define clans or chakra-nature sets in Naruto, and give bonus effects when multiple clan members appear.

Counterplay and Balance (Rock–Paper–Scissors)

Balanced idle RPGs include counterplay so no single archetype dominates. Common patterns:
- **Faction or Elemental Counters:** e.g. Class Advantage: Warriors > Mages > Rogues > Warriors, or Chakra Nature triangles (Water beats Fire, etc.). AFK Arena uses a Light–Dark–Beast cycle internally (Light defeats Hypogean, Hypogean defeats Mauler, etc.), enforcing a simple rock-paper-scissors dynamic. We should implement a similar loop (perhaps *Taijutsu beats Ninjutsu beats Genjutsu*), ensuring every team has a weakness. This prevents "always use Team 7 no matter what."
- **Diverse Counters:** Tanks and high-DEF units counter pure glass-cannon teams, while multi-hit AoE DPS counters teams of isolated squishies. E.g. Gaara's shield nullifies one big hit, so stacking a monster boss might fail against him. If boss can stun backline, having a healer or cleanse (like Ino) is needed. This encourages players to switch or rotate teams versus different challenges.

Counters should be clear but not absolute. We can include a simple visible marker (e.g. "fire vs water vs wind") or implicitly via skills.

Duplication, Awakening, and Summoning Pity

Idle RPG collection revolves around duplicates. Key design elements:

1. **Awakening/Ascension:** Duplicate heroes are typically used to rank-up or awaken a hero. For example, Idle Heroes lets you "convert [warriors] into Spirit material for evolving" stronger heroes【81†L64-L69】. In a Naruto game, duplicates might become scroll fragments or "chakra stones" needed for 5★→6★ ascension, unlocking new skills. Balancing question: How many duplicates to need per star-level? A typical formula is something like:

1. Hero Star Upgrade: require N shards where N = base_shards * (star_level)^2.

2. For instance, 4★→5★ might need 100 shards, 5★→6★ 150 shards (higher cost as power jumps). We provide a sample table below.

3. **Pity and Summoning:** Gacha systems should include pity to avoid endless bad luck. A design example: *AFK Arena's "Get All Heroes" event gave 10 free summons daily, up to 1,000 total, explicitly stating "No more diamond*

worries. Just summon to your heart's content!"【95†L67-L70】. This showed extreme generosity. In general, consider mechanics like:

1. A guaranteed 5★ hero every 80-100 pulls (pity counter resets on 5★ draw).
2. Periodic free multi-summons (e.g. monthly event gives 10 free ten-pulls).
3. Use duplicate pulls to grant pity points (e.g. every spare drops adds +1 to a pity meter).

4. **Duplicate Value:** Not all duplicates need to be full waste. Options: feed duplicates for XP, convert to shards, or use in a "heroic salvage" system. The Idle Heroes resonator idea (mentioned in user reviews【61†L51-L59】) lets players transfer levels between heroes, effectively repaying some cost of building weaker copies. We could include a "reincarnation" feature letting players refund part of a duplicate's XP/gold if they switch builds.
5. **Summoning Currency Economy:** Manage the flow of premium currency (gems) vs free currency. Incorporate daily logins, chest rewards, and ad-watches to earn pulls. For example, the AFK Arena description promises *"10 summons every day"* in an event【95†L67-L70】. Even if not daily long-term, aim for generous *"free-to-play-friendly"* summon opportunities.

Overall, define clear formulas. For instance:

Example Pity Formula: Let *P* be base 5★ pull rate (say 0.5%). Guarantee a 5★ by 100 pulls by tracking pity += 1-(1-P). At 100 pulls force a 5★ if none occurred.

Awakening Costs (sample):

Star Level: 4★→5★ - **Fragments Needed:** 100 shards - **Gold Needed:** 200,000 yen (gold) - **Notes:** Unlock level cap +10; 5★ skill

Star Level: 5★→6★ - **Fragments Needed:** 150 shards - **Gold Needed:** 500,000 - **Notes:** +Another cap; elite skin (Naruto Sage Mode)

Star Level: 6★ (max) - **Fragments Needed:** — - **Gold Needed:** — - **Notes:** —

(Naruto example: shards themed like "Tailed Beast Chakra")

These numbers can be tuned, but illustrate that higher ascensions cost more.

Progression Pacing & F2P Balance

Making a generous economy keeps players engaged. Design patterns:

1. **Idle Rewards:** Continuously give offline XP/gold. For example, Idle Heroes' tagline is "Idle System – Set your heroes training while you're away… they will be stronger"【81†L62-L69】. Implement auto-battle rewards and periodic offline chests so progress isn't stalled.
2. **Daily/Weekly Quests & Events:** Provide routine tasks granting summons or high-value items. E.g. AFK Arena's daily/weekly missions should always be fully completable by F2P. Special limited events (festival draws, login events) can give bonus shards/summons. The AFK Arena "Companions" event giving 1,000+ summons【95†L67-L70】 is an extreme case; smaller versions (e.g. 50-100 free pulls) each month could be sustainable.
3. **Catch-up Mechanisms:** Ensure old players can catch up with newbies. Ideas: VIP-less "reincarnation" as above, or treasure that converts old currency to new, special "Resonator" resets, or cooperative play bonuses (e.g. guild shop discounts) that help F2P scale over time.
4. **Difficulty Curve:** Pace content so normal F2P gear can clear it without spending. Excessive paywall spikes (as one AFK review noted) erode trust. We must avoid steep "difficulty walls" that *force* spending. Level caps and gear tiers should be placed so that reasonably focused free play can eventually break them.
5. **Monetization Focus:** Pivot pay options toward *pay-for-convenience/cosmetics*, not raw power. Subscriptions or battle passes (giving small VIP-like bonuses but still accessible to F2P) can work. For example, a weekly sub

might give +10% idle gold【60†L71-L80】, but not skip critical progression. Avoid gating core heroes behind pay-only (instead use pity or guaranteed event heroes).

6. **Reward Scale:** Scale rewards with player level. (Idle Heroes players complained late-game rewards felt like early-game【81†L62-L69】; solve by ensure activities grow in value).

PvP Matchmaking and Fairness

In PvP (Arena, ranked), match teams by power or tier. Anti-pay-to-win measures include:
- **Separate Leagues:** Whales face whales in upper tiers, free players in lower tiers.
- **Roster Restrictions:** E.g., only heroes up to 5★ allowed in beginner PvP, unlocking 6★ in higher divisions.
- **Normalization Events:** Some games offer a "light" PvP mode where stats are normalized to level or gear, focusing on deck composition.
- **Match Decay:** If a player goes inactive, gradually match them with weaker opponents.

While detailed PvP design is beyond scope, any scheme that lets skill or strategy matter more than who paid wins is key (e.g. AFK Arena's *King's Tower* often matches by Combat Power).

Design Patterns from Top Idle RPGs

1. **Idle Heroes (DHGames):** Advertises "Idle system" and **400+ faction-based heroes**【81†L62-L69】. Rich guild/arena content means players form *balanced parties* to tackle many modes. It uses spirit shards and evolution for duplicates.

2. **AFK Arena (Lilith Games):** Emphasizes passive progress ("AFK farming") with automated battles【95†L67-L70】. It has seven factions and a rock-paper-scissors counter system (noted across the web). AFK Arena has events like "Get All Heroes" ensuring free summons【95†L67-L70】 and a hero-level sharing "resonator" system【61†L51-L59】.

3. **Legends of Glory (Idle Arena):** Features an AFK idle model ("Even while offline, your heroes generate gold and experience"【86†L145-L152】) and calls out "Strategic Team Building" with "exclusive skills and faction synergies"【86†L153-L156】. It suggests a large roster where combining heroes is crucial.

4. **Other Anime Idle Games (Naruto/One Piece titles):** While specifics vary, many mimic these designs: turn-based auto-battles, story-based squads (forming iconic teams), and currency funnels into character stars. Official sources (e.g. App Store pages) often tout hundreds of characters and guild battles – reinforcing the need for **balanced team roles**.

From these patterns, we adopt: **multiple currencies** (XP/gold/shards), **multi-step hero growth** (level→ascend→awaken), and **diverse game modes** (campaign, arena, guild) requiring different team builds.

Naruto-Themed Team Examples

Below are illustrative Naruto squads showing roles and synergy. Each team covers key roles (Tank, DPS, Healer, Support, CC) with a thematic twist. *(These are conceptual examples.)*

Stage: Early (Level 1–30)**** - **Composition (Roles):** 1 DPS, 1 Healer, 1 Support - **Example Heroes (Naruto): Naruto (DPS)** – wind-nature attack**Sakura (Healer)** – medical nin**Kakashi (Support)** – attack buff/utility - **Strategy/Synergy:** Basic *Team 7* archetype: Naruto deals damage, Sakura heals, Kakashi provides minor buffs or AoE. Covers offense, sustain, and a bit of speed buff.

Stage: Mid (Level 30–60)**** - **Composition (Roles):** 1 Tank, 2 DPS, 1 Healer, 1 Support - **Example Heroes (Naruto): Gaara (Tank)** – sand shield, high DEF**Sasuke (DPS)** – lightning Rasengan**Naruto (DPS)** – Rasenshuriken**Sakura (Healer)** – big heal**Shikamaru (Support)** – bind/debuff - **Strategy/Synergy:** *Split focus:* Gaara soaks hits; Naruto/Sasuke provide high burst; Shikamaru immobilizes or stat-drains enemies; Sakura keeps team alive. Strong synergy: e.g. a stun from Shikamaru allows Naruto to unleash a boosted jutsu.

Stage: Late (Level 60+, Endgame)**** - **Composition (Roles):** 2 Tanks, 2 DPS, 1 Healer, 1 Support - **Example Heroes (Naruto): Hashirama (Tank)** – wood shield + regen**Madara (Tank)** – wood/technical resilience**Naruto (DPS)** – Six Paths Mode**Sasuke (DPS)** – Rinnegan techniques**Tsunade (Healer)** – massive heal**Shikamaru (Support)** – combo debuffs - **Strategy/Synergy:** High synergy *Kage team:* Double tanks upfront; Naruto/Sasuke with ultimate Ninjutsu and synergy (legendary Team 7 power); Tsunade's clutch heals; Shikamaru's Nara clan combos. This team can execute combined attacks (speculative), e.g. Naruto's Rasenshuriken followed by Sasuke's Amaterasu for huge damage.

Each example covers: **Tank(s)** (Gaara/Hashirama/Madara), **Primary DPS** (Naruto, Sasuke), **Healer** (Sakura/Tsunade), **Support/Debuff** (Kakashi/Shikamaru). We see a progression: early focus on Team 7 trio, mid adding more crowd-control, late forming a full "Hidden Leaf Kage" squad. These roles interlock: tanks protect healers/DPS; healers keep tanks alive; buffers (Shikamaru) amplify damage. (In a final game design, one might add explicit bonuses – e.g., a "Konoha Legacy" passive if Naruto and Hashirama both in team.)

Formulas & Technical Specs

1. **Stat Scaling:** We can set a formula for hero stats. For instance:
   HP(L, S) = BaseHP × [1 + 0.02×(L-1)] × [1 + 0.10×(S-1)]
   where *L* = level (1–max per star) and *S* = star tier (1★ base to 6★ max). (Thus each level adds 2% of base, and each star adds 10% extra.) Similar formulas apply for Attack/Defense. This ensures roughly linear growth with diminishing returns at higher tiers.
2. **Buff Stacking Rule:** We choose additive stacking of like buffs (to keep it simple):
   FinalStat = BaseStat × (1 + Σbuffs).
   E.g., two buffs +20% and +15% yield a total 1.35× multiplier. Distinct types (attack% and speed%) just multiply separately by their sum.
3. **Pity Progression:** If a 5★ hero has base pull chance *p* per draw, guaranteed pity at *N* draws solves for *p* such that 1-(1-p)^N = 1. For example, guarantee at 100 pulls means p = 0.058, but we can instead just hard-stop after *N* draws. A friendly option: grant 5★ at *90th* pull automatically.
4. **Level Up XP/Gold:** Example progression (Naruto game): leveling a hero from 1→60 might require around 200,000 XP and 300,000 gold (per 1→60, inclusive). We can tabulate breakpoints (every 10 levels) for clarity:

Level Range: 1→10 - **Total XP Required:** 5,000 - **Total Gold Required:** 10,000

Level Range: 11→20 - **Total XP Required:** 15,000 - **Total Gold Required:** 30,000

Level Range: 21→30 - **Total XP Required:** 30,000 - **Total Gold Required:** 60,000

Level Range: 31→40 - **Total XP Required:** 60,000 - **Total Gold Required:** 120,000

Level Range: 41→50 - **Total XP Required:** 100,000 - **Total Gold Required:** 200,000

Level Range: 51→60 - **Total XP Required:** 200,000 - **Total Gold Required:** 300,000

(These are illustrative; actual values scale with hero rarity and intended grind.)

1. **Upgrade/Ascension Example Table:**

Ascend Tier: 4★→5★ - **Hero Rarity:** Epic - **Required Fragments:** 100 "Chakra Shards" - **Gold:** 200k - **Level Cap:** +10 (→Lv50) - **Effect:** +1 Skill/Passive

Ascend Tier: 5★→6★ - **Hero Rarity:** Legendary - **Required Fragments:** 150 "Shards" - **Gold:** 500k - **Level Cap:** +10 (→Lv60) - **Effect:** +Major Ult Skill

Ascend Tier: 6★ (max) - **Hero Rarity:** Legendary - **Required Fragments:** — - **Gold:** — - **Level Cap:** 60 - **Effect:** —

This exemplifies how duplicates (fragments) are required to unlock the next star-tier.

1. **Battle Formula Example:** Suppose Hero A has base attack 1,000 and Hero B has +20% attack buff and Hero C adds +15%. Final attack of a damage dealer = 1000×(1+0.20+0.15) = 1,350. A tank's Taunt might have a formula like: Threat = 100 + 0.5×HP to draw aggro. We would specify such formulas in final design docs.

Design Alternatives & Decisions

1. **Buff Rules:** We must choose additive vs. multiplicative stacking. A possible alternative is to cap stacking (e.g., max +100% total), but we lean to **unlimited additive stacking** for simplicity unless playtesting shows abuse.
2. **Pity System:** Instead of an exact pity count, another option is a **"pity currency"**: each 10× summon without rare grants a token; 10 tokens = guaranteed legendary. This is user-friendly and avoids hard counters. Still, we recommend a clear pity counter (as players like transparent goals).
3. **Currency Inflation:** Deciding how much free vs. paid currency to give is key. We resolved to lean generous (e.g., daily free 10× pulls as in AFK Arena's event【95†L67-L70】) because excessive grind frustrates and kills retention. An open question: what daily/hourly logins are sustainable? (We might A/B test: 2×10-pull per month vs. 5×, etc.)
4. **Progression Curve:** Should heroes "diminish" after a point (like Idle Heroes' Transcendence)? Idle Heroes uses a hyper-ascension (Transcend) where elite heroes get even stronger benefits. We could adopt a **Ninja Ascension** late-game upgrade for vets, but keep base progression flatter to avoid hitting walls early.
5. **Matchmaking:** Simplest is **Combat Power brackets**. More sophisticated is Elo/MMR, but that risks "rank faking." A hybrid: tiered arenas (Bronze, Silver, Gold, etc.) with CP caps.
6. **Free-to-Play Focus:** We prioritize *pay-for-progress* (convenience) rather than *must-pay-to-win*. For example, selling XP boosts or currency packs shortens grind, but all heroes remain achievable by patience. Alternate model: cosmetics store (special Ninja skins). This aligns with best practices: let whales excel slightly faster, but do not lock endgame heroes behind a paywall.

Mermaid Diagrams

Team-Role Relationship Diagram:

```
graph LR
  Tank --Protects--> DPS
  Tank --Absorbs--> Debuffer
  DPS --DependentOn--> Support
  DPS --HelpedBy--> Buff
  Healer --Supports--> Tank
  Healer --Sustains--> DPS
  Debuffer --Weakens--> Opponent
  Buff --Enhances--> DPS
  Buffer --Coordinates--> DPS
```

This illustrates typical role interplay: Tanks protect others; Healers sustain everyone; Buffers aid DPS; Debuffers weaken foes to synergize with DPS.

Progression Flow Diagram:

```
graph TD
  Summon["Summon Hero (Gacha)"] --> BaseHero["Hero (Base)"]
  BaseHero --> LevelUp["Level Up (XP/Gold)"]
  LevelUp --> Ascend["Ascend (★ Upgrade via Shards)"]
  Ascend --> Awaken["Awaken (Final Tier)"]
  Duplicate["Hero Duplicate"] -.-> Ascend
  ShardCollect["Collect Hero Shards"] -.-> Ascend
  Awaken --> Transcend["Transcendence (Optional)"]
  Grind["Grind/Afk Progress"] --> LevelUp
  Grind --> Ascend
  Grind --> ShardCollect
```

This flow shows a hero's growth path: summoned heroes gain XP to level up, use shards (from duplicates) to ascend stars, then optionally reach a transcendence stage. Offline/AFK grind feeds all stages.

Recommended Decisions (F2P-Friendly)

1. **Generous Free Summons & Pity:** Implement daily free summons and a clear pity system (e.g. guarantee 5★ by ~100 pulls) as AFK Arena did with 1,000 free summons【95†L67-L70】. This greatly aids retention.

2. **Idle Rewards:** Use constant passive gain; ensure an idle baseline that even F2P players make visible progress every login (similar to Idle Heroes' model【81†L62-L69】).

3. **Balanced Difficulty:** Tune campaign/raid so a moderately powered F2P team can eventually clear it. Avoid sudden difficulty spikes unless optional.

4. **Alternate Currencies:** Include multiple soft currencies (XP, gold, shards) and ensure many earnable sources (quests, shops, events) so the economy isn't bottlenecked by pay-only currency.

5. **Free/paid cosmetic content:** Skins or renames of heroes can monetize without affecting balance.

6. **Ongoing Events:** Regular time-limited events with leaderboard rewards (e.g. "Defeat the Nine-Tails – top damage get goodies") engage players. Ensure even low ranks get modest rewards.

Unresolved Questions and Considerations

1. **Exact Buff Formula:** Should buffs be fully additive or multiplicative? We plan additive for predictability, but need to test if stacking two 50% buffs is overkill. Perhaps cap buffs or use diminishing returns.

2. **Shard Drop Rates:** How many fragments to drop from stages/events to feel fair? Need ratio so collecting 100 shards over weeks isn't too grindy.

3. **Team Synergy Limits:** How many buffs should stack before hitting a diminishing return (to prevent infinite loops)? Possibly limit one stack per buff type.

4. **Paywall Placement:** Determining which heroes or gear should be earnable vs premium-exclusive. We lean earnable for all base heroes, with some special exclusives (e.g. a Legacy skin) for paying.

5. **PvP Incentives:** Should rank resets or seasonal rewards be generous enough that casual players don't feel obligated to climb every season (to avoid P2W pressure)?

Sources

1. Idle Heroes (DHGames) — Official App Store page (game description)【81†L62-L69】.

2. AFK Arena (Lilith Games) — Official website and App Store content【58†L9-L13】【95†L67-L70】.

3. Legends of Glory: Idle Arena — Google Play Store (developer description)【86†L153-L156】【94†L0-L4】.

4. User reviews and forums (e.g. Reddit) informed F2P concerns (cited for context but not used directly in analysis).

Appendix D. Historical report preserved from the Library: Economy & Summoning Mechanics

Archive status: Full saved report text, converted to Word formatting. In-report citations may be tool-local; treat claims and numbers as historical research leads pending verification.

Economy & Summoning Mechanics

In a Naruto-themed idle RPG, use a **dual-currency economy** (free vs. premium) and robust "gacha" summoning with pity to balance progression. Implement safety nets: for example **hard pity** guarantees a top-tier (e.g. 5★) hero after a fixed number of pulls【62†L273-L281】. (Soft pity can gradually raise rates as players near the threshold【62†L273-L281】.) Give daily/weekly login rewards and quest achievements that dole out premium currency or free pulls【62†L248-L252】, so even non-paying players steadily accumulate summons. Consider a "spark" system: each pull grants a special token, and collecting enough tokens lets the player directly purchase a featured hero【62†L273-L281】. Make duplicate heroes useful – for example, convert them into **hero shards or essence** that let players awaken or upgrade other heroes. (In many idle gachas, duplicates auto-upgrade the hero or can be exchanged in a "fusion" system.) This ties together summoning and progression: as players pull more, they both power up existing heroes and unlock new ones.

Free-to-Play Balance & Monetization

To avoid a "pay-to-win" trap, design generous F2P funnels. Provide regular *free* premium currency (login streaks, event milestones, daily bosses) so free players can meaningfully summon over time. For instance, give at least one free "10x pull" per week from events or a rotating store. Use pity and spark to cap the cost of chasing characters【62†L273-L281】. Limit exclusive advantages to **cosmetics or convenience** (XP boosters, auto-battle) rather than raw power, so paying mostly accelerates progress without outright gatekeeping content. Veteran players note that many idle games **start moderately F2P-friendly but become very P2W after updates**【64†L53-L60】; plan updates carefully to keep long-term balance. For example, ensure new heroes are *not* strictly necessary to beat content, and that story/campaign drops and free-event tokens allow free players to gradually assemble competitive teams. In short, aim for steady but fair progression: paying eases the grind, but free players can still advance and unlock top heroes (just more slowly)【62†L248-L252】【64†L53-L60】.

Progression & Upgrade Systems

Each hero should level up (via XP/Gold), rank up (using ascension stones or shards), and equip gear. **Awakening** (or rank-up) typically requires duplicates or special items: e.g. evolve a 4★ hero to 5★ by consuming 20 shards obtained from pulls. Link this to summoning pity: if duplicates yield shards, then pity rates indirectly affect gear progression as well. Balance costs so that early levels and ranks are easy for free players, but higher tiers require significant investment (encouraging both play and potential spending). For example, you might require 10–20 duplicates to fully awaken a hero to max stars, meaning a free player would need to pull ~100–200 times (possibly spread over months) to guarantee one copy, whereas a payer could hit that faster. Use formulas that grow *sub*-exponentially rather than runaway-exponentially, so that steady play always feels like progress.

Team Composition & Synergy

In any RPG, a well-rounded team has **complementary roles**. Define clear roles (e.g. Tank/Defender, Damage Dealer, Healer, Buffer, Crowd-Control) and encourage mixing them. Synergy is key: *synergy* means "the combined effect [of heroes] is greater than the sum of their individual effects"【53†L192-L200】. For example, one hero might buff allies' attack, another shreds enemy armor; together they accomplish more than alone. A balanced team might include:

1. **Tank/Protector:** absorbs damage to shield squishies. (e.g. a sturdy Leaf Village shinobi.)

2. **Damage Dealer (DPS):** high single-target or AOE damage. (e.g. fire-nature ninjas or lightning users.)

3. **Healer/Support:** restores health or shields, keeping the team alive. (A medical ninja like Tsunade.)

4. **Buffer/Debuffer:** increases team stats or weakens enemies (e.g. buffing chakra regen or reducing enemy defense).

5. **Control/Utility:** stuns, sleeps, or crowd-controls enemies to give allies openings.

A good team isn't just a random set of heroes; it's the right **mixture** of these roles. For instance, a "Team 7" synergy might have Naruto (aoe chakra burst + buffs), Sasuke (single-target nukes), and Sakura (heals); their combined kit outperforms a random trio【53†L192-L200】【55†L1-L4】. In design, you can also add **faction or clan bonuses**: e.g. using multiple Leaf Village ninjas grants a bonus (like AFK Arena's faction synergies【57†L9-L11】). Ultimately, the team's strength should grow non-linearly: two heroes that combo well (say one buffs speed and the other explodes on high speed) should outperform two heroes that act completely independently. This team-effectiveness – a classic synergy – means "the whole is greater than the sum of its parts"【53†L192-L200】.

System Integration & Design Notes

1. **Currencies:** Premium gems should be scarce but supplemented by f2p sources (events, ads, dailies). Gold or common currency is plentiful (used for upgrades).

2. **Duplicates & Awakening:** Decide if duplicates auto-upgrade or convert to universal shards. E.g., duplicate Naruto yields Naruto shards, which can be traded for him or used elsewhere at a penalty. This ties awakening costs to summon rates.

3. **Summon Rates & Pity:** Choose an SSR drop rate (e.g. 1%) and set a hard pity (e.g. 100–200 pulls). Possibly implement a 50/50 for featured heroes as a higher-tier pity layer【62†L269-L278】.

4. **Progression Pace:** Track player power curve. Free players should reach cap endgame in perhaps 6–12 months, paying players in ~1 month, not weeks. Tiered content unlocks (e.g. Chapter bosses requiring 5★ heroes) should be beatable at modest gear by multiple team compositions.

5. **Event Design:** Regular events give out character shards or summons. For example, a "Heroic Anniversary" event could give enough shards to unlock a new character at 5★ if collected across time. This rewards loyalty and offsets paywalls.

6. **Balancing Notes:** Keep an eye on feedback: games like Mythic Heroes initially felt rewarding to F2P players but grew frustrating with paywalls【64†L53-L60】. Monitor that free rewards (login bonuses, milestone gifts) scale with inflation.

Unresolved Questions / Next Steps

1. *Shard vs Essence System:* Will duplicates be used for direct star upgrades or a special currency? (Fusion systems help alleviate RNG frustration.)

2. *Pity Thresholds:* Exactly how many pulls to hard-pity? (Industry varies 90–200 for SSR.)

3. *Team Bonuses:* What specific synergies (e.g. Naruto clan, Academy, Elemental) will be implemented to encourage themed squads?

4. *Progression Formula:* Define XP curve, gear stat growth, and gold sink rate so that both grinders and payers feel steady progress.

5. *Balancing F2P Content:* Which endgame bosses or PvP tiers will free teams realistically reach?

Overall, these systems must be designed holistically. Awakening costs affect summoning pace; summoning generosity (pity, shards) impacts player retention and economy health; team roles and buffs influence how much players need to collect. Document each decision and iterate designs (including formulas and thresholds) once we test with target players.

Sources: Core concepts of gacha economy and pity systems are well-documented【62†L273-L281】【62†L248-L252】. Real-player experience (e.g. Mythic Heroes) underscores the dangers of becoming too pay-to-win【64†L53-L60】. Team-synergy principles follow the basic definition of synergy【53†L192-L200】【55†L1-L4】.

Appendix E. Complete project archive, research-session ledger, and file disposition

Archive audit date: 28 October 2026. This appendix records what was actually recovered, what is included in this master, which prior deliverables were merged, and which research requests have not produced a completed report. It exists to prevent future work from being mistaken for completed research.

E.1 What "complete archive" means in this project

The four full saved research reports listed in E.2 are preserved in editable form in Appendices A-D. The body chapters synthesize their reusable ideas, add explicit project decisions and proposed combat/economy rules, and reconcile some contradictions. Previous working-document versions were not appended as verbatim duplicate snapshots because they repeat the same content and sometimes contain stale or conflicting values. Their unique substantive material is carried into the current body; their file names and dispositions are recorded in E.3.

Important boundary: a Deep Research call that returned only a "started" confirmation, or returned a rate-limit error, is not a completed report. Those requests are captured in E.4-E.5 as pending research scope, not represented as findings. No unsupported content has been invented to fill those gaps.

E.2 All recoverable completed research report outputs

Report / location	Status	Approx. source length	Principal coverage	Archive disposition
Appendix A / "Executive Summary"	Completed saved report	About 5.5k	Core idle-RPG loop; auto/turn combat; progression; hero collection; currencies; monetization; retention; pacing; UX; analytics; anti-grind; legal/IP; representative-game comparison.	Full report text preserved in Appendix A; source record retained in Library.
Appendix B / "Executive Summary (1)"	Completed saved report	About 3.2k	AFK Arena, Mythic Heroes and Idle Heroes comparison; summoning/pity; duplicates and awakenings; resource and currency loops; progression; F2P and payer considerations.	Full report text preserved in Appendix B; source record retained in Library.
Appendix C / "Executive Summary (2)"	Completed saved report	About 3.45k	Team roles; composition synergy; counterplay; duplicate and summoning considerations; progression; PvP; design patterns; Naruto-illustrative teams; formulas; open questions and sources.	Full report text preserved in Appendix C; source record retained in Library.
Appendix D / "Economy & Summoning Mechanics"	Completed saved report	About 1.15k	Free/paid currency; pity and fragments; hero upgrades; progression; team composition; system integration; unresolved economy questions.	Full report text preserved in Appendix D; source record retained in Library.
Total recovered report text	4 completed outputs	About 13.25k	These are the completed report outputs actually found in the project Library during this audit.	This total excludes the synthesis chapters, file notes, and tables added to the living master.

Counting note: word counts are approximate and vary slightly with markdown, tables, hyphenation and headings during conversion. The report text itself, rather than a word-count estimate, is the archive of record.

E.3 Prior deliverables and how their content is handled

Prior file / artifact	Purpose	Disposition in this live master
Naruto_Idle_RPG_Master_Design_Bible.docx (first draft / v0.1)	Initial working game concept	Its project brief and first-pass system ideas were carried forward and refined in Sections 1-15 and the historical report appendices. Superseded as a master.
Naruto_Idle_RPG_Master_Design_Bible_v02_base.docx	Pre-consolidation master base	Its main-body design material is carried forward into Sections 1-15. It is not the live file.
consolidation_addendum.md	Markdown consolidation addendum	Its consolidation audit, scope matrix, integrated game concept, fairness decisions, change log and research appendices are incorporated into the live master. The .md is a working intermediate.

Prior file / artifact	Purpose	Disposition in this live master
consolidation_addendum.docx	Word-formatted addendum	Word rendering of the addendum; substantive content is incorporated into the live master. This is not a separate research result.
Naruto_Idle_RPG_Master_Design_Bible_v0.2.docx	Earlier merged master snapshot	Its main-body content and Appendices A-D are incorporated into the live master; it is superseded.
/Game Design/Naruto_Idle_RPG_Master_Design_Bible.docx	The live continuously edited master for the mobile-first Naruto idle RPG design project.	LIVE MASTER — v0.9 after this revision. Continue editing this exact Library file in place; no competing master is to be created.

The file register above distinguishes research reports from drafts used to assemble them. The four completed report outputs are reproduced in full as historical appendices; prior synthesis drafts are represented by their full unique content in the current chapters rather than pasted in again as duplicate whole-document snapshots. This avoids burying the live decisions under repeated, sometimes contradictory copies of the same draft.

E.4 Chronological research-session ledger

ID	Request / research intent	Recorded output	Status / limitation
R-01	Holistic anime idle-RPG theory, using Naruto/One Piece-style games as genre examples. Requested the full system: gameplay loop, battles, collection, shops, progression, rewards, modes, economy and retention.	The broad completed report “Executive Summary” is archived in Appendix A.	Archived as historical baseline; source claims still need re-verification before shipping.
R-02	Expand and deepen the project; study AFK Arena and Mythic Heroes; make the game more generous for F2P while preserving long-term progression and a sustainable payer value proposition.	Completed reports covering the economy/summoning and broad F2P/benchmark topics are archived in Appendices B and D.	Partly answered in the source reports; exact numerical economy still proposed, not validated.
R-03	Study team composition properly: roles, synergy, weakness, counterplay and the difference between a coherent team and five random high-power heroes.	Appendix C; Sections 5-7, 24-27: coherent compositions, role taxonomy, synergy, weaknesses, 19 placeholder kits, 10 archetypes, counter matrix, normalization packet and edge-case audit.	Written theory/spec substantially expanded; all coefficients, archetype thresholds and counter viability require simulation/playtest.
R-04	Broaden benchmarking beyond AFK Arena and Mythic Heroes to popular idle RPGs and adjacent hero collectors; distinguish genre types and compare transferable patterns.	Sections 4, 25 and 27 contain desk-source comparisons across the named idle and adjacent hero-collector games, with source type and transfer lesson.	Not exhaustive, not hands-on every game/current patch. Any future game-specific mechanics must be date-checked and properly sourced.
R-05	Dedicate a research phase specifically to team building, role taxonomy, action economy, formation, turn order, chakra, statuses, synergy families, counterplay, archetypes, sample squads, simulation and onboarding.	Sections 24-27 include action economy, formation/targeting, Chakra, status timing, Stun DR, synergy families, 19 placeholders, 10 shells, matchup counter coverage, normalized test packet and regression cases.	Detailed written specification exists; it is not empirical validation. Reference simulator and playtest gates remain open.

R-06	Resolve "Unresolved Questions and Considerations" fairly for F2P and pay-to-progress: stacking, fragments, team size, paywalls, PvP rewards, fairness red lines and acceptance metrics.	Sections 18, 24 and 27 resolve stacking, fragments, team size, paywalls, PvP, accessibility and combat-edge decisions as proposals.	Numbers require economy and combat testing. No research claim that generous design alone guarantees commercial success.
R-07	When rate-limited, continue the work through a single editable master file, making it the living source of truth.	One live master file; Appendices A-D archive four recovered completed reports, Appendix F preserves the user-supplied source text, Appendix E audits the research/archive boundary.	Workflow settled. Any uncompleted Deep Research session stays marked pending, not an assumed report.
R-08	Continue combat-only refinement until no core rule is unspecified; keep game modes and power-up add-ons deferred.	Sections 24-27 and new v0.9 normalization/edge-case packet, current source correction, and exit-gate redefinition.	Written specification materially hardened. No empirical balance yet; simulator, matchup trials, client readability and hands-on benchmark checks remain.

E.5 Full scope register: what must not get lost in future updates

Scope	Required coverage	Current state
Core loop / idle system	Offline gains, idle cap and cadence, campaign, auto-battle, resource collection, stage progression.	Existing baseline in Appendix A and Section 17.1. Exact rates/caps remain open.
Combat engine	Turn order/speed, turn economy, basic/skill/ultimate, chakra/energy, targeting, rows/positions, intercept/taunt, AoE, damage and defense layers, shields/heals, crit, status effects, cleanse/dispel, revive, clone/decoy rules, combat log.	Written rules in Sections 24-27: turn order, Chakra, target policy, Attack Down/damage calculation, mitigation, Shield, status timing, Stun DR, follow-ups, clone/taunt/intercept and edge cases. Needs implementation/tests.
Roles / composition	Tank, bruiser, carry, burst, AoE, sustained DPS, controller, debuffer, buffer, healer, cleanser, protector, battery, marker, counterattacker, summon/clone, revive and hybrids. Roles must have jobs, weaknesses, overlap and opportunity cost.	19 functional placeholder kits, 10 team archetypes and normalized role-stat/coefficient budgets are specified in Sections 26-27; final character kits and stats await simulation.
Synergy taxonomy	Role-based, condition/status engine, follow-up/action-chain, turn-order, protection/sustain, mark-focus, formation, element/nature, village/faction, clan, relationship/bond, summon/clone and counter-synergy.	Functional synergy families, caps, tag bonuses, synergy budget and substitution rules are written in Sections 24-27; exact impact remains a hypothesis.
Counterplay	At least two practical answers to potent strategies; no single paid/limited unit as a mandatory answer; counter categories must be accessible.	Two counter categories per major strategy and edge-case stress pack are specified in Sections 26-27; accessibility and effectiveness must be measured.
Team archetypes	Protect-the-carry, tempo, burst pickoff, control, DoT/attrition, counterattack, summons/clones, sustain/grind, mark-focus, anti-meta, beginner balance, boss slayer and wave clear.	Ten named compositions and accessible substitute logic exist; do not call them balanced until the

		directed matchup matrix is run.
Benchmarks	AFK Arena, AFK Journey, Mythic Heroes, Idle Heroes, Epic Seven, Summoners War, RAID: Shadow Legends, Honkai: Star Rail, Reverse: 1999, Brown Dust 2, Cookie Run: Kingdom, NIKKE, Mobile Legends: Adventure and other relevant games.	Source-based comparison spans 13 named games across Sections 4, 25, 27. It is not a full hands-on, current-patch audit of every mode.
Collection / progression links	Summoning, pity, selector banners, fragments, duplicates, awakening, gear/signature equipment, role access, roster flexibility and time-to-build viable teams.	Older examples and hypotheses preserved; any rates/amounts must be recalculated and sourced before implementation.
F2P / pay-to-progress fairness	Free players can build multiple archetypes; spending should mainly buy time, convenience, cosmetics or breadth; duplicate-only power, limited-unit locks and unanswerable power must be limited.	This is a settled project requirement. Numerical targets are draft thresholds for testing.
PvP fairness	Rating bands, normalization options, matchmaking, seasonal resets, rewards, daily-grind pressure, power gap measurement and multiple modes.	A two-lane model and sample rewards exist as proposals; no live economy/playtest validates them.
Modes and live operations	Campaign, idle reward stage, wave/trial, boss/raid, tower/endurance, village/faction challenge, guild/co-op, PvP and events. Each mode must ask a distinct tactical question.	Broad conceptual plan in Section 17; full mode specs not yet complete.
Sustainability / monetization	Cosmetics, convenience, optional progress, payer segmentation, live-ops cost, retention and revenue risks; generosity alone does not guarantee business success.	Hypotheses only; no validated audience or revenue model yet.
Onboarding / UX	Role teaching, formation tutorial, visible synergy triggers, team builder tooltips, loss explanation, combat recap, practice and recommendations.	Design direction outlined; acceptance tests need a prototype.
Validation / analytics	Win rate by matchup, archetype usage/dominance, replacement flexibility, duplicate sensitivity, F2P time-to-build, paid power gap, CC uptime, action share, battle duration, first-death timing and player comprehension.	Acceptance thresholds and seed-based tests are described; not run because there is no executable combat prototype yet.
Legal / IP	Naruto character, name, design, story and protected elements require appropriate rights/licensing for commercial use.	License is not assumed. Naruto names in this document are illustrative only.

E.6 Evidence and completeness limits

- Completed and archived: the four saved report outputs in Appendices A-D and the design synthesis in Sections 1-19.
- Not confirmed as completed reports: the subsequent deep-dive prompts on all listed benchmark games, the full detailed combat-engine specification, and the final fairness-question expansion. Tool responses that only say a session started or that a request was rate-limited are not research evidence.
- Historical report claims can be outdated or based on secondary/community sources. The report appendices preserve what was originally returned; Section 15 and future chapters should use freshly verified source links before asserting current rates, pity, rewards or exact mechanics.
- Numbers like 70-pull pity, 5% synergy caps, 5-8% duplicate sensitivity, fragment income, and PvP gap thresholds are proposed game-design targets unless a source and test result explicitly establish otherwise. They are not industry facts.
- The design has not been validated by a combat simulator, live player study, economy simulation, or commercial test. The archive is comprehensive for recovered outputs, not proof that all open research has already been conducted.

E.7 Live master update protocol

- Keep this file as the only live design bible: /Game Design/Naruto_Idle_RPG_Master_Design_Bible.docx.
- When new research arrives, update the relevant main-body chapter, then update the research-session ledger, source list, decision log and change log in this same file. Do not create a new master for each iteration.
- Archive each genuinely completed report in a clearly labelled appendix or an archive section. Record its title, status, approximate length, sources, date, and which design sections it informs.
- Maintain evidence labels: [FACT / VERIFIED MECHANIC], [HISTORICAL REPORT CLAIM], [PROJECT DECISION], [PROPOSAL], [HYPOTHESIS], [OPEN]. Never silently upgrade a proposal to a fact.

- When a new finding conflicts with an older recommendation, preserve the historical report text but update the current main-body decision register with the new value, rationale, source and validation status.
- Do not call an unanswered Deep Research prompt “completed.” If no final report is available, record it as a pending research task and continue with sourced web research or design analysis only when performed.

E.8 Version 0.9 change record

Version	Change	Status
0.1	Initial working design draft.	Superseded; principal project content carried forward.
0.2	Merged four recovered completed research reports, added the fairness/composition synthesis and consolidated the earlier base plus addendum.	Superseded by v0.3; its content remains in the live body and Appendices A-D.
0.3	Added an explicit archive boundary, research-session ledger, prior-file disposition register, full scope coverage register, evidence/completeness limits and a strict single-master update protocol. Updated cover and header to v0.3.	Historical snapshot; the live master has since progressed through v0.9.
0.4	Expanded archive consolidation and scope register; preserved four recovered reports and uploaded source text.	Historical snapshot; relevant substantive content carried forward.
0.5	Added mobile-first combat and F2P fairness architecture.	Superseded; content preserved in Sections 20-24.
0.6	Expanded core-combat specification and reconciled major combat system conflicts.	Superseded; base specification remains in Section 24.
0.7	Added cross-game combat research and resolved fractional hard-control duration ambiguity.	Superseded; analysis remains in Section 25.
0.8	Added 19 placeholder kits, 10 archetype shells, matchup matrix and combat exit gate.	Superseded; roster architecture remains in Section 26.
0.9	Reconciled active combat tables, defined normalized stat/coefficient packet and action edge cases, corrected source attributions and updated research maturity/exit gate.	Latest version. Specification written; simulator, empirical balance and phone usability remain open.
C-026	Defined ally-origin Chakra cap and incoming-damage deduplication by root action and defender.	Closes the multi-hit/follow-up resource-farming edge case.

Archive conclusion: the master now contains all four recoverable completed report outputs in Appendices A-D, the consolidated design material from the previous draft/addendum files, and a traceable register of the research directions that have not yet yielded completed reports. It is the live file; the pending work remains visibly pending instead of being presented as already researched.

E.9 Version 0.4 change record

Change: Imported the latest user-supplied research bundle as Appendix F in full source order. Rationale: ensure the master preserves the exact text the project owner supplied, in addition to the four prior recovered reports. Status: completed source import; the content remains archival source material, not independently validated claims.

Change: Updated the live master version metadata, header/footer and document map. Rationale: make the new single source of truth identifiable. Status: completed.

Change: Added source-integrity metadata (line count, word count, character count and SHA-256). Rationale: make future comparisons and completeness audits reproducible. Status: completed.

Completeness boundary: this version proves that the supplied Pasted text(1).txt content is archived in Appendix F and that the four saved reports in Appendices A-D remain present. It does not prove that a Deep Research session with no retrievable final report has findings available to archive.

Archive intake detail (moved from front matter to keep document flow clean)

This revision imports the complete supplied file “Pasted text(1).txt” as Appendix F in source order. It is preserved as a verbatim archival transcript rather than silently rewritten into the design synthesis. This intentionally creates some overlap with Appendices A–D so the supplied source itself remains traceable and no material is dropped during consolidation. The current main body remains the working design synthesis; historical reports and proposed values are not automatically verified facts.

Archive integrity record: 718 newline-delimited source lines (719 split entries including the terminal blank line); 12800 words; 84694 characters after newline normalization; SHA-256 of original uploaded bytes: bf87b3143d598c8ee8909a3cddf7e5c018ab854ba60fdd8662d8444cbbcbc54d.

E.10 Version 0.5 change record

Change: Added Sections 20-23 with mobile-first combat rules, target selection, Chakra, formula v0.5, status/CC caps, ten team archetypes and their counter categories. Rationale: move the project from general team theory toward an implementable mobile combat prototype. Status: proposed; requires simulator and playtests.

Change: Added a mobile F2P fairness model: core-role access, 60 pull-equivalents per 28 days, 70-pull no-coin-flip guarantee, 100-fragment selectors, duplicate-power bounds, normalized ranked PvP, and paid-product limits. Rationale: connect team construction directly to the project’s fairness constitution. Status: illustrative targets; require economy/business modeling.

Change: Added mobile platform policy sources and a persistent continuity card. Rationale: preserve platform/device requirements, keep project intent consistent across future conversations, and prevent proposals from being misrepresented as verified benchmark facts. Status: completed as a document update; policy and game-mechanic sources must be rechecked before launch.

Revision integrity: prior main-body sections and all historical source appendices A-D and F were retained. The only edits outside Sections 20-23 were version/map metadata and the maintenance protocol. This revision does not claim the game is playtest-validated.

E.11 Version 0.5 mobile-first research note

Added a mobile-first implementation proposal (Sections 20-23) and freshly checked external references [M1]-[M8]. These are new research/design work for this revision. Verified benchmark observations and platform requirements are cited; numerical combat/economy rules remain proposals.

Future continuity: before each update, reread Section 23 and this archive ledger; inspect the current master and compare headings/appendices before editing. Do not rely on an informal claim that “memory” was updated. The durable project record is this master document plus the conversation context; saved-memory availability may vary by product settings.

E.12 Version 0.9.1 research/concept-phase boundary update

Recorded the project owner’s instruction that current work is limited to research, game theory, game concept and hypothetical scenario review. Explicitly deferred prototype implementation, simulator runs, seeded outcomes and live/player testing until the owner authorizes that phase. Clarified that empirical combat exit gates govern future implementation and validation, not ongoing conceptual research on idle-RPG genre, core loop, progression, economy, modes, UX and connected systems. No historical research archive was deleted.

E.13 Version 0.10.0 holistic research workflow and idle-RPG foundation

Added a whole-game system map and a repeatable domain-by-domain research method. Established the first source-grounded research pass on idle-RPG identity, the active/inactive/return loop, genre boundaries, and an initial hypothetical loop for this project. Added source references with confidence notes. Corrected stale active-body wording that implied research outside combat must wait for future testing; the historical combat sequencing is retained in context and marked superseded where necessary. No simulator, seeded matchup, prototype, or live/player test was run.

E.14 Version 0.10.2 idle reward / campaign / team-building integration

Added a source-grounded comparison of AFK Journey and NIKKE progression links; separated direct team influence on campaign success from the basis used to calculate recurring idle income; compared fixed-rate, checkpoint-rate, offline-simulation and rarity-linked models; recommended a provisional hybrid with a baseline floor and account-level checkpoint progression; documented eight hypothetical cases, exploit risks, accrual-accounting rules and open cap/rate decisions. Status: research/theory only. No simulator, prototype or live test was executed.

E.15 Version 0.10.3 idle reward cap / claim cadence / catch-up research

Added a source-grounded comparison of storage-cap approaches, claim cadence, overflow behavior, and returner catch-up. Flagged conflicting patch-sensitive benchmark information rather than treating a single published cap as verified. Recorded a provisional 24-hour full-rate + 48-hour 25%-rate overflow model for conceptual comparison, nine hypothetical scenarios, F2P safeguards, unresolved settings, and the next resource-source mapping task. Status: theory only; no implementation, simulation or live testing performed.

E.16 Version 0.10.4 idle resource basket and source-allocation research

Added a source-aware resource basket and allocation matrix across idle accrual, campaign first-clears, repeatable active modes, recurring objectives, events, PvP, guild, energy and catch-up systems. Compared broad, core-growth and minimal idle-basket models; recorded hypothetical edge cases and F2P guardrails; flagged the internal inconsistency between earlier illustrative recurring-pull targets instead of silently resolving it. Updated Section 32's next-task marker. Status: theory only; no implementation, simulations or live testing performed.

E.17 Version 0.10.5 resource sinks, upgrade-cost structure and progression pacing

Added Section 34 as the next live research chapter. Connected the existing Section 33 resource-source map to a 17-family proposed sink register; documented character-level, skill, promotion and equipment cost architecture; framed early/mid/late progression by player goals and system unlocks; added a transparent paper affordability model, six hypothetical edge cases, a decision/open-question register and current research sources. The chapter deliberately assigns no final numeric costs or rates and keeps the existing conflicting recurring summon-income examples unresolved. Status: research and theory only; no prototype, economy simulator, seeded battles or live/player tests were performed. The historical reports in Appendices A-D and the verbatim user-supplied archive in Appendix F remain preserved.

E.18 Version 0.10.6 cost-versus-income model and roster-progression integration

Added Section 35 to build on the v0.10.5 sink register. Added a resource-by-resource affordability method that separates dependable recurring income from milestone/variable/paid sources; included a purely illustrative two-resource calculation; compared individual, shared and hybrid roster-level models; proposed a shared training baseline plus individual specialization for further study; clarified core upgrade layers and equipment investment-protection options; and added candidate pacing envelopes, edge-case safeguards, open decisions and evidence-qualified sources. Exact prices, income rates, wait times and the roster model remain unapproved. Research/concept phase only. The working file was advanced to v0.10.6; the Library was not updated because the current upload limit remains in effect.

E.19 Version 0.10.7 summon and fragment acquisition-budget reconciliation

Added Section 36 and updated active cover/version metadata. Reconciled the Section 21.2 60-pull illustrative sum (34 routine objective/login subtotal + 6 social/rotating + 10 event + 10 campaign/tower milestone) and proposed interpreting 45-50 as a provisional recurring/periodic gross target while 60 is a gross window including milestone value. Flagged that the feature-eligible portion is still unresolved, so earlier 32.7-day math is valid only if all 60 counted pulls can be spent on the featured banner. Added formula and tables for 70-pull savings time and 4/5/6-week release cadence; benchmarked official HoYoverse and NTE guarantee rules plus a specialist AFK Arena summon guide; calculated 100-fragment selector timing from the existing 15-20 weekly target; and specified separate standard, featured-eligible, fragment and one-time/variable ledgers. Kept all numbers and architecture proposals unapproved; no simulation or live test. Reclassified the old Section 21.2 column as an illustrative acquisition source and added clarifying notes to Sections 8.2 and 21.2 rather than silently deleting prior values. The working copy advances to v0.10.7; Library synchronization remains pending by explicit project workflow.

E.20 Version 0.10.8 acquisition source ledger and battle-ready character budget

Added Section 37 and updated active cover/header/footer and research roadmap. Researched current recruitment-source and banner-eligibility patterns, including official HoYoverse earnable-vs-paid currency and banner-counter descriptions, a current NIKKE patch-note mirror for ordinary/advanced ticket eligibility, featured mileage carry-over, free-pull exclusions and event/login character access, and AFK Journey's official source/cost adjustment and select-equipment announcement. Added source reliability classes; mapped login/weeklies, repeatable PvE, campaign first-clears, events, PvP, guilds, milestones, returner tracks, shops and paid sources to proposed recruitment/fragments outputs; compared flexible Recruitment Seal vs separate standard/featured ticket architectures; clarified that one seal can be spent once and that pity advances only on eligible featured spending. Reconciled existing values 34 + 6 + 10 + 10 = gross 60 into source-reliability classes; calculated 34/40/50/60 cases to a 70-pull guarantee; proposed a ≤7-day readiness hypothesis paired with the existing 100-fragment, 15–20/week target and 4–8 week archetype-build goal. Added UI, failure-case, decision and source sections. All architecture/rates remain proposals, no prototype/simulation/live test performed, Library unchanged. Next task is conservative/ordinary/high-engagement source ledger and readiness paper modelling. Working master advances to v0.10.8.

E.21 Version 0.10.9 acquisition-source reliability audit and F2P budget scenarios

Added Section 38 to audit the prior illustrative recruitment-source rows and their reliability; identified a possible overlap between daily activity and weekly objective rows that blocks final totals until each reward has a unique source ID. Built conservative / ordinary / high-engagement 28-day totals of 18 / 37 / 50 and 42-day linearized totals of 27 / 55.5 / 75, excluding up to 10 one-time milestone value. Calculated 70-pull featured savings sensitivity when saving 100%, spending 25% on standard, or spending 50% on standard; modeled fragment selector time at 10 / 15 / 20 fragments per week plus a proposed 7-day readiness allowance; recommended studying a dependable 15-fragment weekly evergreen floor if an 8-week field-ready counter target is retained. Added source audit metadata fields, new-account/mature-account separation, six current reference records and paper acceptance checks. All figures remain assumptions or proposals, no prototype/simulator/seeded pull/live testing. Working version advanced to v0.10.9; Library synchronization remains pending.

E.22 Version 0.10.10 mode reward architecture, ticket alternatives and account journeys

Added Section 39 and updated cover/version metadata, research roadmap and maintenance protocol. Compared source-supported mode reward patterns in official HoYoverse and AFK Journey developer material with current specialist guides for AFK Arena, AFK Journey and NIKKE. Mapped campaign first-clears, idle income, objectives, repeatable PvE, guild, PvP, events, endgame and catch-up to distinct reward purposes. Compared flexible Recruitment Seal versus separate standard/featured tickets and calculated conditional 42-day new-account and mature-account journeys from the existing 18/37/50 pull-equivalent scenarios. Preserved the flexible seal plus universal fragments as the leading unapproved proposal; confirmed that fixed calendar promises require a reliable allocation and that 15 fragments/week plus ≤7-day preparation remains an unvalidated target. Added unique-source ledger requirements, UI rules, paper acceptance checks, six source notes, and a revised next-task checkpoint. No prototype, simulator, seeded run, or live player test was performed. Library synchronization remains pending.

E.23 Version 0.10.11 canonical reward ledger and 56 day readiness review

30 October 2026. Added Section 40 and updated the active checkpoint, cover and document map. Defined reward-definition versus occurrence identity and separated grants, conversions, refunds and progress. Preserved the unresolved historical 14/12 overlap; introduced CORE01/CORE02 as an explicit unapproved replacement candidate: 34 seals and 60 fragments per 28 days at full core completion. Added a source-category register with unresolved material and mode quantities clearly excluded from numeric floors. Calculated integer 56-day acquisition cases, featured/standard trade-offs, partial weekly completion and weekly selector dates. Linked the existing 24h full-rate plus 48h quarter-rate bank to collected material supply, revealing a sparse-collection preparation constraint. Added a resource-vector readiness method, normalized affordability example and explicit next research brief. Earlier research and Appendices A-D and F remain intact. No simulation, prototype or live testing was performed. Restored the missing dynamic page-number field in the footer.

Continuity note. The owner-supplied v0.10.10 was synchronized to the canonical /Game Design/Naruto_Idle_RPG_Master_Design_Bible.docx on 30 October 2026 and verified against the supplied bytes. Earlier statements that synchronization was pending describe the historical working-copy state. This research revision continues that same master; it does not create a competing design bible.

E.24 Version 0.10.12 research order and baseline counter readiness

31 October 2026. Added Section 41 and updated the research checkpoint, roadmap and document map. Prioritized the cost of fielding an owned counter before shared-growth pricing, timed role access and content design. Specified a candidate Hall unlock, inherited caps, first-copy skill access and two standard equipment slots. Compared early and mature account budgets, two-, four- and seven-visit storage schedules, missed-week and midweek starts, gear gaps and campaign walls. The next packet prices a shared upgrade at each checkpoint. All stage values and equipment rules are proposals. Evidence for encounter effectiveness remains a future task.

E.25 Version 0.10.13 dated shared Hall budgets

31 October 2026. Added Section 42 with early and mature promotion prices, fixed-stage hourly yields, six dated wallets and a comparison of storage and timely material support. Priority 3 now traces when essential roles can answer campaign threats.

E.26 Version 0.10.14 research decision practice

31 October 2026. Section 23 records the owner's direction to make provisional design choices during concept research and document their tradeoffs. Priority 3 remains the next paper study.

Worked decision. Keep the proposed idle bank as the baseline for the first Wound encounter study. It limits total material income, but under Section 42's assumptions a twice-weekly player buys the mature promotion on day 7. Extending full-rate storage to 72 hours would fund the purchase on day 3 and increase every idle resource collected after a long absence. A targeted grant would affect fewer resources but would need to arrive before the encounter. Revisit the choice if both available responses require the higher Hall level before that player can pay. First compare moving the encounter or making an existing counter available earlier.

E.27 Version 0.10.15 research performance recommendations

31 October 2026. Section 23 now requires a performance recommendation for each upcoming research pass, delivered in chat and recorded in the current checkpoint. Appendix G gives the project guide to the five levels. The next Priority 3 study is rated high because it combines role access, Hall progression and combat timing for two responses to Wound.

E.28 Version 0.10.16 first Wound counter-access study

31 October 2026. Added Section 43 as Priority 3A. Distinguished 56-day acquisition time from actual E-stage reach time; closed the starter entitlement gap with proposed START00 core grant and START01 K-09/K-10 flex choice; contrasted cleanse versus front-row source focus under legal Speed/action rules, including Wound persistence on source death; carried earlier CORE01/02, SELECT01 and Hall wallet arithmetic into aligned, Wednesday-join, mature and returner examples; identified first-copy damage and stage positioning as unvalidated. The Priority 3B full mandatory-threat matrix is next at EXTRA HIGH. Appendices A-D, Appendix F source archive and all historical research text are preserved. No simulation, prototype or live test was run.

E.29 Version 0.10.17 mandatory-threat counter coverage

31 October 2026. Added Section 44 as Priority 3B: audited access across eight provisional first mandatory threats; selected three one-time pre-gate story shinobi grants (K-08, K-04, K-17), documented 56-day ownership and one-visit cases, reconciled first-copy target and AI conditions, recorded adversarial failure cases and role-contract dependencies. Discovered and explicitly reopened one-visit E checkpoint Ryo affordability under unchanged 24h-full + 48h-quarter storage. The Appendix F original-source body and earlier research sections remain intact. No simulator, live test, gameplay implementation or observed win-rate analysis was performed.

E.30 Version 0.10.18 weekly growth and four-encounter paper reconciliation

31 October 2026. Added Section 45: audited once-weekly E/M wallets and chose optional rather than compulsory recurring other-core spending; E Hall21 day7 and M Hall81 day14 remain arithmetic, not playtest outcomes. Wrote eight base-copy kit cards, resolved the 90%-status application limit for anti-heal, and documented W-E/T-E/R-E/S-E noncritical action-order packets with failure branches and clear boundaries. Added explicit Hall20-to-normalized-stat conversion blocker, preserved old rates and source archive, and handed off Priority 3D at EXTRA HIGH. No simulation, prototype or live user test performed.

E.31 Version 0.10.19 Hall-reference bridge and four threat windows

31 October 2026. Added Section 46 before Appendix A. Introduced explicit Hall20/Band I to Section27 reference mapping and Hall21 +2% stat example without fabricating levels22–80. Paper-reviewed four remaining threat categories: burst, Silence control, bounded decoys and two Wound sources. Recorded significant C-E auto-AI counterexample and full-fight blockers, retained original combat economy and no-testing phase boundary. Updated checkpoint, cover and navigation; preserved historical appendices and source text.

E.32 Version 0.10.20 automatic-combat, Wound-tick and protected-rear reconciliation

31 October 2026. Added Section 47 as Priority 3E, preserving all earlier combat/economy/archive sections. Reaffirmed strict automatic Ultimate→eligible Technique→Basic; rejected scripted Basic suppression; exposed C-E round-2 Silence during existing Technique cooldown as non-instructive. Corrected same-round Wound ticks at K-03/K-19 action-slot start, which limits what slower K-09/K-10 can prevent. Authored eight bounded enemy action cards and two-route T-E two-round source-only fight. Replaced R-E with an active Taunt/Intercept guard and demonstrated why K-04's row access is real but its claimed one-round advantage is not universal after all five deployed actions are counted. Kept Hall80→81 symbolic, registered five working decisions and Priority 3F Extra High handoff. No simulation, implementation or player testing.

E.33 Version 0.10.21 control tutorial, K-12 first copy and status-survival reconciliation

1 November 2026. Added Section 48 based on v0.10.20, while keeping the prior all-section prose and historical source archive. Completed K-12 first-copy Mark skill and fallback (subject to 80% status luck); replaced the misleading cooldown-timed Silence tutorial with a previewed round-1 front-right-slot example; showed how formation avoids losing K-03's ready Technique; counted both starter supports in W-E and revised WW-E with the exact same-round Wound clock. Added an honest two-rear R-E comparison and a higher-level encounter acceptance envelope without fabricating Hall22–80. Research only. Priority 3G remains next. Cover/checkpoint/Appendix G updated. No code, simulator, playtests or claims of tested balance.

E.34 Version 0.10.22 first-copy Ultimate and eight-threat integrity research

1 November 2026. Added Section 49 to the existing live master, including first-copy Ultimates for nine early-access kits, a 100-Chakra legality/carry rule, K-03 maximum-buff one-shot rejection, same-victim two-source Wound proportional cap allocation, and a carefully qualified eight-threat encounter audit. No previously completed combat or economy chapter was silently deleted, and the historical source archive remains preserved. Next research: Priority 3H at Extra High. Document revision is theoretical only.

E.35 Version 0.10.23 full burst, decoy and sustain paper certificates

1 November 2026. Added Section 50: full legal five-slot and enemy-action traces for B-E-v2, D-E, and S-E-v2, including K-09/K-10 starter choices, K-12 Mark failure, K-17 Healing Suppression failure, Chakra/Technique cooldown timing, legal protection and all survival totals. Superseded the old 1,850-HP burst and 2,400-HP healer as mandatory mechanic-teaching encounter inputs (historical values retained). Kept two-decoy D-E at its earlier HP but rejected a fabricated K-04 speed advantage. Next: Priority 3I, Extra High; all work remains hypothetical.

E.36 Version 0.10.24 eight-threat opening reconciliation and account-gate policy

1 November 2026. Added Section 51: five remaining W-E/C-E-v2/T-E/R-E-v2/WW-E-v2 legal five-member desk action ledgers, including support-choice sensitivity, status failure/success branches, action end clocks, and price/grant constraints. Consolidated all eight provisional threat teaching gates G1-G8 and reasserted one-time persistent STORY01/02/03 grants before entry. Marked the Hall1->20 from-new-account affordability path as NOT YET PRICED; next

Priority 3J prices it before actual narrative stage authoring. Kept historical §45-50 numbers for traceability and all original archive text. This is research without programmed simulation or playtesting.

E.37 Version 0.10.25 first-account Hall1–20 costs and tutorial funding

1 November 2026. Added Section 52 prior to Appendix A; defined 19 shared Hall1→20 purchases totaling272 Ryo/408 EXP without essence/gear prerequisites; proposed ONB01–04 durable noncombat funding280/420 with 8/12 remainder, E0 idle unlocking at ONB01, and shop protection before ONB04. Audited first seven daily, twice-weekly, weekly and midweek visits and returning accounts using the existing 24h full +48h quarter bank. First G1 Hall20 baseline can be reached on the completed first tutorial session; eight encounter results still conditional and actual stage graph unpriced. All earlier research and historical appendices retained. Next Priority4A at Extra High.

E.38 Version 0.10.26 campaign graph and reward-source architecture

1 November 2026. Section 53 establishes a proposed 15-node/three-chapter/8-battle/7-scene linear teaching arc, stable FC/SCN IDs, gate-relative STORY01/02/03 receipts, Chapter-1 FIRST02 payment location and fixed E0 idle-tier treatment. It adds no extra seals/fragments or unpriced mandatory Hall growth; later stage material vectors remain OPEN and excluded from current floor. Includes replay/return integrity, alternative decisions and official source cautions. No playable game, simulator or live testing. Next Priority4B at Extra High.

E.39 Version 0.10.27 capstone boss and optional solo tactical mastery

1 November 2026. Added Section 54: authorial G8-BC01 Medical Supply Commander 4,700 HP/1,020 heal and one telegraphed 1,000 shield on its qualifying second own slot; R1–R3 complete five-member Hall20/Band-I ledgers for K-09/K-10 and K-17 suppression success/failure; optional zero-resource/zero-reset Tactical Archive TA-01 formation forecasting with one cosmetic; and nonnumeric Chapter2/3 closure IDs. Preserved §53 graph, original §50 historical healer, 10 introductory seals, weekly 34 seals/60 fragments candidate and E0 rate. All values remain theoretical. Next 4C at Extra High.

E.40 Version 0.10.28 campaign pacing and bounded first-clear value

1 November 2026. Added Section 55: preserved 15 logical nodes and seven scene receipt identities while selecting nonblocking, skippable presentation; compared no added materials, one G8-only 80 Ryo/120 EXP grant, and equal Chapter2/3 grants. Computed eight dated no-spend 28/56-day wallet snapshots, acquisition source separation and optional Hall21 timing sensitivity. Selected provisional FIRST03 at G8 only; retained Chapter2 dossier, G8 cosmetic title, core seal/fragment income, fixed E0 and source archive. Recommended next Priority 4D / EXTRA HIGH. No game, simulator, playtest or backend was implemented.

E.41 Version 0.10.29 narrative-first-session integration

1 November 2026. Added Section 56 after §55: original-IP-safe three-chapter “Broken Supply Route” wrapper, two presentation paths through identical ONB01–04/START00/FIRST01 accounting, explicit START01 support choice, comprehensive 15-stage story/teaching matrix, skip/replay and interruption-safe receipt requirements, accessibility and future measurement checks, and the next Priority 5A mobile UI handoff. No source amounts, early combat packets, owned recruit grant dates, game implementation or empirical validation changed. Existing Section 55 and Appendix F were preserved.

E.42 Version 0.10.30 mobile information architecture and state integrity

2 November 2026. Added Section 57 after Section 56, before Appendix A. Chose provisional four-root compact navigation; mapped nested Hall, Idle, Preflight, Battle, Recap, Archive, Wallet and Dossier routes; authored guided/compact/returner/skip paths, notification priorities, actual source explanations, transaction receipt and interruption guards, phone accessibility envelopes, trade-offs and next 5B research boundary. Existing §56 receipts, all campaign and encounter values, training cost, recruitment sources and historical appendices preserved. This is research only; no mockup/prototype or actual usability testing.

2 November 2026. Version 0.10.31. Added Section 58 (Priority 5B): reference small-screen hierarchy, nine screen content contracts, exact START01/FIRST01/02/03/idle/receipt messages, five encounter preflight/recap content cases, six

returning/first-session scenarios, accessibility failure gates and next Priority 5C handoff. No existing entitlement, summon, stamina, idle, acquisition, combat, or Hall amounts changed. Preserved all earlier research, tables and verbatim Appendix F archive. No app, prototype, simulation, human/device test or actual compliance certification was performed.

E.44 Version 0.10.32 accessible recovery and state-integrity research

2 November 2026. Added Section 59 (Priority 5C): state-machine proposal for posted, unknown and rejected reward results; internal source-ID and request idempotency; separate paid purchase verification; 10 ranked interruption scenarios; accessibility focus/announcement matrix; precise recovery microcopy; paper walkthroughs for FIRST03, summons, first loss, 72h idle and 200% text; trade-offs, failure gates and sources. Kept §§1–58, 253 earlier tables, all source amounts and Appendix F untouched. No real backend, app, purchase, screen-reader session, performance check or live validation. Next Priority 5D / HIGH. The source file used here was Library version 36 / document v0.10.31.

Appendix F. User-supplied consolidated research text (full source archive)

Source filename: Pasted text(1).txt

Import date: 27 October 2026

Preservation method: all source lines are appended in their original order. Textual formatting from markdown/plain-text exports (including “mermaid”, “Copy”, table-like spacing, and citation tokens) is retained as text; diagrams are not silently reconstructed or replaced. This appendix intentionally overlaps earlier report appendices where the user-supplied bundle repeats content.

Source integrity: 718 newline-delimited source lines (719 split entries including terminal blank); 12800 words; 84694 characters after newline normalization; SHA-256 (original bytes): bf87b3143d598c8ee8909a3cddf7e5c018ab854ba60fdd8662d8444cbbcbc54d

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Executive Summary

Anime-themed idle RPGs blend incremental progression with collectible heroes and turn-based combat. Players form teams of characters (e.g. Naruto ninjas), engage in auto-battles or simple turn-based fights, and earn resources continuously – even when offline. The core loop typically cycles through accrue idle rewards → upgrade heroes/gear → auto-battle stages → repeat (occasionally prestiging). Progress is driven by leveling characters, unlocking gear, and resetting (ascension) for permanent bonuses, fostering a satisfying sense of growth. Collection (“gacha”) mechanics supply a steady stream of new characters of varying rarity, encouraging team-building and strategic synergies.

A Naruto-themed idle RPG must include robust fighting mechanics (emphasizing ninja skills), a layered progression system (levels, gear tiers, awakening, prestige), and a multi-currency economy with sinks to control inflation. Monetization would center on gacha draws (banner rates ~1% for top-tier heroes as seen in similar games), battle passes, and optional ads, while ensuring fairness (e.g. pity systems). Retention strategies include daily login rewards, timed events (festivals, story chapters), guilds/clans, and push notifications. Balance hinges on smooth power curves and catch-up mechanics so new players aren’t hopelessly behind. The UX/UI should cater to “idle” play: clear feedback of offline gains, simple skill activation, and convenient auto-battle options. Analytics should track DAU/MAU, retention (D1/D7), ARPU, progression speed, etc. Anti-grind designs (e.g. automated leveling) and anti-pay-to-win safeguards (frequent free draws, generous events) are essential. Finally, using the Naruto IP legally requires licensing from rights holders (Kishimoto/Shueisha/Pierrot/Bandai Namco), adhering to brand guidelines (no unauthorized characters, etc.).

The following report dives deeply into each area – core loop, combat, progression, collection, economy, monetization, retention, balancing, UX, analytics, anti-grind/anti-P2W, and legal – with examples and recommendations, supported by industry sources. A comparative table of 6–10 anime-style idle RPGs (e.g. AFK Arena, Idle Heroes, Saint Seiya, ML Adventure) highlights differences in loop, monetization, gacha rates, retention features, pacing and session length.

Core Gameplay Loop

At the heart of an idle RPG is a low-pressure loop of gradual progress. A typical cycle is: (1) Team Setup → (2) Auto-Battle/Idle Combat → (3) Collect Rewards → (4) Upgrade Progress → (repeat). Players recruit and arrange a team of characters, then start battles (often auto-resolved) against waves of enemies. Victory yields experience, gold, loot and character shards. These resources are spent on leveling heroes, equipping gear, enhancing skills, and unlocking new abilities. After upgrading, the player returns to battle more powerful enemies on higher stages. The process runs continuously – notably offline, as most idle RPGs grant rewards even while the player is away. This "offline accumulation" is a defining feature: for example, IdleOn advertises that "your heroes keep fighting even when you're gone", and AFK Arena explicitly promises auto-battles and offline rewards.

Once players hit major thresholds (e.g. max level or a key boss), they perform a prestige/ascension reset: they lose short-term progress (characters revert to level 1) but gain permanent boosts (e.g. a multiplier on future gains). These prestige loops spark long-term engagement by allowing exponential growth over multiple runs. The core loop thus has two layers: a short loop of continuous upgrades, and a long loop of periodic resets for permanent advancement.

The following flowchart depicts this loop: an idle rewards accumulation leads to upgrades, which enable auto-battles, yielding more rewards. Boss defeats trigger optional resets that feed back into upgrades.

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Copy

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flowchart TD
 A[Idle Resource Accumulation\n(offline gains, quest rewards)] --> B[Claim & Spend Resources]
 B --> C[Upgrade Heroes/Gear & Team Setup]
 C --> D[Auto-Battle Begins]
 D --> E[Gain XP, Gold, Loot]
 E --> F{Boss Defeated?}
 F -- No --> A
 F -- Yes --> G[Boss Rewards & Milestone]
 G --> H[Optional Prestige/Ascension]
 H --> I[Permanent Bonuses (XP multipliers, etc.)]
 I --> A
 G --> A
```

Examples from existing games: IdleOn features "unlimited idle gains" and allows hiring defeated foes as passive damage dealers. AFK Arena's description highlights 1,000 free hero summons and "auto-battle your way to victory and earn rewards even while offline". These illustrate how games market the loop of continuous progress with minimal player input.

Recommended parameters: Victory rewards should scale so players see steady progress over ~seconds/minutes of play. Boss or map milestones might occur every few hours of cumulative progress. A prestige reset (with reasonable timing) every 1–3 days of idle play can keep long-term motivation – more frequently leads to grind, less makes progression dull. Upgrade costs typically escalate (double/triple per level), enforcing cadence. For Naruto-themed flavor, prestige could be called "Rebirth" and grant, say, a +5% XP or currency bonus per reset.

Combat Mechanics

Combat in an idle turn-based RPG is usually automated or semi-automated to support the "idle" premise. In each stage or battle, characters perform actions in sequence without direct player input (beyond initial setup or occasional skill triggers). Key mechanics include:

Turn-based Auto-Battle: Characters take turns attacking based on speed or initiative, using basic attacks and skills automatically. The player typically cannot control actions mid-battle, although some games allow manual activation of ultimate skills. AFK Arena, for example, is known for its completely auto-resolved combat (players choose heroes and formation only).

Skill Timing / Ultimates: To retain engagement, games often give players a button to trigger each character's ultimate or special skill. Skill cooldowns might be visible, requiring the player to time activations (e.g. pop Naruto's Rasengan when available). Some idle RPGs bypass this and automate all skills (simplest design), but adding one or two manual abilities per combat can add a layer of strategy.

AI Difficulty Scaling: Enemies grow stronger each stage. Boss enemies use stronger AI patterns. A modern Gacha game often normalizes stats for PvP (as in Naruto Mobile's equalized PvP), but for idle PvE content, enemy stats just scale with level. We recommend graduated

difficulty so early stages breeze through, mid-game requires strategy, and late-game feels challenging. Stage bosses (every 10–20 stages) should have unique mechanics or resistances.

Auto-battle features: Many titles include a 2x/4x speed toggle, “sweep” functions (instant clear of stages once maxed), or an “AFK Mode” showing expected rewards if left idle. AFK Arena has a “AFK Combat Rating Boost” that auto-battles and even gains rewards while offline. Ensure clear UI for these (a toggle or an “Auto” indicator).

Example: IDLE RPG (Steam, 2026) describes “classic turn-based battles” where you “hire defeated enemies as assistants” and use prestige points. IdleOn even lets you unlock passive worker roles (e.g. “miners”) to simulate continuous battles. While Naruto RPG won’t literally hire defeated ninjas, analogous features could be “Assistants” or “Shadow clones” that continue fighting or gathering.

Pros/Cons:

Pros: Auto-battle minimizes grind and keeps users engaged in upgrades rather than twitch play. It’s accessible to casual players and suits mobile “AFK” style.

Cons: Too much automation can feel passive. Always-on auto may bore players who enjoy interaction. Balancing purely idle content is harder, since strategy only lies in pre-battle preparation. Adding limited interactive skills or formations adds strategic depth. Also, fully automated pacing means tuning rewards carefully to avoid either stagnation or pay-to-win rush.

Implementation Notes: Use simple turn queues and states (attack, use skill if ready). Provide visual feedback (damage numbers, hit animations) to make progression satisfying. Keep performance in mind: avoid heavy 3D when hundreds of fights occur (use stylized or 2D effects if needed). Bug reports from trailers indicate performance issues (lag) can sour impressions – ensure smooth animations and loading.

Recommended parameters: Base attack speed such that a full team round takes ~3–5 seconds at normal speed. Ultimates recharge every ~20–60 seconds. Boss fights might last ~30–60 seconds. Auto-battle speed toggle (2× or 4×). Allow offline progress for at least 8–12 hours (IdleOn uses unlimited, but typical games cap 8h–12h). Skill damage and AI aggression should scale so that small increments in hero power noticeably shorten battle time, maintaining dopamine.

Progression Systems

Progression in an idle RPG is multi-dimensional. Common elements include:

Hero Levels: Characters gain XP from battles and level up, increasing base stats. Early levels (1–50) might be fast, later levels (50–100+) slower, requiring more XP per level (common exponential growth).

Equipment/Gear: Heroes equip weapons, armor, accessories, etc. Gear has its own levels or star ranks. Upgrading gear (with resources) boosts hero stats. Often gear tiers increase over time (Tier 1 → Tier 2 → etc). For Naruto, gear could be “ninja tools” like kunai, scrolls, chakra enhancers with quality tiers (Bronze, Silver, Gold, etc).

Hero Rarity & Awakening: Gacha characters come in rarities (e.g. 1★ common to 5★ legend). Higher rarity starts with better base stats. Awakening (or “limit break”) is a secondary progression: using special materials or duplicate shards to unlock character passives or higher potential. For instance, Idle Heroes has “transcendence” (E1–E7) for top-tier heroes; AFK Arena has Lightbearer / Mauler factions, not explicit awaken but equips. In Naruto terms, “Awakening” could be unlocking a character’s next form or hidden potential (e.g. Kakashi’s second Sharingan).

Ascension/Prestige: At key points (e.g. after stage 1-100, or when progression plateaus), players can reset progress. This is often called “Prestige” (Idle RPG), “Transcendence” (Idle Heroes), or “Rebirth.” The reset strips heroes back to level 1 (sometimes keeping one copy if dupes were used), but awards permanent resource bonuses or stat multipliers. This dramatically accelerates subsequent runs. The design here must balance short-term sacrifice for long-term gain.

Meta-Progression: Over multiple ascensions, players unlock higher “new game plus” tiers. Possibly a second currency for meta-upgrades (like “prestige points” invested in global boosts). Example: IDLE RPG (Steam) lets players invest Prestige Points into parameters.

Currencies/Resources: Besides XP/Gold, include upgrade currencies like Shards (for gacha/awakening), Upgrade Ore (for gear), and premium gems. Multiple currencies create choice and complexity.

Examples: Idle Heroes differentiates between items like Scrolls/Orbs (for summoning) and materials like Rune Shards or Upgrade Materials. AFK Arena uses “Hero Scrolls” and “Diamonds.” Generally, progression is gated by activity: stages, events, and elite challenges yield scarce upgrade materials.

Pros/Cons:

Pros: Layered progression (level → gear → awaken → prestige) provides many growth avenues and prevents stagnation. Reset mechanics (prestige) keep players engaged long-term and mitigate power creep.

Cons: Can overwhelm new players with too many systems. Requires careful pacing so players hit each feature at the right time. Risk of pay-to-win if late-game gear/upgrades are too scarce without spending.

Implementation Notes: Plan stat growth curves so no single parameter dominates. Show clear requirements for next milestone (e.g. “Reach level 50 to awaken”). Use daily logins and events to grant small amounts of premium resources (shards, gems) to support progression. Consider “skip” content (sweep tickets or auto-clear) once a stage is repeatedly cleared, to avoid boredom.

Recommended parameters (example values):

Hero Level Cap: Increase gradually – e.g. unlock up to level 50 within first week’s play, then cap at 50 until heroes are transcended or new content released. Maximum level could be 100+ as game matures.

Gear Upgrades: Tiered, with each tier requiring exponentially more gold. Tier progression could unlock every ~50 hero levels.

Prestige Reset: Offer voluntary prestige after boss levels (e.g. every 100 levels). Provide a flat 10–20% permanent bonus to XP or gold per prestige, stacking multiplicatively.

Awakening/Ascension Materials: Tied to events or high-level challenges – e.g. 3–5 duplicates or special shards needed per star upgrade.

Collection Mechanics

Heroes/Characters: The core collectible in a gacha RPG are the heroes. Characters vary by rarity (e.g. 1★–5★), type/class (attacker, defender, healer), faction or element, and have unique skillsets. Designing a large roster of diverse heroes encourages players to experiment with teams. For a Naruto theme, include all fan-favorites across the timeline (Naruto, Sasuke, Sakura, etc) at various incarnations, each classified by village or role (Striker, Tank, Support, etc).

Rarity Distribution: Typical rarity pools might be: 5★ (legendary) ~0.5–1.5% pull rate, 4★ (epic) ~5–10%, 3★ (rare/common) ~80%. This skew means players collect many low-tier heroes and few top-tier. To avoid frustration, include a pity or guaranteed system (e.g. every 40 summons guarantees a 5★).

Fusions/Evolution: Allow combining characters or shards to create higher-tier ones. For example, Idle Heroes lets players merge lower-star heroes into a higher-star version. Possibly implement a “fusion” system (e.g. combining 5 of the same rank to ascend). This uses up duplicates or shards and prevents hoarding too many weak heroes.

Team Composition and Synergies: A key strategic layer: some heroes boost others of same faction/village. E.g. having a Uchiha or Konoha ninja team gives passive buffs. Many idle RPGs use faction synergy: AFK Arena has 7 factions, Idle Heroes has multiple tiers and sets. Naruto idle could use Konoha vs. Akatsuki synergies or Taijutsu vs. Genjutsu types. Encouraging balanced team-building (tank/frontline, DPS, support) ensures players try different combinations.

Recommended design: Start with small teams (3–5 units) that scale to larger (6–9) as game grows. Each hero has 2–4 skills (normal+2 ultimates). Provide a “skill-up” or mastery system so players invest duplicates to improve skill damage or reduce cooldown.

Examples: Mobile Legends Adventure (a known hero-collector RPG) uses a 5-hero formation with synergies; Idle Heroes uses 6-hero teams with formation bonuses. The Naruto blueprint could allow, say, 4 ninjas (2 front, 2 back) plus 1 support. Synergies: e.g. full Naruto Team (Naruto, Sasuke, Sakura, Kakashi) gets a massive team buff.

Pros/Cons:

Pros: Collection drives engagement and monetization. Evolution/fusion gives purpose to extra heroes. Synergies reward thoughtful composition.

Cons: Collectible fatigue (too many heroes) can occur. Overpowered synergies may force specific team meta (hard to balance all heroes). Rarity drops being too rare can frustrate free players.

Implementation Notes: Implement an in-game codex of heroes, categorize by rarity. Provide multiple sources for characters: standard banners, event banners, loyalty rewards. Ensure even 1★ heroes have eventual utility (e.g. feeding into fusion or as awaken fodder). Example: Idle Heroes uses a "Summon Tokens" currency for generic draws, plus faction-specific summons for targeted pulls.

Recommended parameters:

Base hero pool: Start ~30 characters, expanding monthly by 5–10.

Rarity pull rates: ~0.7–1.0% for 5★ (legendary), ~7–10% for 4★.

Team size: unlock 3 slots at start, expand to 5 by mid-game.

Synergy bonus: define clear set bonuses (e.g. 3* Konoha ninjas = +10% all stats). Test to avoid one synergy dominating.

Economy and Resource Flows

Idle RPGs use multiple currencies and resources. A balanced economy channels player engagement and controls inflation. Key components:

Currencies: Usually a free soft currency (e.g. Gold, Coins) earned in every battle, and a premium currency (Gems, Crystals) usually bought with real money or granted sparingly (logins, events). Additional currencies include summon tokens (for gacha pulls), stamina/energy (to limit attempts), and special mats (forge ores, awaken fragments, etc).

Income Sources: Main sources are stage rewards (idle/autobattle yields soft currency and items), daily quests, weekly events, login bonuses, and ads. Most games award a flat amount per hour offline (AFK Arena grants rewards per offline minute). Resource refreshes: energy refills or extra daily currency if players watch ads.

Spending Sinks: Upgrades create sinks to prevent inflation. Examples: hero leveling costs (growing exponentially), equipment crafting/upgrading, skill upgrades, and event shops (players trade tokens for items). Idle Heroes players advise saving resources for events rather than blowing them on minor upgrades, indicating event currencies are major sinks. Sinks also include limited-time gacha spins.

Inflation Control: To avoid runaway wealth, costs ramp up exponentially. New currency types (e.g. Legendary Shards) can be introduced with level expansions. Impose level caps or diminishing returns. Events and bosses should require ever-increasing amounts of resources, matching exponential income growth.

Examples: The Gacha Game wiki notes many games "feature several in-game currencies with intricate conversion methods, obscuring the value of non-premium currencies". For instance, one might have Coins (infinite earnings), Medals (rare currency from PvP), and Crystals (premium shop money). Idle Heroes uses gear upgrade stones, rune stones, and scrolls as intermediate currencies.

Recommended resources flow: Ensure soft currency (e.g. Gold) is plentiful from early play (so upgrades feel satisfying), but require hard currencies (gems/shards) for top-tier pulls and awakening, which come slower. Example split: each stage clear gives 100–1000 Gold and 1 XP orb. Daily quests or events might give 50–200 premium Gems per week. Limit refund (catch-up) to major resets (e.g. bonus currency on prestige).

Pros/Cons:

Pros: Multi-currency lets designers fine-tune motivation. Sinks give players goals to spend.

Cons: Too many currencies confuse players. Poor balancing can either stall progression (if costs too steep) or trivialize currency (if too easy).

Implementation Notes: Track currency supply in analytics. If players amass gold quickly, raise costs or introduce a new sink (e.g. gear upgrades). If demand outstrips supply (players stuck), add more giveaways or reduce costs. Offer ways to convert surplus (sell common shards for gold). Visualize flows for players in UI (e.g. dashboards showing how to earn each currency).

Monetization

Most anime idle RPGs are free-to-play with gacha monetization. This typically includes:

Gacha (Loot Boxes): The primary draw is character banners. Common models: "10-pull" with a chance for high-rarity heroes. Rates ~1% for top-tier. Some games use multiple banners (e.g. "featured Naruto characters" vs general pool). Consider a guaranteed pity: e.g. every 30–50 pulls guarantees a ★5 hero.

Shops and Direct Purchases: In-game shop sells premium currency packs, resource bundles, and exclusive heroes. Often there's a "limited shop" with rotating deals. Bundle examples: "Naruto Starter Pack" giving shards and currency at a discount. Pricing tiers from ~$1.99 up to ~$99.99. Ensure value packs give a clear edge in progress, but also have smaller packs for casual spenders.

Battle Passes: Season or monthly passes (free + premium track) are common. Rewards: minor heroes, shards, currency. Encourages regular play. Example: a "Ninja Pass" that over 30 days grants enough currency for 10 summons plus some gear. Battle passes are accepted by players when they give predictable rewards.

Ads: Optional ads can accelerate progress. E.g. watch an ad to double offline gold for the next hour, or get a free summon ticket. This engages non-spenders and can drive retention without mandatory paywalls.

Fairness/Anti-P2W: Balance must be mindful. As reviews note, many gacha games are perceived as pay-to-win. To mitigate, include generous free aspects: regular free summons (AFK Arena offers daily 10 free pulls), pity on banners, and events that guarantee rare heroes if objectives met. Purchasable currency should only accelerate progression, not gate core content. Value packaging (e.g. "best value" bundles that promise big items) can feel fair if clearly marked. Implement spending limits or exclusive cosmetics (i.e. allow whales to gain vanity or small boosts, but avoid exclusive unbeatable heroes).

Ads/Rewards balance: Since user instructions said "non-pushy", ensure ads are rewarded (player opts-in for a meaningful bonus). No forced ads.

Example Mechanics: Naruto Mobile (Tencent) reportedly uses an aggressive VIP system (though not cited), which may not suit all markets. Instead, a Naruto idle RPG could adapt a soft VIP: small bonuses for spending but not gating content (e.g. every $ spent gives loyalty points for freebies). Character sales or skins could complement gacha (exclusive costumes).

Pros/Cons:

Pros: Gacha draws and shop items directly monetize. Battle passes and ads increase ARPU (average revenue per user) and retention respectively.

Cons: High prices or poor drop rates cause backlash. As the reddit points out, pricing balanced for one region may offend another. Ensuring perceived fairness is critical.

Implementation Notes: Clearly communicate drop rates (many regions now require it). Offer small refunds (soft currency) for duplicate characters. Track whales vs. regular spenders for balance. For Naruto, consider tie-ins (e.g. selling "chakra refills" which restore stamina, a common Pokémon-like model) and check legal restrictions for loot boxes (some regions regulate gacha).

Retention & Engagement

Key retention drivers in idle games include daily login rewards, events, and social features. Specific tactics:

Daily/Monthly Logins: Reward players for each consecutive day played (with bonuses increasing over a 7- or 30-day cycle). Common rewards: premium currency, summon tickets, or exclusive characters after long streaks. This forms a basic habit loop.

Time-Limited Events: Frequent events (weekly/monthly) keep content fresh. Examples: Raid/World Boss events (players team up to slay a boss), gacha festivals (increase drop rates on certain heroes for a week), challenge towers (progressively harder stages with leaderboard). Such events often award rare materials or even limited characters. Idle Heroes has many periodic events where certain currencies can be spent for exclusive gear. Naruto idle could run events like "Village Battle Event" or new story arcs when anime episodes release.

Meta-Progression Systems: Implement systems like guilds/clans (collectively earn guild points or unlock guild shop items), daily achievements, and a sense of overall progression (e.g. an "Adventure Rank" that unlocks new features as it increases). AFK Arena, for instance, has player levels and guild buildings that grant buffs.

Notifications & Reminders: Idle games often use push notifications for "your offline rewards are full" or "energy refilled" to bring players back. Use these sparingly and with valuable info (not spam).

Community/Social: Incorporate friend lists, friend bonuses (e.g. borrow friend heroes in battle, or send gifts), chat and social media integration. Given Naruto's fanbase, social sharing (e.g. "My Uzumaki team just defeated the boss!") could help organic growth.

Metagame Progress: A “Campaign map” that visually shows progress through the Naruto story (e.g. Konoha exams, Chunin Exams, Great Ninja War chapters) provides a narrative long-term engagement. Completing chapters could unlock new game modes or characters.

Examples: Idle RPG Wikipedia notes such games often see “very high player retention” due to steady progression and frequent feedback. The Gacha wiki observes that limited banners and events drive retention and marketing (limited-time availability creates urgency). Naruto idle should follow these patterns: e.g. have “character banners” for seasonal village festivals, and special login weeks after big anime releases.

Recommended cadence:

Daily quests (small tasks) reset every day (e.g. do 10 stages, win 3 PvPs).

Weekly events with cumulative rewards (e.g. “Complete 100 stages for an SSR shard”).

Monthly story event unlocking new chapters and an exclusive hero.

Seasonal events tied to holidays or Naruto anniversaries.

Balancing & Pacing

A well-paced idle RPG avoids steep power spikes or stalls:

Power Curve: Early game should feel fast and rewarding. Mid-game slows down to lengthen engagement. Late-game (post-prestige) opens new strata. Use soft caps (e.g. slowdown of stat growth) to make progression feel meaningful. After each prestige, reset power curve so player feels a new ramp. Common trick: diminishing returns on stats at high levels, making additional levels less impactful, then cured by ascending.

Soft Caps: Impose level caps or gear tiers. For example, hero level cap might be 50 until certain upgrade is purchased. Or cap gear rarity at Bronze for a while, then Silver gear unlocks later. These ensure players pivot to other tasks (like ascending or new characters) rather than brute-forcing one hero.

Hard Caps: A true max-level or gear cap can exist (e.g. max level 100). If so, ensure there is next progression path (promote characters, new mode) to keep playing. Idle games usually have effectively no permanent cap, only prestige resets.

Catch-up Mechanics: Implement ways for new or returning players to reach near-parity. Examples: “paywall dust” – free summons if you skip for a week, or friendship summons. Pity systems (guaranteed rare after X pulls) also help mid-spenders. Many idle RPGs (AFK Arena, Idle Heroes) give an initial hero pool (clearing initial campaign grants most heroes) so newbies start with a variety.

Fairness: As noted, avoid “whale traps” late-game. Ensure free players can still access endgame content, maybe at slower pace. Provide occasional tier boosts (like “new player missions” or bonus XP boosters for first week). The reddit guide for Idle Heroes stresses saving resources for events, implying events are intentionally generous catch-up opportunities.

Implementation Notes: Regularly review progression speed via analytics: track how many days on average to reach key milestones (first SSR, first prestige). Adjust drop rates or earnings if the average strays too far from intended. For Naruto idle, consider giving a small buff (e.g. a set % XP boost) to players who haven’t logged in in 7 days, as a friendly catch-up.

UX/UI Considerations for Idle Play

Design UI for glanceability and minimal input:

Progress Indicators: Clearly display offline gains accrued, next rewards threshold, and idle counters. An “Idle Reward” button (like AFK Arena’s “Claim Offline Rewards”) should summarize time played and multiplier from equips/buffs.

Auto-Controls: Provide one-touch auto or manual battle mode. Allow players to queue stages. Visible toggles for speed. If interactive skills exist, place big icons/buttons. Show skill cooldown timers graphically.

Consolidated Upgrades: Bulk upgrade options (e.g. “Level Up 10 Times” or “Max Level Up to 1-star”) reduce tedium. A single hero can have all upgrade options on one screen.

Minimal Clicks: Minimize unnecessary navigation. Idle players log in sporadically; reward immediate gratification. For example, after battle or login, pop a summary of gained loot and a prominent “Upgrade” suggestion.

Performance: As one player review noted, poor optimization (lag) severely hurts user experience. Use lightweight graphics for battles. A Naruto idle should still feel lively (use flashy jutsus in moderation), but mobile-optimized.

Idle-specific UI: Show "Offline Progress" bar. Indicate when game is stopped earning (e.g. cap reached). Mobile push notifications should mention key info ("Offline reward full!").

Inspirations: Many idle games use a main hub screen with buttons for Story, Arena, Guild, Shop, Team, Events, etc. The idle RPG's main UI should show currency balances (soft money, gems) at top, quick access to upgrade menus. Use Naruto-themed art for icons (Kunai icon for challenges, Ramen bowl for stamina, etc).

Analytics & KPIs to Track

To ensure design goals are met and to balance live operations:

Retention Metrics (D1/D7/D30): Essential for all mobile games. Idle RPGs aim high: consistent login rewards and passive progression can boost retention. Track how many return after day 1, 7, 30; target improvements via events or UI tweaks.

Engagement Metrics: Daily Active Users (DAU), session length, sessions per day. Idle games often have short sessions (players check rewards/upgrades 2–3 times a day) so average session might be 3–10 minutes. Monitor if session length drifts too short (unengaged) or too long (complexity or grind too high).

Progression Velocity: Track how many stages or levels players clear per day/week. If average progress stalls (power gap to new content), adjust difficulty or drop rates.

Monetization Metrics: ARPU/ARPPU, conversion rates (percentage who spend), average purchase. Track gacha spending and pitfall (e.g. high spending concentrated among few whales). If whales dominate, consider introducing more mid-tier spend incentives.

Item Flow: Track currency inflow/outflow and inventory. If economy is unbalanced (players hoard too much currency, or can't spend enough), tune costs or sources.

Feature Usage: Measure which features/events players engage with. For example, do most players participate in daily dungeons? Is PvP used? This guides which new content to build.

Churn Triggers: Use funnels to see where players quit. E.g. if new players repeatedly fail a boss or run out of energy, they may leave. Check where big drop-offs occur (UI confusion? insufficient tutorials?).

No direct sources here, but these are industry-standard KPIs. Analytics tools (Firebase, GameAnalytics, etc.) should be integrated.

Anti-Grind and Anti-Pay-to-Win Patterns

Anti-Grind: Idle design inherently reduces grind by automating battles. Further anti-grind strategies:

Automatically Collecting: Let players queue up many fights at once (e.g. "Simulate 100 battles"). Reward players for logging in (collecting 8h of offline gold in one tap).

Catch-up Systems: Offer generous initial boosts: newbies might get a "first-time double XP" potion, or "new player bundle". Weekly milestone rewards for maintaining login can soften the grind.

Event Funneling: Schedule events that reward common resources to meet upgrade demands (as Idle Heroes guide emphasized, events supply the hard-to-get Awakening Crystals, suggesting the game expects players to wait for events).

Reduce Repetition: Include auto-skip cinematics, auto-complete lower difficulty levels.

Anti-Pay-to-Win: To avoid P2W backlash (as AFK Arena reviews note):

Balance Pay vs F2P: Ensure top-tier heroes can be earned without paying (via pity, shop, or events). Provide marginal differences rather than absolute power gaps (e.g. paying might get a unique skin or slight % boost, not a totally OP hero).

Transparency: Clearly show drop rates (regulations often require it), and assure players that persistence or skill (in team building) can compete with spenders.

Whale Protection: Consider "hard cap per character", meaning even spending more can only get so many resources per cycle, to avoid infinite strengthening.

Inclusive Rewards: Every player should occasionally feel rewarded by play, not just spend. This includes free daily/weekly summons and rewarding event attendance (log-in five days to get a mythic shard, etc).

Implementation Notes: Continually survey player sentiment. If many call the game P2W or too grindy (as reddit threads on idle games often do), recalibrate. For example, reduce cost of an upgrade or increase freebies. Balance patches should address any hero or combo that trivializes content.

Legal/IP Considerations

Using the Naruto license means obtaining a formal agreement with IP holders (Masashi Kishimoto, Shueisha, TV Tokyo, Pierrot, and Bandai Namco Entertainment, as credited on the official site). Key points:

Licensing: Acquire rights for character likenesses, story elements, music, etc. This likely involves negotiations with Shueisha/Pierrot and Bandai Namco (who appear on Naruto Mobile's credits). Budget for high licensing fees.

Content Restrictions: Follow Naruto's brand guidelines. Characters must act "in character" (e.g. Naruto can't have a gunslinging modern move out of nowhere). Avoid infringing crossovers (no, say, Dragon Ball cameos).

Age Rating: Naruto contains ninja violence but is generally family-friendly. The game should reflect the same tone (likely a "Teen" or "PG-13" style).

Legal Disclaimer: The game should include copyright attributions (as seen in [31]). Use official art assets or commissions approved by the licensor.

Regional Sensitivities: The reddit discussion pointed out that pricing acceptable in China may offend Western users. Similar caution with content (Naruto is popular globally, but certain story arcs might not be suitable for younger audiences without context).

In summary: Plan on an expensive license and ongoing compliance (e.g. approvals for new characters, storylines). Emulate successful licensed games (Bandai Namco has experience with Naruto games) by contacting rightsholders early.

Comparative Table of Representative Anime Idle RPGs

Game	Core Loop	Monetization	Gacha Rate (top-tier)	Retention Features	Progression Speed	Typical Session Length
AFK Arena	Team collect → auto-battle → earn rewards → upgrades	Premium bundles, battle pass, event packs	~1.0% legend, 5–10% epic	Daily login, 10 free summons/day, guild tech, events	Moderate; first prestige ≈1 week	~5–10 min (very idle-focused)
Idle Heroes	Similar idle loop with offline gains	Premium summon scrolls, VIP packs, events	~1.0% for Transcend 5★	Daily chests, clan bosses, month-long events	Slow midgame; major breaks at events	~5–15 min
Mobile Legends: Adventure	Collect heroes from ML universe → auto-battle	Hero summons, gems shop, BP	~0.3–1.5% for legend	Daily & weekly tasks, guild war, 72-hour battle royale event	Fast early; slows after 2nd reset	~5–20 min
Saint Seiya Awakening	Auto-turn combat with awaken/upgrade heroes	Character draws, gear packs, tickets	~1.5% for top SS	Time-limited banners, daily sign-ins, VIP system	Moderate; punctuated by chapter releases	~10–20 min
One Punch Man: The Strongest	Idle stamina battles → collect and upgrade Saitama-esque heroes	Character draw tickets, coin packs, crates	~0.8% for SSR hero	Daily login, dim sum events, alliances	Fast (1st prestige ~3–5 days)	~5–15 min
Honkai: Star Rail (anime-style RPG)	Manual/auto turn battles, gear and XP → reset+Skyfold	Gacha banners, light cones, supply packs	~0.6% 5★ character	Daily commissions, resonators, logins	Steady (no idle, so slower effective)	~10–30 min

(Gacha rates are illustrative. Features like "idle" vs "manual" affect session length.)

Sources: Gameplay and feature descriptions are based on official game sites and storefronts, Wikipedia on idle/incremental games, and gacha mechanics. Community insights (player guides, reviews) inform best practices. Licensing information from the Naruto official site. All referenced content is up-to-date as of late 2026.

Economy & Summoning Mechanics

In a Naruto-themed idle RPG, use a dual-currency economy (free vs. premium) and robust "gacha" summoning with pity to balance progression. Implement safety nets: for example hard pity guarantees a top-tier (e.g. 5★) hero after a fixed number of pulls. (Soft pity can gradually raise rates as players near the threshold.) Give daily/weekly login rewards and quest achievements that dole out premium currency or free pulls, so even non-paying players steadily accumulate summons. Consider a "spark" system: each pull grants a special token, and collecting enough tokens lets the player directly purchase a featured hero. Make duplicate heroes useful – for example, convert them into hero shards or essence that let players awaken or upgrade other heroes. (In many idle gachas, duplicates auto-upgrade the hero or can be exchanged in a "fusion" system.) This ties together summoning and progression: as players pull more, they both power up existing heroes and unlock new ones.

Free-to-Play Balance & Monetization

To avoid a "pay-to-win" trap, design generous F2P funnels. Provide regular free premium currency (login streaks, event milestones, daily bosses) so free players can meaningfully summon over time. For instance, give at least one free "10x pull" per week from events or a rotating store. Use pity and spark to cap the cost of chasing characters. Limit exclusive advantages to cosmetics or convenience (XP boosters, auto-battle) rather than raw power, so paying mostly accelerates progress without outright gatekeeping content. Veteran players note that many idle games start moderately F2P-friendly but become very P2W after updates; plan updates carefully to keep long-term balance. For example, ensure new heroes are not strictly necessary to beat content, and that story/campaign drops and free-event tokens allow free players to gradually assemble competitive teams. In short, aim for steady but fair progression: paying eases the grind, but free players can still advance and unlock top heroes (just more slowly).

Progression & Upgrade Systems

Each hero should level up (via XP/Gold), rank up (using ascension stones or shards), and equip gear. Awakening (or rank-up) typically requires duplicates or special items: e.g. evolve a 4★ hero to 5★ by consuming 20 shards obtained from pulls. Link this to summoning pity: if duplicates yield shards, then pity rates indirectly affect gear progression as well. Balance costs so that early levels and ranks are easy for free players, but higher tiers require significant investment (encouraging both play and potential spending). For example, you might require 10–20 duplicates to fully awaken a hero to max stars, meaning a free player would need to pull ~100–200 times (possibly spread over months) to guarantee one copy, whereas a payer could hit that faster. Use formulas that grow sub-exponentially rather than runaway-exponentially, so that steady play always feels like progress.

Team Composition & Synergy

In any RPG, a well-rounded team has complementary roles. Define clear roles (e.g. Tank/Defender, Damage Dealer, Healer, Buffer, Crowd-Control) and encourage mixing them. Synergy is key: synergy means "the combined effect [of heroes] is greater than the sum of their individual effects". For example, one hero might buff allies' attack, another shreds enemy armor; together they accomplish more than alone. A balanced team might include:

Tank/Protector: absorbs damage to shield squishies. (e.g. a sturdy Leaf Village shinobi.)

Damage Dealer (DPS): high single-target or AOE damage. (e.g. fire-nature ninjas or lightning users.)

Healer/Support: restores health or shields, keeping the team alive. (A medical ninja like Tsunade.)

Buffer/Debuffer: increases team stats or weakens enemies (e.g. buffing chakra regen or reducing enemy defense).

Control/Utility: stuns, sleeps, or crowd-controls enemies to give allies openings.

A good team isn't just a random set of heroes; it's the right mixture of these roles. For instance, a "Team 7" synergy might have Naruto (aoe chakra burst + buffs), Sasuke (single-target nukes), and Sakura (heals); their combined kit outperforms a random trio. In design, you can also add faction or clan bonuses: e.g. using multiple Leaf Village ninjas grants a bonus (like AFK Arena's faction synergies). Ultimately, the team's strength should grow non-linearly: two heroes that combo well (say one buffs speed and the other explodes on high speed) should outperform two heroes that act completely independently. This team-effectiveness – a classic synergy – means "the whole is greater than the sum of its parts".

System Integration & Design Notes

Currencies: Premium gems should be scarce but supplemented by f2p sources (events, ads, dailies). Gold or common currency is plentiful (used for upgrades).

Duplicates & Awakening: Decide if duplicates auto-upgrade or convert to universal shards. E.g., duplicate Naruto yields Naruto shards, which can be traded for him or used elsewhere at a penalty. This ties awakening costs to summon rates.

Summon Rates & Pity: Choose an SSR drop rate (e.g. 1%) and set a hard pity (e.g. 100–200 pulls). Possibly implement a 50/50 for featured heroes as a higher-tier pity layer.

Progression Pace: Track player power curve. Free players should reach cap endgame in perhaps 6–12 months, paying players in ~1 month, not weeks. Tiered content unlocks (e.g. Chapter bosses requiring 5★ heroes) should be beatable at modest gear by multiple team compositions.

Event Design: Regular events give out character shards or summons. For example, a "Heroic Anniversary" event could give enough shards to unlock a new character at 5★ if collected across time. This rewards loyalty and offsets paywalls.

Balancing Notes: Keep an eye on feedback: games like Mythic Heroes initially felt rewarding to F2P players but grew frustrating with paywalls. Monitor that free rewards (login bonuses, milestone gifts) scale with inflation.

Unresolved Questions / Next Steps

Shard vs Essence System: Will duplicates be used for direct star upgrades or a special currency? (Fusion systems help alleviate RNG frustration.)

Pity Thresholds: Exactly how many pulls to hard-pity? (Industry varies 90–200 for SSR.)

Team Bonuses: What specific synergies (e.g. Naruto clan, Academy, Elemental) will be implemented to encourage themed squads?

Progression Formula: Define XP curve, gear stat growth, and gold sink rate so that both grinders and payers feel steady progress.

Balancing F2P Content: Which endgame bosses or PvP tiers will free teams realistically reach?

Overall, these systems must be designed holistically. Awakening costs affect summoning pace; summoning generosity (pity, shards) impacts player retention and economy health; team roles and buffs influence how much players need to collect. Document each decision and iterate designs (including formulas and thresholds) once we test with target players.

Sources: Core concepts of gacha economy and pity systems are well-documented. Real-player experience (e.g. Mythic Heroes) underscores the dangers of becoming too pay-to-win. Team-synergy principles follow the basic definition of synergy.

Executive Summary

This report analyzes the design of anime-style idle/RPGs (e.g. AFK Arena, Idle Heroes, Mythic Heroes, etc.) with the goal of creating a Naruto-themed idle game that is truly F2P-friendly and rewards synergy. We compare core mechanics across these games, derive concrete formulas/tables for progression and summoning, and propose team-synergy rules (with Naruto-themed examples). We pay special attention to summoning/pity systems, duplicate handling, awakenings, economy/currency flows, and F2P vs P2W monetization. To sustain free players, we recommend generous pity/compensation and robust offline/idle gains, while still retaining engaging pay options (cosmetics, convenience). Key findings include: an event-driven "1000 summons" model (as in AFK Arena's recent Companions update), clear hard-pity thresholds (guaranteed rare hero after N pulls), extensive daily rewards/events, and explicit role synergies (e.g. ninja-village and team-7 buffs) that make team building meaningful. We also outline alternative monetization models (e.g. low-cost subscriptions, battle passes, extra currency sales) balanced with strong free rewards. Finally, we include mermaid diagrams showing system interactions and a player-progression flowchart, plus a "design log" of open questions and proposed A/B tests.

Aspect	AFK Arena (Lilith Games)	Mythic Heroes (IGG)	Idle Heroes (DroidHang)	Naruto Idle (Concept)
Core Loop	Idle offline gain + auto-battles ⇒ Campaign progression ⇒ gather rewards	Similar: idle & AFK battles in Greek/Norse myth settings	Idle training ("Idle Master") and auto-combat	Idle/offline training plus turn-based battles; village quests and boss raids
Summoning/Gacha	Legendarium (universal banner): 10-summon costs diamonds; occasional events give free pulls (e.g. "1000 summons" login). Hero shards fuse (5→1★, 10→2★ etc.) for awakening.	Mythic "Stargazer Summon": Hero shards + gold for pulls. No built-in pity noted, but community "fair SSR rates". Rares (SSR/UR) drop ~1–2%. Excess shards exchange for banner currency.	Heroic Scrolls from event/shop or crystals; no official pity. 5★ heroes can transcendence-ascend after 5 dupes (Spirit). Transcendence "rebirth" gives stat boost. CR shards convert to gold or other shards.	Summon ninja characters by banner (e.g.

Village Summon, Legendary). Hard pity at ~100 draws for top rarity (mirroring common gacha standards). Duplicates → spirit shards or automatic limit-break.

Pity/Guarantee None in base game, but massive login events effectively act as pity (AFK's "10/day" for 1000). Could add: hard pity = SSR after ~80–100 pulls, soft pity gradually increasing odds. No official hard pity reported; player feedback suggests summoning feels "fair for SSR" at start. To ensure fairness: implement hard pity (e.g. 90+1 system). No hard pity (long-time players report low event rewards relative to level). Many players suggest adding a pity or increased free pulls to aid late-game. Use hard pity (e.g. guarantee an S-rank ninja every 100 pulls, with a "spark" currency on each pull for a bonus). Offer daily/weekly free summons (e.g. 2 free 10x pulls/week or accumulating scroll tokens) to ensure F2P progress.

Duplicates/Upgrade Excess 5★ copies → faction bonuses; extra low-rarity auto-sell to tokens. Heroic traces (ether) for breakthrough. 5 copies → 1★; 10→2★; etc. Commonly: 5 duplicate = 1-star upgrade (limit break); extra = artifact/special currency. 5 duplicate = Transcendence (new Hero "reborn" at 1★, more powerful). Extra rank-up stones convert to gold or training XP. Proposed: Shard System: e.g. each SSR hero needs 5 copies to awaken +1★. Additional copies give "S-Rank upgrade stones" to permanently improve stats. Duplicate SR/SSR can convert to "Spirit points" for power-ups. Nurturing a hero could also require generic "training points" earned in quests.

Progression Hero levels 1–160; gear (Lightbearer Crest, etc. up to +20). Power spikes at 30/60/80☆. Campaign stage XP + gold; faction trials unlock. 7 factions give formation buffs. Prestige through faction 'Majesty' resets. Levels up to a cap (e.g. 100); gear tiers with limited upgrade mats; Ascension (limit-break) at level cap using Hero-specific tokens. Many side systems (pet companions, trinkets). Level cap 200; gear Enhancements require gold and shards. Transcendence resets level (prestige) for bonus. Frequent soft-caps (Progress resets like "Outer Ring 2000th floor") force new mechanics. Levels & Gear: Experience from battles (XP can scale ~100×L^1.2) and daily missions. Level cap (e.g. 100) then "Promotion" needing special materials. Skill Upgrades: Karma scrolls (rare currency from dailies) raise ninja skills. Prestige/Ascension: "Inner Peace" system: rebirth at cap resets level to 1 but grants permanent % stat bonus.

Economy & Currency Multiple currencies: Diamonds (premium), Gold (levels), Hero Shards/faction coins (faction trials, Hero union), Ether/Traces (breakthrough), feathers (cosmetics). Inflation managed by sinks (gear costs, Ascension). Premium gems, Gold, Hero Soulstones. Events yield "Relics" or "Gold Pieces" that buy limited equipment. Market/shop rotates. Many paid bundles for currency. Late-game inflation noted in reviews. Premium "Crystals"; common Gold/Glory; Hero Soulstone shards; Guild coins, Arena Medals. High gold inflation – sellers for gold (e.g. purchase gold with cash) and gold boosters exist. Subscription doubles idle gold. Currencies: Premium: Ryo (wind-style currency for summons, sold cheaply in small packs). Regular: Coins (level-up gear), Scrolls (skill-ups), Chakra (rare hero token used in limited shops). Special: Stamps (exchange for specific hero shards in Shop). Sinks: leveling gear/skills, upgrading village HQ (buffs), purchasing NPC tutor boosts. Control inflation via multiple currencies requiring conversion (coins → scrolls, etc.) so no single sink.

Monetization Gacha-centric. Small convenience subs (e.g. AFK Arena's $2.99/month for passive gold bonus). Cosplay skins. Events encourage spending (e.g. gem packs for summons). Classic AFK criticized as Pay-to-Win, but new "Companions" shifted to F2P focus (1000 summons). Standard gacha IAP: gem packs, hero bundles, cosmetic skins. "Starter pack" deals. No heavy subscription model noted; likely whales only competitive in PvP. Late-game dead zone made whales reluctant. Summon scroll packs, gold packs, XP boosts, premium sets. $0.99 weekly "Golden Dragon Avatar" sub for +100% idle gold. Many players say heavy spending accelerates progression but is not strictly required. F2P-First Model: Emphasize small purchases & cosmetics. Options: low-cost "Daily Nindo" ($1/wk for bonus stamina and modest free summons), Battle Pass (rewards including rare shards), and cosmetic skins only via cash or event achievements. Avoid paywalls on core progression. Maximize free gems/runes via logins & achievements. Consider "gacha filler" (e.g. cheap single-summon tickets) and generous pity (hard cap) to reduce frustration.

Retention Daily login (gifts for 1–30 days, culminating in rare hero shards). Recurring events (Labyrinth, Shadow Faction Wars) and timelines. Season battles (PvP ladder resets) to reward active play. Special story chapters and co-op challenges. Weekly/monthly events tied to holidays (Greek festival, etc.). Login calendar grants month-long rewards (coins, gems). Arena/Guild boss rewards. "Treasure trials" mini-game. Daily Quests (Skirmish, Campaign, Arena). Weekly events (e.g. Ura/Valor Compass). Guild Wars, Arena with seasonal resets. Cumulative login (avatars and scrolls). Many complaint (in 2021) that late-game daily rewards didn't scale with level. Daily/Weekly: Login streak bonus (increasing Ryo, summon tokens). Weekly Ninja Exam (challenge tower) with leaderboard. Special Village Festivals (limited-time raids giving gear/rares). Battle Pass track (free and premium tiers with hero shards). Login/Idle: Offline training yields "Will/Spirit" tokens over time, incentivizing daily collection. Avoid grind walls by letting players accumulate offline rewards even when not powerful.

PvP & Matchmaking Ranked Arena with "Combat Rating" matchmaking. Shuffled rosters (ban phase) to avoid stale meta. Guild Wars (real-time co-op boss battles). PvP rewards are mostly cosmetic/leaderboard. Arena with staged tiers; uneven match-ups at high rank. Friendly duels. Limited-time "God Arena" vs themed teams. Random Arena rank with tickets; players advance through leagues. Guild vs guild real-time battles. Many players find higher tiers dominated by whales. Soft reset (Season return currency). Balanced PvP: Use hidden MMR/battle rating to match fairly, not absolute power. Seasonal ladder with rank rewards. Team-ban drafting to force strategy over brute force. Cosmetic rewards (titles, skins). For F2P catch-up: "Weekly Rival" boss battles that let mid-tier players challenge top-tier NPC teams for gems (like challenge tower rewards).

Team Roles/Synergy Heroes have roles (Tank, Warrior, Ranger, Mage, Support) and factions (Lightbearer, etc.) with faction-buffs (e.g. +ATK if ≥3 same faction). Formations (offense/defense frontlines) matter. Synergy examples: e.g. two support heroes can amplify

healing. Character classes (e.g. Guardian, Berserker, Mage). Some synergy combos (e.g. all-Greek heroes get special bonus on certain stage). Requires mix of roles (at least 1 frontline tank, backline DPS). Classes (Fighter, Priest, Mage, Ranger, Support). Each with passive group buffs (e.g. all Fighters +crit chance). Synergy: matching 3 of a class gives team bonus. Role needed: one healer/priest recommended per team. Naruto Roles: e.g. Atacker (Naruto, Sasuke – AOE damage), Tank (Kakashi, Guy – block/damage soak), Healer/Support (Tsunade/Sakura heal or buff, Shikamaru crowd-control). Synergy Rules:

Village Bonds: Naruto characters from same village (Konoha, Suna, etc.) grant each other bonuses (+5% ATK/DEF for 2, +10% for 3).

Team Combos: Specific trios (e.g. Team 7 = Naruto+Sasuke+Sakura) unlock extra effects like a one-time Team Attack (massive damage).

Clan Buffs: Uchiha clan heroes together boost Critical Rate, etc.

Chakra Types: Matching chakra-nature (2+ Fire ninjas = +fire damage).

Role Synergy: Ensure at least one Tank, one Healer, two DPS. E.g.: Naruto (Shadow Clone passive spawns at 50% HP, plus rubidy synergy), Sasuke (crit buff for Uchiha ally), Sakura (group heal+red buff), Kakashi (slash all enemies on death), Gaara (ground ATK to stun).

Example Teams:

Balanced: Naruto, Sakura, Kakashi, Rock Lee, Hinata (Healer/Buff). Roles cover DPS, heal, defense, speed buff.

Susanoo Uchiha Team: Itachi, Sasuke, Obito, Kakashi, Tobi – all Uchiha; they gain a big crit/ATK buff.

Medical Squad: Sakura, Shizune, Tsunade, Kabuto, Healing Jutsu: powerful healing synergy (e.g. combined ultimate).

Madara Clan: Madara, Hashirama, Sasuke, Itachi, Minato – unstoppable but balanced with fewer buffs.

Counters: Teams need edges (e.g. Susanoo Uchiha vs sharingan-proof characters like Pain/Ten) to avoid rock-paper-scissors breakdown. |

Formulas & Tables (Examples). To concretize progression, we propose typical formulas: e.g. XP to next level = $100\times L^{1.2}$, Gold for next level = $50\times L^{1.5}$ (escalating cost). Summon currency: 10 Ryo (premium) for 1 summon; pity triggers guaranteed SSR at 90 summons (hard pity). Heroes gain +10% stat per Promotion (ascension). Duplicate payout: each extra SSR beyond 5 → 1 full awaken star; beyond that, add 10% to spark gauge. For example tables:

Level (L)	XP needed		Gold cost	Cumulative Gold
1→2	100	50		50
2→3	120	71		121
3→4	139	98		219
…	…	…		…
99→100	~6300	~5000		…

(Numbers illustrative; actual tuning may differ.)

Summons		SSR Chance	Guaranteed SSR?	Notes
1–80	1% each	No	Standard rate	
81–89	+0.5% each		Soft pity boost	Up to ~5%
90	N/A	Yes	100% SSR at 90	

Currency Flow: e.g., 10 minions defeated = 1 Hero XP stone; daily quest = 1000 Gold + 5 Chakra Ryo. Inflation control: use multiple sinks (hero gear costs gold and shards; skill upgrades use rare scrolls; prestige resets use a new currency).

Balancing F2P vs P2W (Monetization Models): We outline several options:

Cosmetics-First: Sell skins and optional cosmetics (headbands, costumes) only. All combat power obtainable via play. Pro: Purely skill-based/strategy. Con: Fewer direct revenue if players ignore cosmetics.

Small Bundles & Subscriptions: Like Idle Heroes, offer a cheap weekly sub ($0.99–$2.99) giving +50–100% idle gold and daily bonuses. Include small packs (e.g. “Daily Ryo Bundle” for $0.99 that yields 500 Ryo usable on summons). Pro: steady revenue, low entry. Con: Could slow F2P progress if poorly tuned.

Battle Pass: Offer a season pass (free+premium track). Premium track (e.g. $5/month) unlocks extra shards, currencies. Keep most essential items in free track (e.g. a 4★ hero shard at tier 10, full SSR hero at tier 50). Pro: predictable revenue and rewards; Con: needs continuous content.

Limited Deals: Occasional time-limited package (e.g. $1 for 100 Ryo, $5 for 550 Ryo). Pro: common tactic, players expect it. Con: Must ensure freebies are comparable; risk bad PR if too expensive.

Pros/Cons Table (F2P-Friendly vs P2W):

Model	Pros	Cons
Hard Pity (guarantee)	Reduces frustration; F2P knows maximum spend. Good marketing.	Less chance-driven revenue; whales may feel capped.
Soft Pity (increasing odds)	Encourages players to keep pulling; less guaranteed whales.	Less predictable F2P outcome; still feels RNG.
All-F2P events (like AFK 1000 summons)	Great goodwill; attracts players; kills P2W narrative.	Potential revenue loss if whales reduce spend.
Small Subscription	Reliable monthly income; minimal annoyance.	Could upset hardcore F2P if power too strong (keep buff mild).
Gacha Packs	Target whales; easy to implement.	Widely seen as P2W; must be optional, not mandatory.
In-Game Ads/Videos	Can reward gems for watching ads.	Affects immersion; low revenue.

Team Building & Synergy: A well-crafted team system is crucial. Roles (Tank/DPS/Healer/Support) must complement each other, and thematic "hero synergies" should reward creative composition. In Naruto terms, we envision:

Role Definitions:

Tanks: High defense/HP (e.g. Gaara, Rock Lee, Kakashi when shielded).

DPS (Attackers): High damage (e.g. Naruto, Sasuke, Madara).

Healers/Support: Provide healing or buffs (e.g. Sakura heals, Shikamaru stuns crowd).

Debuffers: (e.g. Kurenai genjutsu reduces enemy accuracy, Haku slows).

Synergy Rules: Use qualifiers (faction/village, signature combos): see table above. For example, any three Konoha ninjas might grant a flat +10% ATK buff to the team. A team with Naruto & Sasuke & Sakura (Team 7) gains a special "Resonance" effect: once per battle unleash a group buff or all-target nuke. Villages act like factions; using Team 8 (Hinata+Kiba+Shino) could give a speed/stealth bonus.

Counters and Balance: Make sure no one team is "always best." For instance, Uchiha-dominant teams might struggle against Yin-Yang Discord counters (Hinata/Neji type healers). A team full of attackers but no healer should be vulnerable to attrition in long battles.

Example Compositions:

Balanced Ninja Team: Naruto (Atk), Sakura (Heal/Support), Kakashi (Tank/DPS), Rock Lee (DPS/Rage), Hinata (Debuff). Covers offense, defense, heal, control. This team would synergize off-season (all Konoha for +stat buff) and use Shuriken/Clone combos.

Sharingan Squad: Itachi, Sasuke, Kakashi, Obito, Kiba. Three Uchiha grants big crit buff; Kakashi's "Double Sharingan" skill triggers extra bonuses. Countered by clones/misdirection (Naruto/Shino).

All-Out Attack: Madara, Hashirama, Sasuke, Kabuto, Deidara. Nearly all heavy hitters. High base damage but very fragile. They'd smash non-tanks but struggle if a healer like Sakura is present (she can outlast them).

Healing Circle: Sakura, Shizune, Tsunade, Tenten, Ino. Focus on healing and buffing each other; little direct damage. Surprising niche: they can out-sustain tougher teams, but fail if enemy one-shots a healer early.

Teams should feel like real ninja squads, not random collections. By giving clear bonuses for Naruto-specific combos (e.g. "Team 7" or "Hokage Council"), we encourage players to experiment with themed squads. We also recommend a UI that highlights known synergies (e.g. tooltip: "Naruto+Sasuke: Ultimate Duo – +5% crit").

Analytics & Telemetry Needs: To tune F2P balance, the game should track engagement vs spend (segmented by free vs paying users), pull rates, pity outcomes, event conversion rates, and Gini coefficient of power distribution. A/B tests could vary pity thresholds or daily resource caps. For example, test whether offering 5 vs 10 daily free summons increases long-term retention. Track which team comps players use and win with most, to adjust power levels or buff synergies.

Design Log & Open Questions:

Pity Level: Hard pity at 90? Soft vs hard? Experiment: some players get guaranteed SSR at 80, others at 100, measure retention.

Duplicate Use: Should 5 or 6 copies awaken a hero? Try both and survey satisfaction.

Offline Gain: How much idle XP/coins should accrue per minute? Too high kills progression; too low bores player. Start at 1x speed (normal mode), upgradable by prestige bonuses.

Energy Stamina: Will we have a spending-limited energy (like 10 tries/campaign per day)? If yes, compensate free by offering a small amount of "free energy refill" coupons via quests. Could A/B test with vs without purchase refill.

Sub vs Battle Pass: Try launching a $1/wk sub (bonus gold) vs a $4-month battle pass; which gives steadier revenue?

Team Synergy Discovery: We might not reveal all synergies upfront—players could experiment or learn on wiki. But ensure at least 5–10 well-known combos (e.g. core Naruto teams) for early game enjoyment.

Balancing: Will Naruto characters have destiny ties (like Naruto & Kurama) or just regular synergy? Perhaps special "bonded" abilities if using exactly the original Team 7. If too complex, maybe skip Kurama integration.

Mermaid Diagrams: Below are illustrative system flows:

Summoning (Ryo → Hero)

Heroes Unlocked

Team Composition

Battles/Quests

Rewards (Gold, Shards, XP)

Currencies Earned

Level/Gear Upgrades

Event Progression

Show code

Player Journey

Win

Lose

Login / Collect Idle Gains

Choose Stage / Battle

Battle Results

Earn XP/Gold/Shards

Streak Bonus?

Upgrade Heroes / Summon Heroes

Stronger Team → Return to B

Check Events / Daily Tasks

Get Extra Rewards → D

Show code

Conclusion: By synthesizing mechanics from top idle RPGs and biasing toward player-friendly design, our Naruto idle game can offer deep strategy and satisfaction without punishing F2P players. Generous summoning (event pity, free pulls) and transparent progression (no pay-locked content) will build goodwill. At the same time, synergistic teams and a rich anime setting (Naruto clans, jutsus) provide the strategy and fan-service to keep both free and paying players engaged. The recommendations above (formulas, tables, designs) should serve as a concrete blueprint for development and tuning, ensuring the final game feels both authentic to Naruto and fair to all players.

Sources: Authoritative sources on idle/RPG mechanics, F2P/gacha design, and the games themselves were used. Key references include official descriptions for AFK Arena, Mythic Heroes, Idle Heroes, as well as gacha system analyses and player reviews.. Full links are listed below.

Source URLs:

AFK Arena – App Store description (Lilith Games)
AFK Arena – Google Play page
AFK Arena Official Site (Lilith Games)
Idle Heroes – App Store description & reviews
Mythic Heroes – Google Play page (IGG)
Gacha game – Wikipedia (pity system, daily login)

Executive Summary

Anime-style idle RPGs (like AFK Arena, Idle Heroes, etc.) rely on team composition and synergy to create depth. Teams typically balance archetypes (tank, physical DPS, magical DPS, healer, buffer, debuffer), and leverage explicit synergy mechanics (faction bonuses, combo triggers, leader buffs, positioning) to reward smart builds. We catalog patterns from top idle RPGs (e.g. Idle Heroes, AFK Arena, Legends of Glory) and show how these inform a Naruto-themed design. Key points: idle RPG heroes fall into classic RPG roles, and synergy often comes from matching factions or enabling combo chains. Cumulative duplicate systems (shards/awakening) plus pity timers shape progression (e.g. 10 daily summons “for a total exceeding 1,000” in AFK Arena). A balanced F2P economy uses generous daily rewards and catch-up mechanics (offline training, free summons, “resonator” or similar) so free players progress steadily without being forced to pay. We outline design options (stat growth formulas, buff-stacking rules, resource requirements) with tables and diagrams. Finally, Naruto-specific team templates illustrate these principles (e.g. combining chakra-users, medical-nin, and taijutsu fighters into cohesive squads). Unresolved issues (ideal buff stacking rules, exact F2P/pity balance) and alternatives (e.g. optional battle passes, cosmetic currencies) are identified.

Roles/Archetypes in Idle RPGs

Idle RPGs generally divide heroes into familiar roles (with synergy-defined archetypes):

Tanks/Frontline Defenders: High HP/defense units that absorb damage and protect allies. (E.g. Gaara-like “Shield” heroes or armored samurai.)
Physical DPS/Bruisers: Melee attackers dealing sustained physical damage (e.g. Naruto as taijutsu DPS).
Magic/Ninjutsu DPS: Ranged or area attackers using chakra or magic (e.g. Sasuke’s lightning jutsu).
Healers/Clerics: Restore team HP and remove debuffs (e.g. Sakura/Tsunade-type medical nin).
Buffers: Grant positive buffs (attack%, crit%, speed) to allies (e.g. Kakashi boosting chakra regen).
Debuffers/Controllers: Inflict negative statuses (stun, slow, defense-down) on enemies (e.g. Shikamaru’s trap to immobilize foes).
Utility/Support: Other roles like “summoners” (e.g. Kabuto summons), disablers, or hybrid fighters.
Each role interlocks: tanks soak damage so squishies survive, healers sustain the tanks, buffers amplify DPS, and debuffers weaken bosses. For example, Idle Heroes advertises “more than 400 Heroes in different factions with specific skills,” encouraging players to cover roles. The chosen roles must complement each other: a team with only DPS and no healer or tank will suffer in sustained fights, while too many healers and no damage leads to slow clears. Good idle-RPG design explicitly defines these roles, often in guild raids or arena where balanced team-building is key.

Synergy Mechanics

Idle RPGs enrich team-building with explicit synergies:

Faction/Alignment Bonuses: Placing heroes from the same faction or theme yields buffs. For example, Legends of Glory notes each hero has “exclusive skills and faction synergies” and urges players to “build the strongest team” by combining them. In a Naruto game, alliances like Konohagakure clan or Akatsuki team could grant collective bonuses when mixed.

Leader Skills & Tags: Some heroes provide passive buffs to teams (e.g. “Konoha Leader: +10% attack to all Konoha ninja”). These tag-based skills reward specific compositions.

Combo Chains/Triggers: Abilities can trigger follow-up actions in others. For instance, one hero’s taunt can force enemies onto a tank, enabling high-DPS allies to focus them. Or a stunlock from one hero lets a second hero land critical hits.

Buff Stacking: Multiple heroes may add the same buff (e.g. two allies each granting +15% attack). The stacking rule (additive vs multiplicative) matters. A simple formula is additive stacking: Total attack buff = BaseAttack × (1 + 0.15 + 0.15) = Base×1.30 if two heroes each give +15%. Alternatively, some games multiply buffs for diminishing returns (final attack = Base×1.15×1.15). Design choice: additive stacking keeps synergies linear, while multiplicative stacking penalizes heavy stacking.

Positional Strategy: Many idle RPGs allow front/back row placement. Example: a front-line tank (Naruto in Sage Mode, Gaara) stands in front, while fragile DPS (Sasuke, Shikamaru) stay back. Some games have row bonuses (e.g. “front row heroes get +X% defense”). This adds strategic depth: e.g. Tsunade is placed back to heal, while Hatake Kakashi stands front to engage.

Tag/Genre Synergy Examples: Although less common, one could include Naruto-themed combos. For instance, a buff like “If Naruto and Sasuke are both on team, grant +20% group attack for 6 seconds” (hypothetical). These special combos encourage collecting related heroes.

In sum, strong synergy ties units together. As one example, Legends of Glory’s marketing explicitly advises summoning and team-building to exploit faction synergies. We adopt a similar approach: define clans or chakra-nature sets in Naruto, and give bonus effects when multiple clan members appear.

Counterplay and Balance (Rock–Paper–Scissors)

Balanced idle RPGs include counterplay so no single archetype dominates. Common patterns:

Faction or Elemental Counters: e.g. Class Advantage: Warriors > Mages > Rogues > Warriors, or Chakra Nature triangles (Water beats Fire, etc.). AFK Arena uses a Light–Dark–Beast cycle internally (Light defeats Hypogean, Hypogean defeats Mauler, etc.), enforcing a simple rock-paper-scissors dynamic. We should implement a similar loop (perhaps Taijutsu beats Ninjutsu beats Genjutsu), ensuring every team has a weakness. This prevents “always use Team 7 no matter what.”

Diverse Counters: Tanks and high-DEF units counter pure glass-cannon teams, while multi-hit AoE DPS counters teams of isolated squishies. E.g. Gaara’s shield nullifies one big hit, so stacking a monster boss might fail against him. If boss can stun backline, having a healer or cleanse (like Ino) is needed. This encourages players to switch or rotate teams versus different challenges.

Counters should be clear but not absolute. We can include a simple visible marker (e.g. “fire vs water vs wind”) or implicitly via skills.

Duplication, Awakening, and Summoning Pity

Idle RPG collection revolves around duplicates. Key design elements:

Awakening/Ascension: Duplicate heroes are typically used to rank-up or awaken a hero. For example, Idle Heroes lets you “convert [warriors] into Spirit material for evolving” stronger heroes. In a Naruto game, duplicates might become scroll fragments or “chakra stones” needed for 5★→6★ ascension, unlocking new skills. Balancing question: How many duplicates to need per star-level? A typical formula is something like:

mathematica

Copy

Hero Star Upgrade: require N shards where N = base_shards * (star_level)^2.

For instance, 4★→5★ might need 100 shards, 5★→6★ 150 shards (higher cost as power jumps). We provide a sample table below.

Pity and Summoning: Gacha systems should include pity to avoid endless bad luck. A design example: AFK Arena’s “Get All Heroes” event gave 10 free summons daily, up to 1,000 total, explicitly stating “No more diamond worries. Just summon to your heart’s content!”. This showed extreme generosity. In general, consider mechanics like:

A guaranteed 5★ hero every 80-100 pulls (pity counter resets on 5★ draw).

Periodic free multi-summons (e.g. monthly event gives 10 free ten-pulls).

Use duplicate pulls to grant pity points (e.g. every spare drops adds +1 to a pity meter).

Duplicate Value: Not all duplicates need to be full waste. Options: feed duplicates for XP, convert to shards, or use in a "heroic salvage" system. The Idle Heroes resonator idea (mentioned in user reviews) lets players transfer levels between heroes, effectively repaying some cost of building weaker copies. We could include a "reincarnation" feature letting players refund part of a duplicate's XP/gold if they switch builds.

Summoning Currency Economy: Manage the flow of premium currency (gems) vs free currency. Incorporate daily logins, chest rewards, and ad-watches to earn pulls. For example, the AFK Arena description promises "10 summons every day" in an event. Even if not daily long-term, aim for generous "free-to-play-friendly" summon opportunities.

Overall, define clear formulas. For instance:

Example Pity Formula: Let P be base 5★ pull rate (say 0.5%). Guarantee a 5★ by 100 pulls by tracking pity += 1-(1-P). At 100 pulls force a 5★ if none occurred.

Awakening Costs (sample):

Star Level	Fragments Needed	Gold Needed	Notes
4★→5★	100 shards	200,000 yen (gold)	Unlock level cap +10; 5★ skill
5★→6★	150 shards	500,000	+Another cap; elite skin (Naruto Sage Mode)
6★ (max)	—	—	—

(Naruto example: shards themed like "Tailed Beast Chakra")

These numbers can be tuned, but illustrate that higher ascensions cost more.

Progression Pacing & F2P Balance

Making a generous economy keeps players engaged. Design patterns:

Idle Rewards: Continuously give offline XP/gold. For example, Idle Heroes' tagline is "Idle System – Set your heroes training while you're away... they will be stronger". Implement auto-battle rewards and periodic offline chests so progress isn't stalled.

Daily/Weekly Quests & Events: Provide routine tasks granting summons or high-value items. E.g. AFK Arena's daily/weekly missions should always be fully completable by F2P. Special limited events (festival draws, login events) can give bonus shards/summons. The AFK Arena "Companions" event giving 1,000+ summons is an extreme case; smaller versions (e.g. 50-100 free pulls) each month could be sustainable.

Catch-up Mechanisms: Ensure old players can catch up with newbies. Ideas: VIP-less "reincarnation" as above, or treasure that converts old currency to new, special "Resonator" resets, or cooperative play bonuses (e.g. guild shop discounts) that help F2P scale over time.

Difficulty Curve: Pace content so normal F2P gear can clear it without spending. Excessive paywall spikes (as one AFK review noted) erode trust. We must avoid steep "difficulty walls" that force spending. Level caps and gear tiers should be placed so that reasonably focused free play can eventually break them.

Monetization Focus: Pivot pay options toward pay-for-convenience/cosmetics, not raw power. Subscriptions or battle passes (giving small VIP-like bonuses but still accessible to F2P) can work. For example, a weekly sub might give +10% idle gold, but not skip critical progression. Avoid gating core heroes behind pay-only (instead use pity or guaranteed event heroes).

Reward Scale: Scale rewards with player level. (Idle Heroes players complained late-game rewards felt like early-game; solve by ensure activities grow in value).

PvP Matchmaking and Fairness

In PvP (Arena, ranked), match teams by power or tier. Anti-pay-to-win measures include:

Separate Leagues: Whales face whales in upper tiers, free players in lower tiers.

Roster Restrictions: E.g., only heroes up to 5★ allowed in beginner PvP, unlocking 6★ in higher divisions.

Normalization Events: Some games offer a “light” PvP mode where stats are normalized to level or gear, focusing on deck composition.

Match Decay: If a player goes inactive, gradually match them with weaker opponents.

While detailed PvP design is beyond scope, any scheme that lets skill or strategy matter more than who paid wins is key (e.g. AFK Arena’s King’s Tower often matches by Combat Power).

Design Patterns from Top Idle RPGs

Idle Heroes (DHGames): Advertises “Idle system” and 400+ faction-based heroes. Rich guild/arena content means players form balanced parties to tackle many modes. It uses spirit shards and evolution for duplicates.

AFK Arena (Lilith Games): Emphasizes passive progress (“AFK farming”) with automated battles. It has seven factions and a rock-paper-scissors counter system (noted across the web). AFK Arena has events like “Get All Heroes” ensuring free summons and a hero-level sharing “resonator” system.

Legends of Glory (Idle Arena): Features an AFK idle model (“Even while offline, your heroes generate gold and experience”) and calls out “Strategic Team Building” with “exclusive skills and faction synergies”. It suggests a large roster where combining heroes is crucial.

Other Anime Idle Games (Naruto/One Piece titles): While specifics vary, many mimic these designs: turn-based auto-battles, story-based squads (forming iconic teams), and currency funnels into character stars. Official sources (e.g. App Store pages) often tout hundreds of characters and guild battles – reinforcing the need for balanced team roles.

From these patterns, we adopt: multiple currencies (XP/gold/shards), multi-step hero growth (level→ascend→awaken), and diverse game modes (campaign, arena, guild) requiring different team builds.

Naruto-Themed Team Examples

Below are illustrative Naruto squads showing roles and synergy. Each team covers key roles (Tank, DPS, Healer, Support, CC) with a thematic twist. (These are conceptual examples.)

Stage Composition (Roles) Example Heroes (Naruto) Strategy/Synergy

Early (Level 1–30) 1 DPS, 1 Healer, 1 Support Naruto (DPS) – wind-nature attack
Sakura (Healer) – medical nin
Kakashi (Support) – attack buff/utility Basic Team 7 archetype: Naruto deals damage, Sakura heals, Kakashi provides minor buffs or AoE. Covers offense, sustain, and a bit of speed buff.

Mid (Level 30–60) 1 Tank, 2 DPS, 1 Healer, 1 Support Gaara (Tank) – sand shield, high DEF
Sasuke (DPS) – lightning Rasengan
Naruto (DPS) – Rasenshuriken
Sakura (Healer) – big heal
Shikamaru (Support) – bind/debuff Split focus: Gaara soaks hits; Naruto/Sasuke provide high burst; Shikamaru immobilizes or stat-drains enemies; Sakura keeps team alive. Strong synergy: e.g. a stun from Shikamaru allows Naruto to unleash a boosted jutsu.

Late (Level 60+, Endgame) 2 Tanks, 2 DPS, 1 Healer, 1 Support Hashirama (Tank) – wood shield + regen
Madara (Tank) – wood/technical resilience
Naruto (DPS) – Six Paths Mode
Sasuke (DPS) – Rinnegan techniques
Tsunade (Healer) – massive heal
Shikamaru (Support) – combo debuffs High synergy Kage team: Double tanks upfront; Naruto/Sasuke with ultimate Ninjutsu and synergy (legendary Team 7 power); Tsunade’s clutch heals; Shikamaru’s Nara clan combos. This team can execute combined attacks (speculative), e.g. Naruto’s Rasenshuriken followed by Sasuke’s Amaterasu for huge damage.

Each example covers: Tank(s) (Gaara/Hashirama/Madara), Primary DPS (Naruto, Sasuke), Healer (Sakura/Tsunade), Support/Debuff (Kakashi/Shikamaru). We see a progression: early focus on Team 7 trio, mid adding more crowd-control, late forming a full “Hidden Leaf Kage” squad. These roles interlock: tanks protect healers/DPS; healers keep tanks alive; buffers (Shikamaru) amplify damage. (In a final game design, one might add explicit bonuses – e.g., a “Konoha Legacy” passive if Naruto and Hashirama both in team.)

Formulas & Technical Specs

Stat Scaling: We can set a formula for hero stats. For instance:

HP(L, S) = BaseHP × [1 + 0.02×(L-1)] × [1 + 0.10×(S-1)]

where L = level (1–max per star) and S = star tier (1★ base to 6★ max). (Thus each level adds 2% of base, and each star adds 10% extra.) Similar formulas apply for Attack/Defense. This ensures roughly linear growth with diminishing returns at higher tiers.

Buff Stacking Rule: We choose additive stacking of like buffs (to keep it simple):

FinalStat = BaseStat × (1 + Σbuffs).

E.g., two buffs +20% and +15% yield a total 1.35× multiplier. Distinct types (attack% and speed%) just multiply separately by their sum.

Pity Progression: If a 5★ hero has base pull chance p per draw, guaranteed pity at N draws solves for p such that 1-(1-p)^N = 1. For example, guarantee at 100 pulls means p = 0.058, but we can instead just hard-stop after N draws. A friendly option: grant 5★ at 90th pull automatically.

Level Up XP/Gold: Example progression (Naruto game): leveling a hero from 1→60 might require around 200,000 XP and 300,000 gold (per 1→60, inclusive). We can tabulate breakpoints (every 10 levels) for clarity:

Level Range	Total XP Required	Total Gold Required
1→10	5,000	10,000
11→20	15,000	30,000
21→30	30,000	60,000
31→40	60,000	120,000
41→50	100,000	200,000
51→60	200,000	300,000

(These are illustrative; actual values scale with hero rarity and intended grind.)

Upgrade/Ascension Example Table:

Ascend Tier	Hero Rarity	Required Fragments		Gold	Level Cap	Effect
4★→5★	Epic	100 “Chakra Shards”		200k	+10 (→Lv50)	+1 Skill/Passive
5★→6★	Legendary	150 “Shards”		500k	+10 (→Lv60)	+Major Ult Skill
6★ (max)	Legendary	—	—	60	—	

This exemplifies how duplicates (fragments) are required to unlock the next star-tier.

Battle Formula Example: Suppose Hero A has base attack 1,000 and Hero B has +20% attack buff and Hero C adds +15%. Final attack of a damage dealer = 1000×(1+0.20+0.15) = 1,350. A tank’s Taunt might have a formula like: Threat = 100 + 0.5×HP to draw aggro. We would specify such formulas in final design docs.

Design Alternatives & Decisions

Buff Rules: We must choose additive vs. multiplicative stacking. A possible alternative is to cap stacking (e.g., max +100% total), but we lean to unlimited additive stacking for simplicity unless playtesting shows abuse.

Pity System: Instead of an exact pity count, another option is a “pity currency”: each 10× summon without rare grants a token; 10 tokens = guaranteed legendary. This is user-friendly and avoids hard counters. Still, we recommend a clear pity counter (as players like transparent goals).

Currency Inflation: Deciding how much free vs. paid currency to give is key. We resolved to lean generous (e.g., daily free 10× pulls as in AFK Arena’s event) because excessive grind frustrates and kills retention. An open question: what daily/hourly logins are sustainable? (We might A/B test: 2×10-pull per month vs. 5×, etc.)

Progression Curve: Should heroes “diminish” after a point (like Idle Heroes’ Transcendence)? Idle Heroes uses a hyper-ascension (Transcend) where elite heroes get even stronger benefits. We could adopt a Ninja Ascension late-game upgrade for vets, but keep base progression flatter to avoid hitting walls early.

Matchmaking: Simplest is Combat Power brackets. More sophisticated is Elo/MMR, but that risks “rank faking.” A hybrid: tiered arenas (Bronze, Silver, Gold, etc.) with CP caps.

Free-to-Play Focus: We prioritize pay-for-progress (convenience) rather than must-pay-to-win. For example, selling XP boosts or currency packs shortens grind, but all heroes remain achievable by patience. Alternate model: cosmetics store (special Ninja skins). This aligns with best practices: let whales excel slightly faster, but do not lock endgame heroes behind a paywall.

Mermaid Diagrams

Team-Role Relationship Diagram:

Protects

Absorbs

DependentOn

HelpedBy

Supports

Sustains

Weakens

Enhances

Coordinates

Tank

DPS

Debuffer

Support

Buff

Healer

Opponent

Buffer

Show code

This illustrates typical role interplay: Tanks protect others; Healers sustain everyone; Buffers aid DPS; Debuffers weaken foes to synergize with DPS.

Progression Flow Diagram:

Summon Hero (Gacha)

Hero (Base)

Level Up (XP/Gold)

Ascend (★ Upgrade via Shards)

Awaken (Final Tier)

Hero Duplicate

Collect Hero Shards

Transcendence (Optional)

Grind/Afk Progress

Show code

This flow shows a hero's growth path: summoned heroes gain XP to level up, use shards (from duplicates) to ascend stars, then optionally reach a transcendence stage. Offline/AFK grind feeds all stages.

Recommended Decisions (F2P-Friendly)

Generous Free Summons & Pity: Implement daily free summons and a clear pity system (e.g. guarantee 5★ by ~100 pulls) as AFK Arena did with 1,000 free summons. This greatly aids retention.

Idle Rewards: Use constant passive gain; ensure an idle baseline that even F2P players make visible progress every login (similar to Idle Heroes' model).

Balanced Difficulty: Tune campaign/raid so a moderately powered F2P team can eventually clear it. Avoid sudden difficulty spikes unless optional.

Alternate Currencies: Include multiple soft currencies (XP, gold, shards) and ensure many earnable sources (quests, shops, events) so the economy isn't bottlenecked by pay-only currency.

Free/paid cosmetic content: Skins or renames of heroes can monetize without affecting balance.

Ongoing Events: Regular time-limited events with leaderboard rewards (e.g. "Defeat the Nine-Tails – top damage get goodies") engage players. Ensure even low ranks get modest rewards.

Unresolved Questions and Considerations

Exact Buff Formula: Should buffs be fully additive or multiplicative? We plan additive for predictability, but need to test if stacking two 50% buffs is overkill. Perhaps cap buffs or use diminishing returns.

Shard Drop Rates: How many fragments to drop from stages/events to feel fair? Need ratio so collecting 100 shards over weeks isn't too grindy.

Team Synergy Limits: How many buffs should stack before hitting a diminishing return (to prevent infinite loops)? Possibly limit one stack per buff type.

Paywall Placement: Determining which heroes or gear should be earnable vs premium-exclusive. We lean earnable for all base heroes, with some special exclusives (e.g. a Legacy skin) for paying.

PvP Incentives: Should rank resets or seasonal rewards be generous enough that casual players don't feel obligated to climb every season (to avoid P2W pressure)?

Sources

Idle Heroes (DHGames) — Official App Store page (game description).

AFK Arena (Lilith Games) — Official website and App Store content.

Legends of Glory: Idle Arena — Google Play Store (developer description).

User reviews and forums (e.g. Reddit) informed F2P concerns (cited for context but not used directly in analysis).

--- END VERBATIM SOURCE TEXT ---

Appendix G. Live-master maintenance protocol

The single live master is /Game Design/Naruto_Idle_RPG_Master_Design_Bible.docx. Source retrieved from verified Library version 36 / document v0.10.31. Research revision v0.10.32 adds Section 59 (Priority 5C): transaction uncertainty, accessibility states and first-loss/returning recovery without changing economy or combat. Next proposed packet is Priority 5D with HIGH recommended. All research is paper-only; no app, server, seeded simulation, device/user test or compliance certification. On save, verify the actual committed Library version and the attachment bytes.

For the Section 23 handoff, choose the lowest level that can cover the question thoroughly. Use light for a focused check, medium for one bounded comparison, high for a question connecting several systems, extrahigh for reconciling conflicting rules across many cases, and max for a whole-project audit or foundational redesign.

On each revision, verify that the previous master’s full body and appendices remain present, compare source archive checksums where available, update the version and change log, and render the DOCX to inspect page layout. Mark requests that only produced a start confirmation or a rate-limit response as pending rather than completed research.

Version 0.6 change record — 28 October 2026: Expanded the combat engine into a closure specification; filled the status-effect and archetype sections; added role budget, exact initiative/Chakra/targeting rules, damage/healing formulas, bounded synergy and clone/follow-up rules, 10 sample squads, accessible counter matrix, mobile readability criteria, simulation acceptance gates, updated current references and deferred progression add-ons/game-mode scope until the combat exit gate is met. Final layout QA: changed cross-section numbered lists to locally numbered text, set new table rows to avoid splitting across pages, and replaced the static footer number with a dynamic PAGE field.